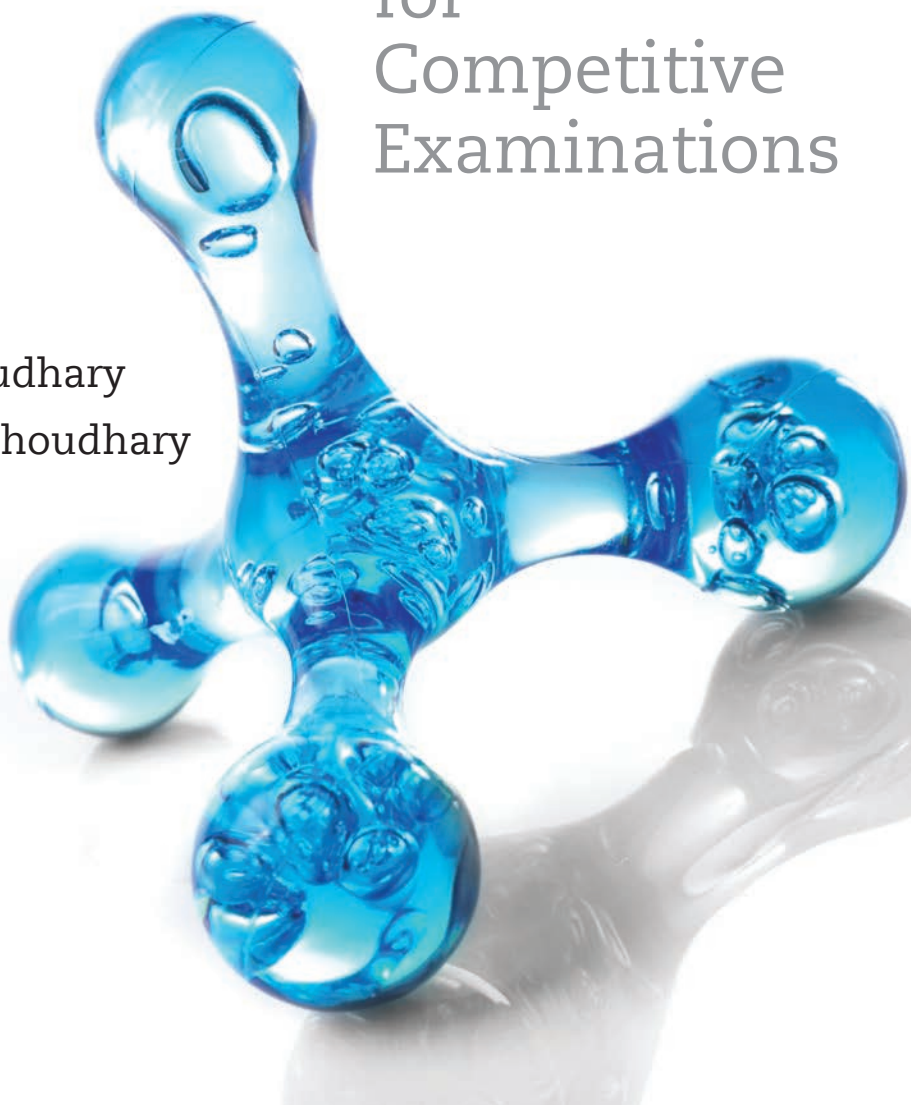


Advanced Problems in

ORGANIC CHEMISTRY

for
Competitive
Examinations

Akshay Choudhary
Mandakini Choudhary



Advanced Problems in Organic Chemistry

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Competitive Examinations**

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Competitive Examinations**

Akshay Choudhary
Mandakini Choudhary

PEARSON

Delhi • Chennai

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Dedicated to my Parents
Shri Om Prakash Choudhary
and
Smt. Sita Devi

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Preface

In order to understand a subject, small concept plays a significant role. Theory is only beneficial when it has been successfully applicable to solve problems. To clear each and every concept, different types of problems are enumerated herein. So, there remain no doubts or misconception rather a strong hold over the subject is developed. Following the famous saying, 'practice makes a man perfect,' we are presenting this problem practice book to clear the conceptual basics of the subject where a systematic arrangement of practicing problem is done.

This book includes all level of problems in organic chemistry that every JEE aspirant needed. Lastly, not to say more, but the book itself signifies its name. Hoping this book will help the students to achieve their targets.

Readers may send their worthwhile suggestions at: akshaymandakini24@gmail.com.

Akshay Choudhary
Mandakini Choudhary

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Acknowledgements

The editorial team at Pearson Education has been pivotal in pushing the project, while my wife has very ably and diligently done the proof-reading of the complete manuscript. I am indebted to my family members (Om Prakash Choudhary, Sita Devi, Hemant Choudhary and Kamla Choudhary) and my uncle (Shyam Lal Choudhary) has played a prime role in making me more focused. I am very grateful to my daughter (Sagarika) and son (Aatish) whose joyful faces give me patience to work.

I am very thankful to my friends, P.B. Saxena, P.C. Reddy, M.S. Chauhan, Ramashis Paul, Sunil Jangid, S. Kothari, S.K. Mishra, Navneet Jethwani, V. Sharma, Kumud Ranjan, Jitendra Chandwani, Yogesh Malav, Deepak Chaturvedi, Rakesh Sharma, Piyush Meshawari, Rajesh Kumar, Manoj Agarwal and Subkaran Choudhary for their valuable feedback and suggestions.

Suggestions, queries, and criticisms will always be welcomed from one and all to improve the quality of this book.

Akshay Choudhary

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About the Authors



Akshay Choudhary, a renowned faculty of organic chemistry, is offering his service to IIT-JEE aspirants to achieve their goal. Many students received success under his guidance and achieved top ranks in IIT-JEE exam. The author is M.Sc. in chemistry as well as JRF from NCL, Pune. He is NET and GATE qualified also.

His keen interest in the subject and continuous efforts have been successfully elaborated in the form of this book.



Mandakini Choudhary, B.Sc, B.Ed—the co-author of this book—has four years of teaching experience in some renowned schools. She has special command on reaction mechanism and intermediates. Throughout her teaching career, she has helped a number of students to succeed in the board examinations.

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General Organic Chemistry

1

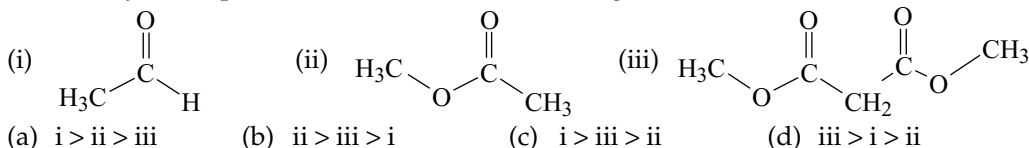
Question Bank

LEVEL 1

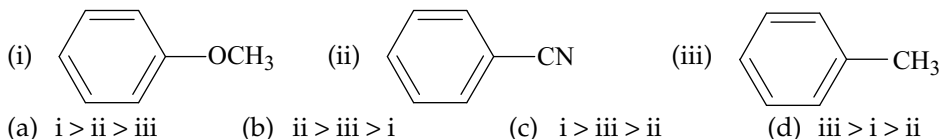
Arrange the items in Questions 1–38 in **DECREASING ORDER** (i.e., greatest, most etc. first) with respect to the indicated property.

Use the following code to indicate your answers.

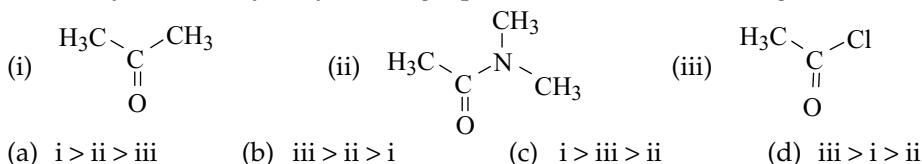
1. The acidity of the protons H in each of the following is



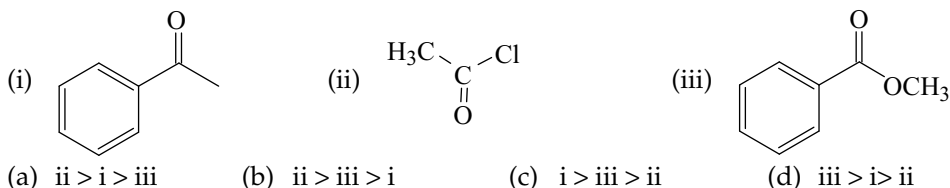
2. Rate of reaction of $\text{HNO}_3/\text{H}_2\text{SO}_4$ with each of the following is



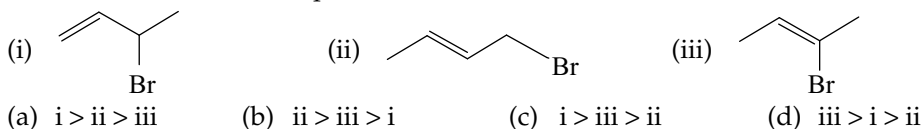
3. Reactivity towards hydrolysis using aqueous acid of the following is



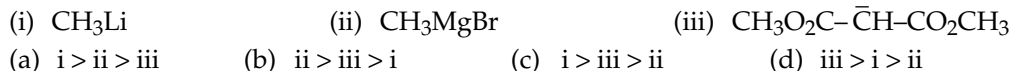
4. Reactivity of the following towards reaction with LiAlH_4 is



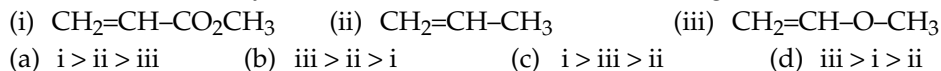
5. The relative yield of the following alkenyl bromides from the reaction of 1,3-butadiene with HBr (dark, N_2 atmosphere) at -15°C is



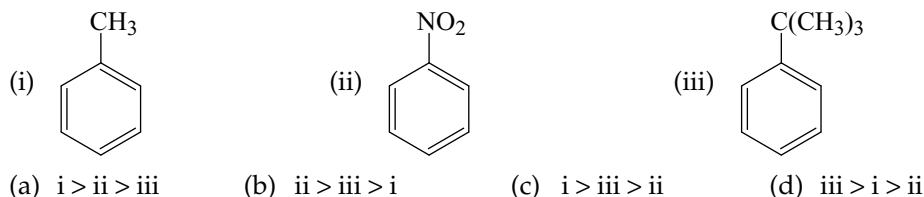
6. The amount of conjugate addition obtained in the reaction of the following with 3-butenone is



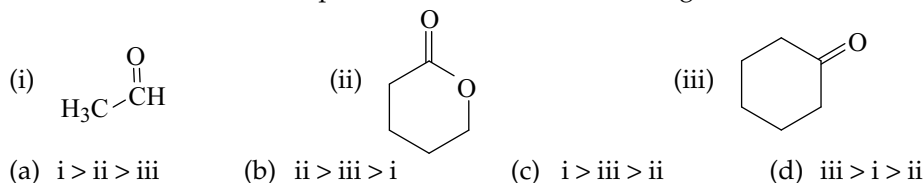
7. The relative reactivity towards Br_2 in CHCl_3 of the following is



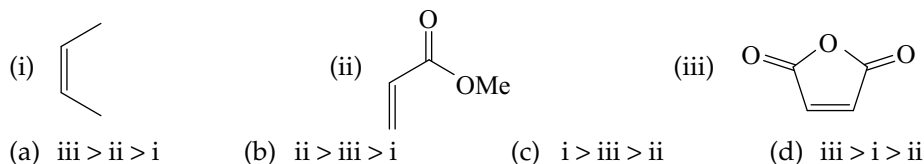
8. The % of the para product produced in the reaction of $\text{Br}_2/\text{FeBr}_3$ with each of the following is



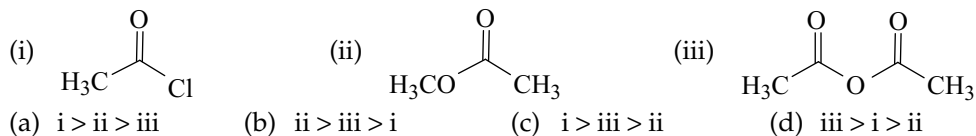
9. The number of enolizable protons in each of the following is



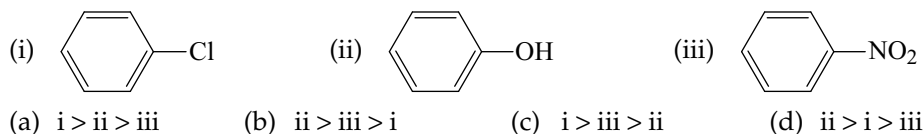
10. The relative reactivity towards 1,3-cyclopentadiene of each of the following is



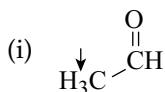
11. The relative rate of hydrolysis using dilute aq. NaOH of the following is



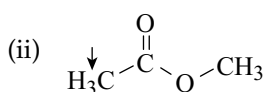
12. The relative rate of reaction of isopropyl chloride/ AlCl_3 with each of the following is



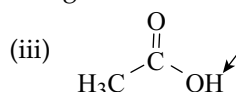
13. The relative acidity of the indicated H in each of the following is



(a) $i > ii > iii$



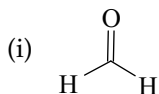
(b) $ii > iii > i$



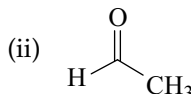
(c) $i > iii > ii$

(d) $iii > i > ii$

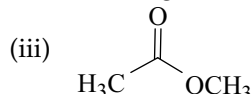
14. The relative reactivity towards reaction with MeMgBr of the following is



(a) $i > ii > iii$



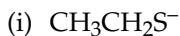
(b) $ii > iii > i$



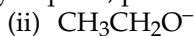
(c) $i > iii > ii$

(d) $iii > i > ii$

15. The relative nucleophilicity in polar, protic solvents of the following is



(a) $i > ii > iii$



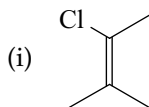
(b) $ii > iii > i$



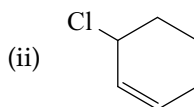
(c) $i > iii > ii$

(d) $iii > i > ii$

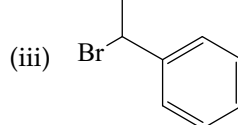
16. The relative rate of reaction with aq. EtOH/AgNO₃ of the following is



(a) $i > ii > iii$



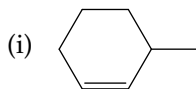
(b) $iii > ii > i$



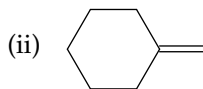
(c) $i > iii > ii$

(d) $iii > i > ii$

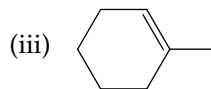
17. The relative yield of the following products produced by the reaction of conc. H₂SO₄ with 1-methylcyclohexanol is



(a) $i > ii > iii$



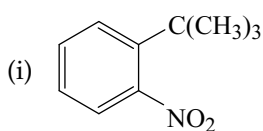
(b) $ii > iii > i$



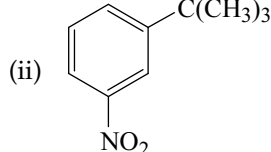
(c) $iii > ii > i$

(d) $iii > i > ii$

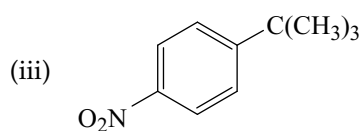
18. The relative yield of the following products produced in the nitration reaction of t-butylbenzene is



(a) $i > ii > iii$



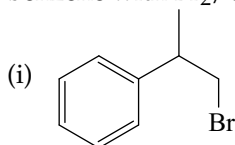
(b) $ii > iii > i$



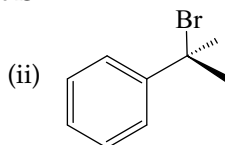
(c) $i > iii > ii$

(d) $iii > i > ii$

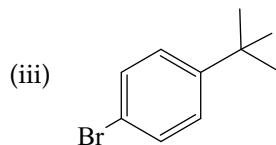
19. The relative yield of the following products produced by the reaction of isopropylbenzene with Br₂/UV light is



(a) $i > ii > iii$



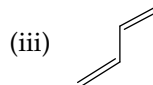
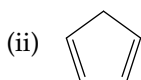
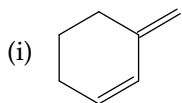
(b) $ii > i > iii$



(c) $i > iii > ii$

(d) $iii > i > ii$

20. The relative reactivity towards dimethyl cis-butendioate (also called dimethyl maleate) of the following is



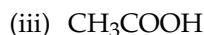
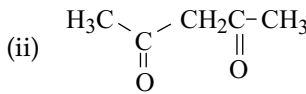
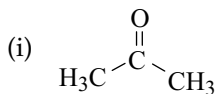
(a) $i > ii > iii$

(b) $ii > iii > i$

(c) $i > iii > ii$

(d) $iii > i > ii$

21. The acidity of the protons H in each of the following is



(a) $iii > ii > i$

(b) $ii > iii > i$

(c) $i > iii > ii$

(d) $iii > i > ii$

22. The relative nucleophilicity in polar protic solvents of each of the following is



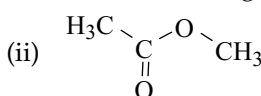
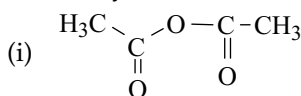
(a) $i > ii > iii$

(b) $ii > iii > i$

(c) $i > iii > ii$

(d) $iii > i > ii$

23. Reactivity towards NH_3 of each of the following is



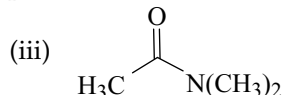
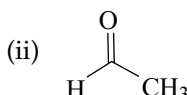
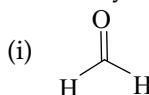
(a) $i > ii > iii$

(b) $ii > iii > i$

(c) $i > iii > ii$

(d) $iii > i > ii$

24. Reactivity of the following towards reaction with NaBH_4 is



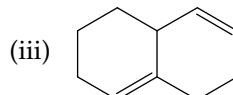
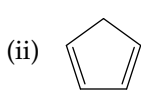
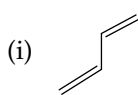
(a) $i > ii > iii$

(b) $ii > iii > i$

(c) $i > iii > ii$

(d) $iii > i > ii$

25. The relative reactivity towards 1-buten-3-one of each of the following is



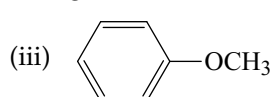
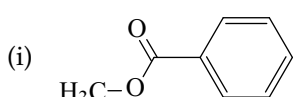
(a) $i > ii > iii$

(b) $ii > iii > i$

(c) $i > iii > ii$

(d) $ii > i > iii$

26. Rate of reaction of $\text{CH}_3\text{COCl}/\text{AlCl}_3$ with each of the following is



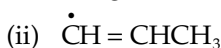
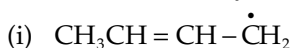
(a) $iii > ii > i$

(b) $ii > iii > i$

(c) $i > iii > ii$

(d) $iii > i > ii$

27. The relative stability of the following radicals is



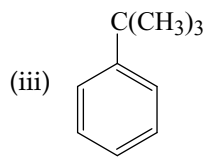
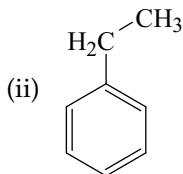
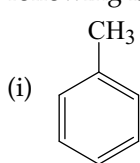
(a) $i > ii > iii$

(b) $ii > iii > i$

(c) $i > iii > ii$

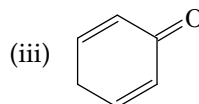
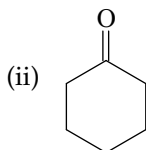
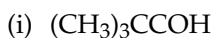
(d) $iii > i > ii$

28. The ortho/para product ratio produced in the reaction of $\text{Br}_2/\text{FeBr}_3$ with each of the following is



- (a) $i > ii > iii$ (b) $ii > iii > i$ (c) $i > iii > ii$ (d) $iii > i > ii$

29. The amount of the enol form present at equilibrium for each of the following is



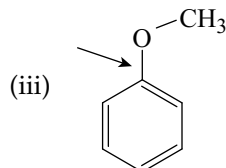
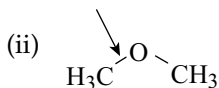
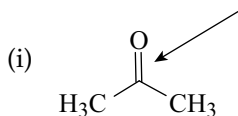
- (a) $i > ii > iii$ (b) $ii > iii > i$ (c) $i > iii > ii$ (d) $iii > ii > i$

30. The relative yield of the following alkenes produced by the reaction of trans-1-chloro-2-methylcyclohexane with KOH/heat is



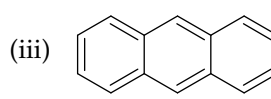
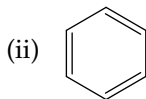
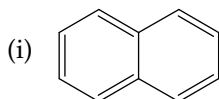
- (a) $i > ii > iii$ (b) $ii > iii > i$ (c) $i > iii > ii$ (d) $iii > i > ii$

31. Identify correct C–O bond length order



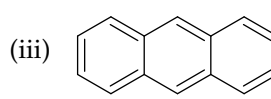
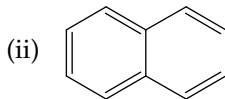
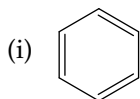
- (a) $i > ii > iii$ (b) $ii > iii > i$ (c) $i > iii > ii$ (d) $iii > i > ii$

32. The resonance energy of each of the following is



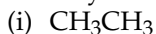
- (a) $i > ii > iii$ (b) $ii > iii > i$ (c) $i > iii > ii$ (d) $iii > i > ii$

33. Identify order of per ring resonance energies of each of the following



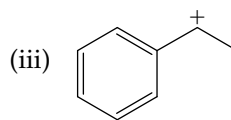
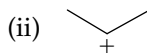
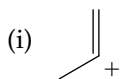
- (a) $i > ii > iii$ (b) $ii > iii > i$ (c) $i > iii > ii$ (d) $iii > ii > i$

34. Identify the correct boiling point order of each of the following



- (a) $i > ii > iii$ (b) $ii > iii > i$ (c) $i > iii > ii$ (d) $iii > ii > i$

35. The relative stability of the following carbocations is



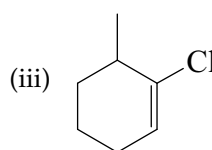
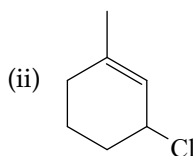
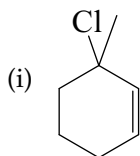
(a) $i > ii > iii$

(b) $ii > iii > i$

(c) $i > iii > ii$

(d) $iii > ii > i$

36. The relative yield of the following products from the reaction of HCl with 1-methyl-1,3-cyclohexadiene at 50°C is



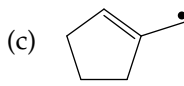
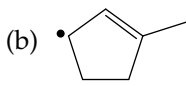
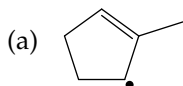
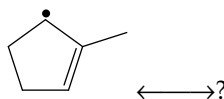
(a) $i > ii > iii$

(b) $ii > i > iii$

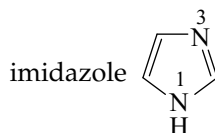
(c) $i > iii > ii$

(d) $iii > ii > i$

37. Which of the following systems are resonance contributors of the radical shown below? (select all that apply)



38. Imidazole has a $pK_a = 7$ with respect to its conjugate acid. Which N is protonated in this conjugate acid and why?



- (a) N1 because imidazole is an aromatic heterocycle where $n = 1$ as per Huckel's rule.
- (b) N1 is protonated because it is sp^3 hybridised.
- (c) N3 is protonated because it is sp^2 hybridised.
- (d) N1 is protonated because the lone pair is part of the aromatic π system.

39. Cyclopentadiene has a $pK_a = 15$, whereas cyclopentane has a $pK_a > 50$. This is because

- (a) Cyclopentadiene is particularly unstable.
- (b) Cyclopentane contains no lone pairs.
- (c) Cyclopentadiene is a 4π anti-aromatic compound.
- (d) Cyclopentadiene is a 4π non-aromatic compound and after deprotonation it is aromatic.

40. The α -hydrogens of esters typically have a $pK_a = 25$, whereas for ketones $pK_a = 20$. This is because:

- There is no resonance stabilisation of the enolates of esters.
- The inductive effect of the oxygen in the ester destabilises the ester enolate.
- The electron donating alkoxy group in esters destabilises the enolate.
- The electron donating alkoxy group in esters stabilises the enolate.

41. Which of the following is most likely to undergo a favorable hydride shift?

-
-
-
-

42. Relative stabilities of the following carbocations will be in the order

- $\text{CH}_3\text{O}-\text{C}_6\text{H}_4-\text{CH}_2^+$
 - $\text{C}_6\text{H}_5-\text{CH}_2^+$
 - $\text{H}_3\text{C}-\text{C}_6\text{H}_4-\text{CH}_2^+$
 - CH_3CH_2^+
- $\text{I} < \text{II} < \text{III} < \text{IV}$
 - $\text{IV} < \text{III} < \text{II} < \text{I}$
 - $\text{IV} < \text{II} < \text{III} < \text{I}$
 - $\text{II} < \text{IV} < \text{III} < \text{I}$

43. Which of the following forms most stable carbocation upon removal of OH^- ?

- $(\text{CH}_3)_3\text{C}-\text{CH}_2\text{OH}$
- $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
- $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$
- $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{OH}$

44. Which of the following carbonium ion is most stable?

- $\text{CH}_3-\text{C}^+(\text{CH}_3)_2$
- $\text{C}_2\text{H}_5-\text{C}^+(\text{C}_2\text{H}_5)_2$
- $\text{CH}_3-\text{C}^+(\text{H})(\text{CH}_3)$
- $\text{C}_2\text{H}_5-\text{C}^+(\text{H})(\text{C}_2\text{H}_5)$

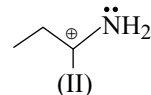
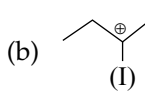
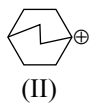
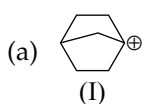
45. Which of the following carbocations would not likely rearrange to a more stable carbocation?

-
-
-
-

46. Which carbocation is the most stabilised?

-
-
-
-
-

51. Which of the following two carbonium ions is more stable? Explain why.



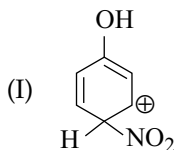
(a) a-II, b-II

(b) a-I, b-I

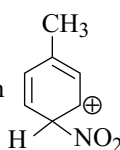
(c) a-I, b-II

(d) a-II, b-I

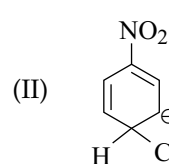
52. Consider the following statements:



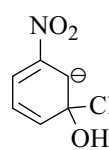
is more stable than



(II)



is more stable than



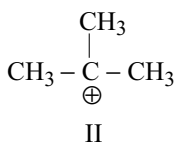
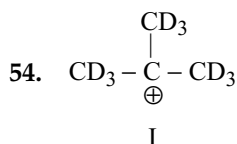
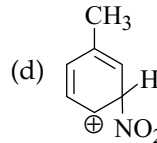
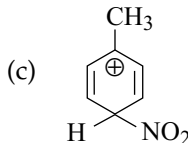
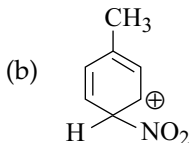
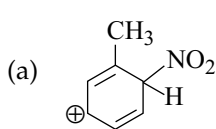
(a) I and II both are correct

(b) I and reverse of II are correct

(c) II and reverse of I are correct

(d) I and II both are incorrect

53. Which is the most stable arenium carbocation?



Which of the following statements is correct?

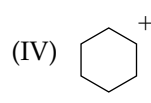
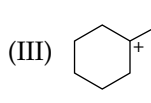
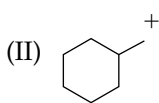
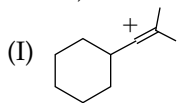
(a) I is more stable than II

(b) II is more stable than I

(c) Both are equally stable

(d) Stability criterion cannot be applied in this case

55. List the following carbocations in order of decreasing stability (starting with the most stable)



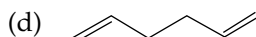
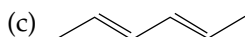
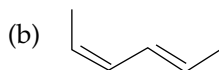
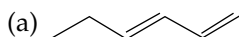
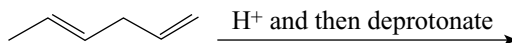
(a) II, III, I, IV

(b) III, IV, II, I

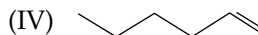
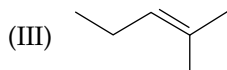
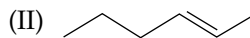
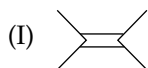
(c) III, IV, I, II

(d) I, II, IV, III

56. Under thermodynamic control, which of the following products would predominate?

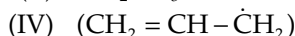


57. Rank the following in order of stability (lowest to highest)



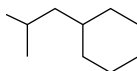
- (a) $\text{IV} < \text{II} < \text{III} < \text{I}$ (b) $\text{IV} < \text{III} < \text{II} < \text{I}$ (c) $\text{I} < \text{III} < \text{II} < \text{IV}$ (d) $\text{IV} < \text{II} = \text{III} < \text{I}$

58. Rank, from the most stabilised to the least stabilised, the following free radicals according to their stabilisation energies



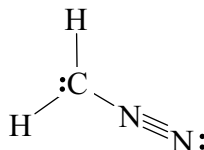
- (a) $\text{IV} > \text{III} > \text{II} > \text{I}$ (b) $\text{I} > \text{IV} > \text{III} > \text{II}$ (c) $\text{III} > \text{IV} > \text{I} > \text{II}$ (d) $\text{III} > \text{IV} > \text{II} > \text{I}$

59. How many tertiary hydrogen(s) are (is) there in the following structure?



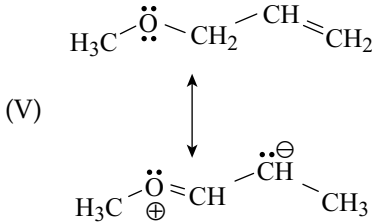
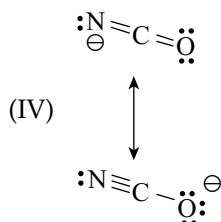
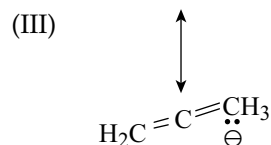
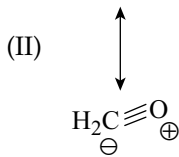
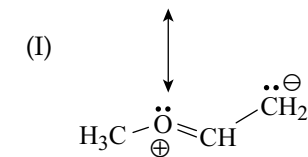
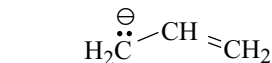
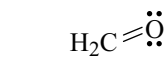
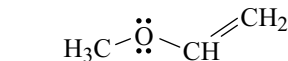
- (a) 1 (b) 2 (c) 3 (d) 6

60. For the following incomplete Lewis structure, what are the correct formal charges for the carbon and the nitrogen attached to the carbon?



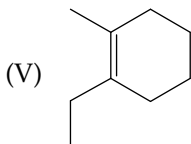
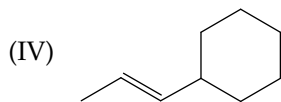
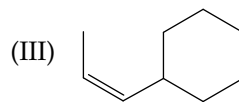
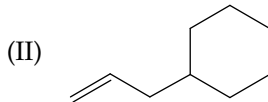
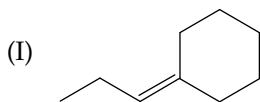
- (a) C:0, N:0 (b) C:-1, N:0 (c) C:-1, N:+1 (d) C:+1, N:-1

61. Which of the following are pairs of resonance structures?



- (a) I, II, III (b) I, IV (c) II, IV, V (d) I, III, IV

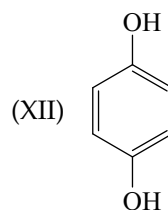
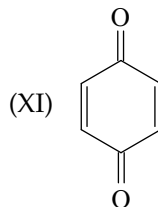
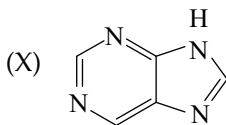
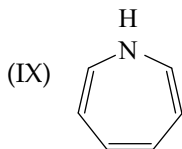
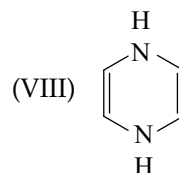
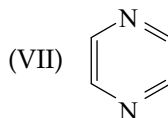
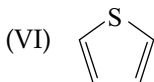
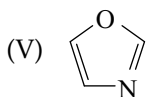
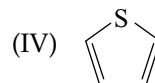
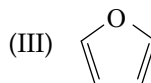
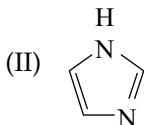
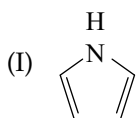
62. Arrange the following alkenes in order of their stability (most to the least)



- (a) $V > II > III > IV > I$
 (c) $V > II > III > I > IV$

- (b) $V > II > IV > III > I$
 (d) $V > I > IV > III > II$

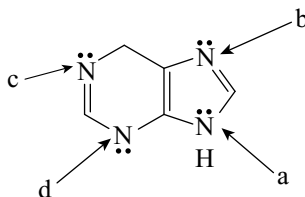
63. Which of the following compounds are aromatic compounds?



- (a) I, II, III, IV, V, VI, VIII, X, XII
 (c) I, II, III, IV, V, VI, VII, X, XII

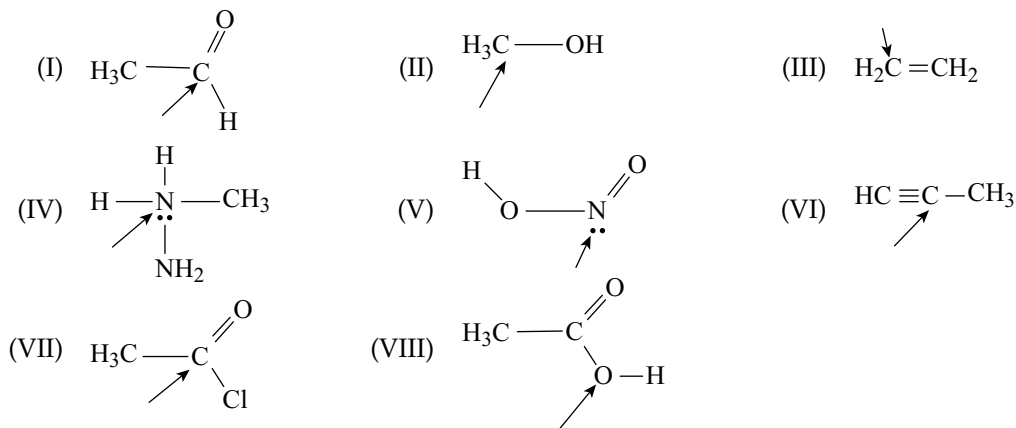
- (b) I, II, III, VI, VIII, X, XI, XII
 (d) I, II, III, V, VII, VIII, IX, X, XII

64. For the following compounds, which nitrogen is the least tendency to be protonated?



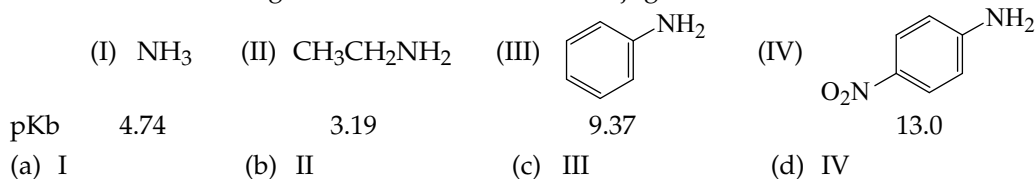
- (a) Nitrogen indicated by arrow "b"
 (b) Nitrogen indicated by arrow "a"
 (c) Nitrogen indicated by arrow "c"
 (d) Nitrogen indicated by arrow "d"

65. Using the VSEPR model, predict which atoms pointed by an arrow have sp^2 hybridisation. (Note: not all the lone pair electrons are displayed)

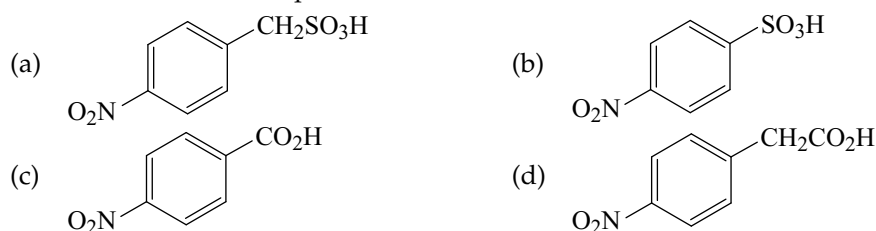


- (a) I, II, IV, VIII (b) I, III, V, VII (c) II, III, V, VII (d) II, IV, V, VIII
66. Arrange the following compounds in order of their acidity. (most to least)
- (I) CH_3CH_2OH (II) CFH_3CO_2H (III) CF_2HCO_2H
 (IV) CF_3COOH (V) CH_3CO_2H
- (a) $IV > III > II > V > I$ (b) $IV > III > II > I > V$
 (c) $V > II > III > IV > I$ (d) $V > III > IV > II > I$
67. Rank the following intermediates according to the stability (most stable first). Explain your choices.
- (a) $CH_3CH_2CH_2\dot{C}H_2$, $CH_3\dot{C}HCH_2CH_3$, $(CH_3)_2\dot{C}CH_2CH_3$, $(CH_3)_3\dot{C}$
 (b) $CH_3CH_2CH_2\dot{C}H_2$, $CH_3\dot{C}HCH_2CH_3$, $(CH_3)_2\dot{C}CH_2CH_3$, $(CH_3)_3\dot{C}$
 (c) $CH_3CH_2CH_2\dot{C}H_2$, $CH_3\dot{C}HCH_2CH_3$, $(CH_3)_2\dot{C}CH_2CH_3$
 (d) $CH_2\cdot$, $CH_2CH\cdot$, $C_6H_5CH\cdot$, $(C_6H_5)_2C\cdot$

68. Which of the following base has the most acidic conjugate acid?



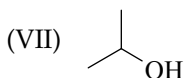
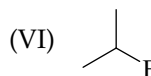
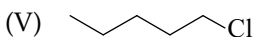
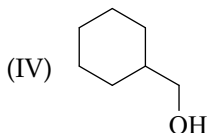
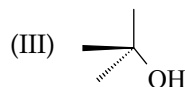
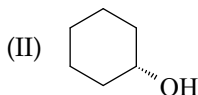
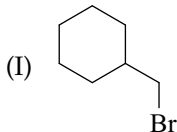
69. What is the structure of p-nitrobenzenesulfonic acid?



70. Which of the following compound has the highest boiling point?

- (a) CH_3OCH_3 (b) CH_3COCH_3 (c) $\text{CH}_3\text{CH}_2\text{OH}$ (d) $\text{CH}_3\text{CO}_2\text{H}$

71. Which are secondary alkyl alcohols among the following?

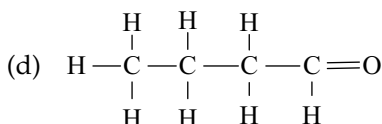
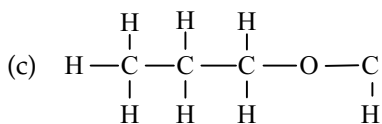
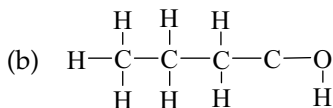
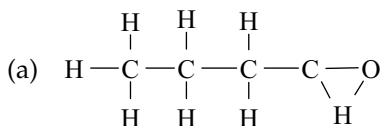


- (a) VII (b) II, IV, VI (c) I, VI, X (d) II, VII

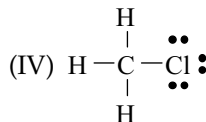
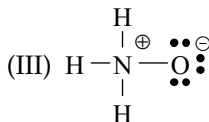
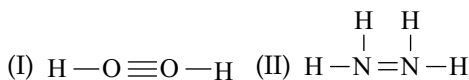
72. Which is the electronic configuration that describes Na^+ ?

- (a) $1s^2, 2s^2, 2p^6$ (b) $1s^2, 2s^2, 2p^6, 3s^2, 3p^6$
(c) $1s^2, 2s^2$ (d) $1s^2, 2s^2, 2p^6, 3s^2$

73. What is the Lewis structure of $\text{CH}_3 - \text{CH}_2 - \text{Cl}_3 - \text{CHO}$?



74. Which Lewis structure(s) is(are) correct?



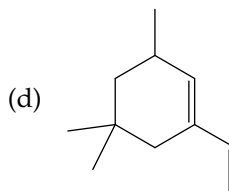
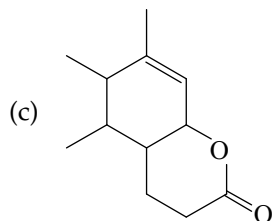
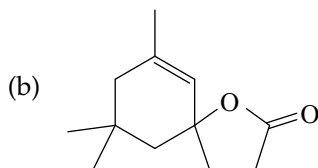
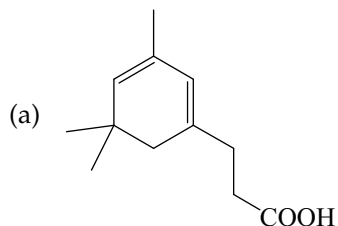
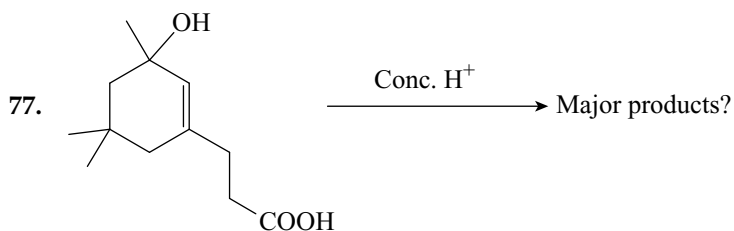
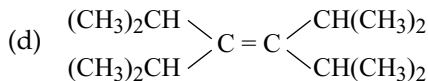
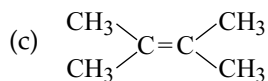
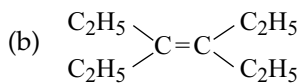
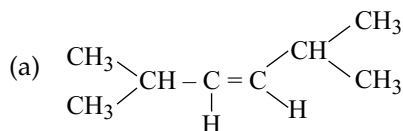
- (a) I, II (b) II, IV (c) III, IV (d) I, III

75. Which molecules are non-polar?

- (I) NH_3 (II) CO_2 (III) H_2O
(IV) CF_4 (V) Br_2 (VI) BF_3
(VII) CH_2Cl_2 (VIII) H_2O_2

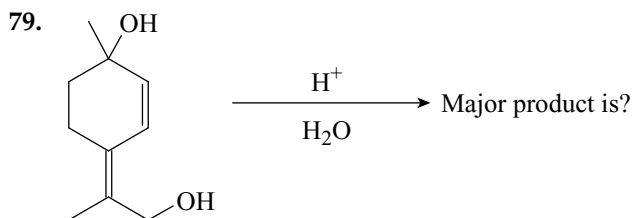
- (a) II, IV, V, VI (b) I, IV, VI, VIII (c) III, IV, V, VI (d) I, III, VII, VIII

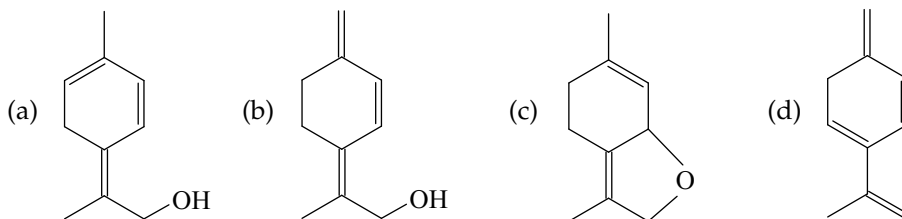
76. Which of the following is the most stable alkene?



78. Hyperconjugation is best described as:

- (a) delocalisation of p electrons into a nearby empty orbital
- (b) delocalisation of σ electrons into a nearby empty orbital
- (c) the effect of alkyl groups donating a small amount of electron density inductively into a carbocation
- (d) the migration of a carbon or hydrogen from one carbocation to another

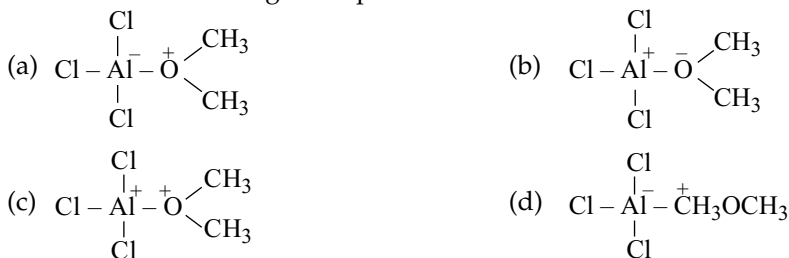




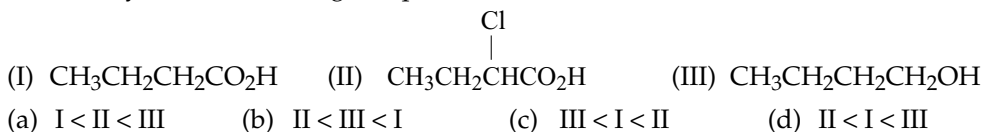
80. Which of the following statements best explains why 1-propyne can be deprotonated by the ethyl anion (CH_3CH_2^-)?
- The acetylide anion is a stronger base than the ethyl anion.
 - Ethane has a lower pK_a than acetylene.
 - The lone-pair orbitals in acetylide anions have more s character than those in alkyl anions.
 - The acetylide anion is a weaker base than an alkyl anion.

81. Which of the following statements is NOT true for ethylene ($\text{CH}_2 = \text{CH}_2$) molecule?
- Both carbons are sp^2 hybridised.
 - $\text{C} = \text{C}$ bond length is shorter than the $\text{C}-\text{C}$ bond length in ethane.
 - The two $\text{C} = \text{C}$ bonds are equally strong. (one is π and the other is σ .)
 - The entire molecule has a planar geometry.

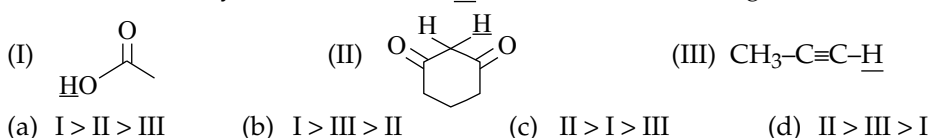
82. Which of the following is the product of the reaction between AlCl_3 and CH_3OCH_3 ?



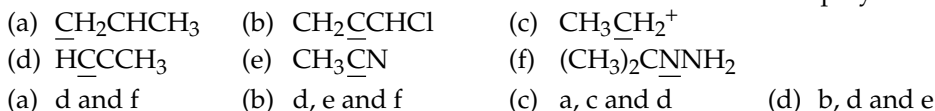
83. The acidity for the following compounds increases in the order



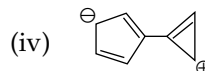
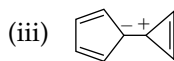
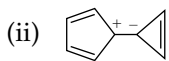
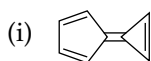
84. The relative acidity of the underlined H in each of the following is in the order



85. Which of the underlined atoms in the molecules shown below have sp hybridisation?



86. A compound  shows a large dipole moment. Which of the following resonance structures can be used to adequately explain this observation?



(a) i

(b) iii and iv

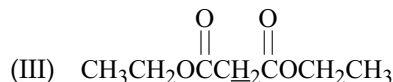
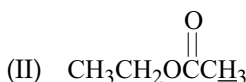
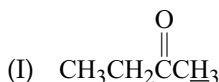
(c) ii and iii

(d) iv only

87. The nitrogen in trimethylamine is

(a) sp^2 hybridised (b) sp^3 hybridised (c) sp hybridised (d) sp^3d^2 hybridised

88. The acidity of the protons H:



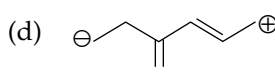
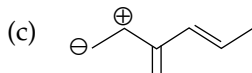
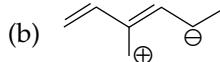
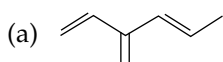
(a) I > II > III

(b) I > III > II

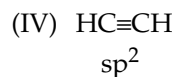
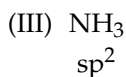
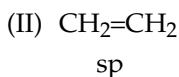
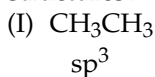
(c) II > I > III

(d) III > I > II

89. Which of the following is not a valid resonance structure of the others?



90. Which is(are) the correct orbital hybridisation (s) for the C and N atoms in the following structures?



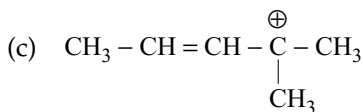
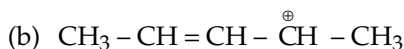
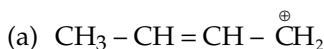
(a) I

(b) II, III

(c) III, IV

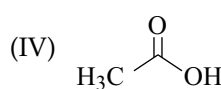
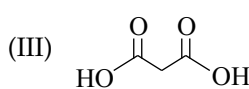
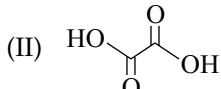
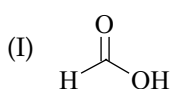
(d) I, III

91. Which allylic carbocation is the most stable carbocation?



(d) All have same stability

92. Rank the following molecules in order of decreasing acidity (increasing pK_a)



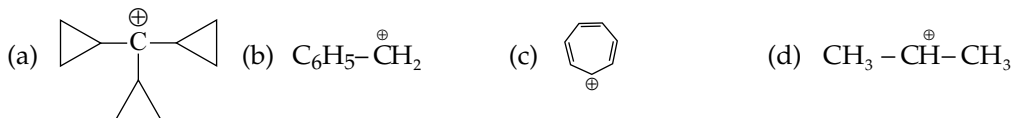
(a) II > III > I > IV

(b) III > II > I > IV

(c) I > IV > II > III

(d) IV > III > I > II

93. Which among the following carbocations is most stable?



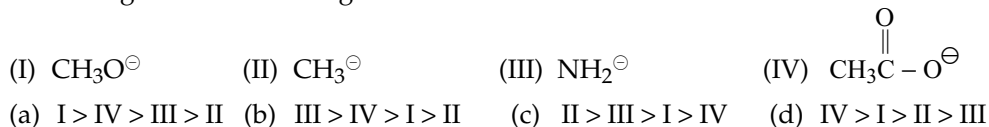
94. Which of the following statements about resonance structures is false?

- (a) Individual resonance structures are imaginary, not real.
- (b) Resonance forms differ only in the placement of their π - or non-bonding electrons or unpaired electron.
- (c) Different resonance structures of a substance do not have to be equivalent.
- (d) In valid resonance structures, all atoms from the second row of the periodic table must have an octet of electrons.

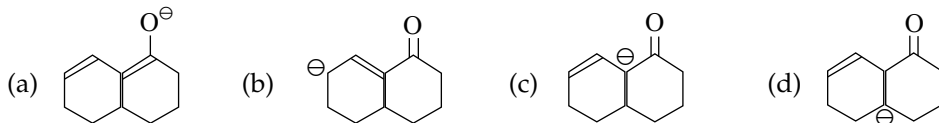
LEVEL 2

Single and Multiple Choice Type

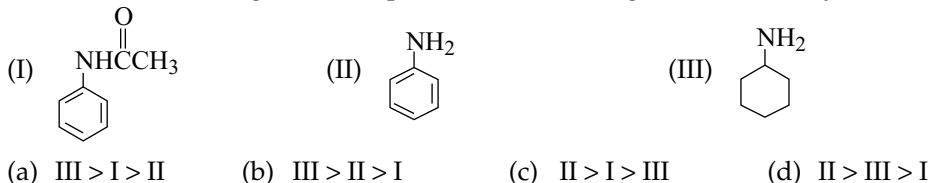
1. The strength of the following bases decreases in the order



2. Which of the following is not a resonance structure of the others?



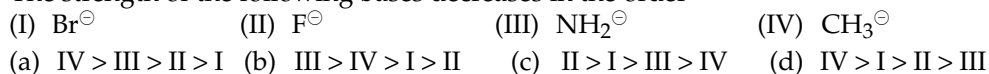
3. Rank of the following three compounds in decreasing order of basicity is



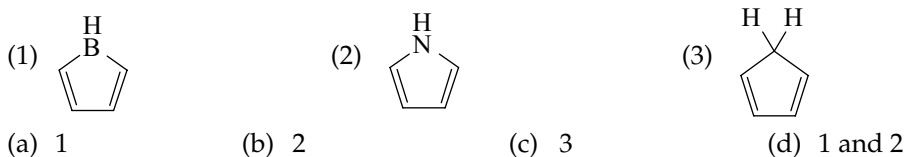
4. Compare the hybridisation of the central carbon atoms in carbon dioxide (CO_2) and allene ($\text{H}_2\text{C}=\text{C}=\text{CH}_2$). Which statement is correct?

- (a) The hybridisation types of these two carbons cannot be compared because of large electronegativity difference between carbon and oxygen in CO_2 that does not exist in allene.
 (b) The hybridisation of carbon in CO_2 cannot be determined, because the lone electron pairs on oxygen do not allow for angle measurements that are necessary to determine the hybridisation involved.
 (c) In CO_2 the carbon is sp^2 hybridised, but in allene the central carbon is sp hybridised.
 (d) In CO_2 the carbon is sp hybridised but in allene the central carbon is sp^2 hybridised.

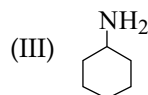
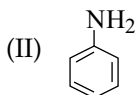
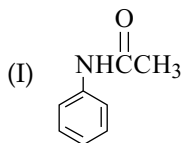
5. The strength of the following bases decreases in the order



6. Choose the following species that would be predicted to be aromatic according to Hückel's rule.

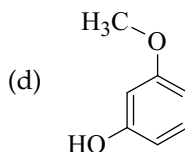
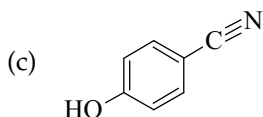
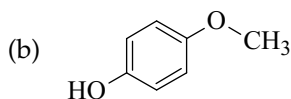
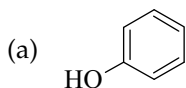


7. Rank of the following three compounds in decreasing order of basicity is

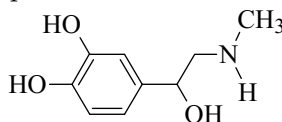


- (a) III > I > II (b) III > II > I (c) II > I > III (d) II > III > I

8. Which of the following phenols would be the most acidic?

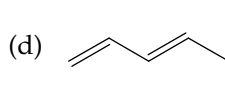
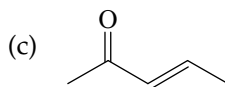
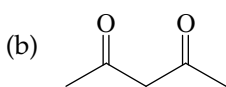
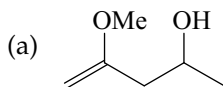
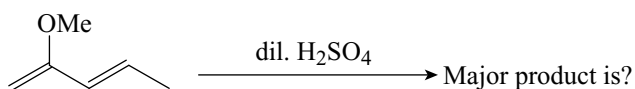


9. Choose the correct formula for epinephrine.

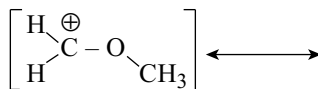


- (a) C₁₀H₁₃NO₃ (b) C₉H₁₂NO₃ (c) C₁₀H₁₄NO₃ (d) C₉H₁₃NO₃

10.



11. Which is an acceptable resonance structure for the following drawing?

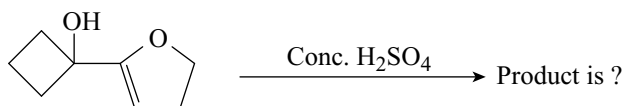


- (a) CH₃CH⁺ - OH (b) CH₂ = ⁺O - CH₃ (c) CH₂CH₂O⁺ (d) CH₃CH=O⁺H

12. In allene (H₂C = C = CH₂), the terminal carbons are sp² hybridised. Each of the two terminal H₂C groups are situated in such a manner that the two "terminal" planes are 90° from each other. Other compounds with double bonds on successive carbons may also exist. They are called cumulenes. What is the relationship between the two terminal H₂C groups in a cumulene containing three consecutive double bonds (H₂C = C = C = CH₂)?

- (a) They are in perpendicular planes.
 (b) They are in two planes 60° from each other.
 (c) They are in the same plane.
 (d) They are in two planes 120° from each other.

13.



- (a) (b) (c) (d) None of these

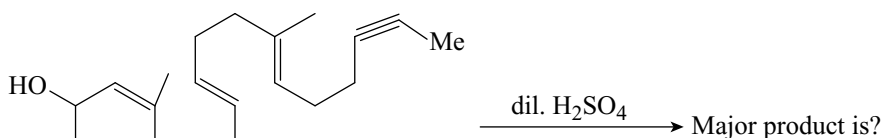
14. Alkyne hydrogens are more acidic than alkene or alkane hydrogens because

- (a) The alkyne carbon has higher 's' character.
 (b) The anion formed is more stable.
 (c) The electrons in the sp orbital are closer to the nucleus.
 (d) All of the above.

15. Which of the following is expected to be the least basic?

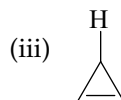
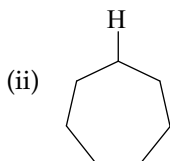
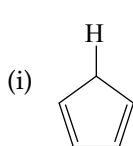
- (a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$ (b)
 (c) (d) $(\text{CH}_3)_3\text{N}$

16.



- (a) (b)
 (c) (d)

17. The acidity of the protons H in each of the following is



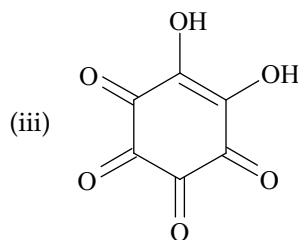
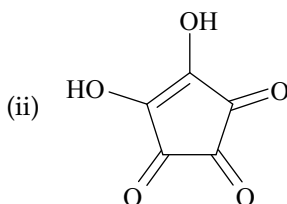
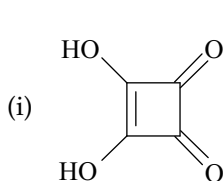
(a) $i > ii > iii$

(b) $ii > iii > i$

(c) $i > iii > ii$

(d) $iii > i > ii$

18. The acidity of the protons H in each of the following is



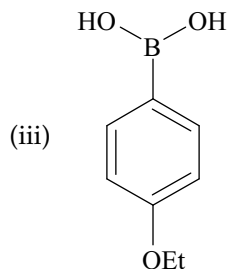
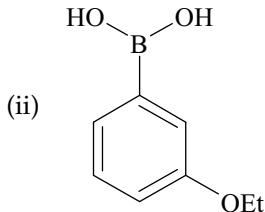
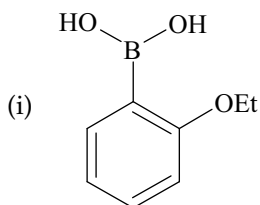
(a) $i > ii > iii$

(b) $iii > ii > i$

(c) $i > iii > ii$

(d) $iii > i > ii$

19. Identify correct acidic strength order in the following compounds



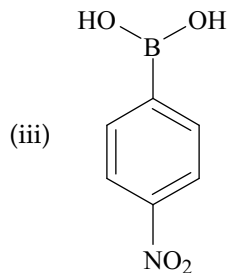
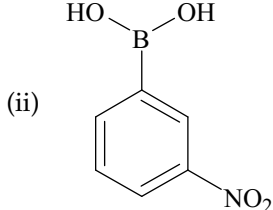
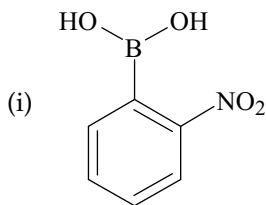
(a) $i > ii > iii$

(b) $ii > iii > i$

(c) $ii > i > iii$

(d) $iii > i > ii$

20. Identify correct acidic strength order in the following compounds



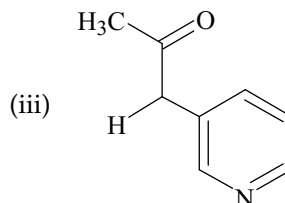
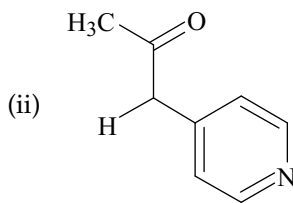
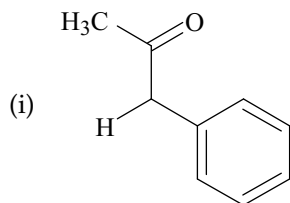
(a) $i > ii > iii$

(b) $iii > ii > i$

(c) $i > iii > ii$

(d) $iii > i > ii$

21. Identify correct acidic strength order in the following compounds



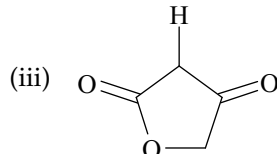
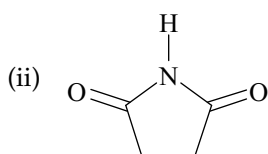
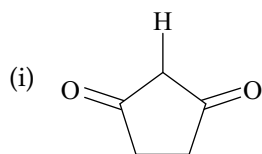
(a) $i > ii > iii$

(b) $ii > iii > i$

(c) $i > iii > ii$

(d) $iii > i > ii$

22. Identify correct acidic strength order in the following compounds



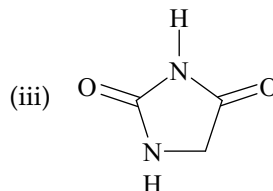
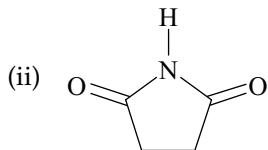
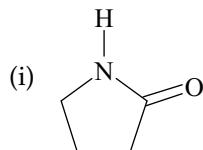
(a) $ii > i > iii$

(b) $ii > iii > i$

(c) $i > iii > ii$

(d) $iii > i > ii$

23. Identify correct acidic strength order in the following compounds



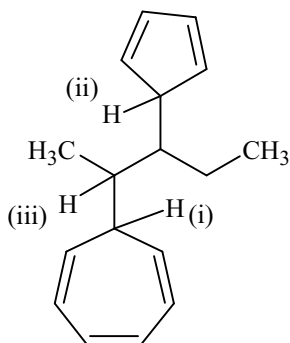
(a) $i > ii > iii$

(b) $ii > iii > i$

(c) $i > iii > ii$

(d) $iii > i > ii$

24. Identify correct acidic strength order in the following compounds



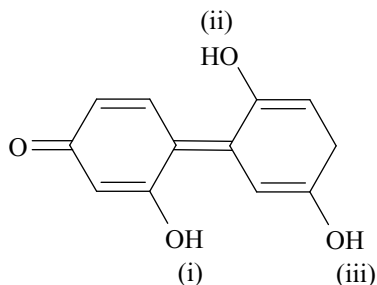
(a) $i > ii > iii$

(b) $ii > iii > i$

(c) $i > iii > ii$

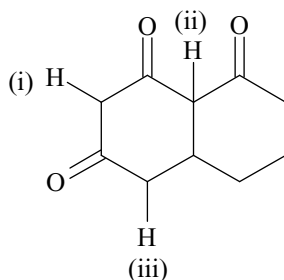
(d) $iii > i > ii$

25. Identify correct acidic strength order in the following compounds



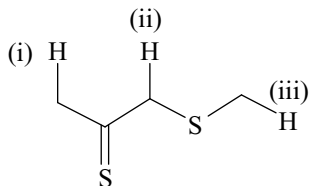
- (a) $i > ii > iii$ (b) $ii > iii > i$ (c) $i > iii > ii$ (d) $iii > i > ii$

26. Identify correct acidic strength order in the following compounds



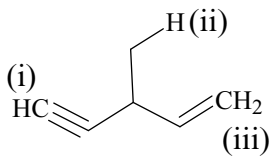
- (a) $i > ii > iii$ (b) $ii > iii > i$ (c) $i > iii > ii$ (d) $iii > i > ii$

27. Identify correct acidic strength order in the following compounds



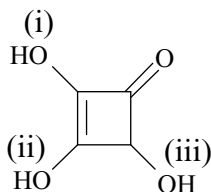
- (a) $i > ii > iii$ (b) $ii > iii > i$ (c) $i > iii > ii$ (d) $ii > i > iii$

28. Identify correct acidic strength order in the following compounds



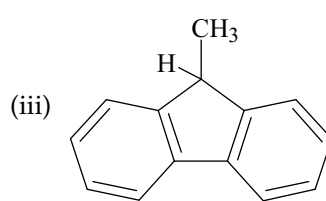
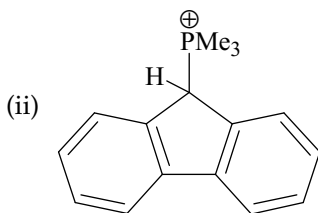
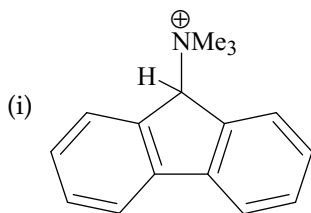
- (a) $i > ii > iii$ (b) $ii > iii > i$ (c) $i > iii > ii$ (d) $iii > i > ii$

29. Identify correct acidic strength order in the following compounds



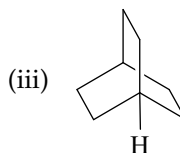
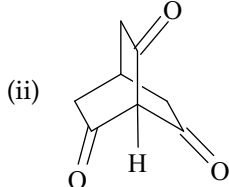
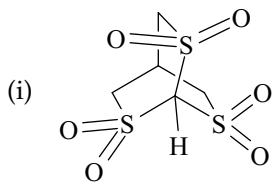
- (a) $ii > i > iii$ (b) $ii > iii > i$ (c) $i > iii > ii$ (d) $iii > i > ii$

30. The acidity of the protons H in each of the following is



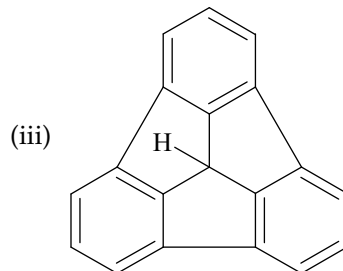
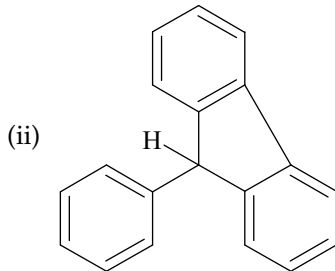
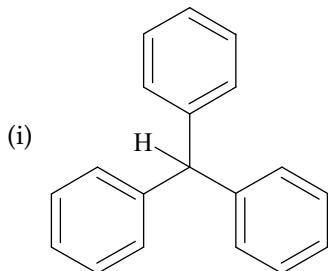
- (a) $i > ii > iii$ (b) $ii > iii > i$ (c) $ii > i > iii$ (d) $iii > i > ii$

31. The acidity of the protons H in each of the following is



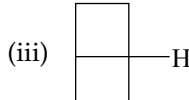
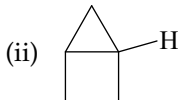
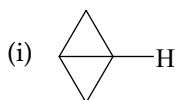
- (a) $i > ii > iii$ (b) $ii > iii > i$ (c) $i > iii > ii$ (d) $iii > i > ii$

32. Identify correct acidic strength order in the following compounds



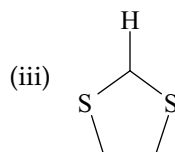
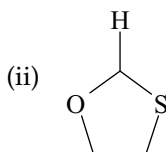
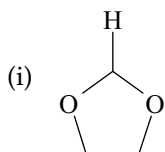
- (a) $iii > ii > i$ (b) $ii > iii > i$ (c) $i > iii > ii$ (d) $iii > i > ii$

33. Identify correct acidic strength order in the following compounds



- (a) $i > ii > iii$ (b) $ii > iii > i$ (c) $i > iii > ii$ (d) $iii > i > ii$

34. Identify correct acidic strength order in the following compounds



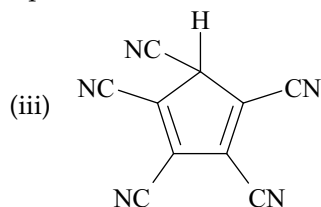
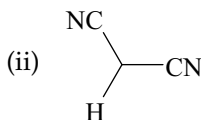
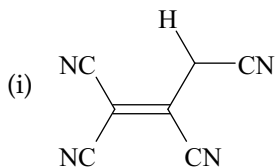
(a) $i > ii > iii$

(b) $iii > ii > i$

(c) $i > iii > ii$

(d) $iii > i > ii$

35. Identify correct acidic strength order in the following compounds



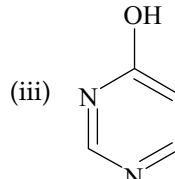
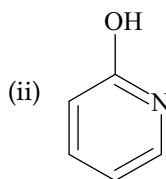
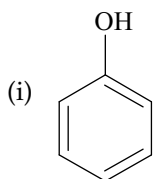
(a) $i > ii > iii$

(b) $ii > iii > i$

(c) $i > iii > ii$

(d) $iii > i > ii$

36. Identify correct acidic strength order in the following compounds



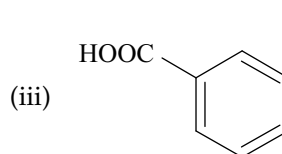
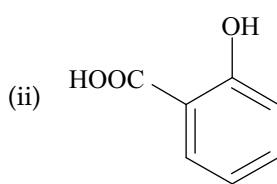
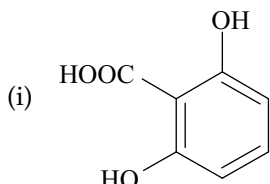
(a) $iii > ii > i$

(b) $ii > iii > i$

(c) $i > iii > ii$

(d) $iii > i > ii$

37. Identify correct acidic strength order in the following compounds



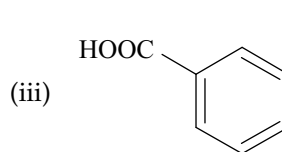
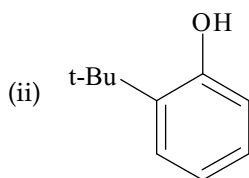
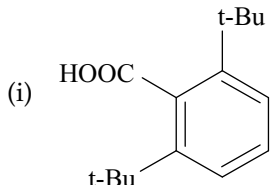
(a) $i > ii > iii$

(b) $ii > iii > i$

(c) $i > iii > ii$

(d) $iii > i > ii$

38. Identify correct acidic strength order in the following compounds



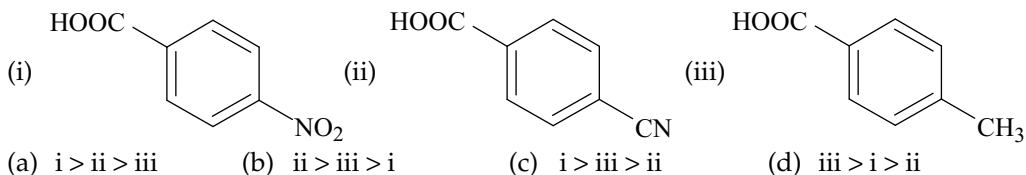
(a) $i > ii > iii$

(b) $ii > iii > i$

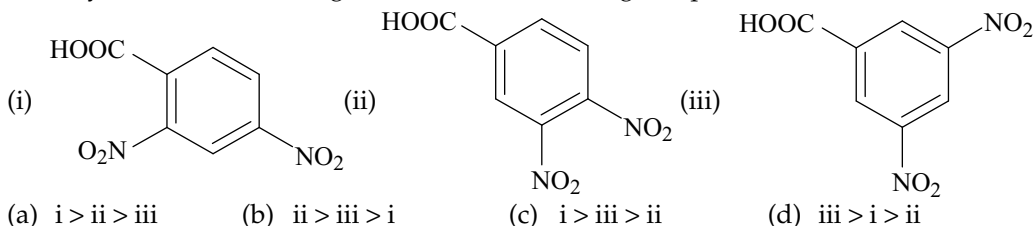
(c) $i > iii > ii$

(d) $iii > i > ii$

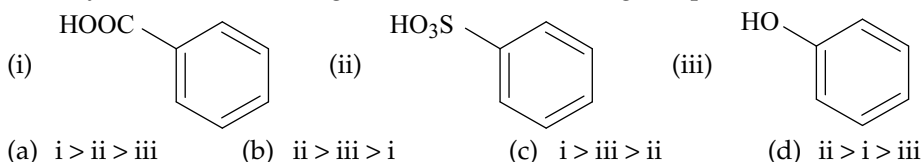
39. Identify correct acidic strength order in the following compounds



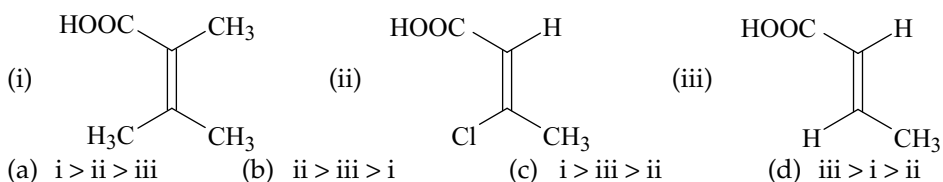
40. Identify correct acidic strength order in the following compounds



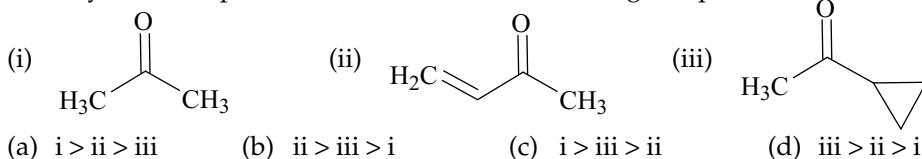
41. Identify correct acidic strength order in the following compounds



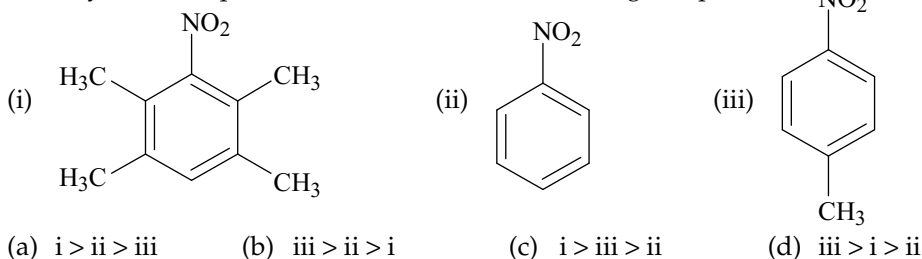
42. Identify correct acidic strength order in the following compounds



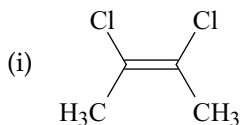
43. Identify correct dipole moment order in the following compounds



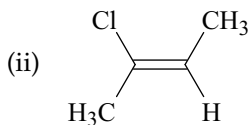
44. Identify correct dipole moment order in the following compounds



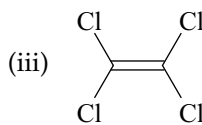
45. Identify correct dipole moment order in the following compounds



(a) $i > ii > iii$



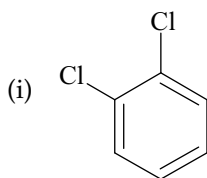
(b) $ii > iii > i$



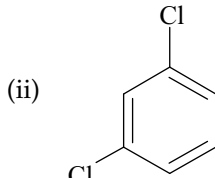
(c) $i > iii > ii$

(d) $iii > i > ii$

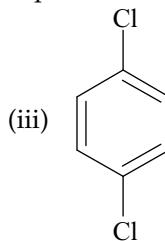
46. Identify correct dipole moment order in the following compounds



(a) $i > ii > iii$



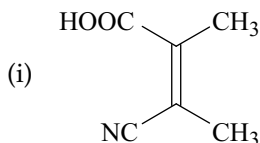
(b) $ii > iii > i$



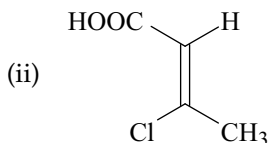
(c) $i > iii > ii$

(d) $iii > i > ii$

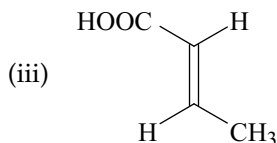
47. Identify correct acidic strength order in the following compounds



(a) $i > ii > iii$



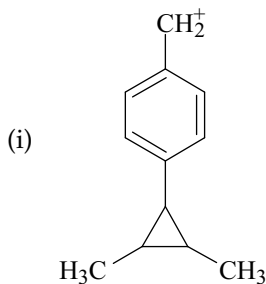
(b) $ii > iii > i$



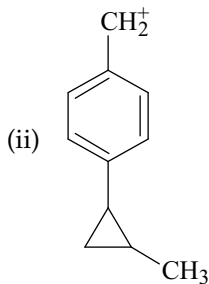
(c) $i > iii > ii$

(d) $iii > i > ii$

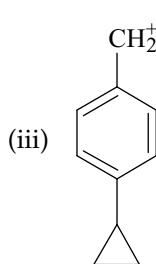
48. Identify correct stability order in the following compounds



(a) $i > ii > iii$



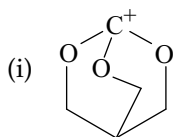
(b) $ii > iii > i$



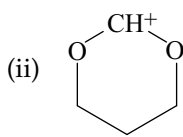
(c) $i > iii > ii$

(d) $iii > i > ii$

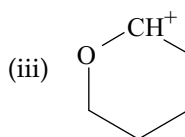
49. Identify correct stability order in the following compounds



(a) $i > ii > iii$



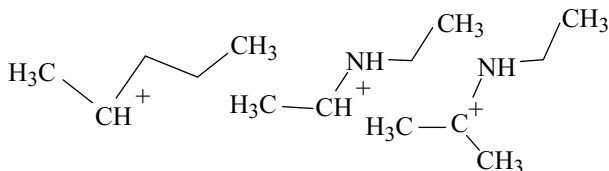
(b) $ii > iii > i$



(c) $i > iii > ii$

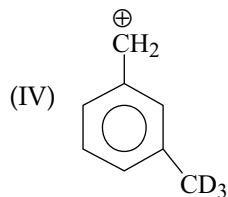
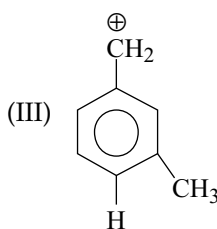
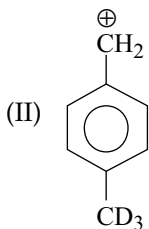
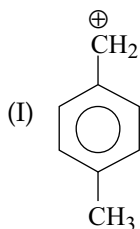
(d) $iii > i > ii$

50. Identify correct stability order in the following compounds



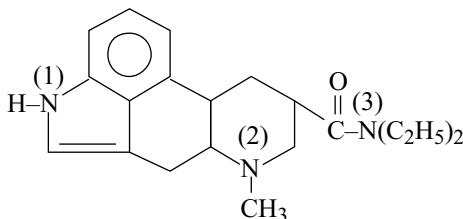
- (a) i > ii > iii (b) ii > iii > i (c) iii > ii > i (d) iii > i > ii

51. Correct order of stability of the following carbocation is



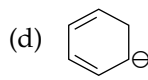
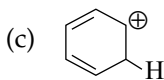
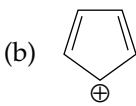
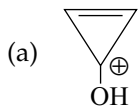
- (a) I > II > III > IV (b) I > II > IV > III (c) II > I > IV > III (d) I > IV > II > III

52. Which nitrogen in LSD is most basic?

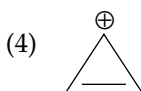
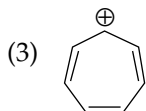
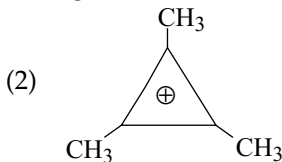
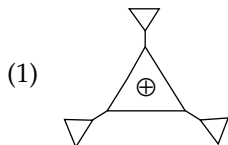


- (a) 1 (b) 2 (c) 3 (d) All are equally basic

53. Which one of the following ion is aromatic?

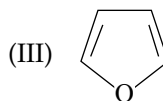
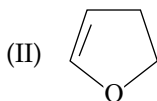
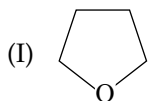


54. Arrange the following carbocations in decreasing order



- (a) 1, 2, 3, 4 (b) 1, 2, 4, 3 (c) 1, 3, 2, 4 (d) 4, 3, 2, 1

55. Arrange the following in decreasing order of their solubility in water or extent of hydrogen bonding with H_2O



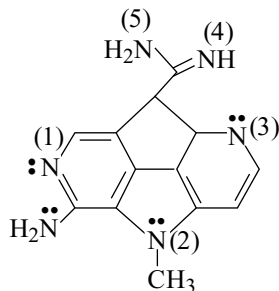
(a) I, II, III

(b) III, II, I

(c) II, III, I

(d) III, I, II

56. The decreasing order of basic strength is



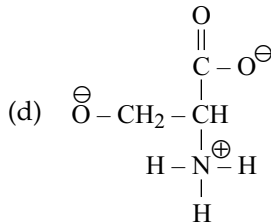
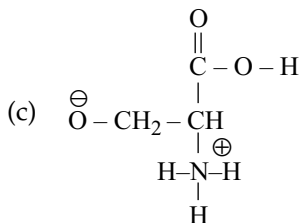
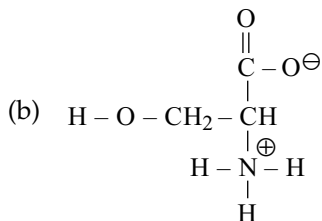
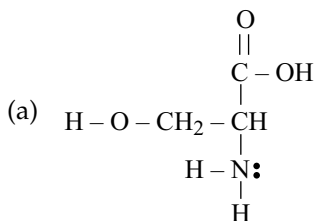
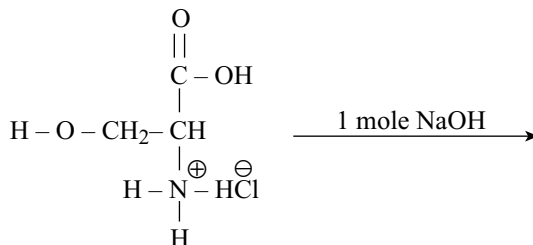
(a) $1 > 5 > 3 > 4 > 2$

(b) $4 > 1 > 5 > 3 > 2$

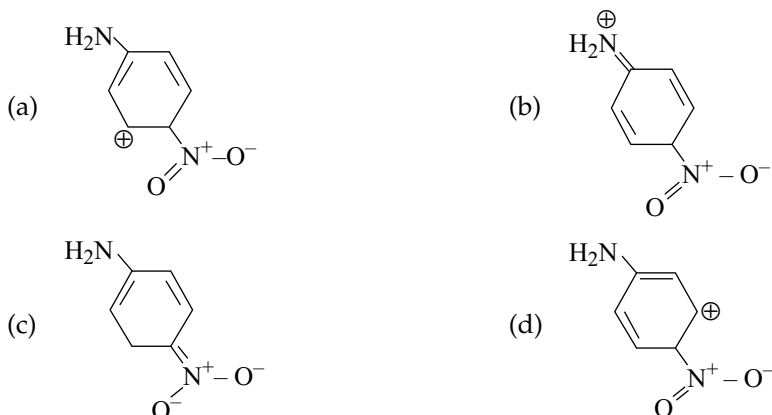
(c) $5 > 4 > 1 > 2 > 3$

(d) $4 > 5 > 3 > 1 > 2$

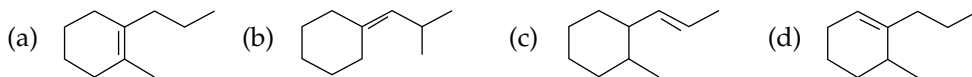
57. What is the major product obtained from the following reaction?



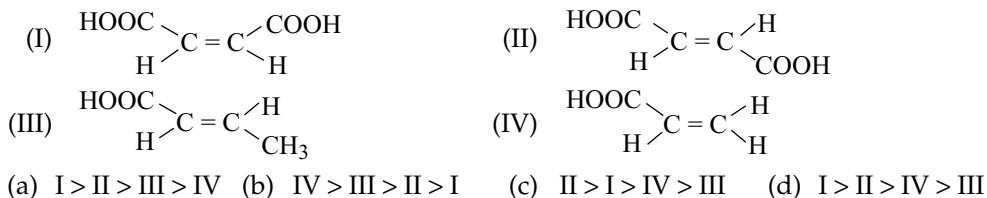
58. All the following are the resonance structure of one another except



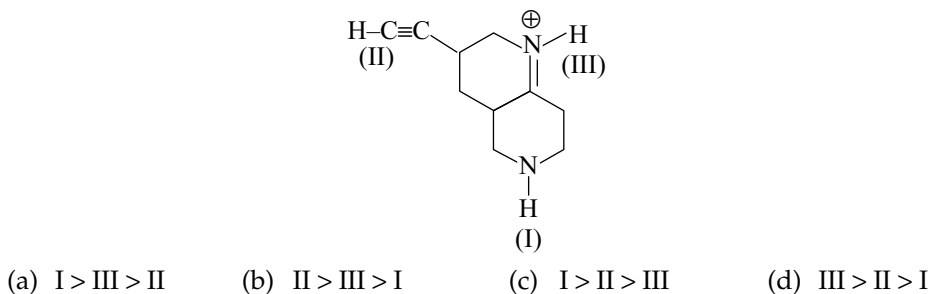
59. Which of the following shows minimum heat of combustion?



60. Arrange the following in the decreasing order of their acidic strength



61. Arrange the following hydrogens in the order of their acidic behaviour



62. Which of the following acid gives evolution of CO_2 with NaHCO_3 ?



- (III) (IV) (V)
- (a) I, III, IV, V (b) III, IV, V (c) I, II, III, IV, V (d) I, III, IV

63. Maximum resonance energy is known for

- (a) (b) (c) (d)

64. Arrange the following in the order of their heat of hydrogenation, when all of them are converted to n-pentane

- (I) (II) (III) (IV)
- (a) II > III > I > IV (b) I > IV > II > III (c) IV > II > I > III (d) II > IV > III > I

65. Most acidic hydrogen among the following is

-
- (a) a (b) b (c) c (d) d

66. In which of the following molecule the mesomeric effect does not operate?

- (a) (b) (c) (d)

67. Which is not carbene out of the following?

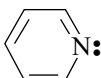
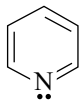
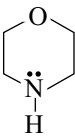
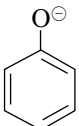
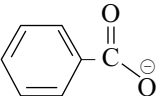
- (a) :CH₂ (b) :CCl₂ (c) (d) :CCl₃

68. Out of the following reagents, pure electrophiles are

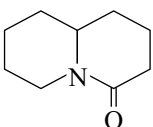
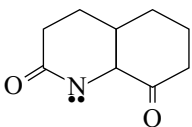
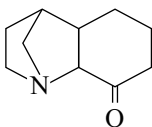
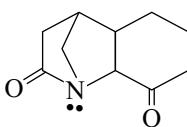
(P) R_3N ; (Q) $:CCl_2$; (R) CH_3^+ ; (S) H_2O ; (T) H_3O^+ ; (U) Na^+

(a) Q, R, T, U (b) Q, R (c) Q, R, S, T (d) Q, R, S, T, U

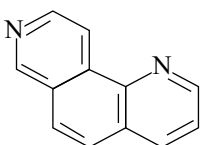
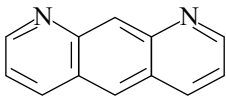
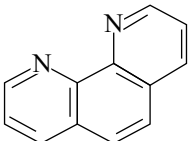
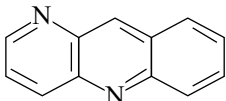
69. Select the correct statement

- (a)  is more basic than  (b)  is more basic than 
- (c)  is more basic than  (d) All of them

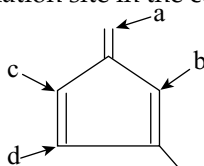
70. Which of the following is most basic?

- (a)  (b)  (c)  (d) 

71. Which of the following is most basic?

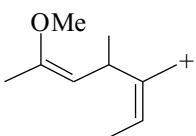
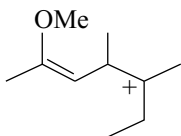
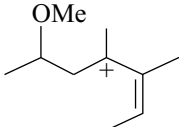
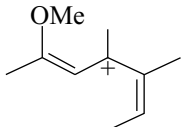
- (a)  (b) 
- (c)  (d) 

72. Which is the least likely protonation site in the conjugated alkene shown below?

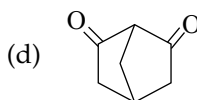
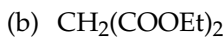
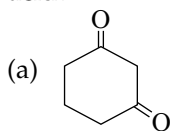


- (a) a (b) b (c) c (d) d

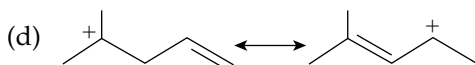
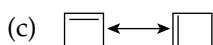
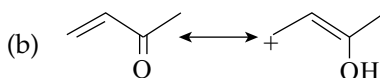
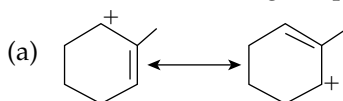
73. Which of the following is the most stabilised carbocation?

- (a)  (b) 
- (c)  (d) 

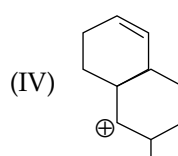
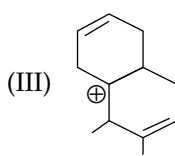
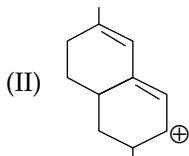
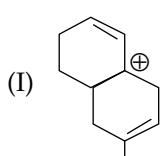
74. Which one of the following compounds would you expect to be the strongest carbon acid?



75. Which of the following is a pair of resonance structure?

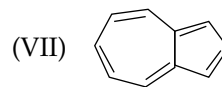
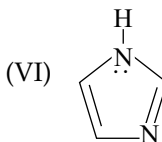
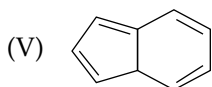
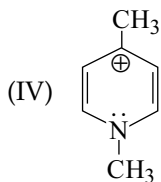
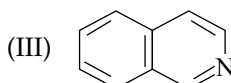
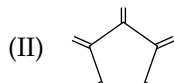
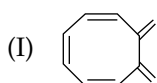


76. Rank the following carbocations in increasing order of stability



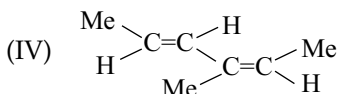
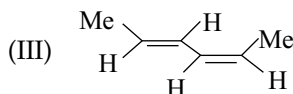
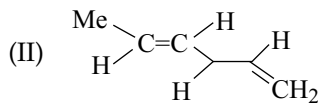
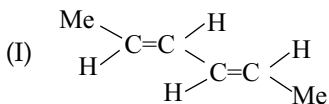
(a) $\text{IV} < \text{III} < \text{I} < \text{II}$ (b) $\text{IV} < \text{I} < \text{III} < \text{II}$ (c) $\text{III} < \text{II} < \text{I} < \text{IV}$ (d) $\text{I} < \text{III} < \text{II} < \text{IV}$

77. In which of the following option correctly follow aromaticity?



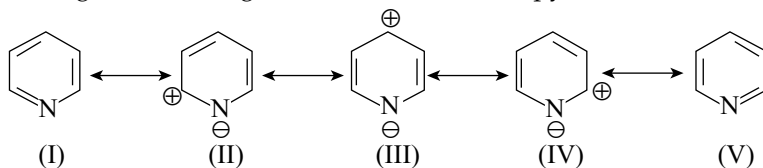
(a) I, III, IV, VII (b) III, IV, VI, VII (c) I, III, IV, VII (d) III, IV, VI, VII

78. Which of the following is the correct order for decreasing order of heat of hydrogenation (magnitude)?



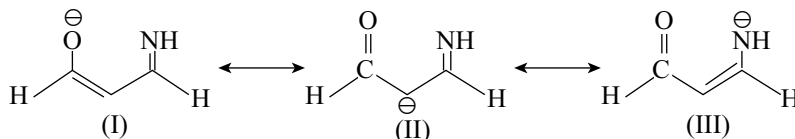
(a) $\text{I} > \text{II} > \text{III} > \text{IV}$ (b) $\text{II} > \text{I} > \text{III} > \text{IV}$ (c) $\text{IV} > \text{III} > \text{I} > \text{II}$ (d) $\text{II} > \text{III} > \text{I} > \text{IV}$

79. Among the following canonical structures of pyridine, the correct order of stability is



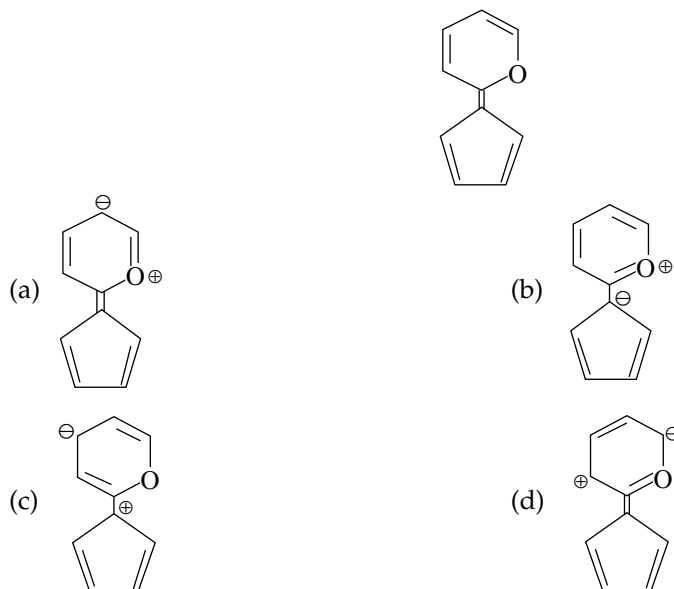
- (a) $(I = V) > (II = IV) > III$ (b) $(II = IV) > (I = V) > III$
 (c) $(I = V) > III > (II = IV)$ (d) $III > (II = IV) > (I = V)$

80. The correct order of stability among the following canonical structures is

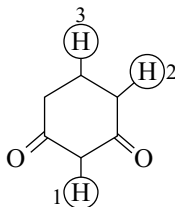


- (a) $I > II > III$ (b) $I > III > II$ (c) $II > I > III$ (d) $III > I > II$

81. The most stable canonical structure of the given molecule is

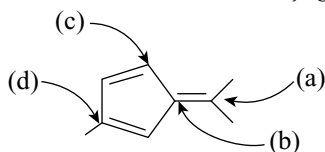


82. Which of the following order is correct for the acidity of indicated H-atoms?



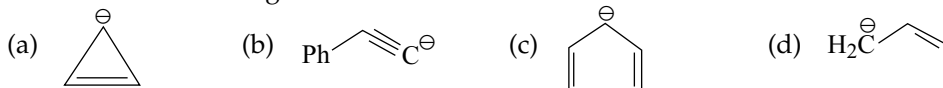
- (a) $(H)1 > (H)2 > (H)3$ (b) $(H)3 > (H)2 > (H)1$ (c) $(H)2 > (H)1 > (H)3$ (d) $(H)1 > (H)3 > (H)2$

83. Which is the least likely protonation site in the conjugated alkene shown below?

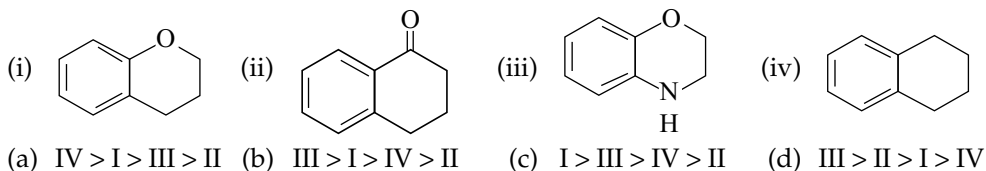


- (a) a (b) b (c) c (d) d

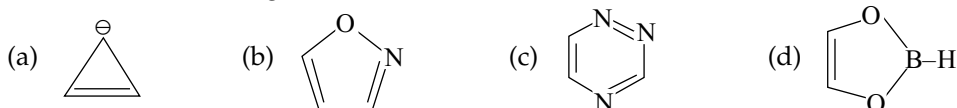
84. Which of the following carbanion is most stable?



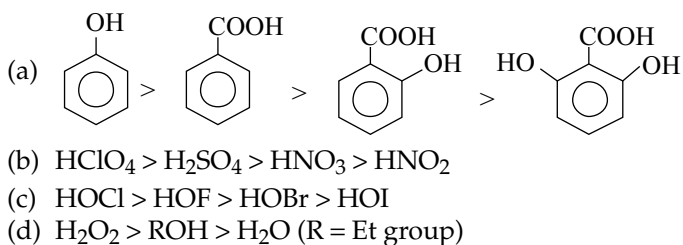
85. Identify correct order of electron cloud in benzene ring for the following compounds



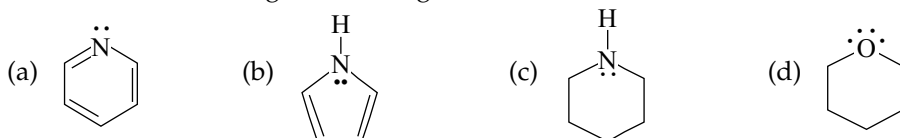
86. Which of the following is antiaromatic?



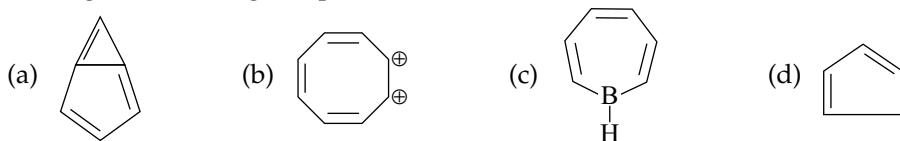
87. The correct order of pK_a is



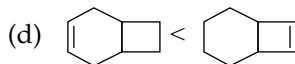
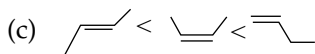
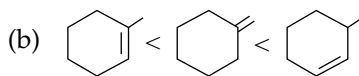
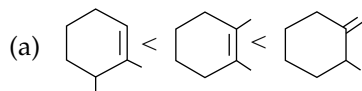
88. Which of the following is the strongest base?



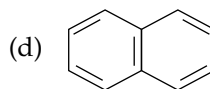
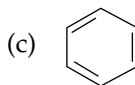
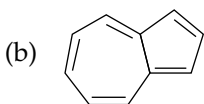
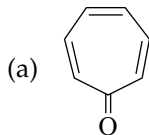
89. Among the following compounds which nonaromatic?



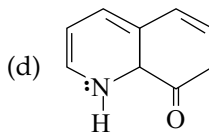
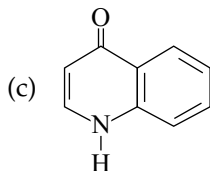
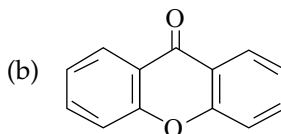
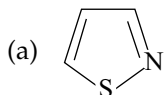
90. Identify correct order of heat of hydrogenation



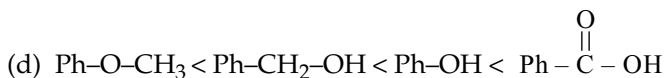
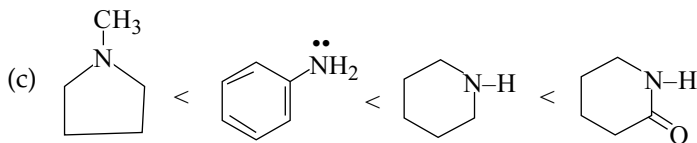
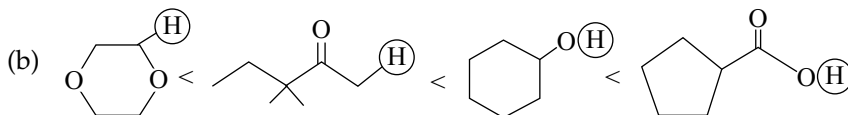
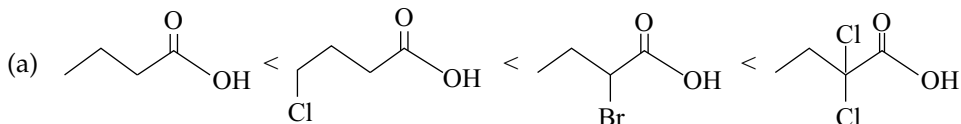
91. Dipole moment of which compound(s) is/are not zero?



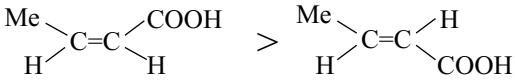
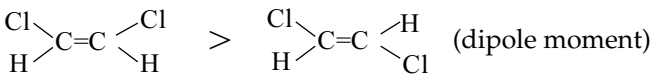
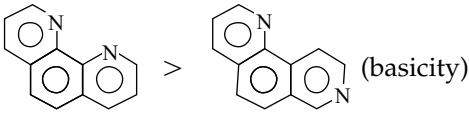
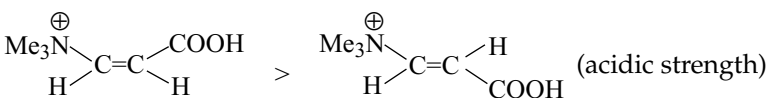
92. Which of the following is/are aromatic?



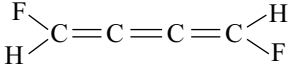
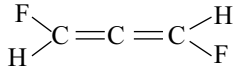
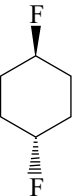
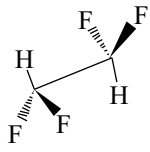
93. Which of following represent the correct order of acidity?



94. Which of the following is correct?

- (a)  (acidic strength)
- (b)  (dipole moment)
- (c)  (basicity)
- (d)  (acidic strength)

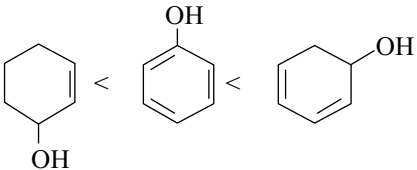
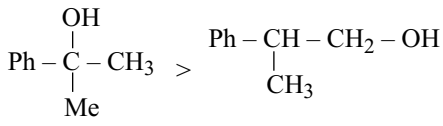
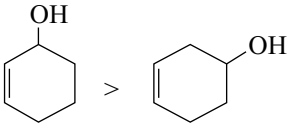
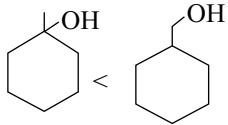
95. Which of the following molecules have dipole moment?

- (a)  (b) 
- (c)  (d) 

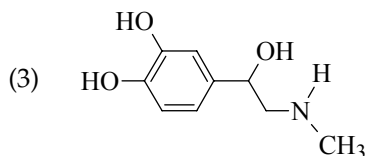
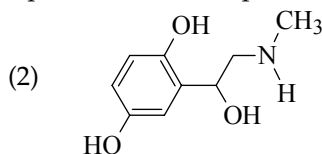
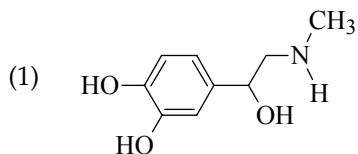
96. $\text{CH}_3 - \text{CH}_2 - \text{NH}_2 \xrightarrow[\text{HCl}]{\text{NaNO}_2} [\text{P}]$; Products of the reaction are:

- (a) $\text{CH}_3 - \text{CH}_2 - \text{OH}$ (b) $\text{Et} - \text{Cl}$ (c) $\text{Et} - \overset{\text{O}}{\parallel} \text{C} - \text{H}$ (d) $\text{Et} - \text{ONO}$

97. Identify correct order of rate of dehydration

- (a)  (b) 
- (c)  (d) 

98. Choose the constitutional isomer(s) of epinephrine (see above problem)



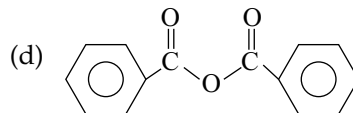
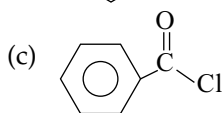
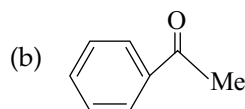
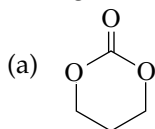
(a) 1

(b) 2

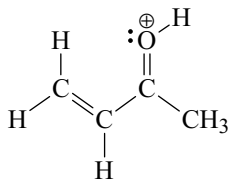
(c) 3

(d) 1 & 2

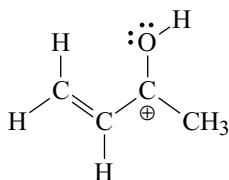
99. 2-phenyl-2-propanol can be prepared by treating which of these compounds with MeMgBr followed by hydrolysis?



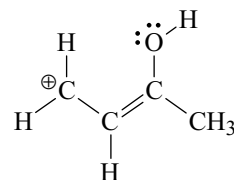
100. Choose those that are resonance structures of protonated methyl vinyl ketone.



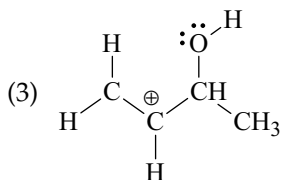
(1)



(2)



protonated methyl
vinyl ketone



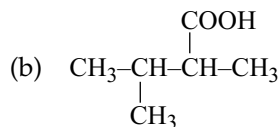
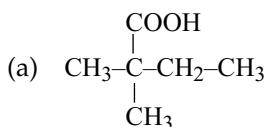
(a) 1

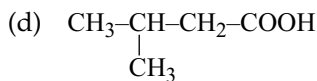
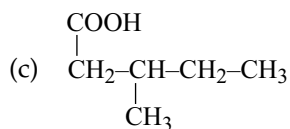
(b) 2

(c) 3

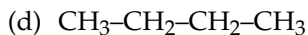
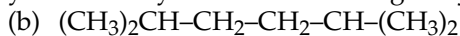
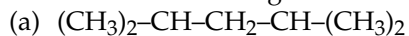
(d) 1 & 2

101. Which of the following acid will give isopentane on decarboxylation with soda lime?

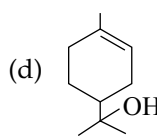
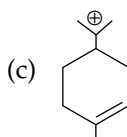
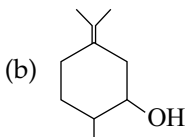
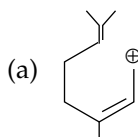




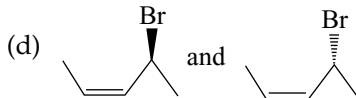
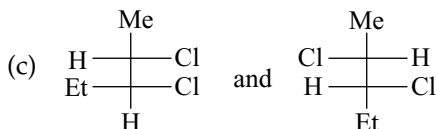
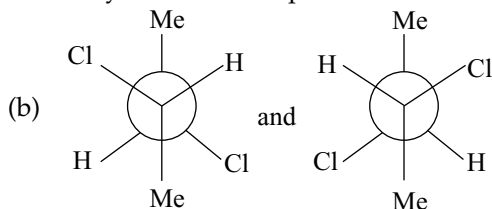
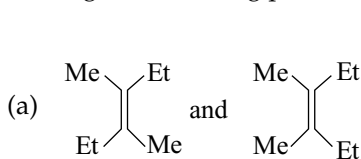
102. Which of the following alkane can not be synthesised by Wurtz reaction in good yield?



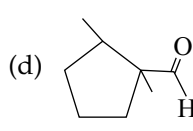
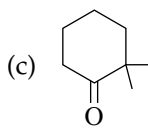
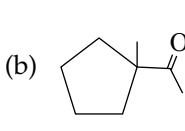
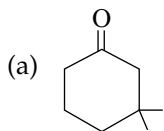
103. $\text{HO}-\text{CH}_2-\text{CH}=\text{C}(\text{CH}_3)-\text{CH}_2-\text{CH}=\text{C}(\text{CH}_3)_2 \xrightarrow[\text{H}_2\text{O}]{\text{H}^+}$ product and intermediate form is/are:



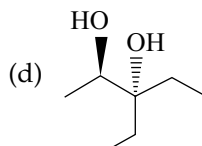
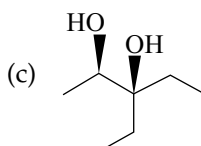
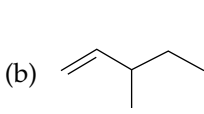
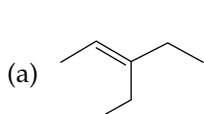
104. Among the following pair(s) of compounds identify enantiomeric pairs



105. $\xrightarrow{\text{H}^+}$ possible products obtained is/are



106. $\text{CH}_3\text{CH}_2\text{COCl} \xrightarrow[\text{H}_3\text{O}^+]{\text{EtMgBr (excess)}} (\text{A}) \xrightarrow[\Delta]{\text{H}_2\text{SO}_4} (\text{B}) \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) Br}_2/\text{H}_2\text{O}/\text{NaOH}} (\text{C})$; Products B and C are



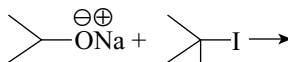
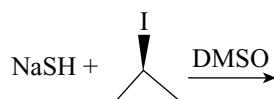
107. Which statement is/are correct in the following?

- (a) Allyl bromide gives S_N1 reaction but vinyl chloride does not.
 (b) Primary alkyl halides can give S_N1 or S_N2 reaction. This depends on the structure of the substrate and nature of solvent.
 (c) $\text{CH}_3\text{-CH=CH-CH}_2\text{Cl}$ reacts with KCN to give mixture of two isomeric products.
 (d) $\text{CH}_3\text{-CH}_2\text{-O-CH}_2\text{Br}$ is less reactive than tert. butyl bromide for S_N1 reaction.

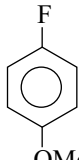
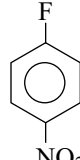
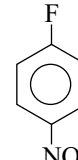
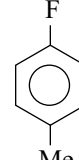
108. Test for identification of But-2-ene and benzene is

- (a) Tollen's Reagent test (b) 1% Alkaline KMnO_4
 (c) Iodoform test (d) $\text{Br}_2 + \text{H}_2\text{O}$ test

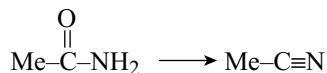
109. Identify the reaction that follows S_N2 path

- (a)  (b) $\text{NaI/Acetone} + \text{isopropyl chloride} \rightarrow$
 (c)  (d) $\text{NaN}_3 + \text{1-iodobutane} \xrightarrow{\text{DMF}}$

110. Among the following compounds, which liberates F^- on reaction with MeO^- ?

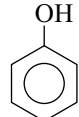
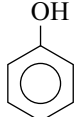
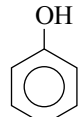
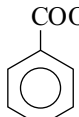
- (a)  (b)  (c)  (d) 

111. Identify reagents used for the following conversions

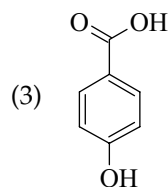
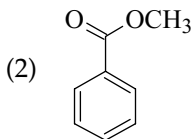
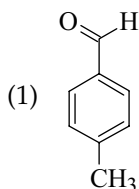


- (a) $\text{P}_2\text{O}_5/\Delta$ (b) NaCl/Δ (c) POCl_3/Δ (d) $\text{Al}_2\text{O}_3/\Delta$

112. Among the following reactions, which form salicylic acid (after acidification)?

- (a)  + $\text{CHCl}_3 + \text{NaOH} \longrightarrow$ (b)  + $\text{CCl}_4 + \text{NaOH} \longrightarrow$
 (c)  + $\text{CO}_2 + \text{NaOH} \longrightarrow$ (d)  + $\text{NaOH} \longrightarrow$

113. Choose the order for increasing boiling point of the following compounds.



(1) (2) (3)
 increasing
 (a) ——— boiling ———>
 point

(2) (1) (3)
 increasing
 (b) ——— boiling ———>
 point

(1) (3) (2)
 increasing
 (c) ——— boiling ———>
 point

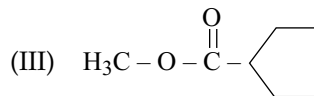
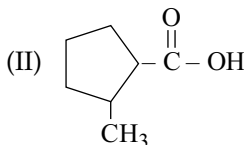
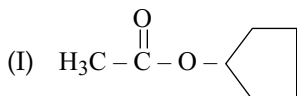
(2) (3) (1)
 increasing
 (d) ——— boiling ———>
 point

114. ; identify reagent which reduces both groups

- (a) NaBH_4
 (c) Zn-Hg/HCl

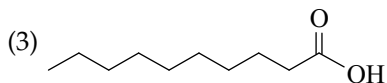
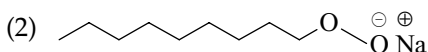
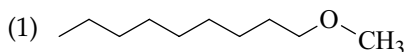
- (b) LAH
 (d) $\text{NH}_2\text{-NH}_2/\text{OH}^-/\Delta$

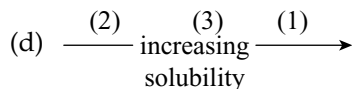
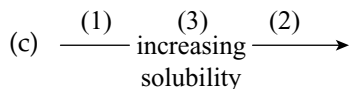
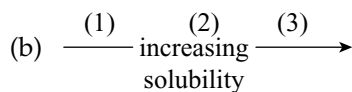
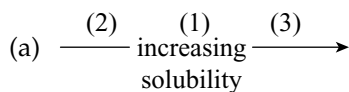
115. For the given compounds, correct statement is/are



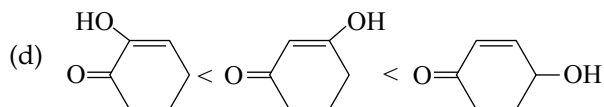
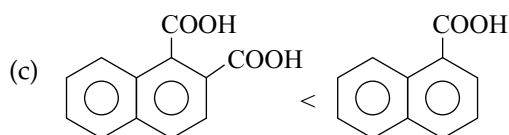
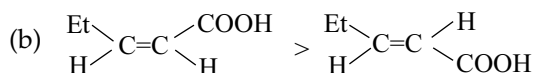
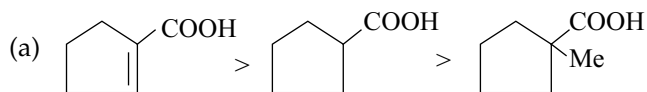
- (a) I and II are positional isomers
 (b) II and III are functional group isomers
 (c) I and III are metamers
 (d) I and III are positional isomers

116. Choose the order that has the following compounds correctly arranged with respect to increasing solubility in water.

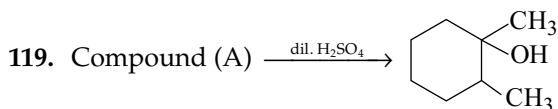
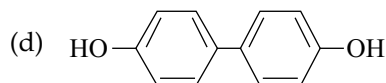
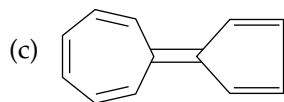
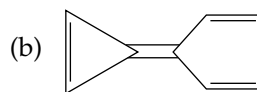
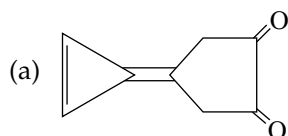




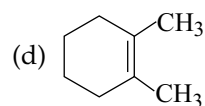
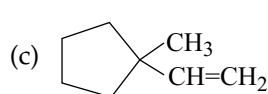
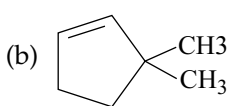
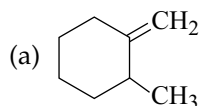
117. Which of following order is/are correct?



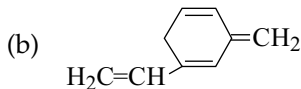
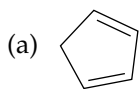
118. Which has non-zero dipole moment?



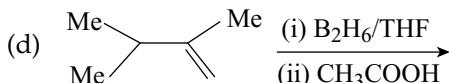
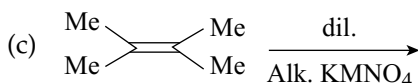
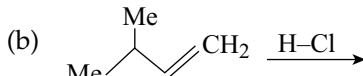
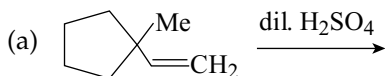
Compound (A) can be



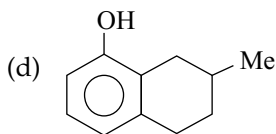
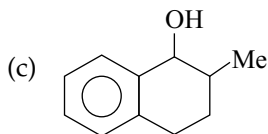
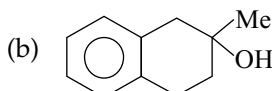
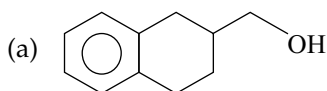
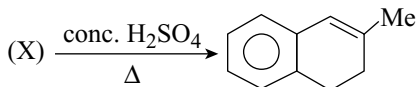
120. Which of the following gives glyoxal as one of the product on ozonolysis?



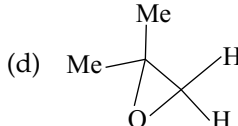
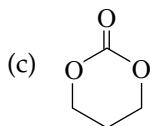
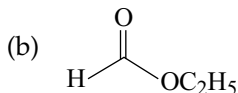
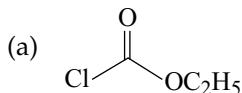
121. In which of the following reactions the rearrangement of carbocation is involved?



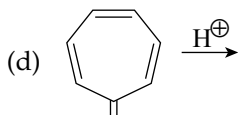
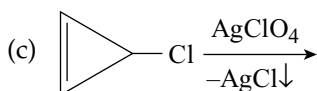
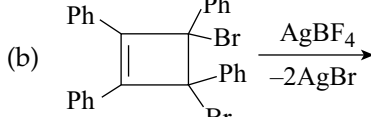
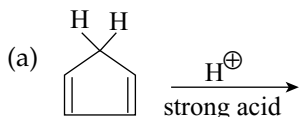
122. In the given reaction, the possible structure of compound (X) is



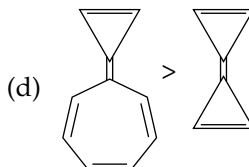
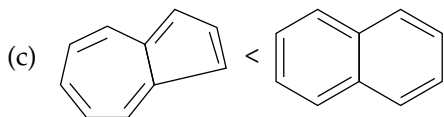
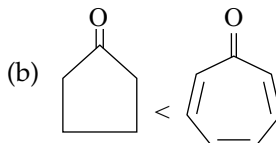
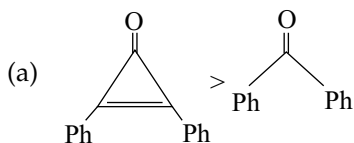
123. Which of the following compounds will give or tertiary alcohol as a resultant product when they are treated with either 1 mole of MeMgBr or with excess MeMgBr followed by H₂O?



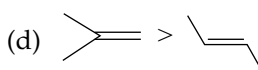
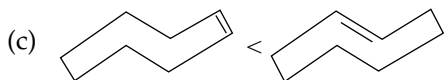
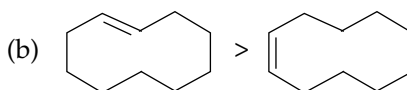
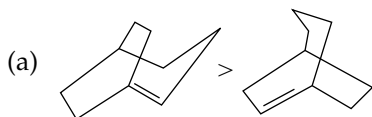
124. Which of the following reactions give an aromatic product?



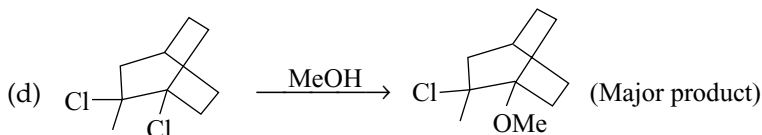
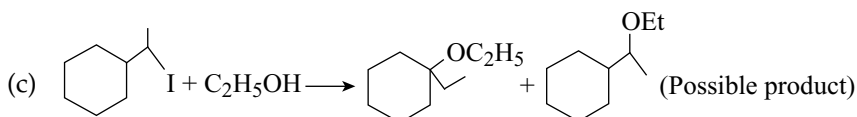
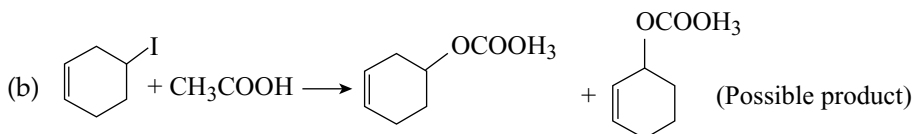
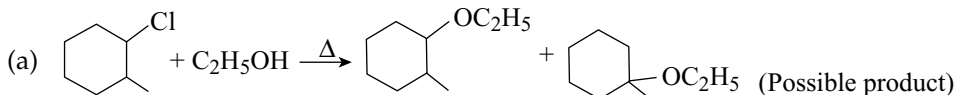
125. Identify the correct order of dipole moment



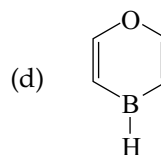
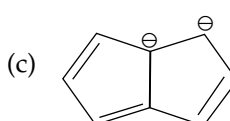
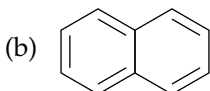
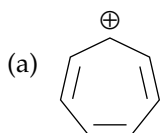
126. Identify the correct order of heat of hydrogenation



127. Which of the following reactions give the correct product?

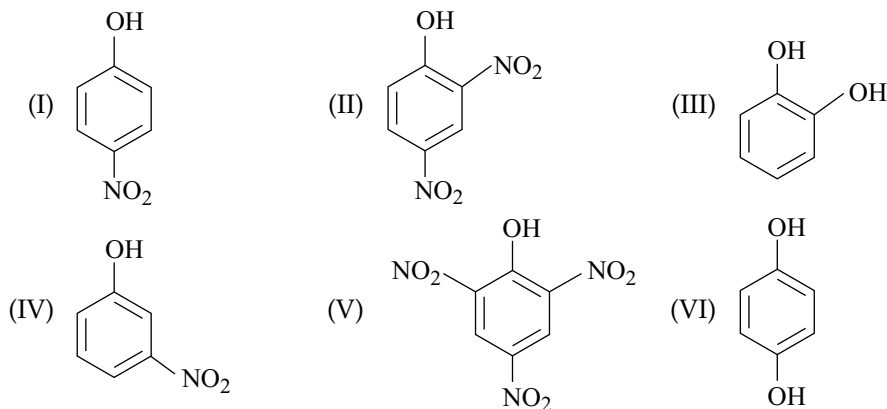


128. From the compound shown below, choose which is aromatic



Comprehension Type

Passage 1



129. The correct order of K_a value is:
 (a) I > III > IV (b) IV > III > VI (c) V > VI > III (d) II > I > IV
130. Which of the following gives effervescence with NaHCO_3 ?
 (a) II (b) VI (c) III (d) IV
131. Which of the following pairs of phenol derivative are stronger acid than phenol?
 (a) I, II, III (b) II, IV, VI (c) I, II, VI (d) V, III, VI

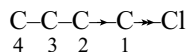
Passage 2

In a covalent single bond between unlike atoms, the electron pair forming the σ bond is never shared absolutely equally between the two atoms; it tends to be attracted a little more towards the more electronegative atom of the two.

This is generally represented as



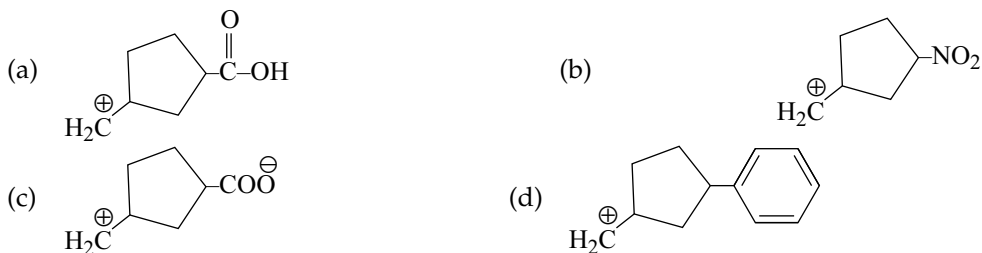
If the carbon atom bonded to chlorine is itself attached to further carbon atoms, the effect can be transmitted further as



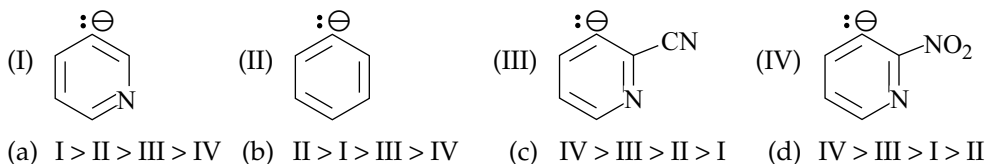
The effect of Cl on C_2 is less than the effect of Cl on C_1 ; however, the transmission quickly dies away in a saturated chain, usually being too small to be noticeable beyond C_2 . These influences on the electron distribution in σ bonds are known as **inductive effects**.

Electron releasing groups w.r.t. the hydrogen atom are known to have +I effect and electron withdrawing groups are known to have -I effect. Electron donating group increases the stability of carbocation and withdrawing group increases the stability of carbanion.

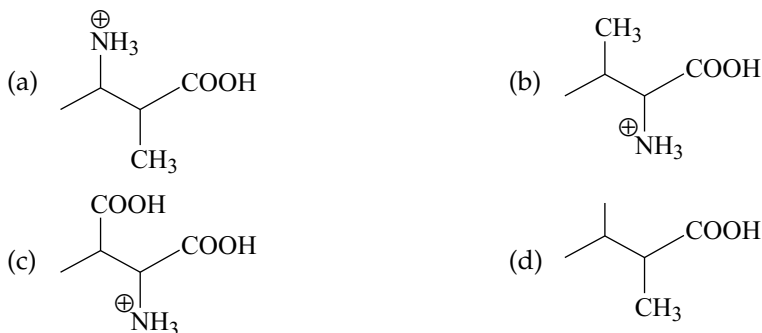
132. Which of the following carbocation is expected to be most unstable?



133. Correct order of the stability of the given carbanion is



134. Most acidic compound in aqueous medium is



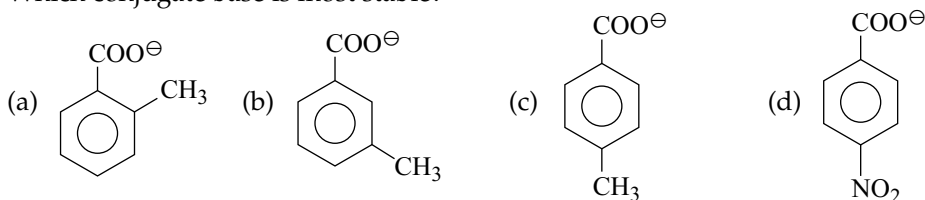
Passage 3

Benzoic acid is more acidic than acetic acid. Acidity of formic acid is more than the benzoic acid. Among monosubstituted benzoic acid derivatives, the ortho derivative is most acidic due to ortho effect. Acidity of any acid can be explained by the stability of conjugate base of the acid.

135. Which of the following is most acidic in character?

- (a) o-nitrobenzoic acid (b) p-nitrobenzoic acid
(c) m-nitrobenzoic acid (d) Benzoic acid

136. Which conjugate base is most stable?



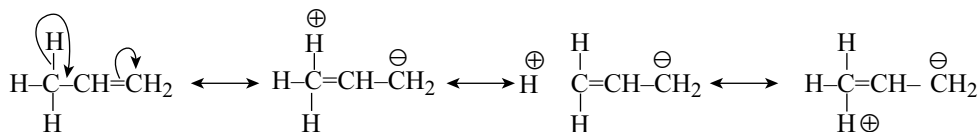
137. Arrange acidity of the given compounds increasing order

- (I) p-nitrophenol (II) p-fluorophenol (III) p-chlorophenol
 (a) (I), (II), (III) (b) (II), (I), (III) (c) (II), (III), (I) (d) (III), (II), (I)

Passage 4

When (C–H) sigma electrons are in conjugation with pi bond, this conjugation is known as σ (C–H) π conjugation, excessive conjugation or hyperconjugation.

- (i) Compound should have at least one sp^2 -hybrid carbon of either alkene, alkyl carbocation or alkyl free radical.
 (ii) α -carbon with respect to sp^2 hybrid carbon should have at least one hydrogen.
 (iii) Resonating structures due to hyperconjugation may be written involving “no bond” between the alpha carbon and hydrogen atoms.



In the above resonating structures there is no covalent bond between carbon and hydrogen, and from this point of view, hyperconjugation may be regarded as “no bond resonance”. Actually the hydrogen atom is not free from the carbon. These resonating structures only suggest that: (a) there is some ionic character between C–H bond and (b) carbon–carbon double bond acquires some single bond character.

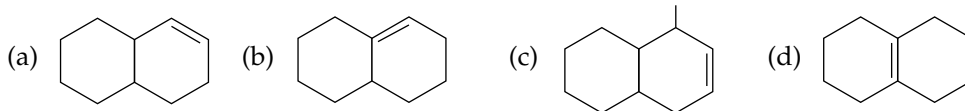
We can explain the stability of alkene, carbocation and carbon free radical on the basis of hyperconjugation.

$$\text{Stability of alkene} \propto \text{number of } \alpha\text{-H} \propto \frac{1}{\text{Heat of hydrogenation}}$$

138. Which of the following statements are correct for $C_6H_5-CCl_3$?

- (a) CCl_3 group is electron withdrawing due to the –I effect and reverse hyperconjugation.
 (b) CCl_3 group is meta directing due to the –M effect.
 (c) CCl_3 group is o, p-directing because it is +R group.
 (d) CCl_3 group can exert +M effect.

139. Which of the following has the lowest heat of hydrogenation?



140. Carbon–carbon double bond length will be maximum in which of the following compounds?

- (a) $CH_3-CH=CH_2$ (b) $CH_3-CH=CH-CH_3$
 (c) $CH_3-\overset{\overset{CH_3}{|}}{C}=C-\overset{\underset{CH_3}{|}}{CH_3}$ (d) $CH_2=CH_2$

Passage 5

For a compound to be classified as aromatic, it must fulfill both of the following criteria.

- (i) It must have an uninterrupted cyclic cloud of π electrons above and below the plane of the molecule (often called a π cloud).

For the π cloud to be cyclic, the molecule must be cyclic.

For the π cloud to be uninterrupted, every atom in the ring must have a p orbital.

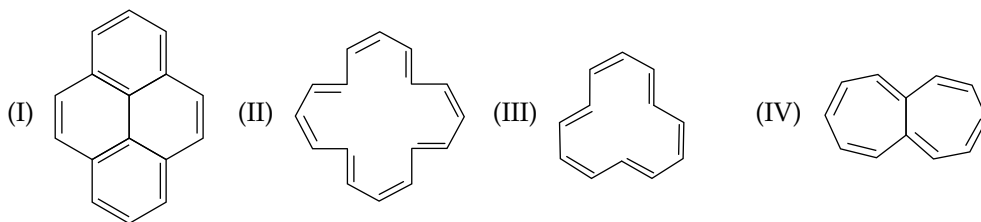
For the π cloud to form, each p orbital must be able to overlap with the p orbitals on either side of it. Therefore, the molecule must be planar.

- (ii) The π cloud must contain an odd number of pairs of π electrons.

Benzene is an aromatic compound because it is cyclic and planar, every carbon in the ring has a p orbital and the π cloud contains three pairs of π electrons.

The German chemist Erich Huckel was the first to recognise that an aromatic compound must have an odd number of pairs of π electrons. He described this requirement by what has come to be known as Huckel's rule, or the $4n + 2$ rule. The rule states that for a planar, cyclic compound to be aromatic, its uninterrupted π cloud must contain $(4n + 2)$ π electrons, where n is any whole number. According to Huckel's rule, then an aromatic compound must have 2 ($n = 0$), 6 ($n = 1$), 10 ($n = 2$), 14 ($n = 3$), 18 ($n = 4$), etc. π electrons. Because there are two electrons in a pair, Huckel's rule requires that an aromatic compound must have 1, 3, 5, 7, 9, etc. pairs of π electron. Thus, Huckel's rule is just a mathematical way of saying that an aromatic compound must have an odd number of pairs of π electrons. For an anti-aromatic system a planar, cyclic compound must contain $(4n)$ π electrons.

141. Which one of the following compounds is non-aromatic?

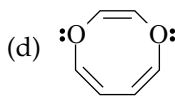
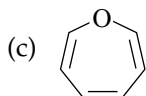
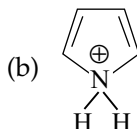
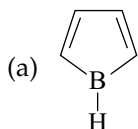


- (a) only I (b) I and IV (c) II, III, IV (d) I and II

142. Which of the following will show aromatic behaviour?

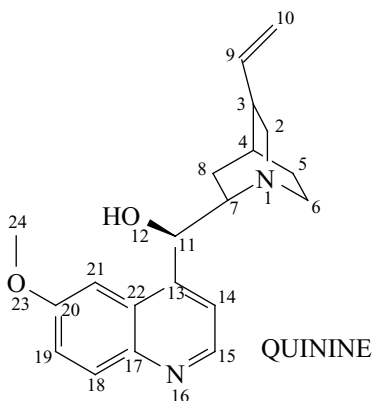


143. Which of the following is a non-aromatic system?



Passage 6

For each of the questions 149–154 about QUININE (shown right), select the answer from those provided.



144. What is the oxidation state of C11?

- (a) +1 (b) 0 (c) -1 (d) -2

145. Of the following list, which atom(s) is (are) sp^3 hybridised?

- (a) O12 (b) N16 (c) C21 (d) N1

146. Which of the following carbon atoms are tertiary?

- (a) C24 (b) C19 (c) C20 (E) C22

147. Which carbon atom(s) is (are) ortho to a methoxy group?

- (a) C5 (b) C6 (c) C7 (d) C21

148. Which of the following functional groups are found in QUININE?

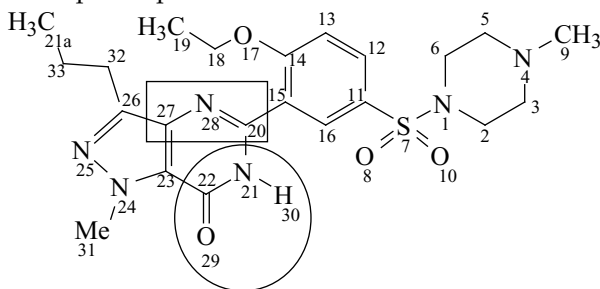
- (a) Alcohol (b) Amide
(c) Ester (d) Ether

149. How many units of unsaturation are there in QUININE? (units of unsaturation are the same as the index of hydrogen deficiency or IHD)

- (a) 7 (b) 8 (c) 9 (d) 10

Passage 7

In the year of its launch, VIAGRA (below) was used by over three million satisfied customers. Each of the questions below refers to the structure of VIAGRA. Now select the answer(s) from the options provided.



150. What are the oxidation states of C3 and N25?
 (a) -1, -2 (b) +1, -2 (c) -1, +3
 (d) -1, -3 (e) +1, -3
151. What is the functional group in the circle?
 (a) amide (b) amine (c) aniline (d) nitrile
152. What is the functional group in the rectangular box?
 (a) epoxide (b) amine (c) ester (d) ether
153. How many units of unsaturation are there in VIAGRA?
 (a) 7 (b) 8 (c) 11
 (d) 12 (e) 13
154. In VIAGRA, what are the hybridisations of O10, N4 and N28, respectively?
 (a) sp^3 , sp^3 , sp^3 (b) sp^2 , sp^3 , sp^3 (c) sp^3 , sp^2 , sp^3 (d) sp^2 , sp^3 , sp^2
155. In VIAGRA, what are the hybridisations of C2, C3 and N24, respectively?
 (a) sp^3 , sp^3 , sp^3 (b) sp^2 , sp , sp^3 (c) sp^3 , sp^2 , sp^3 (d) sp^3 , sp^3 , sp^2
156. How many carbon chiral centres are there in VIAGRA?
 (a) 0 (b) 1 (c) 2 (d) 3

Matrix Type

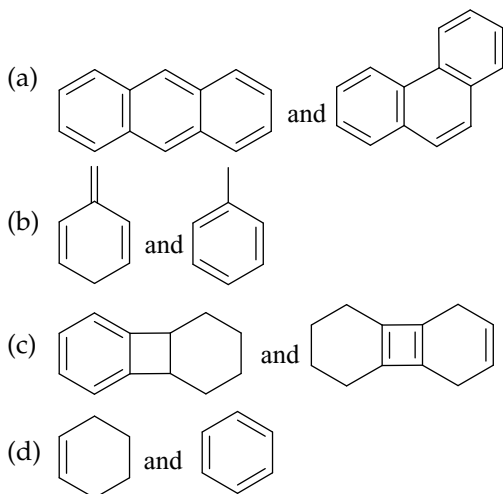
157. Column I (Groups)

- (a) $-\overset{\oplus}{N}Me_3$
 (b) $-\overset{\oplus}{P}Me_3$
 (c) $-OMe$
 (d) $-CH_3$

Column II (Effect)

- (p) +I
 (q) -I
 (r) +R
 (s) -R
 (t) +H

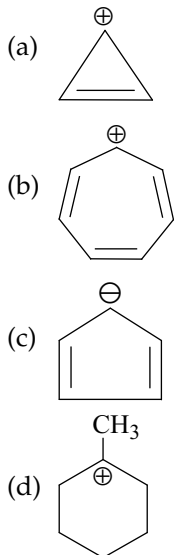
158. Column I



Column II

- (p) (I) compound has more heat of hydrogenation than (II)
- (q) (I) compound has more resonance energy than (II) compound
- (r) (I) compound is more stable than (II)
- (s) (II) compound has more heat of hydrogenation than (I)
- (t) (II) compound has more degree of unsaturation than (I)

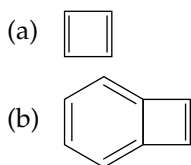
159. Column I



Column II

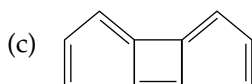
- (p) Hyperconjugation
- (q) All carbon atoms are sp^2 -hybridised
- (r) Aromatic
- (s) Resonance
- (t) Diamagnetic

160. Column I



Column II

- (p) Aromatic
- (q) Antiaromatic



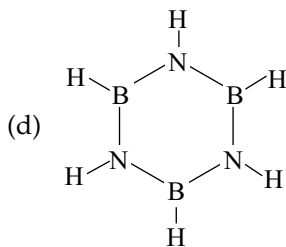
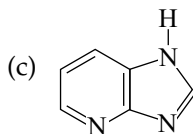
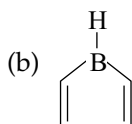
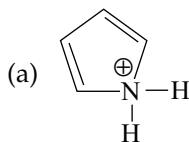
(d) C_8H_8

(r) Readily reacts with metal to form stable dianion

(s) Paramagnetic in nature due to presence of unpaired electrons in molecular orbitals

(t) Diamagnetic in nature due to presence of even electrons in molecular orbitals

161. Column I



Column II

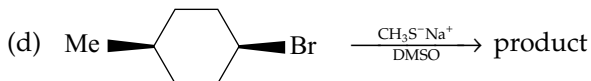
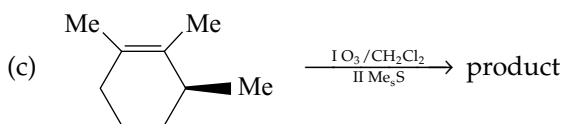
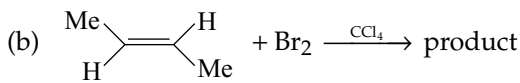
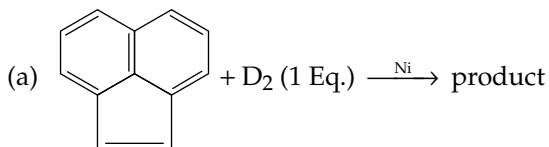
(p) Aromatic

(q) Antiaromatic

(r) Nonaromatic

(s) Degree of unsaturation = 3

162. Column I (Reaction)



Column II (Product)

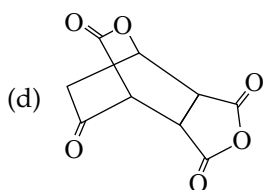
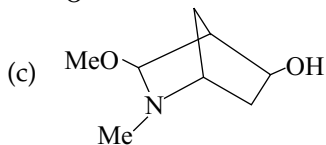
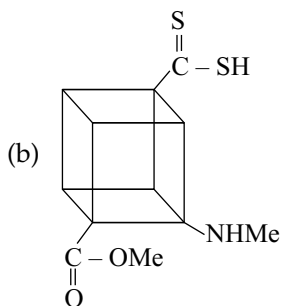
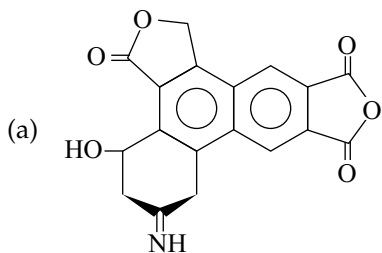
(p) Contains even chiral carbons

(q) Contains odd chiral carbon

(r) Optically inactive

(s) Contains plane of symmetry

163. Column I



Column II

(p) Total number of functional groups in the compound is even. [excluding $>C=C<$ bond]

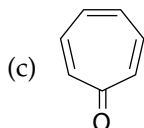
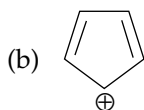
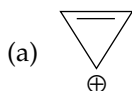
(q) Total number of functional groups in the compound is odd. [excluding $>C=C<$ bond]

(r) Degree of unsaturation in the compound is even

(s) Degree of unsaturation in the compound is odd

(t) Lactone is present as a functional group in the molecule

164. Column I

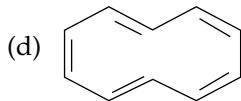


Column II

(p) Antiaromatic

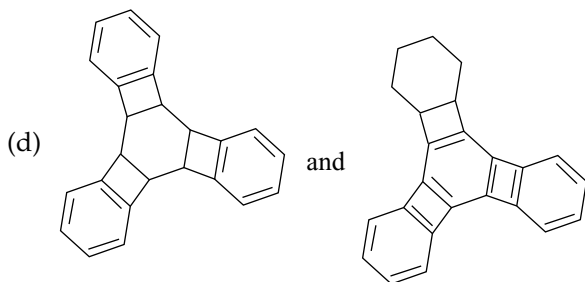
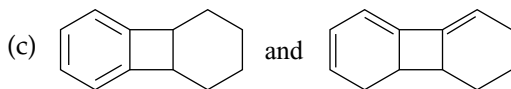
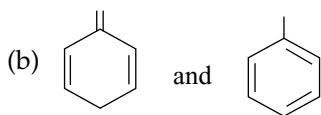
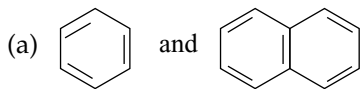
(q) Aromatic

(r) Nonaromatic



(s) Degree of unsaturation (D.O.U) = 6

165. Column I (group)



Column II (effect of group)

(p) Ist compound has more heat of hydrogenation than the IInd

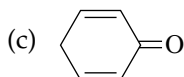
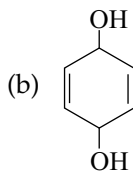
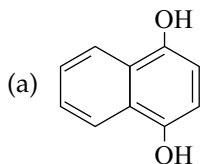
(q) Ist compound has more resonance energy than the IInd

(r) Ist compound is more stable than the IInd

(s) IInd compound has more heat of combustion than the Ist

(t) IInd compound has more degree of unsaturation than the Ist

166. Column I (compound)

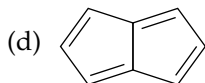


Column II (relationship)

(p) Aromatic

(q) Nonaromatic

(r) Dipolemoment is nonzero

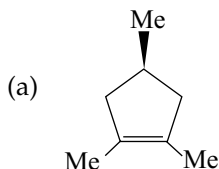


(s) Antiaromatic

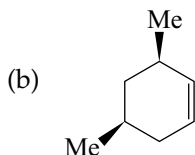
(t) Dipolemoment is zero

167. Column I

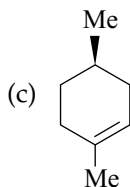
Column II



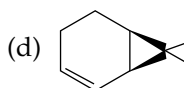
(p) Compound which is optically active



(q) Compound which is optically inactive due to the presence of plane of symmetry and gives optically inactive hydrogenation product (major product)



(r) Compound which is optically active but gives optically inactive hydrogenation product (major product) with 1 equivalent of H_2 and metal

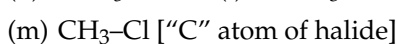
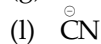
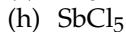
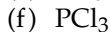
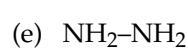
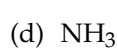
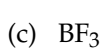
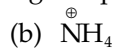
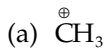


(s) Degree of unsaturation in compound = 2

(t) Compound which give optically active product on ozonolysis

Integer Type

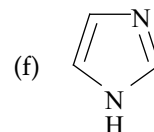
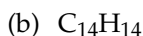
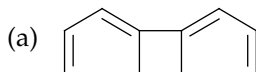
168. From the following compounds/ions:

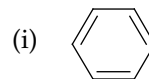
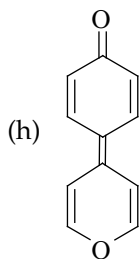
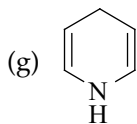


Identify value of "X".

Where "X" is the total number of E^+ /electrophiles.

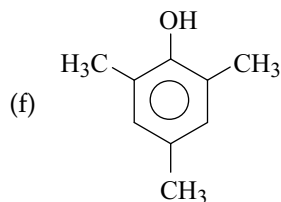
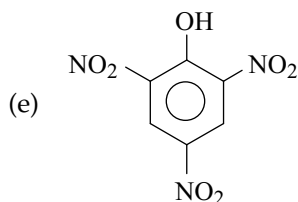
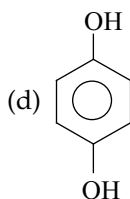
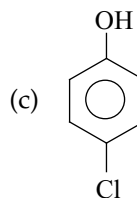
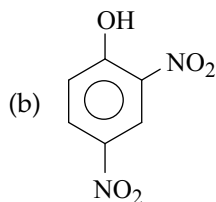
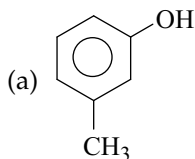
169. From the following compound





Identify the total number of aromatic compounds.

170. The following compounds are phenol derivative

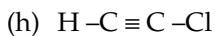
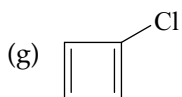
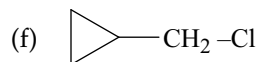
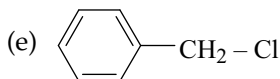
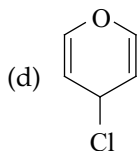
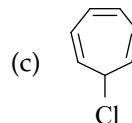
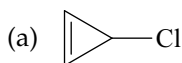


If number of phenol derivative = x.

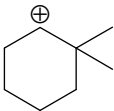
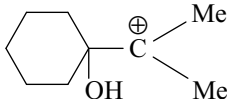
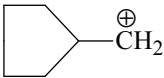
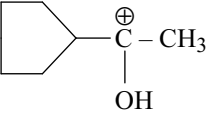
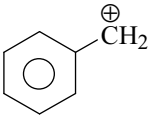
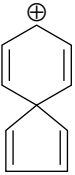
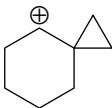
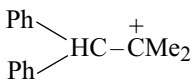
Number of phenol derivative gives +ve test with $\text{NaHCO}_3 = y$.

Then, find value of $x + y$.

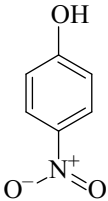
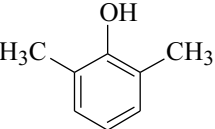
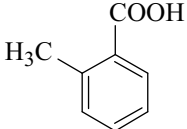
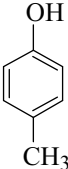
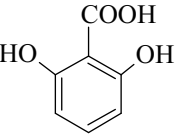
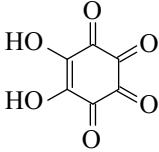
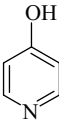
171. How many products will be aromatic ion when AgNO_3 reacts with the following compounds?



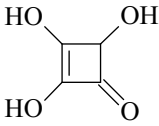
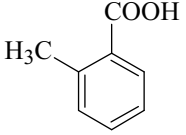
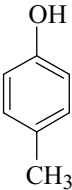
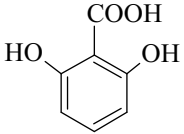
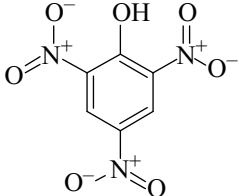
172. How many carbocations undergo rearrangements?

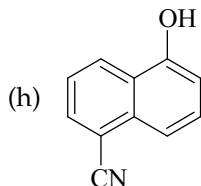
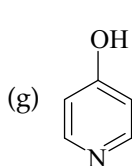
- (i)  (ii)  (iii) 
- (iv)  (v)  (vi) $\text{Me}_3\text{C}-\text{CH}_2-\text{CH}_2^+$
- (vii)  (viii) $\text{CH}_2=\text{CH}^+$ (ix) 
- (x)  (xi) $\text{CH}_2=\text{CH}-\text{CH}_2^+$ (xii) 

173. How many compounds are soluble in aqueous NaHCO_3 ?

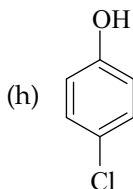
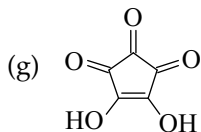
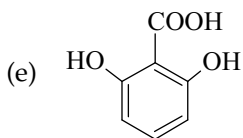
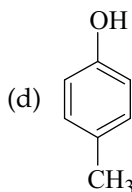
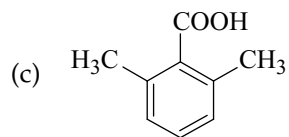
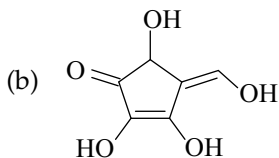
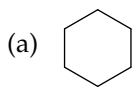
- (a)  (b)  (c) 
- (d)  (e)  (f) 
- (g)  (h) $\text{H}-\text{C}\equiv\text{C}-\text{H}$

174. How many compounds are soluble in aqueous NaOH ?

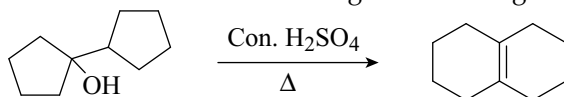
- (a)  (b)  (c) 
- (d)  (e)  (f) 



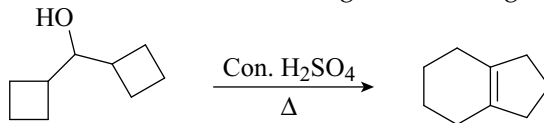
175. How many compounds liberate hydrogen gas on reaction with Na metal?



176. How many transition states are formed during the following reaction?

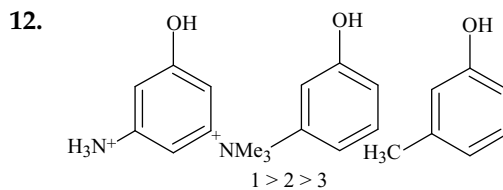
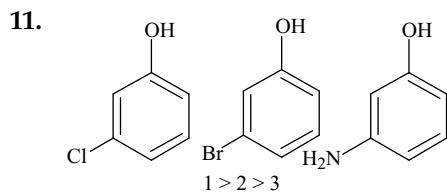
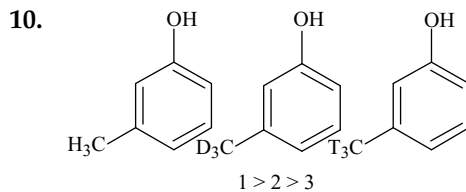
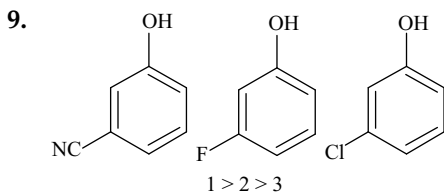
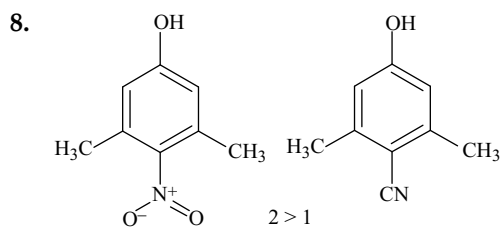
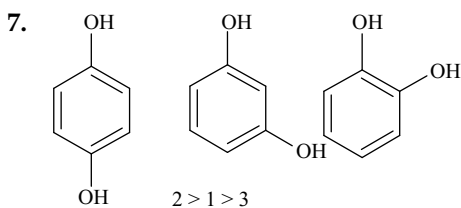
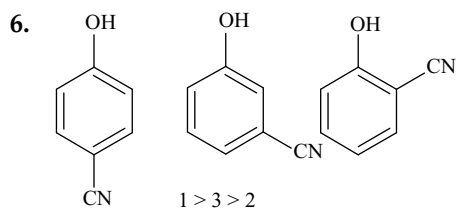
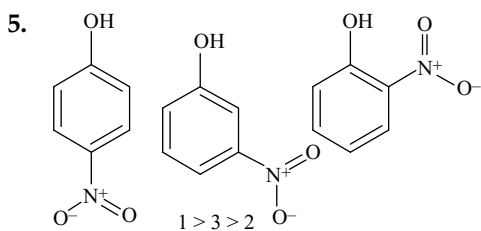
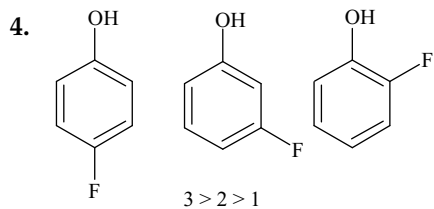
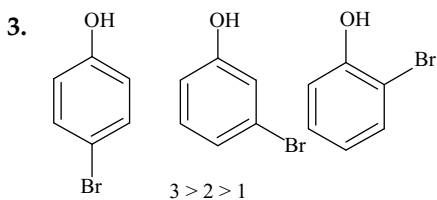
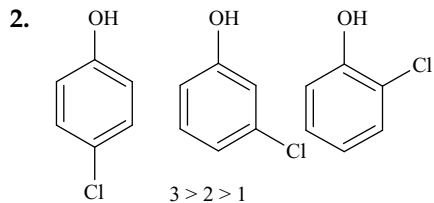
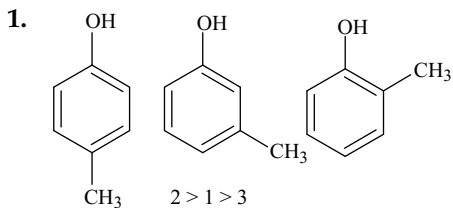


177. How many transition states are formed during the following reaction?



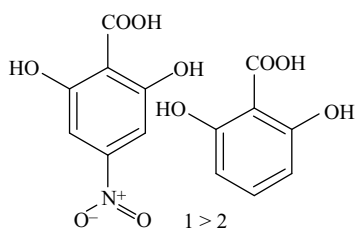
WORKBOOK EXERCISE 1

Correct Acidic Strength of Compounds

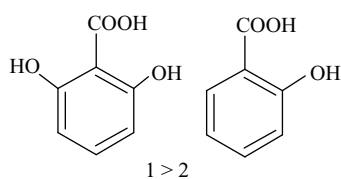


13.
3 > 1 > 2
14.
3 > 2 > 1
15.
2 > 1 > 3
16.
1 > 2 > 3
17.
1 > 2 > 3
18.
1 > 2 > 3
19.
1 > 2 > 3
20.
1 > 2 > 3
21.
3 > 1 > 2
22.
1 > 3 > 2
23.
1 > 2 > 3
24.
1 > 3 > 2
25.
2 > 1
26.
2 > 1

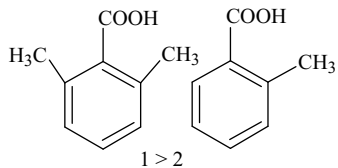
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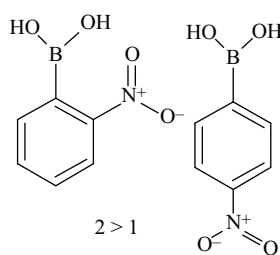
28.



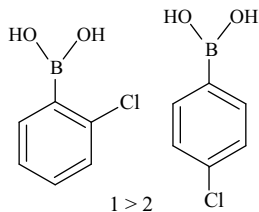
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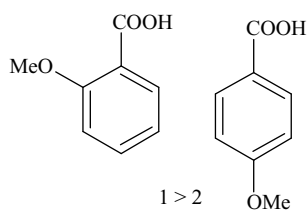
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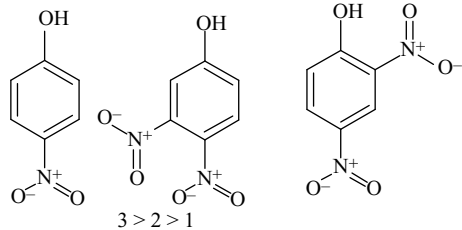
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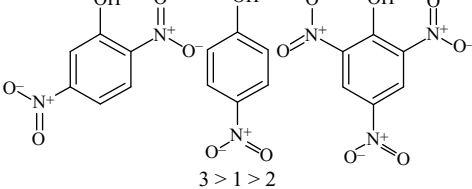
32.



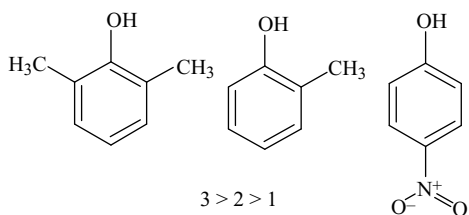
33.



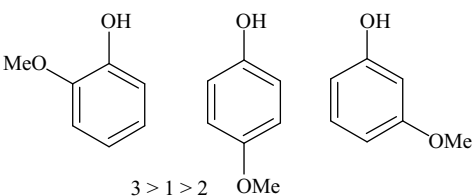
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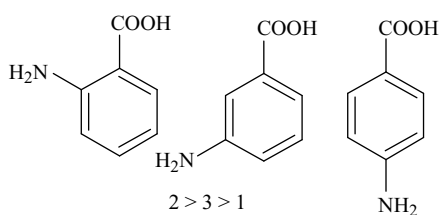
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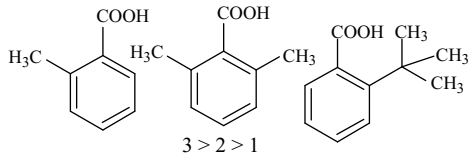
36.



37.

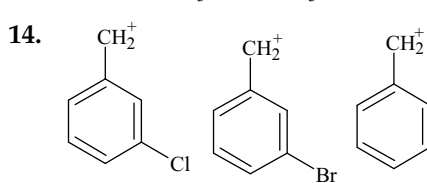
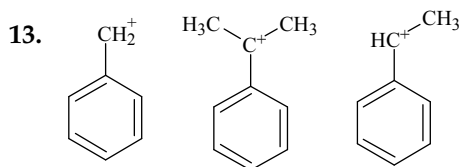
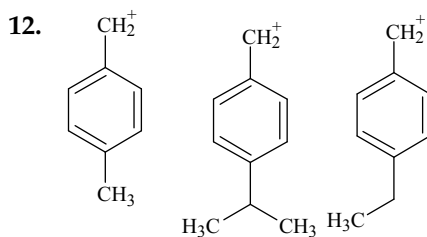
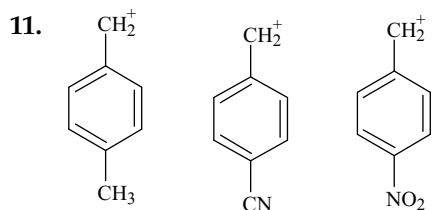
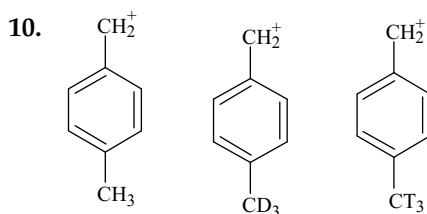
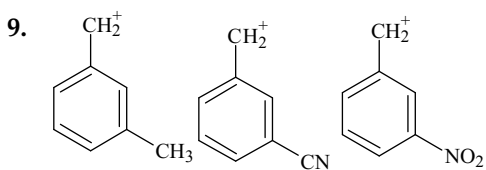
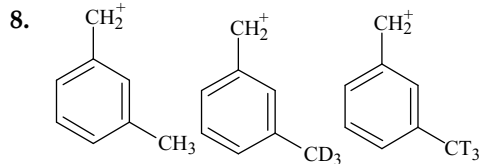
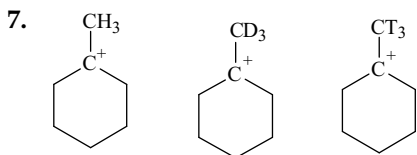
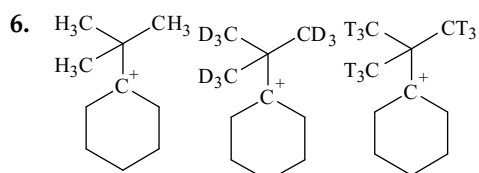
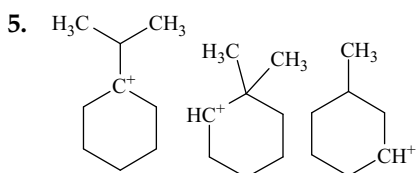
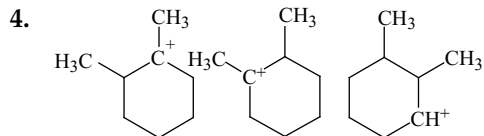
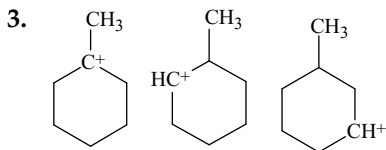
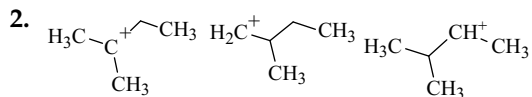
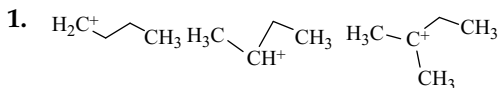


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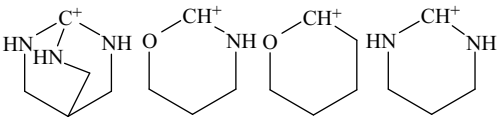
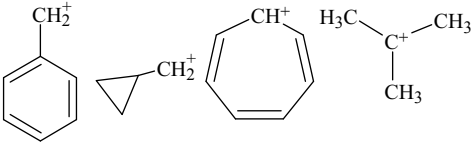
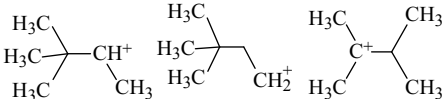
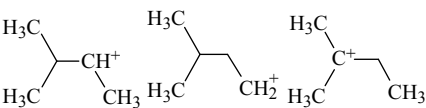
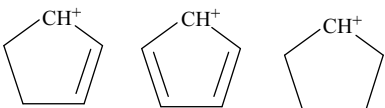
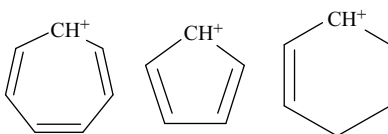
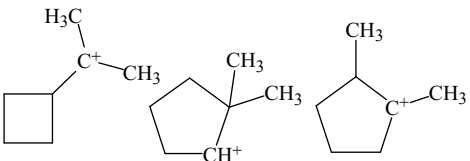
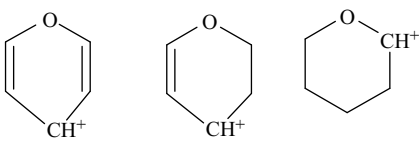
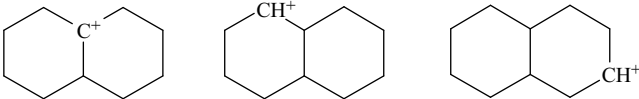


Correct Carbocation Stability Order

Correct Carbocation Stability Order



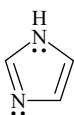
- 15.
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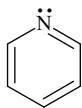
WORKBOOK EXERCISE 3

Identify localised and delocalised lp in the following examples

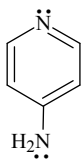
1.



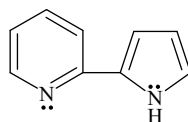
2.



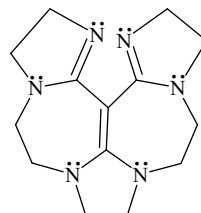
3.



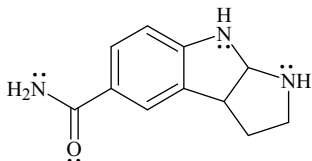
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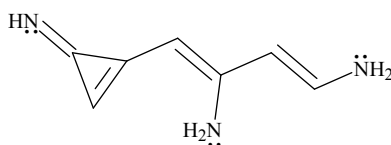
5.



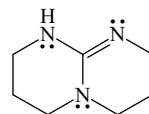
6.



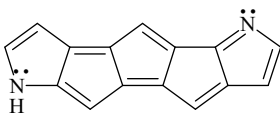
7.



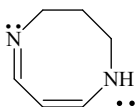
8.



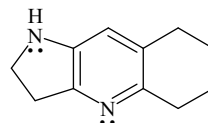
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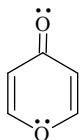
10.



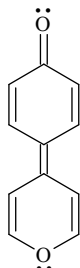
11.



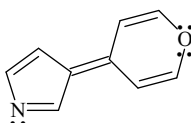
12.



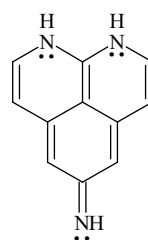
13.



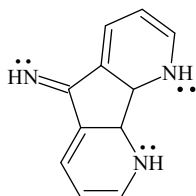
14.



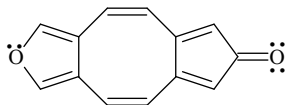
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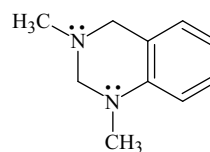
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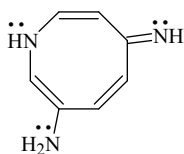
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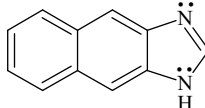
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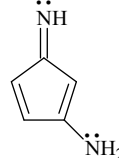
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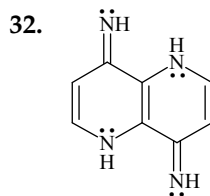
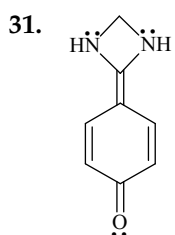
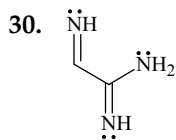
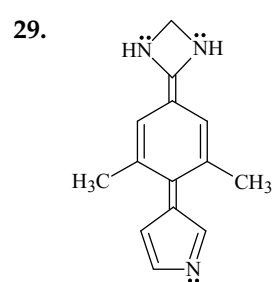
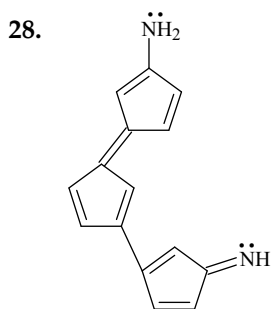
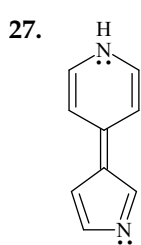
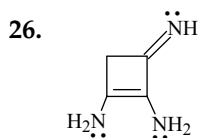
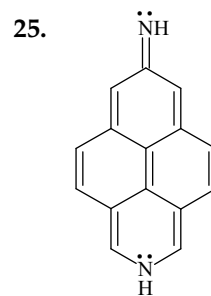
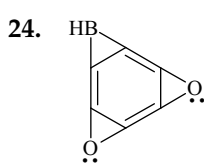
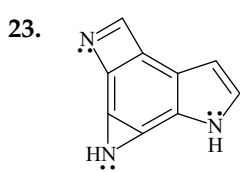
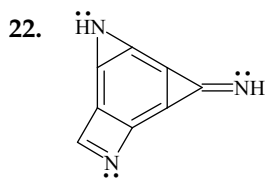


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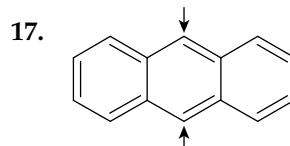
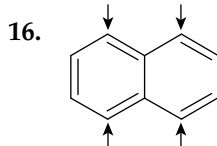
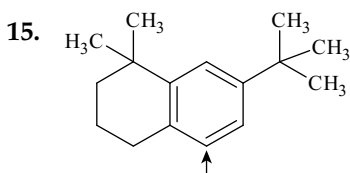
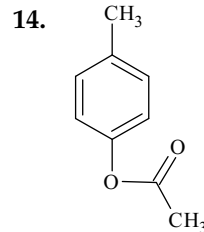
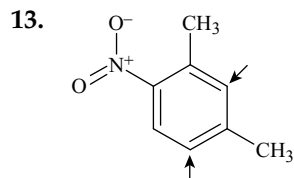
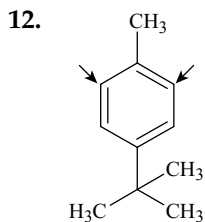
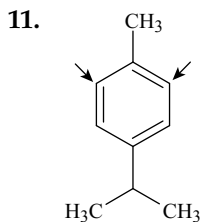
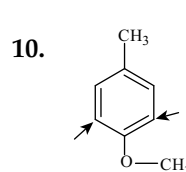
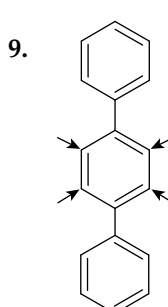
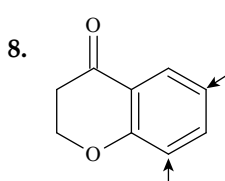
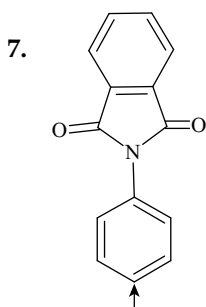
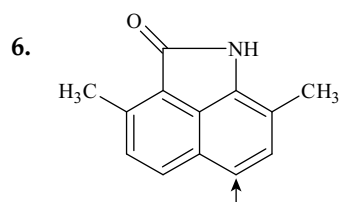
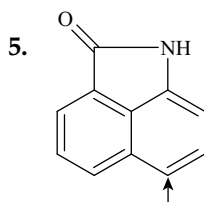
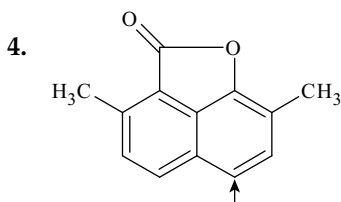
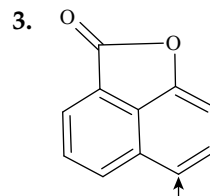
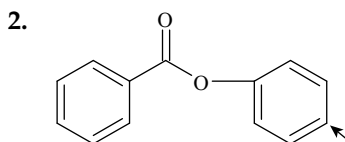
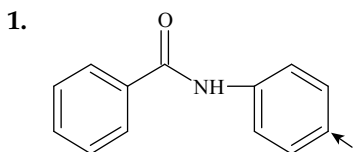
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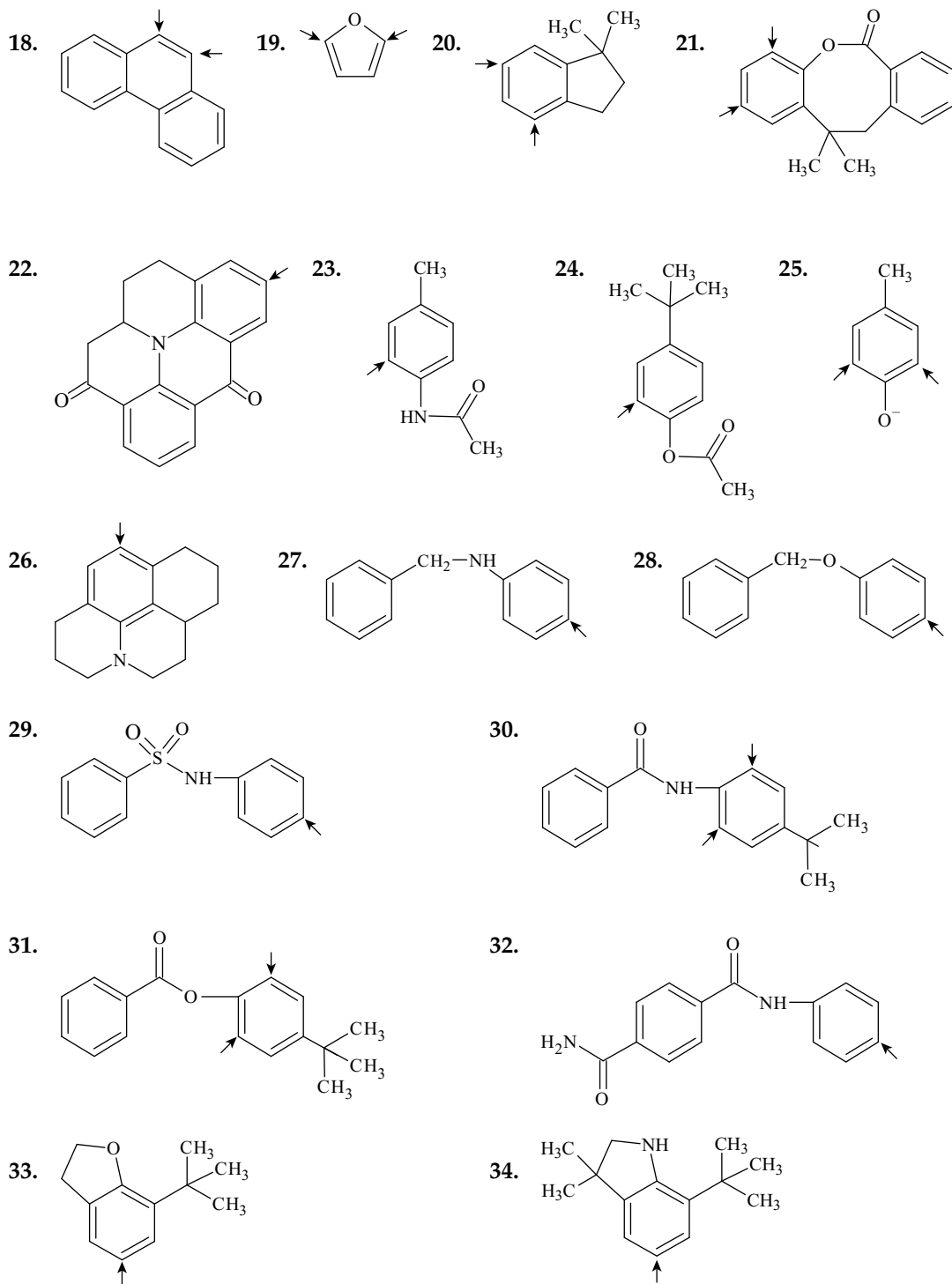




WORKBOOK EXERCISE 4

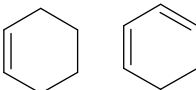
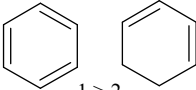
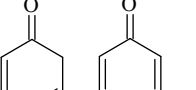
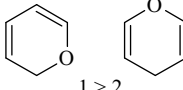
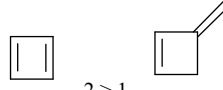
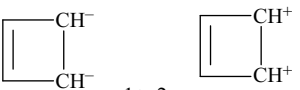
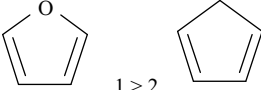
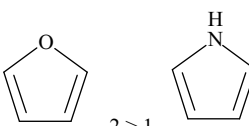
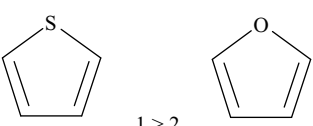
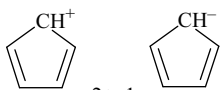
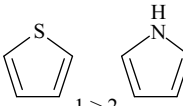
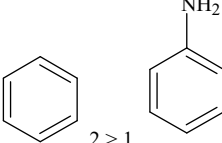
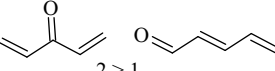
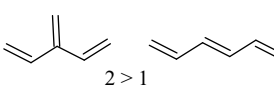
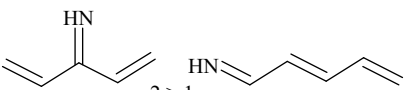
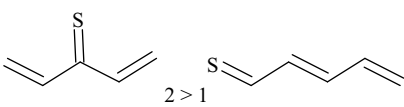
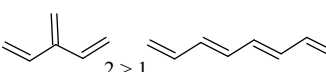
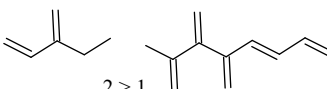
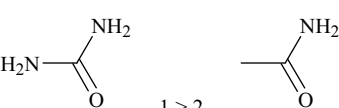
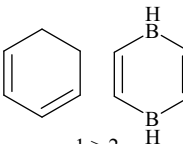
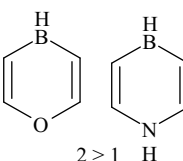
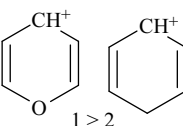
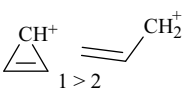
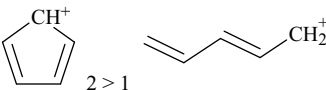
Identify the site of electrophilic attack at benzene ring

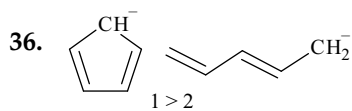
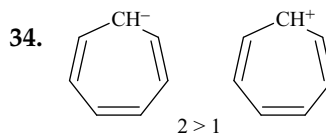
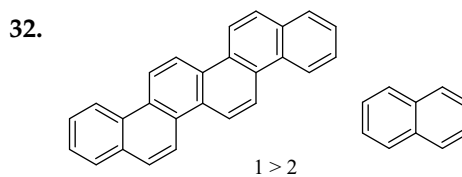
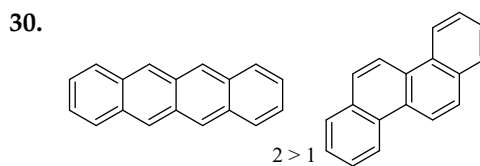
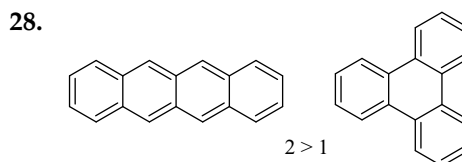
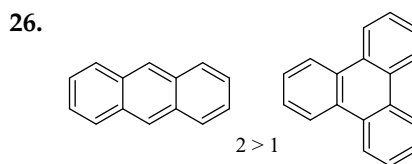
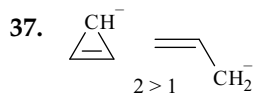
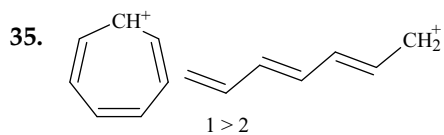
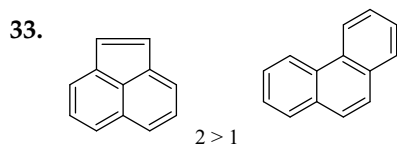
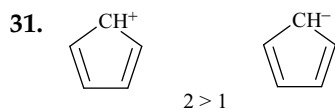
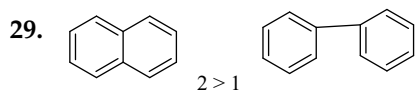
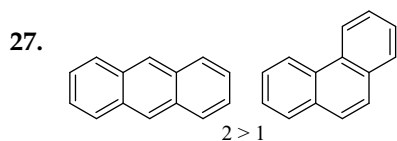
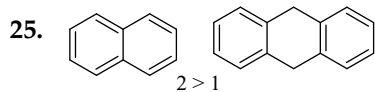




WORKBOOK EXERCISE 5

Identify the correct order of resonance energy for the following pairs of compounds

1.  2 > 1
2.  1 > 2
3.  1 > 2
4.  1 > 2
5.  2 > 1
6.  1 > 2
7.  1 > 2
8.  2 > 1
9.  1 > 2
10.  2 > 1
11.  1 > 2
12.  2 > 1
13.  2 > 1
14.  2 > 1
15.  2 > 1
16.  2 > 1
17.  2 > 1
18.  2 > 1
19.  1 > 2
20.  1 > 2
21.  2 > 1
22.  1 > 2
23.  1 > 2
24.  2 > 1



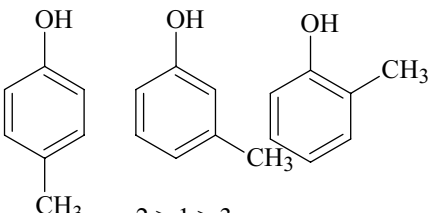
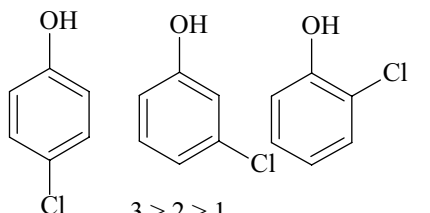
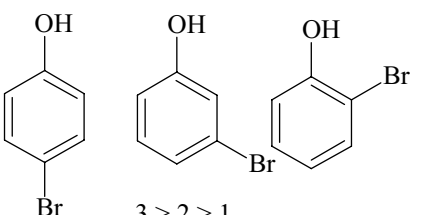
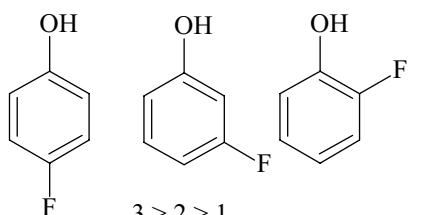
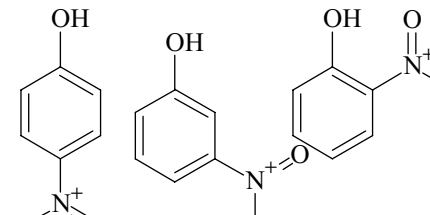
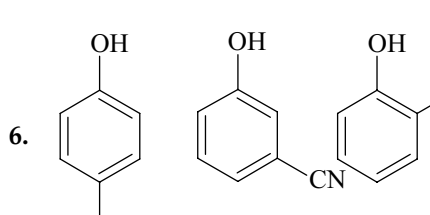
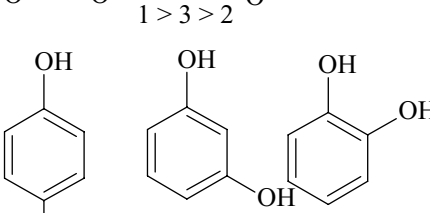
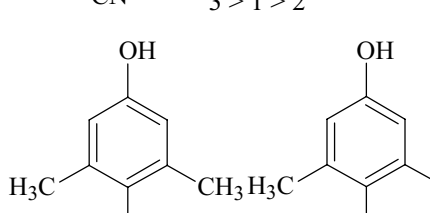
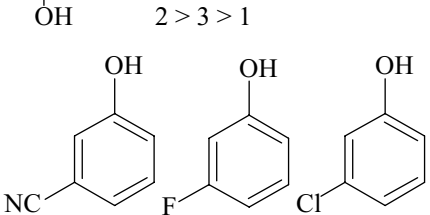
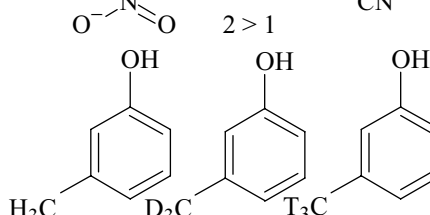
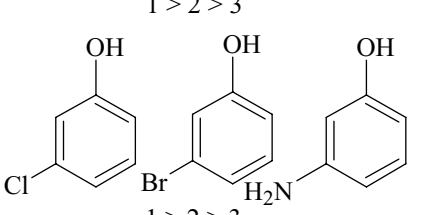
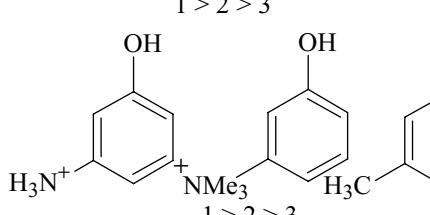
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
d	c	b	a	a	d	b	c	d	a	c	d	d	a	a
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
b	c	d	b	b	a	b	d	a	d	d	c	c	d	d
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
b	d	a	b	d	b	a	c	d	c	d	c	c	a	bc
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
d	ac	b	c	a	a	a	c	b	b	c	a	a	b	c
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
d	d	c	b	b	a	c	d	b	d	d	a	d	R	a
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
c	b	b	c	cd	c	a	c	b	d	b	b	d	d	a
91	92	93	94											
c	a	a	c											

LEVEL 2

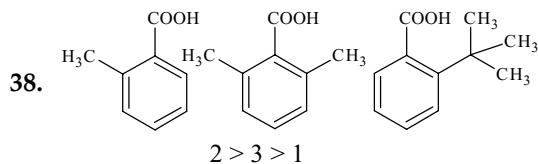
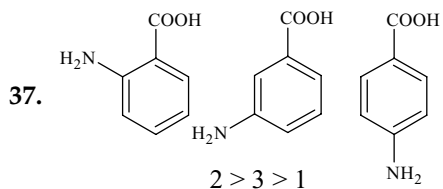
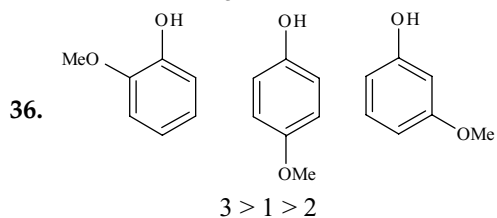
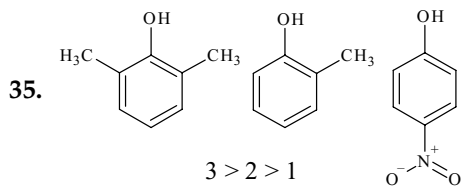
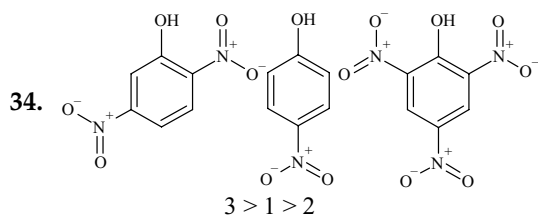
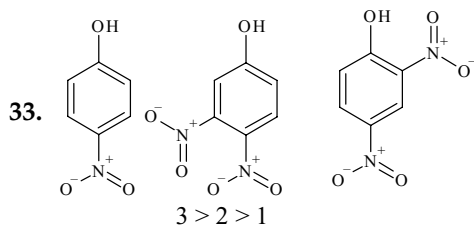
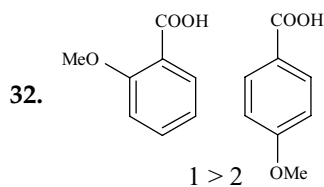
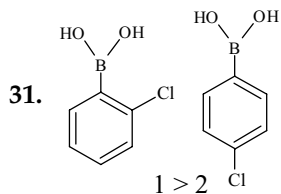
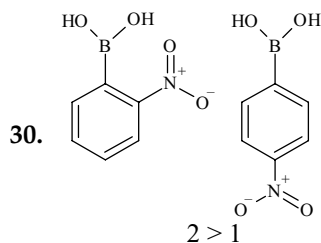
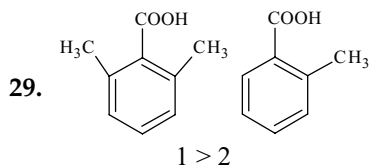
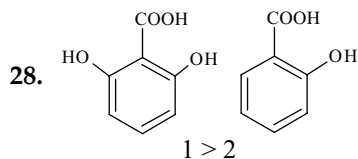
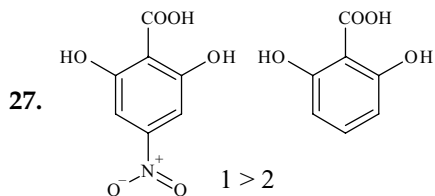
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
c	d	b	d	a	b	b	c	b	c	b	c	a	abcd	b
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
d	a	b	c	b	b	a	b	b	d	a	d	c	a	c
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
a	a	a	b	d	a	a	a	a	a	d	b	d	b	a
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
a	a	a	b	c	b	b	a	a	a	b	b	c	a	d
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
d	c	d	d	c	a	d	b	c	c	c	b	d	a	a
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
a	b	d	a	b	b	a	c	c	b	a	ab	c	d	cd
91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
ab	abc	abd	abcd	bcd	abd	bc	b	bcd	d	abc	ac	acd	cd	bc
106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
ad	abc	bd	bcd	bc	acd	bc	b	bc	bc	c	ab	abcd	ac	acd
121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
ab	abc	acd	abcd	abd	bc	abcd	abcd	d	a	a	b	d	c	a
136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
d	c	a	d	c	c	b	b	a	ad	cd	ad	d	d	c
151	152	153	154	155	156	157(a)	157(b)	157(c)	157(d)	158(a)	158(b)	158(c)	158(d)	159(a)
a	b	c	d	d	a	q	qs	qr	pt	p	p	qrs	st	qrst
159(b)	159(c)	159(d)	160(a)	160(b)	160(c)	160(d)	161(a)	161(b)	161(c)	161(d)	162(a)	162(b)	162(c)	162(d)
qrst	qrst	p	qrs	qrs	pt	r	rs	qs	p	p	prs	prs	q	rs
163(a)	163(b)	163(c)	163(d)	164(a)	164(b)	164(c)	164(d)	165(a)	165(b)	165(c)	165(d)	166(a)	166(b)	166(c)
pst	qs	qr	qst	q	p	p	rs	rst	p	qrs	qrs	pr	qr	qr
166(d)	167(a)	167(b)	167(c)	167(d)	168	169	170	171	172	173	174	175	176	177
rs	qs	prs	prst	prst	8	5	8	4	8	4	7	3	5	7

WORKBOOK EXERCISE 1

Acidic Strength of Compounds

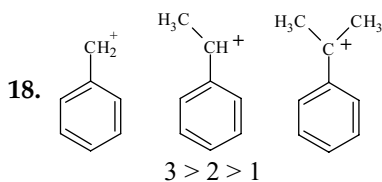
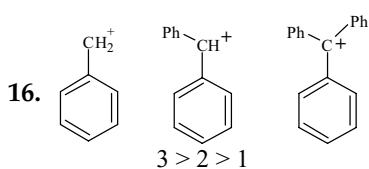
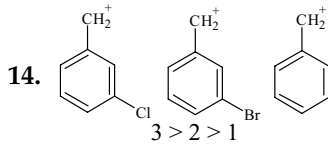
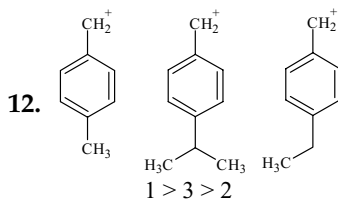
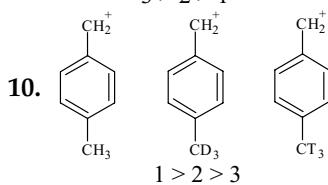
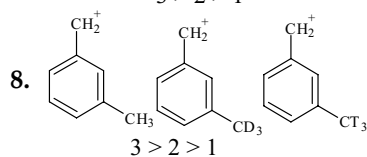
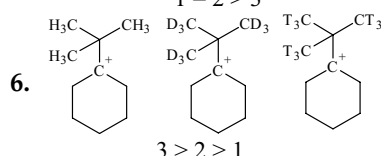
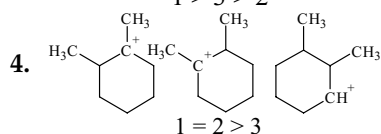
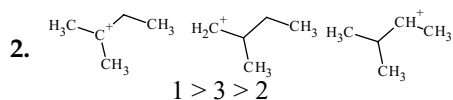
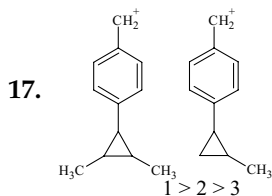
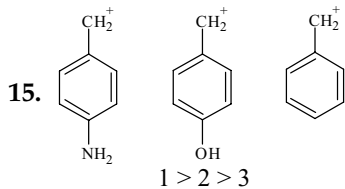
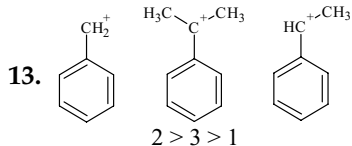
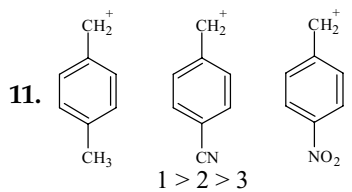
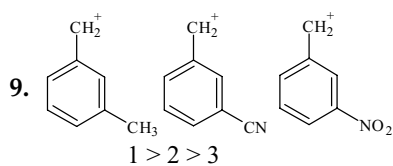
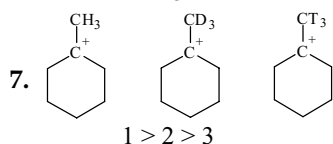
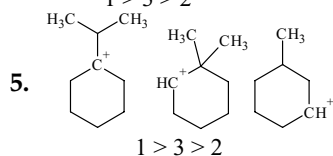
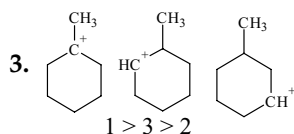
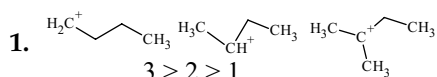
1. 
 $2 > 1 > 3$
2. 
 $3 > 2 > 1$
3. 
 $3 > 2 > 1$
4. 
 $3 > 2 > 1$
5. 
 $1 > 3 > 2$
6. 
 $3 > 1 > 2$
7. 
 $2 > 3 > 1$
8. 
 $2 > 1$
9. 
 $1 > 2 > 3$
10. 
 $1 > 2 > 3$
11. 
 $1 > 2 > 3$
12. 
 $1 > 2 > 3$

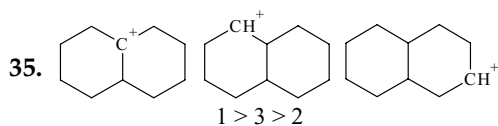
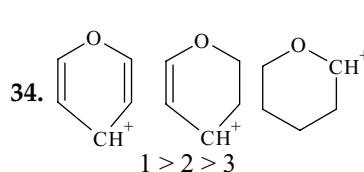
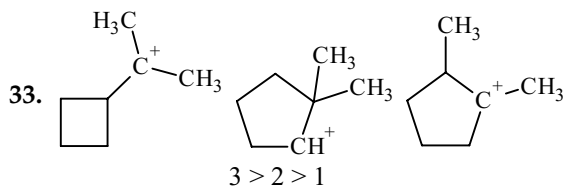
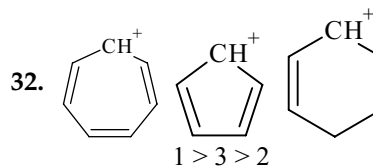
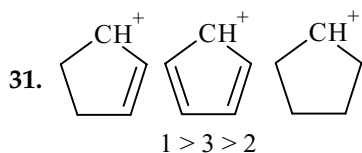
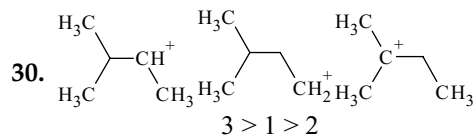
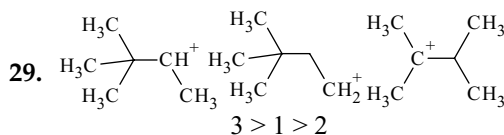
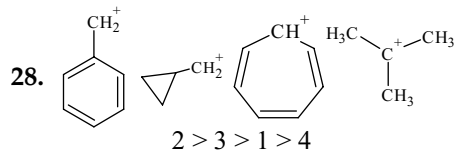
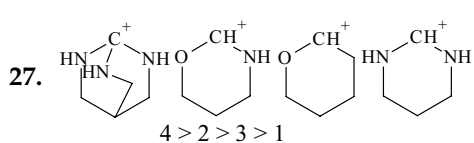
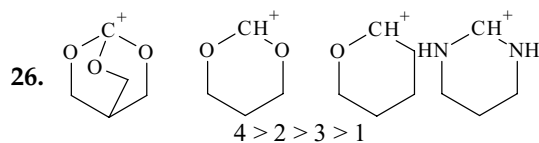
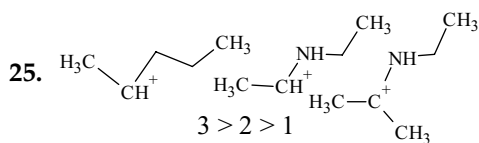
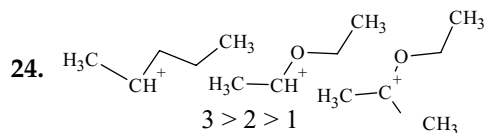
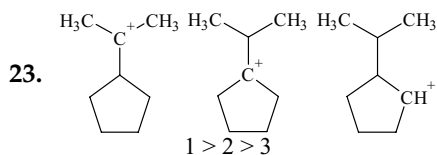
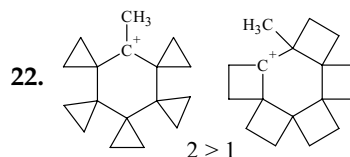
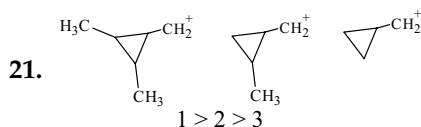
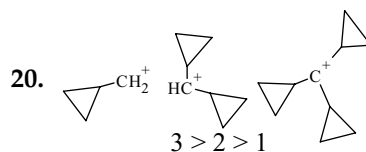
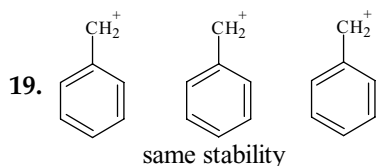
13.
3 > 1 > 2
14.
3 > 2 > 1
15.
1 > 2 > 3
16.
1 > 2 > 3
17.
1 > 2 > 3
18.
1 > 2 > 3
19.
1 > 2 > 3
20.
1 > 2 > 3
21.
3 > 1 > 2
22.
1 > 3 > 2
23.
1 > 2 > 3
24.
1 > 3 > 2
25.
2 > 1
26.
2 > 1



WORKBOOK EXERCISE 2

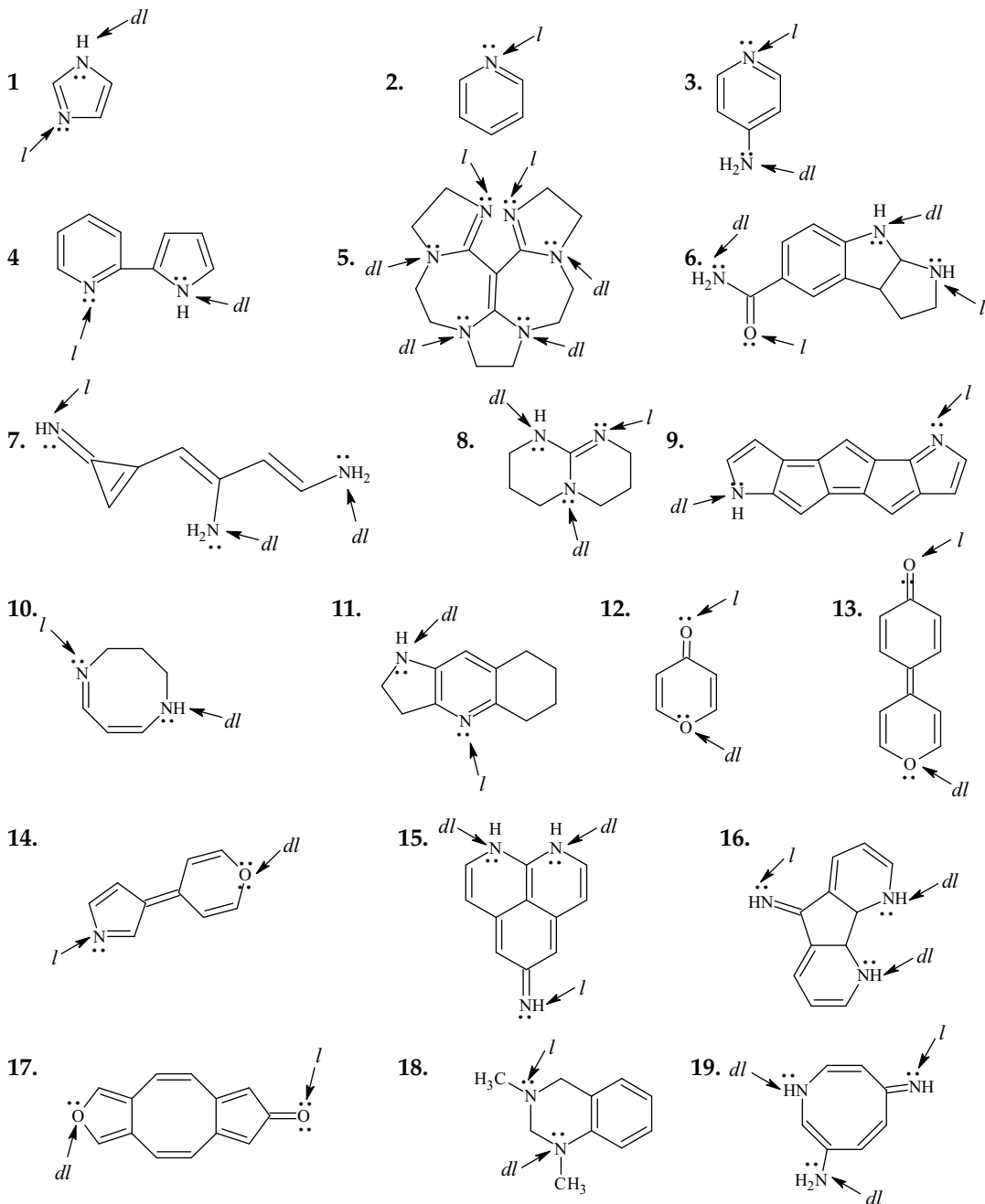
Carbocation Stability order

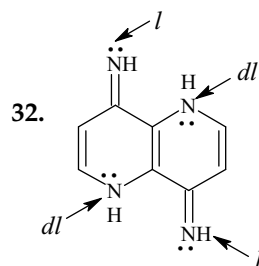
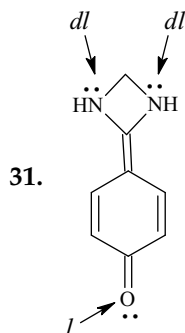
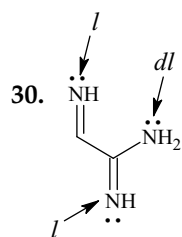
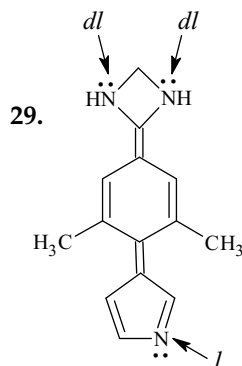
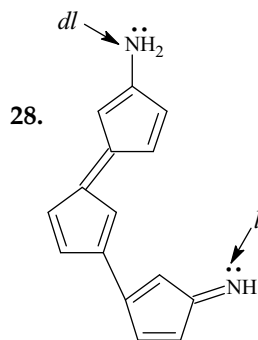
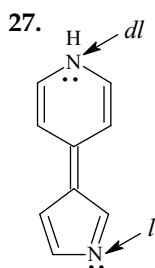
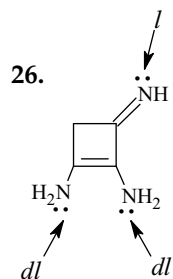
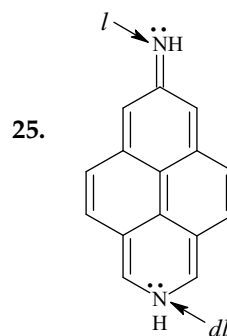
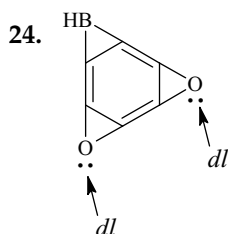
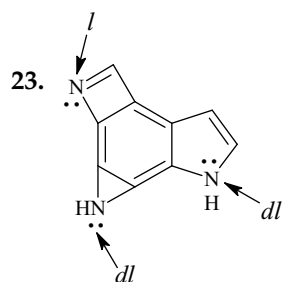
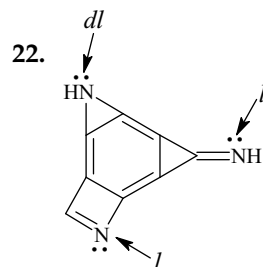
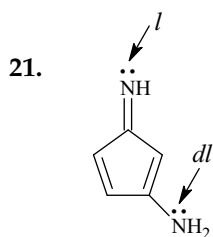
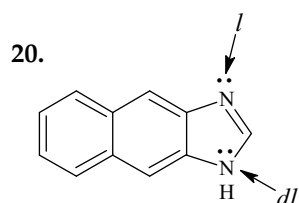




WORKBOOK EXERCISE 3

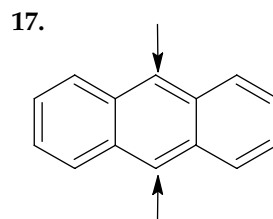
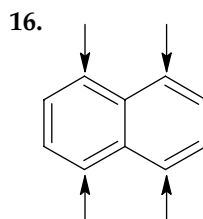
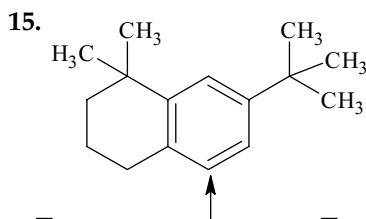
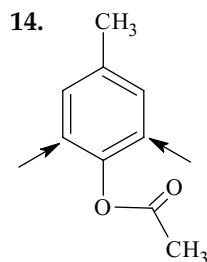
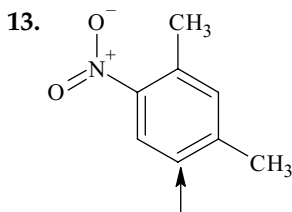
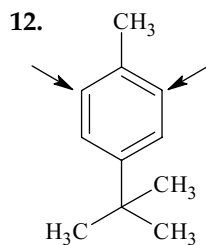
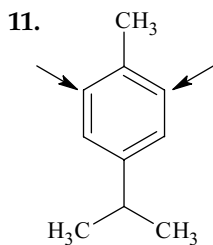
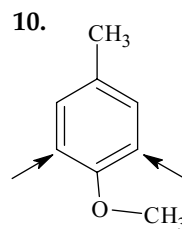
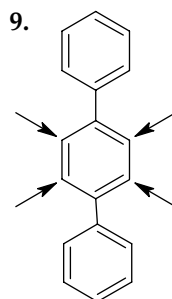
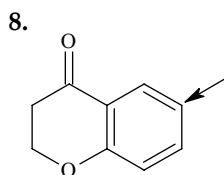
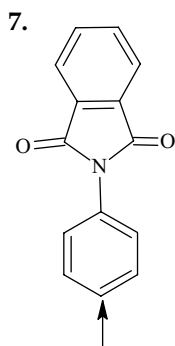
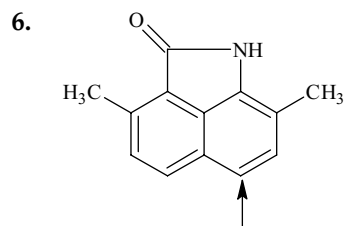
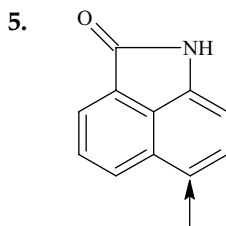
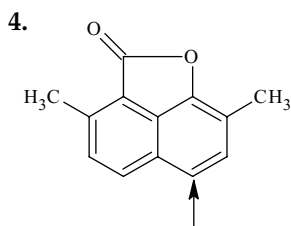
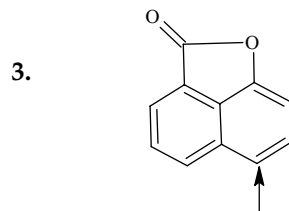
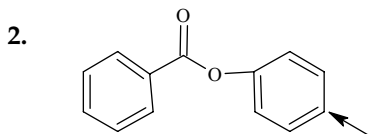
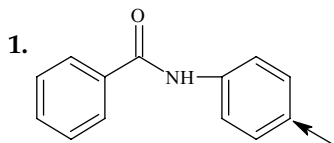
Identify localised and delocalised 1p in the following examples



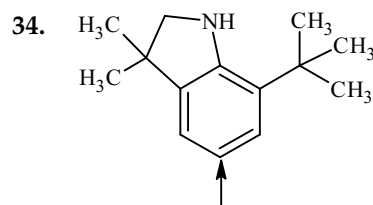
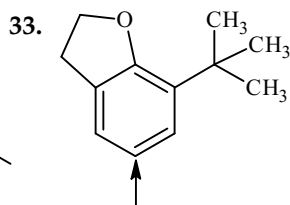
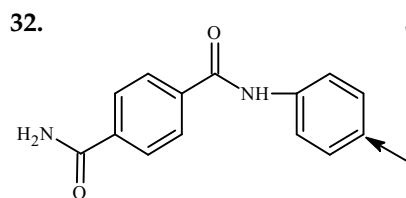
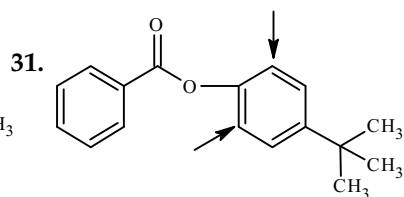
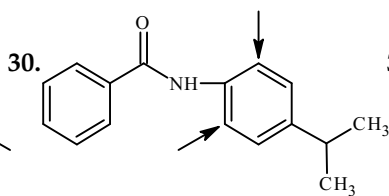
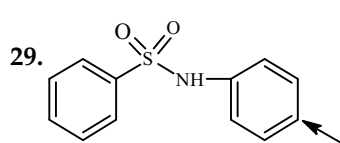
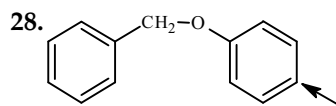
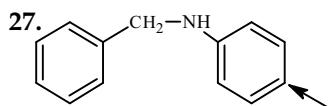
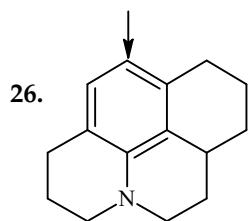
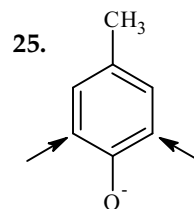
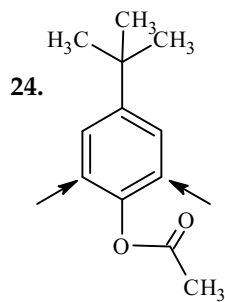
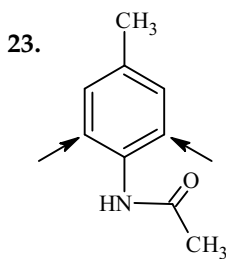
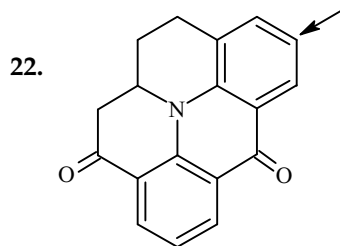
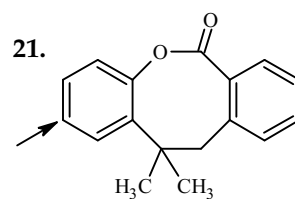
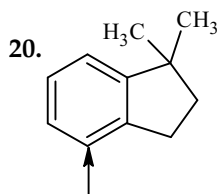
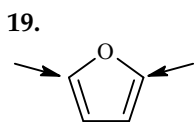
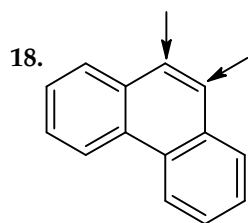


WORKBOOK EXERCISE 4

Identify the site of electrophilic attack at benzene ring

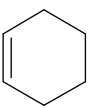
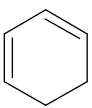
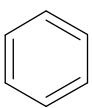
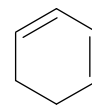
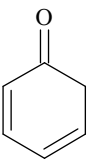
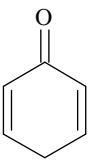
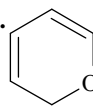
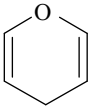
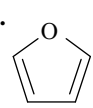
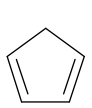
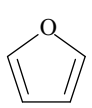
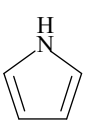
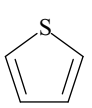
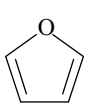
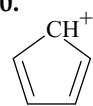
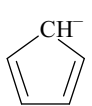
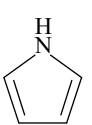
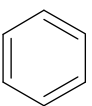
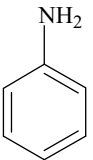
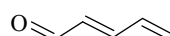
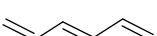
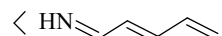
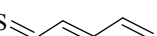
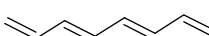
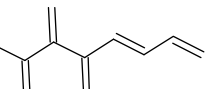
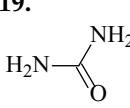
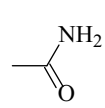
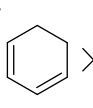
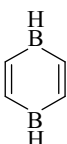
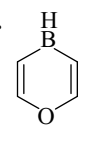
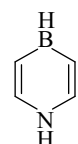
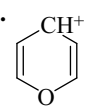
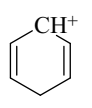
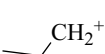
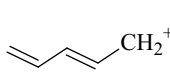


[Note : Major attack represents by arrow (→)]

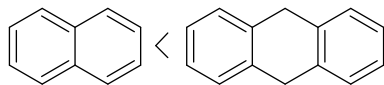


WORKBOOK EXERCISE 5

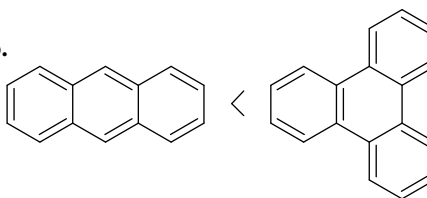
Identify the correct order of resonance energy for the following pairs of compounds

1.  < 
2.  > 
3.  > 
4.  > 
5.  < 
6.  > 
7.  > 
8.  < 
9.  > 
10.  < 
11.  > 
12.  < 
13.  < 
14.  < 
15.  < 
16.  < 
17.  < 
18.  < 
19.  > 
20.  > 
21.  < 
22.  > 
23.  > 
24.  < 

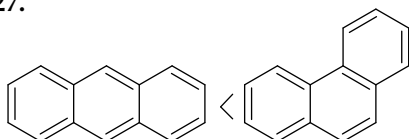
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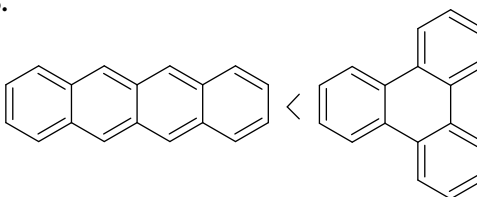
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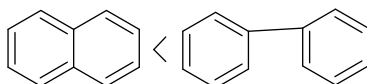
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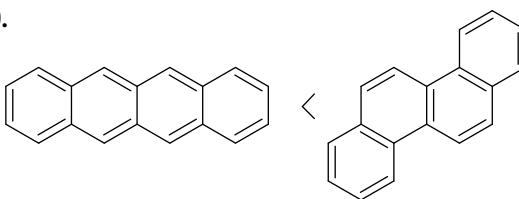
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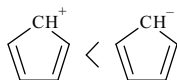
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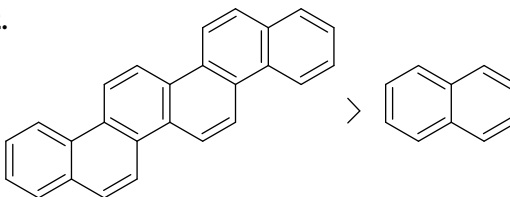
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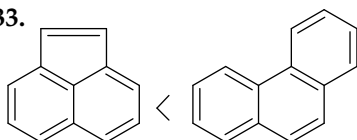
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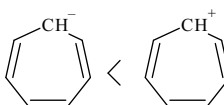
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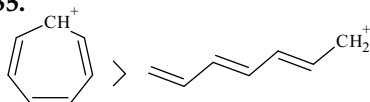
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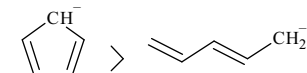
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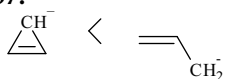
35.



36.



37.

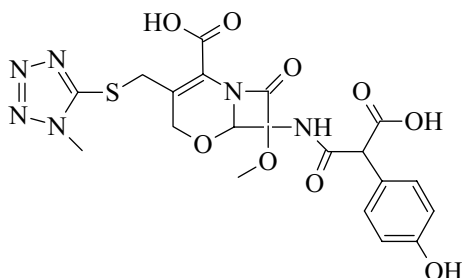


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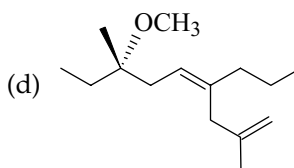
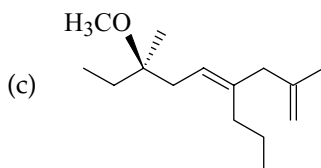
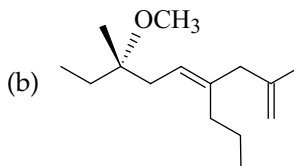
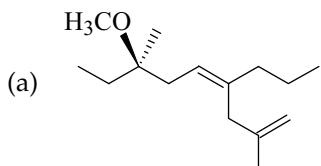
Question Bank

LEVEL 1

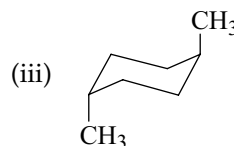
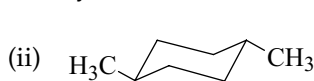
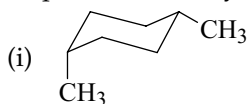
1. How many stereocenters does latomoxef (an oxacephem antibiotic) have?



- (a) 2 (b) 3 (c) 4 (d) 5
2. Which molecule is (R, Z)-7-methoxy-2, 7-dimethyl-4-propylnona-1, 4-diene?



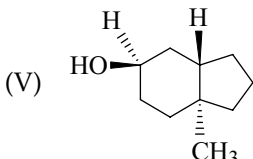
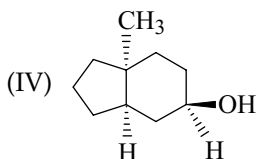
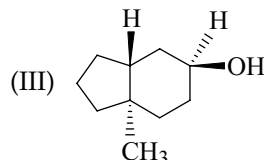
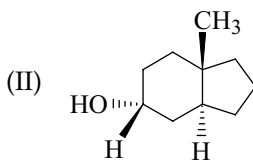
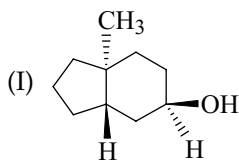
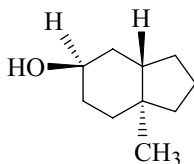
3. Choose the correct order that has the following compounds correctly arranged with respect to thermodynamic stability.



- (a) ii < i < iii
(c) i < iii < i

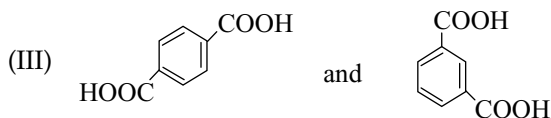
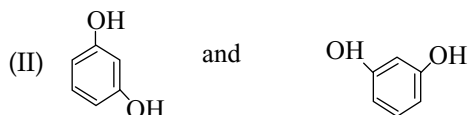
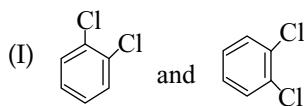
- (b) i < ii < ii
(d) iii < i < ii

4. Which of the structures below is a diastereomer of A?



- (a) I (b) III (c) II and IV (d) IV and V
5. The total number of structural isomers of $C_4H_{11}N$ would be
(a) 4 (b) 8 (c) 5 (d) 10
6. Which of the following pair is the chain isomer?
(a) (b) (c) (d)
7. How many geminal dichloride with different formula are possible for $C_3H_6Cl_2$?
(a) only one (b) two (c) three (d) four
8. What is the relation between 3-ethylpentane and 3-methylhexane?
(a) Chain isomers (b) Position isomers
(c) Functional isomers (d) Metamers
9. How many isomers are possible for methyl anthracene?
(a) 2 (b) 3 (c) 4 (d) 5
10. The compounds $(CH_3)_3N$ and $CH_3CH_2CH_2NH_2$ represent
(a) chain isomerism (b) position isomerism
(c) functional isomerism (d) all of the above
11. An isomer of ethanol is
(a) methanol (b) acetone (c) diethyl ether (d) dimethyl ether
12. The number of primary, secondary and tertiary amines possible with the molecular formula C_3H_9N respectively
(a) 1, 2, 2 (b) 1, 2, 1 (c) 2, 1, 1 (d) 3, 0, 1

13. Examine the relation between the following pairs of compounds

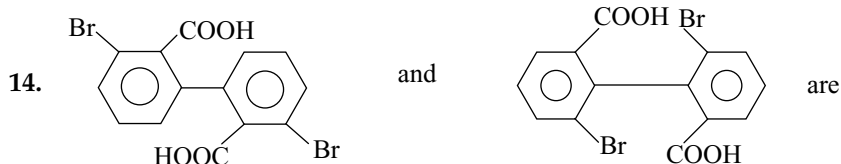


(a) All I, II, III are identical

(b) All I, II, III are isomers

(c) I, II are identical, III is isomer

(d) I is identical and II, III are isomers



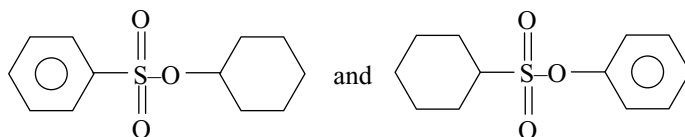
(a) Positional

(b) Chain

(c) Geometrical

(d) Functional

15. Given compound shows which type of isomerism?

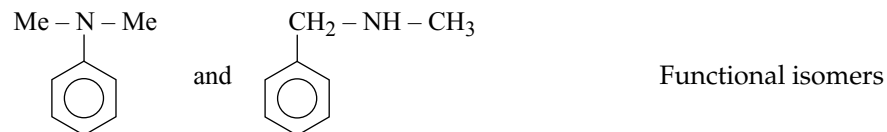
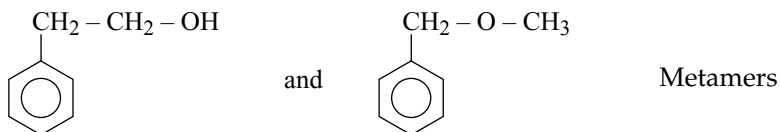
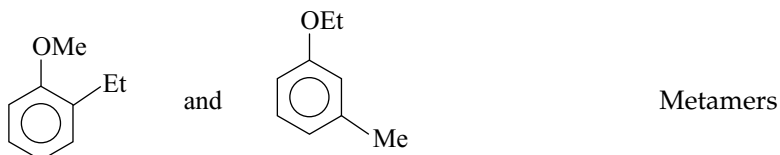
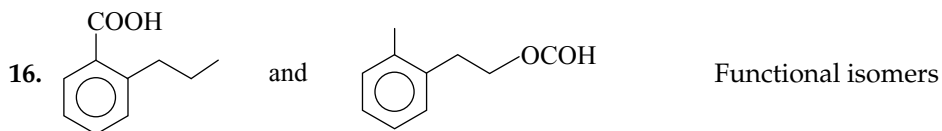


(a) Chain isomerism

(b) Positional isomerism

(c) Metamerism

(d) Functional group isomerism



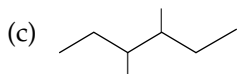
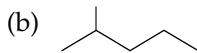
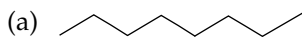
(a) TTFT

(b) FTTF

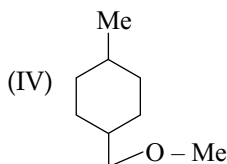
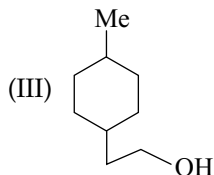
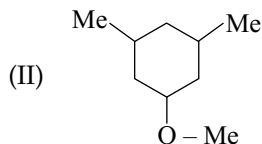
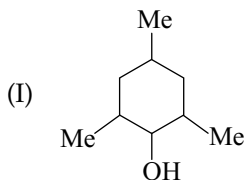
(c) TTFT

(d) TTFT

17. Which compound is not the isomer of 3-ethyl-2-methylpentane?



18. Which of the following is not the correct relationship?



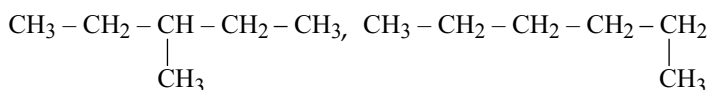
(a) II and IV are metamers

(b) I and II are functional isomers

(c) I and III are chain isomers

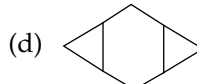
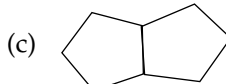
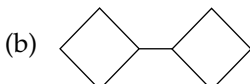
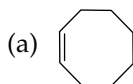
(d) I and IV are positional isomers

19. What is the correct relationship between the following compounds?



(a) Chain isomers (b) Position isomers (c) Functional isomers (d) Identical

20. Which one of the compound is not isomer of others?



21. $\text{CH}_3 - \text{CH}_2 - \text{NH} - \text{CHO}$; $\text{CH}_3 - \underset{\text{NH}_2}{\text{CH}} - \text{CHO}$

I

II

Which type of isomerism is observed between I and II?

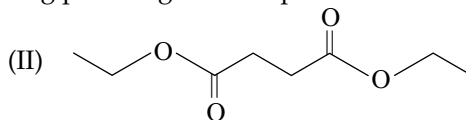
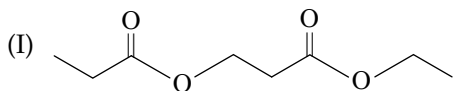
(a) Chain isomers

(b) Position isomers

(c) Functional isomers

(d) Metamers

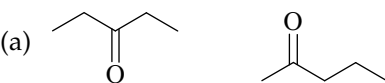
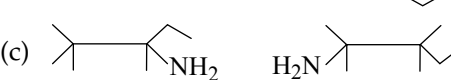
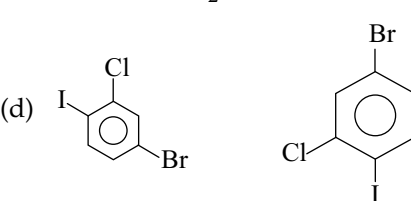
22. The correct relationship among the following pairs of given compounds is



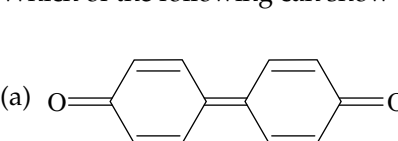
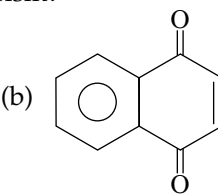
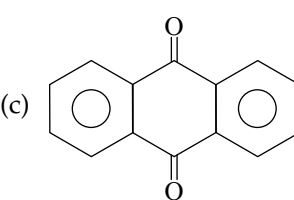
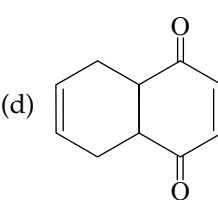
(a) Chain isomer (b) Positional isomer (c) Metamer

(d) Functional isomer

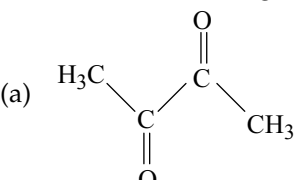
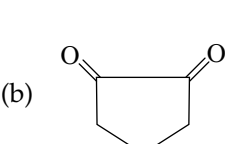
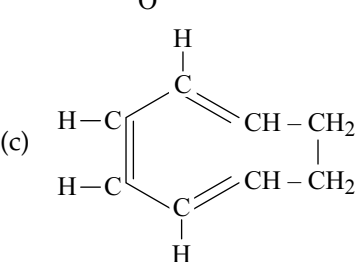
23. Which of the following is a pair of metamers?

- (a) 
- (b) 
- (c) 
- (d) 

24. Which of the following can show tautomerism?

- (a) 
- (b) 
- (c) 
- (d) 

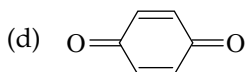
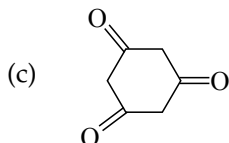
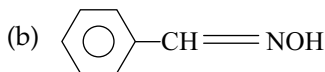
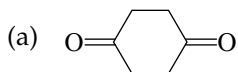
25. Which of the following can show tautomerism?

- (a) 
- (b) 
- (c) 
- (d) all of these

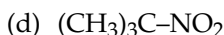
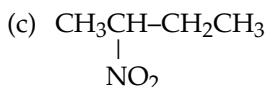
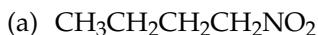
26. Which of the following can show tautomerism?

- (a) $(\text{CH}_3)_3\text{CNO}$ (b) $(\text{CH}_3)_2\text{NH}$ (c) R_3CNO_2 (d) RCH_2NO_2

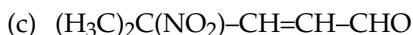
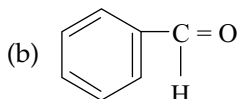
27. Which will not show tautomerism?



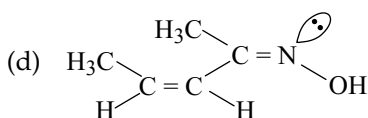
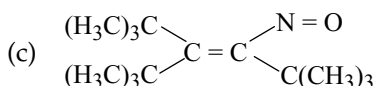
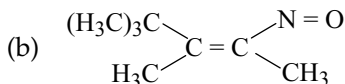
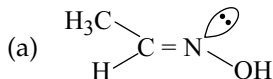
28. Which will not show tautomerism?



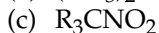
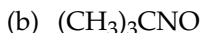
29. Which of the following compound show tautomerism?



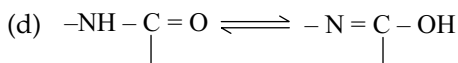
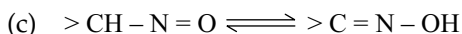
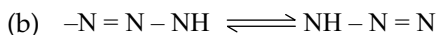
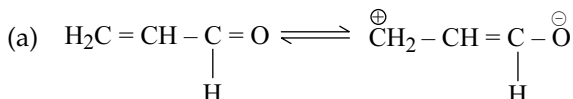
30. Which of the following will not show tautomerism?

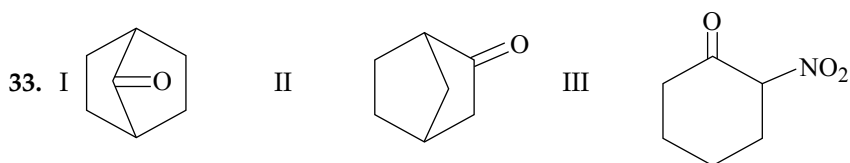


31. Tautomerism will be exhibited by



32. Which of the following is not an example of tautomeric equilibrium?

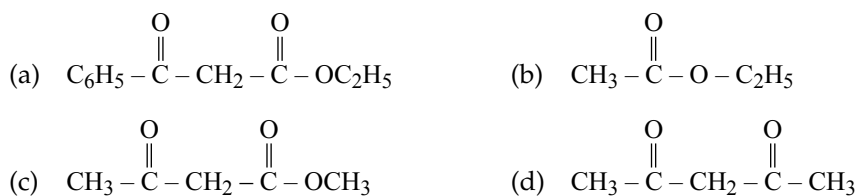




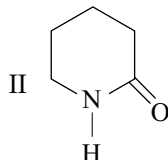
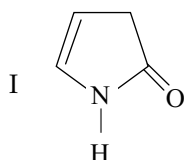
Among these compounds, which of the following is the correct order of % enol content?

- (a) I > II > III (b) III > II > I (c) II > III > I (d) I > III > II

34. Which of the following will have highest percentage of enol content?

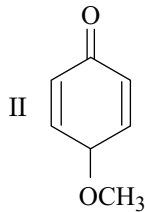
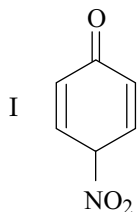


35. Which of the following compounds have less enol content?



- (a) I (b) II (c) both (a) and (b) (d) none of these

36. Which of the following compounds have higher enol content?



- (a) I (b) II (c) I = II (d) none of these

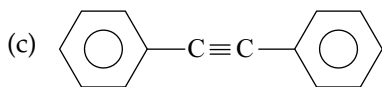
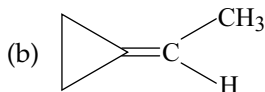
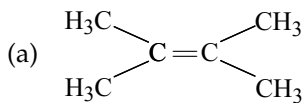
37. Geometrical isomers differ in

- (a) position of functional group (b) spatial arrangement of atoms
 (c) position of atoms (d) length of carbon chain

38. Stereoisomers have different

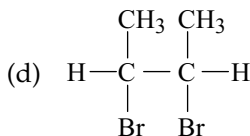
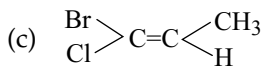
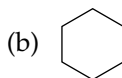
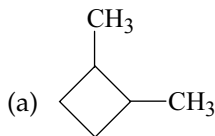
- (a) Molecular formula (b) Structural formula
 (c) Configuration (d) Molecular mass

39. Which of the following compounds has restricted rotation?

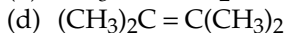
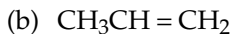
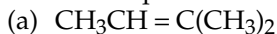


(d) All of these

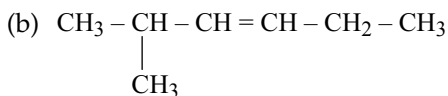
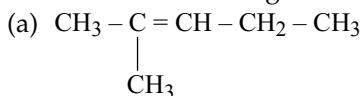
40. Which of the following compounds does not have restricted rotation?



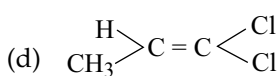
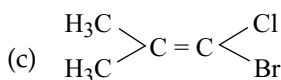
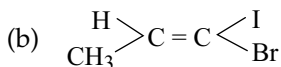
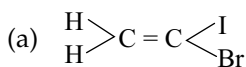
41. Which compound can show geometrical isomerism?



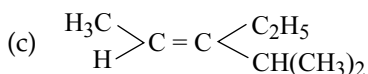
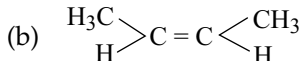
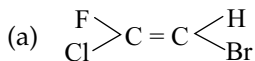
42. Which of the following will not show cis-trans isomerism?



43. Geometrical isomerism is shown by

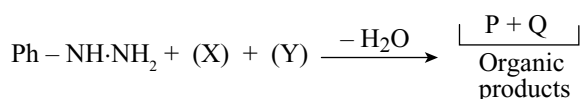


44. The "E"-isomer is



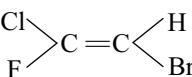
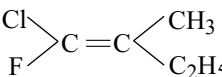
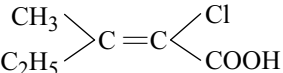
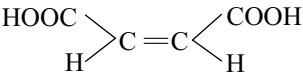
(d) none of the above

45. The compounds X and Y shown in the below reaction can be

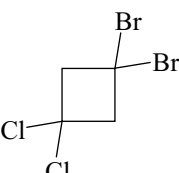
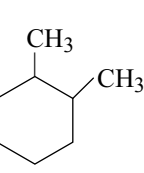
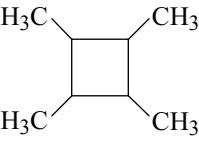
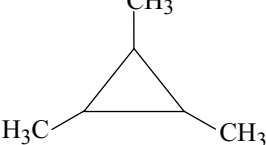


- (a) $\text{CH}_3 - \text{CH}_2 - \underset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{CH}_3 + \text{CH}_3 - \underset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{Ph}$ (b) $\text{Ph} - \underset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{CH}_3 + \text{CH}_3\text{CHO}$
- (c) $\text{CH}_2 = \text{O} + \text{CH}_3\text{CHO}$ (d) $\text{CH}_2 = \text{O} + \text{CH}_3 - \underset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{CH}_3$

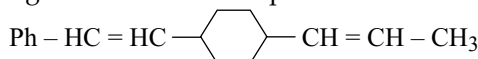
46. The "Z"-isomer is

- (a)  (b) 
- (c)  (d) 

47. Which of the following compounds cannot show geometrical isomerism?

- (a)  (b) 
- (c)  (d) 

48. The total number of geometrical isomers possible in following compound is



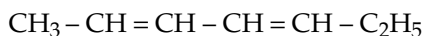
- (a) 2 (b) 1 (c) 6 (d) 8

49. The total number of geometrical isomers possible in following compound is



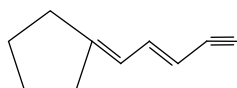
- (a) 4 (b) 6 (c) 3 (d) 2

50. The number of geometrical isomers in the following compound is



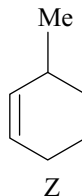
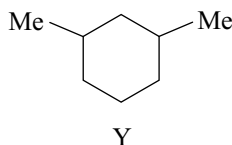
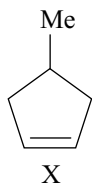
- (a) 4 (b) 3 (c) 2 (d) 5

51. The number of cis-trans isomer possible for the following compound is



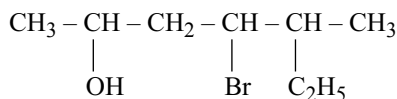
- (a) 2 (b) 4 (c) 6 (d) 8

52. What characteristic is the best common to both cis-2-butene and trans-2-butene?
 (a) B.P. (b) Dipole moment
 (c) Heat of hydrogenation (d) Product of hydrogenation
53. Number of chiral carbon atoms in the compound x, y and z respectively would be

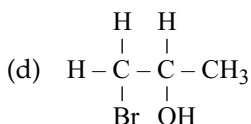
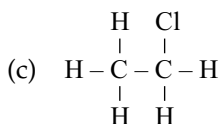
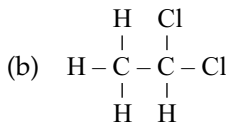
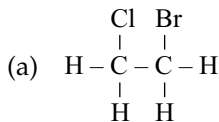


- (a) 0, 2, 1 (b) 1, 0, 1 (c) 1, 2, 1 (d) 1, 2, 0

54. Number of chiral carbon present in the following compound is

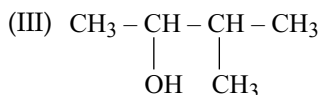
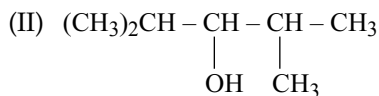
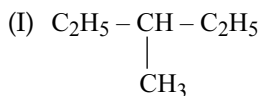


- (a) 2 (b) 3 (c) 4 (d) 5
55. Which of the following have asymmetric carbon atom?



56. Meso-tartaric acid and d-tartaric acid are
 (a) positional isomers (b) enantiomers
 (c) diastereomers (d) racemic mixture

57. Observe the following structures I to III

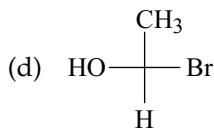
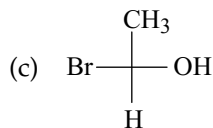
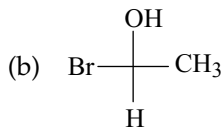
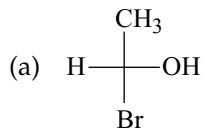


Correct statement is

- (a) All three are chiral compounds (b) I and II are chiral
 (c) Only II is chiral (d) Only III is chiral

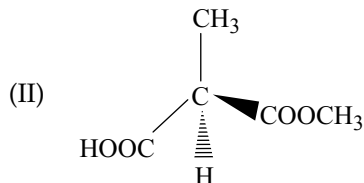
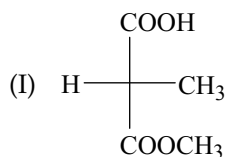
58. The number of optically active compounds in the isomers of C_4H_9Br is
 (a) 1 (b) 2 (c) 3 (d) 4

59. Which of the following compound has "S" configuration?

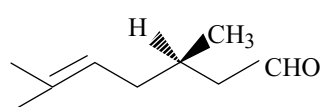
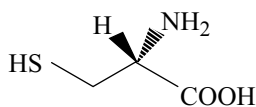
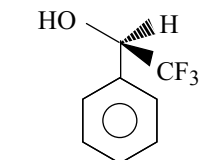


60. The number of optically active isomers observed in 2,3-dichlorobutane is
 (a) 0 (b) 2 (c) 3 (d) 4

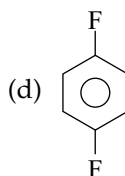
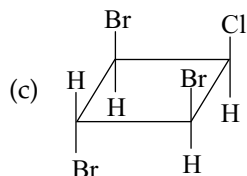
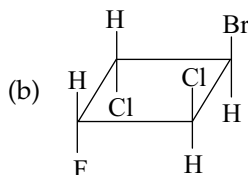
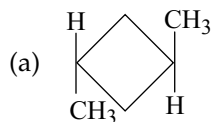
61. The correct configuration assigned for compound (I) and (II) respectively is



62. The R/S configurations of these compounds are respectively

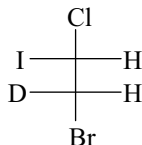
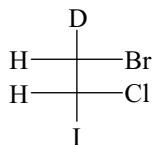


63. Which of the following compound has plane of symmetry (POS) but not centre of symmetry (COS)?

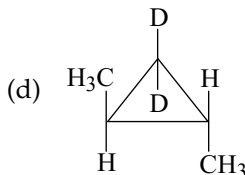
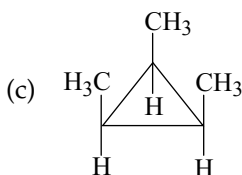
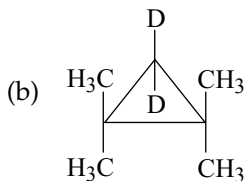
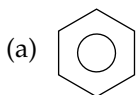


64. The instrument which can be used to measure optical activity, i.e., specific rotation
 (a) Refractometer (b) Photometer (c) Voltmeter (d) Polarimeter

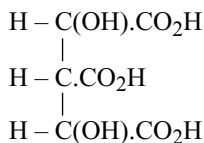
65. The two compounds given below are



- (a) Enantiomer (b) Identical (c) Meso compound (d) Diastereomers
66. Which of the following compounds do not possess a C_2 axis of symmetry?



67. How many stereoisomers can exist for the following acid?



- (a) Two (b) Four (c) Eight (d) Sixteen
68. $\text{CH}_3 - \underset{\text{Br}}{\text{CH}} - \underset{\text{Br}}{\text{CH}} - \underset{\text{Br}}{\text{CH}} - \text{CH}_3$

Total number of stereoisomers in the above compound is

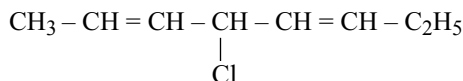
- (a) 6 (b) 4 (c) 8 (d) 16
69. Total number of stereoisomers of the compound is given below
- $$\text{CH}_3 - \underset{\text{OH}}{\text{CH}} - \underset{\text{Br}}{\text{CH}} - \text{CH}_3$$
- (a) 2 (b) 4 (c) 6 (d) 8

70. How many stereoisomers of the following molecule are possible?

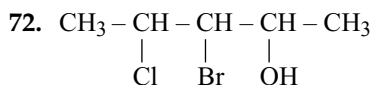


- (a) two optical isomers
- (b) two geometrical isomers
- (c) two optical and two geometrical isomers
- (d) none

71. Total number of stereoisomer of compound is given below



- (a) 2
- (b) 4
- (c) 6
- (d) 8



Total number of stereoisomers in the above compound is

- (a) 6
- (b) 4
- (c) 8
- (d) 16

73. The enantiomeric excess and observed specific rotation of a mixture containing 6 g of (+)-2-butanol and 4 (g) of (–)-2-butanol are respectively (If the specific rotation of enantiomerically pure (+)-2-butanol is +13.5 units)

- (a) 80%, +2.7 unit
- (b) 20%, –27 unit
- (c) 20%, +2.7 unit
- (d) 80%, –27 unit

74. Which of the following pair of isomers cannot be separated by fractional crystallisation or fractional distillation?

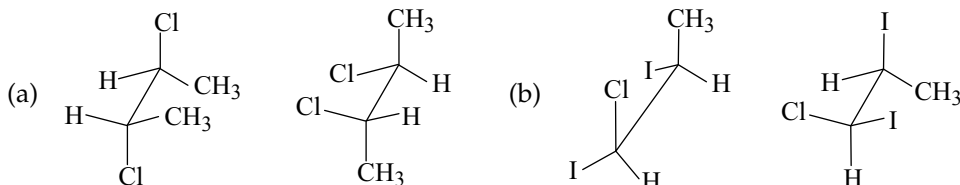
- (a) Maleic acid and fumaric acid
- (b) (+)-Tartaric acid and meso-tartaric acid
- (c) $\text{CH}_3 - \underset{\text{NH}_2}{\text{CH}} - \text{COOH}$ and $\text{H}_2\text{N} - \text{CH}_2 - \text{CH}_2 - \text{COOH}$

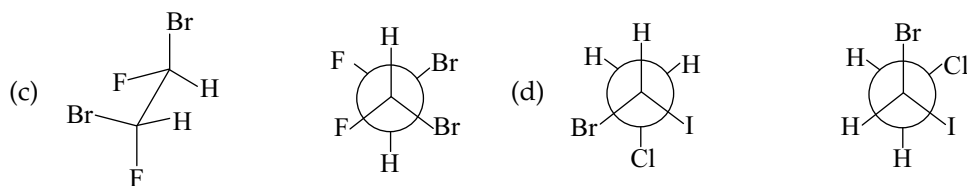
(d) (+)-lactic acid and (–)-lactic acid

75. Increasing order of stability among the three main conformation (i.e., eclipse, anti, gauche) of ethylene glycol is

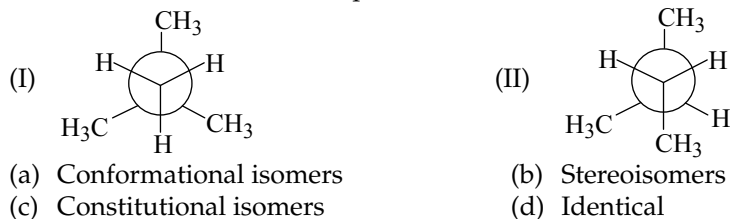
- (a) Eclipse, gauche, anti
- (b) Gauche, eclipse, anti
- (c) Eclipse, anti, gauche
- (d) Anti, gauche, eclipse

76. Which of the following pairs of compound is/are identical?

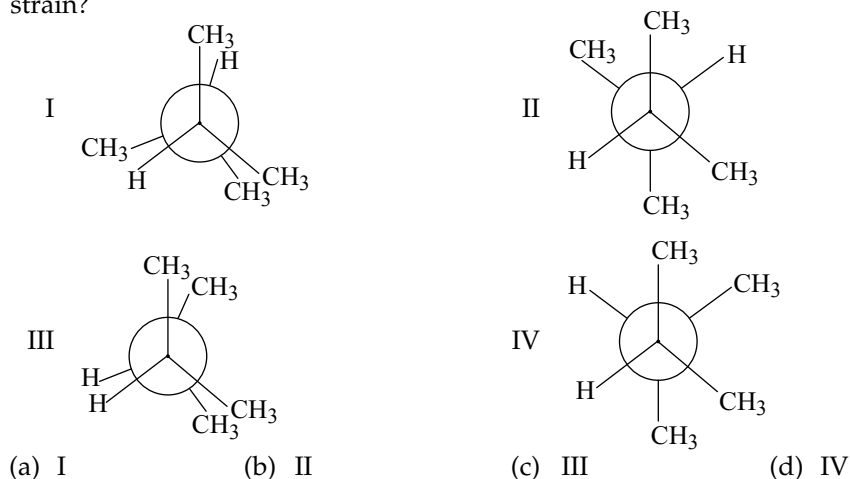




77. The two structures I and II represent



78. In which of the following has minimum torsional strain and minimum Vander waal strain?

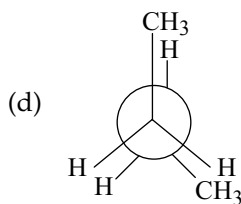
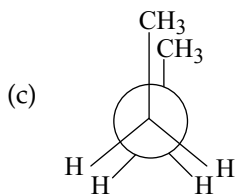


79. In the Newman projection formula of the least stable staggered form of n-butane, which of the following reasons is the causes of its instability?

- (a) Vander Waal's strain
(b) Torsional strain
(c) Combination of both
(d) None of these

80. Which of the following represent the staggered conformation with dihedral angle $\phi = 60^\circ$?

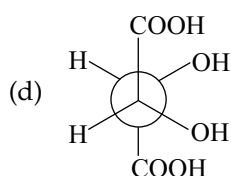
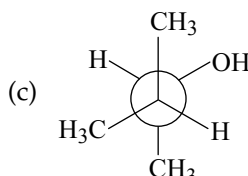
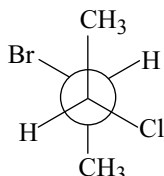
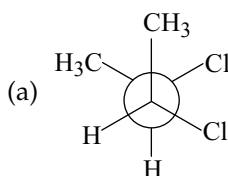




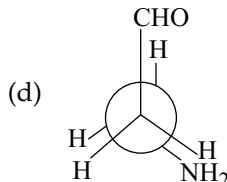
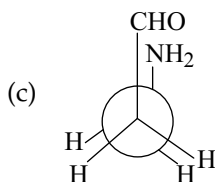
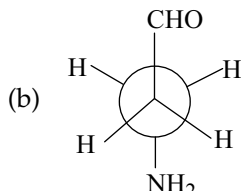
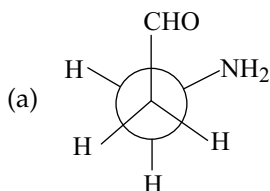
81. The dihedral angle between two methyl groups in partially eclipsed conformation of n-butane is

- (a) 180° (b) 120° (c) 90° (d) $109^\circ 28'$

82. Which of the following is an achiral molecule?



83. Which of the following is most stable?



84. Evaporation of an aqueous solution of ammonium cyanate gives urea. This reaction follows the class of

- (a) Polymerisation (b) Isomerisation
(c) Association (d) Dissociation

85. The possible number of alkynes with the formula C_5H_8 is

- (a) 2 (b) 3 (c) 4 (d) 5

86. How many isomers of $C_5H_{11}OH$ will be primary alcohols

- (a) 2 (b) 3 (c) 4 (d) 5

87. Number of isomeric forms of C_7H_9N having benzene ring will be

- (a) 7 (b) 6 (c) 5 (d) 4

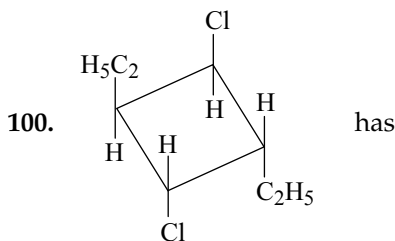
88. Which of the following is an isomer of diethyl ether
 (a) $(\text{CH}_3)_3\text{COH}$ (b) CH_3CHO (c) $\text{C}_3\text{H}_7\text{OH}$ (d) $(\text{C}_2\text{H}_5)_2\text{CHOH}$
89. Total number of isomeric alcohols with the formula $\text{C}_4\text{H}_{10}\text{O}$ is
 (a) 1 (b) 2 (c) 3 (d) 4
90. The molecular formula of a saturated compound is $\text{C}_2\text{H}_4\text{Cl}_2$. The formula permits the existence of two
 (a) functional isomers (b) position isomers
 (c) optical isomers (d) cis-trans isomers
91. The type of isomerism found in urea molecule is
 (a) Chain (b) Position (c) Tautomerism (d) None of these
92. An alkane can show structural isomerism if it has number of minimum carbon atoms
 (a) 1 (b) 2 (c) 3 (d) 4
93. How many chain isomers can be obtained from the alkane C_6H_{14} ?
 (a) 4 (b) 5 (c) 6 (d) 7
94. Keto-enol tautomerism is observed in
 (a) $\text{C}_6\text{H}_5-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$ (b) $\text{C}_6\text{H}_5-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$
 (c) $\text{C}_6\text{H}_5-\overset{\text{O}}{\parallel}{\text{C}}-\text{C}_6\text{H}_5$ (d) $\text{C}_6\text{H}_5-\overset{\text{O}}{\parallel}{\text{C}}-\underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}}-\text{C}_6\text{H}_5$
95. The number of geometrical isomers in case of a compound with the structure
 $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}=\text{CH}-\text{C}_2\text{H}_5$ is
 (a) 4 (b) 3 (c) 2 (d) 5
96. Which one of the following will show geometrical isomerism?
 (a) $\text{CH}_2\text{Cl} \begin{array}{c} \diagup \\ \text{C} \\ \diagdown \end{array} \begin{array}{c} \text{CH}_3 \\ \text{CH}=\text{C}(\text{CH}_3)_2 \end{array}$
 (b) $\text{CH}_3 \begin{array}{c} \diagup \\ \text{C} \\ \diagdown \end{array} \begin{array}{c} \text{H} \\ \text{CH}=\text{C}(\text{CH}_3)_2 \end{array}$
 (c) $\text{CH}_2=\text{CH} \begin{array}{c} \diagup \\ \text{C} \\ \diagdown \end{array} \begin{array}{c} \text{CH}_2\text{Cl} \\ \text{CH}=\text{CH}_2 \end{array}$
 (d) $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_2\text{CH}_3$
97. In the reaction: $\text{CH}_3\text{CHO} + \text{HCN} \longrightarrow \text{CH}_3\text{CH}(\text{OH})\text{CN}$
 a chiral centre is produced. The number of stereoisomers of the product is
 (a) 3 (b) 2 (c) 4 (d) none of these
98. The molecule 3-penten-2-ol can exhibit
 (a) Optical isomerism (b) Geometrical isomerism
 (c) Metamerism (d) Tautomerism

The correct answer is

- (a) (a) and (b) (b) (a) and (c) (c) (b) and (c) (d) (a) and (d)

99. Find the total number of isomers (including stereo isomers) in dimethyl cyclopropane and dimethyl cyclobutane

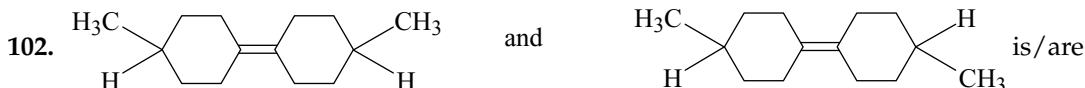
- (a) 6, 8 (b) 5, 6 (c) 4, 5 (d) 4, 6



- (a) plane of symmetry (b) centre of symmetry
(c) C_3 axis of symmetry (d) C_4 axis of symmetry

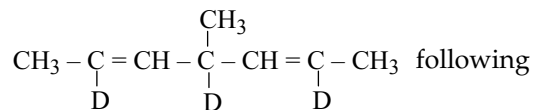
101. How many optically active cycloalkanones are possible with the molecular formula C_5H_8O ?

- (a) 8 (b) 6 (c) 5 (d) 4



- (a) Geometrical isomers (b) Position isomers
(c) Chain isomers (d) Enantiomers

103. For the given compound which of the following statement(s) are correct?



- (a) It has 4 stereoisomers (b) It has 3 stereoisomers
(c) It has 2 chiral centres (d) The compound does not show G.I.

104. Racemic mixture is formed by mixing two

- (a) Isomeric compounds (b) Chiral compounds
(c) Meso compounds (d) Enantiomers

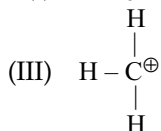
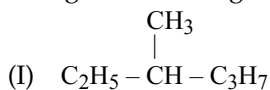
105. Which of the following does not show geometrical isomerism?

- (a) 1,2-dichloro-1-pentene
(b) 1,3-dichloro-2-pentene
(c) 1,1-dichloro-1-pentene
(d) 1,4-dichloro-2-pentene

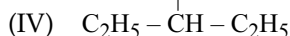
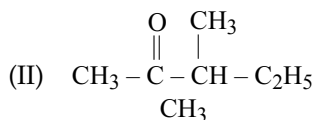
106. The general formula $C_nH_{2n}O_2$ could be for open chain

- (a) Diketones (b) Carboxylic acids
(c) Diols (d) Dialdehydes

107. Among the following four structures I to IV



it is true that



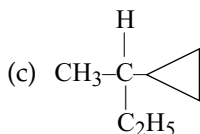
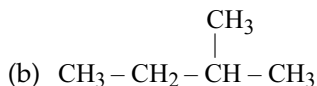
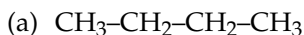
(a) All four are chiral compounds

(b) Only I and II are chiral compounds

(c) Only III is a chiral compound

(d) Only II and IV are chiral compounds

108. Amongst the following compounds, the optically active alkane having lowest molecular mass is



109. Which of the following compounds is not chiral?

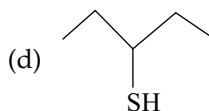
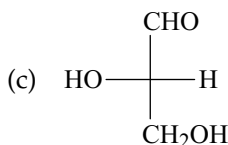
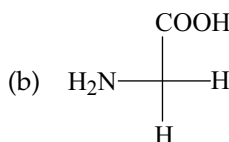
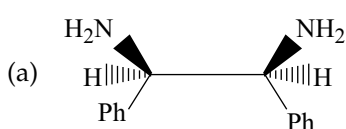
(a) 1-chloropentane

(b) 2-chloropentane

(c) 1-chloro-2-methylpentane

(d) 3-chloro-2-methylpentane

110. Which of the following molecules is expected to rotate the plane of polarised light?



111. The correct decreasing order of priority for the functional groups of organic compounds in the IUPAC system of nomenclature is

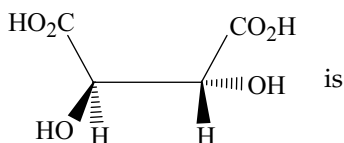
(a) $-\text{SO}_3\text{H}$, $-\text{COOH}$, $-\text{CONH}_2$, $-\text{CHO}$

(b) $-\text{CHO}$, $-\text{COOH}$, $-\text{SO}_3\text{H}$, $-\text{CONH}_2$

(c) $-\text{CONH}_2$, $-\text{CHO}$, $-\text{SO}_3\text{H}$, $-\text{COOH}$

(d) $-\text{COOH}$, $-\text{SO}_3\text{H}$, $-\text{CONH}_2$, $-\text{CHO}$

112. The absolute configuration of



(a) R, R

(b) R, S

(c) S, R

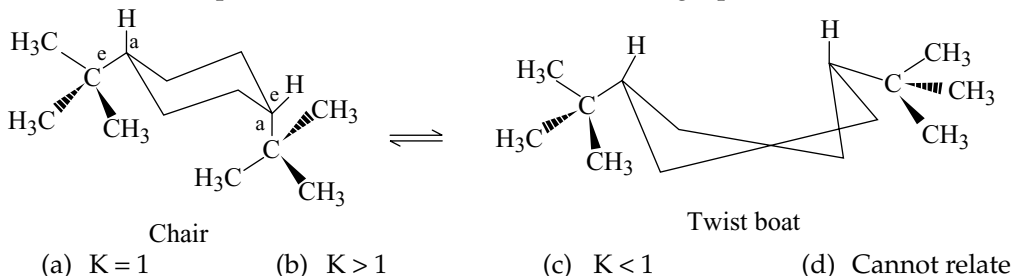
(d) S, S

113. The alkene that exhibits geometrical isomerism is
(a) 2-methyl propene (b) 2-butene
(c) 2-methyl-2-butene (d) propene
114. The number of stereoisomers possible for a compound of the molecular formula $\text{CH}_3\text{-CH=CH-CH(OH)-Me}$ is
(a) 2 (b) 4 (c) 6 (d) 3
115. Out of the following, the alkene that exhibits optical isomerism is
(a) 3-methyl-2-pentene (b) 4-methyl-1-pentene
(c) 3-methyl-1-pentene (d) 2-methyl-2-pentene
116. Identify the compound that exhibits tautomerism
(a) 2-butene (b) Lactic acid (c) 2-pentanone (d) Phenol

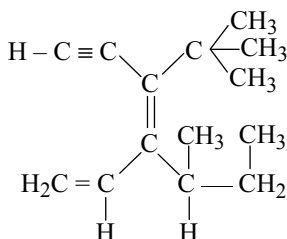
LEVEL 2

Single and Multiple-choice Type

1. The value of equilibrium constant (K) for the following equilibria

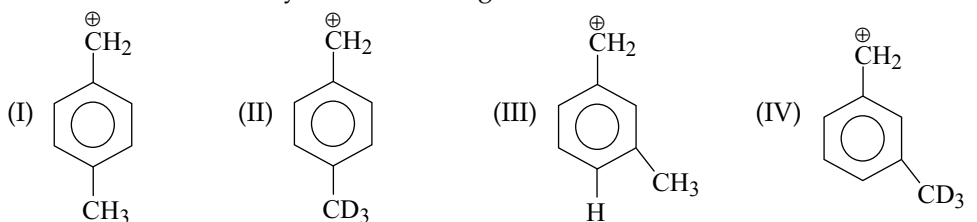


2. E/Z nomenclature of the following compound is



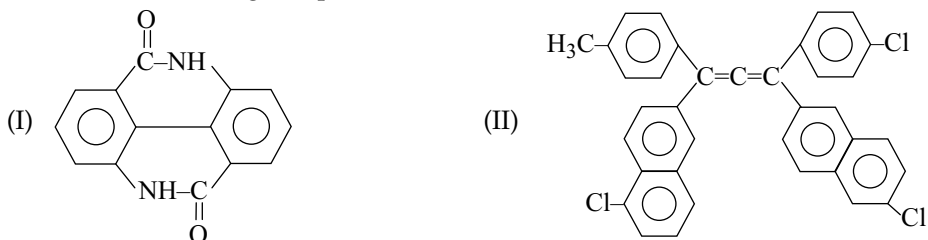
- (a) E (b) Z
(c) Does not show geometrical isomerism (d) None

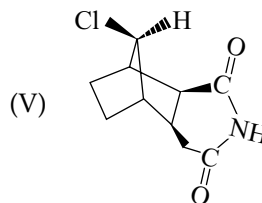
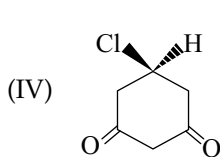
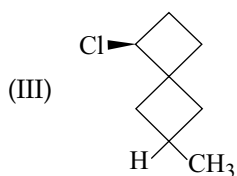
3. Correct order of stability of the following carbocations is



- (a) $I > II > III > IV$ (b) $I > II > IV > III$ (c) $II > I > IV > III$ (d) $I > IV > II > III$

4. Which of the following compounds are chiral?





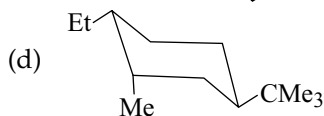
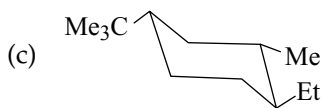
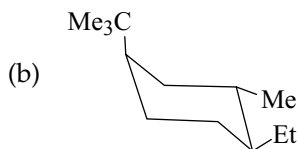
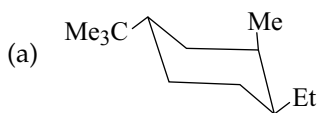
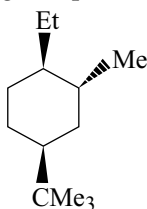
(a) I, II, III, V

(b) II, III, V

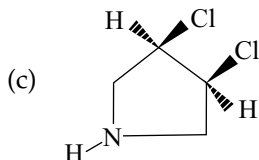
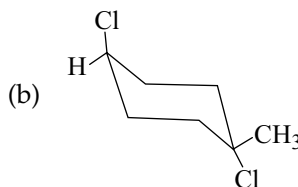
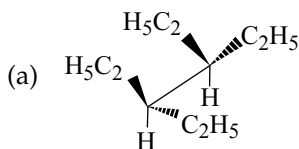
(c) I, II, III, V

(d) I, IV, V

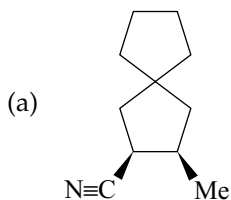
5. The conformation of the following compound is



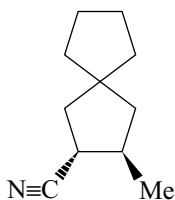
6. Which of the following represent the meso compound?

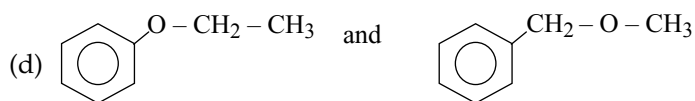
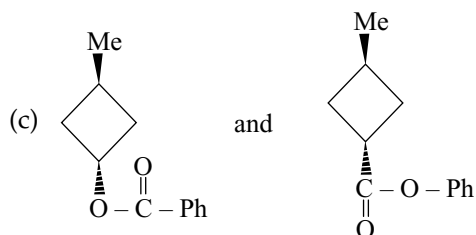
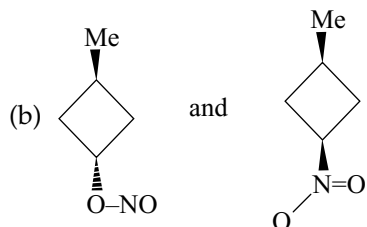


7. Among the following pair of compounds functional isomer is

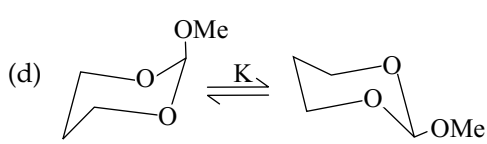
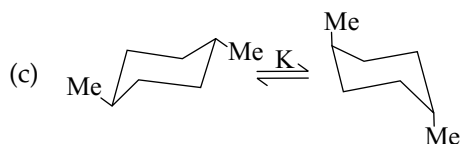
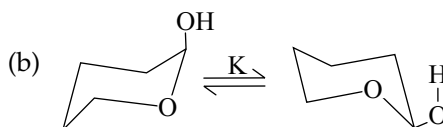
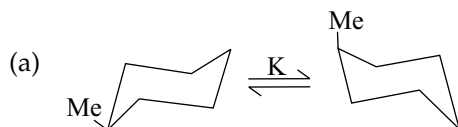


and

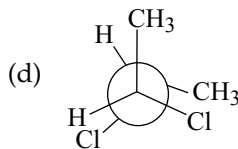
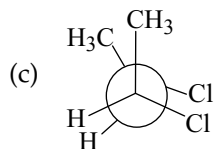
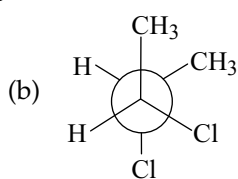
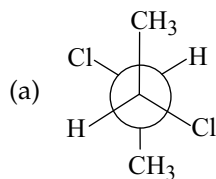
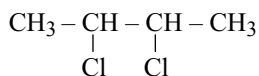




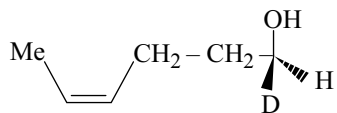
8. Identify equilibrium which has maximum K (equilibrium constant)



9. The most stable Newmann projection of the 2,3-dichlorobutane, whose Fischer projection is given below is

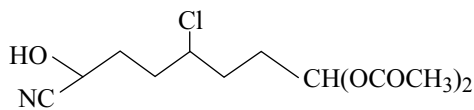


10. The correct stereochemical descriptions for the structure given below are



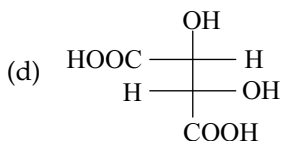
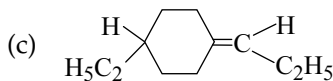
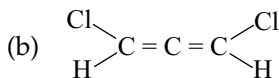
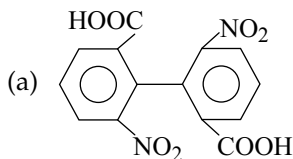
- (a) 1S, 4E (b) 1R, 4E (c) 1R, 4Z (d) 1S, 4Z

11. The number of optical isomers possible for the compound is

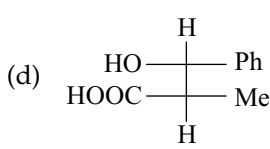
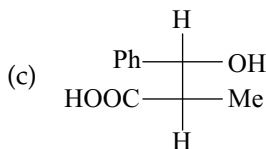
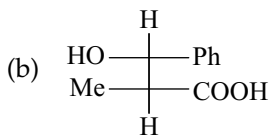
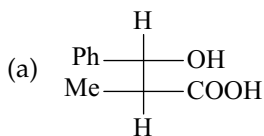
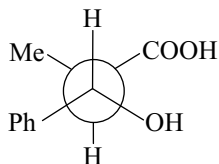


- (a) 2 (b) 4 (c) 8 (d) 6

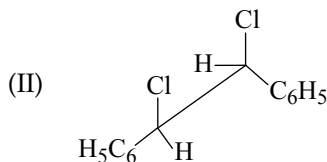
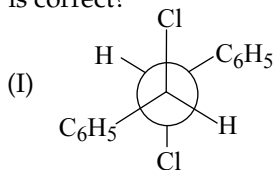
12. Which one of the following compounds will not rotate the plane of plane polarised light?



13. The correct Fischer projection formula of the Newman projection representation is

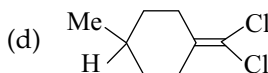
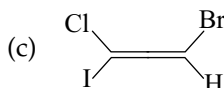
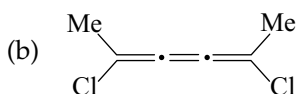
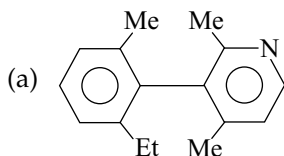


14. Which one of the following statements regarding the projections shown below (I and II) is correct?



- (a) Both the projections represent the same configuration
 (b) Both (I) and (II) are optically active
 (c) Only (I) is optically active
 (d) Only (II) is optically active

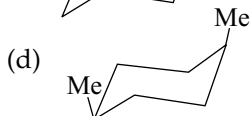
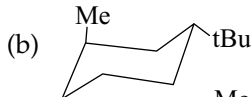
15. Among the following compounds which is used for resolution of racemic mixture?



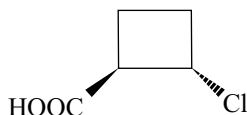
16. Identify specific rotation of a mixture have 5 g of (+)2-butanol and 7 g of (-)2-butanol (If specific rotation of 100% pure compound is 13.5°)

- (a) -2.25° (b) -1.6° (c) 12.2° (d) 4.45°

17. Identify compound having highest heat of combustion

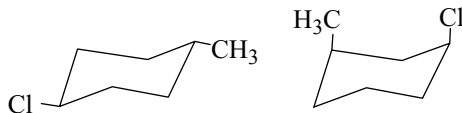


18. Identify R and S configurations of chiral centres in the following compound



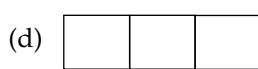
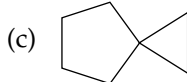
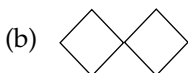
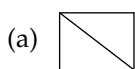
- (a) (1)-R (2)-S (b) (1)-S (2)-R (c) (1)-S (2)-S (d) (1)-R (2)-S

19. What is the relationship between the two structures shown?

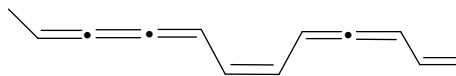


- (a) Different drawings of the same conformation of the same compound
 (b) Stereoisomers
 (c) Constitutional isomers
 (d) Different conformations of the same compound

20. C_8H_{12} has 3° of unsaturation. Thus, it can have the structure

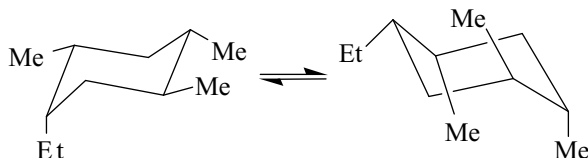


21. Calculate the total number of geometrical isomers for the following compound



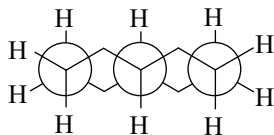
- (a) 2 (b) 4 (c) 8 (d) 16

22. Identify the value of K_C for the following equilibrium

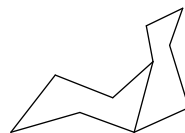


- (a) $K_C > 1$ (b) $K_C < 1$ (c) $K_C = 1$ (d) None of these

23. Identify the relationship between the following pairs of compound

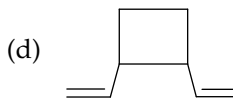
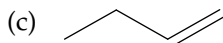
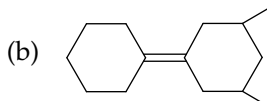
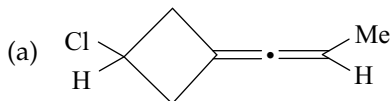


and



- (a) Positional isomers (b) Geometrical isomers
(c) Functional isomers (d) Identical compounds

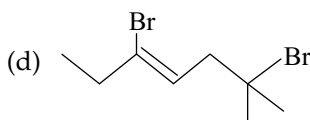
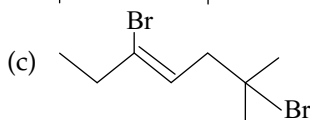
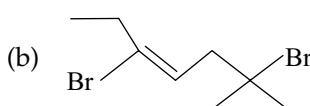
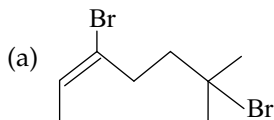
24. Identify compounds that give geometrical isomerisation



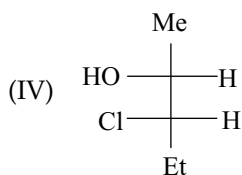
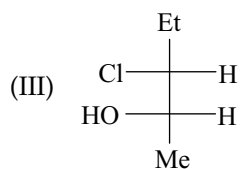
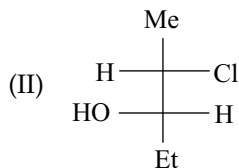
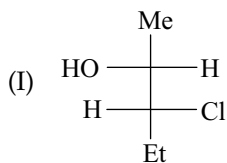
25. Which conformation has a C_3 axis of symmetry?

- (a) Boat (b) Twist boat (c) Chair (d) Enveloped

26. Which of following represent (E)-3,6-dibromo-6-methyl-3-heptene?



27. Which of the following represent a pair of enantiomers?

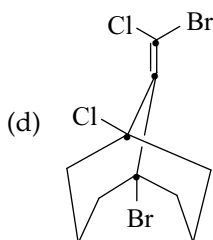
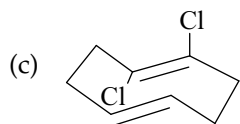
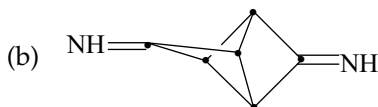
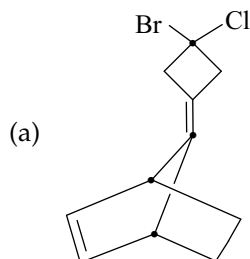


- (a) I and II (b) III and IV (c) I and IV (d) II and III

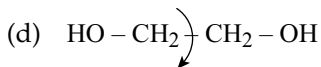
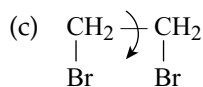
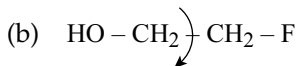
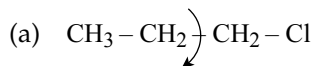
28. Choose the total number of constitutional isomers with the formula $\text{C}_4\text{H}_{10}\text{O}$.

- (a) 9 (b) 7 (c) 5 (d) 3

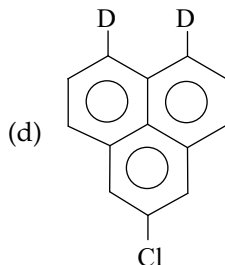
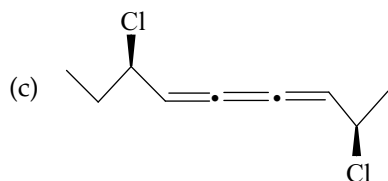
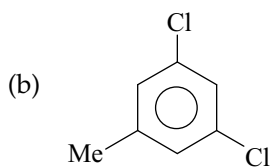
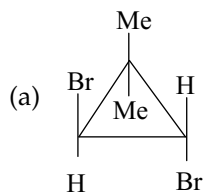
29. Identify optically active molecules



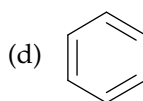
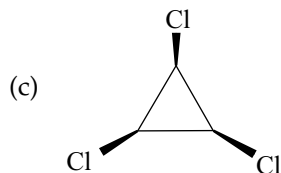
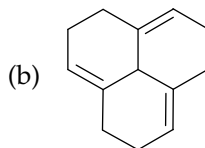
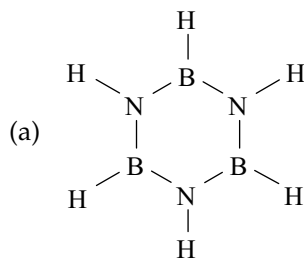
30. In which structure Gauche form has less potential energy than anti form



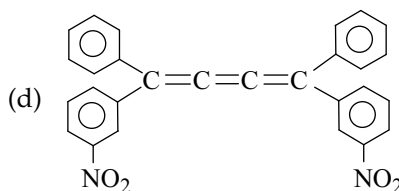
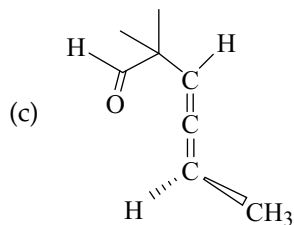
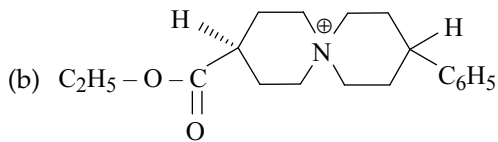
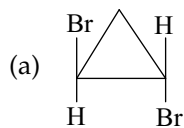
31. Which of the following molecule/s show the plane of symmetry as well as axis of symmetry?



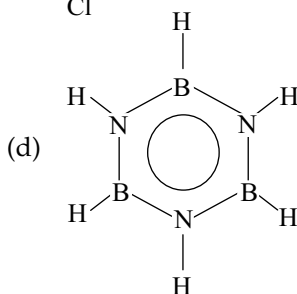
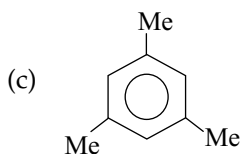
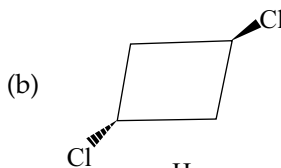
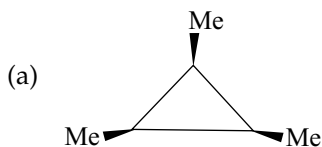
32. The C_3 axis of symmetry is present in which of the following compounds?



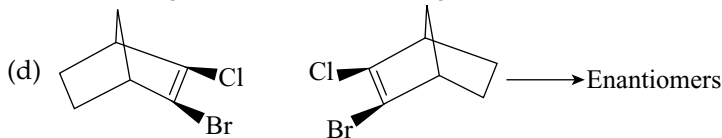
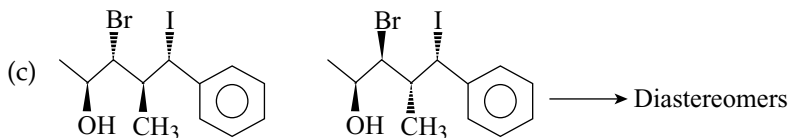
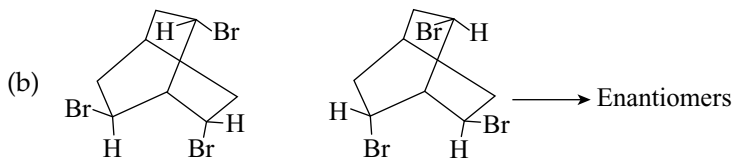
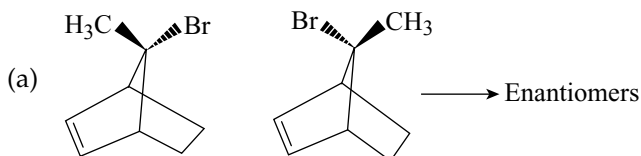
33. Which of the following compounds are chiral?



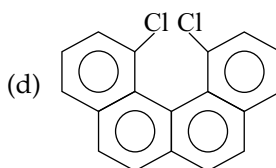
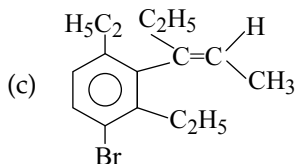
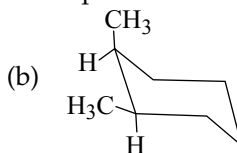
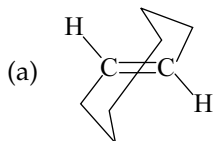
34. Which of the following molecule(s) has C_3 axis of symmetry and plane of symmetry?



35. In which of the following case correct relationship is given?



36. Which of the following is an optically active compound?



37. Which of the following options is correct?

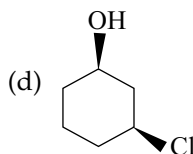
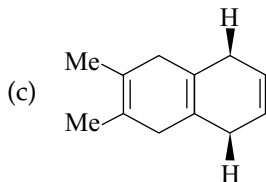
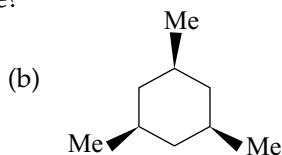
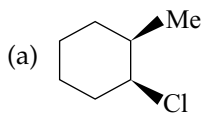
(a) $\text{p}K_1$ Cis but-2-ene dioic acid < Trans but-2-ene dioic acid

(b) Dipole moment $\begin{array}{c} \text{NC} \\ \diagup \\ \text{C} = \text{C} \diagdown \\ \text{H} \end{array} \begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{H} \end{array} < \begin{array}{c} \text{H} \\ \diagup \\ \text{C} = \text{C} \diagdown \\ \text{NC} \end{array} \begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{H} \end{array}$

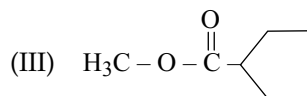
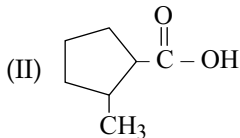
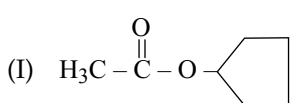
(c) Stability $\begin{array}{c} \text{Cl} \\ \diagup \\ \text{C} = \text{C} \diagdown \\ \text{H} \end{array} \begin{array}{c} \text{Cl} \\ \diagdown \\ \text{H} \end{array} < \begin{array}{c} \text{H} \\ \diagup \\ \text{C} = \text{C} \diagdown \\ \text{Cl} \end{array} \begin{array}{c} \text{Cl} \\ \diagdown \\ \text{H} \end{array}$

(d) Basicity $\begin{array}{c} \text{H} \\ \diagup \\ \text{C} = \text{C} \diagdown \\ \text{Cl} \end{array} \begin{array}{c} \text{NH}_2 \\ \diagdown \\ \text{H} \end{array} < \begin{array}{c} \text{H} \\ \diagup \\ \text{C} = \text{C} \diagdown \\ \text{Cl} \end{array} \begin{array}{c} \text{CH}_2\text{NH}_2 \\ \diagdown \\ \text{H} \end{array}$

38. Which of the following are optically active?

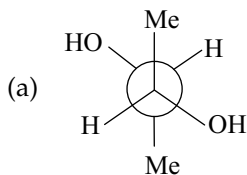


39. For the given compounds, correct statement is/are

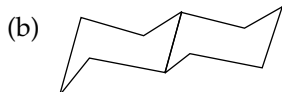
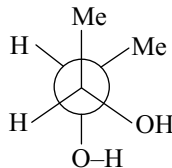


- (a) I and II are positional isomers
 (b) II and III are functional group isomers
 (c) I and III are metamers
 (d) I and III are positional isomers

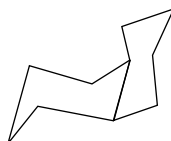
40. Among the following pair of compounds or conformers, identify pair(s) in which the Ist compound has more stability than the IInd

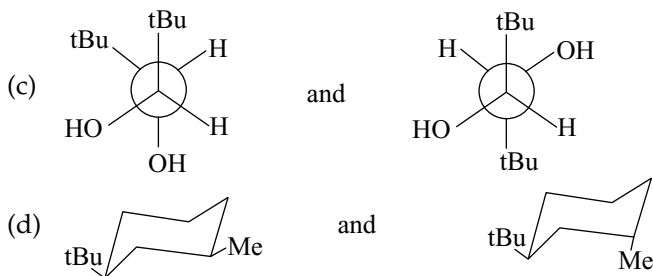


and

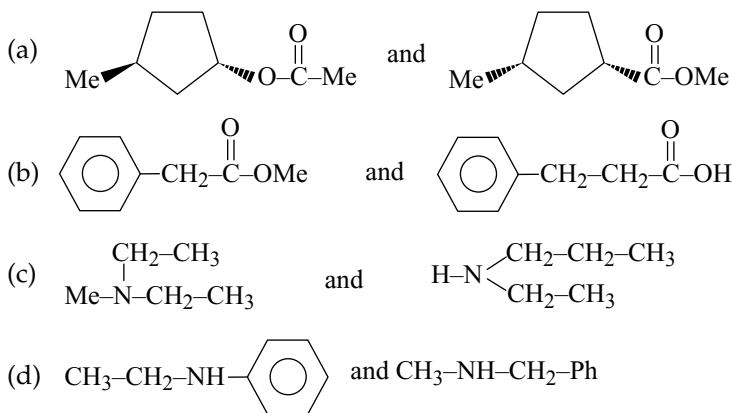


and

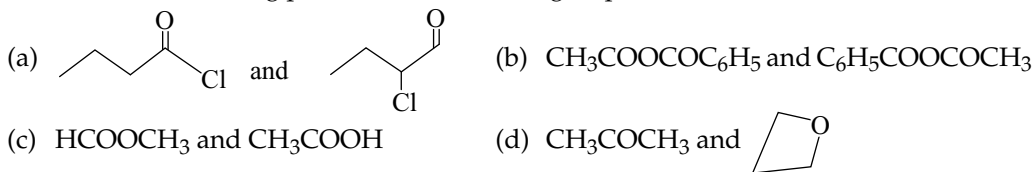




41. Among the following pair of compounds identify metamers



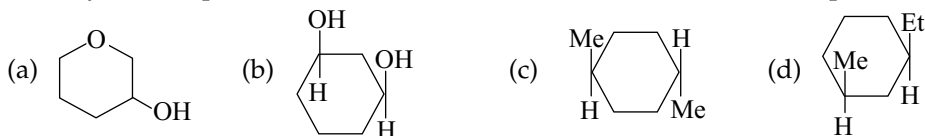
42. Which of the following pairs show functional group isomerism?



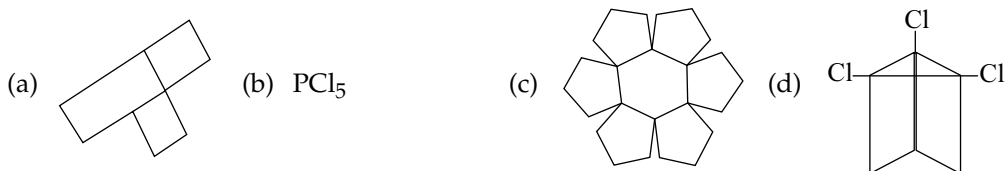
43. Identify compound(s) in which gauche conformer is more stable than staggered

- (a) Ethylene diamine (b) Succinic acid
(c) Ethylene glycol (d) n-butane

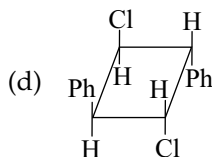
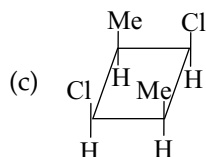
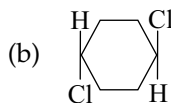
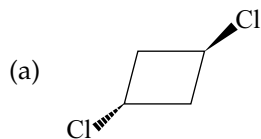
44. Identify the compound which has axial conformer more stable than equatorial



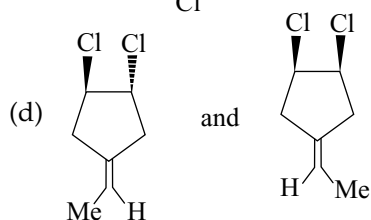
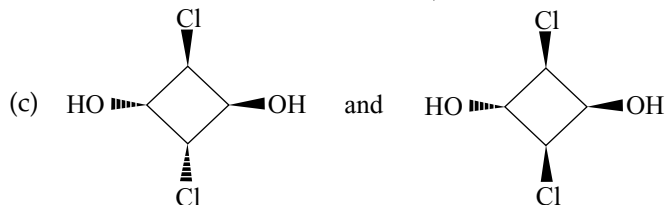
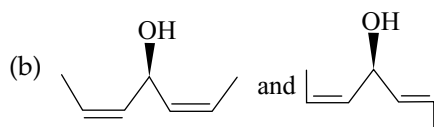
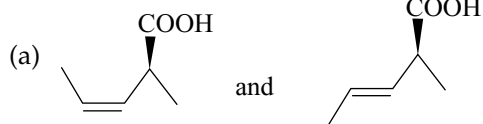
45. Among the following compounds, C_3 axis is present in



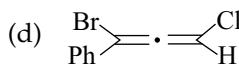
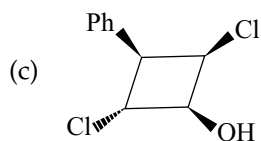
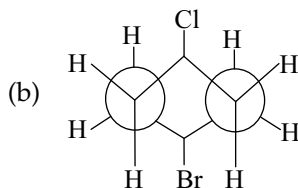
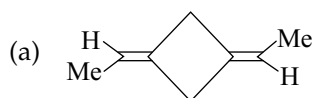
46. Identify among the following compounds having plane of symmetry, centre of symmetry and axis of symmetry



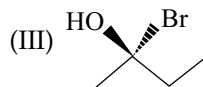
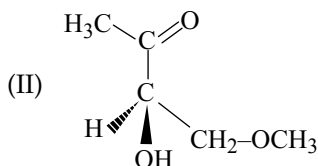
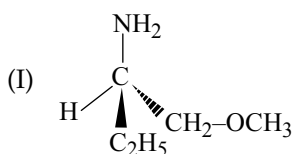
47. Identify diastereo isomeric pairs



48. Identify the optically active compound

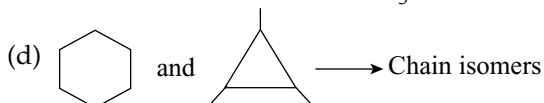
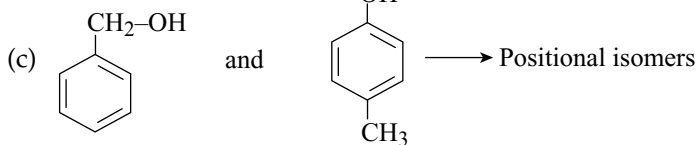
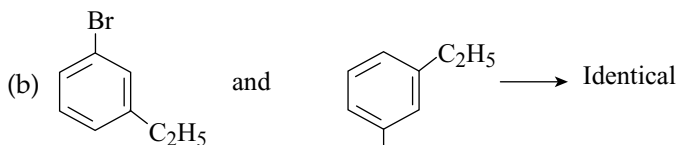
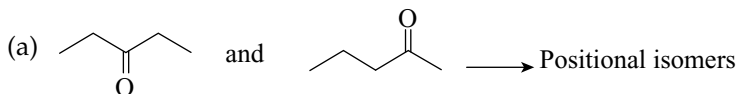


49. Which of the following statements is correct?

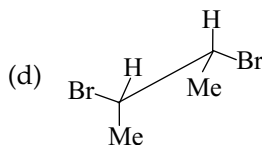
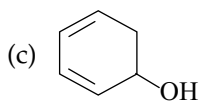
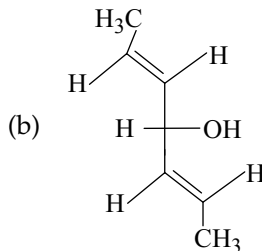
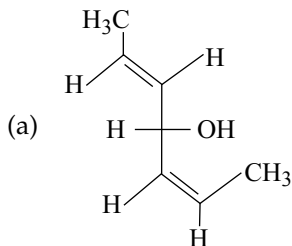


- (a) I and II have R-configuration (b) I and III have R-configuration
(c) Only III has R-configuration (d) I and III have S-configuration

50. Which of the following relationships are correctly matched?



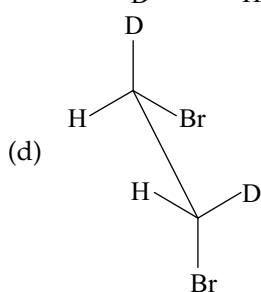
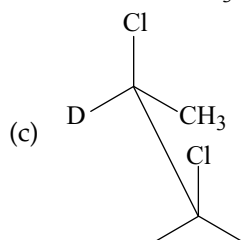
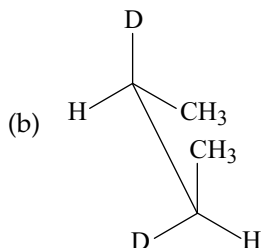
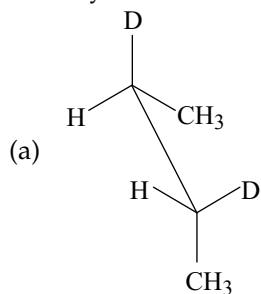
51. Which of the following molecules is/are optically active?



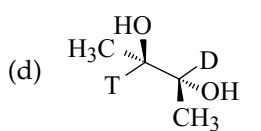
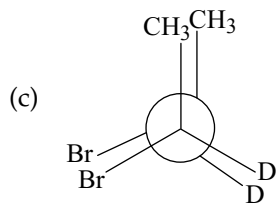
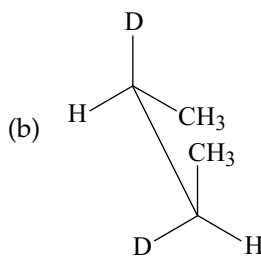
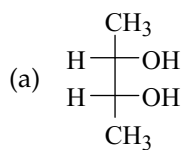
52. Identify compound(s) in which gauche conformer is more stable than staggered

- (a) 1, 2-difluoroethane (b) Chloropropane
(c) Ethylene glycol (d) Succinic acid

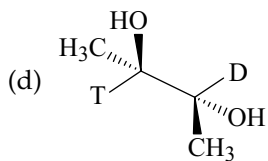
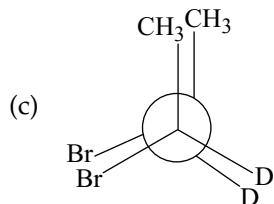
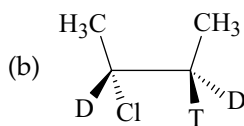
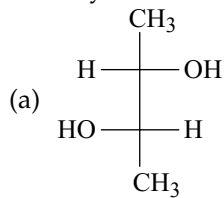
53. Identify the structure of Erythro



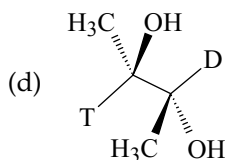
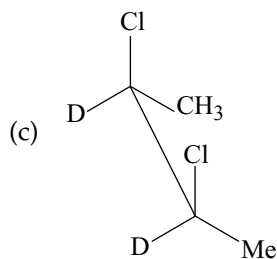
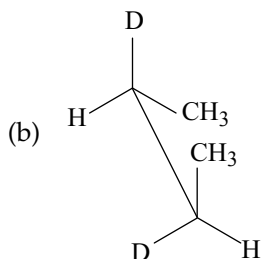
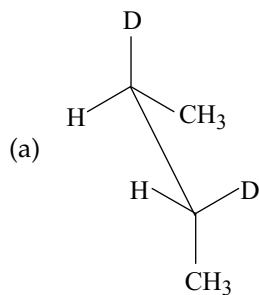
54. Identify the structure of Erythro



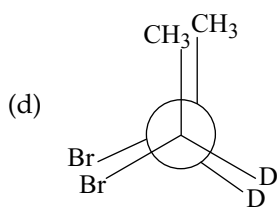
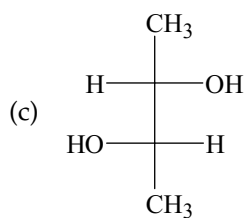
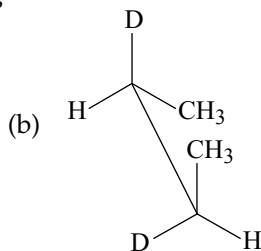
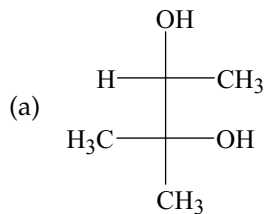
55. Identify the structure of Threo



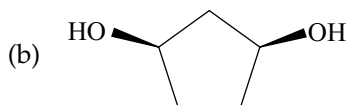
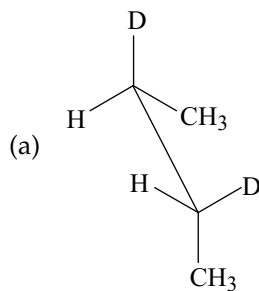
56. Identify the structure of meso compounds

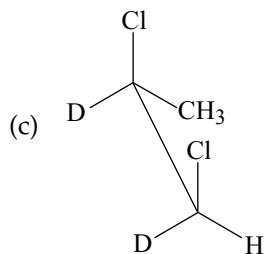


57. Identify the structure of meso compounds

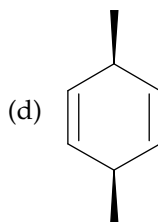
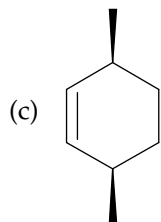
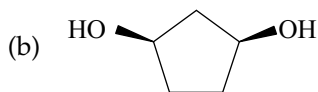
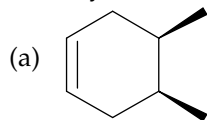


58. Identify the structure of optically inactive compounds

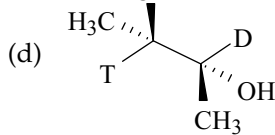
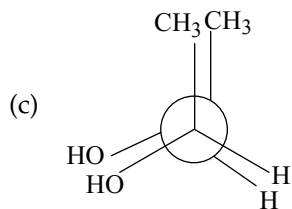
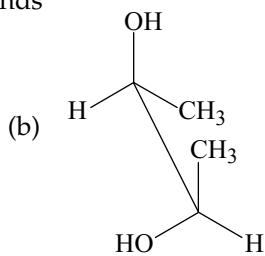
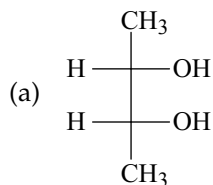




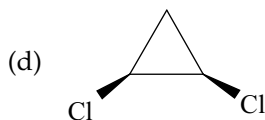
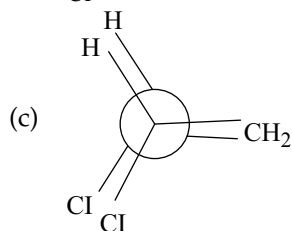
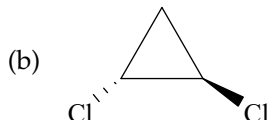
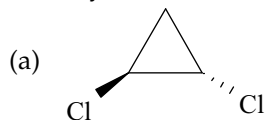
59. Identify the structure of Meso



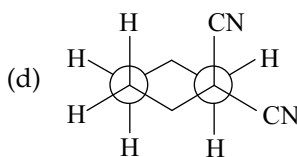
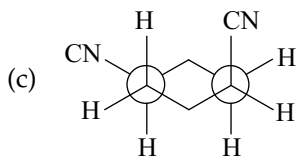
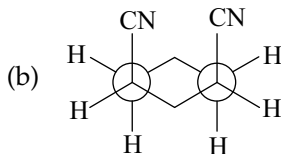
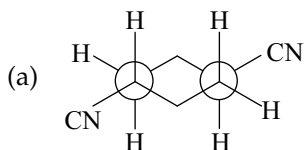
60. Identify the structure of identical compounds



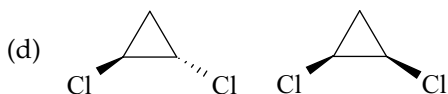
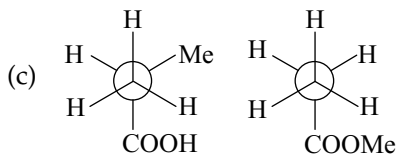
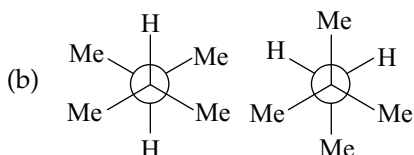
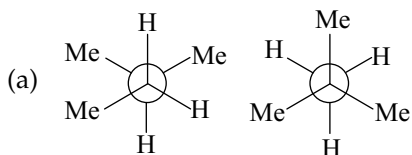
61. Identify the structure of identical compounds



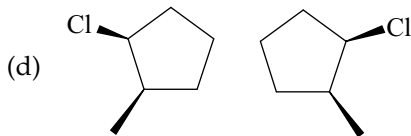
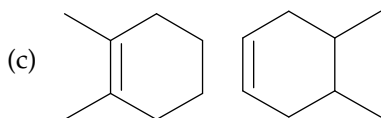
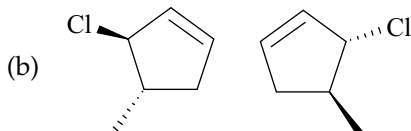
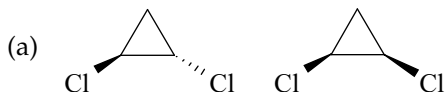
62. Identify the geometrical isomers



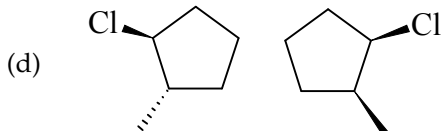
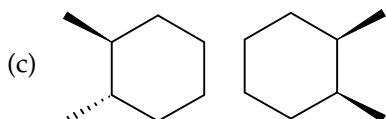
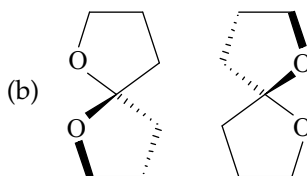
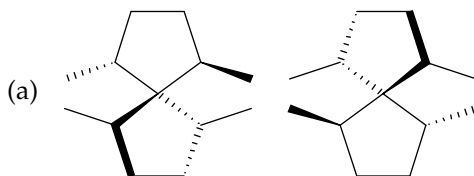
63. Identify the pair of chain isomers



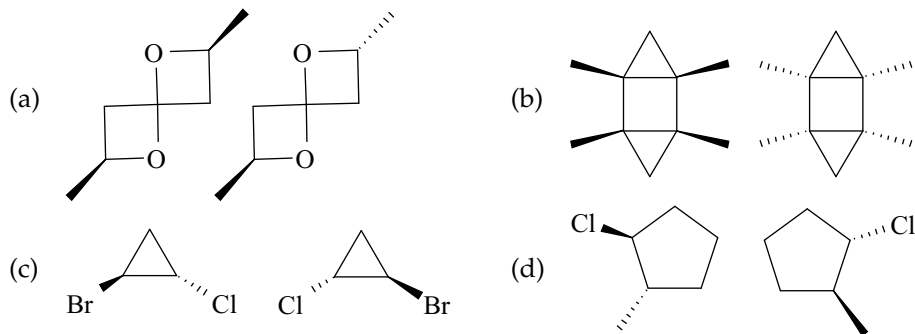
64. Identify the pair of diastereoisomers



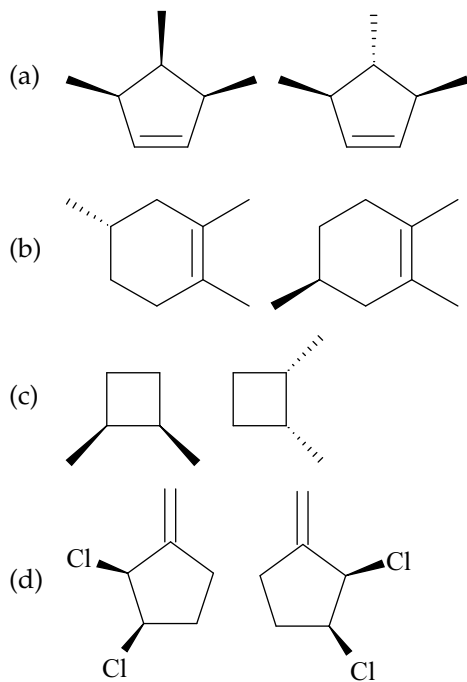
65. Identify the pair of diastereoisomers



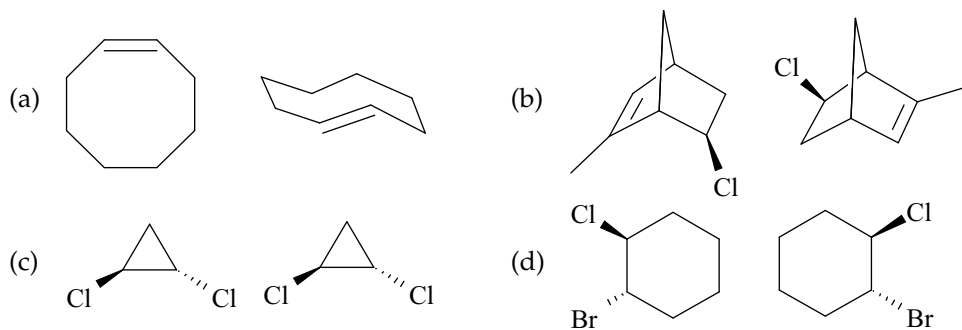
66. Identify the pair of diastereomers



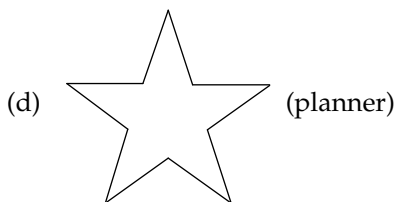
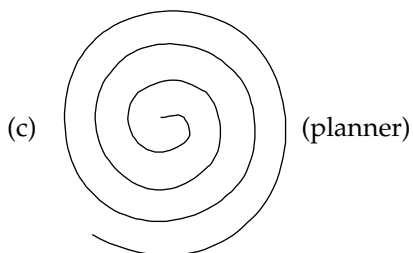
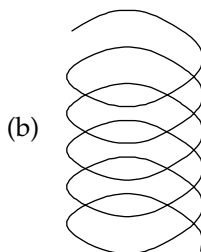
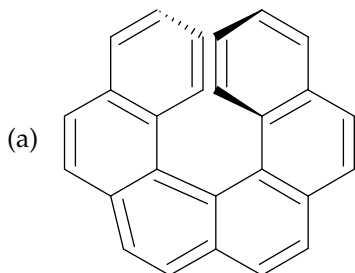
67. Identify the pair of diastereoisomers



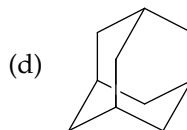
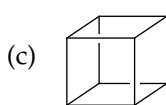
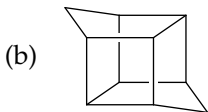
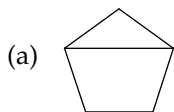
68. Identify the pair of enantiomers



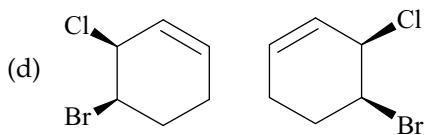
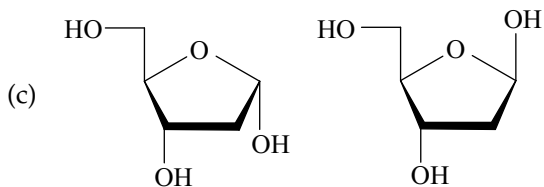
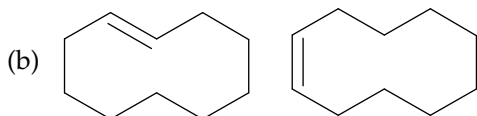
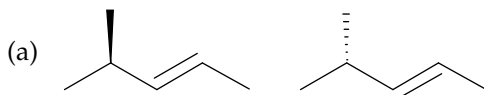
69. Identify the optically active shape



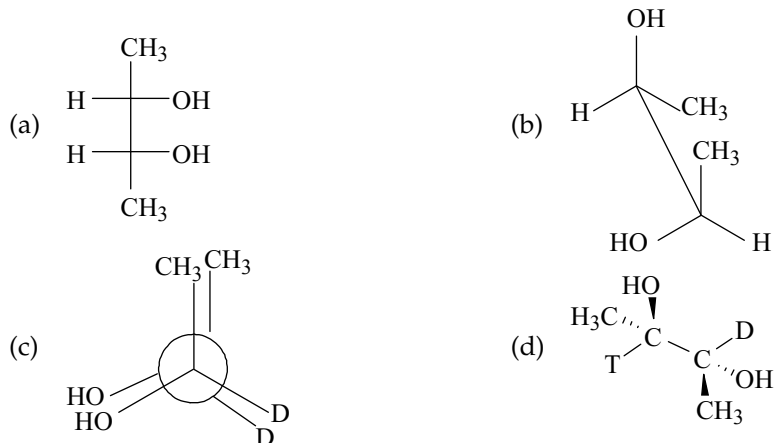
70. Identify the structures having centre of symmetry



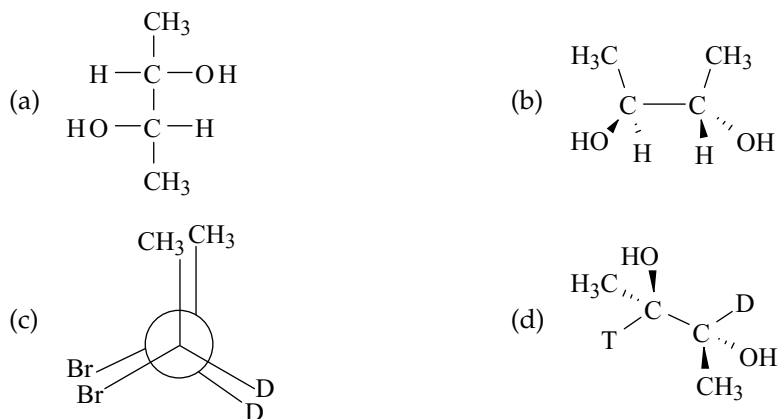
71. Identify the pair of enantiomers



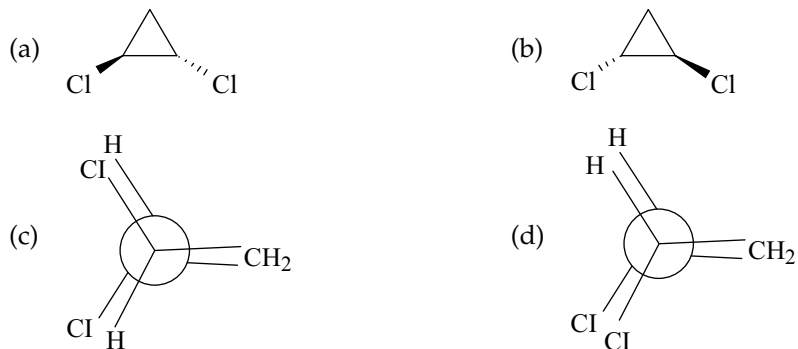
72. Identify the structure of Erythrobutane-2,3-diol



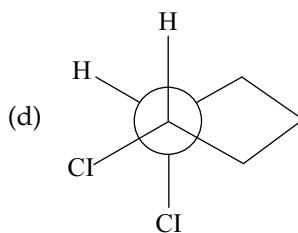
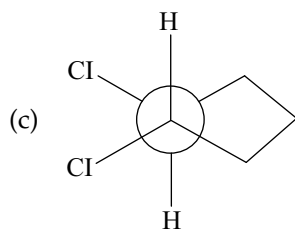
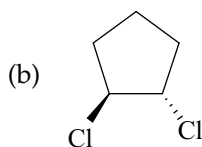
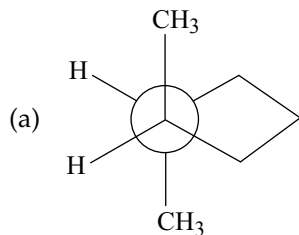
73. Identify the structure of Threobutane-2,3-diol



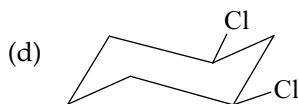
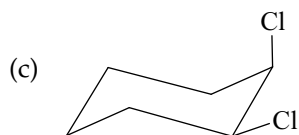
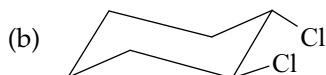
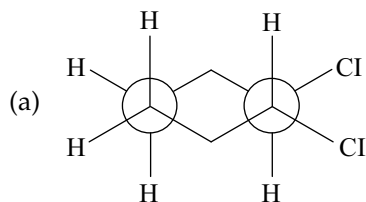
74. Identify the structure of identical compounds



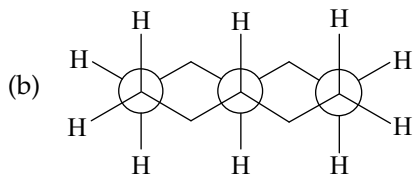
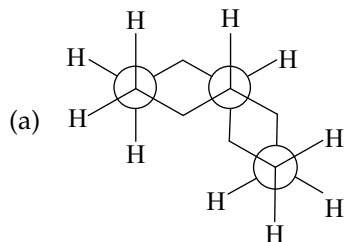
75. Identify the structure of identical compounds



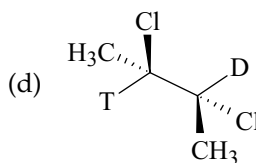
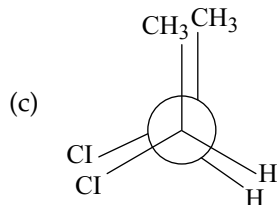
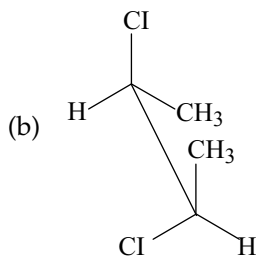
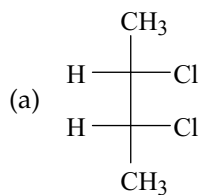
76. Identify the structure of enantiomeric compounds



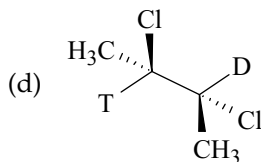
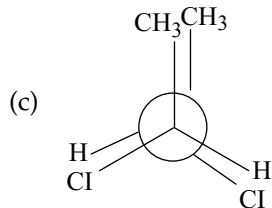
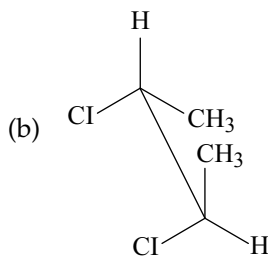
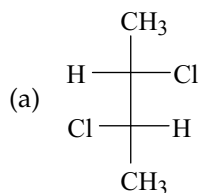
77. Identify the structure of transdecaline



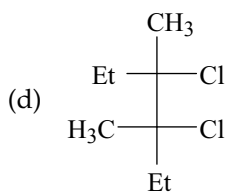
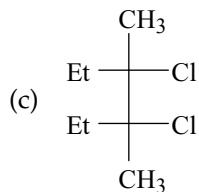
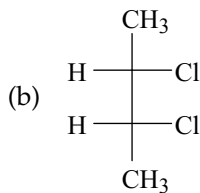
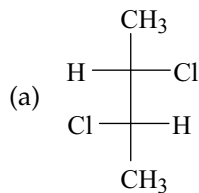
78. Identify the compounds that give trans product on reaction with Zn dust and heat



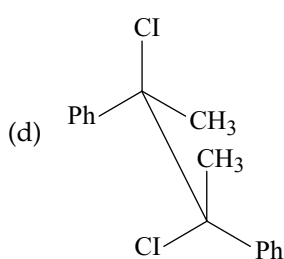
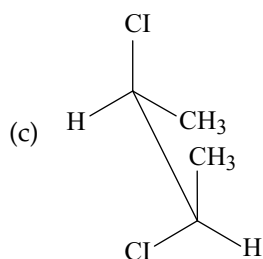
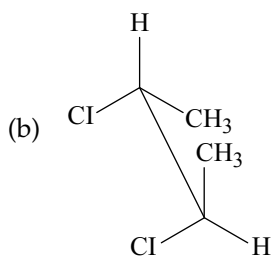
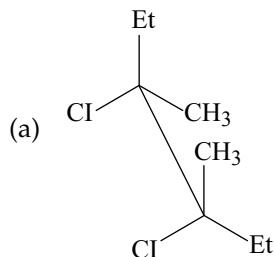
79. Identify the compounds that give cis product on reaction with Zn dust and heat



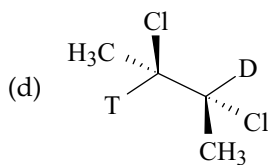
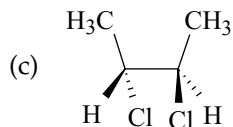
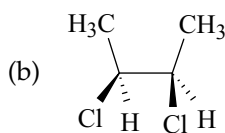
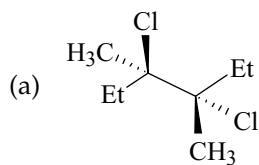
80. Identify the compounds that give cis product on reaction with Zn dust and heat



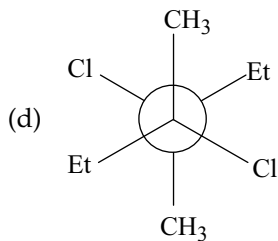
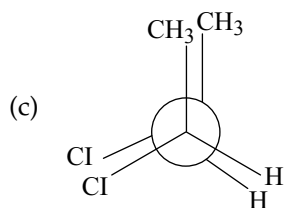
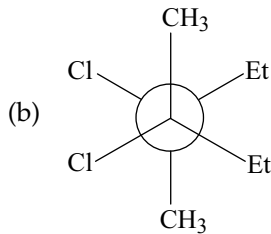
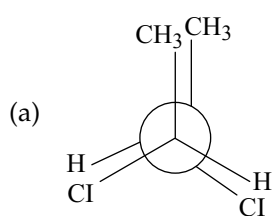
81. Identify the compounds that give cis product on reaction with Zn dust and heat



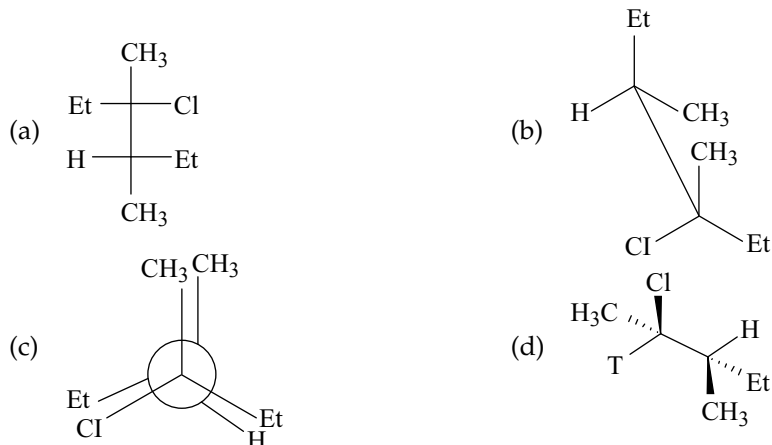
82. Identify the compounds that give trans product on reaction with Zn dust and heat



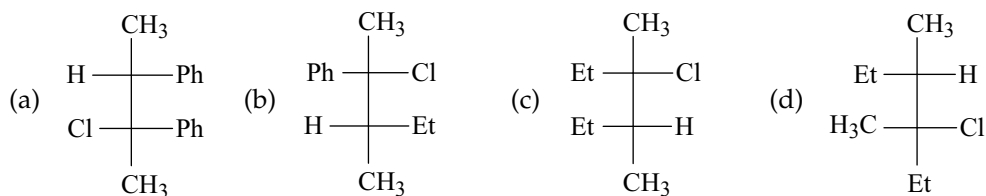
83. Identify the compounds that give trans product on reaction with Zn dust and heat



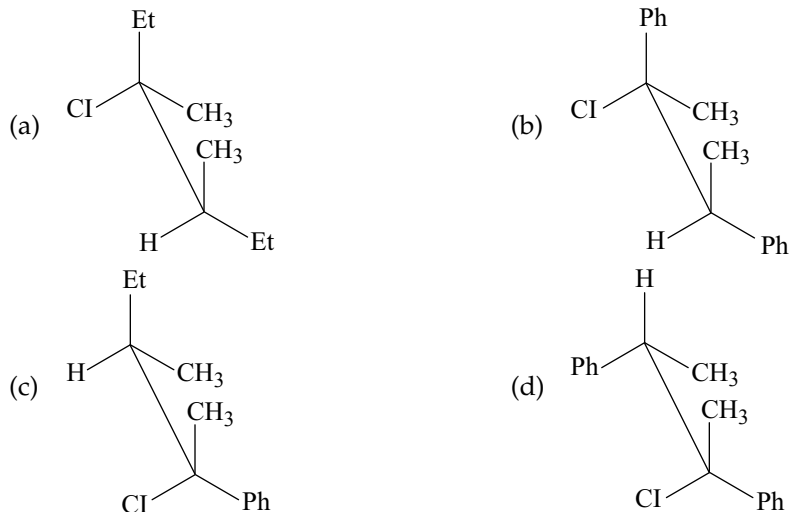
84. Identify the compounds that give trans product on reaction with alcoholic KOH and heat



85. Identify the compounds that give trans product on reaction with alcoholic KOH and heat



86. Identify the compounds that give trans product on reaction with alcoholic KOH and heat

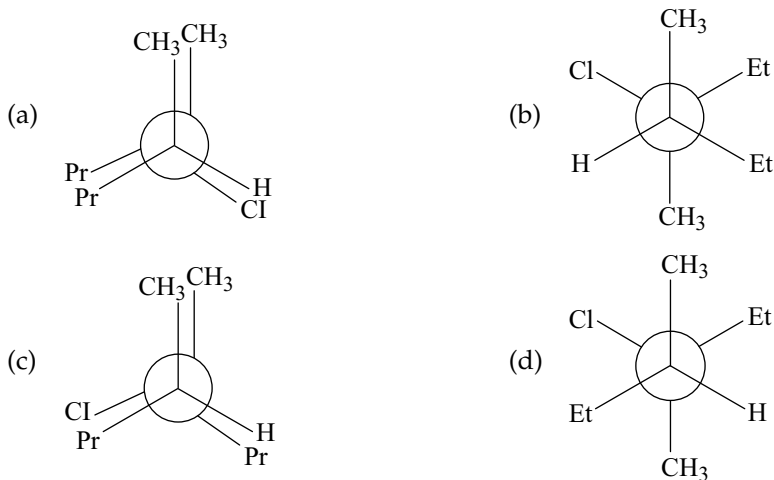


87. Identify the compounds that give trans product on reaction with alcoholic KOH and heat





88. Identify the compounds that give trans product on reaction with alcoholic KOH and heat



Comprehension Type

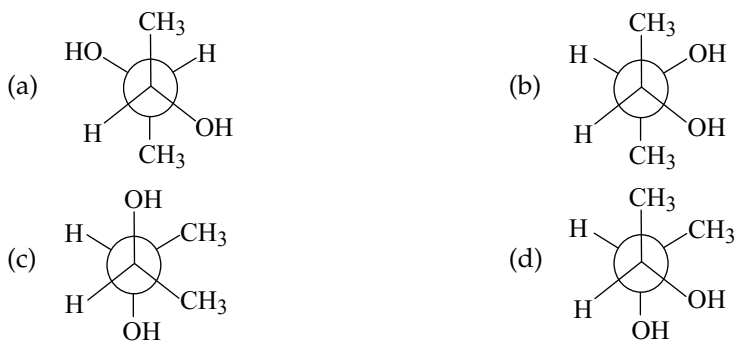
Passage 1

Different spatial arrangements of the atom that result from restricted rotation about a single bond are conformers. The general stability order of these conformer are as follows.

Anti > Gauch > Partially eclipsed > Fully eclipsed

Although anti is more stable than gauche but in some cases gauche is more stable than anti.

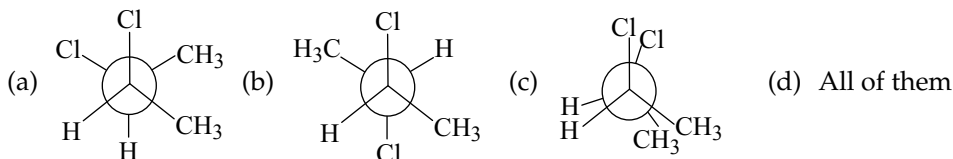
89. Which one of the following is the most stable conformer?



90. Number of possible conformers of butane is

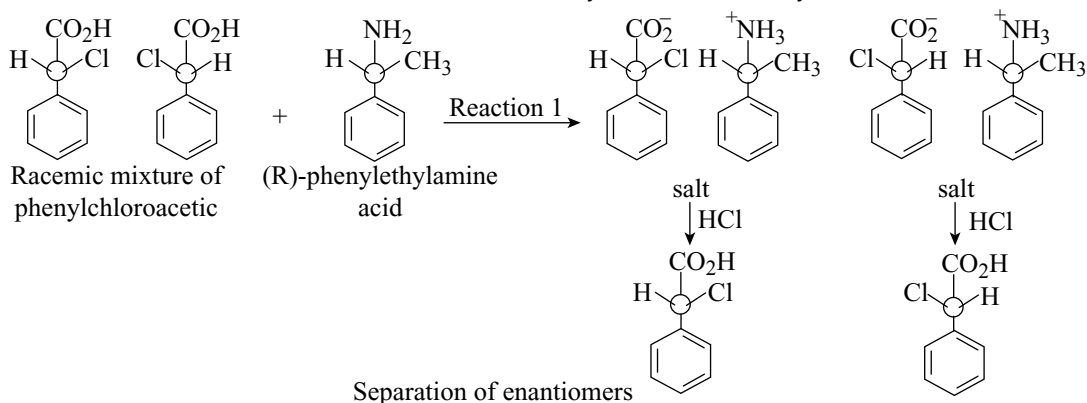
- (a) 2 (b) 4 (c) 6 (d) Infinite

91. Which of the following are more stable conformers?



Passage 2

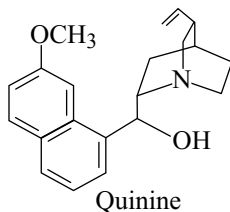
With few exceptions, enantiomers cannot be separated through physical means. When in racemic mixtures, they have the same physical properties. Enantiomers have similar chemical properties as well. The only chemical difference between a pair of enantiomers occurs in reactions with other chiral compounds. Thus resolution of a racemic mixture typically takes place through a reaction with another optically active reagent. Since living organisms usually produce only one of two possible enantiomers, many optically active reagents can be obtained from natural sources. For instance, (S)-(+)-lactic acid can be obtained from animal muscle tissue and (S)-(-)-2-methyl-1-butanol from yeast fermentation.



In the resolution of a racemic acid, a solution of (R)-phenylethylamine is reacted with a racemic mixture of phenylchloroacetic acid to form the corresponding salts. The salts are then separated by careful fractional crystallisation. Hydrochloric acid is added to the separated salts, and the respective acids are precipitated from their solutions.

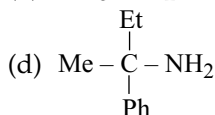
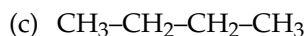
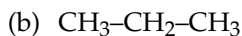
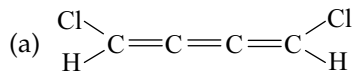
Resolution of a racemic base can be accomplished in the same manner with tartaric acid.

92. Quinine, a natural antimalarial, is commonly used as an optically active reagent to resolve acidic enantiomers. How many chiral carbons exist in the quinine molecule drawn below?

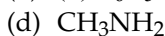
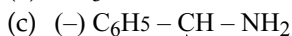
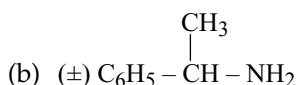
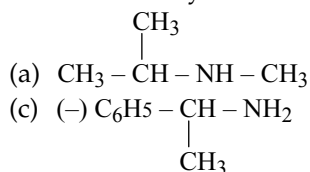


- (a) 5 (b) 2 (c) 3 (d) 4

93. Which of the following compounds might be used to resolve a racemic mixture of acidic enantiomers?



94. Which of the following amines could in principle be used as a resolving agent for a racemic carboxylic acid?



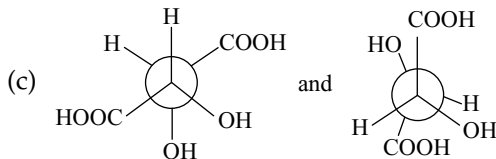
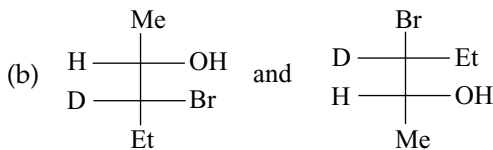
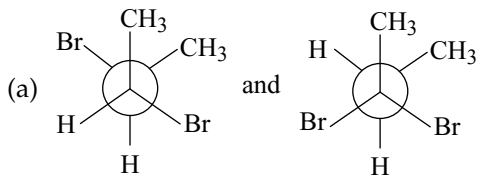
Passage 3

Stereoisomers are compounds that have same sequence of covalent bonds but differ in the relative dispositions of their atoms in space. Geometrical and optical isomers are the two important types of configurational isomers.

The compound with double bonds or ring structure has restricted rotation, so exists in two geometrical forms. The double bonds in larger rings can also cause geometrical isomerism. The optical isomers rotate the plane of plane-polarised light. A sp^3 -hybridised carbon atom bearing four different types of substituents is called an asymmetric centre or chiral centre. A chiral object or molecule cannot be superimposed on its mirror image. Stereoisomers that are mirror images of each other are called enantiomers. The stereoisomers that are not mirror images of each other are called diastereomers. Diastereomers have different physical properties.

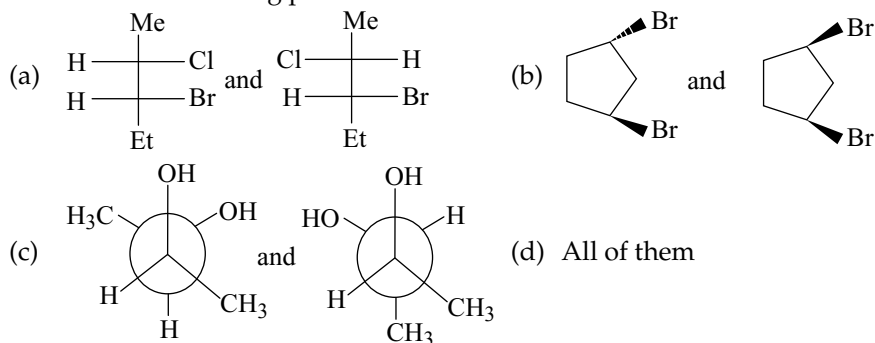
A racemic mixture is optically inactive and contains equal amounts of both the enantiomers. Resolution refers to method of separating a racemic mixture into two pure enantiomers. A meso compound is an optically inactive stereoisomer, which is achiral due to the presence of an internal plane of symmetry or centre of symmetry within the molecule.

95. The pair showing identical species is



(d) none of these

96. Which of the following pairs are diastereomers?



97. The following two compounds are



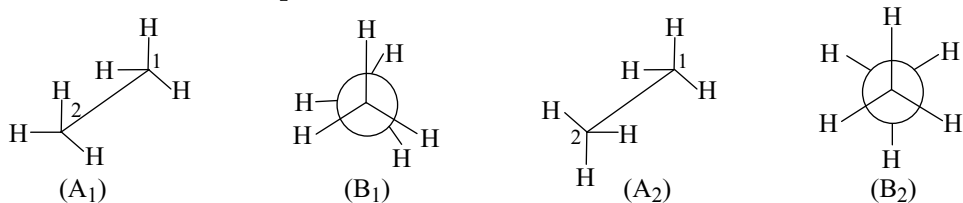
Passage 4

Isomers that have the same skeletons (that is, with component atoms attached in the same sequence) but differ from each other with relative positions of some atoms in three-dimensional space by virtue of rotation about σ bonds are called **conformational isomers or conformers**.

A quantitative description that relates relative atomic positions to the changes in potential energy during rotation about a σ bond describes the energetics of conformational interconversion, a process known as **conformational analysis**.

Rotation about (C^2-C) bond in ethane can give various conformers:

In structure A_1 or B_1 , C-H flagpoles at C-1 completely overlap each other at C-2. This form is called **eclipsed conformation**.



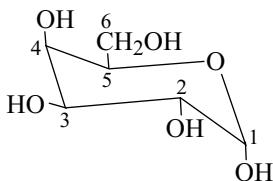
In structure A_2 or B_2 , C-H flagpoles at C-1 is fixed at 60° dihedral angle so that it is exactly between two C-H flagpoles at C-2. This is called **staggered conformation**.

- Various other structures in between eclipsed and staggered conformations are called **skew conformation**.
- Structures B_1 and B_2 show the orientations of the hydrogens on the front carbon relative to those on the back carbon, these are called **Newman projections**.
- Structures A_1 and A_2 are called **Sawhorse projections**.

98. Among the following conformers, which has highest potential energy for n-butane (along C_2-C_3 bond rotation)?

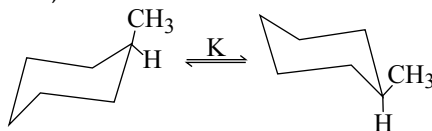
- (a) Skew (b) Fully eclipsed (c) Staggered (d) Partially eclipsed

99. In the following chair conformer correct orientation of $-OH$ groups is/are



- (a) OH groups at C_1 , C_2 and C_4 are axial
 (b) OH groups at C_1 and C_2 are axial
 (c) OH groups at C_1 and C_4 are axial
 (d) OH groups at C_1 , C_2 and C_4 are equatorial

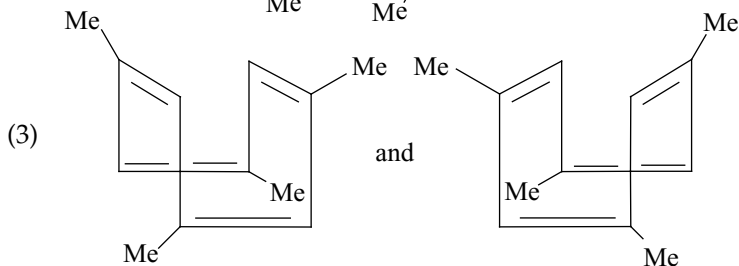
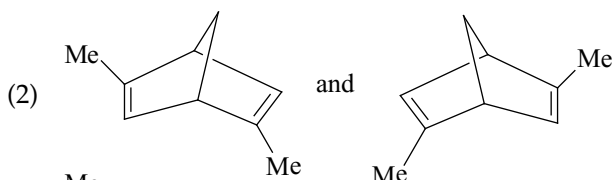
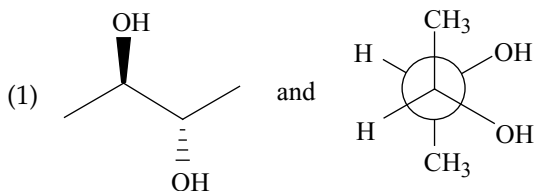
100. In the following equilibrium, the value of "K" is

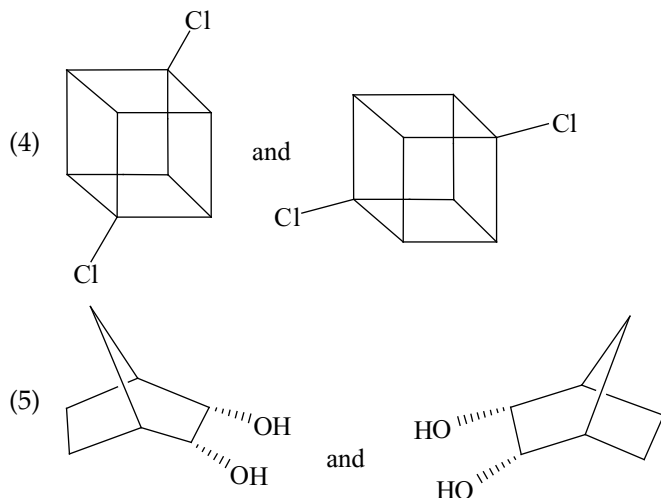


- (a) $K < 1$ (b) $K > 1$ (c) $K = 1$ (d) None of these

Passage 5

Consider the following pairs of compounds



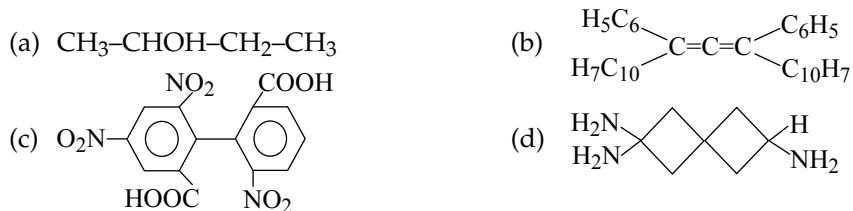


101. Identical pair of compounds is/are
 (a) 1 and 2 (b) 2 and 3 (c) 3, 4 and 5 (d) 1 and 5
102. Both compounds in above pair are meso
 (a) 1st pair (b) 2nd pair (c) 5th pair (d) 4th pair
103. Diastereoisomeric pair is
 (a) 1st pair (b) 2nd pair (c) 3rd pair (d) 4th pair

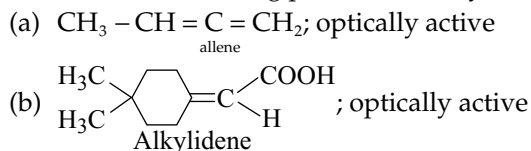
Passage 6

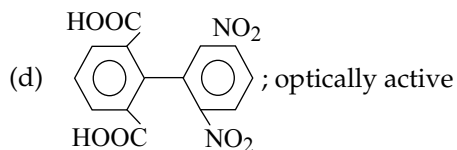
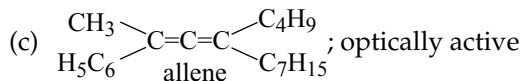
Presence of chiral carbon in organic compound is neither a necessary nor a sufficient condition showing optical activity. The chirality, i.e., dissymmetry of a molecule as a whole is the necessary condition for optical activity.

104. Which of these compounds will NOT show optical activity?

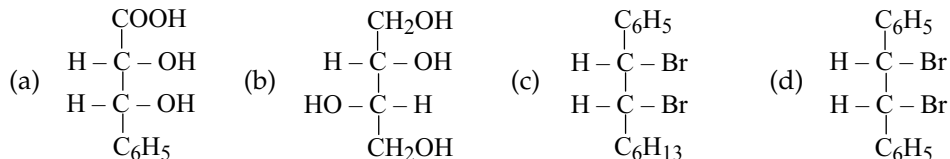


105. Which of the following pairs is correctly matched?



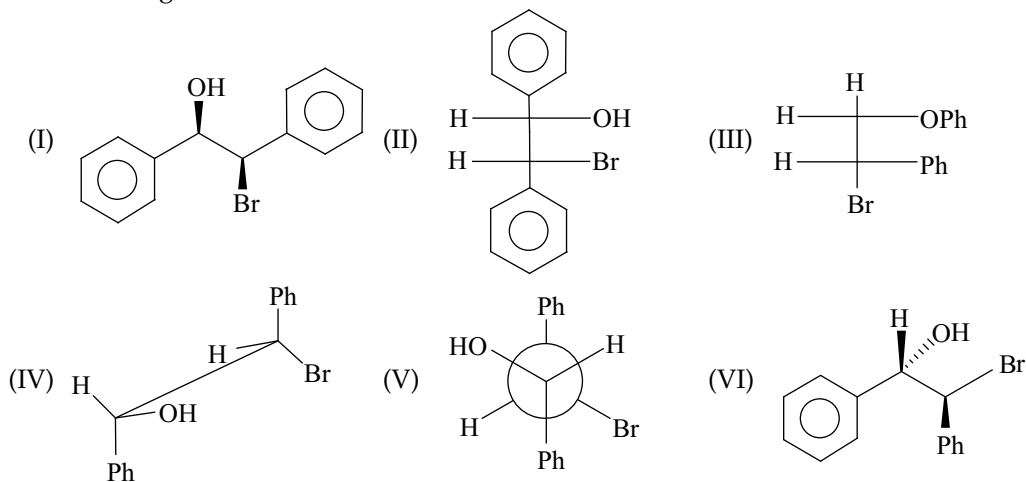


106. Which one of the following is an achiral molecule?



Passage 7

Observe the given molecule/s



107. Correct statement is

- (a) I and II are enantiomers (b) III and V are diastereomers
(c) I and IV are identical (d) I and VI are diastereomers

108. Diastereomeric pair is

- (a) I and II (b) I and VI (c) II and V (d) III and VI

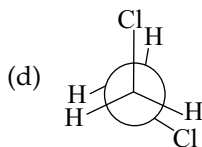
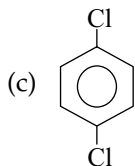
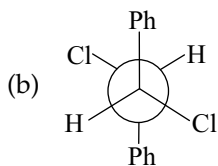
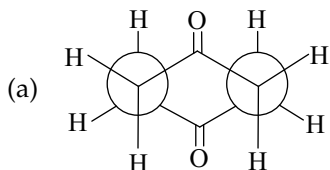
109. If Br is attached in place of -OH group in structure "V" (with same stereochemistry) then resultant structure is

- (a) Meso
(b) Threo
(c) Structure has odd degree of unsaturation
(d) Structure is identical to (I) compound

Matrix Type

Match the columns

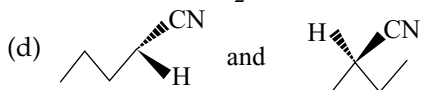
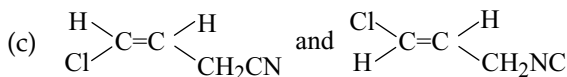
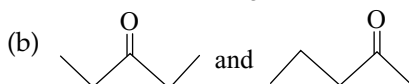
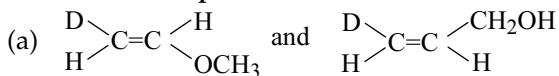
110. Column I



Column II

- (p) Compounds give positional isomerisation with 1,2-dichloro benzene.
- (q) Degree of unsaturation in the given compound is even [excluding zero value]
- (r) Given compound or conformer has zero dipole moment.
- (s) Net dipole of one mole of compound is nonzero.
- (t) Compound has ketone functional group.

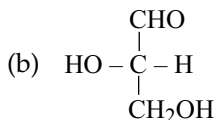
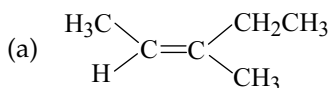
111. Column I (Compound)



Column II (Isomerism)

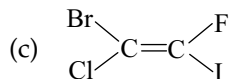
- (p) Functional isomers
- (q) Geometrical isomers
- (r) Position isomers
- (s) Chain isomers
- (t) Metamer

112. Column I (Structure)

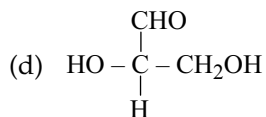


Column II (Configuration)

- (p) E
- (q) Z



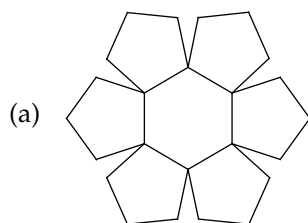
(r) R



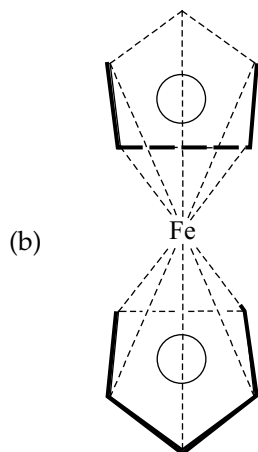
(s) S

(t) Plane of symmetry

113. Column I

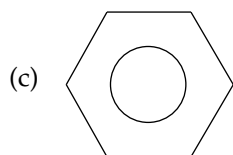


(p) Plane of symmetry absent

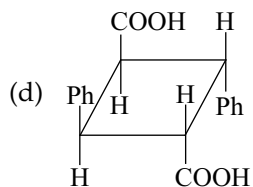


(q) C_6

Ferrocene (staggered)



(r) S_6

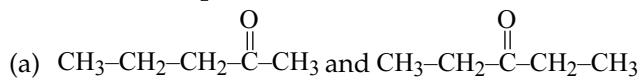


(s) C_i/COS

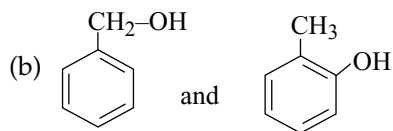
(t) C_5

114. Column I (Compounds)

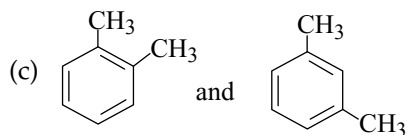
Column II (Type of isomerism)



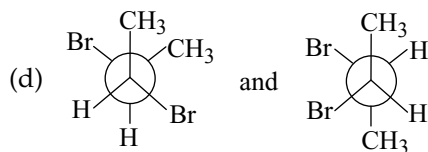
(p) Positional



(q) Conformational



(r) Metamerism

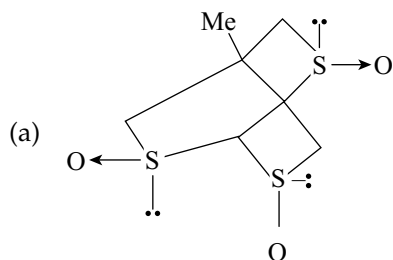


(s) Functional

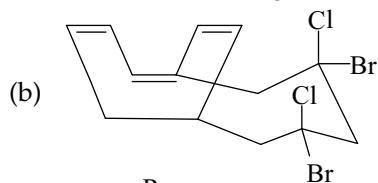
(t) Optical isomerism

115. Column I

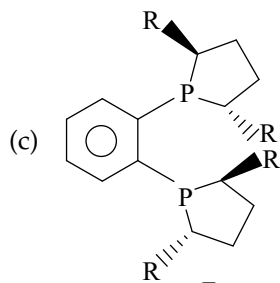
Column II



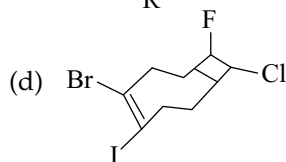
(p) Optically inactive compound



(q) Optically active compound



(r) Meso compound

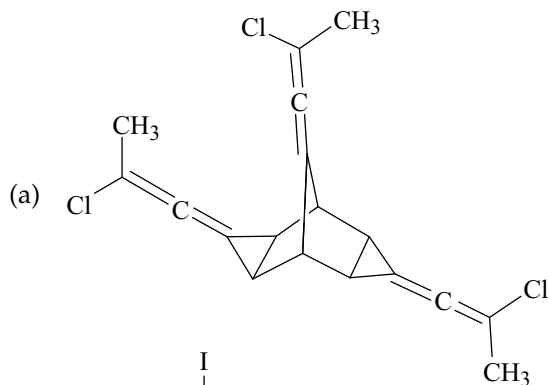


(s) Axis of symmetry present

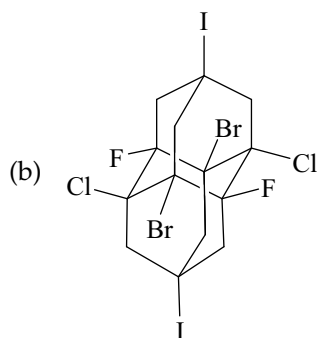
(t) Asymmetric compound

116. Column I

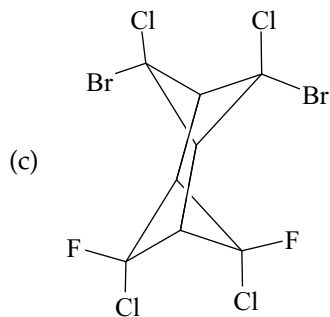
Column II



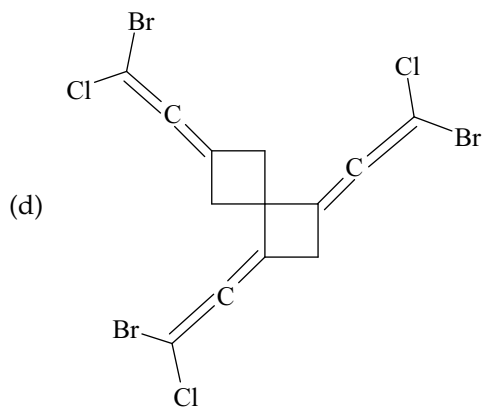
(p) Optically inactive compound



(q) Optically active compound



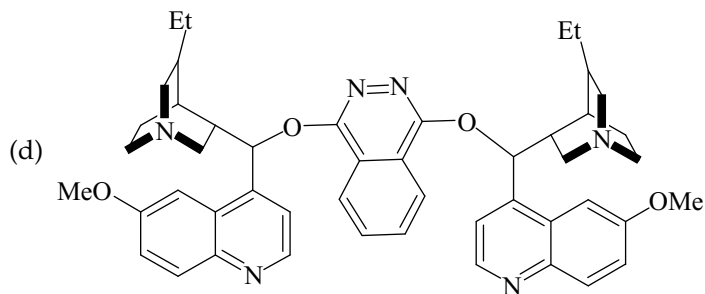
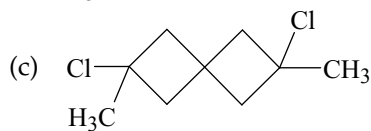
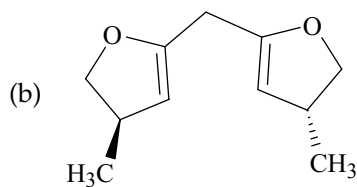
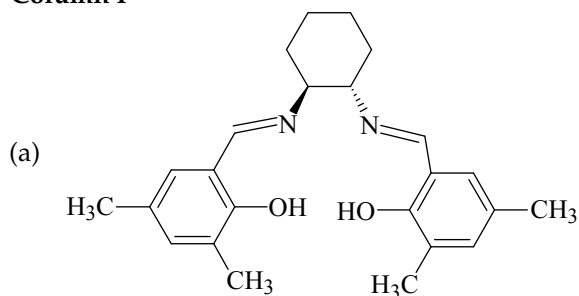
(r) Meso compound



(s) Axis of symmetry present

(t) Asymmetric compound

117. Column I



Column II

(p) Optically inactive compound

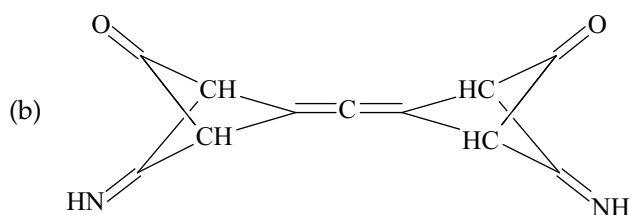
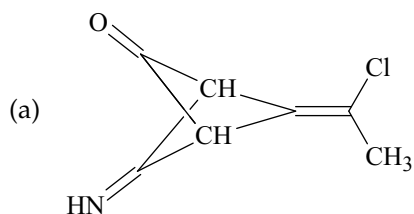
(q) Optically active compound

(r) Meso compound

(s) Axis of symmetry present

(t) Dissymmetric compound

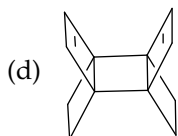
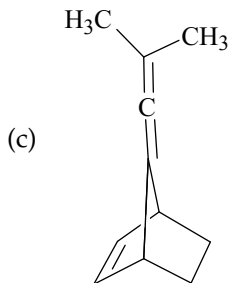
118. Column I



Column II

(p) Optically inactive compound

(q) Optically active compound

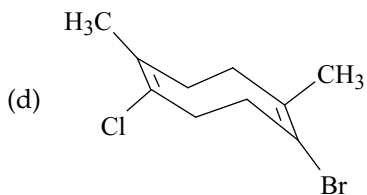
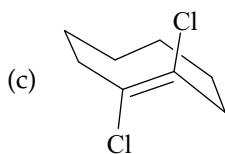
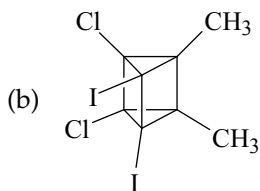
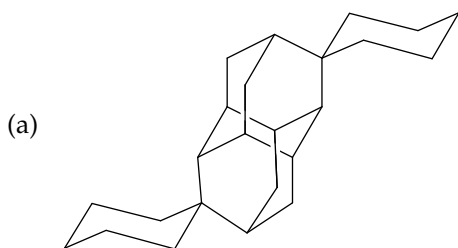


(r) Meso compound

(s) Axis of symmetry present

(t) Asymmetric compound

119. Column I



Column II

(p) Optically inactive compound

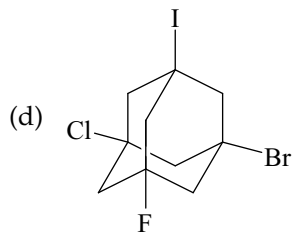
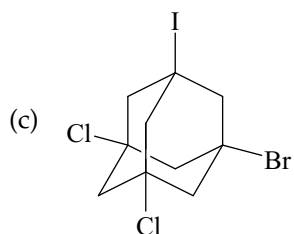
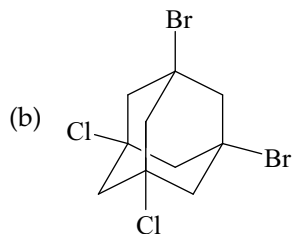
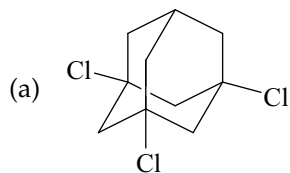
(q) Optically active compound

(r) Meso compound

(s) Axis of symmetry present

(t) Asymmetric compound

120. Column I



Column II

(p) Optically inactive compound

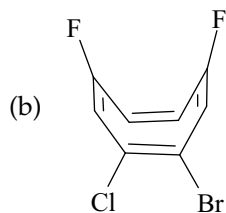
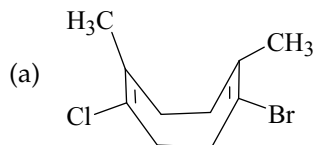
(q) Optically active compound

(r) Dissymmetric compound

(s) Axis of symmetry present

(t) Asymmetric compound

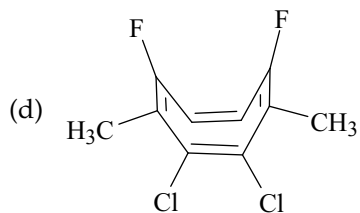
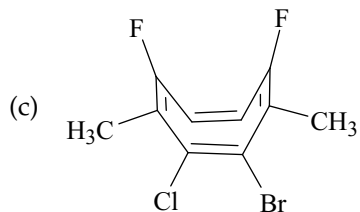
121. Column I



Column II

(p) Optically inactive compound

(q) Optically active compound

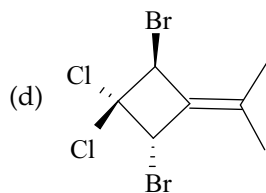
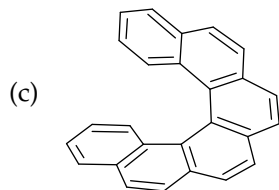
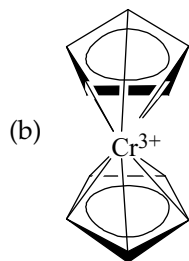
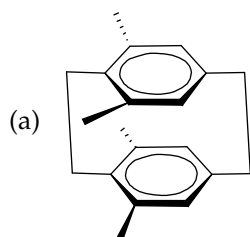


(r) Degree of unsaturation is odd

(s) Axis of symmetry present

(t) Asymmetric compound

122. Column I



Column II

(p) Optically inactive compound

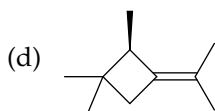
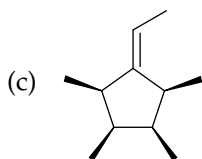
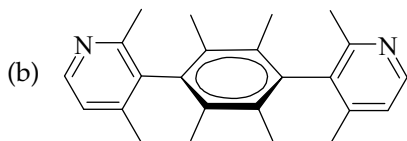
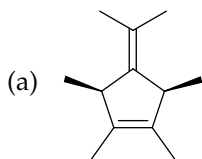
(q) Optically active compound

(r) Dissymmetric compound

(s) Axis of symmetry present

(t) Asymmetric compound

123. Column I



Column II

(p) Optically inactive compound

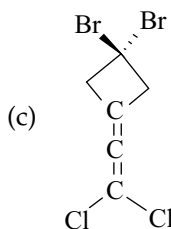
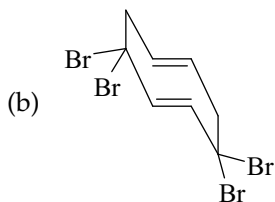
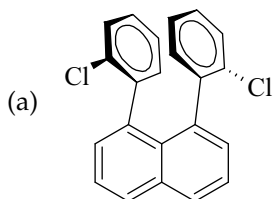
(q) Optically active compound

(r) Dissymmetric compound

(s) Axis of symmetry present

(t) Asymmetric compound

124. Column I

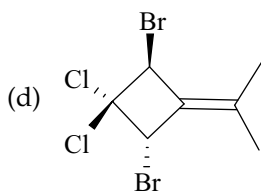


Column II

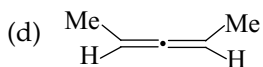
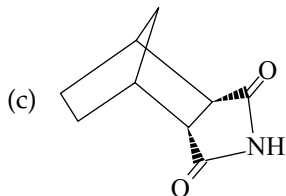
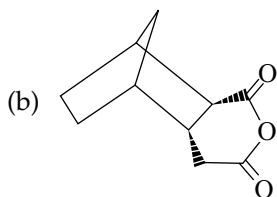
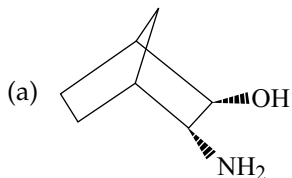
(p) Optically inactive compound

(q) Optically active compound

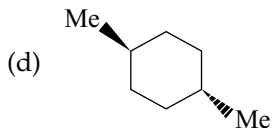
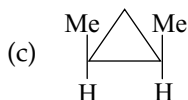
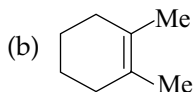
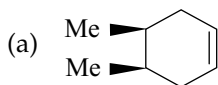
(r) Dissymmetric compound



125. Column I



126. Column I



(s) Axis of symmetry present

(t) Asymmetric compound

Column II

(p) Optically active molecules without chiral centres

(q) Optically active molecules with chiral centres.

(r) Compounds have even number of chiral centres

(s) Optically inactive molecules

Column II

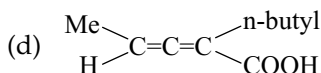
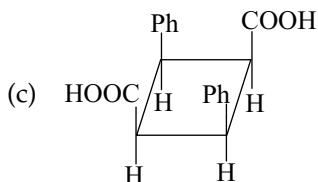
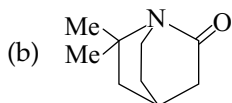
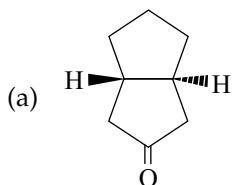
(p) Optically active

(q) Optically inactive

(r) Compounds show geometrical isomerism

(s) Plane of symmetry

127. Column I



Column II

(p) Molecule has chiral centre

(q) Molecule is asymmetric

(r) Molecule is dissymmetric

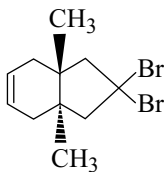
(s) Molecule is optically inactive

Integer Type

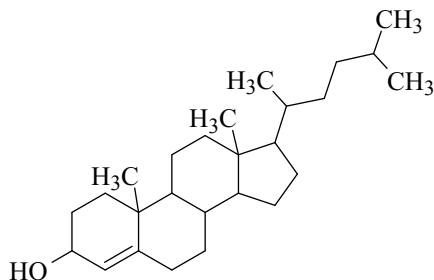
128. $\mu_{\text{obs}} = \sum \mu_i x_i$; where μ_i is the dipole moment of stable conformer and x_i is the mole fraction of that conformer of $Z\text{-CH}_2\text{-CH}_2\text{-Z}$ in Newmann's projection. If $\mu_{\text{solution}} = 1$ D and mole fraction of anti form = 0.82, find μ_{Gauche} .

129. Number of primary amine possible for $\text{C}_4\text{H}_{11}\text{N}$ is

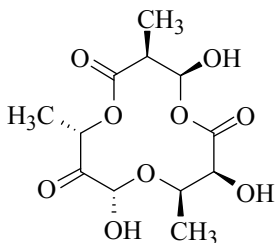
130. How many chiral centres are present in the following compound?



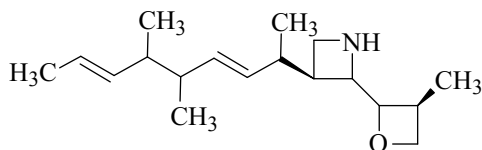
131. How many chiral centres are present in the following compound?



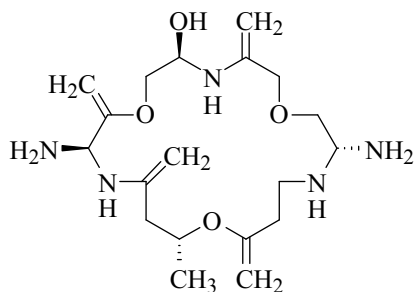
132. How many stereoisomers are possible for the following compound?



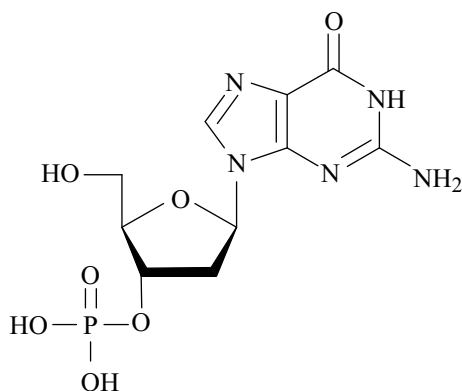
133. How many stereoisomers are possible for the following compound?



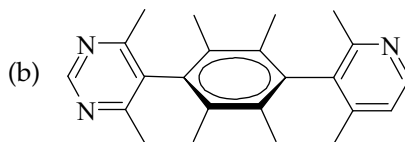
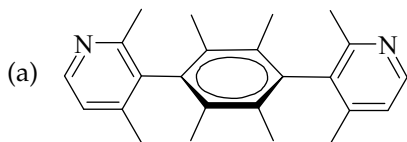
134. How many stereoisomers are possible for the following compound?

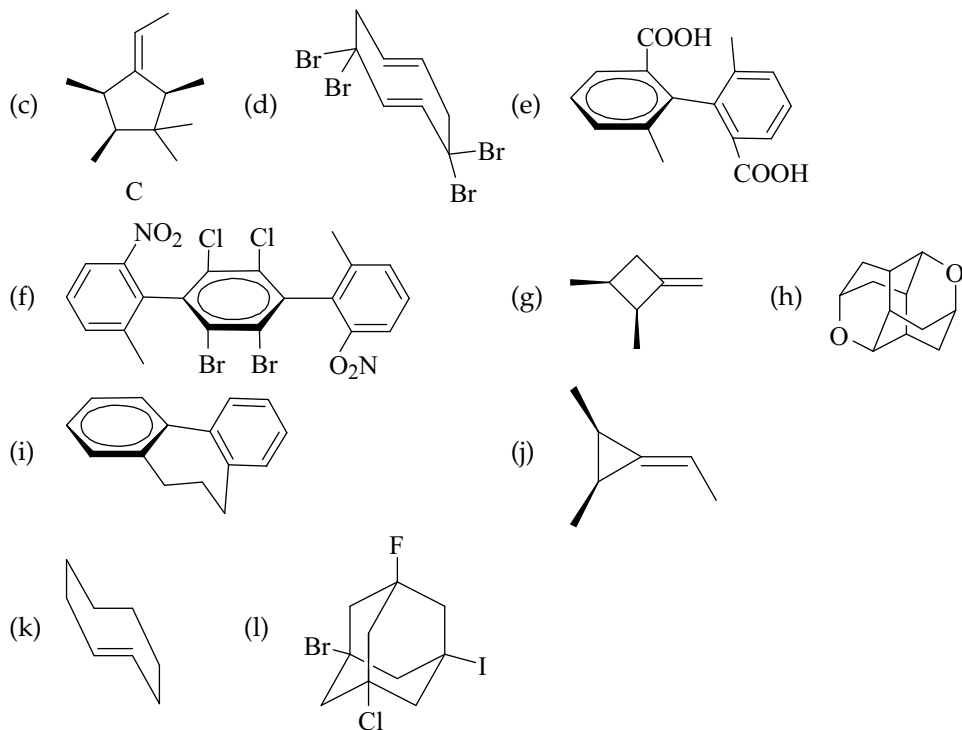


135. How many stereoisomers are possible for the following compound?

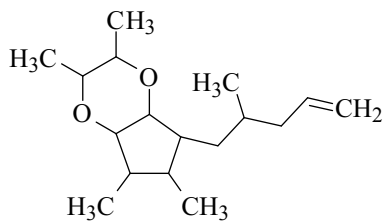


136. How many compounds are optically active?

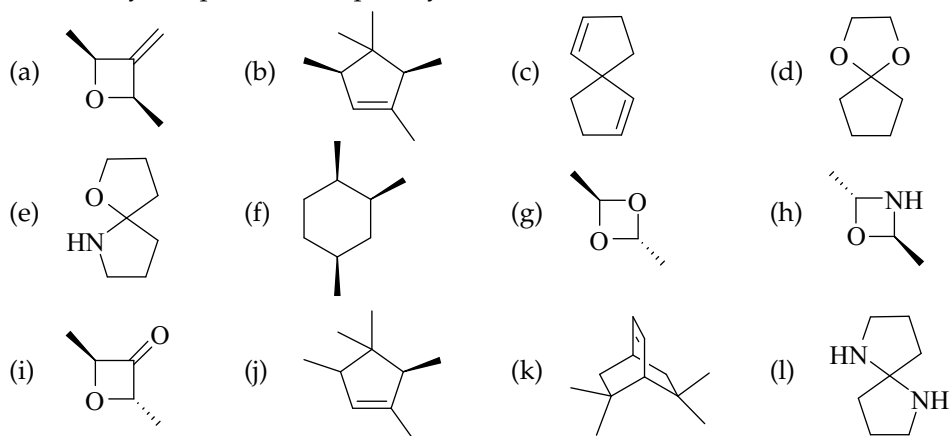


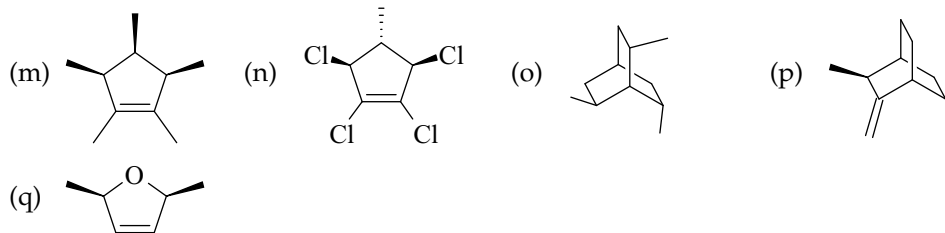


137. How many stereocentres are possible for the following compound?

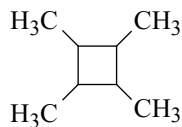


138. How many compounds are optically inactive?

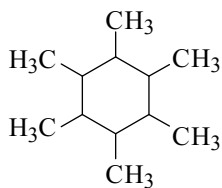




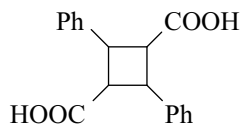
139. How many geometrical isomers are possible for the following compound?



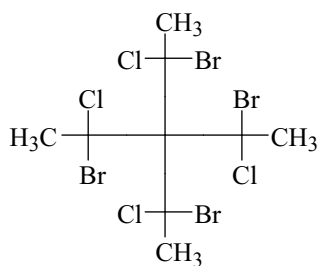
140. How many geometrical isomers are possible for the following compound?



141. How many geometrical isomers are possible for the following compound?

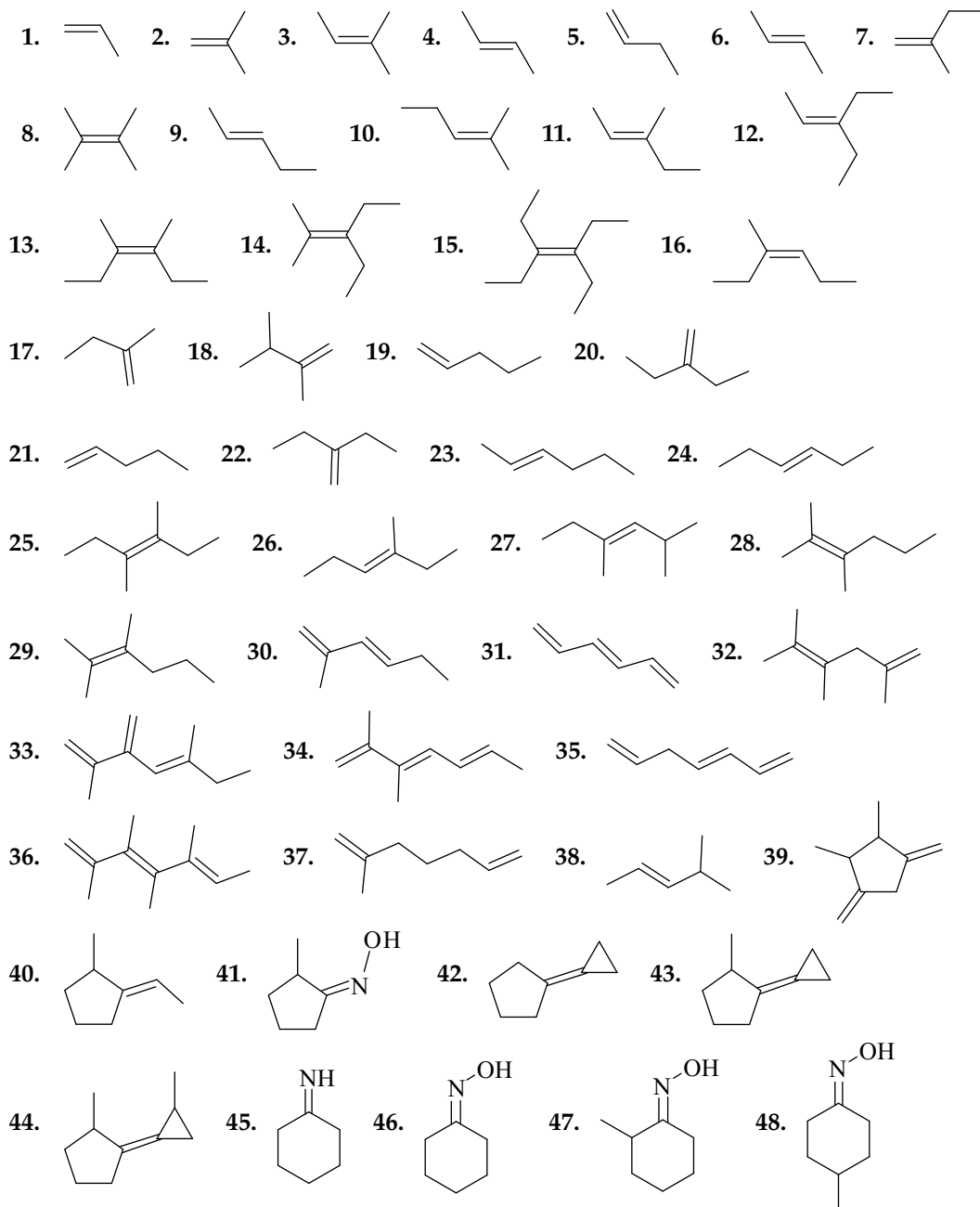


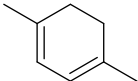
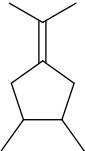
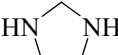
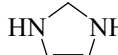
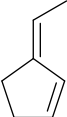
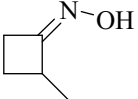
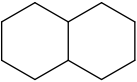
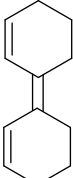
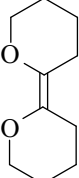
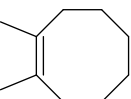
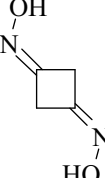
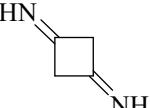
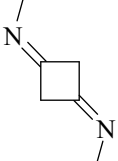
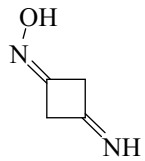
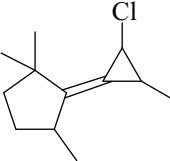
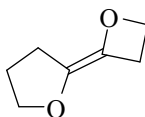
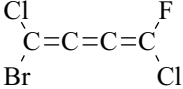
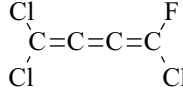
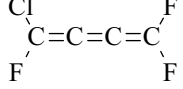
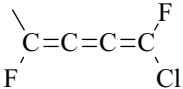
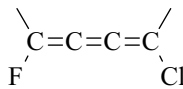
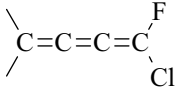
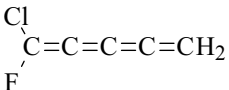
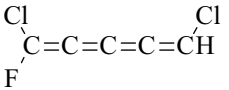
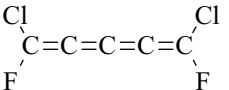
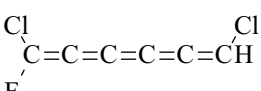
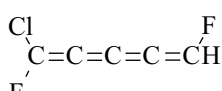
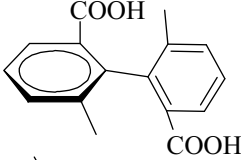
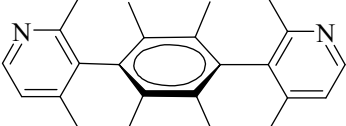
142. How many stereoisomers are possible for the following compound?

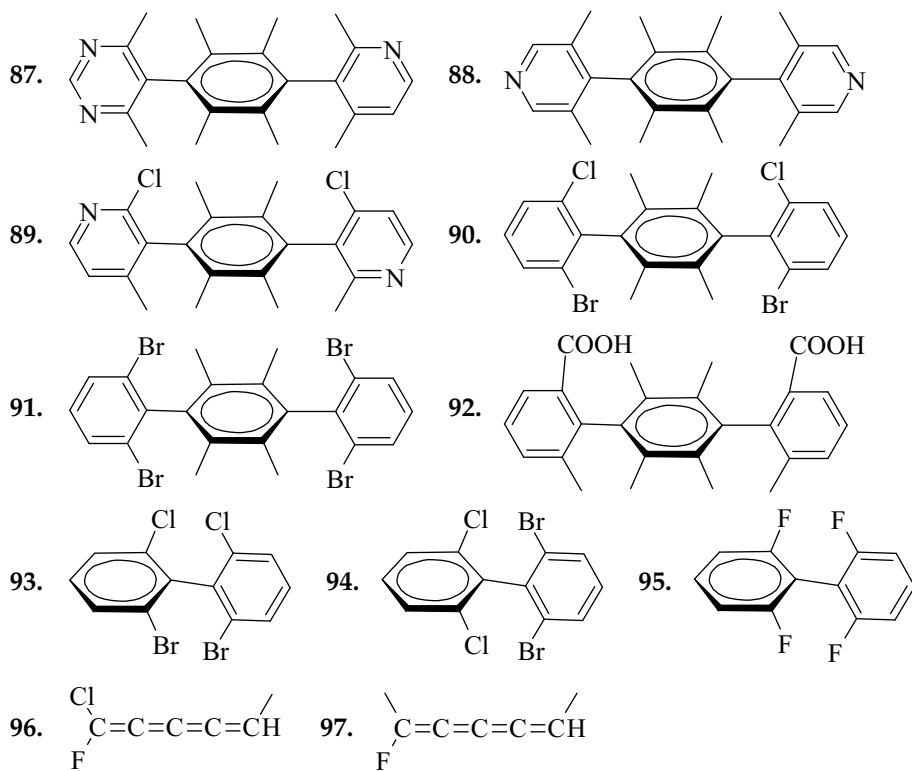


WORKBOOK EXERCISE 1

Identify molecules that show Geometrical Isomerism

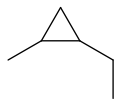
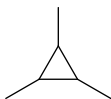
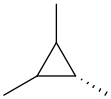
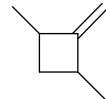
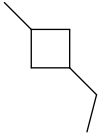
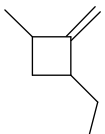
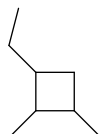
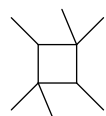
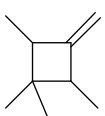
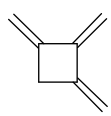
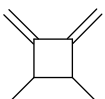
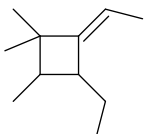
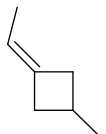
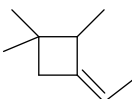
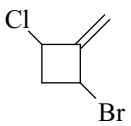
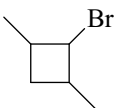
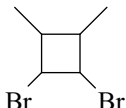
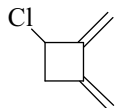
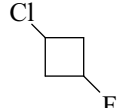
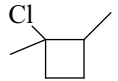
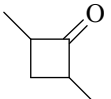
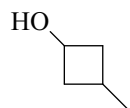
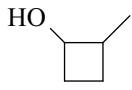
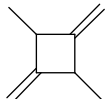
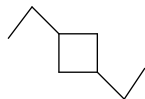
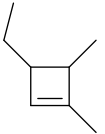
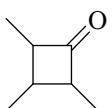
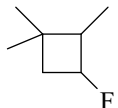
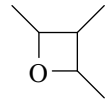
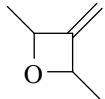
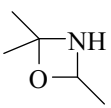
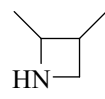
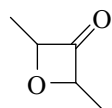
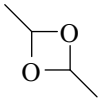
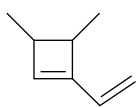
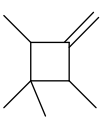
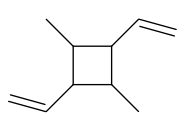
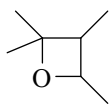
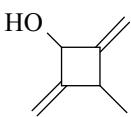
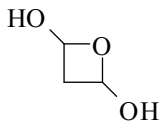


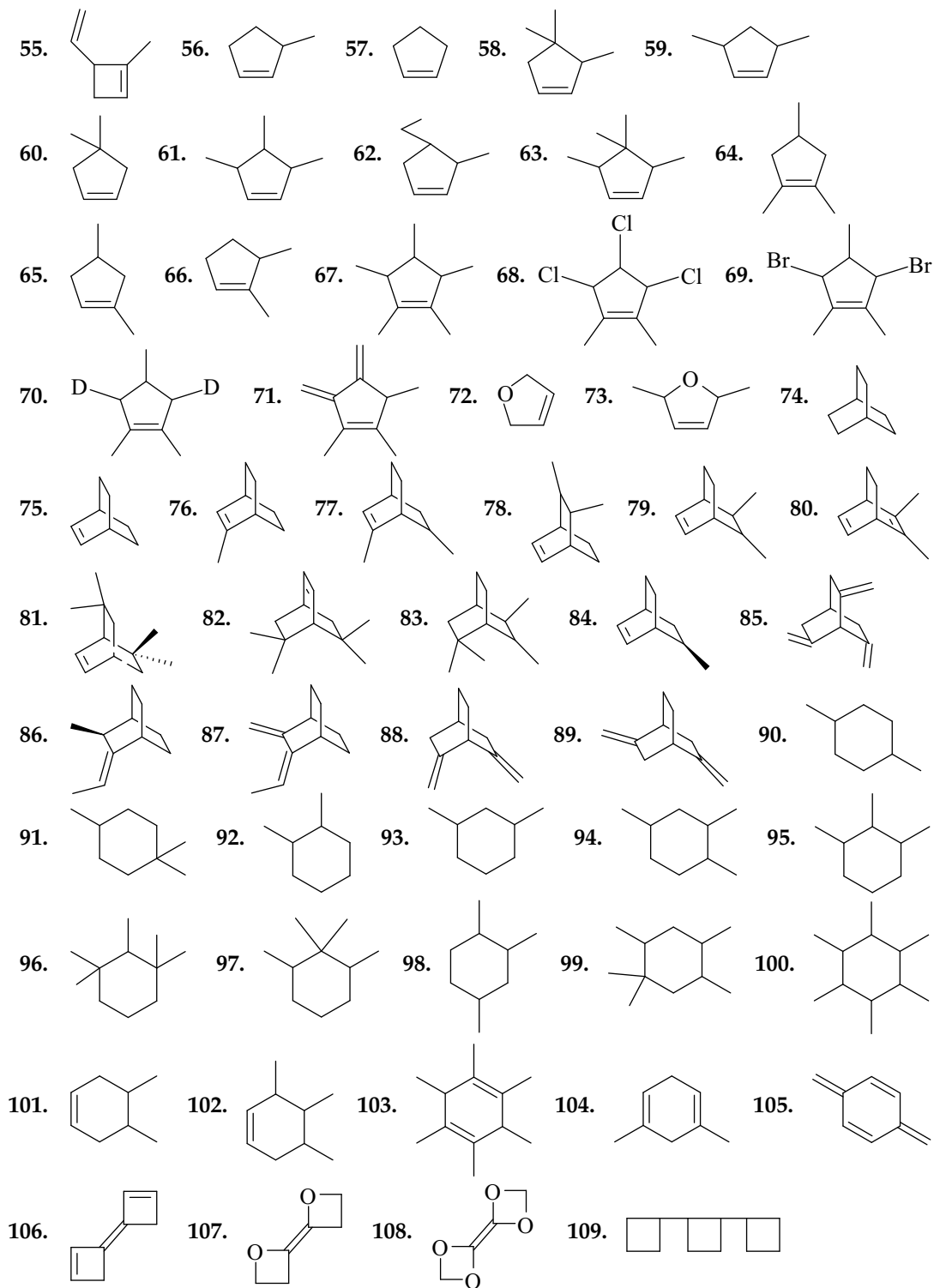
49.  50.  51.  52.  53. 
54.  55.  56.  57.  58. 
59.  60.  61.  62.  63. 
64.  65.  66.  67.  68. 
69.  70.  71.  72. 
73.  74.  75. 
76.  77.  78. 
79.  80.  81. 
82.  83.  84. 
85.  86. 

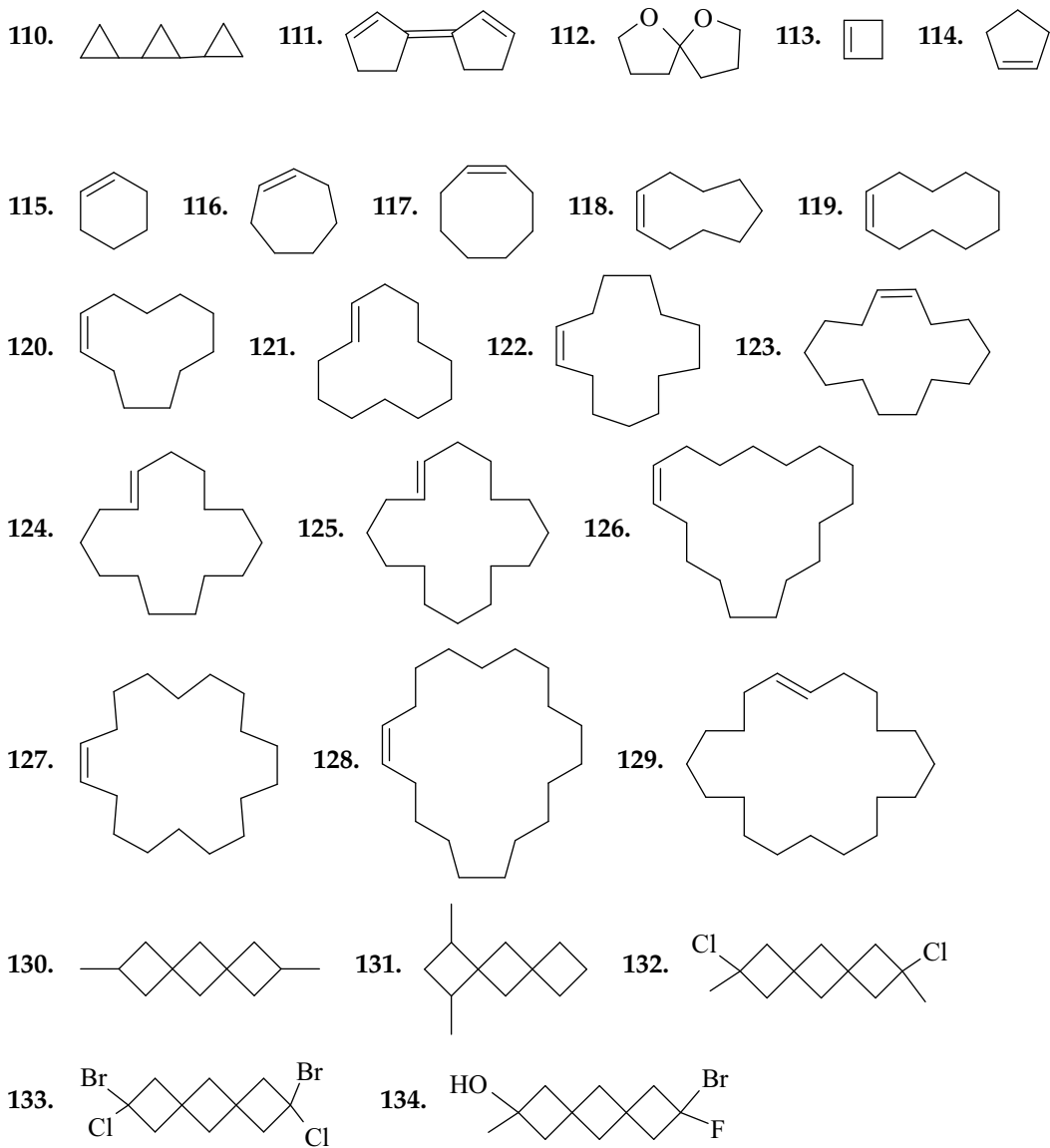


WORKBOOK EXERCISE 2

Identify molecules that show Geometrical Isomerism

1.  2.  3.  4.  5.  6.  7.  8.  9.  10.  11.  12.  13.  14.  15.  16.  17.  18.  19.  20.  21.  22.  23.  24.  25.  26.  27.  28.  29.  30.  31.  32.  33.  34.  35.  36.  37.  38.  39.  40.  41.  42.  43.  44.  45.  46.  47.  48.  49.  50.  51.  52.  53.  54. 

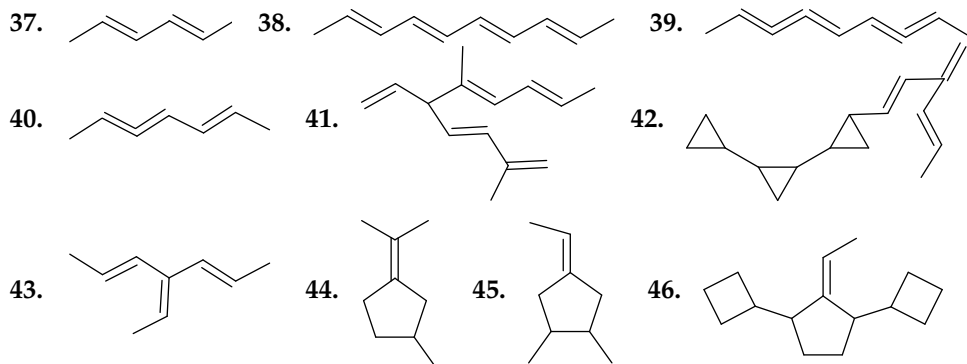




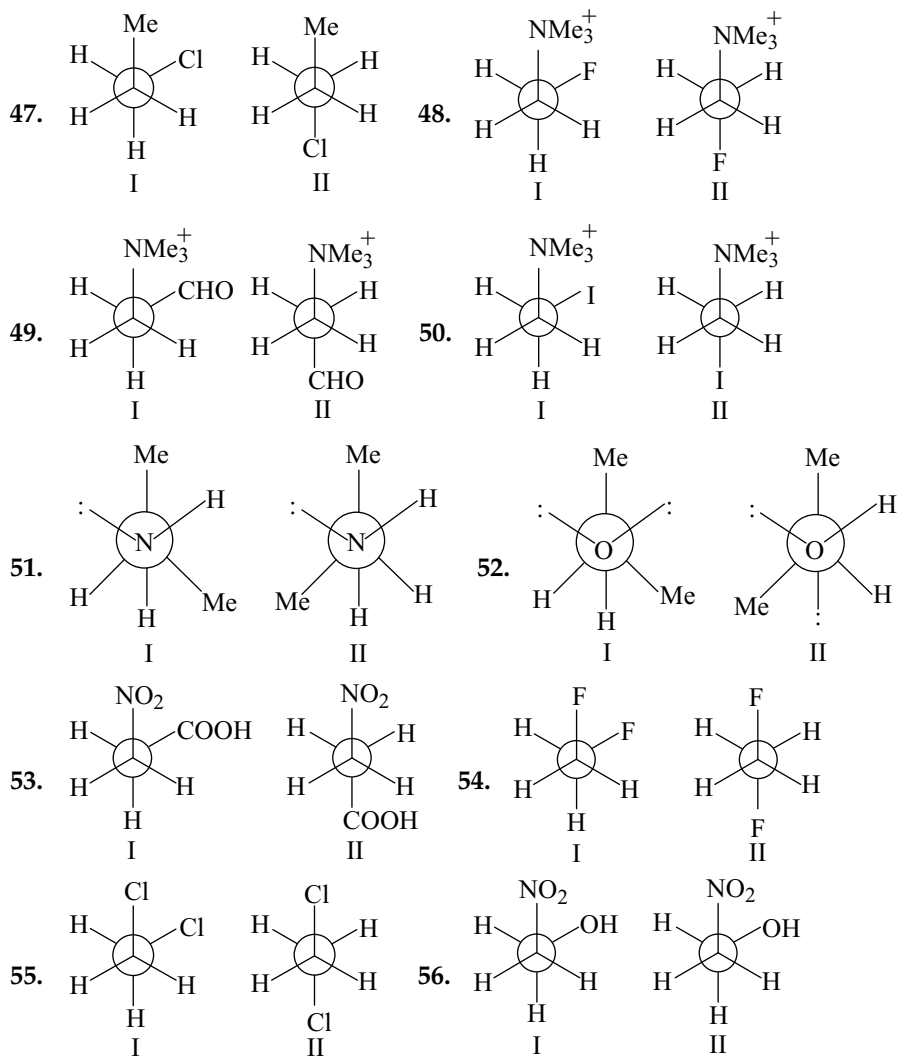
WORKBOOK EXERCISE 3

Identify molecule give G.I.

- 1.
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- 35.
- 36.

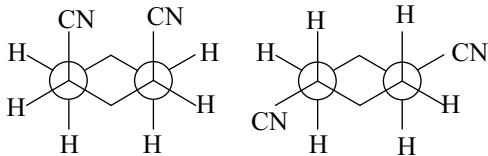
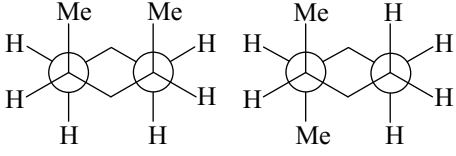
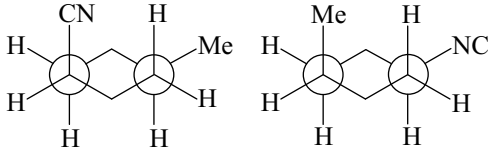
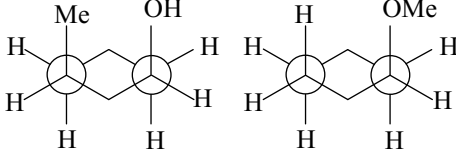
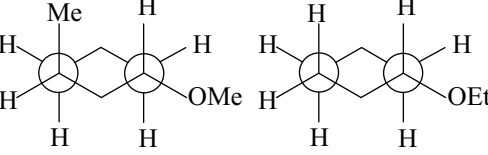
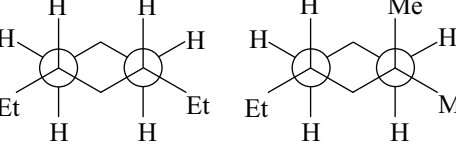
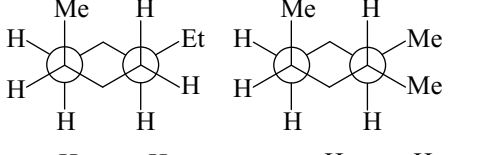
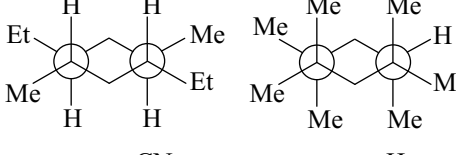
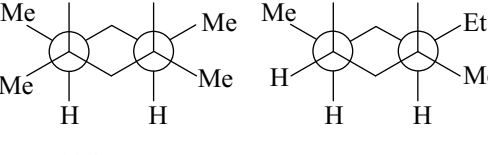
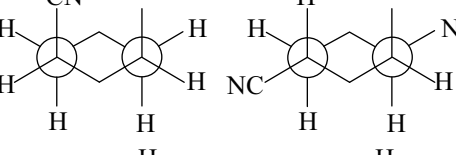
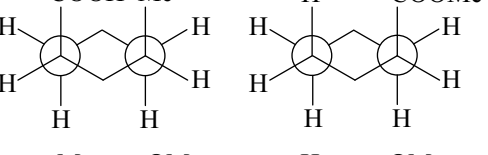
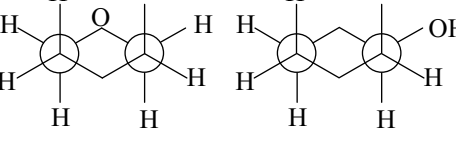
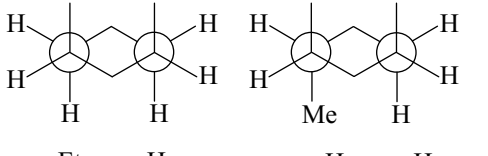
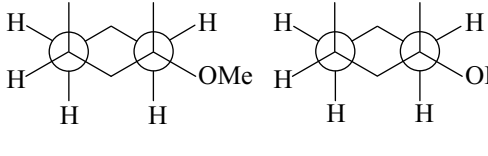


Identify stable conformer



WORKBOOK EXERCISE 4

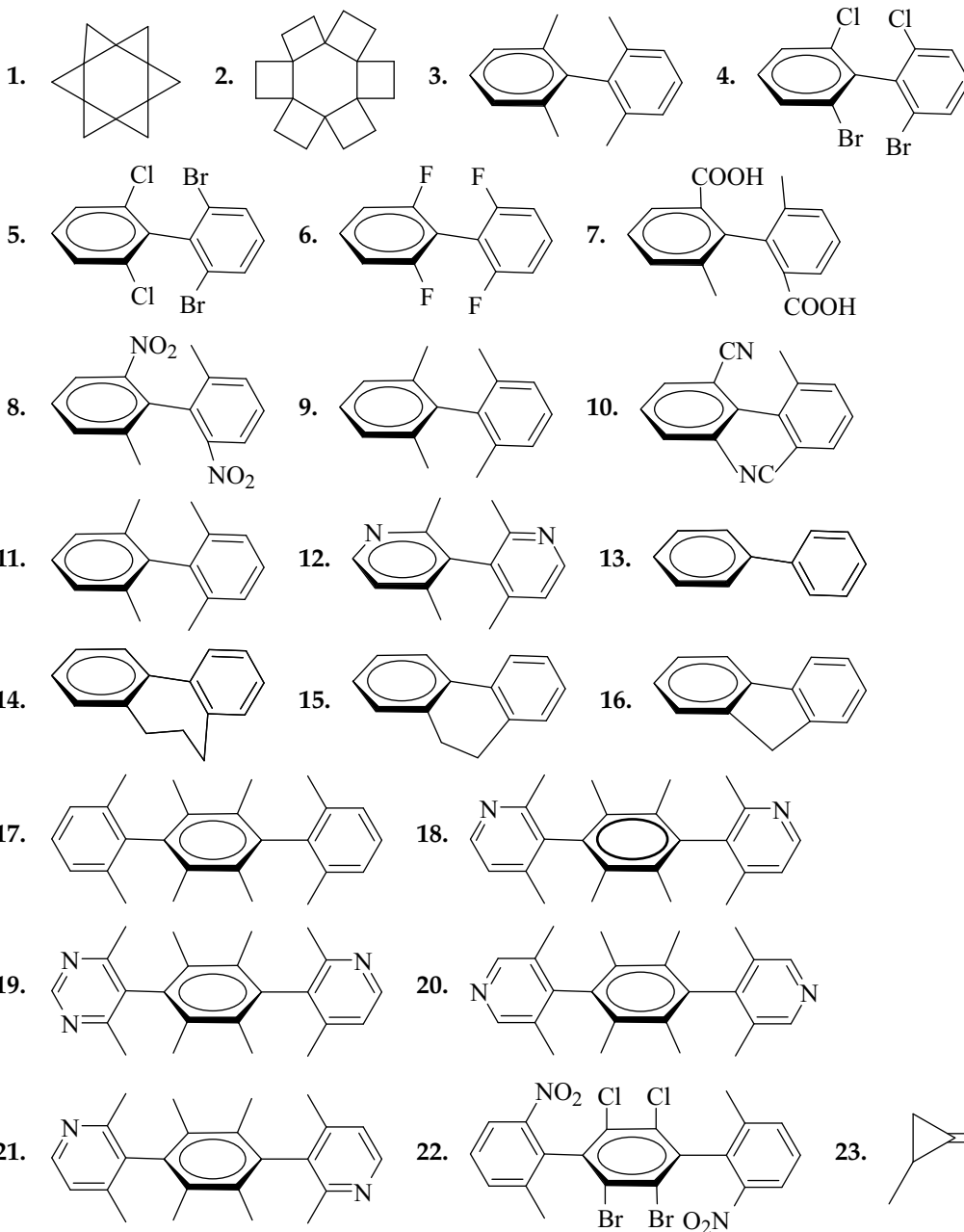
Identify Isomeric Relationship between pair of Compounds

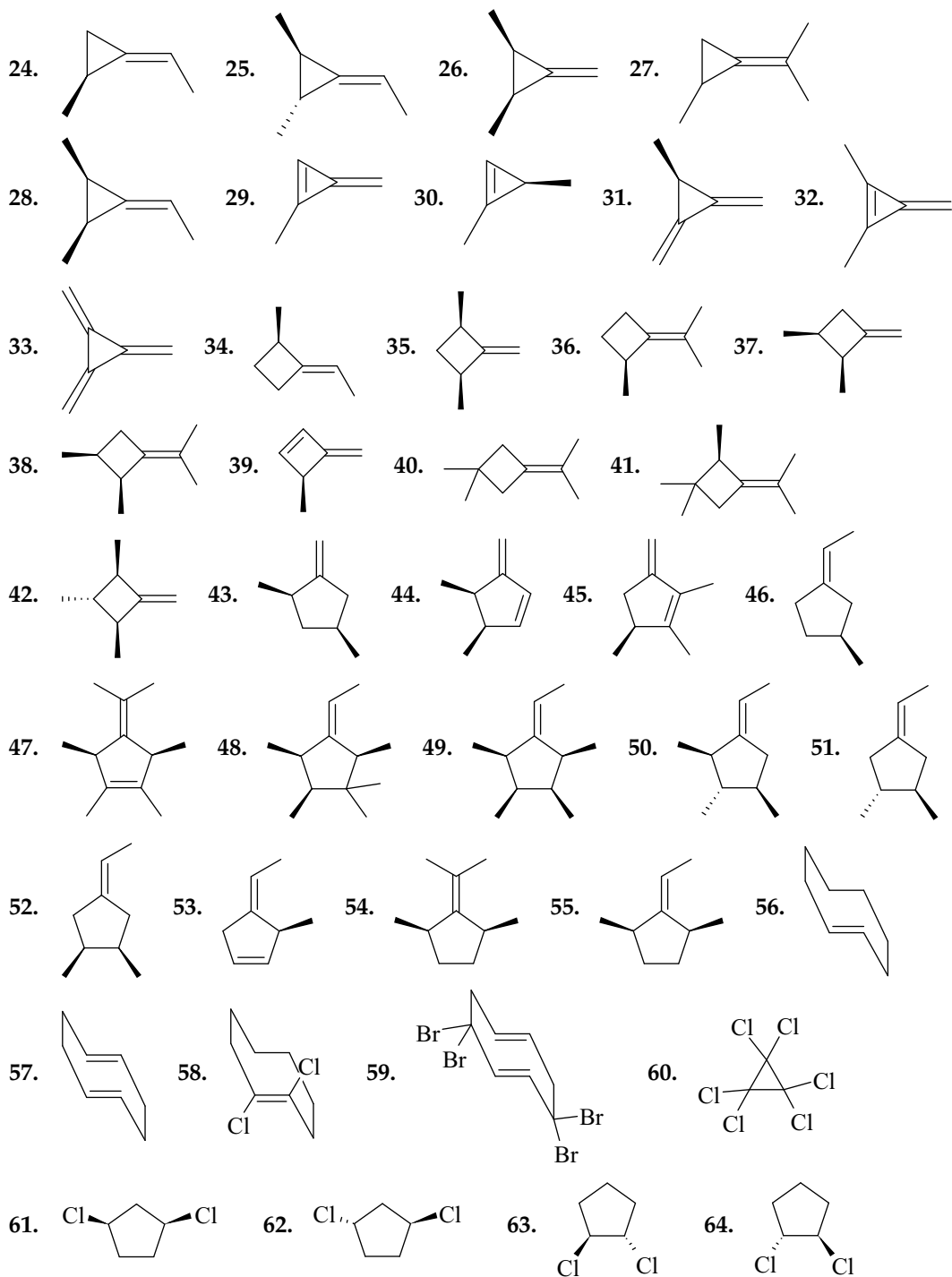
1. 
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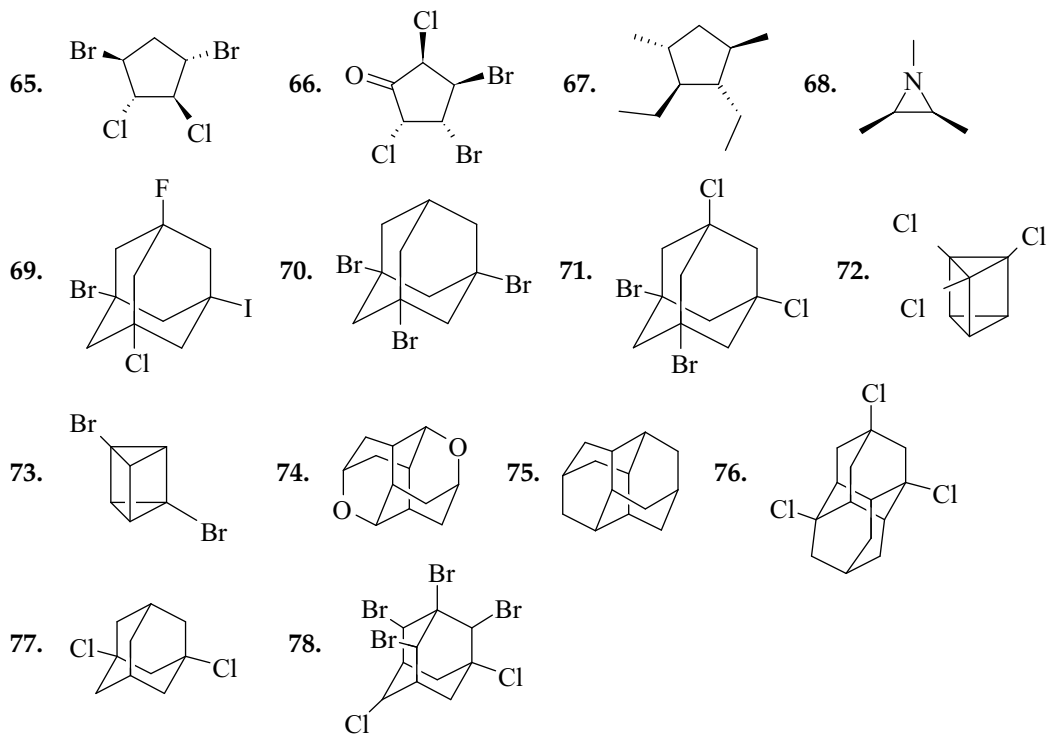
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- 25.
- 26.
- 27.

WORKBOOK EXERCISE 5

Identify optically active compounds







Answer Keys and Solutions to Workbook Exercises

LEVEL 1

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
b	a	d	d	b	b	b	a	c	b	d	c	c	a	c
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
c	d	b	d	a	c	ab	c	d	d	d	d	d	a	d
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
c	a	d	b	a	b	c	b	d	d	c	b	a	c	d
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
d	a	d	a	c	d	a	d	a	b	d	c	b	c	c
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
b	a	d	c	c	a	b	b	b	a	c	d	d	c	a
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
c	d	c	b	a	b	a	a	b	b	c	c	a	d	b
91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
c	d	b	b	a	d	b	a	d	b	c	a	a	d	c
106	107	108	109	110	111	112	113	114	115	116				
b	b	c	a	c	a	a	b	b	c	c				

LEVEL 2

[illegible]

WORKBOOK EXERCISE 1

Molecules that show Geometrical Isomerism

4, 6, 9, 11, 13, 16, 23, 24, 25, 26, 27, 30, 31, 33, 34, 35, 36, 38, 39, 40, 41, 44, 47, 50, 51, 52, 53, 54, 55, 56, 58, 59, 60, 61, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 76, 77, 82, 86, 89, 90, 92.

WORKBOOK EXERCISE 2

Molecules that show Geometrical Isomerism

6, 9, 11, 12, 13, 14, 17, 18, 19, 20, 21, 22, 23, 25, 26, 28, 29, 30, 31, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 59, 61, 62, 63, 67, 68, 69, 70, 73, 78, 79, 86, 87, 90, 92, 93, 94, 95, 97, 98, 99, 100, 101, 102, 103, 106, 107, 109, 110, 111, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134.

WORKBOOK EXERCISE 3

Molecules that show Geometrical Isomerism

1, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 17, 18, 19, 20, 21, 22, 23, 25, 27, 28, 29, 30, 31, 32, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 45, 46.

Stable Conformer

47(I), 48(I), 49(I), 50(II), 51(II), 52(I), 53(I), 54(I), 55(I), 56(I).

WORKBOOK EXERCISE 4

Isomeric Relationship between pair of compounds

- | | |
|------------------------|------------------------|
| 1. Positional Isomers | 2. Positional Isomers |
| 3. Functional Isomers | 4. Functional Isomers |
| 5. Metamers Isomers | 6. Chain Isomers |
| 7. Chain Isomers | 8. Chain Isomers |
| 9. Chain Isomers | 10. Functional Isomers |
| 11. Functional Isomers | 12. Not Isomers |
| 13. Positional Isomers | 14. Metamers Isomers |
| 15. Chain Isomers | 16. Metamers Isomers |
| 17. Chain Isomers | 18. Functional Isomers |
| 19. Chain Isomers | 20. Positional Isomers |
| 21. Functional Isomers | 22. Functional Isomers |
| 23. Positional Isomers | 24. Positional Isomers |
| 25. Chain Isomers | 26. Metamers Isomers |
| 27. Functional Isomers | |

WORKBOOK EXERCISE 5

Optically active compounds

4, 7, 8, 10, 12, 14, 15, 22, 23, 24, 25, 27, 28, 30, 34, 36, 37, 38, 39, 41, 43, 44, 45, 46, 48, 49, 50, 51, 52, 53, 55, 56, 58, 59, 62, 63, 64, 65, 66, 67, 69, 73.

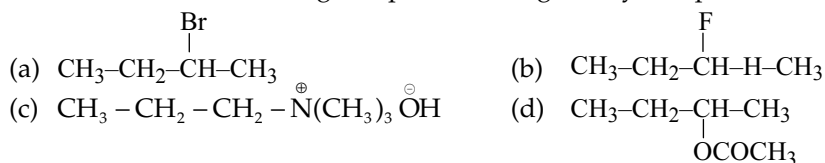
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Question Bank

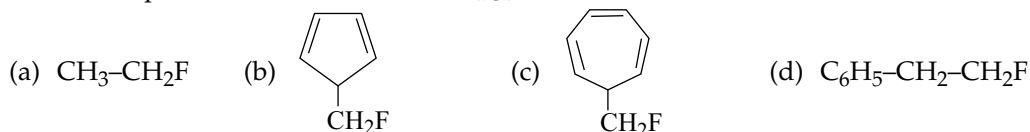
LEVEL 1

1. When the trans-2-pentene is treated with Br_2 in the presence of CCl_4 , then the number of stereoisomers formed is
 (a) 1 (b) 2 (c) 3 (d) 4

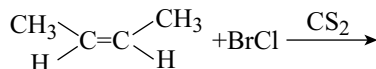
2. Which one of the following compounds will give Saytzeff product in E_2 reaction?



3. Which compound is most reactive for $\text{E}_{1\text{CB}}$ reaction?

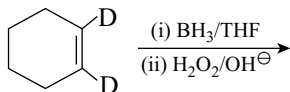


4. In the given reaction

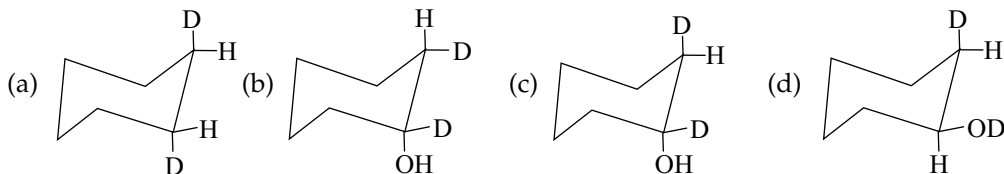


The correct option about the product is

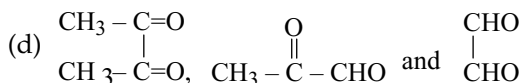
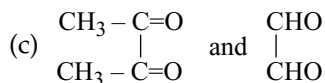
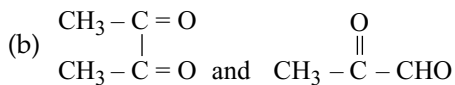
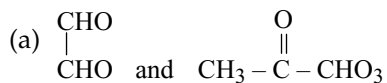
- (a) (+)-2-bromo-3-chlorobutane (b) ($\pm$)-threo-2-bromo-3-chlorobutane
 (c) (+)-threo-2-bromo-3-chlorobutane (d) (-)-threo-2-bromo-3-chlorobutane
5. In the given reaction



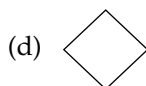
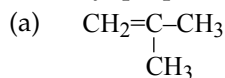
the product is



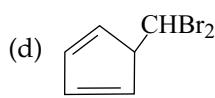
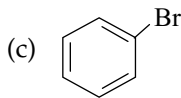
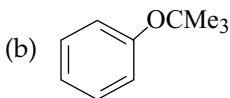
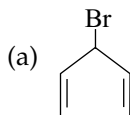
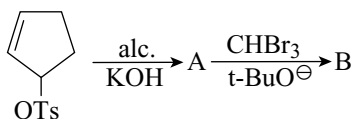
6. o-xylene on ozonolysis may give



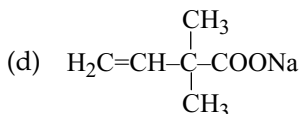
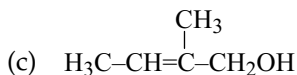
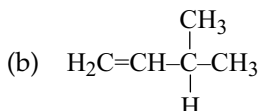
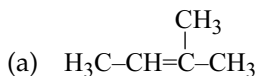
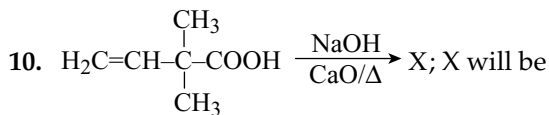
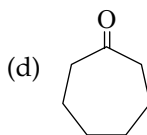
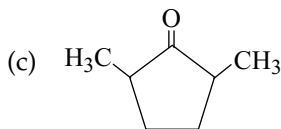
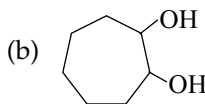
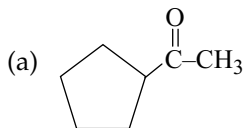
7. A hydrocarbon (C_4H_8) on reaction with m-chloro perbenzoic acid (MCPBA) gives (X). (X) on reaction with KOH (aq.) gives (Y), which on treatment with conc. H_2SO_4 forms 2-methyl propanal. The hydrocarbon is



8. The final product of the following sequence is



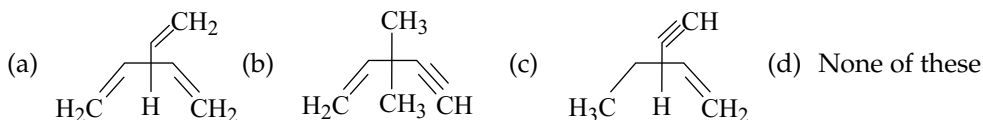
9. Major product of reaction between cycloheptyne and H_2O , $\text{H}_2\text{SO}_4/\text{HgSO}_4$ is



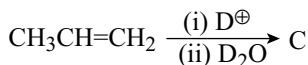
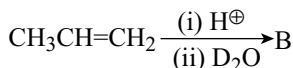
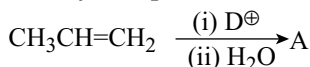
11. $\text{CH}_3-\overset{\text{CH}_3}{\underset{|}{\text{C}}}=\text{CH}_2 \xrightarrow[\text{CH}_2\text{Cl}_2]{\text{MCPBA}} [\text{X}] \xrightarrow[\text{(ii) H}_2\text{O}^{18}]{\text{(i) H}^+} [\text{Y}] \xrightarrow{\text{Conc. H}^+} [\text{Z}]$; The major product (Z) can be



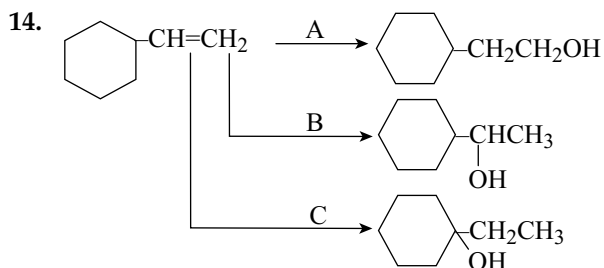
12. An optically active organic compound has the molecular formula C_7H_{10} (A). On reaction with $\text{H}_2 + \text{Pt}$ it forms an optically inactive compound (B). Then, compound (A) will be



13. Identify end products A, B and C of the following

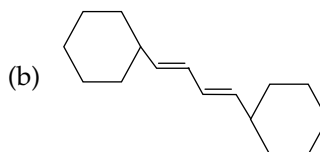
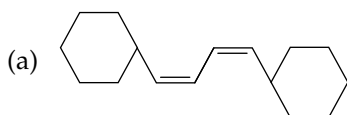
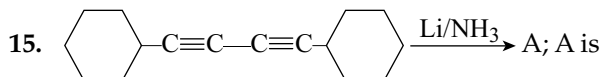


- (a) CH_3CHCH_3 in all case
 OH
 (b) $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{D}$, $\text{CH}_3\text{CH}(\text{OD})\text{CH}_3$, $\text{CH}_3\text{CH}(\text{OD})\text{CH}_2\text{D}$
 (c) CH_3CHCH_3 in all cases
 OD
 (d) $\text{CH}_3\text{CHCH}_2\text{D}$ in all cases
 OD



Schemes A, B and C are

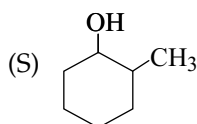
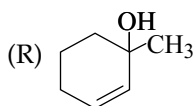
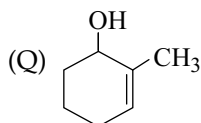
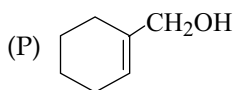
- (a) simple acid catalysed hydration
 (b) hydroboration, mercuration-demercuration, acid-catalysed hydration
 (c) acid-catalysed hydration, hydroboration, mercuration-demercuration
 (d) mercuration-demercuration, acid-catalysed hydration, hydroboration



(c) both are correct

(d) none is correct

16. Rate of dehydration when given compounds are treated with conc. H_2SO_4 is

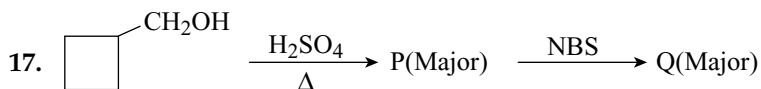


(a) $\text{P} > \text{Q} > \text{R} > \text{S}$

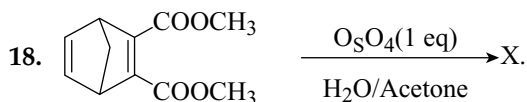
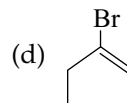
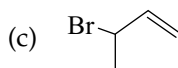
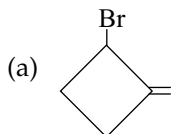
(b) $\text{Q} > \text{P} > \text{R} > \text{S}$

(c) $\text{R} > \text{Q} > \text{P} > \text{S}$

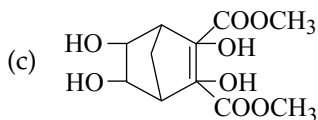
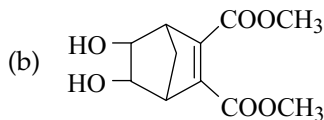
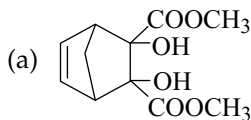
(d) $\text{R} > \text{Q} > \text{S} > \text{P}$



The structure of Q is



Identify "X":



(d) reaction will not occur

19. 1-Penten-4-yne reacts with bromine at -80°C to produce

(a) 4, 4, 5, 5-Tetrabromopentene

(b) 1, 2-Dibromo-1, 4-pentadiene

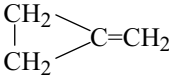
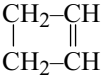
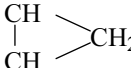
(c) 1, 1, 2, 2, 4, 5-hexabromopentane

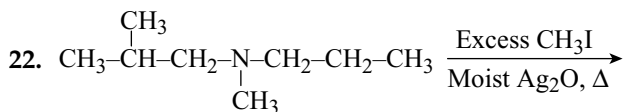
(d) 4, 5-dibromopentyne

20. Which of the following reagents cannot be used to locate the position of triple bond in $\text{CH}_3\text{-C}\equiv\text{C-CH}_3$?

- (a) Br_2 (b) O_3 (c) Cu_2^{2+} (d) KMnO_4

21. An organic compound of molecular formula C_4H_6 , (A) forms a precipitate with ammonical silver nitrate and ammonical cuprous chloride. "A" has an isomer "B", one mole of which reacts with 1 mole of Br_2 to form 1, 4-dibromo-2-butene. Another isomer of A is "C", one mole of C reacts with only 1 mole of Br_2 to give vicinal dibromide. A, B and C are

- (a) $\text{CH}_3\text{-CH}_2\text{-C}\equiv\text{CH}$ and $\text{CH}_2=\text{CH-CH=CH}_2$; 
- (b) $\text{CH}_3\text{-C}\equiv\text{C-CH}_3$ and $\text{CH}_3\text{-CH=C=CH}_2$; $\text{CH}_3\text{-C}\equiv\text{C-CH}_3$
- (c)  and ; $\text{CH}_2=\text{CH-CH=CH}_2$
- (d) $\text{CH}_3\text{-C-C}\equiv\text{CH}_3$ and ; $\text{CH}_2=\text{CH-CH=CH}_2$

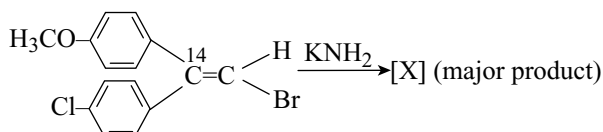


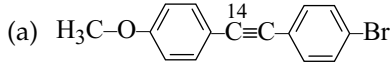
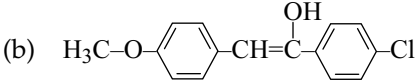
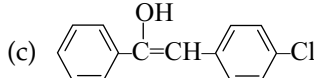
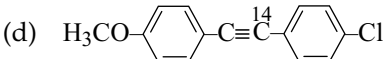
Product mixture $\xrightarrow{\text{O}_3/\text{Zn-H}_2\text{O}}$ Final product mixture

The final product mixture contains:

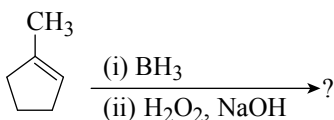
- (a) $\text{CH}_3\text{CHO} + \text{CH}_3\text{COCH}_3$ (b) $\text{CH}_3\text{CHO} + \text{CH}_3\text{CH}_2\text{CH}_3 + \text{HCHO}$
 (c) $\text{CH}_3\text{CHO} + \text{HCHO}$ (d) $\text{CH}_3\text{CHO} + \text{CH}_3\text{COCH}_3 + \text{HCHO}$

23. Choose the correct major product

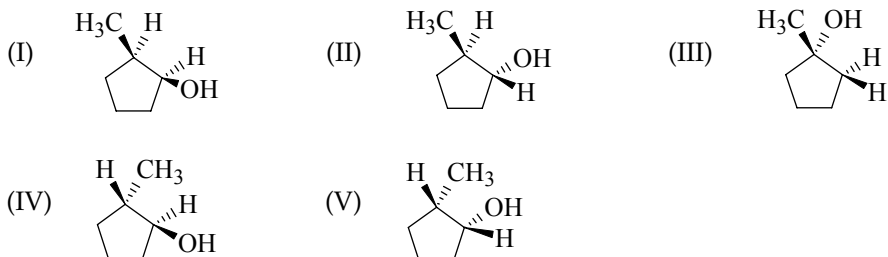


- (a)  (b) 
- (c)  (d) 

24. Supposed you carried out the hydroboration of 1-methylcyclopentene:

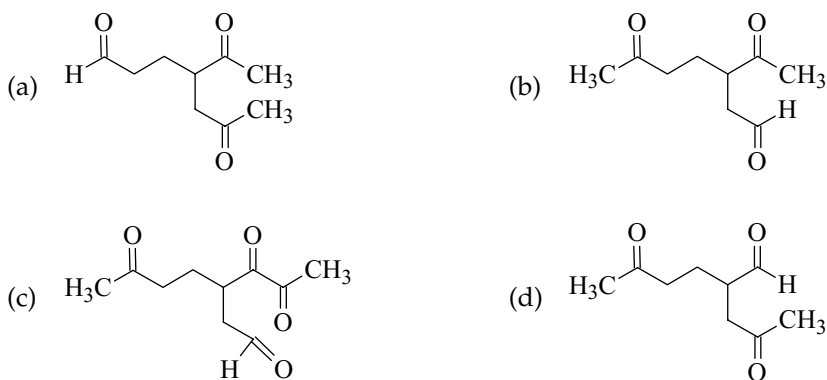
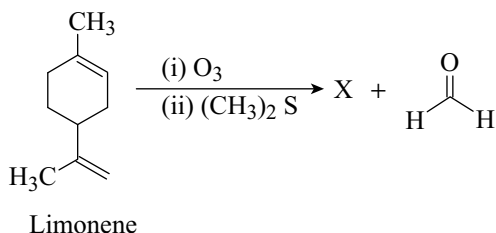


Choose the correct answer for the products formed in the above reaction.

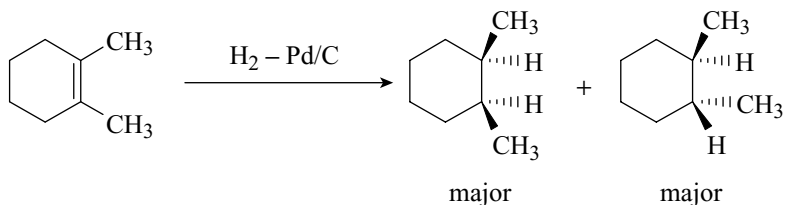


- (a) An equal mixture of 1 and 5 (b) An equal mixture of 1 and 2
(c) An equal mixture of 2 and 4 (d) An equal mixture of 4 and 5

25. The ozonolysis of limonene (oil of lemons) give compound X plus formaldehyde. Choose the correct structure for X.

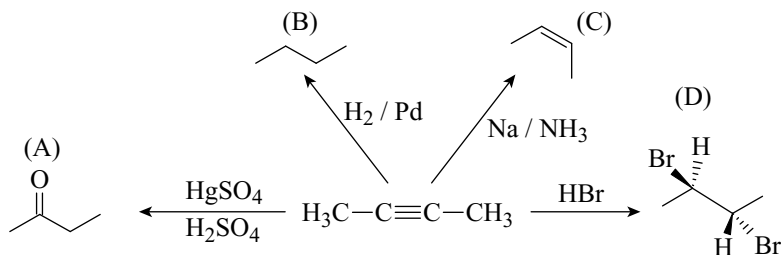


26. Choose the incorrect statement about the following catalytic hydrogenation.



- (a) The minor product occurs as the result of a catalyzed isomerization of the reactant.
 (b) The minor trans isomer is actually present as a racemic mixture.
 (c) The syn addition of hydrogen gives the cis isomer as the major product.
 (d) The catalyst (Pd/C) speeds up the reaction by stabilizing the major product of the reaction.

27. Choose the incorrect reaction.



28. Product of the reaction $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_3 \xrightarrow[-78^\circ\text{C}]{\text{O}_3/\text{CH}_2\text{Cl}_2}$ will be

- (a) $\text{CH}_3\text{-CHO}$ (b) $\text{CH}_3\text{-COOH}$
 (c) $\text{CH}_3\text{-CH(OH)-CH(OH)-CH}_3$ (d) $\text{CH}_3\text{-CH(OOCH}_3\text{)-CH(OOCH}_3\text{)-CH}_3$

29. $\text{H}_2\text{C}(\text{OH})\text{-CH=CH}_2 \xrightarrow[\text{(ii) NaSO}_3\text{H}]{\text{(i) OsO}_4}$ (A) $\xrightarrow{\text{KHSO}_4}$ (B); compound (B) is

- (a) $\text{H}_2\text{C=CH-CH}_2\text{-OH}$ (b) $\text{H}_2\text{C=CH-COOH}$
 (c) $\text{H}_2\text{C=CH-CHO}$ (d) $\text{H}_2\text{C=C=CH}_2$

30. $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_3 \xrightarrow[600^\circ\text{C}]{\text{CrO}_3/\text{Al}_2\text{O}_3}$ [P]; Product is

- (a) Mixture of 1-butene and 2-butene (b) Cyclobutene
 (c) 1,3-cyclobutadiene (d) None of these

31. In the presence of peroxide, HCl and HI do not give anti-Markownikov's addition to alkenes because

- (a) All the steps are endothermic in both the cases
 (b) One is oxidising and other is reducing
 (c) One of the steps is endothermic in both the cases
 (d) All the steps are exothermic in both the cases

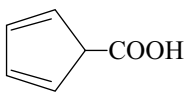
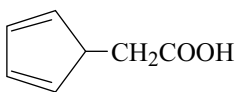
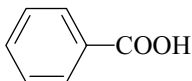
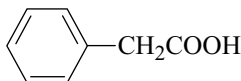
32. Hydrogenolysis is a process of

- (a) Addition of H_2 across $\text{C}-\text{C}$ multiple bond
 (b) Elimination of H_2 in $\text{CH}-\text{CH}$ bonds
 (c) Cleavage of a single bond by H_2
 (d) Cleavage of $\text{C}-\text{C}$ multiple bonds by H_2

33. In the reaction with Tollen's reagent acetylene shows
 (a) Oxidising property (b) Reducing property
 (c) Basic property (d) Acidic property
34. Arrange the following halogenating agents in order of decreasing selectivity in free radical reactions
 (1) NBS (2) NCS (3) Cl_2 (4) F_2
 (a) $4 > 3 > 2 > 1$ (b) $1 > 2 > 3 > 4$ (c) $2 > 1 > 3 > 4$ (d) $1 > 2 > 4 > 3$

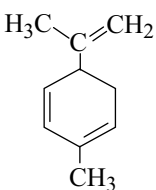
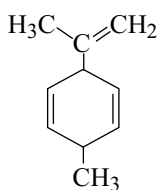
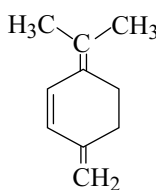
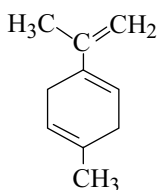


The product P_3 is

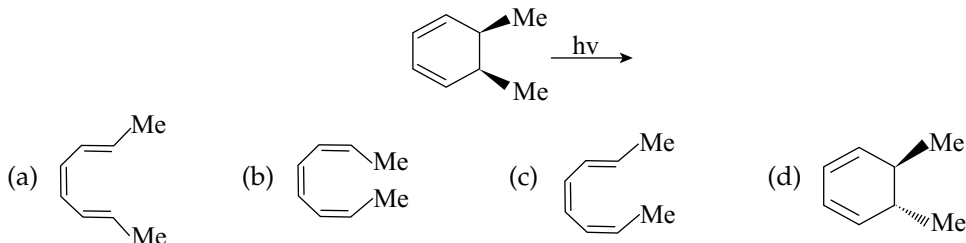
- (a)  (b) 
 (c)  (d) 

36. A compound having the molecular formula $\text{C}_{10}\text{H}_{14}$ is hydrogenated with H_2/Pd to give 1-isopropyl-4-methyl cyclohexane and on reductive ozonolysis it gives $\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$, $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$ and $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$

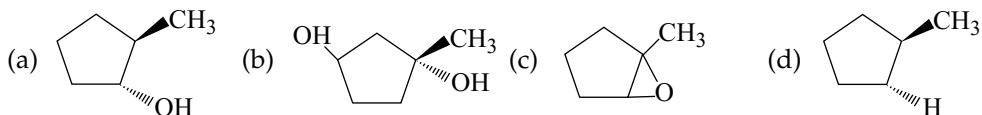
The structure of the compound would be

- (a)  (b)  (c)  (d) 

37. The product obtained during the following photochemical reaction is



38. The major product formed on hydroboration oxidation of 1-methylcyclopentene is



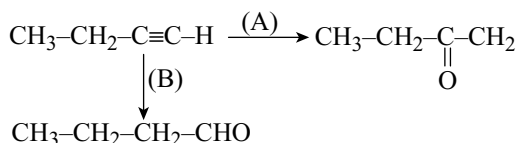
39. Compound (A) on bromination gives (B), which gives (C) with alcoholic KOH. (C) decolourises 1% alkaline KMnO_4 solution and on ozonolysis, it gives two molecules of smallest carbonyl compound. Compound (A) will be

- (a) C_2H_2 (b) C_2H_4 (c) C_2H_6 (d) $\text{C}_2\text{H}_5\text{Cl}$

40. $\text{CH}_2=\text{CH}-\text{C}\equiv\text{CH}$ on reaction with 1 mole of DBr gives

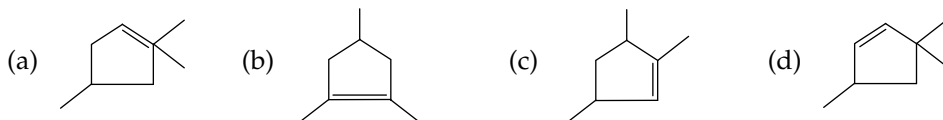
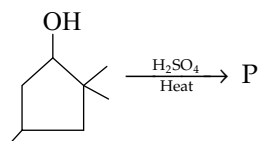
- (a) $\text{CH}_2=\text{CH}-\text{CBr}=\text{CHD}$ (b) $\text{CH}_2(\text{Br})-\text{CHDC}\equiv\text{CH}$
(c) $\text{DCH}_2-\text{CHBrC}\equiv\text{CH}$ (d) $\text{CH}_2=\text{CH}-\text{CD}=\text{CHBr}$

41. In the given reaction, A and B respectively are



- (a) $(\text{Sia})_2\text{BH}/\text{H}_2\text{O}_2/\text{HO}^-$ and $\text{H}_2\text{O}/\text{HgSO}_4/\text{H}^+$
(b) $\text{H}_2\text{O}/\text{HgSO}_4/\text{H}^+$ and $(\text{Sia})_2\text{BH}/\text{H}_2\text{O}_2/\text{HO}^-$
(c) $\text{H}_2\text{O}/\text{HgSO}_4/\text{H}^+$ and $\text{Na}, \text{CH}_3\text{-I}$
(d) None

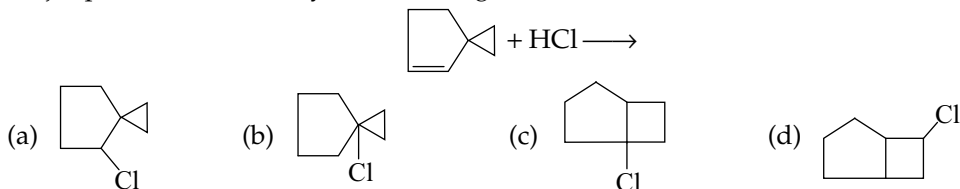
42. Identify the major product P obtained by the reaction



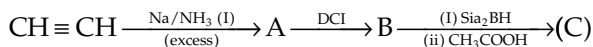
43. During the preparation of ethane by Kolb's electrolytic method using inert electrode the pH of the electrolyte

- (a) Decreases progressively as the reaction proceeds
(b) Increases progressively as the reaction proceeds
(c) Remains constant throughout the reaction
(d) May decrease if concentration of the electrolytes is not very high

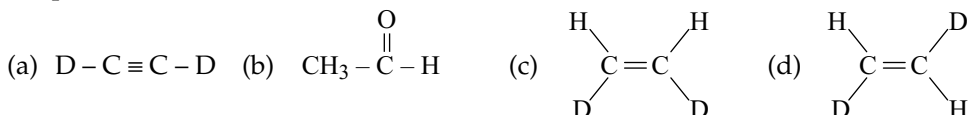
44. Major product obtained by the following reaction is



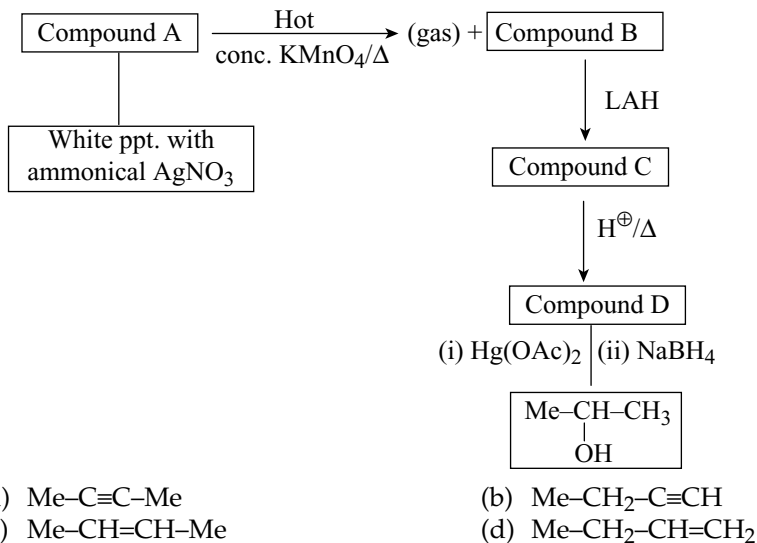
45. In the reaction



the product (C) is



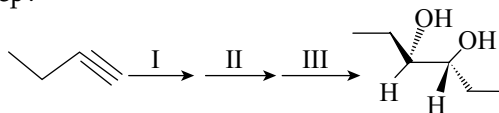
46. Identify structure of compound [A]



47. The reaction of propene with HBr in the presence of ROOR (peroxide) proceeds through which of the following most stable intermediates?

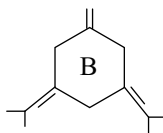


48. The following transformation is carried out in three steps. What is the appropriate reagent for the first step?



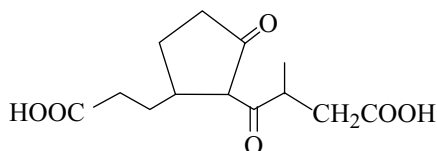
- (a) $\text{H}_2/\text{Lindlar's catalyst}$ (b) $\text{C}_2\text{H}_2/\text{NaNH}_2/\text{CH}_3\text{I}$
 (c) $\text{NaNH}_2/\text{NH}_3: \text{EtBr}$ (d) $\text{H}_2/\text{Pd/C}$

49. Compound A was treated with a large excess of CH_3MgBr . The resulting product was exposed to $\text{POCl}_3/\text{pyridine}$ to give compound B, as one of many products. Which of the following compound can be A?



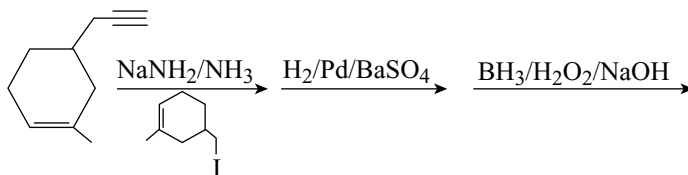
- (a)
- (b)
- (c)
- (d)

50. Which molecule will give the following dicarboxylic acid upon treatment with acidic solution of KMnO_4/Δ ?

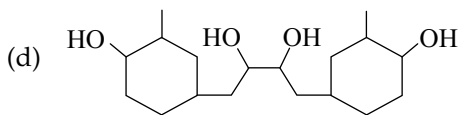
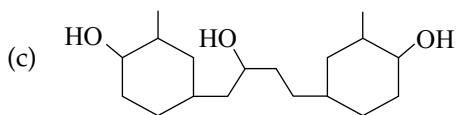


- (a)
- (b)
- (c)
- (d)

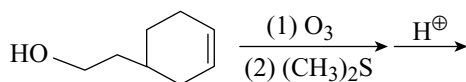
51. What is the product of the following sequence of reaction?



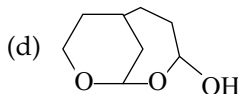
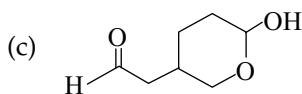
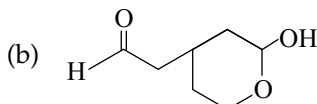
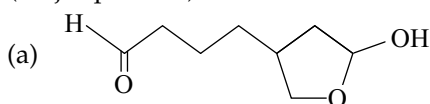
- (a)
- (b)



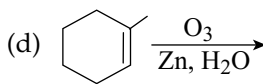
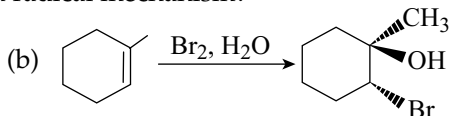
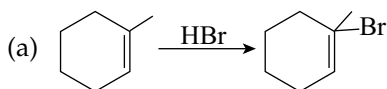
52. Give the major product of the following sequence



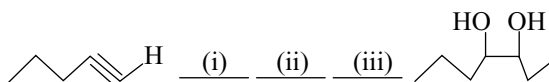
(Major product)



53. Which of the following reactions involve a radical mechanism?



54. For the following multistep reaction, which set of reagents would be more likely to give the desired product in good yield?



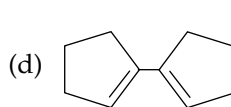
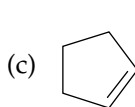
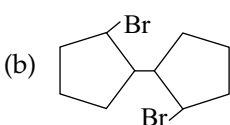
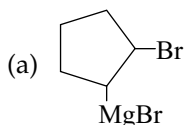
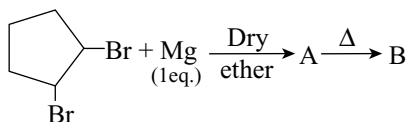
(a) (i) HBr, (ii) O₃/Zn-H₂O, (iii) Li/NH₃

(b) (i) NaNH₂/NH₃/C₂H₅I, (ii) Lindlar's catalyst/H₂, (iii) OsO₄ followed by NaHSO₃

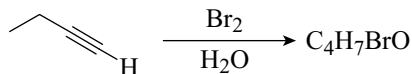
(c) (i) H₂/Pd-C, (ii) NaNH₂/NH₃ followed by C₂H₅I, (iii) KMnO₄/OH⁻

(d) (i) HgSO₄/H₂SO₄, (ii) Lindlar's catalyst/H₂, (iii) OsO₄ followed by NaHSO₃

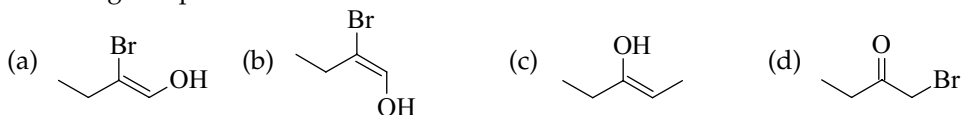
55. In the following reaction, compound (B) is



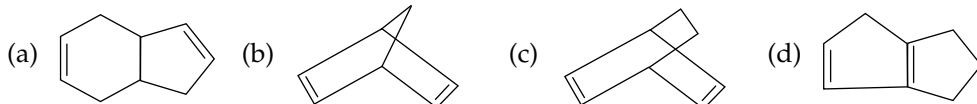
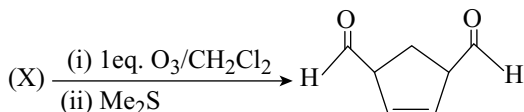
56. Here is a reaction

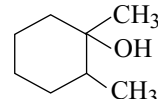


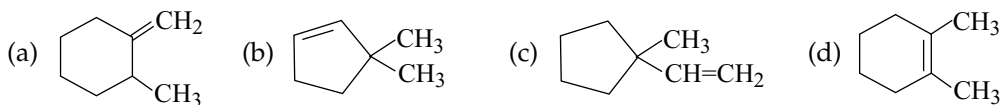
Use your knowledge of mechanisms to choose the most likely product from among the following compounds.



57. Identify the reactant (X) on the given reaction



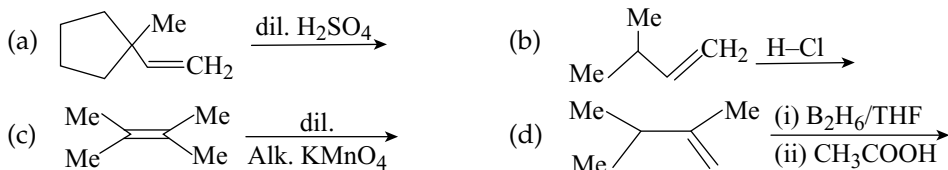
58. Compound (A) $\xrightarrow{\text{dil. H}_2\text{SO}_4}$ 
Compound (A) can be



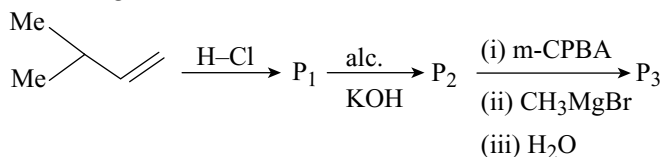
59. Which of the following gives glyoxal as one of the product on ozonolysis?

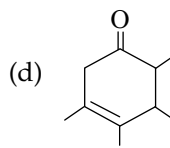
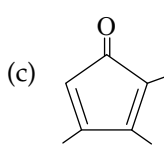
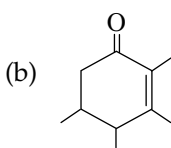
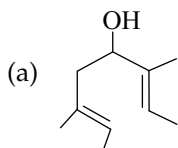
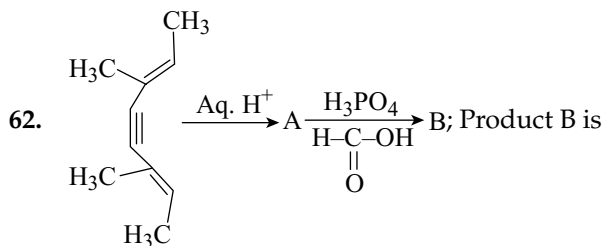
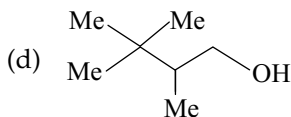
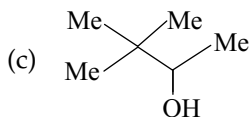
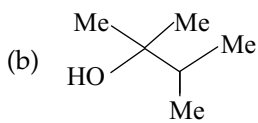
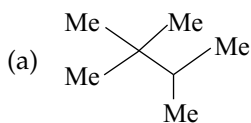


60. In which of the following reaction the rearrangement of carbocation is involved?

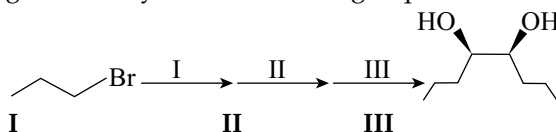


61. The final product of the given reaction is



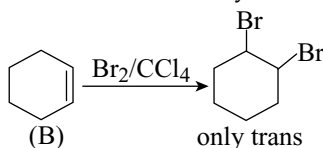
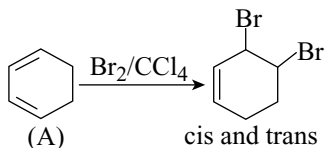


63. Choose the best reagent to carry out the following sequence of reaction



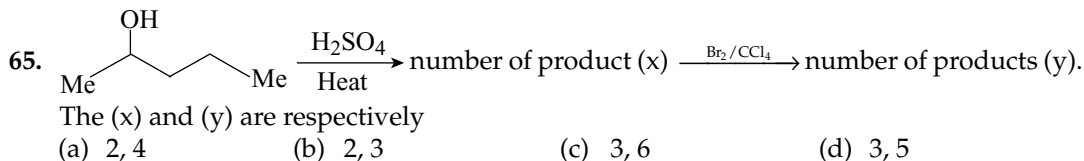
- | | |
|---|--|
| (a) Acetylene/ $\text{NaNH}_2/\text{NH}_3$ H_2 /Pd | $\text{OsO}_4/\text{NaHSO}_3$ |
| (b) Acetylene/ $\text{NaNH}_2/\text{NH}_3$ H_2 /Lindlar's catalyst | $\text{BH}_3/\text{NaOH}/\text{H}_2\text{O}_2$ |
| (c) Pentynes/ $\text{NaNH}_2/\text{NH}_3$ Li/ NH_3 | MnO_4 |
| (d) Pentynes/ $\text{NaNH}_2/\text{NH}_3$ H_2 /Lindlar's catalyst | $\text{OsO}_4/\text{NaHSO}_3$ |

64. When cyclohexadiene (A) reacts with Br_2 , a mixture of cis- and trans-1, 2-addition products is formed (in addition to other products). However, when cyclohexene (B) reacts with Br_2 under identical conditions, only trans product is observed. What is the best explanation for the observed difference in stereochemistry of the addition?

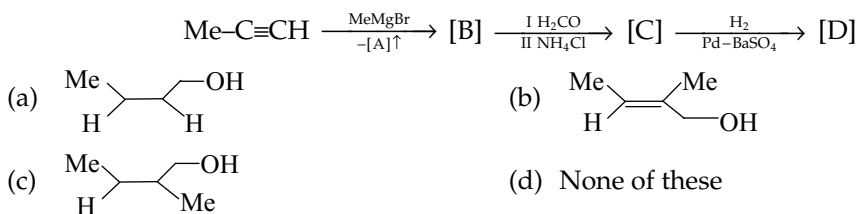


- (a) The cis and trans products are the result of aromaticity in the cyclic TS for reaction of A. In B there are only four electrons in TS, and cyclic TS is destabilised.
- (b) Reaction of A proceeds through an intermediate that has an sp^3 -hybridised carbocation, while the analogous intermediate in reaction of B has sp^2 -hybridised carbocation.

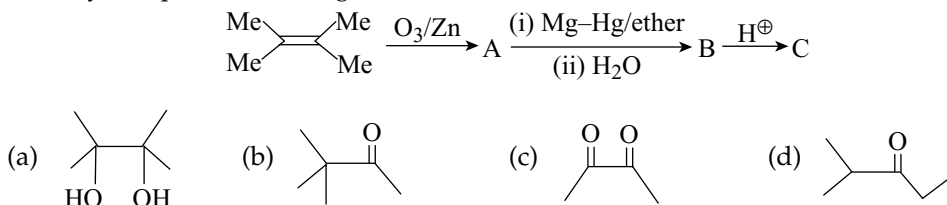
- (c) Both reactions occur through bromonium ions, but because of planarity enforced by neighbouring double bond, cis addition is not sterically hindered in A.
 (d) B reacts through a bromonium ion intermediate, while A does through an allyl cation.



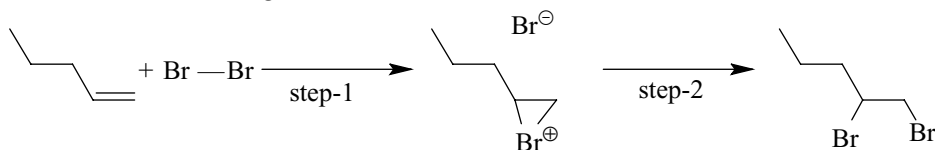
66. Final product in the given sequence is



67. Identify "C" product in the given reaction

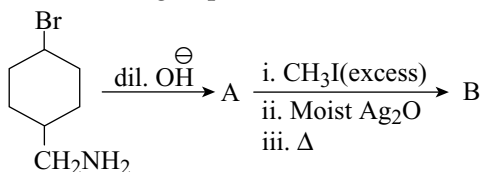


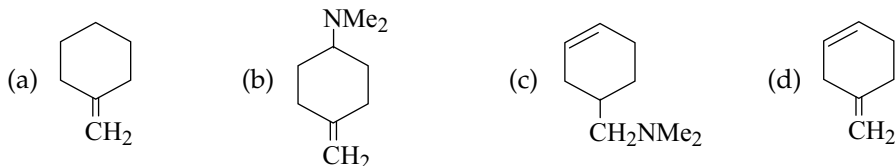
68. Consider the following reaction.



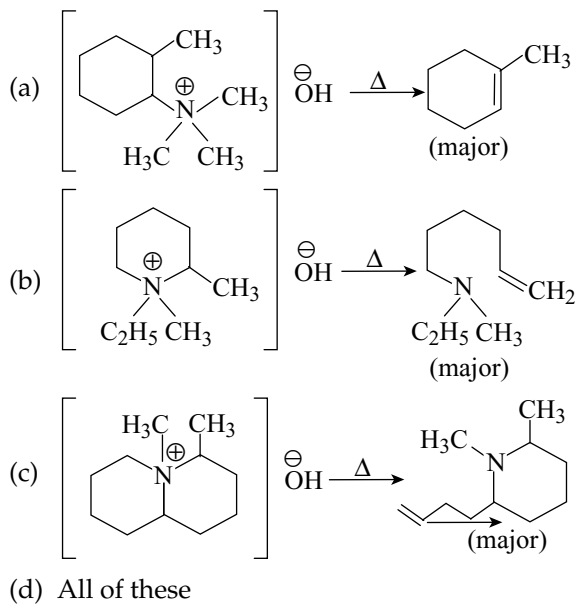
Which one of the following statements is incorrect?

- (a) The Br_2 addition occurs with anti-stereospecificity.
 (b) The final product will be a mixture of enantiomers.
 (c) In step 2 the Br^- anion acts as a Lewis base.
 (d) In step 1 the Br_2 molecule acts as a Lewis base.
69. Identify the product in the following sequence of reaction

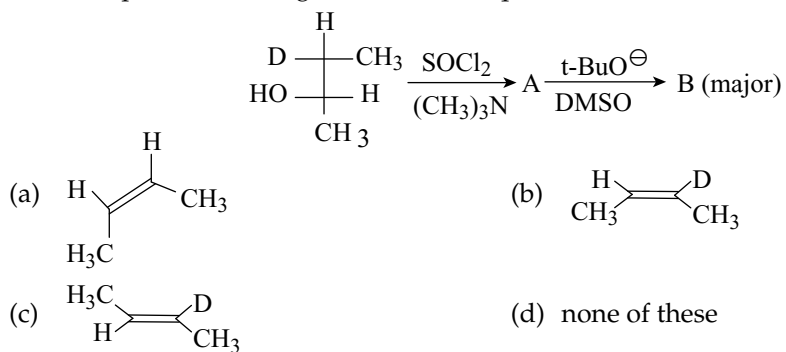




70. Which of the following is correctly matched?



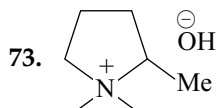
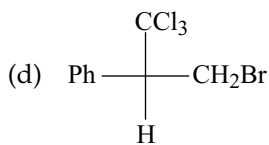
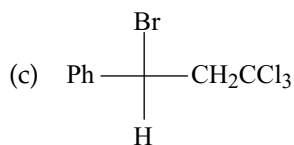
71. The final product of the given reaction sequence is



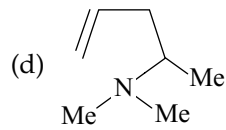
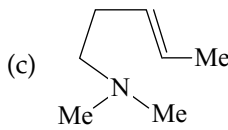
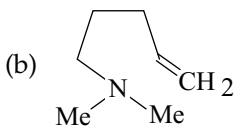
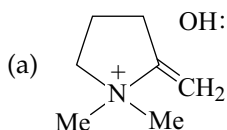
72. $\text{Ph}-\text{CH}=\text{CH}_2 + \text{BrCCl}_3 \xrightarrow{\text{peroxide}}$

Product is





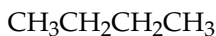
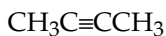
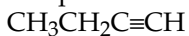
The above compound readily undergoes elimination on heating to yield which of the following products?



74. Select true statement(s)

- Instead of radical substitution, cyclopropane undergoes electrophilic addition reactions in sun light.
- In general, bromination is more selective than chlorination.
- The 2, 4, 6-tri-tert-butylphenoxy radical is resistant to dimerisation.
- The radical-catalysed chlorination, $\text{ArCH}_3 \rightarrow \text{ArCH}_2\text{Cl}$, occurs faster when Ar = phenyl than when Ar = p-nitrophenyl.

75. Which reagent is the most useful for distinguishing compound I from the rest of the compounds?



(I)

(II)

(III)

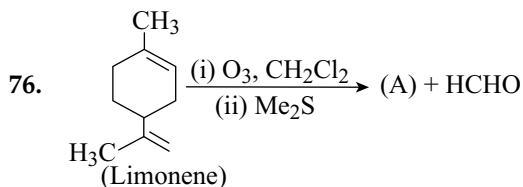
(IV)

(a) alk. KMnO_4

(b) Br_2/CCl_4

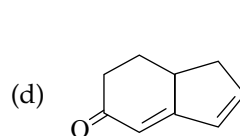
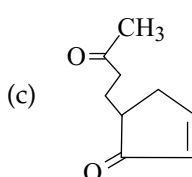
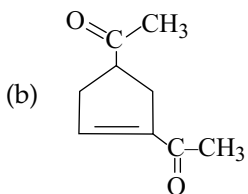
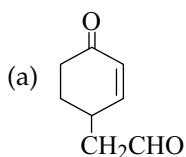
(c) $\text{Br}_2/\text{CH}_3\text{COOH}$

(d) Ammonical AgNO_3



(A) $\xrightarrow[\Delta]{\text{dil. } \text{OH}^\oplus}$ products (mixture)

For the given reaction, the products are



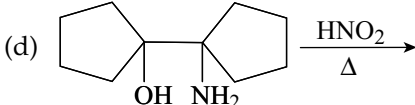
77. Which of the following reaction is correct regarding the formation of major product (alkene)?

- (a) $\left[\begin{array}{c} \text{H}_3\text{C}-\text{CH}_2-\text{N}^+(\text{Me})_2-\text{CH}_3 \end{array} \right] \xrightarrow{\ominus\text{OH}, \Delta} \text{CH}_2=\text{CH}_2$
- (b) $\left[\begin{array}{c} \text{H} \quad \text{Me} \\ | \quad | \\ \text{H}_3\text{C}-\text{C}-\text{N}^+(\text{Me})_2-\text{CH}_3 \\ | \\ \text{H}_3\text{C} \end{array} \right] \xrightarrow{\ominus\text{OH}, \Delta} \text{CH}_3-\text{CH}=\text{CH}_2$
- (c) $\left[\begin{array}{c} \text{Me} \\ | \\ \text{Ph}-\text{CH}_2-\text{N}^+(\text{Me})_2-\text{CH}_3 \end{array} \right] \xrightarrow{\ominus\text{OH}, \Delta} \text{Ph}-\text{CH}=\text{CH}_2$
- (d) $\left[\begin{array}{c} \text{Me} \\ | \\ \text{H}_3\text{C}-\text{CH}_2-\text{N}^+(\text{Me})_2-\text{CH}_2\text{Cl} \end{array} \right] \xrightarrow{\ominus\text{OH}, \Delta} \text{CH}_2=\text{CH}-\text{Cl}$

78. $\begin{array}{c} \text{CH}_3 \quad \text{C}_2\text{H}_5 \\ | \quad | \\ \text{H}_3\text{C}-\text{C}-\text{CH}-\text{CH}-\text{OH} \\ | \quad | \\ \text{CH}_3 \quad \text{CH}_3 \end{array} \xrightarrow[\Delta]{\text{H}^+} \text{Product mixture}$
the product's mixture contains

- (a) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{H}_3\text{C}-\text{CH}-\text{CH}-\text{CH}-\text{C}_2\text{H}_5 \\ | \quad | \\ \text{CH}_3 \quad \text{OH} \end{array}$ (b) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{H}_3\text{C}-\text{C}-\text{C}=\text{C}-\text{H} \\ | \quad | \quad | \\ \text{CH}_3 \quad \text{C}_2\text{H}_5 \quad \text{CH}_3 \end{array}$
- (c) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{H}_3\text{C}-\text{C}-\text{C}=\text{C}-\text{CH}_3 \\ | \quad | \quad | \\ \text{CH}_3 \quad \text{C}_2\text{H}_5 \quad \text{CH}_3 \end{array}$ (d) $\begin{array}{c} \text{CH}_3 \quad \text{H} \\ | \quad | \\ \text{H}_3\text{C}-\text{C}-\text{CH}-\text{C}=\text{CH}_2 \\ | \quad | \\ \text{CH}_3 \quad \text{C}_2\text{H}_5 \end{array}$

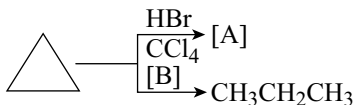
79. Which of the following reaction does/do not takes place by formation and rearrangement of carbocation?

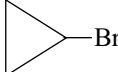
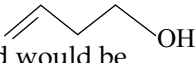
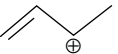
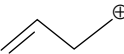
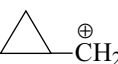
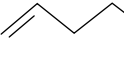
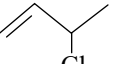
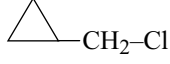
- (a) $\begin{array}{c} \text{R}-\text{CH}-\text{CH}=\text{CH}_2 \\ | \\ \text{CH}_3 \end{array} \xrightarrow[\text{(ii) NaBH}_4]{\text{(i) Hg(OAc)}_2, \text{H}_2\text{O}}$ (b) $\begin{array}{c} \text{R}-\text{CH}-\text{CH}=\text{CH}_2 \\ | \\ \text{CH}_3 \end{array} \xrightarrow[\text{(ii) H}_2\text{O}_2]{\text{(i) BH}_3}$
- (c) $\begin{array}{c} \text{CH}_3-\text{C}=\text{CH}_2 \\ | \\ \text{CH}_3 \end{array} \xrightarrow[\text{Peroxide}]{\text{HBr}}$ (d) 

80. Which alkyl halide will form Wittig reagent with PPh_3 and $\text{C}_6\text{H}_5\text{Li}$?

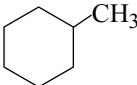
- (a) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{C}-\text{Br} \\ | \\ \text{CH}_3 \end{array}$ (b) $\begin{array}{c} \text{CH}_3-\text{CH}-\text{Br} \\ | \\ \text{CH}_3 \end{array}$ (c) CH_3-Br (d) $\text{C}_6\text{H}_5-\text{Br}$

81. Alkyne can be converted into vic dicarbonyl compound by
 (a) SeO_2 (b) Baeyer's reagent (c) KMnO_4/H^+ (d) $\text{K}_2\text{Cr}_2\text{O}_7$
82. Which of these substrates will give rearranged product in hydration reaction?
 (a) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CH}=\text{CH}_2$ (b) $\text{CH}_3-\ddot{\text{S}}-\text{CH}_2-\text{CH}=\text{CH}_2$
 (c) $\text{CH}_3-\text{CH}=\text{CH}_2$ (d) $\text{Ph}_2\text{C}=\underset{\text{CH}_3}{\underset{\text{CH}_3}{\text{CH}}}-\text{CH}_3$
83. The reagent(s) of choice for conversion of propene to methylcyclopropane is/are
 (a) $\text{CH}_2\text{N}_2/\text{ether}$ (b) CH_2I_2 (c) $\text{CH}_2\text{I}_2/\text{Zn}$ (d) $\text{CH}_3\text{I}/\text{NaOH}$
84. Anti-Markonikov addition is given by which of the following alkenes?
 (a) $\text{CH}_2=\text{CH}-\overset{\oplus}{\text{N}}(\text{CH}_3)_3$ (b) $\text{CH}_2=\text{CH}-\text{CF}_3$
 (c) $\text{CH}_3-\underset{\text{CH}_3}{\text{C}}=\text{CH}_2$ (d) $\text{C}_6\text{H}_5-\text{CH}=\text{CH}_2$
85. Which of the following gives allylic substitution product?
 (a) $\text{CH}_2=\text{CH}-\text{CH}_3 \xrightarrow{\text{NBS}/h\nu} \text{product}$ (b) $\text{CH}_2=\text{CH}-\text{CH}_3 \xrightarrow[\Delta]{\text{SeO}_2} \text{product}$
 (c) $\text{CH}_2=\text{CH}-\text{CH}_3 \xrightarrow{\text{OsO}_4} \text{product}$ (d) $\text{CH}_2=\text{CH}-\text{CH}_3 \xrightarrow[h\nu]{\text{SO}_2\text{Cl}_2} \text{product}$
86. In the given reaction, identify compound [A]

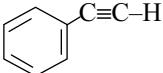


- (a) Compound [A] is -Br (b) Compound A is $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$
 (c) Reagent B is H_2/Ni at 120°C (d) Reagent B is LiAlH_4
87. In which of the following Hoffman's elimination product is more?
- (a) $\text{CH}_3-\underset{\text{H}_3\text{C}}{\overset{\text{CH}_3}{\text{C}}}-\underset{\text{OH}}{\text{CH}}-\text{CH}_3 \xrightarrow{\text{conc. H}_2\text{SO}_4}$ (b) $\text{CH}_3-\underset{\text{H}_3\text{C}}{\overset{\text{CH}_3}{\text{C}}}-\underset{\text{Br}}{\text{CH}}-\text{CH}_3 \xrightarrow[\text{t-butoxide}]{\text{Potassium}}$
 (c) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2-\overset{\oplus}{\text{N}}(\text{CH}_3)_2 \xrightarrow[\text{(ii) } \Delta]{\text{(i) AgOH}}$ (d) $\text{CH}_3-\underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}}-\text{CH}_2-\text{Cl} \xrightarrow{\text{Alc. KOH}}$
88. When -OH is treated with HCl then different carbocations and products formed would be
 (a) 
 (b) 
 (c) 
 (d)   + 

89. Which of the following compounds can give bromination reaction with NBS/hv?

- (a) $\text{CH}_3\text{-C}\equiv\text{CH}$ (b)  (c) $\text{CH}_3\text{-CH(CH}_3\text{)-CH}_3$ (d) $\text{C}_6\text{H}_5\text{-CH}_3$

90. Which of the following alkynes show acidic character?

- (a) $\text{H-C}\equiv\text{C-H}$ (b) $\text{CH}_3\text{-C}\equiv\text{C-H}$ (c)  (d) $\text{CH}_3\text{-C}\equiv\text{C-CH}_3$

91. Aqueous solution of which of the following compounds is electrolysed, when acetylene gas is obtained?

- (a) Sodium fumarate (b) Sodium maleate
(c) Sodium acetate (d) Calcium carbide

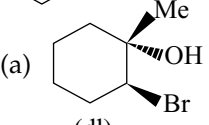
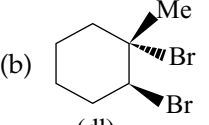
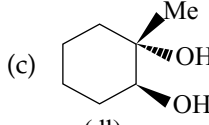
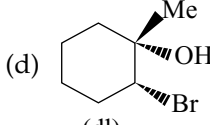
92. Which of the following acid will give isopentane on decarboxylation with soda lime?

- (a) $\begin{array}{c} \text{COOH} \\ | \\ \text{CH}_3\text{-C-CH}_2\text{-CH}_3 \\ | \\ \text{CH}_3 \end{array}$ (b) $\begin{array}{c} \text{COOH} \\ | \\ \text{CH}_3\text{-CH-CH-CH}_3 \\ | \\ \text{CH}_3 \end{array}$
(c) $\begin{array}{c} \text{COOH} \\ | \\ \text{CH}_2\text{-CH-CH}_2\text{-CH}_3 \\ | \\ \text{CH}_3 \end{array}$ (d) $\begin{array}{c} \text{CH}_3\text{-CH-CH}_2\text{-COOH} \\ | \\ \text{CH}_3 \end{array}$

93. Which of the following alkane cannot be synthesised by Wurtz reaction in good yield?

- (a) $(\text{CH}_3)_2\text{CH-CH}_2\text{-CH-(CH}_3)_2$ (b) $(\text{CH}_3)_2\text{CH-CH}_2\text{-CH}_2\text{-CH-(CH}_3)_2$
(c) $\text{CH}_3\text{-CH}_2\text{-C(CH}_3)_2\text{-CH}_2\text{-CH}_3$ (d) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_3$

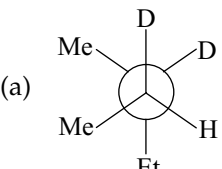
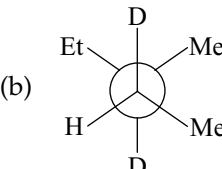
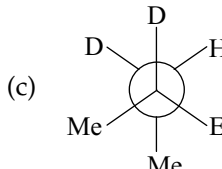
94.  $\xrightarrow{\text{Br}_2/\text{H}_2\text{O}}$ major product of the reaction is

- (a)  (b)  (c)  (d) 

95. Identify per cent yield of 2°-chlorination product of 2-methyl butane (Excluding stereoisomer), If propane on monochlorination gives 1-chloro and 2-chloro propane in 45% and 55% yield respectively and isobutane on monochlorination gives 1°-chloro and 3°-chloro product in 65% and 35% yield, respectively

- (a) 40% (b) 31.28% (c) 54.3% (d) 34.28%

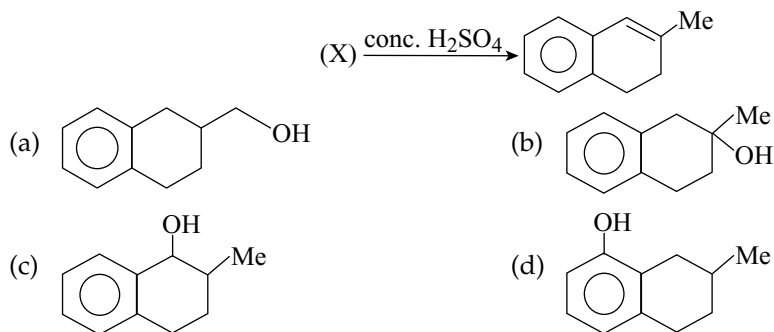
96. Identify major product of reaction of (E)-3-methyl-2-pentene with D_2/Ni

- (a)  (b)  (c)  (d) (a) and (c) both

97. Test for identification of But-2-ene and benzene is

- (a) Tollen's Reagent test (b) 1% Alkaline KMnO_4
(c) Iodoform test (d) $\text{Br}_2 + \text{H}_2\text{O}$ test

98. In the given reaction, the possible structure of compound (X) is



99. Which of the following reactions will give an alkyne?

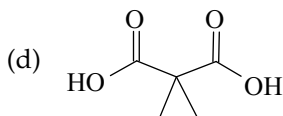
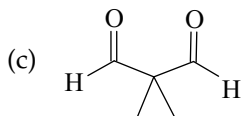
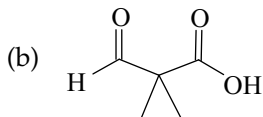
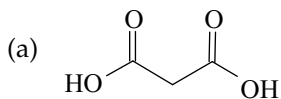
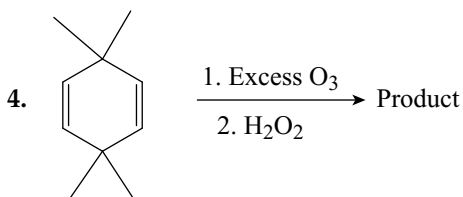
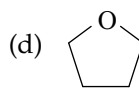
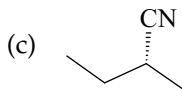
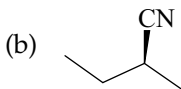
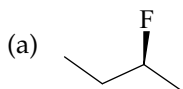
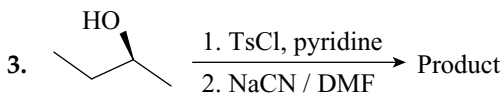
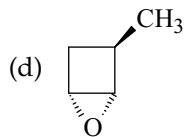
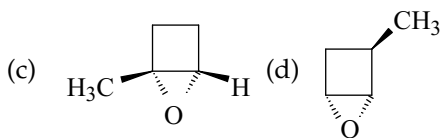
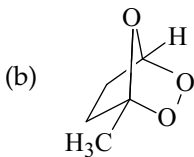
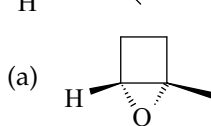
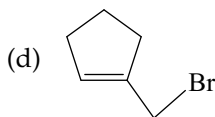
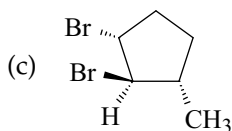
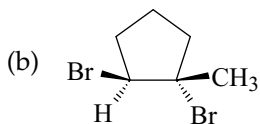
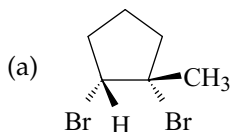
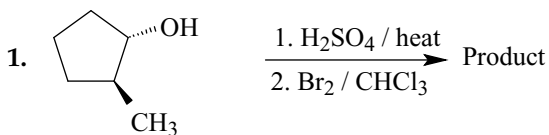
- (a) Potassium fumarate $\xrightarrow{\text{Electrolysis}}$ (b) $\text{CH}_3\text{CBr}_2\text{CHBr}_2 \xrightarrow{\text{Zn dust}}$
(c) $\text{CH}_3\text{CH}_2\text{CHBr}_2 \xrightarrow{\text{alc. KOH/NaNH}_2/\Delta}$ (d) $\text{CH}_3\text{CHBrCH}_2\text{Br} \xrightarrow{\text{NaNH}_2/\Delta}$

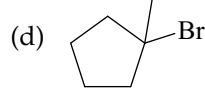
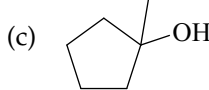
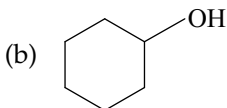
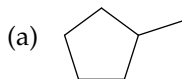
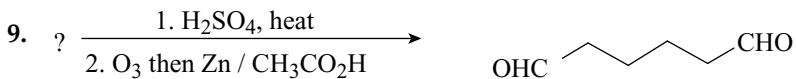
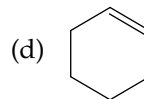
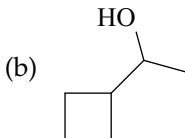
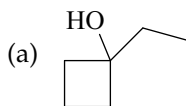
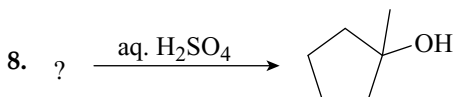
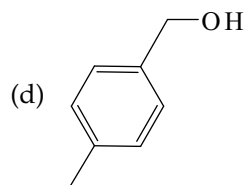
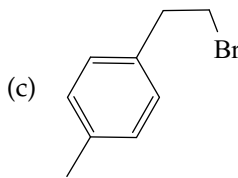
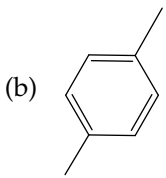
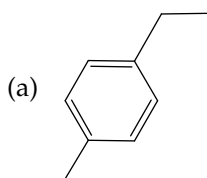
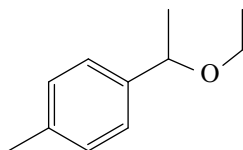
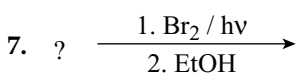
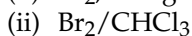
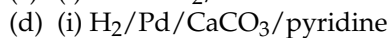
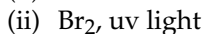
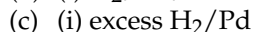
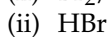
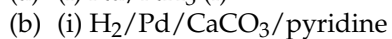
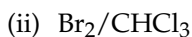
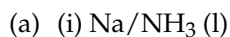
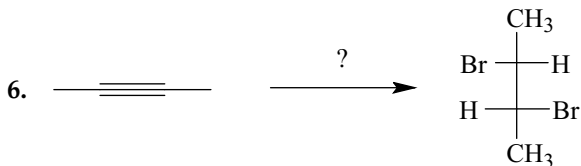
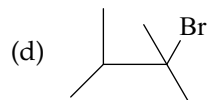
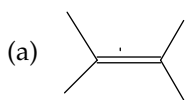
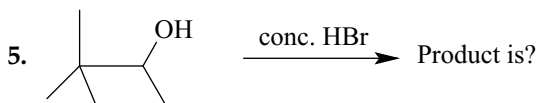
100. $4\text{HC}\equiv\text{CH} \xrightarrow[\text{iron tube}]{\text{red hot}} \text{"X"}$. "X" is

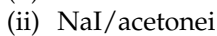
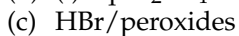
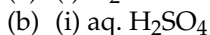
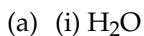
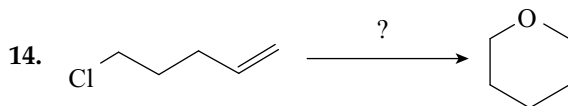
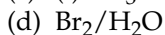
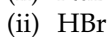
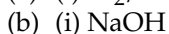
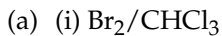
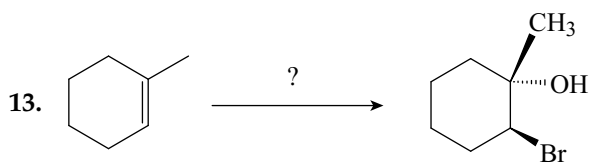
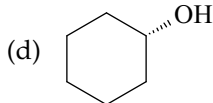
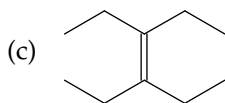
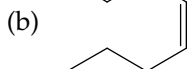
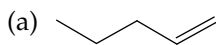
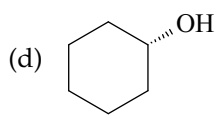
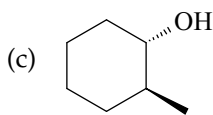
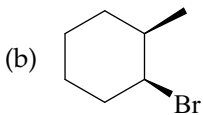
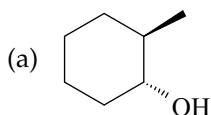
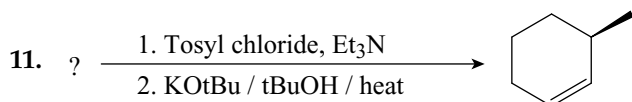
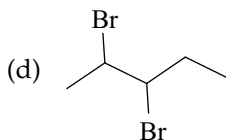
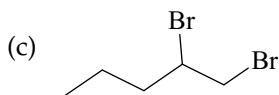
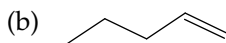
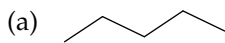
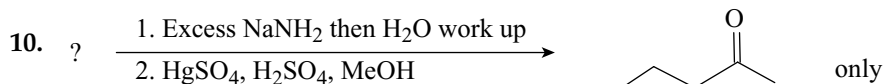
- (a) Benzene (b)
- (c)
- (d)

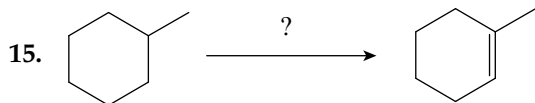
LEVEL 2

Single and Multiple-choice Type



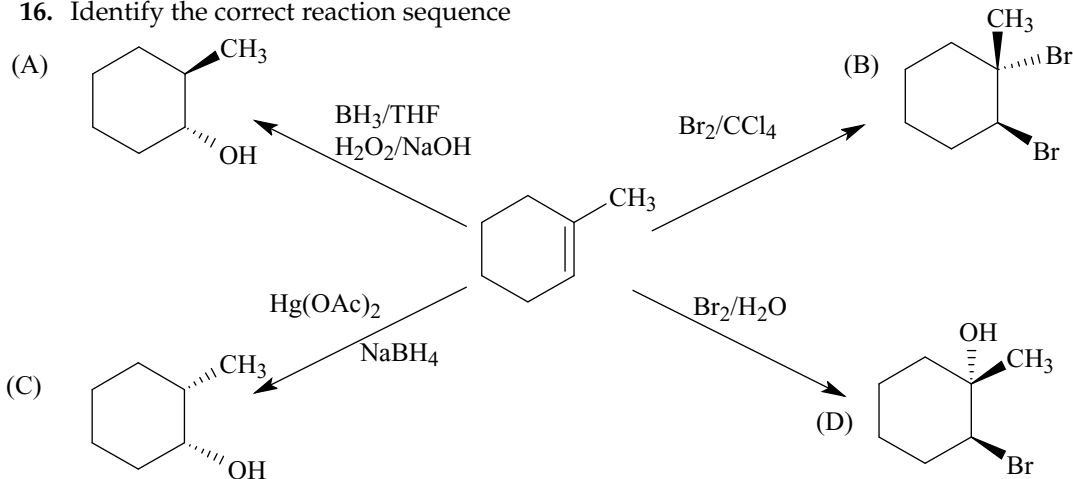






- (a) heat H_2/Pd
 (b) (i) HCl (ii) $\text{KOH}/\text{EtOH}/\text{heat}$
 (c) (i) $\text{SOCl}_2/\text{Et}_3\text{N}$ (ii) $\text{KOH}/\text{EtOH}/\text{heat}$
 (d) (i) $\text{Br}_2/h\nu$ (ii) $\text{KOH}/\text{EtOH}/\text{heat}$

16. Identify the correct reaction sequence



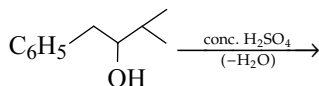
17. Which of the following is most reactive towards aqueous HBr ?

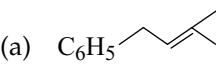
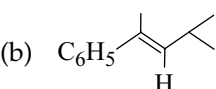
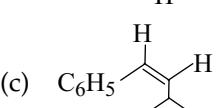
- (a) 1-Phenyl-2-propanol (b) 1-Phenyl-1-propanol
 (c) 3-Phenyl-1-propanol (d) 2-Phenyl-1-propanol

18. Ethylbenzene when treated with chlorine in the presence of light mainly gives

- (a) β -phenylethyl chloride (b) α -phenylethyl chloride
 (c) o-chloroethyl benzene (d) o- and p-chloroethylbenzene

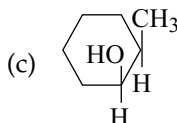
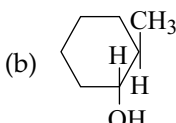
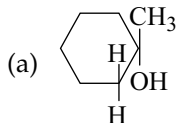
19. When the following alcohol is treated with conc. H_2SO_4 , the major product obtained is



- (a) 
 (b) 
 (c) 

(d) All the three will be formed in equal amounts

20.  X. The compound X is



(d) Both (b) and (c)

21. Give the nature of A and B in the given reaction



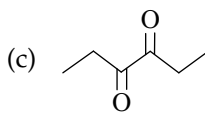
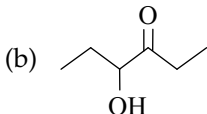
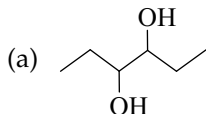
(a) A and B both are $(\text{CH}_3)_2\text{C}=\text{CH}_2$

(b) A and B both are $(\text{CH}_3)_2\text{CO} + \text{CH}_2\text{O}$

(c) A is $(\text{CH}_3)_3\text{COH}$, while B is $(\text{CH}_3)_2\text{C}=\text{CH}_2$ or $(\text{CH}_3)_2\text{CO}$

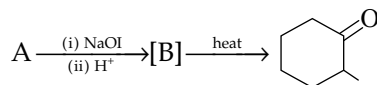
(d) A and B both are $(\text{CH}_3)_3\text{COH}$, i.e., there is no reaction

22. Which of the following is liable to be oxidised by periodic acid?

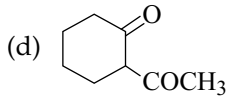
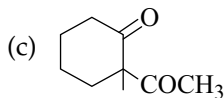
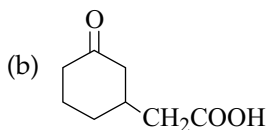
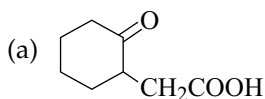


(d) All the three

23. From the given set of reaction



starting compound A corresponds to



24. Methanoic acid is heated with conc. H_2SO_4 to form

(a) CO

(b) CO_2

(c) CH_4

(d) $(\text{COOH})_2$

25. When ethane-1,2-dioic acid is heated with conc. H_2SO_4 , it gives

(a) CO + HCOOH

(b) CO_2 + HCOOH

(c) CO + CO_2 + HCOOH

(d) CO + CO_2 + H_2O

26. When sodium formate is heated with soda lime, we get

(a) CH_4

(b) Ethyne

(c) Sodium oxalate

(d) No action

27. Sodium formate is heated at 360°C to gives

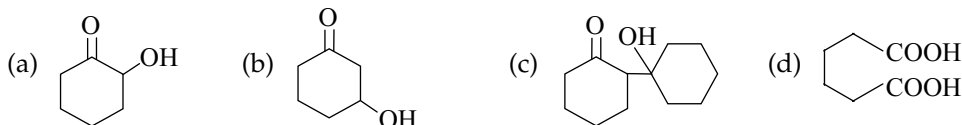
(a) CO

(b) CO_2

(c) Sodium oxalate

(d) No action

28. When cyclohexanone is treated with Na_2CO_3 solution, we get



29. In the given reaction, $\text{CH}_3\text{--CH}_2\text{--C}\equiv\text{C--H} \xrightarrow[\text{(ii) H}_2\text{O}_2/\text{OH}^-]{\text{(i) BH}_3} [\text{X}]$, [X] will be

- (a) Butanal (b) Butanone (c) 2-butanol (d) 1-butanol

30. In the given reaction, $\text{CH}_3\text{--C}\equiv\text{C--CH}_3 \xrightarrow{\text{H}_2/\text{Ni}_2\text{B}/\Delta} [\text{X}]$, [X] will be

- (a) 1-butene (b) trans-2-butene (c) cis-2-butene (d) 1-butyne

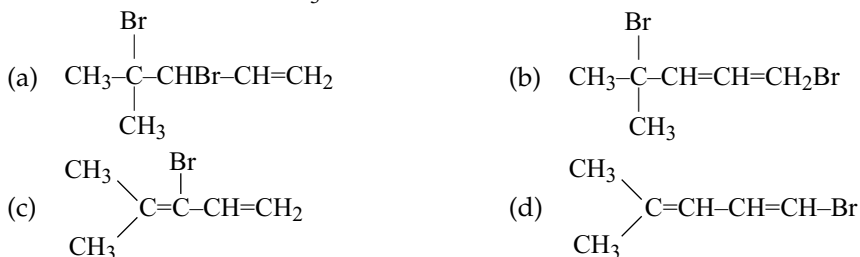
31. In the given reaction, $\text{C}_6\text{H}_5\text{--C}\equiv\text{C--CH}_3 \xrightarrow{\text{Na}/\text{NH}_3(\text{l})} [\text{X}]$, [X] will be

- (a) 1-phenyl propane (b) 1-phenyl propene
(c) trans-1-phenyl propene (d) cis-1-phenyl propene

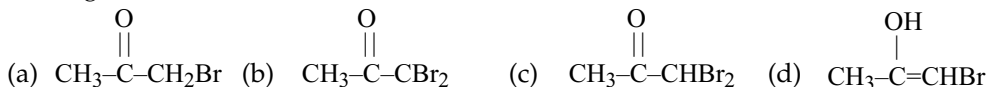
32. In the given reaction, $\text{CH}_3\text{--CH}_2\text{--C}\equiv\text{C--CH}_3 + \text{HOH} \xrightarrow{\text{HOH}/\text{H}^+/\text{Hg}^{++}} [\text{X}]$, [X] will be

- (a) 2-pentanone (b) 3-pentanone
(c) Pentanol (d) Mixture of 2-pentanone and 3-pentanone

33. In the given reaction, $\text{CH}_3\text{--C}(\text{CH}_3)=\text{CH--CH=CH}_2 + \text{Br}_2 \xrightarrow{-80^\circ\text{C}} [\text{X}]$, [X] will be



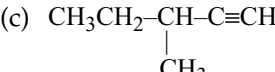
34. In the given reaction, $\text{CH}_3\text{--C}\equiv\text{CH} \xrightarrow{\text{HOBr}} [\text{X}]$, [X] will be



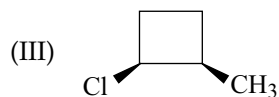
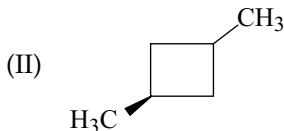
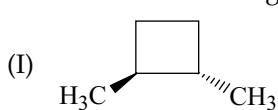
35. 8 mL of a gaseous hydrocarbon needs 40 mL of oxygen for its complete combustion. The hydrocarbon is

- (a) CH_4 (b) C_3H_4 (c) C_3H_8 (d) C_3H_6

36. 0.34 g of a hydrocarbon when heated with methyl magnesium bromide gives 112 mL of CH_4 at STP. Possible structure of the hydrocarbon is

- (a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{C}\equiv\text{CH}$ (b) $\text{CH}_3\text{CH}_2\text{C}\equiv\text{C--CH}_3$
(c)  (d) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{C}\equiv\text{CH}$

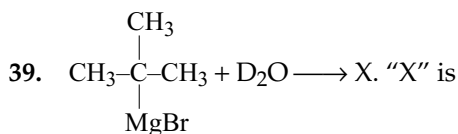
37. Which of the following structures are chiral?



- (a) I and III (b) All of the three (c) I and II (d) II and III

38. The lowest boiling point is expected for

- (a) Isooctane (b) n-octane
(c) 2,2,3, trimethyl butane (d) n-heptane



- (a) (b) (c) (d)

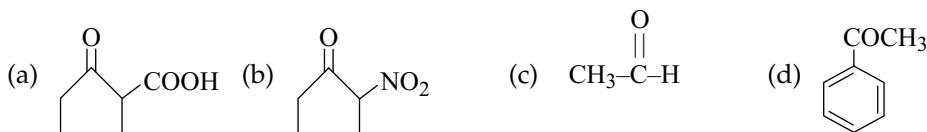
40. Acetic acid, when reacts with excess of HI in the presence of red phosphorus gives

- (a) Ethanol (b) Ethane (c) Acetaldehyde (d) Acetone

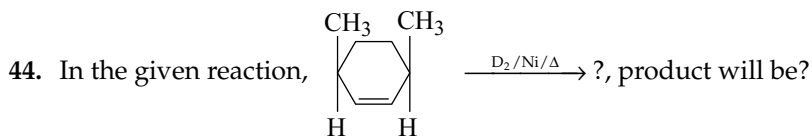
41. Sodium adipate, on electrolysis gives

- (a) Cyclobutane (b) Cyclopropane (c) But-2-ene (d) But-2-yne

42. Clemensen reduction cannot be used in which of the following?

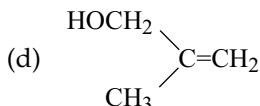
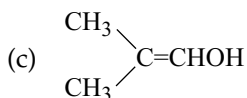
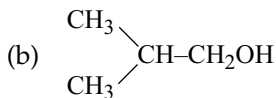
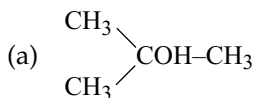


43. Wolff Kishner reduction cannot be used in which of the following?

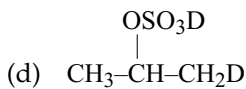
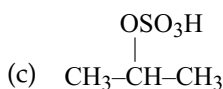
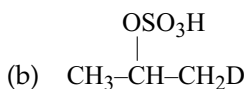
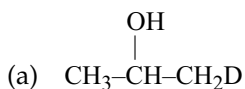


- (a) (b) (c) Both (d) 70% (a) and 30% (b)

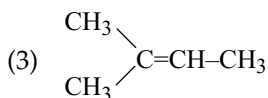
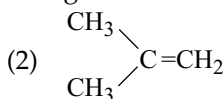
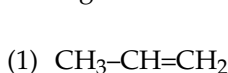
45. In the given reaction, $\begin{array}{c} \text{CH}_3 \\ \diagdown \\ \text{C}=\text{CH}_2 \\ \diagup \\ \text{CH}_3 \end{array} \xrightarrow[\text{(ii) NaOH/H}_2\text{O}_2]{\text{(i) BH}_3} [\text{X}], [\text{X}] \text{ will be}$



46. In the given reaction, $\text{CH}_3-\text{CH}=\text{CHD} \xrightarrow{\text{Conc. H}_2\text{SO}_4} [\text{X}], [\text{X}] \text{ will be}$



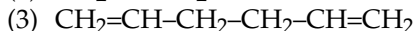
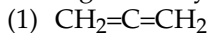
47. Arrange the following compounds in decreasing order of their heat of hydrogenation



Select the correct answer

- (a) 1, 4, 2, 3 (b) 1, 4, 3, 2 (c) 3, 2, 4, 1 (d) 3, 2, 1, 4

48. Arrange stability of the given compounds in decreasing order



Select the correct answer

- (a) 2, 4, 1, 3 (b) 4, 2, 3, 1 (c) 4, 2, 1, 3 (d) 2, 4, 3, 1

49. $2\text{C}_2\text{H}_5\text{Cl} + \text{Zn} \longrightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3 + \text{ZnCl}_2$. The reaction is known as

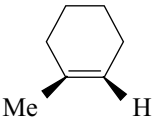
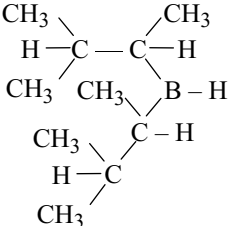
- (a) Frankland reaction (b) Wurtz reaction
(c) Fittig reaction (d) Wurtz-Fittig reaction

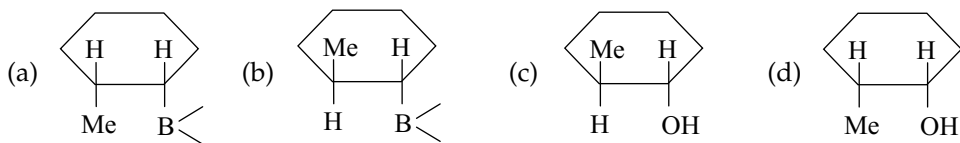
50. When isobutane is chlorinated in the presence of diffused sunlight, then the product formed is

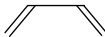
- (a) tertiary butyl chloride in major amount
(b) isobutyl chloride in major amount
(c) both 50% each
(d) n-butyl chloride, isobutyl chloride and sec-butyl chloride are formed

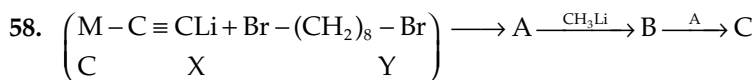
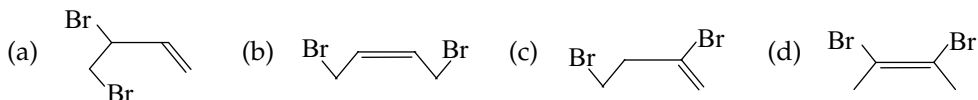
51. By which of the following methods, alkanes containing odd number and even number of carbon atoms can be prepared with good yield?
 (a) Wurtz reaction (b) Frankland reaction
 (c) Riemer Tiemann reaction (d) Groove's process
52. During the halogenation of n-pentane, assuming no regeoselectivity, the ratio in which 1-chloropentane, 2-chloropentane and 3-chloropentane are formed is
 (a) 1:2:3 respectively (b) 3:2:1 respectively
 (c) 9:4:1 respectively (d) 1:1:2 respectively
53. When isobutane is brominated in the presence of diffused sunlight then the product formed is
 (a) exclusively tertiary butyl bromide (b) exclusively isobutyl bromide
 (c) exclusively n-butyl bromide (d) s-butyl bromide
54. Which of the following statements is correct?
 (a) Chlorination of CD_4 is about 12 times faster than the chlorination of CH_4 .
 (b) Chlorination of CH_4 is about 12 times faster than the chlorination of CD_4 .
 (c) Chlorination of CH_4 and CD_4 takes place at the same step.
 (d) C-H and C-D bond energies are the same.
55. When 2-butyne is brominated, A is formed. When 2-butyne is reacted with $\text{HgSO}_4 + \text{H}_2\text{SO}_4$, then B is formed which then gives C. Hence

- (a) A is $\text{Me} \begin{array}{c} \diagup \\ \text{C} \\ \diagdown \end{array} \text{C} \begin{array}{c} \diagdown \\ \text{C} \\ \diagup \end{array} \text{Me}$ and B is $\text{Me} \begin{array}{c} \text{OH} \\ | \\ \text{C} \\ | \\ \text{CH} \end{array} \text{Me}$
- (b) A is $\text{Me} \begin{array}{c} \diagup \\ \text{C} \\ \diagdown \end{array} \text{C} \begin{array}{c} \diagdown \\ \text{C} \\ \diagup \end{array} \text{Br}$ and B is $\text{Me} \begin{array}{c} \text{OH} \\ | \\ \text{C} \\ | \\ \text{CH} \end{array} \text{Me}$
- (c) B is $\text{Me} \begin{array}{c} \text{OH} \\ | \\ \text{C} \\ | \\ \text{CH} \end{array} \text{Me}$ and C is $\text{Me} \begin{array}{c} \text{O} \\ || \\ \text{C} \end{array} \text{CH}_2\text{CH}_3$
- (d) B is $\text{Me} \begin{array}{c} \text{OH} \\ | \\ \text{C} \\ | \\ \text{OH} \end{array} \text{CH}_2\text{Me}$ and C is $\text{Me} \begin{array}{c} \text{O} \\ || \\ \text{C} \end{array} \text{CH}_2\text{CH}_3$

56. When  is reacted with  followed by treatment with $\text{H}_2\text{O}_2/\text{OH}^-$, then the different products formed at different stages are



57. When  is treated with Br_2 (1-equivalent), it would give



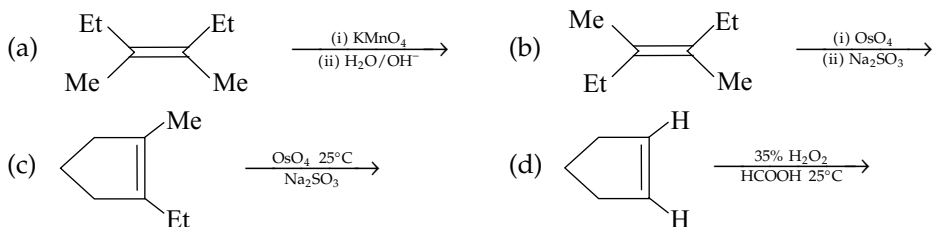
In this reaction, sequences

- (a) A is $\text{H}-\text{C}\equiv\text{C}-(\text{CH}_2)_8-\text{Br}$
 (b) B is $\text{H}-\text{C}\equiv\text{C}-\text{H}$ and C is $\text{Li C}\equiv\text{C}-(\text{CH}_2)_8-\text{Br}$
 (c) B is $\text{Li C}\equiv\text{C}-(\text{CH}_2)_8-\text{Br}$ and C is $\begin{array}{c} \text{CH}_2 - \text{C}\equiv\text{C} - \text{CH}_2 - \text{CH}_2 \\ | \qquad \qquad \qquad | \\ \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 \end{array}$
 (d) B is $\text{Li C}\equiv\text{C}-(\text{CH}_2)_8-\text{Br}$ and C is $\text{Br}(\text{CH}_2)_8\text{C}\equiv\text{C}(\text{CH}_2)_8\text{Br}$

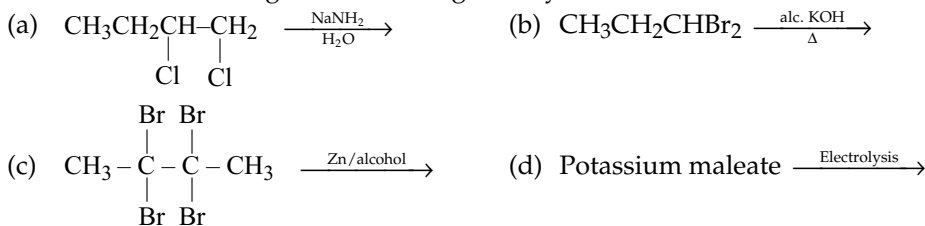
59. When $\text{CH}_2=\text{CH}-\text{Br}$ is reacted with HBr then the product formed is A and when $\text{CH}_2=\text{CH}-\text{COOH}$ is treated with HBr then the product formed is C. Hence here

- (a) A is $\begin{array}{c} \text{CH}_2 - \text{CH}_2 \\ | \quad | \\ \text{Br} \quad \text{Br} \end{array}$ (b) A is $\begin{array}{c} \text{Br} \\ | \\ \text{CH}_3 - \text{CH} \\ | \\ \text{Br} \end{array}$
 (c) C is $\begin{array}{c} \text{CH}_3 - \text{CH} - \text{COOH} \\ | \\ \text{Br} \end{array}$ (d) C is $\begin{array}{c} \text{CH}_2 - \text{CH}_2 - \text{COOH} \\ | \\ \text{Br} \end{array}$

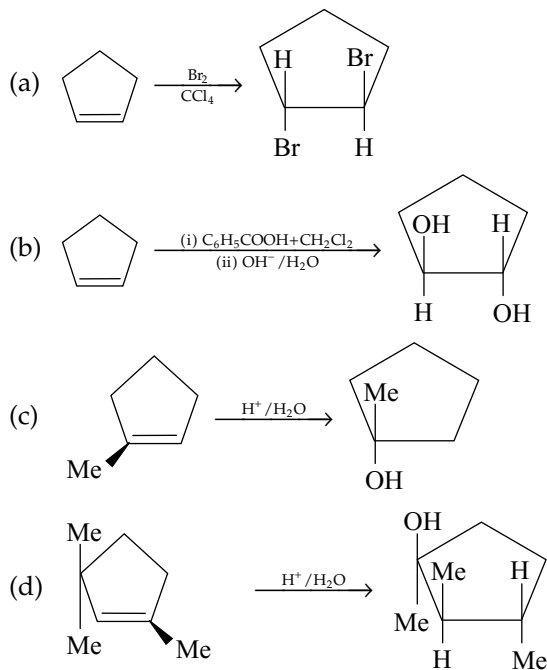
60. Which of the following will give cis diol?



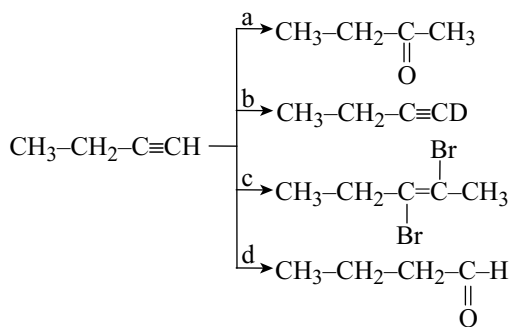
61. Which of the following reactions will give alkyne?



62. Which of the following reactions are correct?



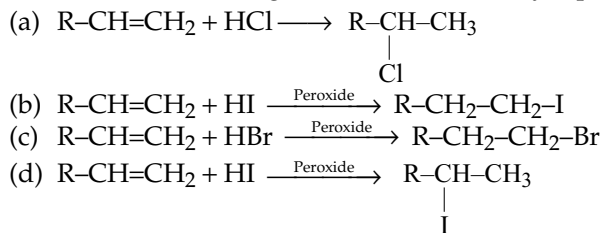
63. In the conversion given below

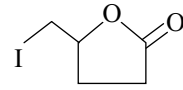


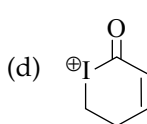
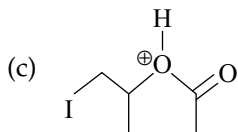
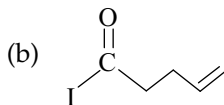
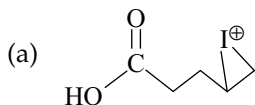
- (a) $\text{H}_2\text{O}/\text{Hg}^{2+}/\text{H}^+$
 (c) Br_2/CCl_4

- (b) (i) one equivalent of LDA, (ii) D_2O
 (d) (i) $(\text{Sia})_2\text{BH}$, (ii) $\text{H}_2\text{O}_2/\text{OH}^-$

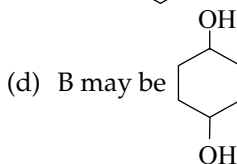
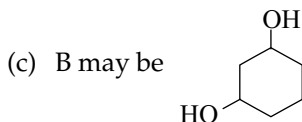
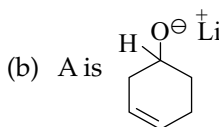
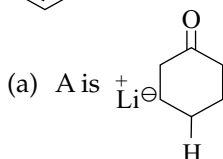
64. Which of the following reactions are correctly represented?



65. The iodo lactisation of  with I_2 gives the compound  + HI. The different intermediates are



66.  $\xrightarrow{LiAlH_4}$ A $\xrightarrow{H^+/H_2O}$ B. Here



67. The reaction of propane with nitric acid in vapour phase gives

- (a) 1-nitropropane (b) 2-nitropropane
(c) Nitromethane (d) Nitroethane

68. n-hexane reacts with Pt at 770 K to give

- (a) Cyclohexane (b) Benzene (c) Isohexane (d) Neohexane

69. Iodination of methane can be carried out in the presence of

- (a) HI (b) HIO_3 (c) HNO_3 (d) NaOH

70. Cracking of alkanes involves

- (a) Homolytic fission (b) Free radical
(c) Heterolytic fission (d) Carbocation

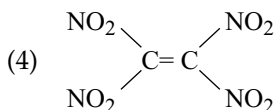
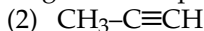
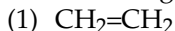
71. Arrange reactivity of the given compounds in decreasing order for electrophilic addition reaction

- (1) $C_6H_5-CH=CH_2$ (2) $C_6H_5-\underset{\substack{| \\ CH_3}}{C}=CH-CH_3$
(3) $C_6H_5-\underset{\substack{| \\ C_6H_5}}{C}=CH-CH_3$ (4) $CH_2=CH-NO_2$

Select the correct answer

- (a) 4, 1, 2, 3 (b) 3, 2, 1, 4 (c) 2, 3, 1, 4 (d) 2, 3, 4, 1

72. Which among the following compounds will give electrophilic addition reaction?



Select the correct answer

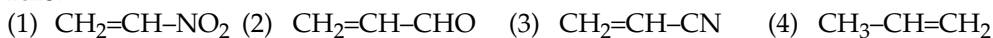
(a) 1, 2 and 3

(b) 1 and 2

(c) 1, 2 and 4

(d) 1, 2, 3 and 4

73. In which compound addition reaction will take place according to anti-Markovnikov's rule?



Select the correct answer

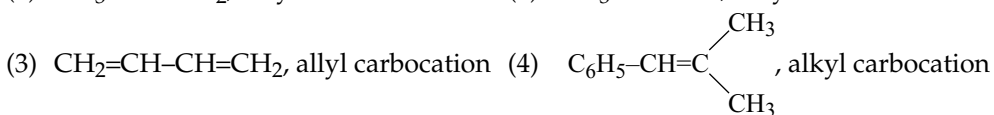
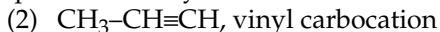
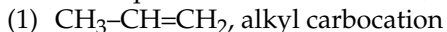
(a) 1, 2 and 3

(b) 1, 2 and 4

(c) 1 and 2

(d) 1 and 3

74. For electrophilic addition with HCl , which pair is correctly matched?



Select the correct answer

(a) 1, 3 and 4

(b) 2, 3 and 4

(c) 1, 2 and 3

(d) 1, 2 and 4

75. Consider the following statements

(1) Alkene is more reactive than alkyne for electrophilic addition reaction.

(2) Alkyne gives nucleophilic as well as electrophilic addition reaction.

(3) Alkyne is more reactive than alkene for nucleophilic addition reaction.

(4) For electrophilic addition reaction, RI of alkyne is alkyl carbocation.

Of these, the correct statements are

(a) Only 1

(b) 1 and 2

(c) 1, 2 and 4

(d) 1, 2 and 3

76. Consider the following statements

(1) Conjugated diene gives direct as well as conjugate addition.

(2) Conjugated diene gives only direct addition.

(3) Conjugated diene gives only conjugate addition.

(4) Thermodynamically controlled product is obtained by less stable reaction intermediate.

Of these, the correct statements are

(a) Only 1

(b) 1 and 4

(c) 2 and 3

(d) Only 3

77. Consider the following statements

(1) Conjugated diene gives 1, 2 and 1, 4 adduct.

(2) Conjugated diene gives kinetically and thermodynamically controlled product.

- (3) Formation of kinetically controlled product takes place by formation of stable RI.
 (4) Formation of thermodynamically controlled product takes place by the formation of stable RI.

Of these, the correct statements are

- (a) 1, 2 and 3 (b) 1, 2 and 4 (c) Only 1 (d) Only 4

78. Which among the following reagents give syn addition with alkenes?

- (1) Br_2 (2) $\text{dil. KMnO}_4/\text{OH}^-$
 (3) $\text{OsO}_4/\text{NaSO}_3\text{H}/\text{HOH}$ (4) $\text{H}_2/\text{Ni}/\Delta$

Select the correct answer

- (a) Only 1 (b) 2 and 3 (c) 2, 3 and 4 (d) Only 4

79. Match List I (reaction) with List II (reagent) and select the correct answer from the codes given below

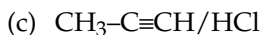
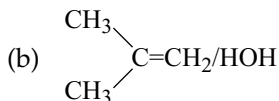
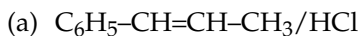
- | List I | | | | List II | | | |
|--------|--|---|---|---------|---|---|---|
| (a) | $\text{CH}_3\text{-CH=CH}_2 \longrightarrow \text{CH}_3\text{-CH}_2\text{-CH}_2\text{-OH}$ | | | (1) | $\text{Na}/\text{NH}_3(l)$ | | |
| (b) | $\text{CH}_3\text{-CH=CH}_2 \longrightarrow \text{CH}_3\text{-}\overset{\text{OH}}{\underset{ }{\text{CH}}}\text{-CH}_3$ | | | (2) | (i) BH_3 (ii) $\text{H}_2\text{O}_2/\overset{\oplus}{\text{OH}}$ | | |
| (c) | $\text{CH}_3\text{-C}\equiv\text{C-CH}_3 \longrightarrow \text{trans-2-butene}$ | | | (3) | Ni_2B | | |
| (d) | $\text{CH}_3\text{-C}\equiv\text{C-CH}_3 \longrightarrow \text{cis-2-butene}$ | | | (4) | (i) $\text{Conc. H}_2\text{SO}_4$ (ii) HOH | | |
| a | b | c | d | a | b | c | d |
| (a) 4 | 2 | 1 | 3 | (b) 4 | 2 | 3 | 1 |
| (c) 2 | 4 | 1 | 3 | (d) 2 | 4 | 3 | 1 |

80. Match List I (substrate/reagent) with List II (product) and select the correct answer

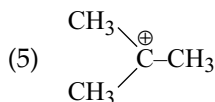
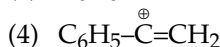
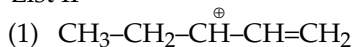
- | List I | | List II | |
|--------|--|---------|---------------------------|
| (a) | $\begin{array}{c} \text{CH}_3 \quad \quad \text{CH}_3 \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{H} \quad \quad \text{H} \end{array} + \text{Br}_2$ | (1) | $(\pm)$ 2,3-dibromobutane |
| (b) | $\begin{array}{c} \text{CH}_3 \quad \quad \text{H} \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{H} \quad \quad \text{CH}_3 \end{array} + \text{Br}_2$ | (2) | $(\pm)$ 2,3-butanediol |
| (c) | $\begin{array}{c} \text{CH}_3 \quad \quad \text{CH}_3 \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{H} \quad \quad \text{H} \end{array} + \text{Baeyer reagent}$ | (3) | Meso-2,3-dibromobutane |
| (d) | $\begin{array}{c} \text{CH}_3 \quad \quad \text{H} \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{H} \quad \quad \text{CH}_3 \end{array} + \text{Baeyer reagent}$ | (4) | Meso-2,3-butanediol |
| a | b | c | d |
| (a) 3 | 1 | 4 | 2 |
| (c) 1 | 3 | 2 | 4 |
| a | b | c | d |
| (b) 1 | 3 | 4 | 2 |
| (d) 3 | 1 | 2 | 4 |

81. Match List I (substrate/reagent) with List II (RI of the reaction) and select the correct answer

List I



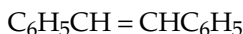
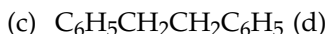
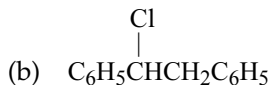
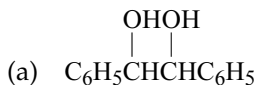
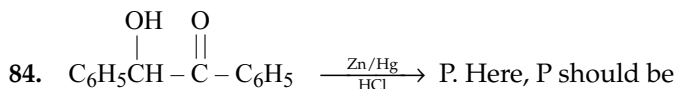
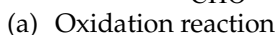
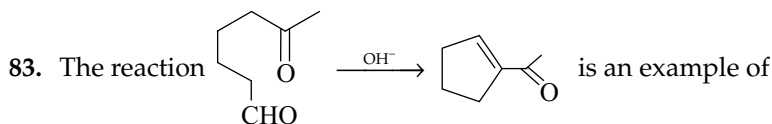
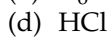
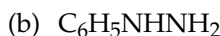
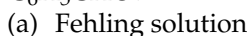
List II



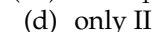
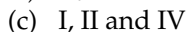
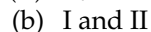
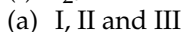
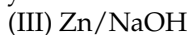
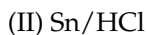
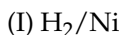
	a	b	c	d
(a)	3	5	4	1
(c)	5	3	4	1

	a	b	c	d
(b)	3	5	4	2
(d)	5	3	1	4

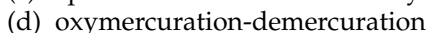
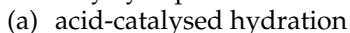
82. Which of the following reagent reacts in different ways with CH_3CHO , HCHO and $\text{C}_6\text{H}_5\text{CHO}$?



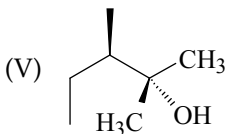
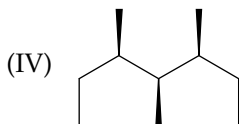
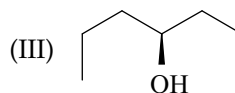
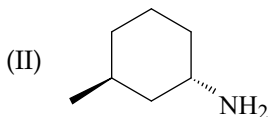
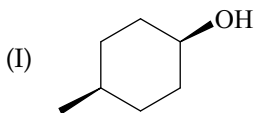
85. Nitrobenzene can be reduced to aniline by



86. 1-methylcyclopentene can be converted into 2-methylcyclohexanol by

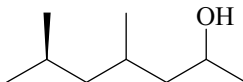


87. 2-methylpropanol-2 can be obtained by the acid-catalysed hydration of
 (a) $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$ (b) $\text{CH}_3\text{CH}=\text{CHCH}_3$
 (c) $(\text{CH}_3)_2\text{C}=\text{CHCH}_3$ (d) either of the three
88. Predict the nature of P in the following reaction: $\text{CH}_3\text{C}\equiv\text{CCH}_3 \xrightarrow[\text{heat}]{\text{NaNH}_2/\text{inert solvent}} \text{P}$
 (a) $\text{CH}_2=\text{CHCH}=\text{CH}_2$ (b) $\text{CH}_2=\text{C}=\text{CH}-\text{CH}_3$
 (c) $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CH}$ (d) No reaction
89. Arrange the following alcohols in order of increasing ease of dehydration
 $\text{CH}_3\text{CH}_2\text{OH}$ $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$ $\text{Cl}_3\text{CCH}_2\text{OH}$ $\text{F}_3\text{CCH}_2\text{OH}$
 I II III IV
 (a) $\text{II} < \text{I} < \text{IV} < \text{III}$ (b) $\text{IV} < \text{III} < \text{II} < \text{I}$ (c) $\text{IV} < \text{III} < \text{I} < \text{II}$ (d) $\text{II} < \text{I} < \text{III} < \text{IV}$
90. Which of the following statements are true?
 I: Structural isomers are compounds with the same molecular formula, but are different in the connectivity (order of attachment) of their atoms.
 II: Stereoisomers are compounds with the same molecular formula and same order of attachment of their atoms, but are different in the orientation of their atoms or groups in space.
 III: Enantiomers are stereoisomers whose molecules are mirror images of each other.
 IV: Diastereomers are stereoisomers whose molecules are not mirror images of each other.
 V: A molecule with two chiral centres designated as (R,R) will have an enantiomer with two chiral centres designated as (S,R).
 VI: Cis-1,2-dichlorocyclopentane and trans-1,2-dichlorocyclopentane are enantiomer to each other.
 (a) I, II, IV, VI (b) I, II, III, IV (c) I, II, III, IV, VI (d) II, III, IV, VI
91. Which of the following are chiral molecules?



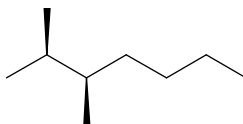
- (a) II (b) I, II, IV (c) II, V (d) II, IV, V

92. How many stereoisomers are possible for the following compound?



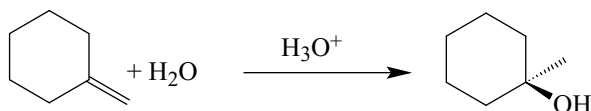
- (a) 2 (b) 3 (c) 4 (d) 8

93. What is the name of the following compound?



- (a) (2R,3S)-2,3-dimethylhexane (b) (R)-2,3-dimethylhexane
(c) (S)-2,3-dimethylhexane (d) (2S,3R)-2,3-dimethylhexane

94. What is the role of H_3O^+ in this reaction?



- (a) Base (b) Nucleophile (c) Catalyst (d) Leaving group

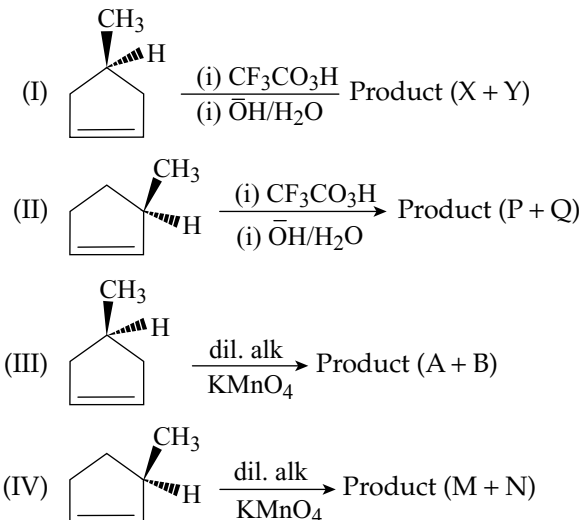
95. Which statements are true for $\text{S}_{\text{N}}2$ reaction of alkyl halides?

- I: Both of the alkyl halide and nucleophile are involved in the transition state.
 II: Reaction proceeds with inversion of configuration at the substitution centre.
 III: Reaction proceeds with retention of configuration at the substitution centre.
 IV: The order of reactivity is $3^\circ > 2^\circ > 1^\circ$.
 V: The nucleophile must have an unshared electron pair and bears a negative charge.
 VI: The greater the nucleophilicity of the nucleophile, the greater the rate of reaction.
- (a) I, II, IV, V (b) I, II, V, VI (c) I, III, V, VI (d) I, II, VI

Comprehension Type

Passage 1

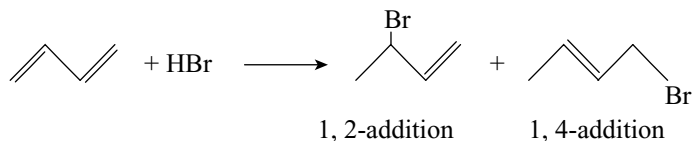
In the given reactions



96. For the given reaction (I), X and Y are
 (a) Meso compound (b) Diastereomers
 (c) Identical (d) Enantiomers
97. For the given reaction (IV), products M and N are
 (a) Enantiomers (b) Diastereomers (c) Identical (d) Meso compound
98. Which of the following is a correct statement?
 (a) Products A and B are Diastereo Isomers (b) Products P and Q are enantiomers
 (c) Products A and B are identical (d) Products P and Q are identical

Passage 2

The reaction of 1, 3-butadiene with HBr is shown below. At 40°C the major product is the 1, 4-addition product; however, at -80°C the major product is the 1, 2-addition product.

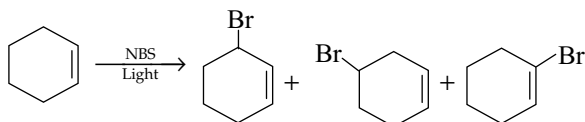


99. Thermodynamically controlled product is
 (a) 1, 2-addition product (b) 1, 4-addition product
 (c) The products have same stability (d) cannot be determined
100. Why are two products formed?
 (a) The carbocation intermediate allows delocalisation of the second double bond.
 (b) There are two double bonds present.
 (c) The fact that the carbocation is planar allows attack from both sides of the plane.
 (d) There are 2 moles of HBr.
101. Which of the two products has a lower activation energy for formation?
 (a) 1, 4-addition product.
 (b) 1, 2-addition product.
 (c) The products have same activation energy.
 (d) The relative activation energy cannot be determined.

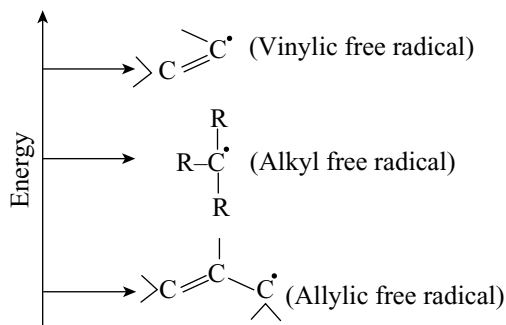
Passage 3

Karl Ziegler reported that alkenes react with N-bromosuccinimide (NBS) in the presence of light to give products resulting from substitution of hydrogen by bromine at the allylic position, i.e., the position next to the double bond.

Let us consider the halogenation of cyclohexene

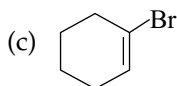
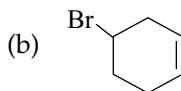
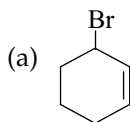


Energy level diagram for allylic, vinylic and alkyl free radicals is given



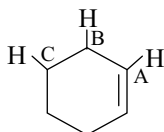
Answer the following questions

102. In the treatment of cyclohexene with NBS, which of the following products will be formed?



(d) cannot be predicted

103. Consider the three types of C—H bonds in cyclohexene



Which of the following is/are correctly matched?

- (a) A-Vinylic C—H bond
 (b) B-Allylic C—H bond
 (c) C-Alkyl C—H bond
 (d) All of these
- 104.

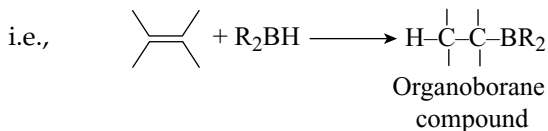
(4, 4-dimethyl cyclohexene)

Above compound on treatment with NBS gives allylic bromides. How many product(s) will be obtained in this reaction (neglecting stereoisomers)?

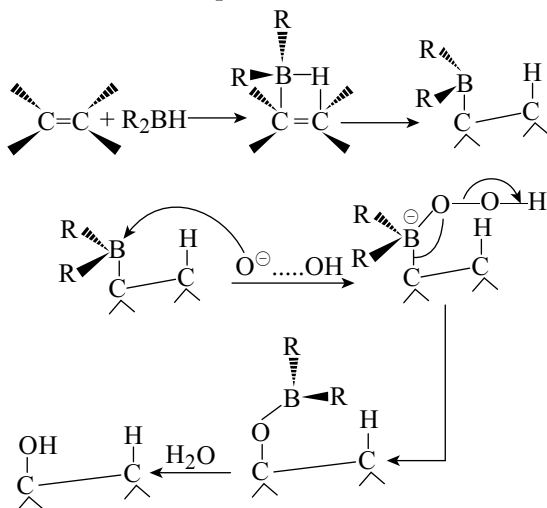
- (a) One (b) Two (c) Three (d) Four

Passage 4

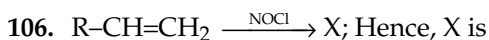
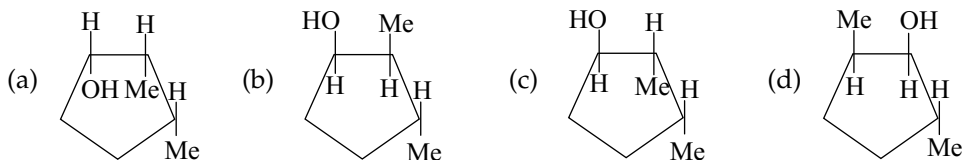
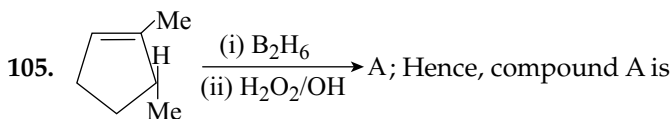
Hydroboration is a reaction in which boron hydride acts as an electrophile. R_2BH adds to a carbon-carbon double bond which acts as a nucleophile



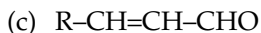
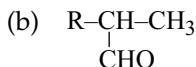
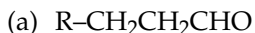
The organoborane compound then is oxidised by treatment with hydrogen peroxide in aqueous medium to form alcohol. The OH-group enters the carbon atom from the same side where the boron atom was present.



Hence this reaction is highly regioselective and the boron atom attaches to that carbon atom which is less sterically hindered.



107. $R-CH=CH_2 + CO + H_2 \xrightarrow{[Co_2(CO)_8]} A$; Hence, A is



(d) Both (a) and (b)

Passage 5

Compound having atleast one π -bond gives addition reaction. Alkene behaves as a nucleophile and hence it gives an electrophilic addition reaction. Electrophilic addition reaction in most of the cases takes place by formation of carbocation as reaction intermediate.

108. Which one of the following is NOT correct for electrophilic addition of alkenes?

- (a) In the first step, alkene reacts with electrophile to form a π -complex.
- (b) π -complex converts into carbocation and the step is a rate-determining step.
- (c) Product formation takes place by formation of most stable reaction intermediate.
- (d) Rearranged product is not formed in addition reaction with HBr.

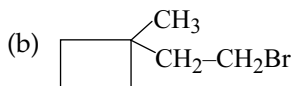
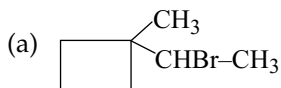
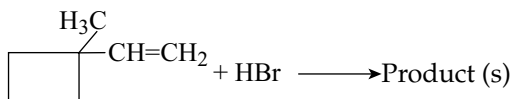
109. Consider the following statements

- (1) Unsymmetrical alkene gives addition product according to Markovnikoff's rule.
- (2) Addition reaction is a regioselective reaction.
- (3) Rearranged product is formed in addition reaction.
- (4) Alkene gives mixed addition product with $NaCl/HOH/H^+$.

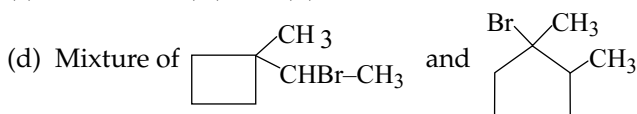
Which one is/are correct?

- (a) 1, 2, 3 and 4 (b) 2, 3 and 4 (c) 1, 3 and 4 (d) 1, 2 and 4

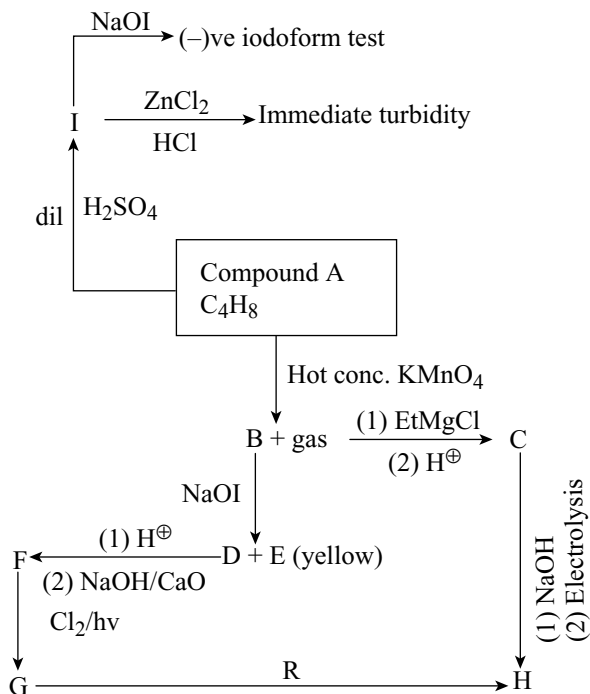
110. In the given reaction



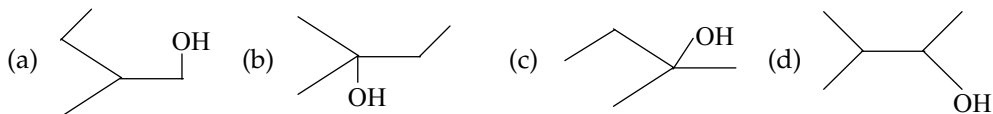
(c) Mixture of (A) and (B)



Passage 6



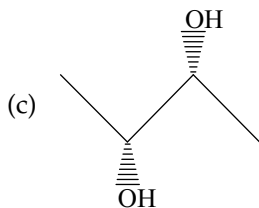
111. Compound B $\xrightarrow{\begin{smallmatrix} \text{CH-CH}_3 \\ || \\ \text{PPh}_3 \end{smallmatrix}}$ X-compound $\xrightarrow{\begin{smallmatrix} (1) B_2H_6-THF \\ (2) NaOH/H_2O_2 \end{smallmatrix}}$ Y-compound
Identify structure of compound Y



112. (Compound E) $\xrightarrow{Ag/\Delta}$ Comp. 1 $\xrightarrow{\begin{smallmatrix} (1) 2\text{eq. NaNH}_2 \\ (2) 2\text{eq. CH}_3\text{I} \end{smallmatrix}}$ Comp. 2 $\xrightarrow{\begin{smallmatrix} \text{Pb/BaSO}_4 \\ H_2 \end{smallmatrix}}$ Comp. 3 $\xrightarrow{\begin{smallmatrix} (1) OsO_4 \\ (2) NaBH_4 \end{smallmatrix}}$ Comp. 4

Identify stereochemistry compound 4

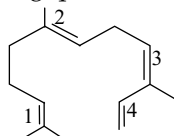




(d) (B) and (C) both

Passage 7

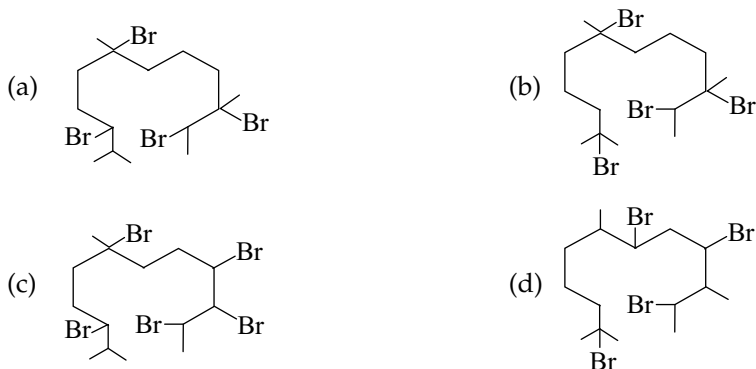
A terpene that is contained in the oil of citronella is α -farnesene. Refer to the structure of α -farnesene to answer the following questions.



113. What reaction conditions could be used to produce acetone from α -farnesene?

- (a) H_2SO_4 and heat
(b) HBr
(c) O_3 and $(\text{CH}_3)_2\text{S}$
(d) dil. acid and cold conditions

114. If α -farnesene is reacted with an excess HBr , what would be the product?



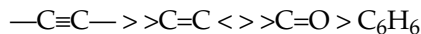
115. In the reaction of α -farnesene with excess HBr , which double bond would be slowest to react?

- (a) The bond labelled 1
(b) The bond labelled 2
(c) The bond labelled 3
(d) The bond labelled 4

Passage 8

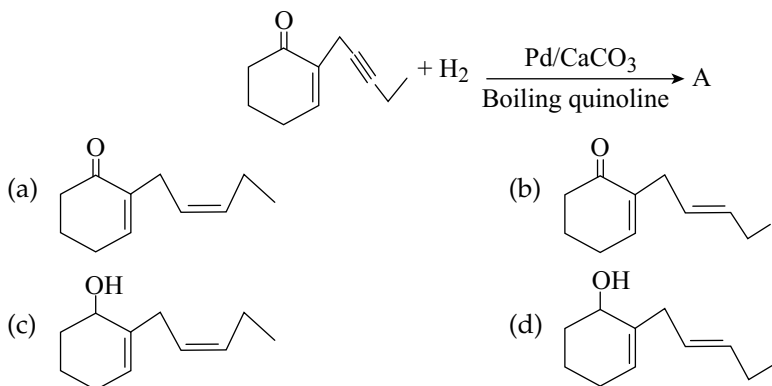
Hydrogenation of alkenes and alkynes takes place in the presence of certain catalysts. In Sabatier Senderen's reaction, the addition of hydrogen takes place in the presence of Raney nickel catalyst. Platinum and palladium can also be used as catalysts in these reactions. These are heterogeneous catalysts and used in finely divided state. Experimentally, it is observed that less crowded alkenes adsorb H_2 with faster rate. Controlled hydrogenation of alkyne in the presence of Lindlar's catalyst yields cis product, i.e.,

"cis" alkene. Thus, in the presence of Lindlar's catalyst "syn" addition takes place. The relative rate of hydrogenation follows the order

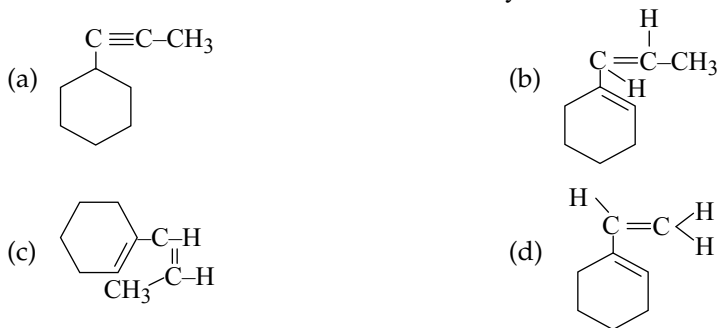


Non-terminal alkynes are reduced in the presence of Na or Li metal dissolved in liquid ammonia. In this reaction, anti addition of hydrogen results into the trans-product.

116. The product of the following reaction is



117. Product (B) is



118. $\text{CH}_3-\text{C}\equiv\text{C}-\text{CH}_3 + \text{H}_2 \xrightarrow[\text{Boiling quinoline}]{\text{Pd/CaCO}_3} (\text{A})$; The product (A) will be



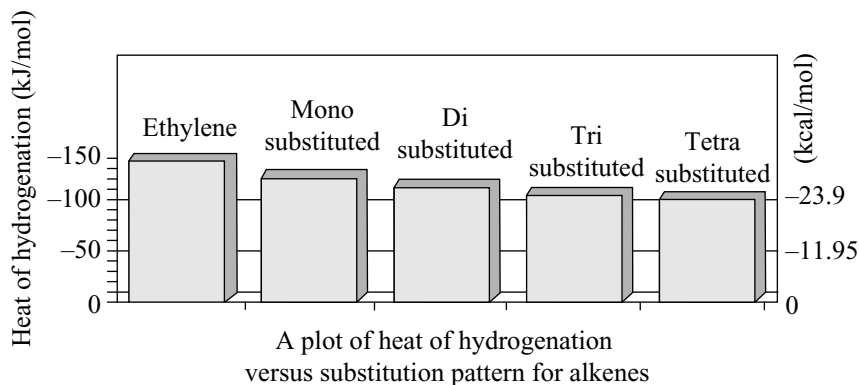
Passage 9

The structure of alkyne is linear. Terminal alkyne is acidic in character. It reacts with base to give acid-base reaction. Alkyne is nucleophile and gives electrophilic as well as nucleophilic addition reaction.

119. Which one of the following compounds forms carbonyl production reaction with 1% $\text{HgSO}_4 + \text{dil. H}_2\text{SO}_4$?
 (a) $\text{CH}_2 = \text{CH}_2$ (b) $\text{CH}_3 - \text{C} \equiv \text{C} - \text{C}_6\text{H}_5$
 (c) $\text{CH}_3 - \text{C} \equiv \text{C} - \text{CH}_3$ (d) Cyclohexene
120. Terminal alkyne will react with which of the following?
 (a) Tollen's reagent (b) $\text{SiI}_2\text{BH}/\text{THE}$
 (c) $\text{H}_2\text{O}/\text{HgSO}_4/\text{H}_2\text{SO}_4$ (d) All of these
121. Which one of the following will react with NaNH_2 ?
 (a) $\text{CH}_3 - \text{CH}_3$ (b) $\text{CH}_2 = \text{CH}_2$ (c) $\text{CH}_3 - \text{C} \equiv \text{CH}$ (d) $\text{CH}_3 - \text{C} \equiv \text{C} - \text{CH}_3$

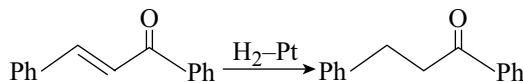
Passage 10

Following figure is given to test analytical ability. Based on it, answer the questions at the end of it.

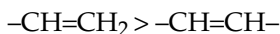


Catalytic hydrogenation is usually a stereospecific reaction called syn addition.

- The $\text{C} \equiv \text{C}$ bond is reduced more readily than $\text{C}=\text{C}$ but other unsaturated groups (except nitro and acid chlorides) are reduced less readily. Catalytic hydrogenation can, therefore, be used for the selective reduction of $\text{C}=\text{C}$ in the presence of aromatic rings and carbonyl groups, whether or not the unsaturated functions are conjugated.

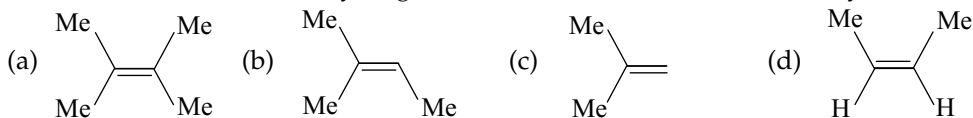


The rate of hydrogenation of olefinic bonds under standard state is

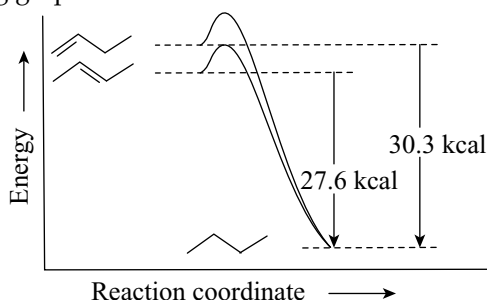


or a ring double bond

122. Base on the data of heat of hydrogenation, which has maximum stability?

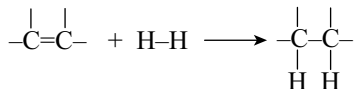


123. Consider following graph



From this, it is clear that

- (a) cis-2-butene is more stable than 1-butene by 2.7 kcal
 - (b) trans-2-butene is more stable than 1-butene by 2.7 kcal
 - (c) trans-2-butene is more stable than cis-2-butene by 11 kJ
 - (d) trans-2-butene is more stable than 1-butene by 11 kJ
124. Bond energies (in kcal mol⁻¹) of different types of bonds have been given as >C=C< (π bond=40); H-H=(104) and >C-H=(87).



Heat of hydrogenation of the above reaction is

- (a) 57 kcal mol⁻¹
- (b) -57 kcal mol⁻¹
- (c) -30 kcal mol⁻¹
- (d) 30 kcal mol⁻¹

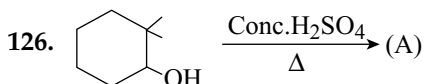
Passage 11

Strictly speaking, then, dehydration is not an E1 reaction of the protonated alcohol. In a true E1 elimination, the rate of reaction depends only upon heterolysis step, since every carbocation formed goes rapidly on to the product, that is, loss of a proton is much faster than regeneration of substrate. Here that is not the case for carbocations are formed reversibly from the protonated alcohol, and every so often one loses a proton to yield an alkene. Where the structure of alkyl group permits, rearrangement takes place. The initially formed carbocation rearranges to a more stable carbocation. The alkenes obtained are those formed by a loss of proton from this rearranged carbocation as well as from the original one.

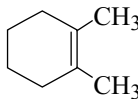
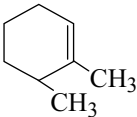
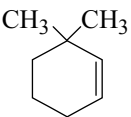
When more than one alkene can be formed the preferred product is the more stable one. Another factor comes in here. Since dehydration is reversible, the composition of the product does not necessarily reflect which alkene is formed faster but depending upon how nearly reaction approaches equilibrium which alkene is more stable.

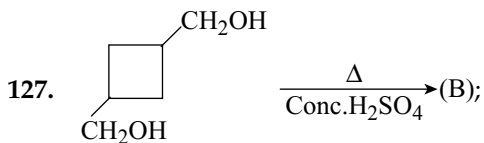
125. When neopentyl alcohol, (CH₃)₃CCH₂OH is heated with an acid, it is slowly converted to a 85:15 mixture of two alkenes of formula C₅H₁₀. The 85% of these alkene is

- (a) $\begin{array}{c} \text{CH}_3-\text{C}=\text{CH}_2 \\ | \\ \text{CH}_3 \end{array}$
- (b) $\begin{array}{c} \text{CH}_3-\text{C}-\text{CH}=\text{CH}_2 \\ | \\ \text{CH}_3 \end{array}$
- (c) $\begin{array}{c} \text{CH}_3-\text{C}=\text{CH}-\text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$
- (d) $\begin{array}{c} \text{CH}_2=\text{C}-\text{CH}_2-\text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$

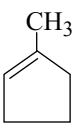
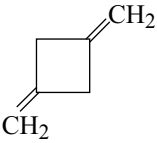


The product (A) is

- (a)  (b)  (c)  (d) None



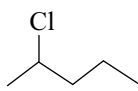
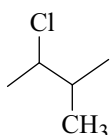
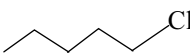
The product (B) is

- (a)  (b)  (c)  (d) None

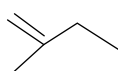
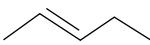
Passage 12

An organic compound A ($\text{C}_5\text{H}_{11}\text{Cl}$) is optically active and on treatment with ethanolic KOH solution yields B (C_5H_{10}) as a major product, which does not show stereoisomerism. Also A on treatment with $(\text{CH}_3)_2\text{CuLi}$ yields C (C_6H_{14}), which is optically inactive. Deduce structures of A to C.

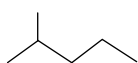
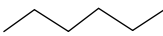
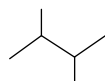
128. Identify structure of "A" compound

- (a)  (b) 
 (c)  (d) 

129. Identify structure of "B" compound

- (a)  (b) $\text{CH}_3\text{CH}=\overset{\text{CH}_3}{\text{C}}-\text{CH}_3$ (c)  (d) 

130. Identify structure of "C" compound

- (a)  (b)  (c)  (d) 

Passage 13

An organic compound $A(C_{13}H_{23}Cl)$ exists as distereomers and decolourise bromine water. A on treatment with ethanolic solution of KOH produces isomeric B and C with their molecular formula $C_{13}H_{22}$. Treatment of either B or C with Rany Nickel produces 4-isopropyl-1-tertiarybutyl cyclohexane. A on oxidative ozonolysis gives acetone as one product. Identify A, B and C considering C to be enantiomeric.

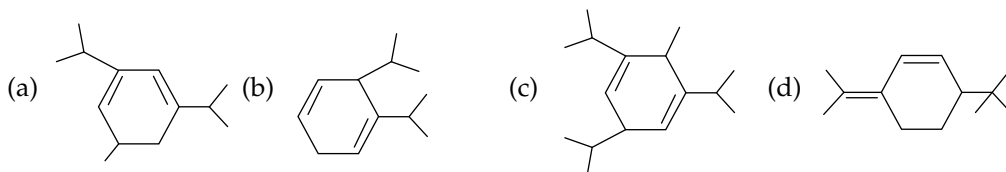
131. According to the information given in the above paragraph compound (A) will be



132. According to the information given in the above paragraph compound (B) will be



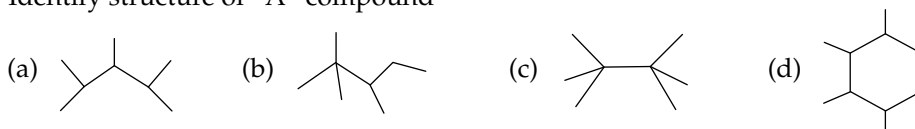
133. According to the information given in the above paragraph compound (C) will be



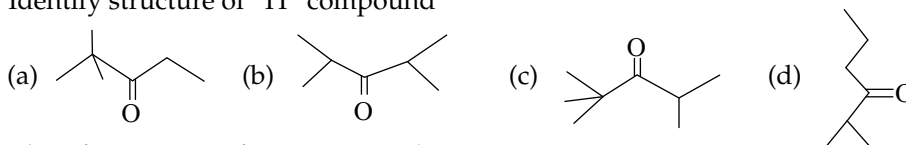
Passage 14

An optically active hydrocarbon A has molecular formula C_8H_{18} . A on monochlorination gives five alkyl halide B to F with their molecular formula $C_8H_{17}Cl$. B does not undergo dehydrohalogenation on treatment with alcoholic solution of KOH. Treatment of either C or D with alcoholic KOH yields same alkene G(C_8H_{16}), which on ozonolysis followed by work-up with Zn-dimethyl sulphide gives an optically inactive compound $C_6H_{12}O$ and ethanol. Also C is enantiomeric, whereas D is distereomeric. E on dehydrohalogenation yields an alkene, which on reductive ozonolysis yields H($C_7H_{14}O$), which is optically inactive. H on treatment with $LiAlH_4$ yields I($C_7H_{16}O$) which can be resolved into enantiomers. F on dehydrohalogenation yields an alkene (C_8H_{16}), which on reductive ozonolysis yields J($C_7H_{14}O$), which is optically active and has same configuration as that of A. Identify A to J explaining the reactions involved.

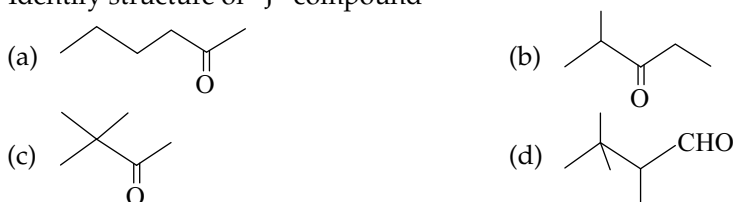
134. Identify structure of "A" compound



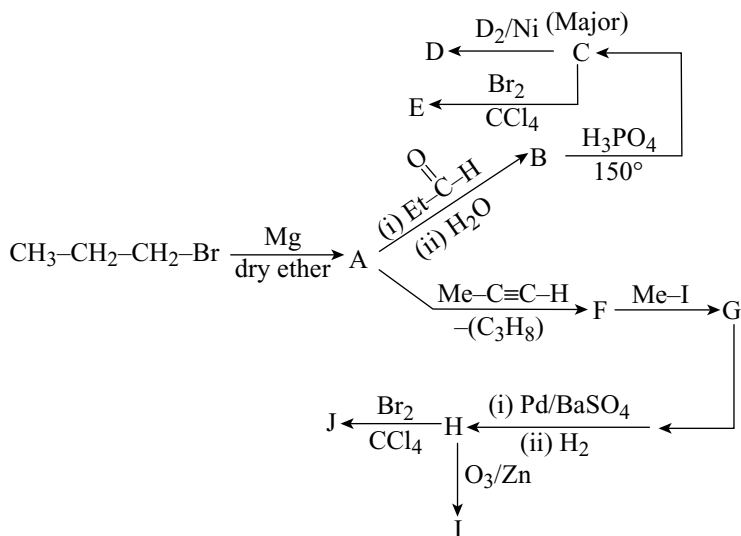
135. Identify structure of "H" compound



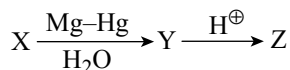
136. Identify structure of "J" compound



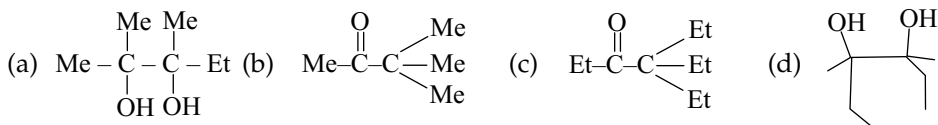
Passage 15



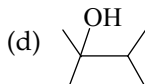
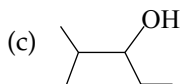
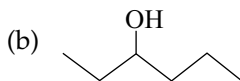
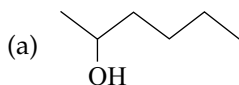
137. Identify "Z" compound



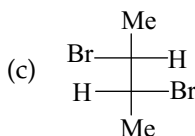
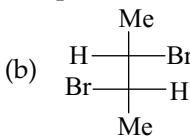
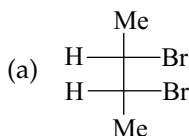
where X is a functional isomer of "W" which is next higher homologue of "I"



138. In the above reaction sequence "B" compound is



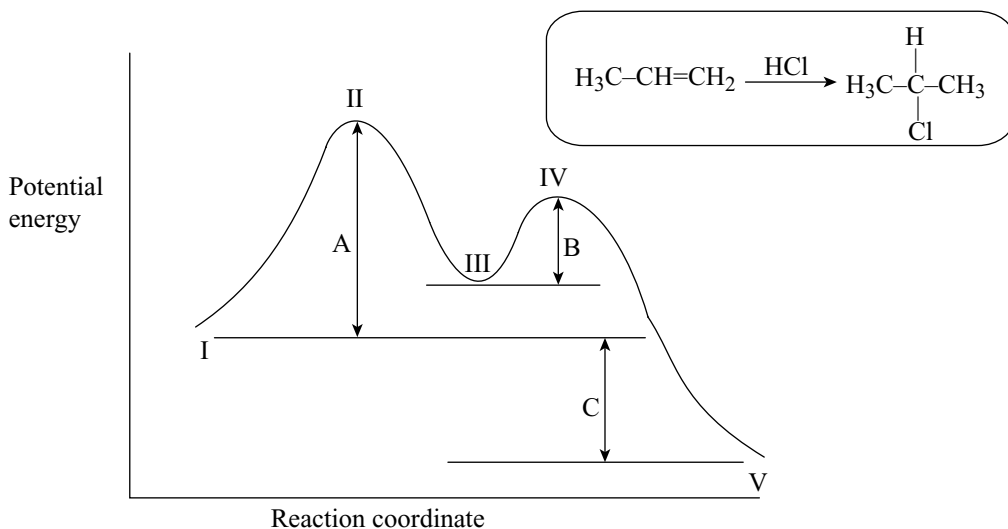
139. In the above reaction sequence "J" compound is



(d) both (b) and (c)

Passage 16

Based on the potential energy diagram for the following reaction



140. Is this an endothermic or exothermic reaction?

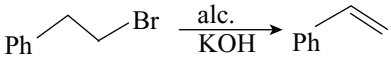
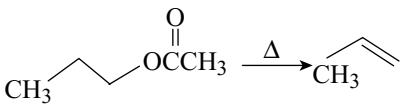
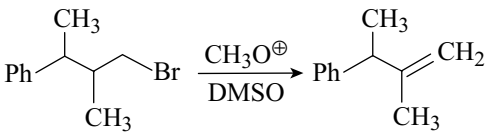
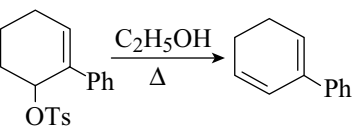
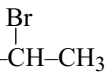
- (a) Endothermic
- (b) Exothermic
- (c) There is not enough information to determine.
- (d) This reaction can be either exothermic or endothermic.

141. What is B representing in this potential energy diagram?

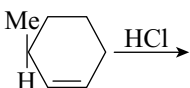
- (a) heat of reaction: the energy required for the reaction to occur
- (b) heat of reaction: the overall energy change for this reaction
- (c) activation energy: the energy required for the reaction to occur
- (d) activation energy: the overall energy change for this reaction

142. What is C representing in this potential energy diagram?
 (a) heat of reaction: the energy required for the reaction to occur
 (b) heat of reaction: the overall energy change for this reaction
 (c) activation energy: the energy required for the reaction to occur
 (d) activation energy: the overall energy change for this reaction
143. What is II representing in this potential energy diagram?
 (a) transition state
 (b) intermediate
 (c) activation energy
 (d) heat of reaction
144. What is III representing in this potential energy diagram?
 (a) transition state
 (b) intermediate
 (c) activation energy
 (d) heat of reaction
145. Which step is the rate-determining (rate-limiting) step?
 (a) from I to V (b) from I to III (c) from II to III (d) from III to V

Matrix Type

- | 146. Column I
(Reaction) | Column II
(Type of reaction) |
|---|---------------------------------|
| (a)  | (p) E ₁ |
| (b)  | (q) E ₂ |
| (c)  | (r) E ₁ CB |
| (d)  | (s) E _i |
-
- | 147. Column I | Column II (Type of reaction) |
|---|------------------------------|
| (a) (CH ₃) ₃ C—Br | (p) E ₁ CB |
| (b)  | (q) First-order kinetics |
| (c)  | (r) E ₁ |
| (d)  | (s) E ₂ |

148. Column I

- (a) $\text{CH}_2=\text{CH}-\text{COOH} + \text{HBr}$
 (b) $\text{cis CH}_3-\text{CH}=\text{CH}-\text{C}_2\text{H}_5 + \text{KMnO}_4 (\text{cold alk.})$
 (c) $\text{cis CH}_3-\text{CH}=\text{CH}-\text{CH}_3 + \text{X}_2 \xrightarrow{\text{CS}_2}$
 (d) 

Column II

- (p) Nonregioselective
 (q) Trans addition
 (r) Primary carbocation
 (s) Optically active

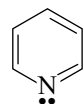
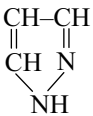
149. Column I
(Reaction)

- (a) $(\text{CH}_3)_3\text{C}-\text{CH}=\text{CH}_2 \longrightarrow (\text{H}_3\text{C})_2\underset{\text{OH}}{\text{C}}-\text{CH}(\text{CH}_3)_2$
 (b) $(\text{CH}_3)_3\text{C}-\text{CH}=\text{CH}_2 \longrightarrow (\text{CH}_3)_3\underset{\text{OH}}{\text{C}}-\text{CH}-\text{CH}_3$
 (c) $\text{H}_5\text{C}_6-\text{CH}=\text{CH}_2 \longrightarrow \text{C}_6\text{H}_5-\text{CHO}$
 (d) $\text{C}_6\text{H}_5-\text{C}\equiv\text{CH} \longrightarrow \text{C}_6\text{H}_5-\text{CH}_2-\text{CHO}$

Column II
(Reagent)

- (p) $\text{B}_2\text{H}_6/\text{H}_2\text{O}_2/\text{OH}^-$
 (q) $\text{H}_2\text{O}/\text{H}^+/\text{MnO}_2$
 (r) $\text{Hg}(\text{OAc})_2/\text{H}_2\text{O}/\text{NaBH}_4$
 (s) $\text{H}_2\text{O}/\text{H}^+$

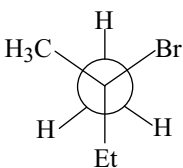
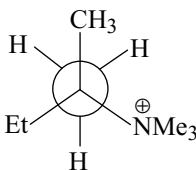
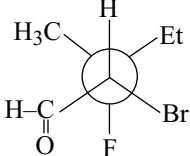
150. Column I

- (a) $\text{HC}\equiv\text{CH} \longrightarrow$ 
 (b) $\text{H}_5\text{C}_2-\text{C}\equiv\text{C}-\text{C}_2\text{H}_5 \longrightarrow \text{H}_5\text{C}_2-\underset{\text{O}}{\underset{\text{||}}{\text{C}}}-\text{CH}_2-\text{C}_2\text{H}_5$
 (c) $\text{HC}\equiv\text{CH} \longrightarrow$ 
 (d) $\text{H}_5\text{C}_2-\text{C}\equiv\text{C}-\text{C}_2\text{H}_5 \longrightarrow 2\text{H}_5\text{C}_2-\text{COOH}$

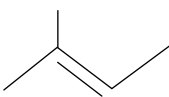
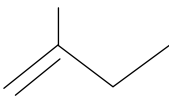
Column II

- (p) $\text{O}_3/\text{H}_2\text{O}$
 (q) CH_2N_2 (cold ether solution)
 (r) HCN (Red hot Fe)
 (s) $\text{H}_2\text{O}/\text{H}_2\text{SO}_4/\text{HgSO}_4$

151. Column I

- (a)  $\xrightarrow[\text{KOH}]{\text{Alc.}}$
 (b)  $\xrightarrow[\text{KOH}]{\text{Alc.}}$
 (c)  $\xrightarrow[\text{Bu}^\oplus \text{O}^\ominus \text{K}^\oplus]{\text{Alc.}}$

Column II

- (p) 
 (q) E^2
 (r) 
 (s) E^1CB

152. Column I

- (a) $R-CH_2-X$
- (b) $\begin{array}{c} R \\ | \\ R-C-X \\ | \\ R \end{array}$
- (c) $CH_2=CH-X$
- (d) $R-COO^- Na^+$

Column II

- (p) Corey-house reaction
- (q) Kolbe electrolysis
- (r) Wurtz reaction
- (s) Frankland reaction

153. Column I

- (a) But-1-yne
- (b) But-2-yne
- (c) Benzene
- (d) $CH_3C\equiv CNa$

Column II

- (p) Reacts with $KMnO_4/KOH$ to give acetic acid
- (q) With Hg^{2+}/H_2SO_4 gives butan-2-one
- (r) Reacts with CH_3Cl to form but-2-yne
- (s) Reacts with $O_3/Zn/H_2O$ to give glyoxal

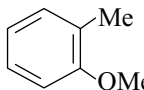
154. Column I

- (a) Oxidative ozonolysis of alkene
- (b) Oxidative ozonolysis of alkyne
- (c) Reductive ozonolysis of alkene
- (d) Reductive ozonolysis of alkyne

Column II

- (p) $R-CHO$ ($R-C\equiv C-R$)
- (q) $R-COOH$ ($R-CH=CH-R$)
- (r) $\begin{array}{c} O \\ || \\ R-C-R \end{array}$ ($R-C\equiv C-R$)
- (s) $\begin{array}{c} O \quad O \\ || \quad || \\ (R-C-C-R) \end{array}$ ($R-CH=C(R)-R$)

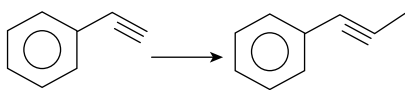
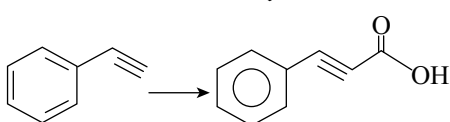
155. Column I

- (a) $Me-C\equiv C-Me \xrightarrow[liq. NH_3]{Na} P_1 \xrightarrow[CCl_4]{Br_2} \text{Final product}$
- (b) $Et-C\equiv C-Et \xrightarrow[liq. NH_3]{Na} P_1 \xrightarrow[Ni]{D_2} \text{Final product}$
- (c) $\text{---} \equiv C - \begin{array}{c} Cl \\ | \\ H \end{array} \xrightarrow[Pd/BaSO_4]{H_2} \text{Final product}$
- (d)  $\xrightarrow[NH_3]{Li/liq.} P_1 \xrightarrow[1 eq. O_3]{metal} \text{Final product}$

Column II

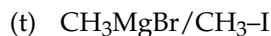
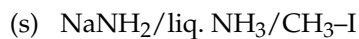
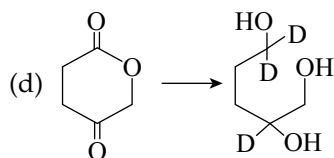
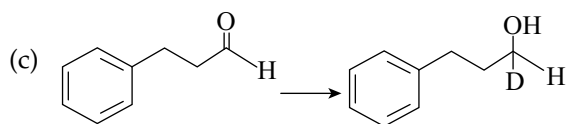
- (p) Final product of reaction is racemic mixture.
- (q) Final product of reaction is meso.
- (r) Net optical rotation of final product is zero.
- (s) Net optical rotation of final product is nonzero.
- (t) Final product has one asymmetric carbon atom.

156. Column I

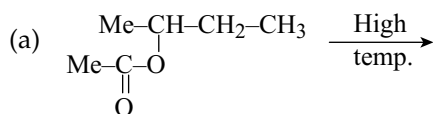
- (a) 
- (b) 

Column II

- (p) $NaBD_4/EtOH$
- (q) $LiAlD_4/dil. HCl$

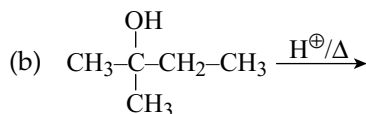


157. Column I
(Substrate for elimination reaction)

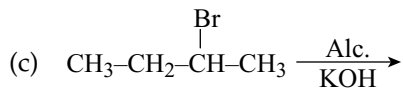


Column II
(Type of elimination)

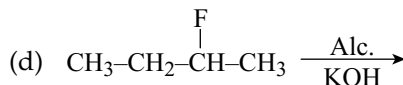
(p) E1



(q) E2



(r) Ei/Pyrolysis

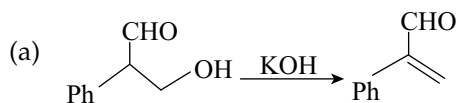


(s) Saytzeff elimination

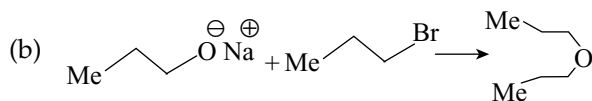
(t) Hoffmann elimination

158. Column I (reaction)

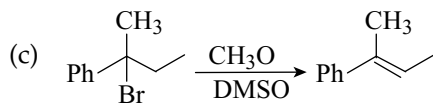
Column II (type of reaction: Major)



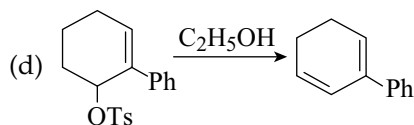
(p) E₁



(q) E₂



(r) E₁CB



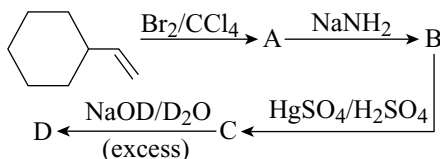
(s) S_N²

Integer Type

159. Number of hydrocarbons formed when $\text{C}_2\text{H}_5\text{Br}$ and $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-Br}$ are treated with Na in the presence of dry ether is:
160. How many number of moles of H_2 is used for the complete hydrogenation of the given compound in the presence of a metal catalyst?



161. Number of products formed by the ozonolysis of 1-4-butadiene is
162. Consider the following reaction scheme

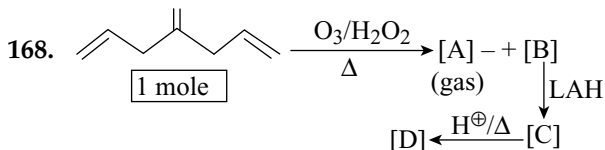


How many oxygens are present in the compound D?

163. What volume of ethane (NTP; 1 bar, 273 K) is formed from 38 g of sodium propionate by fusion with sode lime?
164. $(\text{CH}_3)_2\text{C}=\text{CH}-\text{CH}_3 \xrightarrow[\text{H}_2]{\text{Catalyst}}$ Optical isomers

165. Degree of unsaturation in is

166. Minimum number of C-atoms in alkynes to show optical isomerism is
167. Number of products obtained on ozonolysis of 1, 2-dimethyl benzene is



If X = Number of moles of CO_2

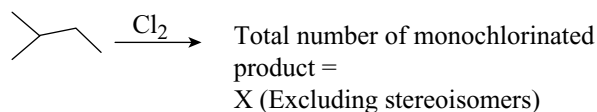
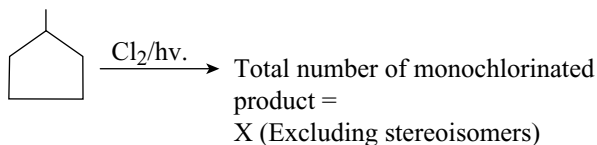
Y = Number of $\alpha\text{-H}$ in "D" compound

$X + Y = Z$

Identify value of "Z".

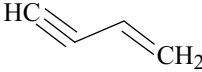
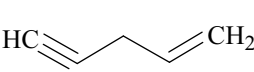
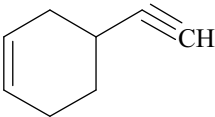
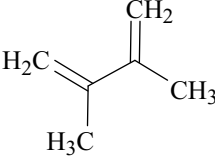
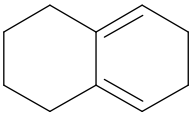
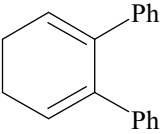
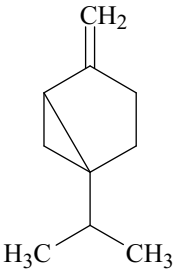
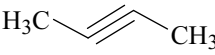
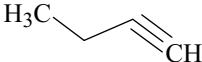
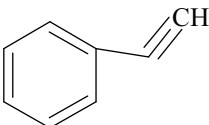
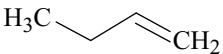
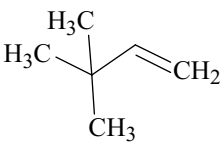
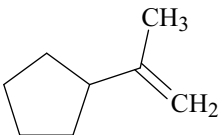
169. An optically active compound A has the molecular formula C_6H_{10} . The compound gives a precipitate when treated with $Ag(NH_3)_2OH$. On catalytic hydrogenation, A yields B(C_6H_{14}), which is optically inactive. Identify total number of " α " "H" in product formed by treatment of A with O_3/H_2O_2 then LAH and then H^+/Δ .

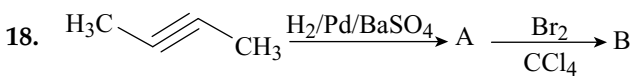
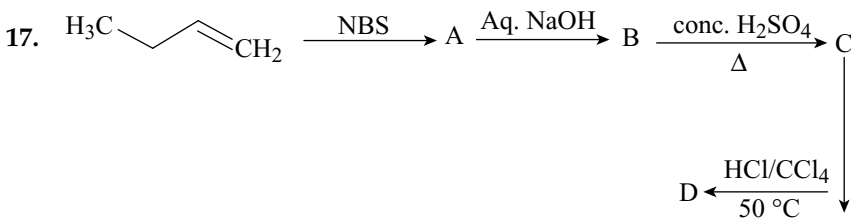
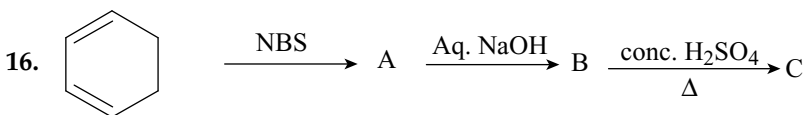
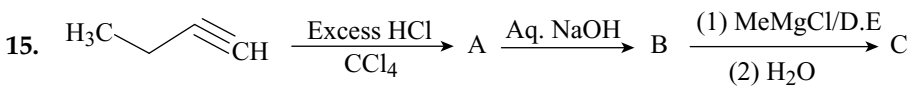
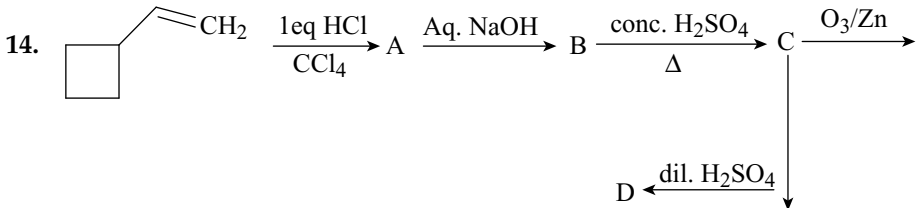
170. Consider the following reactions



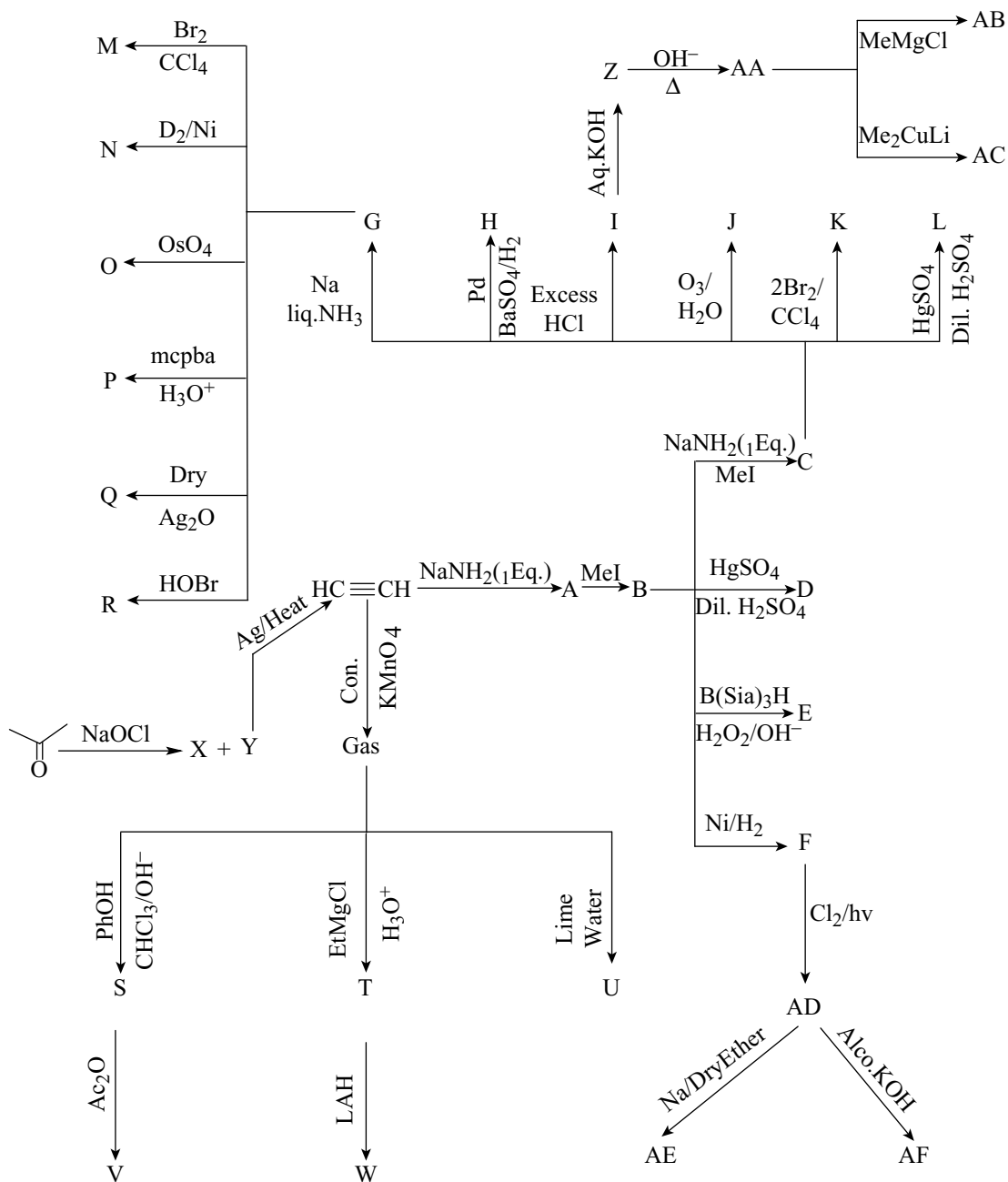
Identify value of $X + Y$.

WORKBOOK EXERCISE 1

1.  $\xrightarrow[\text{CCl}_4]{1\text{eq HCl}}$
2.  $\xrightarrow[\text{CCl}_4]{\text{Excess HCl}}$
3.  $\xrightarrow[\text{CCl}_4]{1\text{eq HCl}}$
4.  $\xrightarrow[\text{CCl}_4]{1\text{eq HCl}}$
5.  $\xrightarrow[\text{CCl}_4]{1\text{eq HCl}}$
6.  $\xrightarrow[\text{CCl}_4]{1\text{eq HCl}}$
7.  $\xrightarrow[\text{CCl}_4]{1\text{eq HCl}}$
8.  $\xrightarrow[\text{CCl}_4]{\text{Excess HCl}}$
9.  $\xrightarrow[\text{CCl}_4]{\text{Excess HCl}}$
10.  $\xrightarrow[\text{CCl}_4]{\text{Excess HCl}}$ A $\xrightarrow[\text{CCl}_4]{\text{Excess NaOH}}$ B
11.  $\xrightarrow[\text{CCl}_4]{1\text{eq HCl}}$ A $\xrightarrow{\text{Aq. NaOH}}$ B $\xrightarrow[\Delta]{\text{conc. H}_2\text{SO}_4}$ C
12.  $\xrightarrow[\text{CCl}_4]{1\text{eq HCl}}$ A $\xrightarrow{\text{Aq. NaOH}}$ B $\xrightarrow[\Delta]{\text{conc. H}_2\text{SO}_4}$ C
13.  $\xrightarrow[\text{CCl}_4]{1\text{eq HCl}}$ A $\xrightarrow{\text{Aq. NaOH}}$ B $\xrightarrow[\Delta]{\text{conc. H}_2\text{SO}_4}$ C



WORKBOOK EXERCISE 2



Answer Keys and Solutions to Workbook Exercises

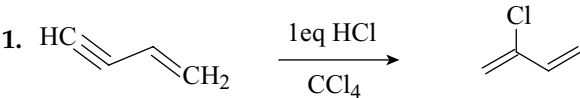
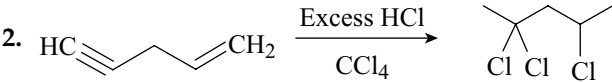
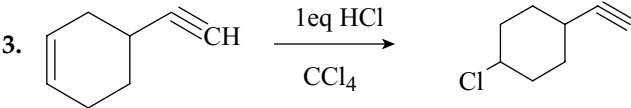
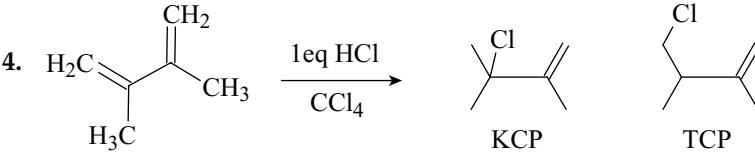
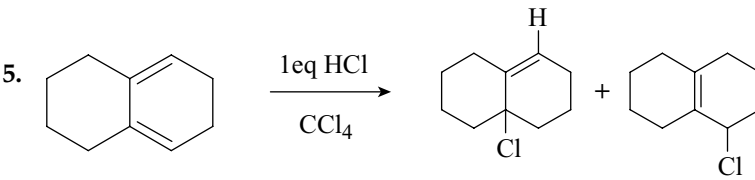
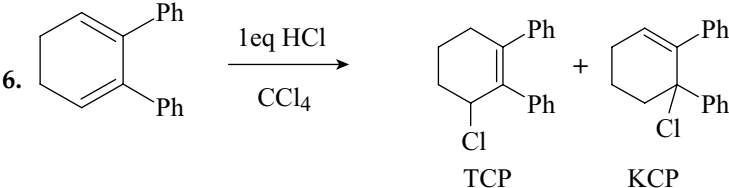
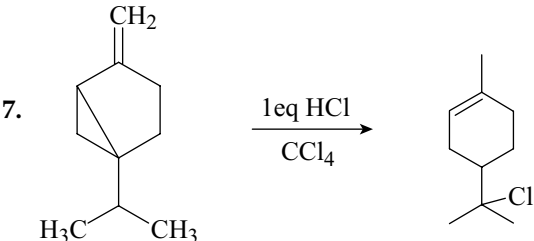
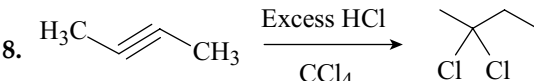
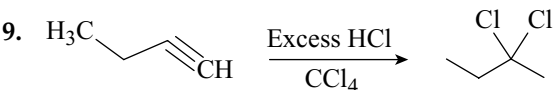
LEVEL 1

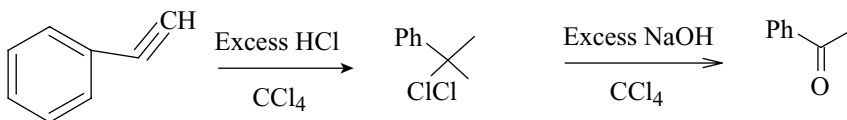
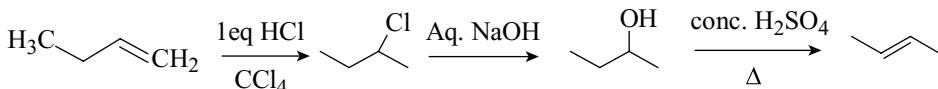
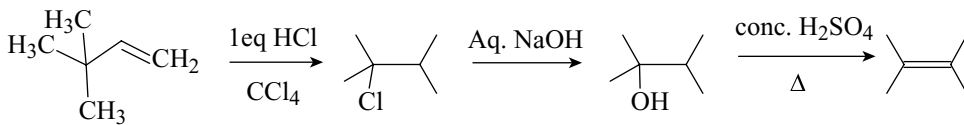
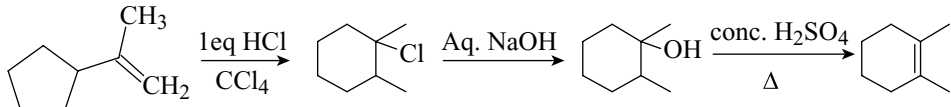
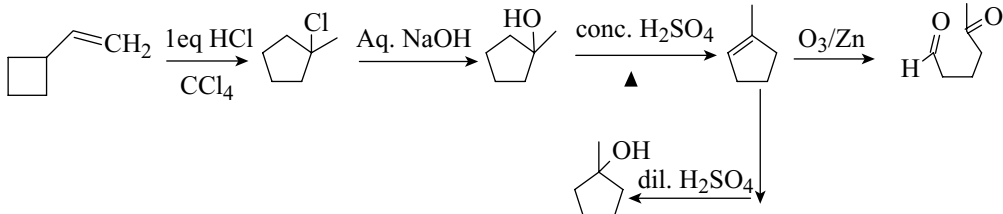
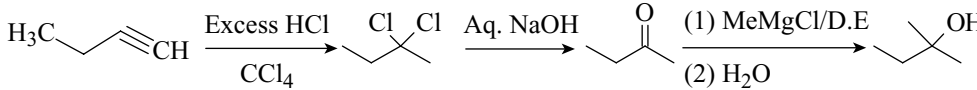
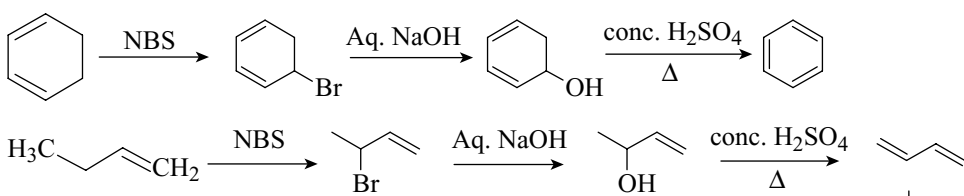
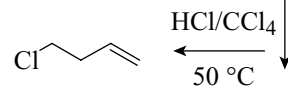
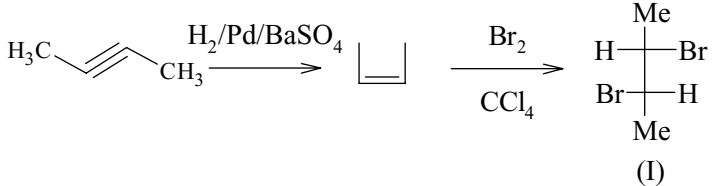
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a	a	b	b	c	d	a	c	d	ab	a	c	b	b	b
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
c	c	b	d	a	a	d	d	d	a	a	c	d	c	a
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
c	b	d	a	c	d	a	a	c	a	b	b	b	c	c
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
b	b	c	b	d	c	b	c	b	c	d	b	acd	acd	ab
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
b	d	d	d	c	d	b	abd	c	b	a	ac	b	abc	d
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
abcd	abcd	abd	abc	bc	a	a	ac	ab	abd	bc	bc	abcd	ad	abc
91	92	93	94	95	96	97	98	99	100					
ab	abc	c	a	b	d	bd	abc	abcd	b					

LEVEL 2

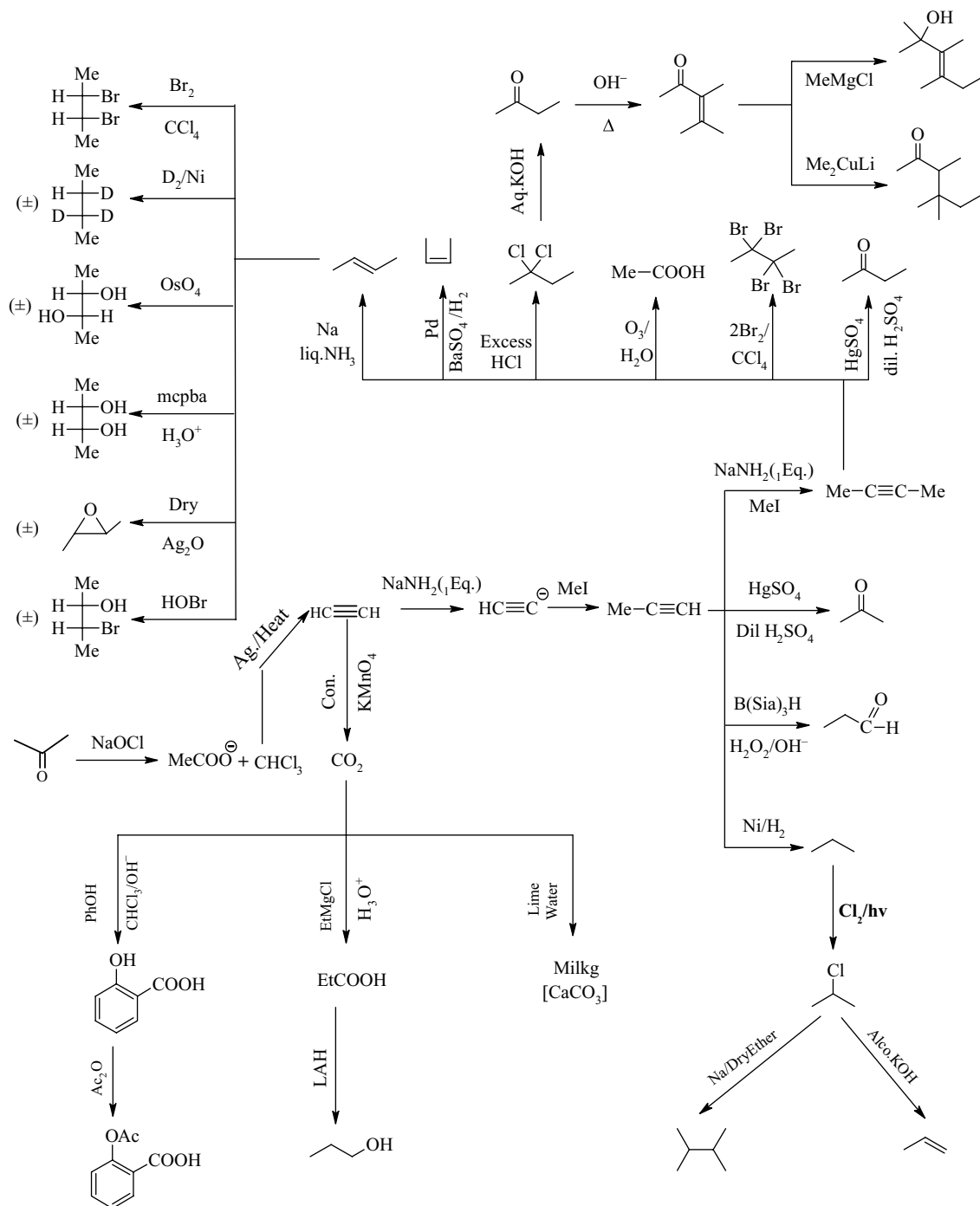
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16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
abd	b	b	b	b	c	d	c	a	d	b	a	c	a	c
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
c	d	a	b	b	a	a	c	b	b	a	b	a	d	b
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
b	a	b	a	b	c	b	a	b	bc	bc	ab	c	bd	c
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
cd	ac	abcd	ac	ac	bcd	abcd	b	bc	ab	b	a	a	c	d
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
b	a	c	c	b	a	a	d	b	c	b	c	c	c	b
91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
c	c	c	c	b	d	b	a	b	a	b	a	d	c	c
106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
d	a	d	a	d	d	a	c	b	d	a	c	b	b	d
121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
c	a	b	b	c	a	a	c	b	d	b	c	d	b	a
136	137	138	139	140	141	142	143	144	145	146(a)	146(b)	146(c)	146(d)	147(a)
d	b	b	d	b	c	b	a	b	b	r	s	q	p	qrs
147(b)	147(c)	147(d)	148(a)	148(b)	148(c)	148(d)	149(a)	149(b)	149(c)	149(d)	150(a)	150(b)	150(c)	150(d)
qrs	qrs	p	r	ps	pqs	s	s	r	q	p	r	s	q	p
151(a)	151(b)	151(c)	152(a)	152(b)	152(c)	152(d)	153(a)	153(b)	153(c)	153(d)	154(a)	154(b)	154(c)	154(d)
pq	qr	s	prs	ps	pr	q	p	pq	s	pr	q	p	s	r
155(a)	155(b)	155(c)	155(d)	156(a)	156(b)	156(c)	156(d)	157(a)	157(b)	157(c)	157(d)	158(a)	158(b)	158(c)
qrt	prt	st	r	s	r	pq	q	rt	ps	qs	qt	r	s	q
158(d)	159	160	161	162	163	164	165	166	167	168	169	170		
p	7	7	3	4	10	0	6	6	3	8	5	8		

WORKBOOK EXERCISE 1

1. 
2. 
3. 
4. 
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9. 

10. 
11. 
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14. 
15. 
16. 
17. 
18. 

WORKBOOK EXERCISE 2



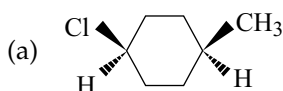
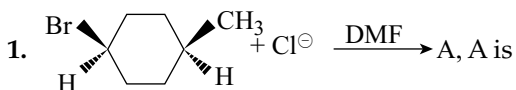
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Alkyl Halides, Alcohols and Ethers

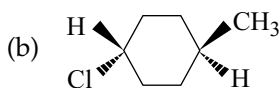
4

Question Bank

LEVEL 1

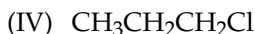
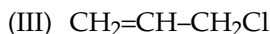
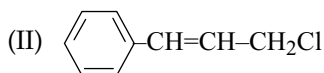
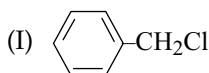


(c) Both are correct



(d) None is correct

2. Arrange the following in the order of their reactivity of S_N2 reaction



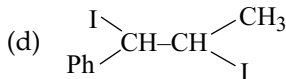
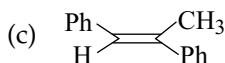
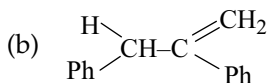
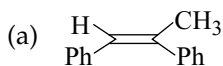
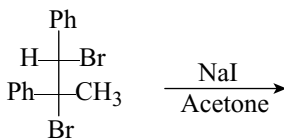
(a) (IV) > (III) > (I) > (II)

(b) (II) > (I) > (III) > (IV)

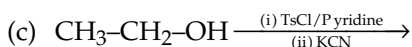
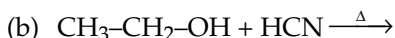
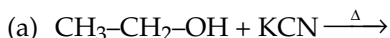
(c) (I) > (II) > (IV) > (III)

(d) (III) > (II) > (I) > (IV)

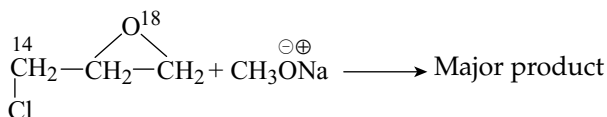
3. Identify the product of the following reaction

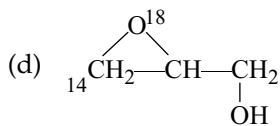
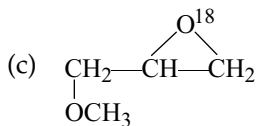
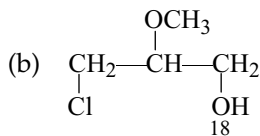
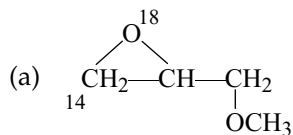


4. The conversion of ethanol to propanenitrite is best done by

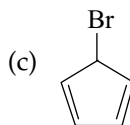
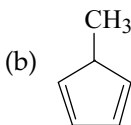
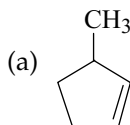
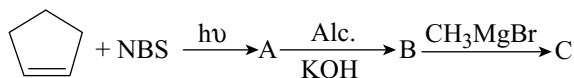


5. Major product of the following reaction is



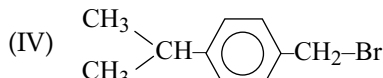
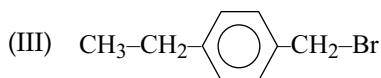
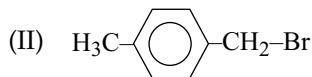
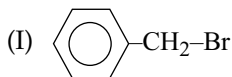


6. The final product of the following reaction is



(d) none of these

7. Arrange the following compounds in order of decreasing rate of hydrolysis for S_N1 reaction



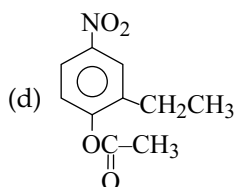
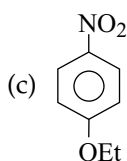
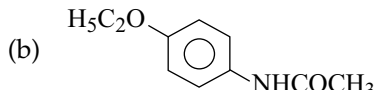
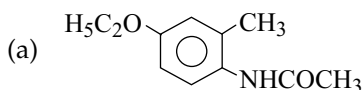
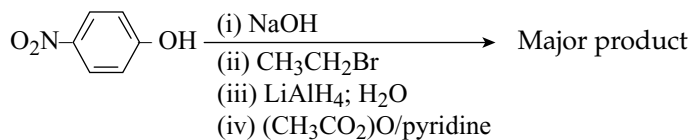
(a) (III) > (IV) > (II) > (I)

(b) (IV) > (III) > (II) > (I)

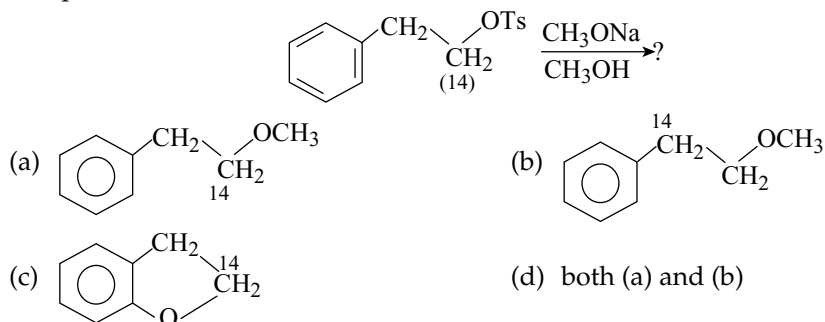
(c) (II) > (III) > (IV) > (I)

(d) (I) > (II) > (III) > (IV)

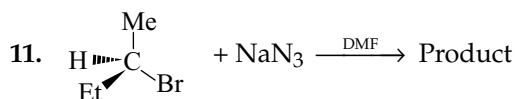
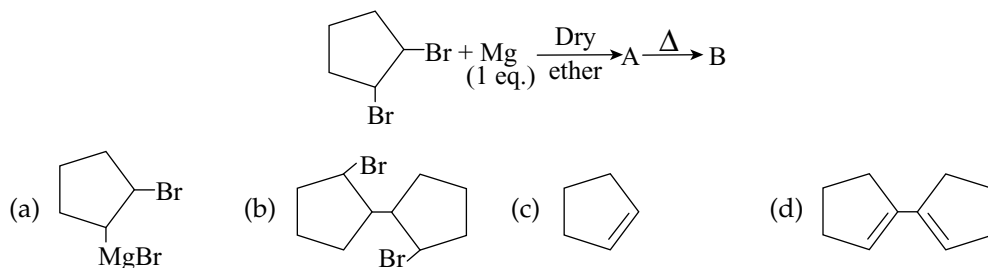
8. For the given reaction major product will be



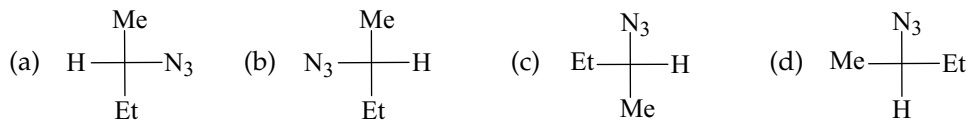
9. The product/s formed is/are



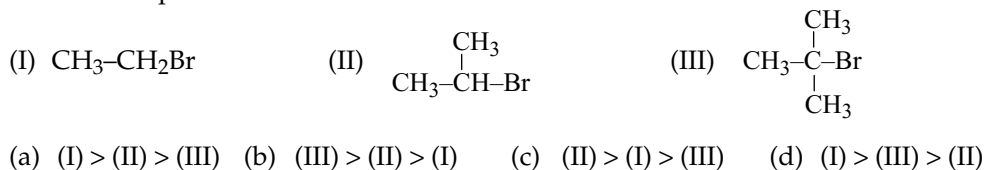
10. In the following reaction, compound (b) is



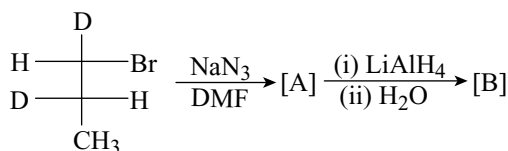
The correct representation of the product is

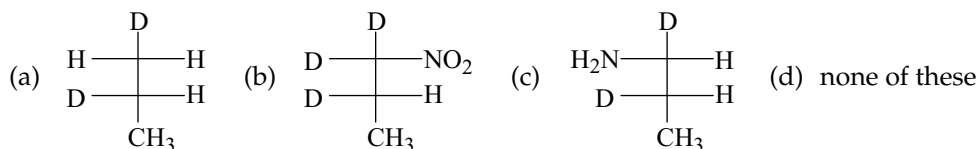


12. Arrange the following in the order of their reactivity with alcoholic AgCN to yield the substitution product



13. The final product is

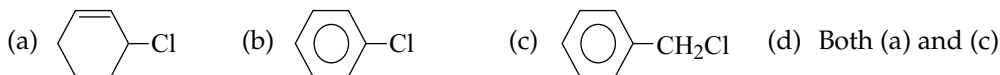




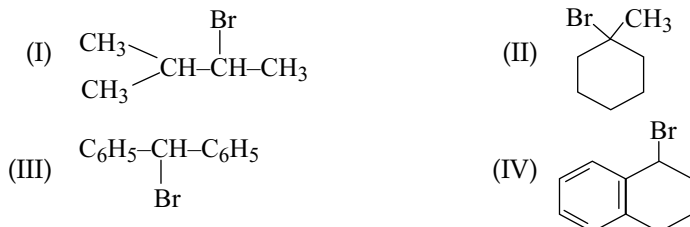
14. An aromatic compound (A), $C_7H_6Cl_2$, gives $AgCl$ on boiling with alcoholic $AgNO_3$ solution and yields C_7H_7OCl on treatment with sodium hydroxide. (a) on oxidation gives a monochlorobenzoic acid which affords only one mononitroderivative. The compound (a) is



15. Which will give white ppt. with $AgNO_3 + NH_4OH$?

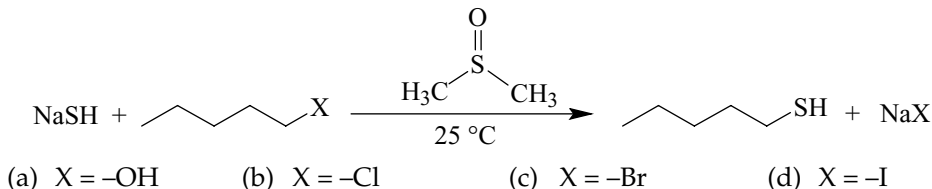


16. Consider the S_N1 solvolysis of the following halides in aqueous formic acid



Which one of the following is correct sequence of the halide given above in the decreasing order of their reactivity?

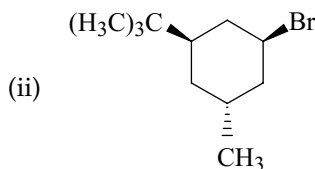
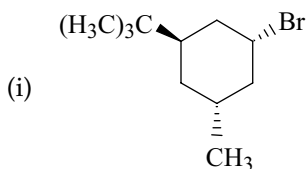
- (a) (III) > (IV) > (II) > (I) (b) (II) > (IV) > (I) > (III)
 (c) (I) > (II) > (III) > (IV) (d) (III) > (I) > (II) > (IV)
17. For which leaving group X would you predict the S_N2 reaction shown below to take place most rapidly?



18. Which of the following order is **correct** for dipole moment?

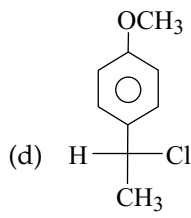
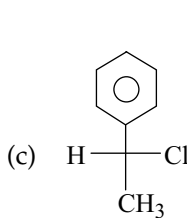
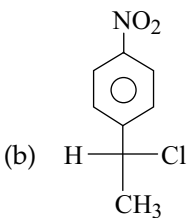
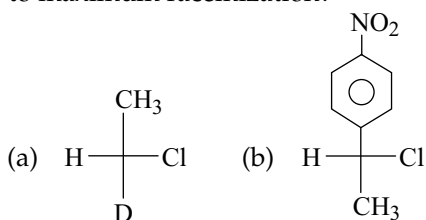


19. Choose the correct statement(s) concerning reactions of the two stereoisomers of 3-tert-butyl-5-methylbromocyclohexane.

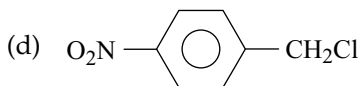
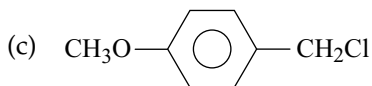
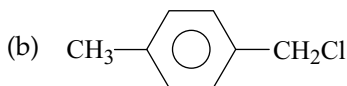
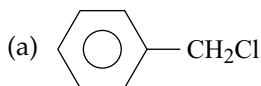


- (i) 1 will undergo S_N1 reactions faster than 2
 (ii) 1 will undergo $E1$ reactions faster than 2
 (iii) 1 and 2 undergo S_N1 reactions at some rate
 (a) i (b) ii (c) iii (d) i & ii

20. Under identical conditions, solvolysis of which of the following substrates would lead to maximum racemization?

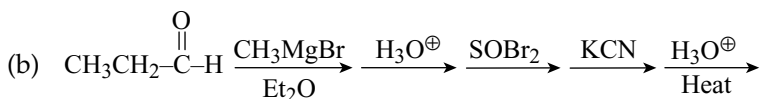
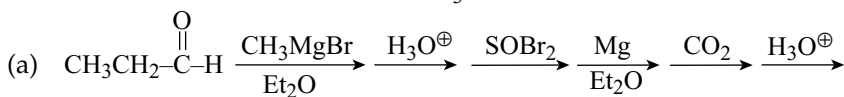
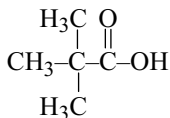


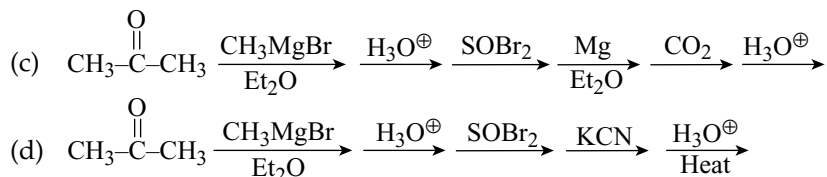
21. Consider the following chlorides



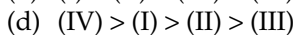
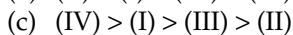
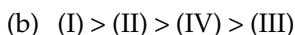
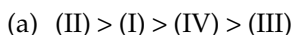
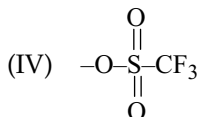
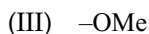
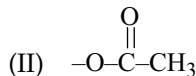
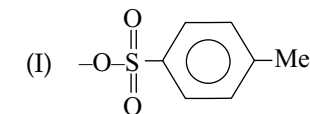
The order of reactivity of (a), (b), (c) and (d) towards hydrolysis by S_N1 mechanism is

- (a) (a) < (b) < (c) < (d) (b) (d) < (c) < (b) < (a)
 (c) (d) < (a) < (b) < (c) (d) (c) < (b) < (a) < (d)
22. Which of the following would be the best synthesis of the acid shown below?

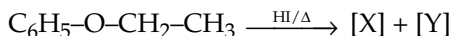




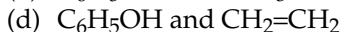
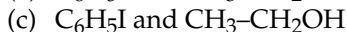
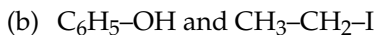
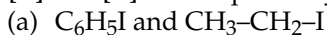
23. Rank the groups in order of decreasing leaving group ability



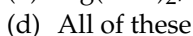
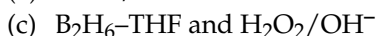
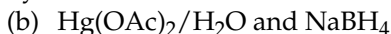
24. In the given reaction



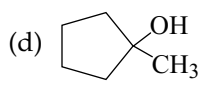
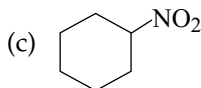
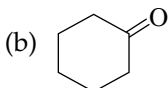
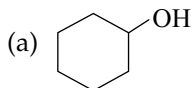
[X] and [Y] will respectively be



25. 1-Propanol can be prepared from propene by



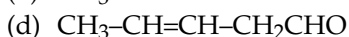
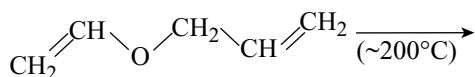
26. $\xrightarrow{(\text{aq.})\text{AgNO}_3}$; Major product will be



27. The decreasing order of reactivity of methyl alcohol (I), isopropyl alcohol (II), tertiary butyl alcohol (III) and ethyl alcohol (IV) for esterification will be



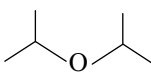
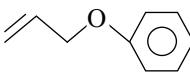
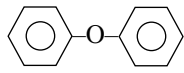
28. The product of the reaction is



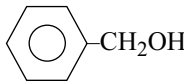
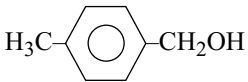
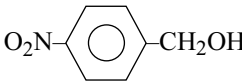
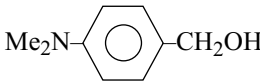
29. Which of the following is correctly matched?

- (a) $\text{CH}_3\text{-CH}_2\text{-O-C(CH}_3)_3 \xrightarrow{\text{HOH/H}^\oplus} \text{S}_{\text{N}}1$ (b) $\text{CH}_3\text{-C(CH}_3)_2\text{-CH}_2\text{-O} \xrightarrow{\text{HOH/H}^\oplus} \text{S}_{\text{N}}1$
- (c) $\text{CH}_3\text{-CH-CH}_2\text{-O} \xrightarrow{\text{CH}_3\text{O}^\ominus/\text{CH}_3\text{OH}} \text{S}_{\text{N}}2$ (d) All of these

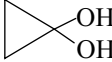
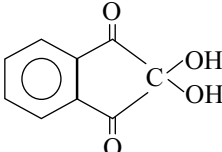
30. Which of the following ether does not give peroxide formation?

- (a)  (b) 
- (c)  (d) 

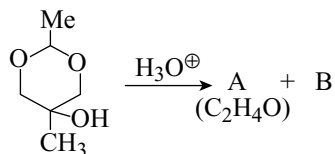
31. Correct order of the reactivity of the given alcohols towards the substitution with (NaBr + H₂SO₄)

- (I)  (II) 
- (III)  (IV) 
- (a) (I) > (II) > (III) > (IV) (b) (IV) > (II) > (I) > (III)
- (c) (III) > (I) > (II) > (IV) (d) (IV) > (III) > (II) > (I)

32. Which of the following is a stable hydrate?

- (a)  (b) $\text{Cl}_3\text{C-CH(OH)}_2$
- (c)  (d) all of these

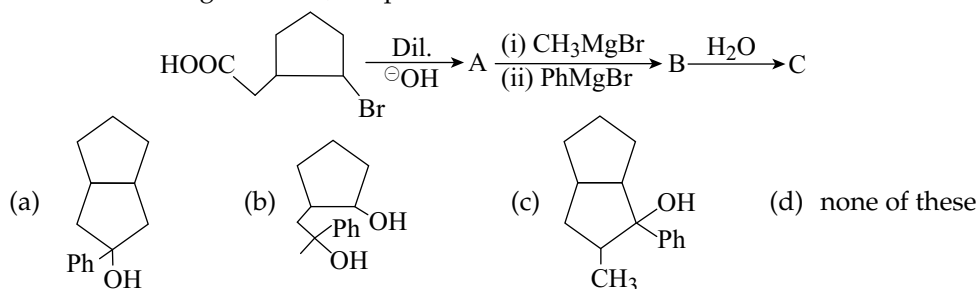
33. The products of the given reactions



(B) $\xrightarrow{\text{HIO}_4}$ products

- (a) CH₃COOH (b) HCHO (c) HCOOH (d) both (a) and (b)

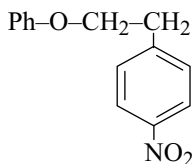
34. For the following reactions, the product C is



35. Compound [X], whose MF is $\text{C}_4\text{H}_8\text{O}$ gives positive haloform test but gives no 2, 4-DNP derivative is



36. When HI is added in excess to the unsymmetrical ether



it results in

- (a) PhCH_2OH

(c) $\text{HOCH}_2\text{CH}_2\text{Ph}-\text{NO}_2$

(b) PhI

(d) $\text{ICH}_2\text{CH}_2\text{PhNO}_2$

37. $\text{CH}_3-\underset{\text{OH}}{\underset{|}{\text{CH}}}-\underset{\text{NH}_2}{\underset{|}{\text{CH}}}-\text{COOR} \xrightarrow[\text{(ii) H}_3\text{O}]{\text{(i) 1mol HIO}_4} \text{Products}$

The product mixture will contain

- (a) $\text{CH}_3\text{CHO} + \text{OHC}-\text{NH}_2 + \text{CO}_2$

(c) $\text{CH}_3\text{CHO} + \text{OHC}-\text{COOR} + \text{NH}_3$

(b) CH_3COOH only

(d) $\text{CH}_3\text{COOH} + \text{NH}_2\text{COOR} + \text{HCOOH}$

38. Williamson's synthesis is applied to prepare ether. Identify the ether from the following which cannot be prepared by it?

- (a) $(\text{CH}_3)_3\text{C}-\text{O}-\text{CH}_2\text{CH}_2\text{CH}_3$

(c) $\text{CH}_2=\text{CH}-\text{CH}_2-\text{O}-\text{CH}_3$

(b) PhOCH_3

(d) $\text{CH}_3-\text{CH}=\text{CH}-\text{O}-\text{CH}=\text{CH}_2$

39. Which of the following predictions regarding the solubility of each pair of compounds in water are correct?

(I) CH_2Cl_2 is more soluble than CH_3OH in water

(II) $\text{H}_3\text{C}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$ is more soluble than $\text{H}_3\text{C}-\overset{\text{CH}_2}{\parallel}{\text{C}}-\text{CH}_3$ in water

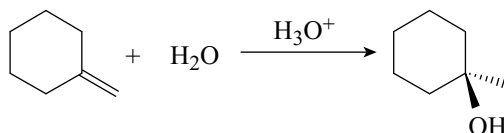
(III) $\text{CH}_3\text{CH}_2\text{Cl}$ is more soluble than NaCl in water

(IV) $\text{CH}_3\text{CH}_2\text{CH}_2\text{SH}$ is more soluble than $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ in water

(V)  is more soluble than  in water

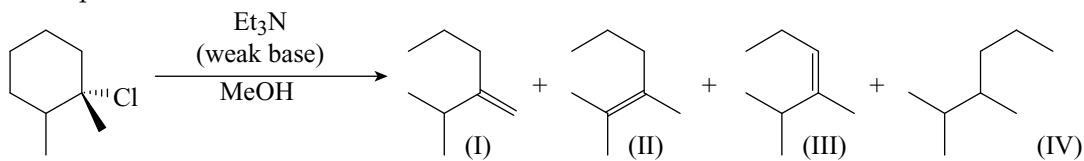
- (a) (III), (IV), (V) (b) (I), (II), (IV) (c) (II), (IV), (V) (d) (II), (V)

40. What is the role of H_3O^+ in this reaction?



- (a) Nucleophile (b) Catalyst (c) Leaving group (d) Base

41. Arrange the yields (high to low) for the products of the following reaction and judge the possible mechanism.



- (a) (II) > (IV) > (III) > (I) via E_1 mechanism
 (b) (II) > (IV) > (III) > (I) via E_2 mechanism
 (c) (II) > (III) > (IV) > (I) via E_1 mechanism
 (d) (IV) > (I) > (II) > (III) via E_2 mechanism

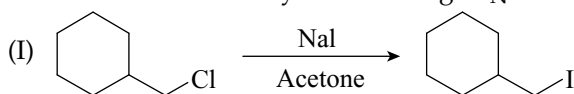
42. What reagent can be used to convert 1-isopropylcyclopentene to 1-isopropylcyclopentanol?

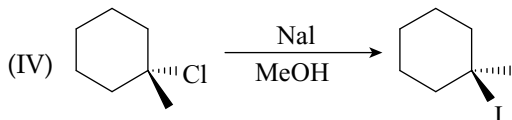
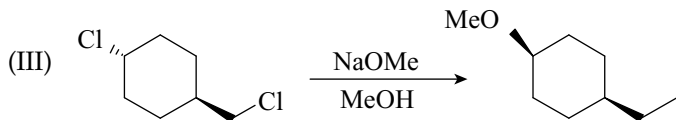
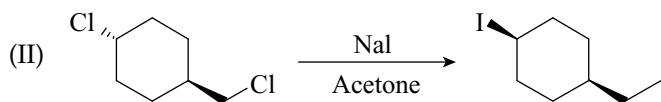
- (a) HBr (b) H_2O , diluted H_2SO_4
 (c) H_2/Pd (d) H_2O , neutral

43. Which statements are true for $\text{S}_{\text{N}}2$ reaction of alkyl halides?

- (I) Both of the alkyl halide and nucleophile are involved in the transition state.
 (II) Reaction proceeds with inversion of configuration at the substitution centre.
 (III) Reaction proceeds with retention of configuration at the substitution centre.
 (IV) The order of reactivity is $3^\circ > 2^\circ > 1^\circ$.
 (V) The nucleophile must have an unshared electron pair and bear a negative charge.
 (VI) The greater the nucleophilicity of the nucleophile, the greater the rate of reaction.
 (a) (I), (II), (V), (VI) (b) (I), (III), (V), (VI)
 (c) (I), (II), (IV), (V) (d) (I), (II), (VI)

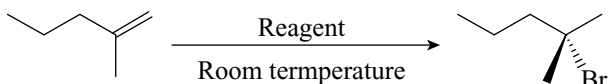
44. What reactions will likely occur through $\text{S}_{\text{N}}1$ mechanism?





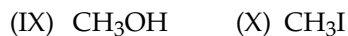
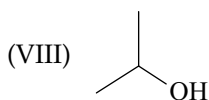
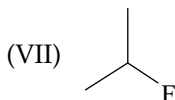
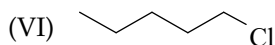
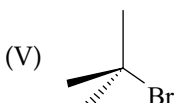
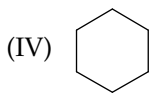
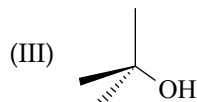
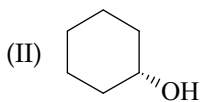
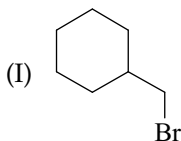
- (a) (II), (III), (IV) (b) (I), (IV) (c) (III), (IV) (d) (II), (III)

45. What is the reagent(s) needed for the following reaction?



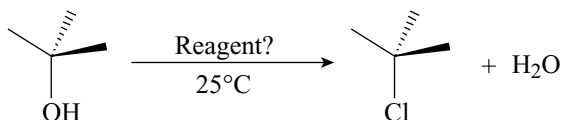
- (a) HCl (b) NaBr (c) HBr (d) Br₂

46. Which are secondary alkyl halides?



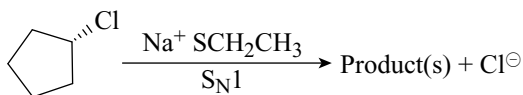
- (a) (I), (VII) (b) (VII) (c) (V), (X) (d) (I), (V), (VI), (X)

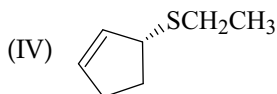
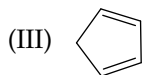
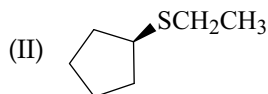
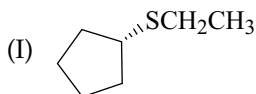
47. What reagent can be used for the following transformation?



- (a) HCl (b) Na₂Cr₂O₇ (c) Cl₂ (d) NaCl

48. What could be the product(s) for the following S_N1 reaction?





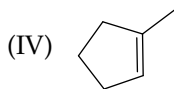
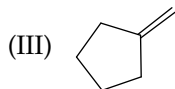
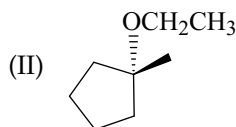
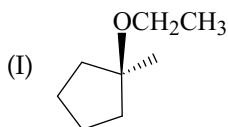
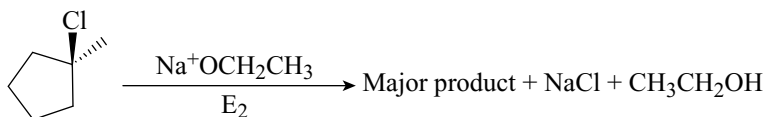
(a) (I), (II)

(b) (I), (III)

(c) (III), (IV)

(d) (II), (III)

49. What could be the major product for the following E_2 reaction?



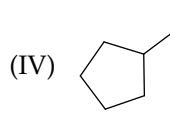
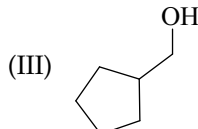
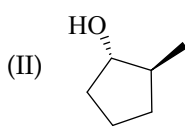
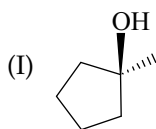
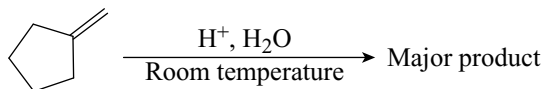
(a) (I)

(b) (II)

(c) (III)

(d) (IV)

50. What could be the major product for the following reaction?



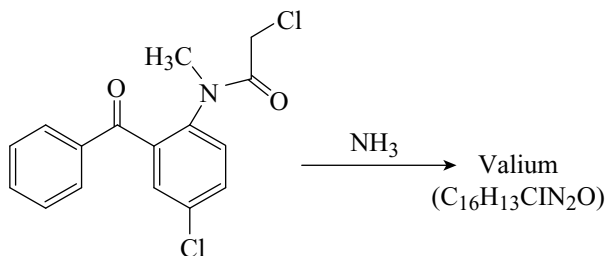
(a) (I)

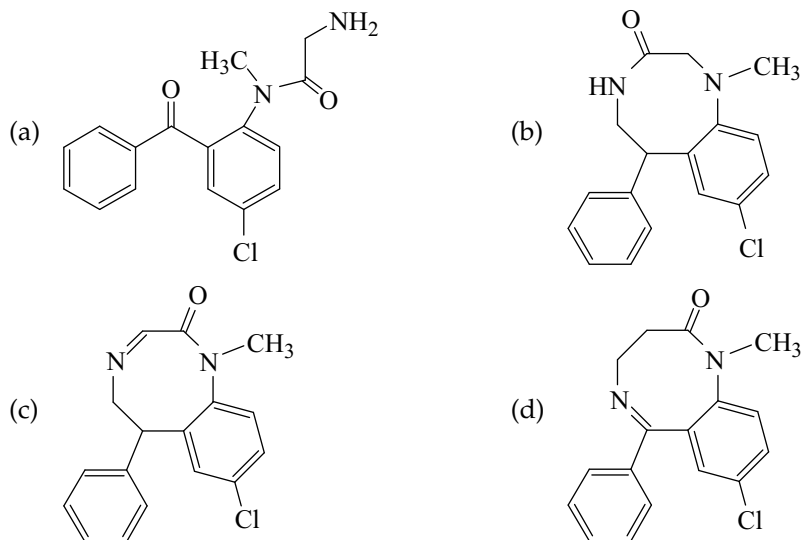
(b) (II)

(c) (III)

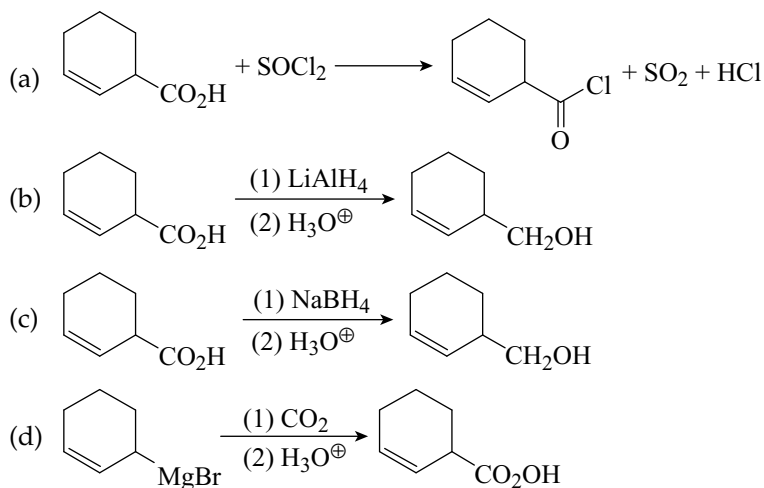
(d) (IV)

51. The last step in the synthesis of Valium[®] is treatment of the following aromatic compound with NH_3 . Choose the correct structure of Valium[®].

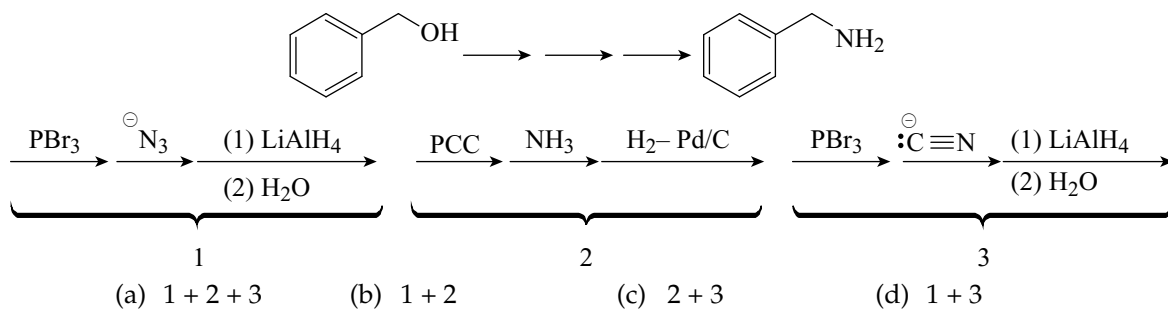




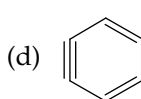
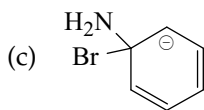
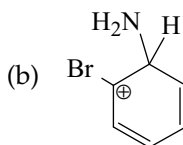
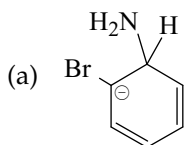
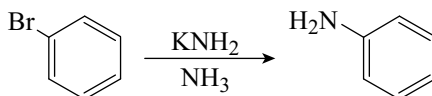
52. Choose the reaction that is not correctly shown.



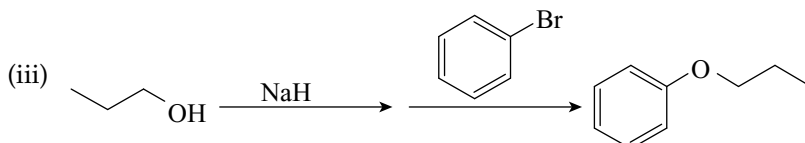
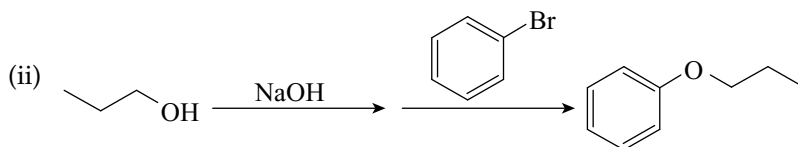
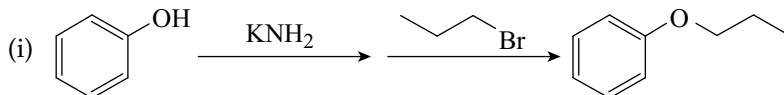
53. Choose those reactions that would give the following amine.



54. Bromobenzene reacts with potassium amide to form aniline. Identify the most important intermediate.

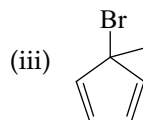
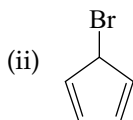
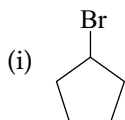


55. Here are three possible syntheses of phenylpropylether. Which one would work?



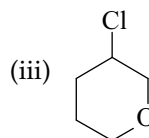
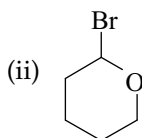
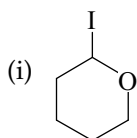
- (a) (i) only (b) (iii) only (c) (ii) or (iii) (d) (i) or (iii)

56. Identify correct reactivity order for $\text{S}_{\text{N}}1$ reaction



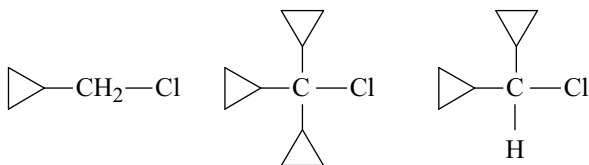
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

57. Identify correct reactivity order for $\text{S}_{\text{N}}1$ reaction



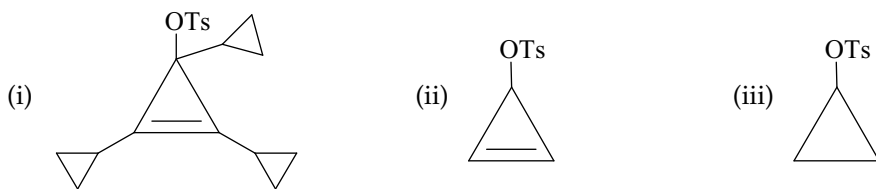
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

58. Identify correct reactivity order for S_N1 reaction.



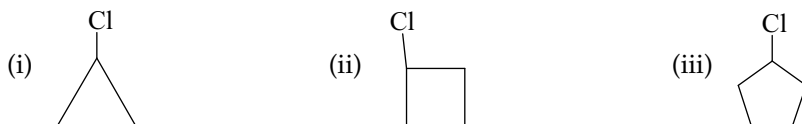
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

59. Identify correct reactivity order for S_N1 reaction



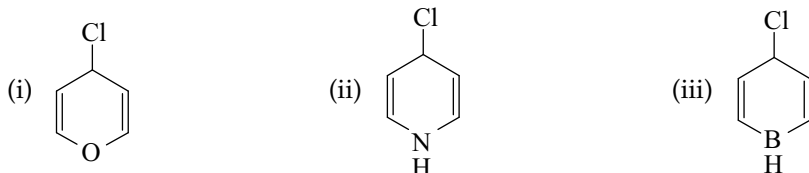
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

60. Identify correct reactivity order for S_N1 reaction



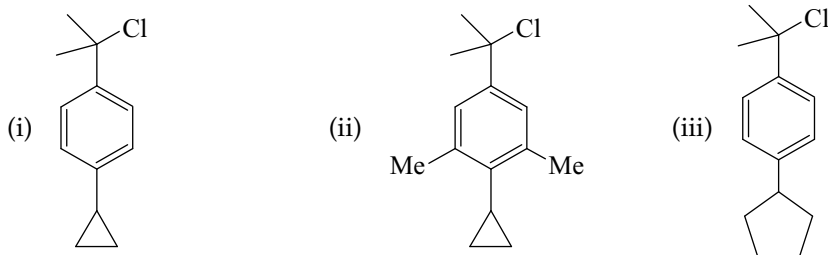
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (ii) > (i)

61. Identify correct reactivity order for S_N1 reaction



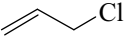
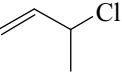
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (ii) > (i) > (iii)

62. Identify correct reactivity order for S_N1 reaction

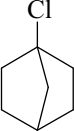
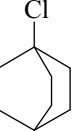
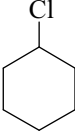


- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

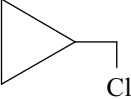
63. Identify correct reactivity order for S_N1 reaction

- (i)  (ii)  (iii) 
 (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (ii) > (i) > (iii)

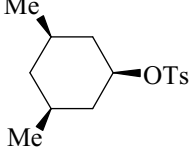
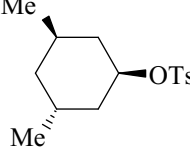
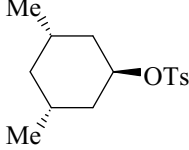
64. Identify correct reactivity order for S_N1 reaction

- (i)  (ii)  (iii) 
 (a) (i) > (ii) > (iii) (b) (iii) > (ii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

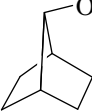
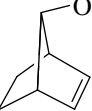
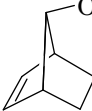
65. Identify correct reactivity order for S_N1 reaction

- (i)  (ii)  (iii) 
 (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

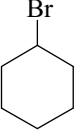
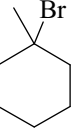
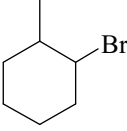
66. Identify correct reactivity order for S_N1 reaction

- (i)  (ii)  (iii) 
 (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

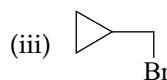
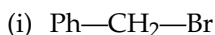
67. Identify correct reactivity order for S_N1 reaction

- (i)  (ii)  (iii) 
 (a) (i) > (ii) > (iii) (b) (iii) > (ii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

68. Identify correct reactivity order for S_N1 reaction

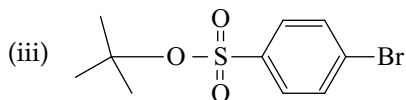
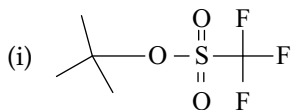
- (i)  (ii)  (iii) 
 (a) (i) > (ii) > (iii) (b) (ii) > (i) > (iii) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

69. Identify correct reactivity order for S_N1 reaction



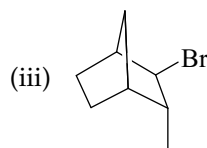
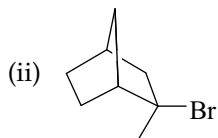
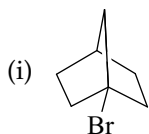
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

70. Identify correct reactivity order for S_N1 reaction



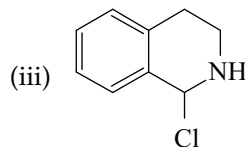
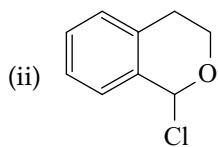
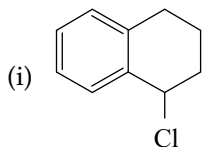
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

71. Identify correct reactivity order for S_N1 reaction



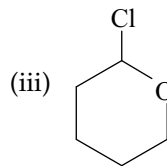
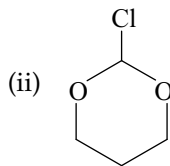
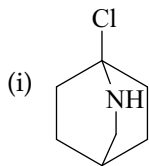
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

72. Identify correct reactivity order for S_N1 reaction



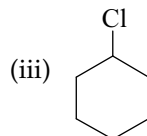
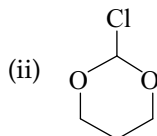
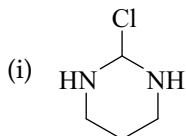
- (a) (i) > (ii) > (iii) (b) (iii) > (ii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

73. Identify correct reactivity order for S_N1 reaction



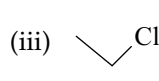
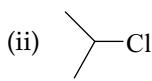
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

74. Identify correct reactivity order for S_N1 reaction



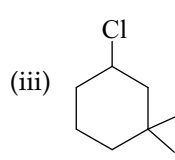
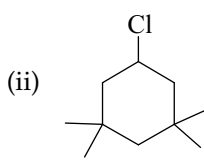
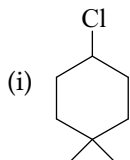
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

75. Identify correct reactivity order for S_N2 reaction



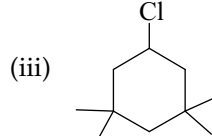
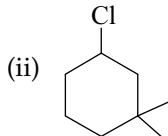
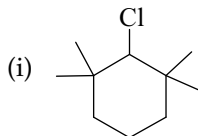
- (a) (iii) > (ii) > (i) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

76. Identify correct reactivity order for S_N2 reaction



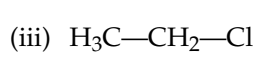
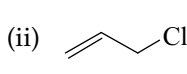
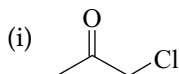
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

77. Identify correct reactivity order for S_N2 reaction



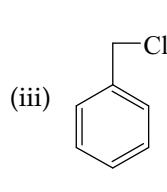
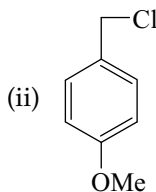
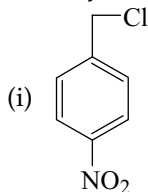
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

78. Identify correct reactivity order for S_N2 reaction



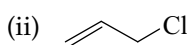
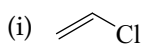
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

79. Identify correct reactivity order for S_N2 reaction



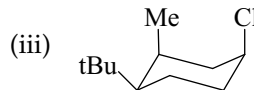
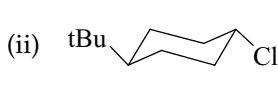
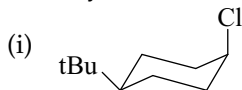
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

80. Identify correct reactivity order for S_N2 reaction



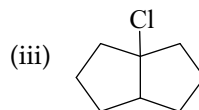
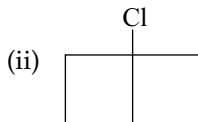
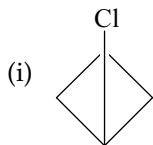
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (ii) > (i)

81. Identify correct reactivity order for S_N2 reaction



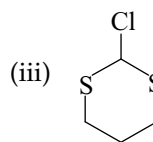
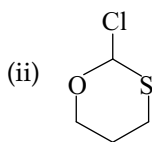
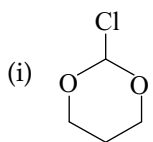
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

82. Identify correct reactivity order for S_N2 reaction



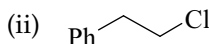
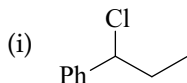
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (ii) > (i)

83. Identify correct reactivity order for S_N2 reaction



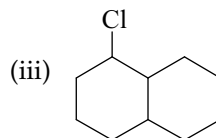
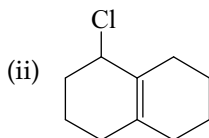
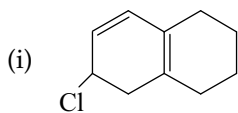
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

84. Identify correct reactivity order for E_2 reaction with alcoholic KOH



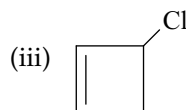
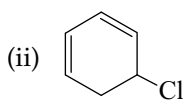
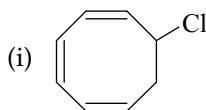
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

85. Identify correct reactivity order for E_2 reaction with alcoholic KOH



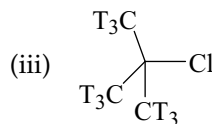
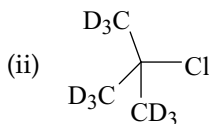
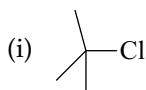
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

86. Identify correct reactivity order for E_2 reaction with alcoholic KOH



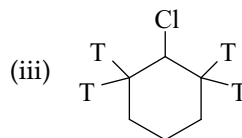
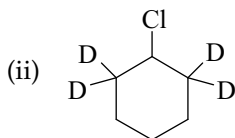
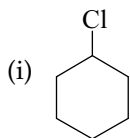
- (a) (i) > (ii) > (iii) (b) (ii) > (i) > (iii) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

87. Identify correct reactivity order for E_2 reaction with alcoholic KOH



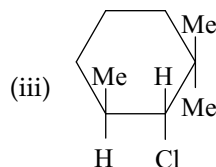
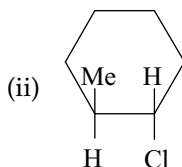
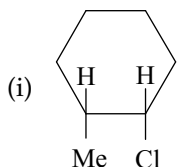
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

88. Identify correct reactivity order for E_2 reaction with alcoholic KOH



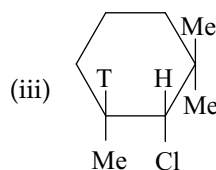
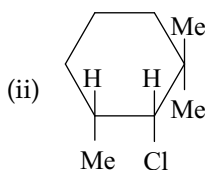
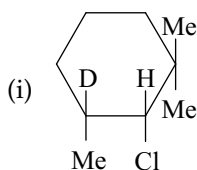
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

89. Identify correct reactivity order for E_2 reaction with alcoholic KOH



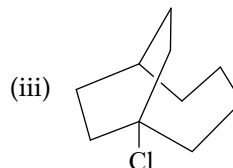
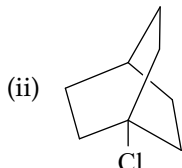
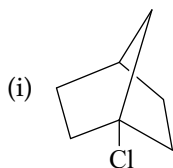
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

90. Identify correct reactivity order for E_2 reaction with alcoholic KOH



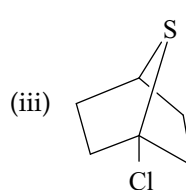
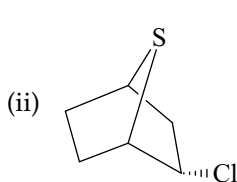
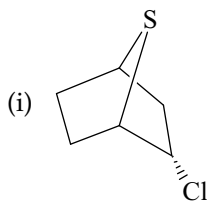
- (a) (i) > (ii) > (iii) (b) (ii) > (i) > (iii) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

91. Identify correct reactivity order for E_2 reaction with alcoholic KOH



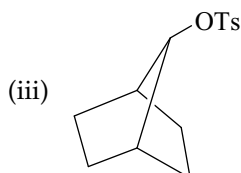
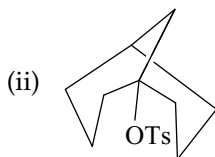
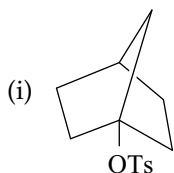
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (ii) > (i)

92. Identify rate of sololysis or hydrolysis



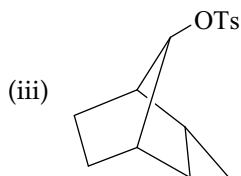
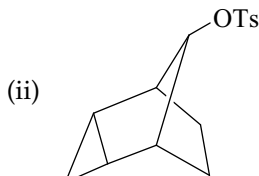
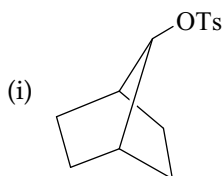
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

93. Identify rate of sololysis or hydrolysis



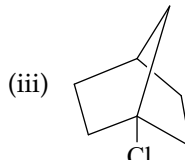
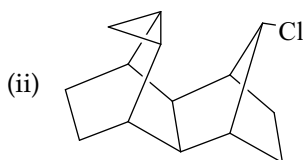
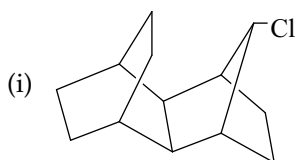
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (ii) > (i)

94. Identify rate of sololysis or hydrolysis



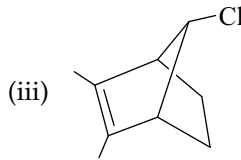
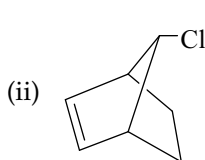
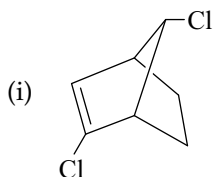
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

95. Identify rate of sololysis or hydrolysis



- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (ii) > (i) > (iii) (d) (iii) > (i) > (ii)

96. Identify rate of sololysis or hydrolysis

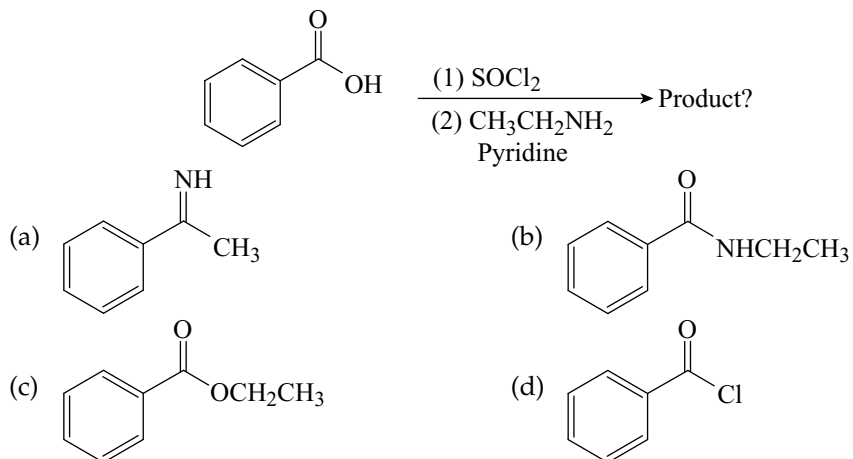


- (a) (iii) > (ii) > (i) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

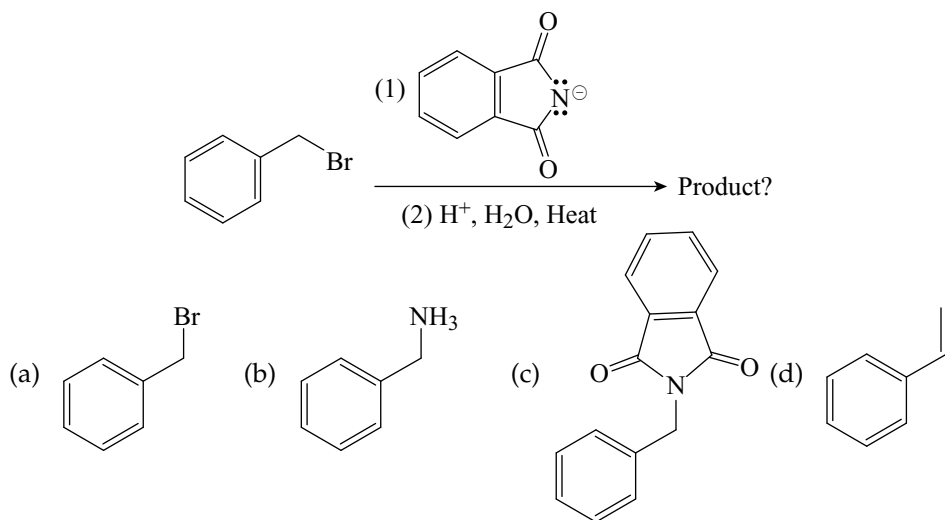
97. Which statements are true for S_N1 reaction of alkyl halides?

- (I) Both of the alkyl halide and nucleophile are involved in the transition state.
 (II) Reaction proceeds with inversion of configuration at the substitution centre.
 (III) Reaction proceeds via the formation of carbocation intermediate.
 (IV) The order of reactivity is $3^\circ > 2^\circ > 1^\circ$.
 (V) The nucleophile must have an unshared electron pair and bear a negative charge.
 (VI) Protic solvents favour S_N1 reaction.
- (a) (III), (IV), (VI) (b) (II), (IV), (V) (c) (I), (II), (VI) (d) (I), (II), (V), (VI)

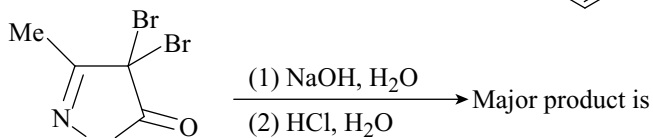
98. What could be the product for the following reaction?



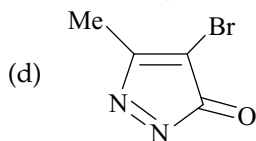
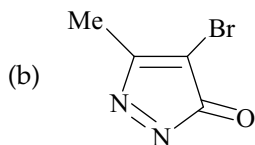
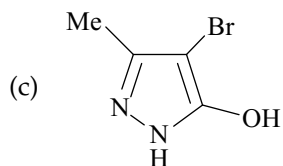
99. What could be the product for the following reaction?

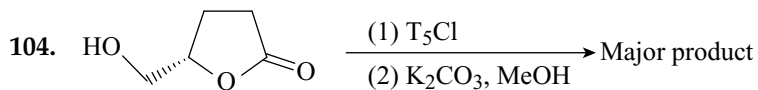
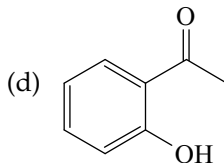
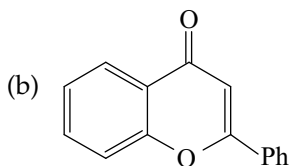
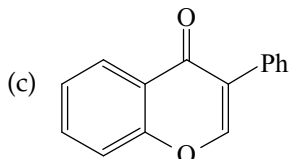
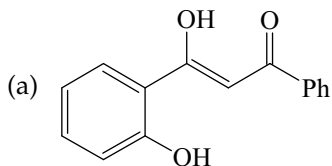
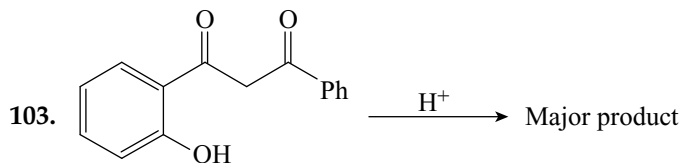
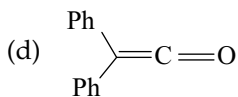
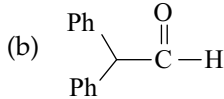
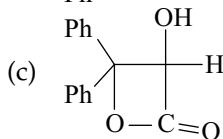
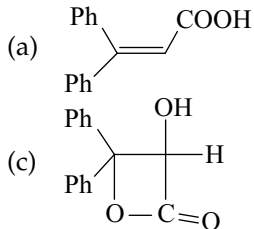
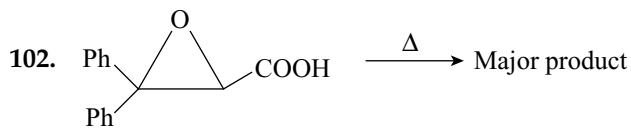
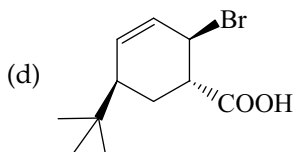
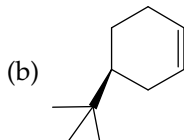
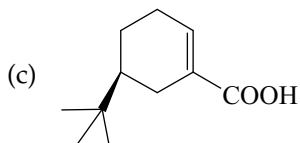
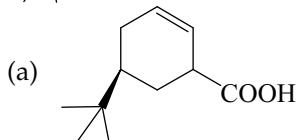
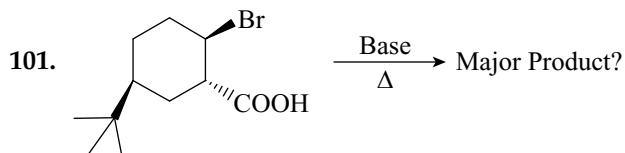


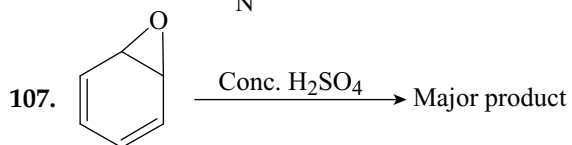
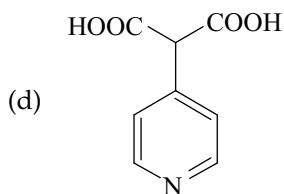
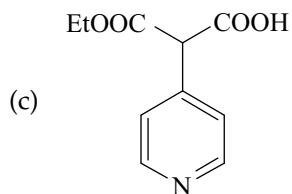
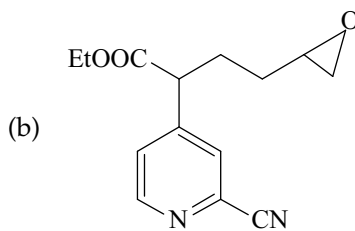
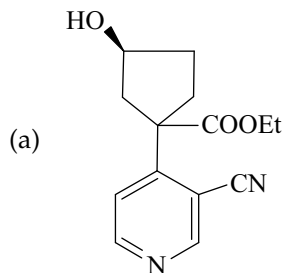
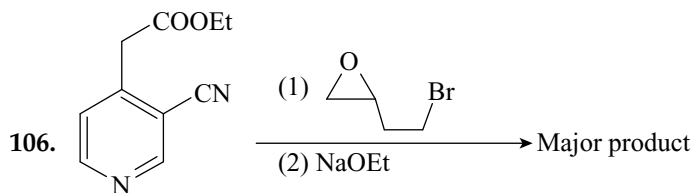
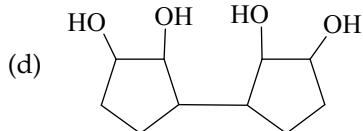
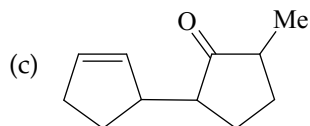
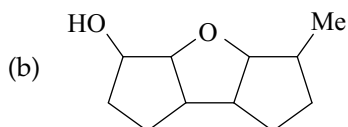
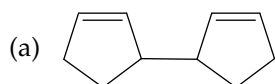
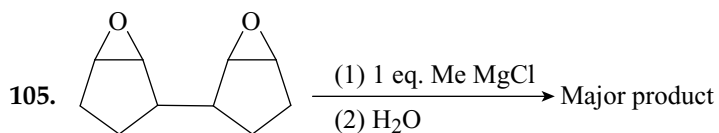
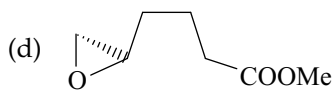
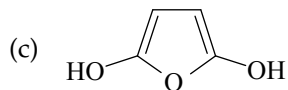
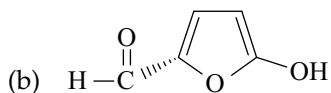
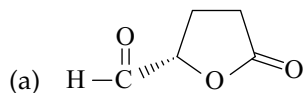
100.

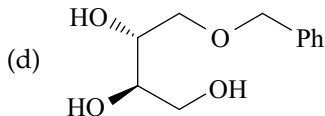
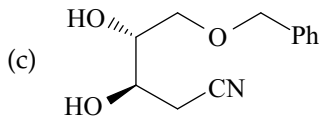
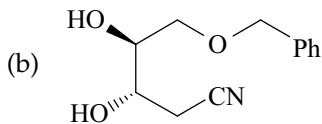
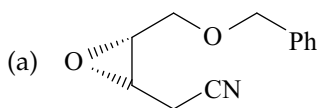
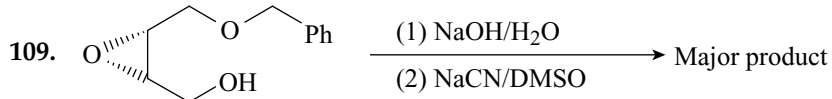
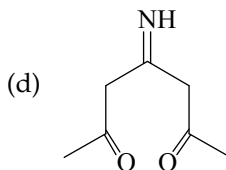
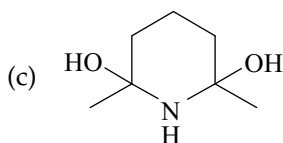
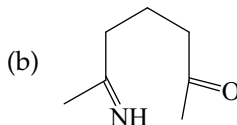
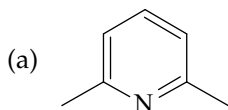
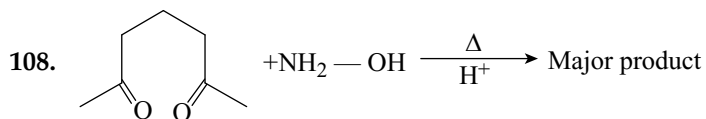
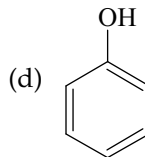
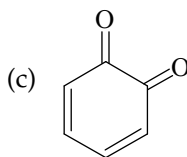
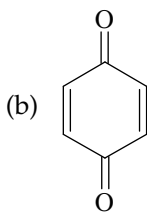
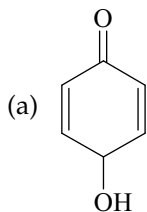


(a) $\text{Me} - \text{C} \equiv \text{C} - \text{COOH}$

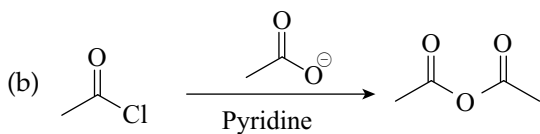
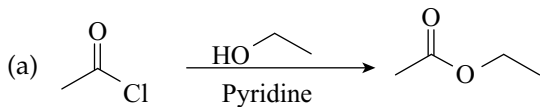


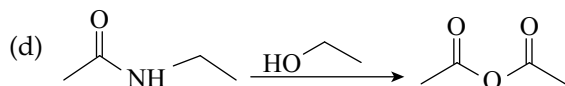
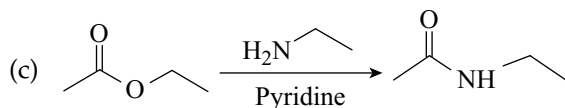




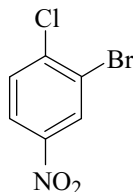


110. Four of the five reactions shown below should proceed as shown to give good yields. One of reaction has problems and will not proceed as indicated. Identify this one reaction which will not proceed as shown.

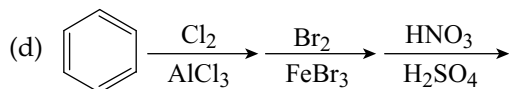
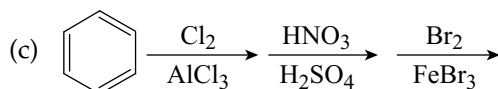
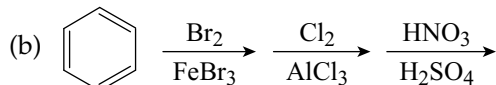
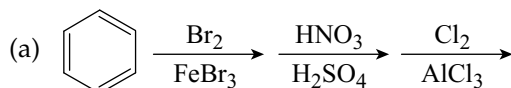




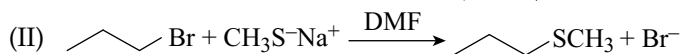
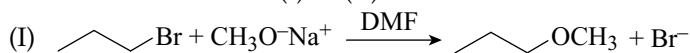
111. Which of the following synthetic schemes would be the best for the synthesis of the compound, 2-bromo-1-chloro-4-nitrobenzene?



2-bromo-1-chloro-4-nitrobenzene

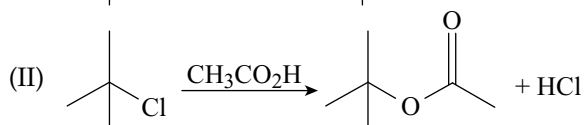
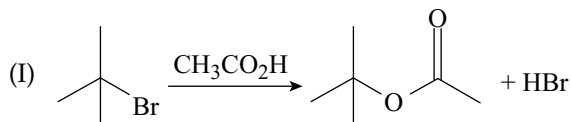


112. Which reaction is faster (I) or (II) and what is its mechanism?



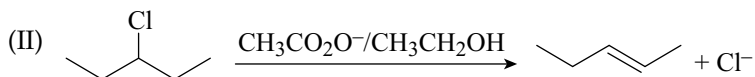
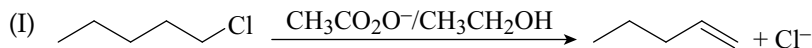
- (a) reaction (I), S_N1 (b) reaction (II), S_N2
(c) reaction (II), S_N1 (d) reaction (I), S_N2

113. Which reaction is faster (I) or (II) and what is its mechanism?



(a) reaction (II), S_N1 (b) reaction (II), S_N2 (c) reaction (I), S_N1 (d) reaction (I), S_N2

114. Which reaction is faster (I) or (II) and what is its mechanism?



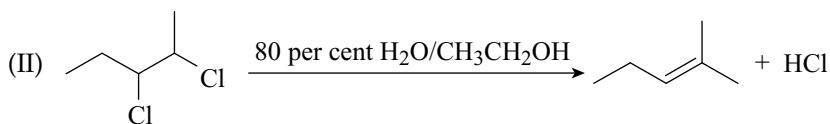
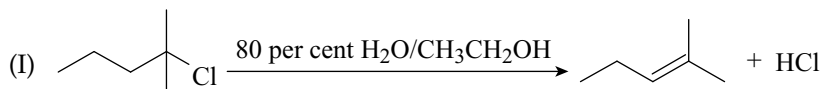
(a) reaction (II), E2

(b) reaction (I), E2

(c) reaction (I), E1

(d) reaction (II), E1

115. Which reaction is faster (I) or (II) and what is its mechanism?



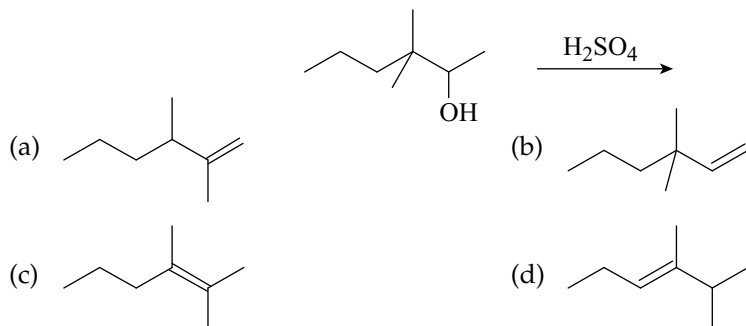
(a) reaction (II), E1

(b) reaction (I), E2

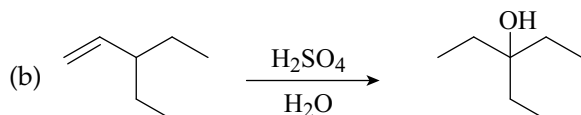
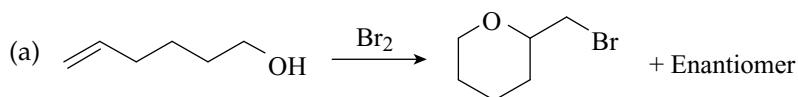
(c) reaction (II), E2

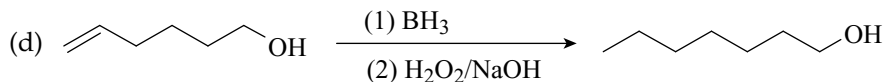
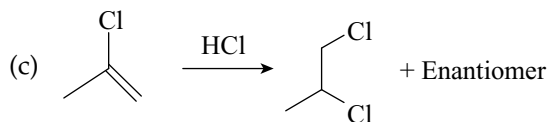
(d) reaction (I), E1

116. Predict the major product of the following reaction.

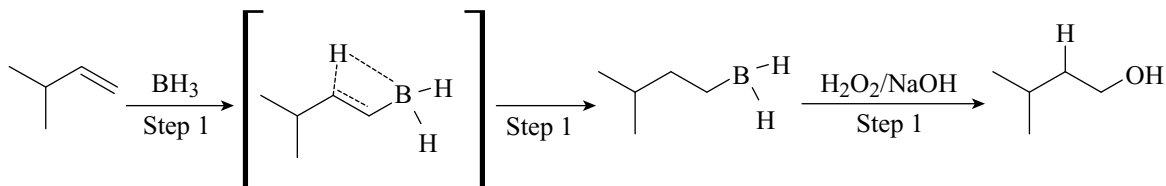


117. Which ONE of the following reactions is NOT correct?





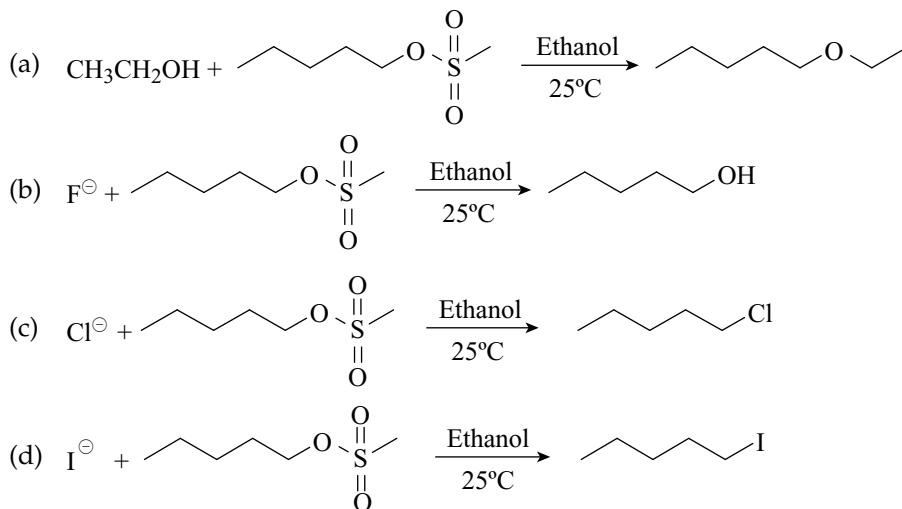
118. Consider the following reaction.



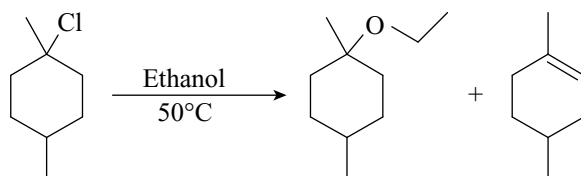
Which ONE of the following statements is INCORRECT?

- (a) In step 1, the boron atom acts as a Lewis base.
- (b) In step 2, the boron atom is oxidized by hydrogen peroxide.
- (c) The borane addition occurs with syn-stereospecificity.
- (d) In step 1, a hydride is added at carbon 2.

119. Which one of the following five S_N2 reactions will take place most rapidly?



120. Which of the following changes will increase the rate of elimination reaction in the following reaction?

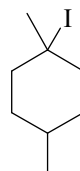


(i) using 100°C instead of 50°C

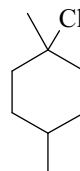
(ii) using water instead of ethanol

(iii) adding ethoxide to the ethanol

(iv) using



instead of



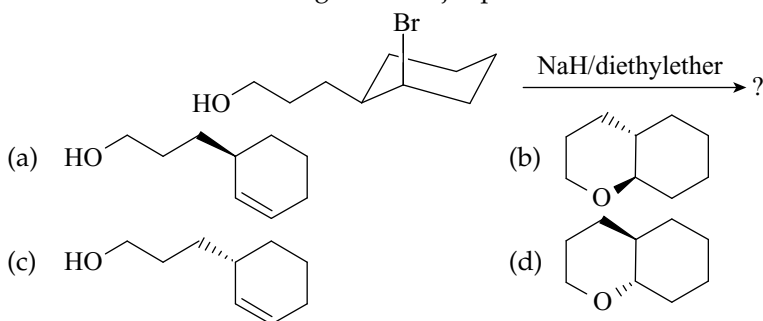
(a) (i) and (ii)

(b) (ii) and (iii)

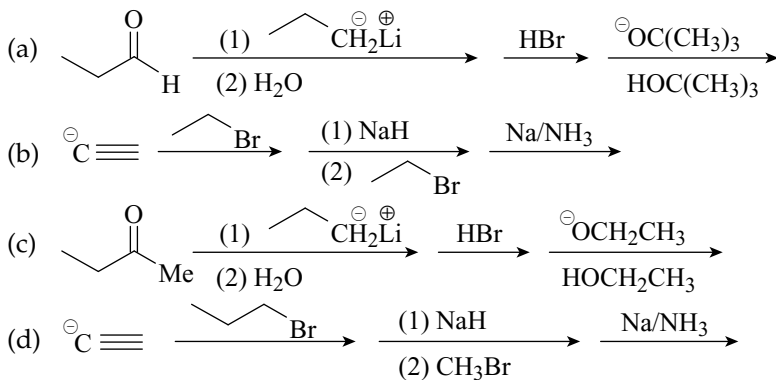
(c) (i), (iii) and (iv)

(d) (iii) and (iv)

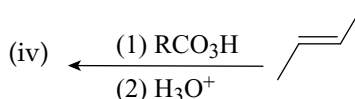
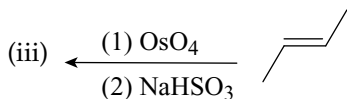
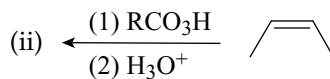
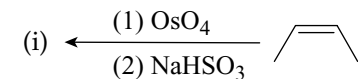
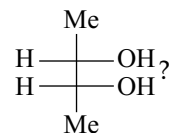
121. Which one of the following is the major product of the reaction below?



122. Which one of the sequences below is the best synthesis of (E)-3-hexene?



123. Which reactions will provide the following diol as the major product



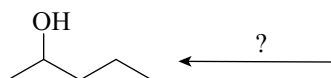
(a) (i)

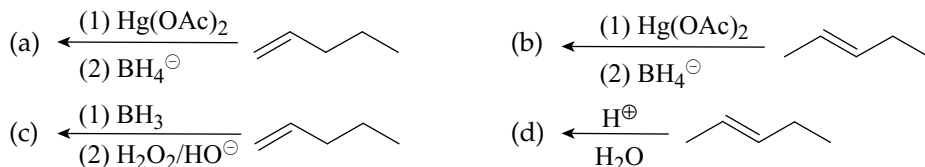
(b) (iii)

(c) (ii) and (iii)

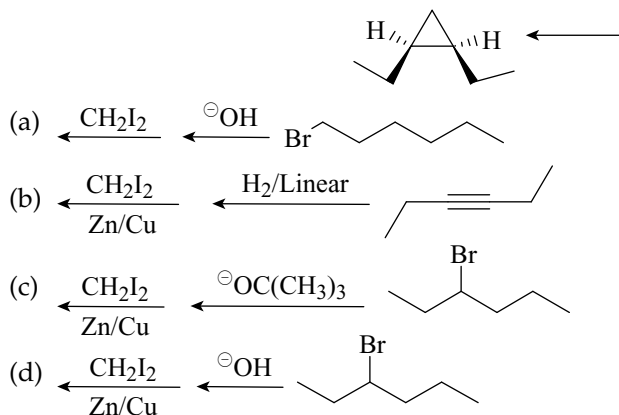
(d) (i) and (iv)

124. Which is the best synthesis of the following alcohol?

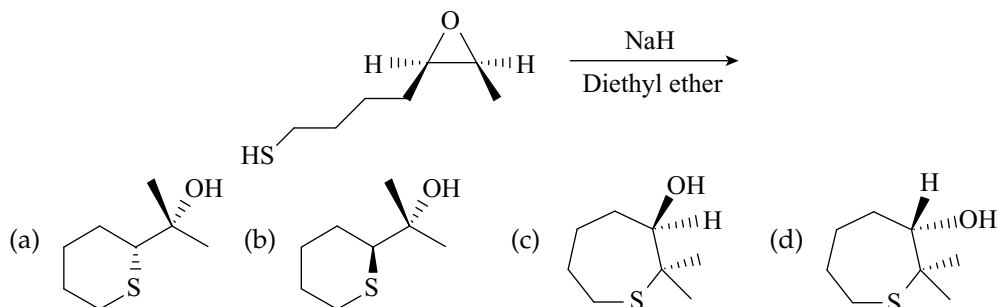




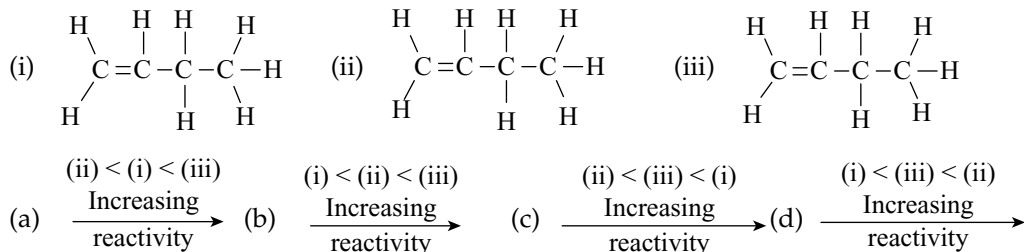
125. Which is the best synthesis of the following cyclopropane?



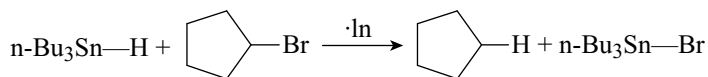
126. Which is the correct product of the following reaction?

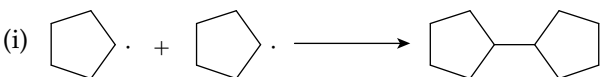
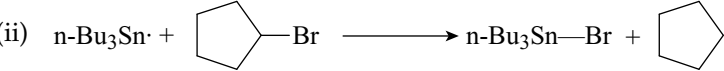


127. Arrange the indicated C—H bonds in order of increasing reactivity towards chlorination.

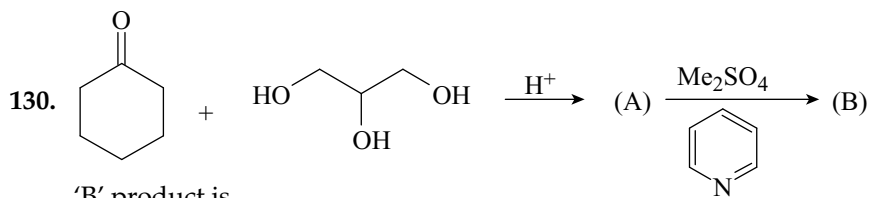
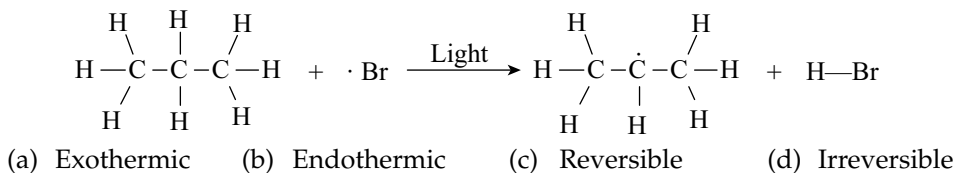


128. Choose those reaction(s) that is (are) the propagation step(s) in the following free radical chain reaction.

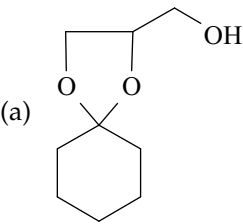
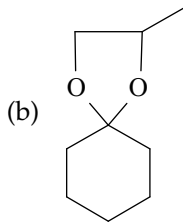
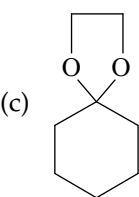
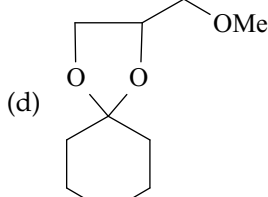


- (i)  (ii) 
 (iii) $\text{n-Bu}_3\text{Sn-H} + \cdot\text{In} \longrightarrow \text{n-Bu}_3\text{Sn}\cdot + \text{H-In}$
 (a) (i) (b) (ii) (c) (iii) (d) (i) and (ii)

129. Choose the answer that is correct for the ΔH° (kJ/mole) of the following step in the bromination of propane.



'B' product is

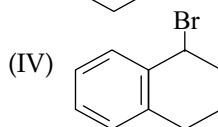
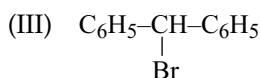
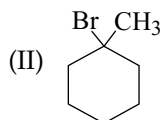
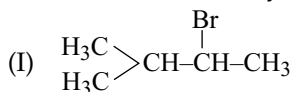
- (a)  (b) 
 (c)  (d) 

LEVEL 2

Single and Multiple-choice Type

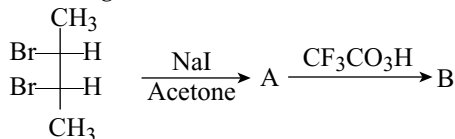
1. Which of the following shows greater reactivity towards S_N2 reaction than $\text{CH}_3\text{—CH}_2\text{Br}$ with sodium methoxide?
- (a) $\text{CH}_3\text{—O—CH}_2\text{Br}$ (b) CH_3Br
 (c) $\text{CH}_2=\text{CH—Br}$ (d) $\text{CH}_2=\text{CH—CH}_2\text{Br}$

2. Consider the S_N1 solvolysis of the following halides in aqueous formic acid?



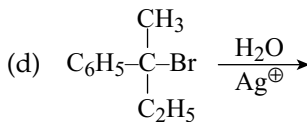
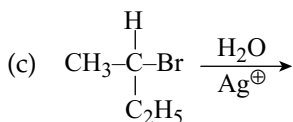
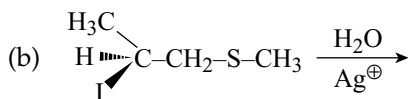
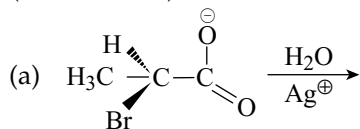
Which one of the following is correct sequence of the halides, given above, in the decreasing order of their reactivity?

- (a) (III) > (IV) > (II) > (I) (b) (II) > (IV) > (I) > (III)
 (c) (I) > (II) > (III) > (IV) (d) (III) > (I) > (II) > (IV)
3. The correct statement for the given reaction is



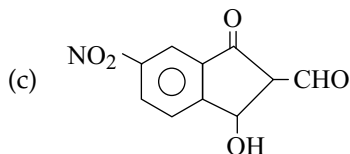
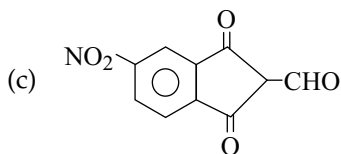
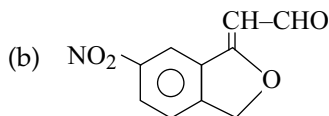
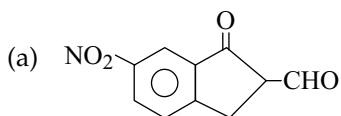
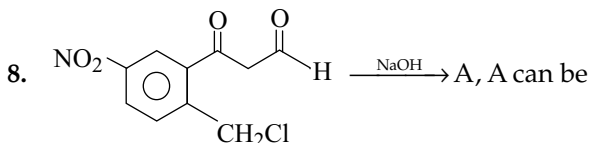
- (a) B is optically inactive due to external compensation
 (b) B is optically inactive due to internal compensation
 (c) A is predominantly cis-alkene
 (d) B does not have chiral centres
4. $\text{CH}_3\text{—CH}_2\text{SH} \xrightarrow[\text{(iii) H}_2\text{O}]{\text{(i) Dil. OH}^\ominus, \text{(ii) Ethylene oxide}} \text{Product, product is}$
- (a) $\text{CH}_3\text{—CH}_2\text{—S—CH}_2\text{—CH}_2\text{—OH}$ (b) $\text{CH}_3\text{—CH}_2\text{O}^\ominus$
 (c) $\text{CH}_3\text{—CH}_2\text{—O—CH}_2\text{—CH}_2\text{—OH}$ (d) $\text{CH}_3\text{—CH}_2\text{—S}^\oplus \begin{array}{l} \diagup \text{CH—OH} \\ \diagdown \text{CH}_2 \end{array}$
5. Which of the following statements are correct?
- (a) The S_N1 reaction obeys the 1st order kinetics
 (b) The solvolysis of ethylbromide in ethanoic solution is first order kinetics
 (c) R configuration of a compound may result in R and S configurations in an S_N2 reaction
 (d) 1-Bromo-1-phenylethane shows greater reactivity towards S_N1 and S_N2 reaction compared to isopropyl bromide

6. Which of the following reactions will proceed with retention of configuration only (no inversion) at the chiral carbon?

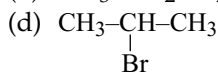
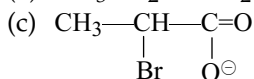
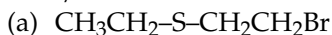


7. Correct statements is

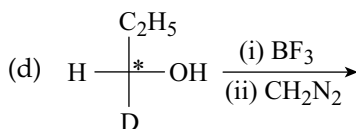
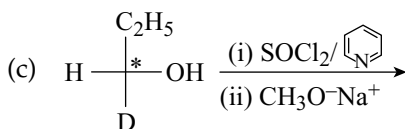
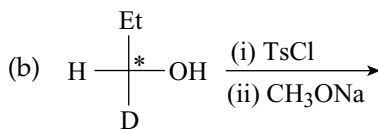
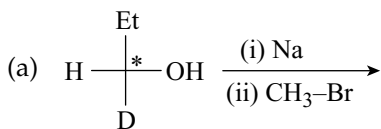
- (a) Nucleophilicity of SH^- is greater than OH^- in dimethyl formamide
 (b) Hydrazine is a better nucleophile than NH_3
 (c) Phenoxide ion is more basic than acetate ion
 (d) CH_3^- is a better nucleophile than OH^- in a non-polar solvent

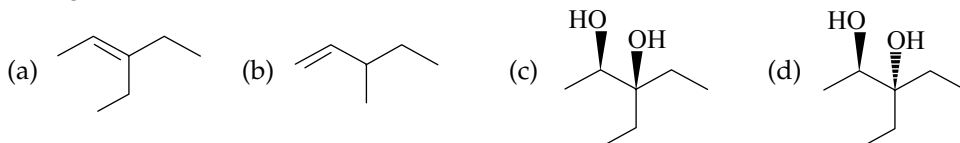
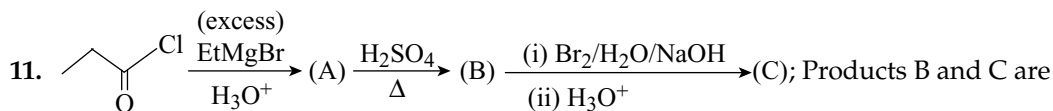


9. Identify the compounds that will undergo substitution reaction faster than $\text{C}_2\text{H}_5\text{Br}$ with OH^-/DMSO .



10. In which of the following cases, the configuration of chiral C is retained in product?

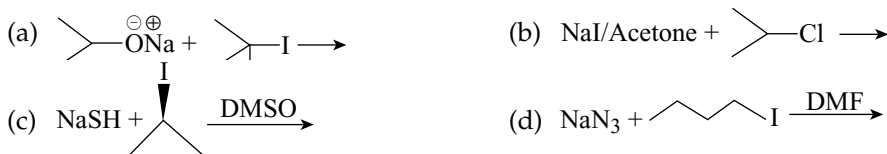




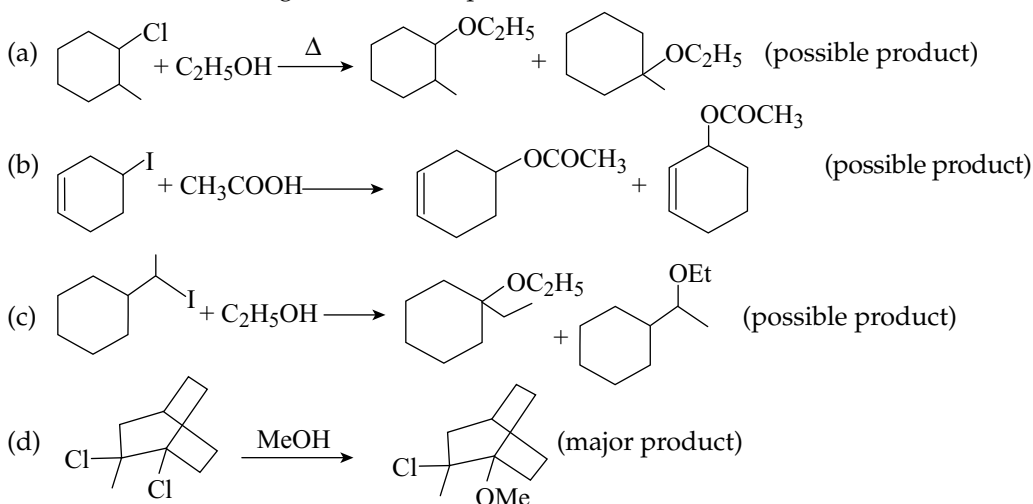
12. Which statement is/are correct in the following statements?

- (a) Allyl bromide gives $\text{S}_{\text{N}}1$ reaction but vinyl chloride does not.
 (b) Primary alkyl halides can give $\text{S}_{\text{N}}1$ or $\text{S}_{\text{N}}2$ reaction. This depends on the structure of the substrate and nature of solvent.
 (c) $\text{CH}_3\text{-CH=CH-CH}_2\text{Cl}$ reacts with KCN to give mixture of two isomeric products.
 (d) $\text{CH}_3\text{-CH}_2\text{-O-CH}_2\text{Br}$ is less reactive than tert. butyl bromide for $\text{S}_{\text{N}}1$ reaction.

13. Identify reaction that follows $\text{S}_{\text{N}}2$ path



14. Which of the reactions give the correct product?



15. Which of the following alcohol gives the white turbidity immediately with $\text{HCl} + \text{ZnCl}_2$ (anhy.)?



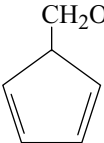
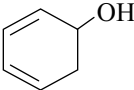
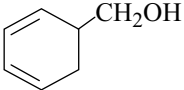
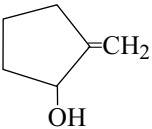
16. $A \xrightarrow{\text{HIO}_4} \text{CH}_3\text{CHO} + 2\text{HCOOH} + \text{CH}_3\text{COOH}$. The structure of A is

- (a) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{C}-\text{OH} \\ | \\ \text{CH}-\text{OH} \\ | \\ \text{CH}-\text{OH} \\ | \\ \text{CHO} \end{array}$ (b) $\begin{array}{c} \text{CH}_3-\text{CH}-\text{OH} \\ | \\ \text{CH}-\text{OH} \\ | \\ \text{CH}_3-\text{C}=\text{O} \\ | \\ \text{CHO} \end{array}$ (c) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{C}=\text{O} \\ | \\ \text{CH}-\text{OH} \\ | \\ \text{CH}-\text{OH} \\ | \\ \text{CH}_2\text{OH} \end{array}$ (d) none of these

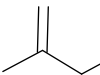
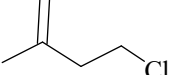
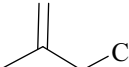
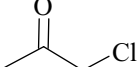
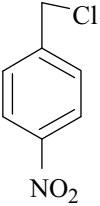
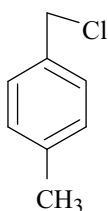
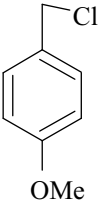
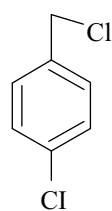
17. Which of the following gives turbidity immediately with Lucas-reagent (anhy. $\text{ZnCl}_2 + \text{HCl}$)?

- (a) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3-\text{C}-\text{OH} \\ | \\ \text{CH}_3 \end{array}$ (b) $\begin{array}{c} \text{Ph}-\text{CH}-\text{OH} \\ | \\ \text{CH}_3 \end{array}$
 (c) $\text{Me}-\text{C}_6\text{H}_4-\begin{array}{c} \text{CH}_3 \\ | \\ \text{C}-\text{OH} \\ | \\ \text{CH}_3 \end{array}$ (d) $\text{CH}_3\text{CH}_2\text{OH}$

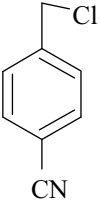
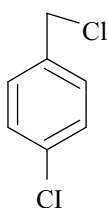
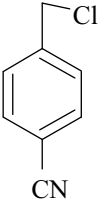
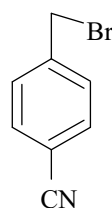
18. The possible structure of the alcohol is

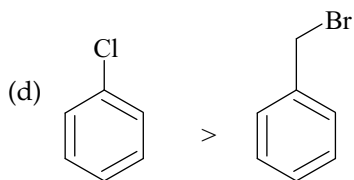
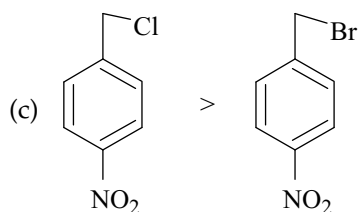
- $\text{ROH (alcohol)} \xrightarrow[\text{Heat}]{\text{H}_2\text{SO}_4} \text{C}_6\text{H}_6$
- (a)  (b)  (c)  (d) 

19. Identify correct reactivity order for $\text{S}_\text{N}2$ reaction

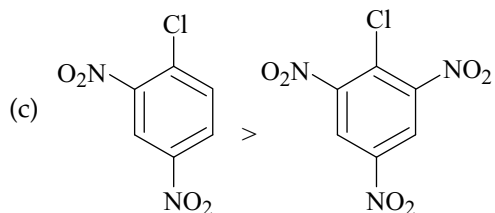
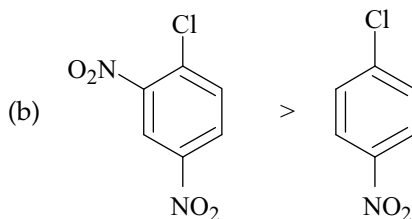
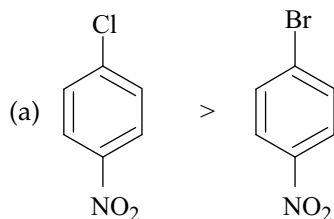
- (a)  $>$  (b)  $>$ 
 (c)  $>$  (d)  $>$ 

20. Identify correct reactivity order for $\text{S}_\text{N}2$ reaction

- (a)  $>$  (b)  $>$ 

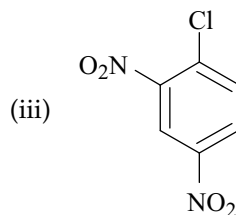
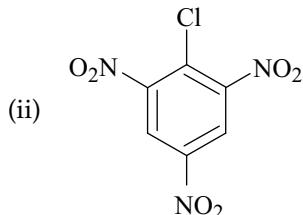
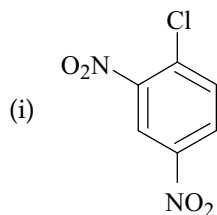


21. Identify correct reactivity order for ArSN reaction



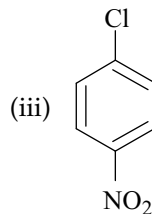
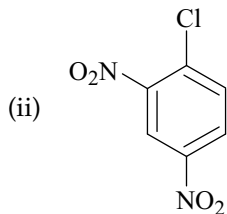
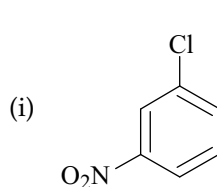
(d) (a) and (b) are correct

22. Identify correct reactivity order for ArSN reaction



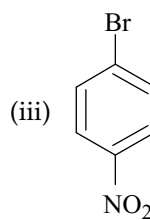
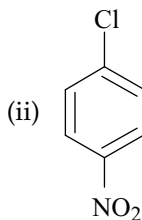
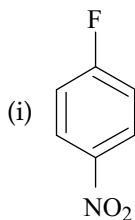
(a) (i) > (ii) > (iii) (b) (ii) > (i) > (iii) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

23. Identify correct reactivity order for ArSN reaction



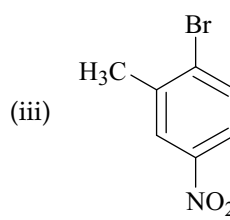
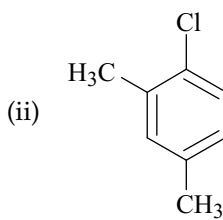
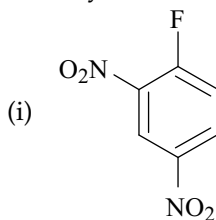
(a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

24. Identify correct reactivity order for ArSN reaction



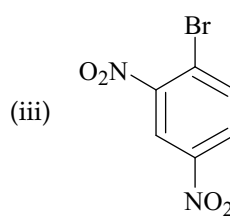
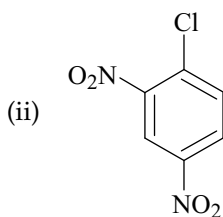
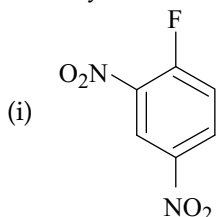
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

25. Identify correct reactivity order for ArSN reaction



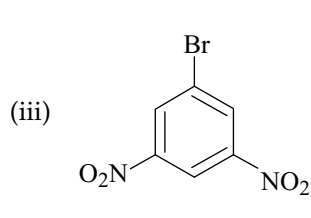
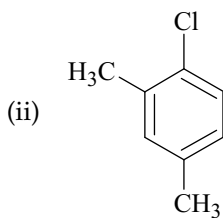
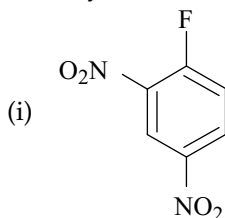
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

26. Identify correct reactivity order for ArSN reaction



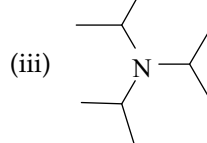
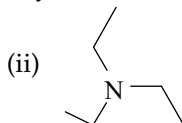
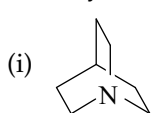
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

27. Identify correct reactivity order for ArSN reaction



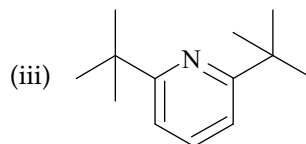
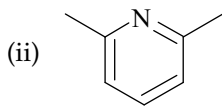
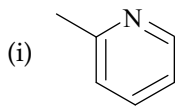
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

28. Identify correct nucleophilicity order



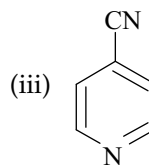
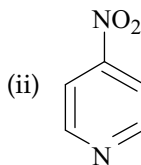
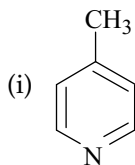
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

29. Identify correct nucleophilicity order



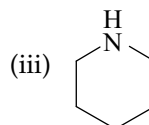
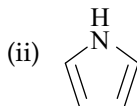
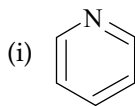
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

30. Identify correct nucleophilicity order



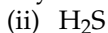
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

31. Identify correct nucleophilicity order



- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

32. Identify correct nucleophilicity order



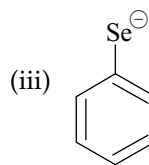
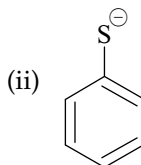
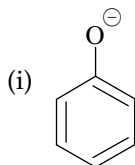
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (ii) > (i)

33. Identify correct nucleophilicity order



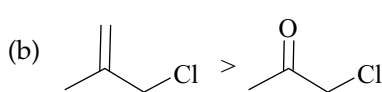
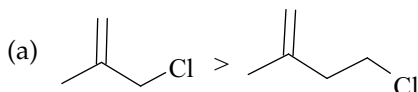
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (i) > (ii)

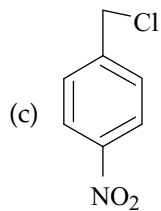
34. Identify correct nucleophilicity order



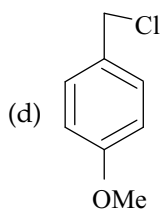
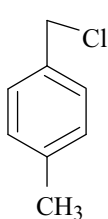
- (a) (i) > (ii) > (iii) (b) (ii) > (iii) > (i) (c) (i) > (iii) > (ii) (d) (iii) > (ii) > (i)

35. Identify correct reactivity order for $\text{S}_{\text{N}}1$ reaction

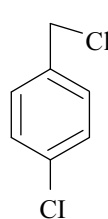




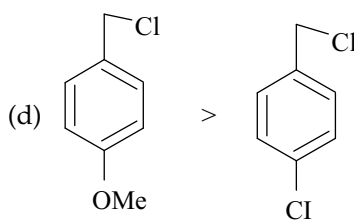
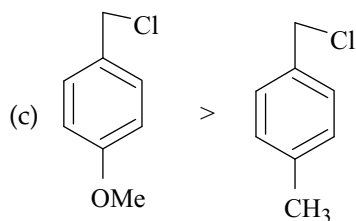
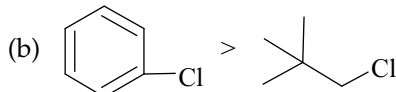
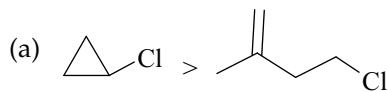
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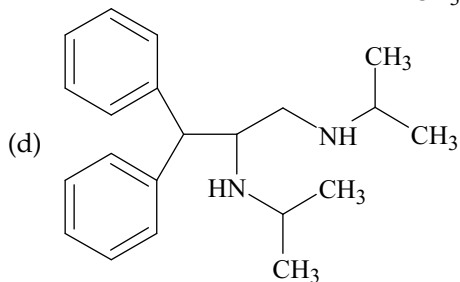
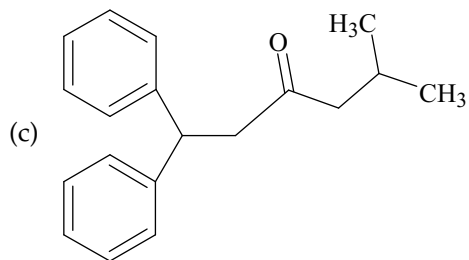
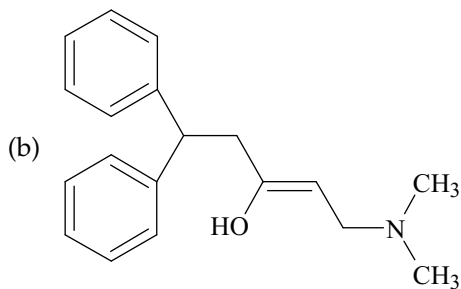
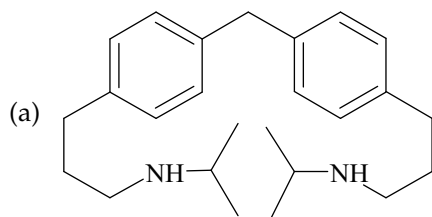
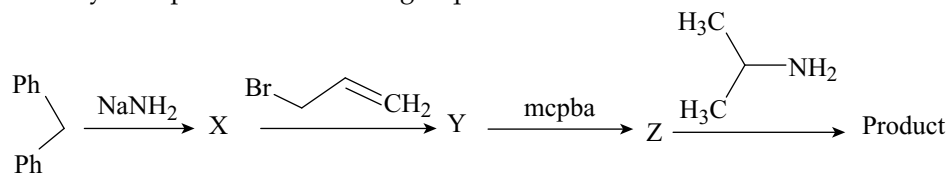
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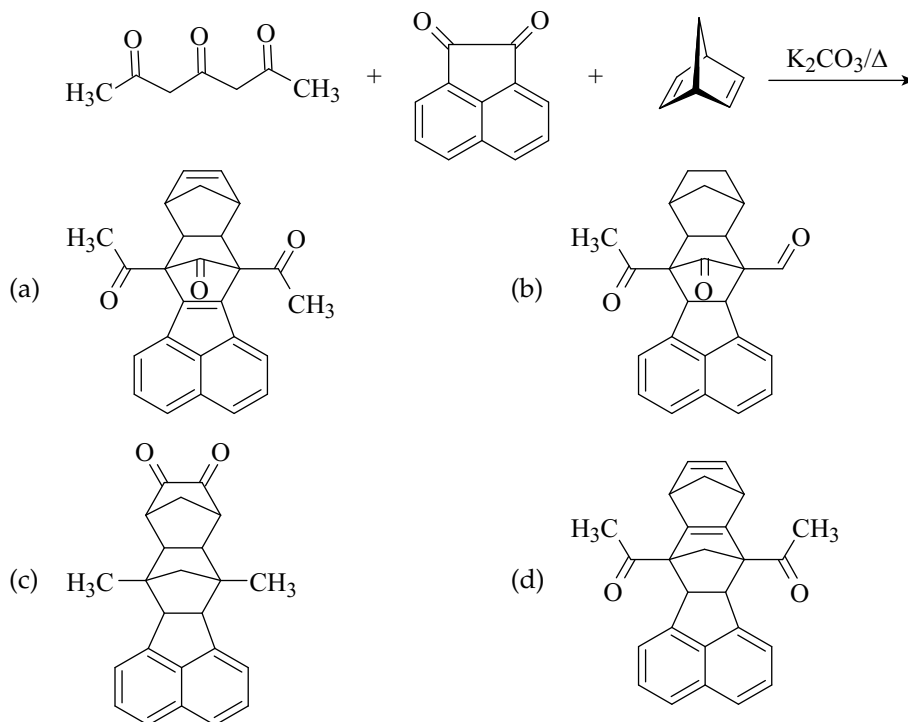
36. Identify correct reactivity order for S_N1 reaction



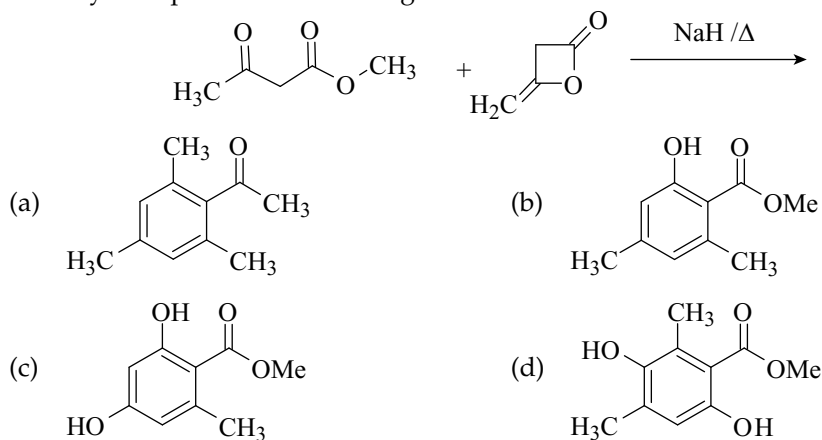
37. Identify final product in following sequence



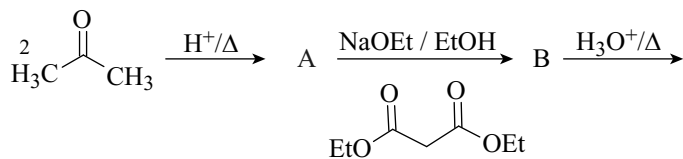
38. Identify final product of following reaction

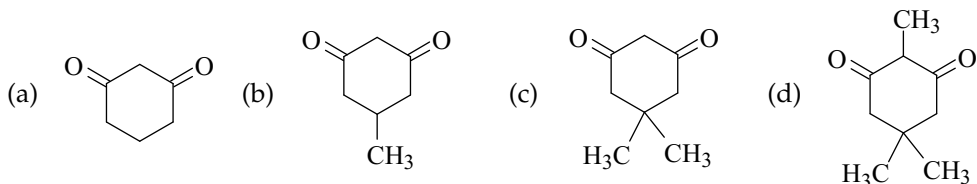


39. Identify final product of following reaction

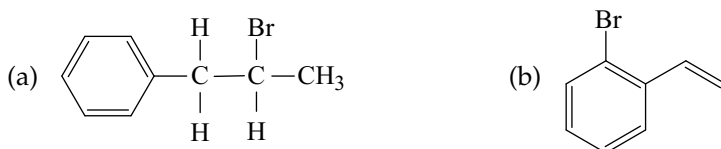
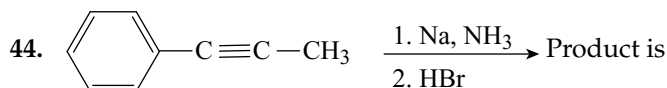
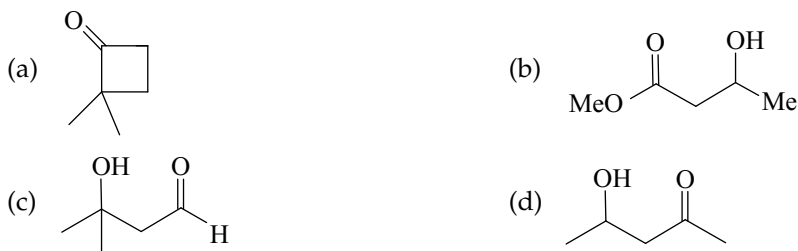
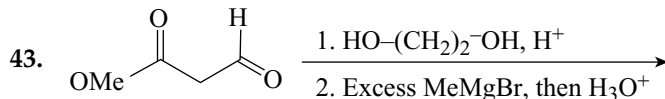
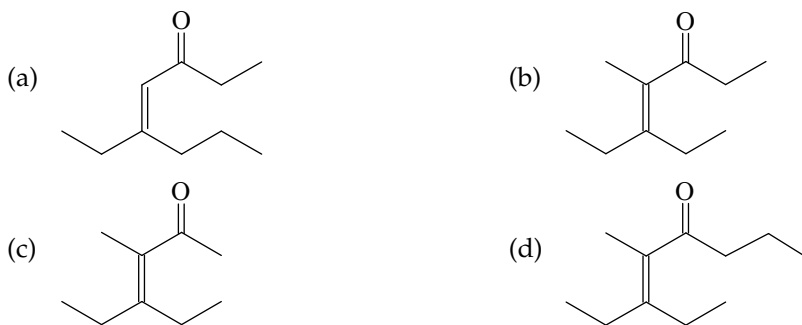


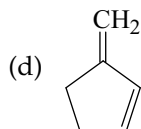
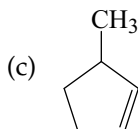
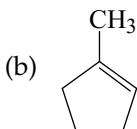
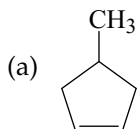
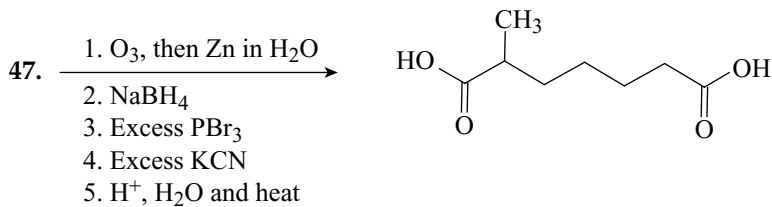
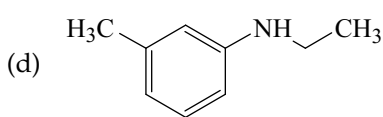
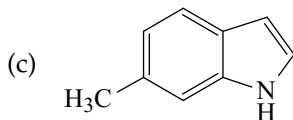
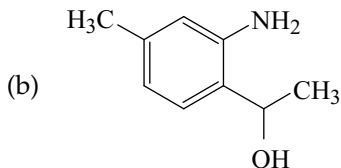
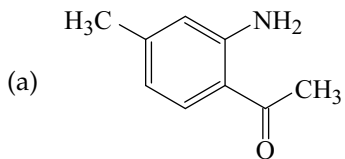
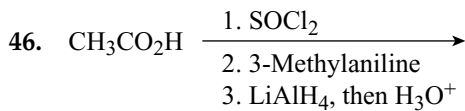
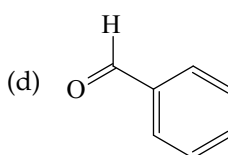
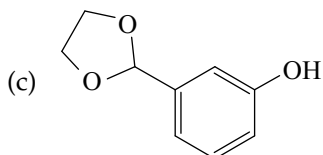
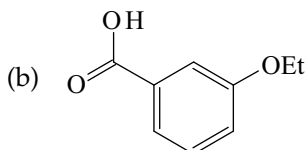
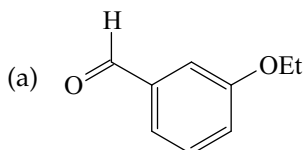
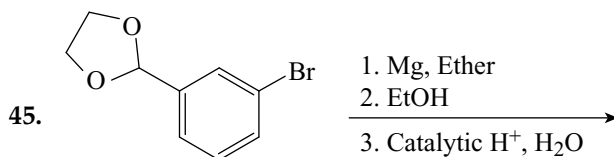
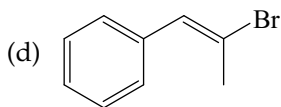
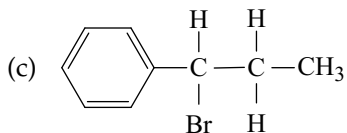
40. Identify final product of following reaction

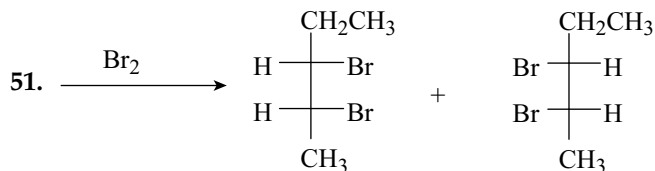
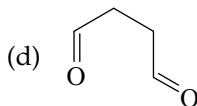
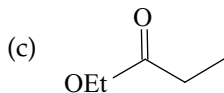
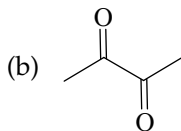
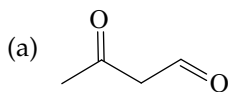
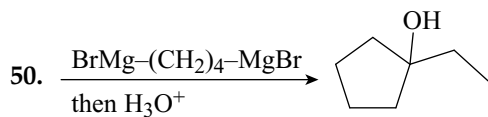
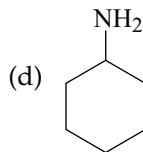
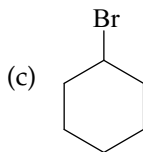
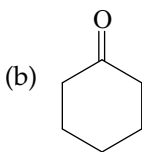
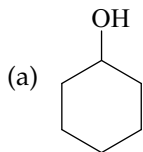
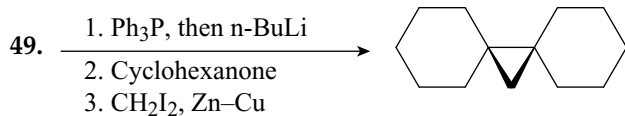
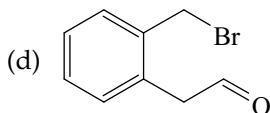
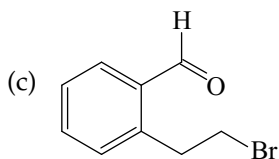
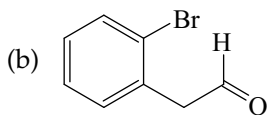
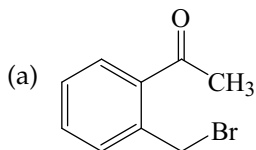
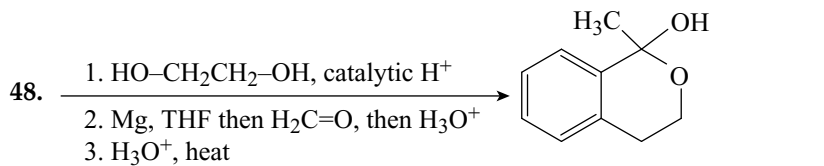




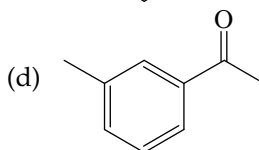
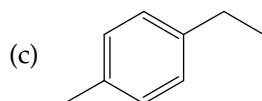
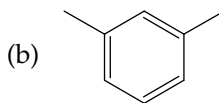
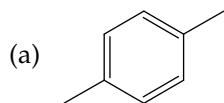
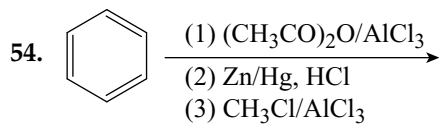
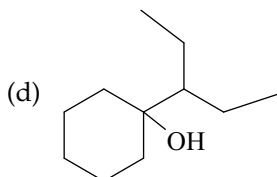
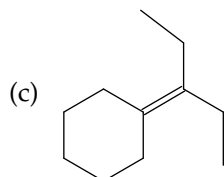
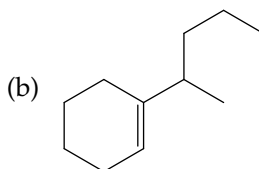
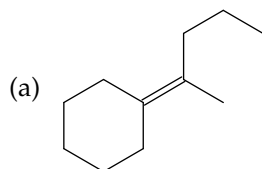
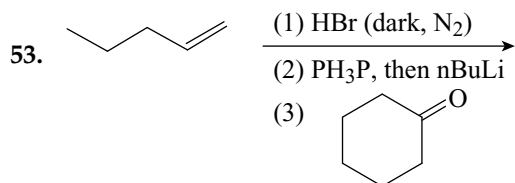
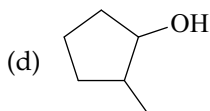
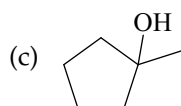
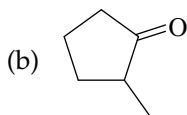
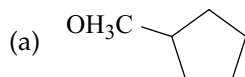
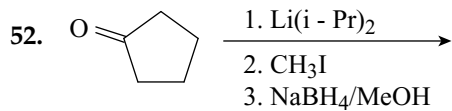
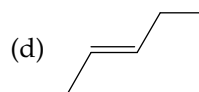
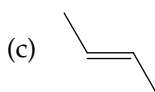
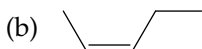
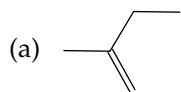
41. Select all of the following statements about the aldol experiment that are true
- The ketone was added to a solution containing the aldehyde and base.
 - The reaction can be catalysed by acid
 - The electrophile is the enolate ion
 - The aldehyde is used in excess
42. Which of the following structures is the product from the self-condensation of two molecules of pentan-3-one?

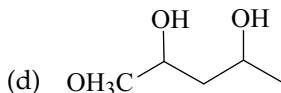
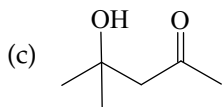
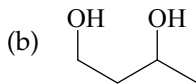
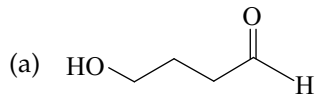
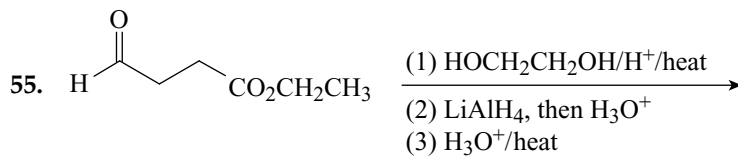




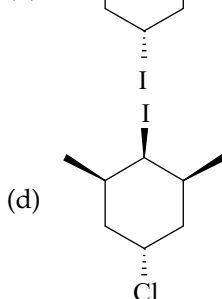
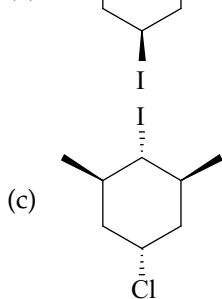
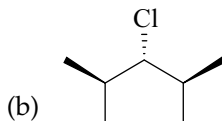
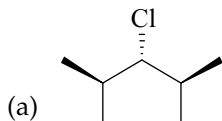
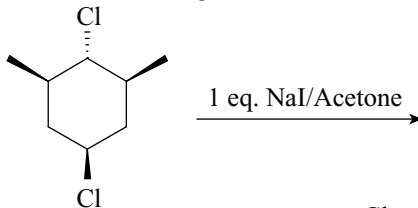


(50:50 of these enantiomers)

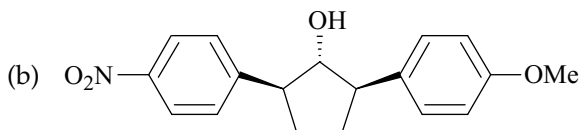
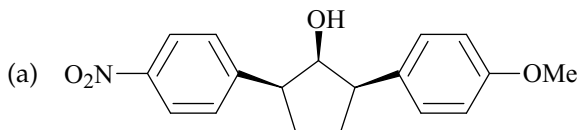
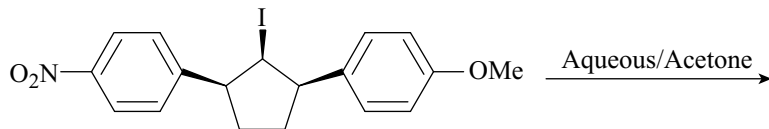


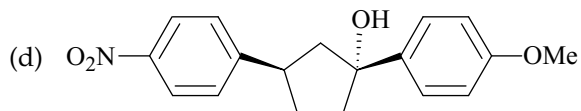
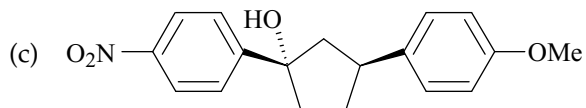


56. Identify major product for the following reaction

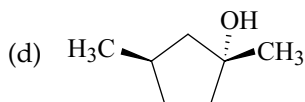
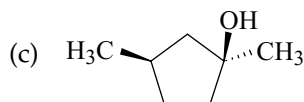
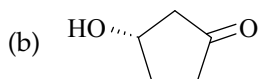
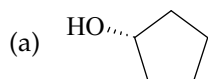
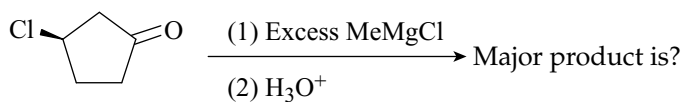


57. Identify major product for the following reaction

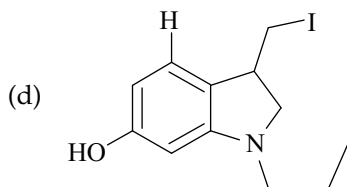
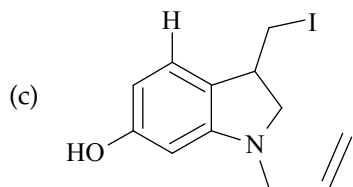
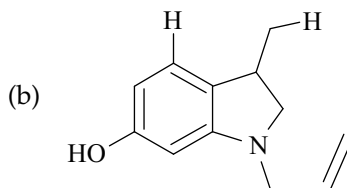
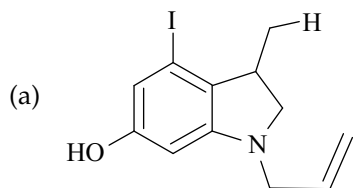
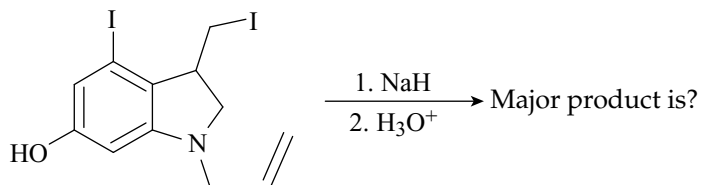




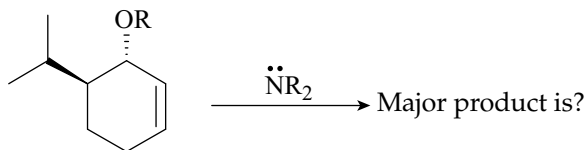
58. Identify major product for the following reaction

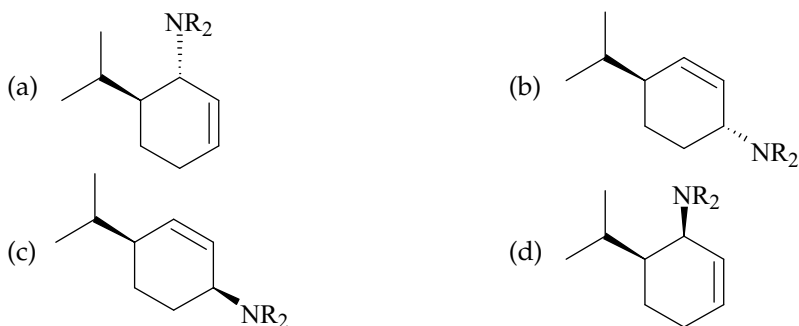


59. Identify major product for the following reaction

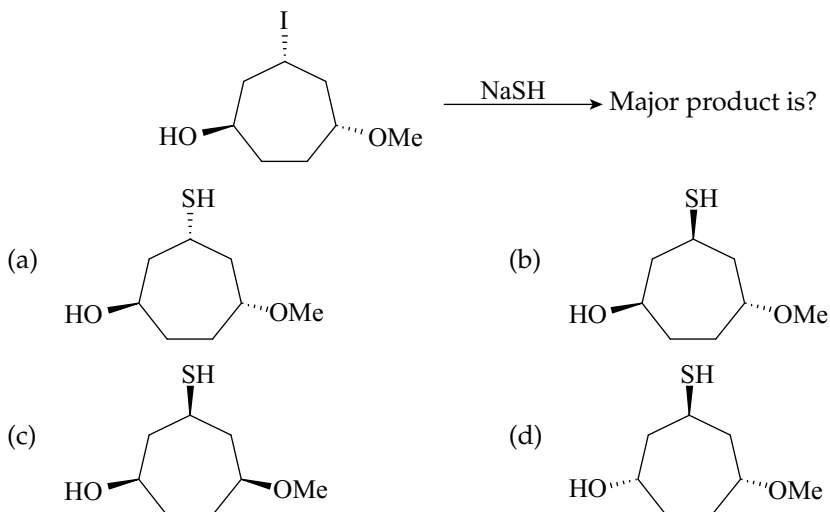


60. Identify major product for the following reaction

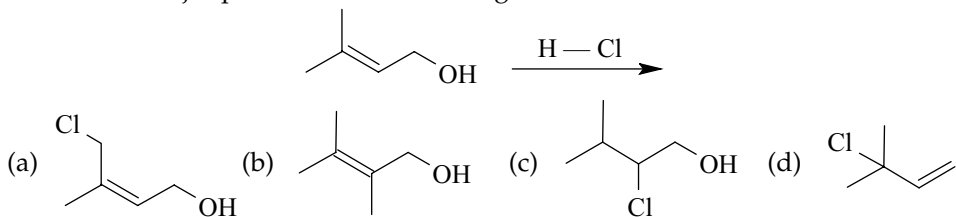




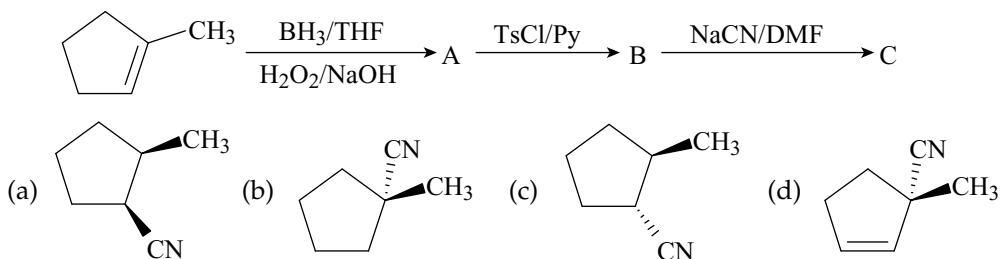
61. Identify major product for the following reaction



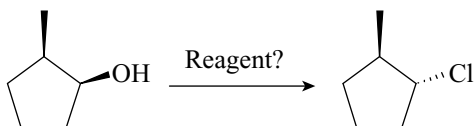
62. Choose the major product of the following reaction.



63. Identify major product of the following reaction

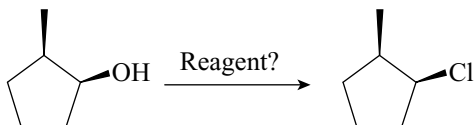


64. Identify best reagent for the following conversion



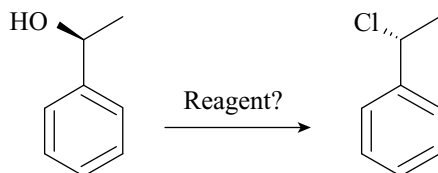
- (a) ZnCl_2/HCl (b) PCl_5 (c) PCl_3 (d) SOCl_2

65. Identify best reagent for the following conversion



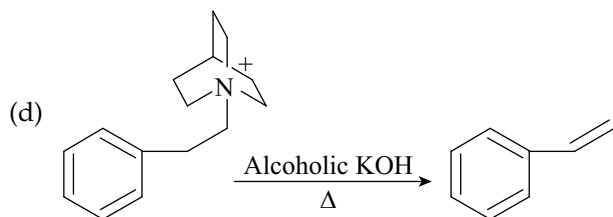
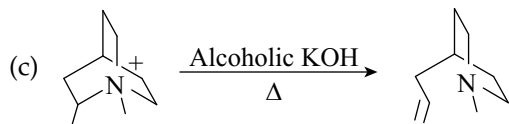
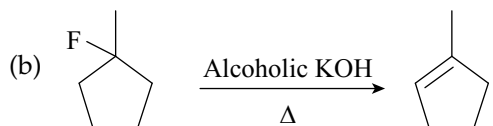
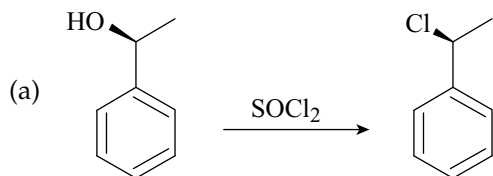
- (a) ZnCl_2/HCl (b) PCl_5 (c) PCl_3 (d) SOCl_2

66. Identify best reagent for the following conversion

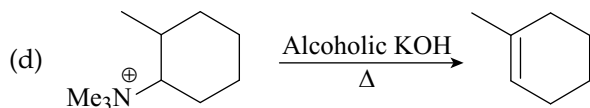
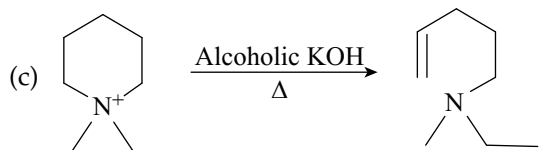
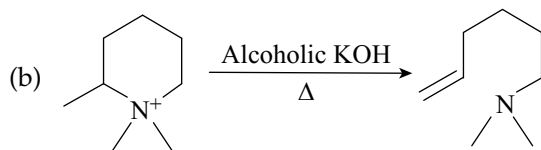
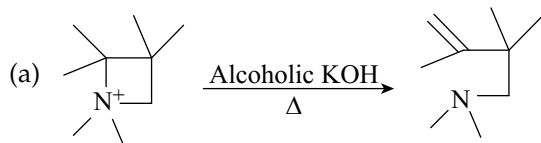


- (a) ZnCl_2/HCl (b) PCl_5 (c) PCl_3 (d) SOCl_2

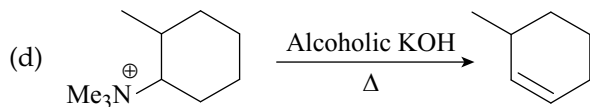
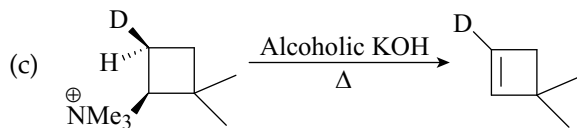
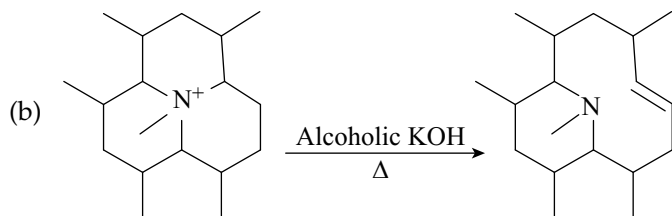
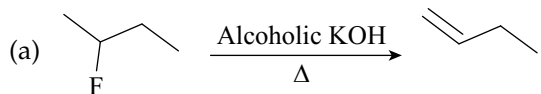
67. Identify reactions that correctly match with their products



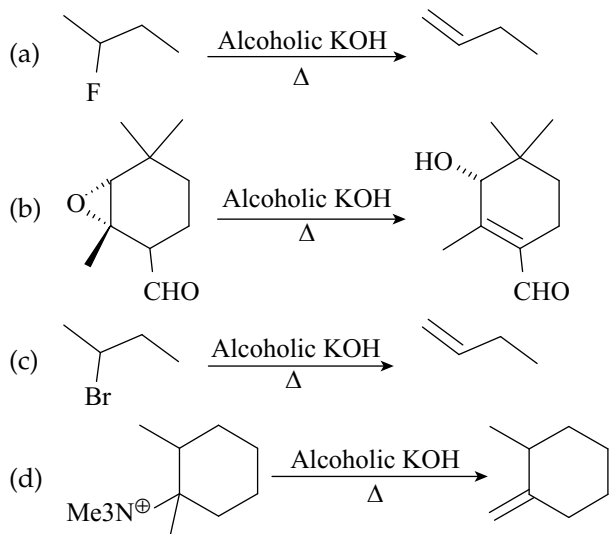
68. Identify reactions that correctly match with their products



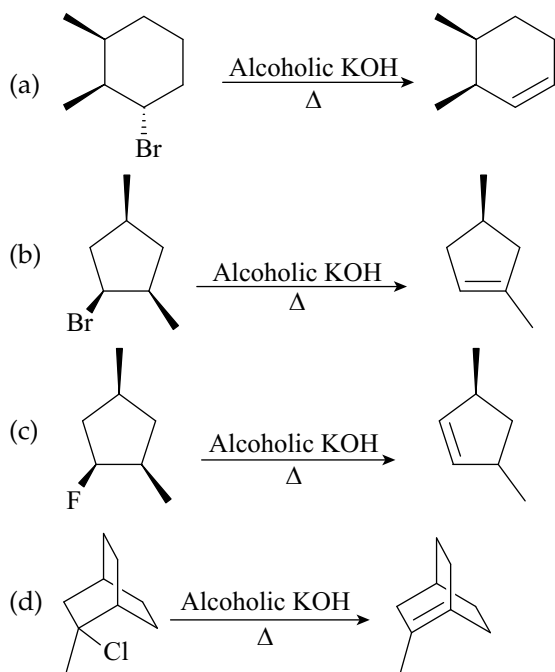
69. Identify reactions that correctly match with their products



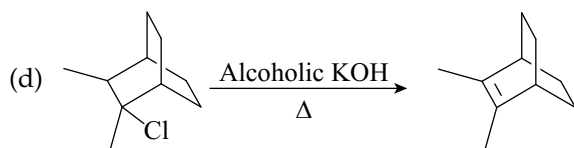
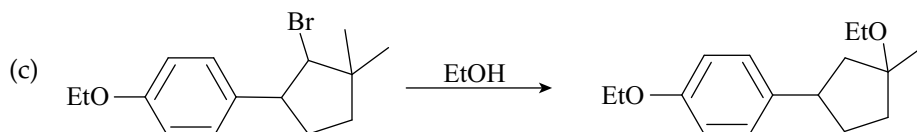
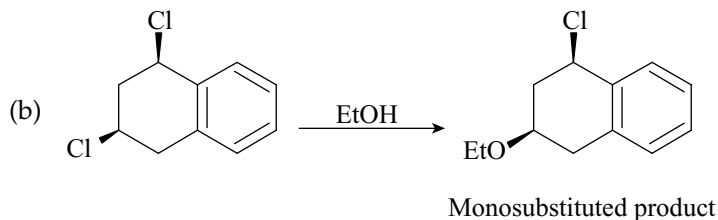
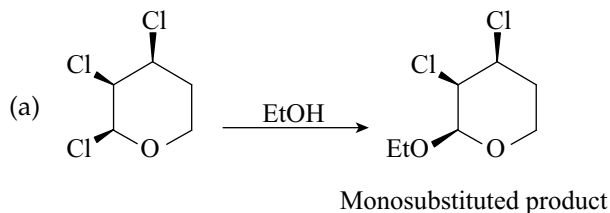
70. Identify reactions that correctly match with their products



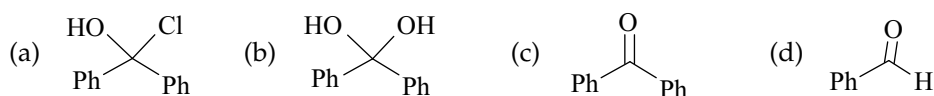
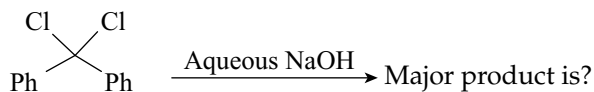
71. Identify reactions that correctly match with their products



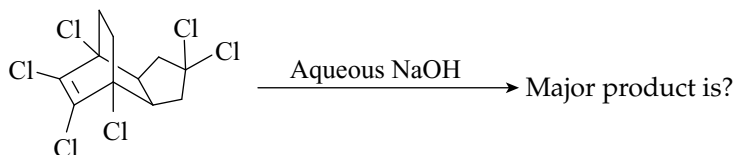
72. Identify reactions that correctly match with their products?

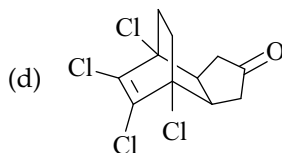
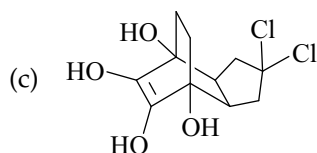


73. Identify major product for the following reaction

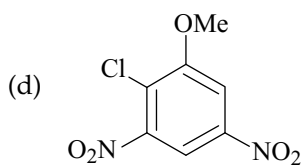
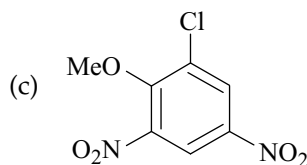
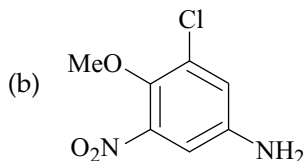
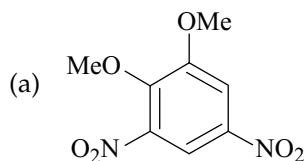
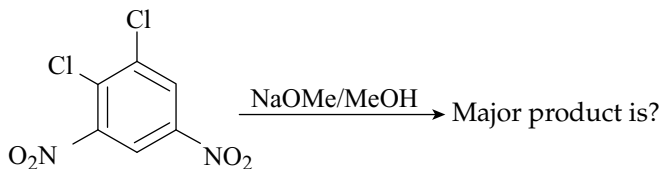


74. Identify major product for the following reaction

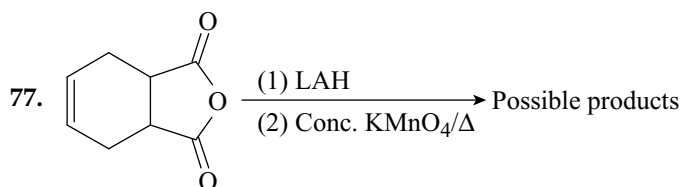
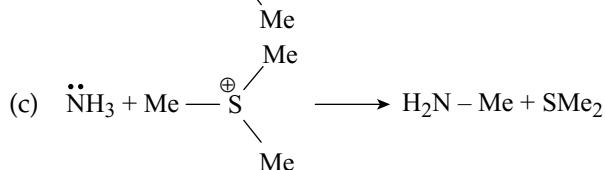
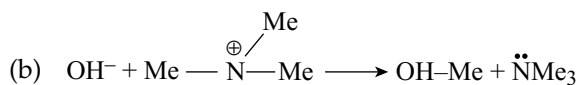
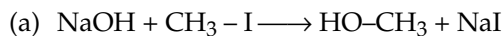


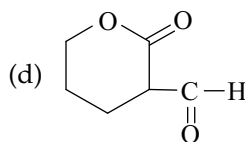
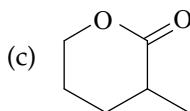
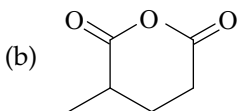
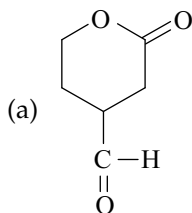
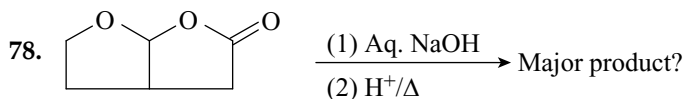
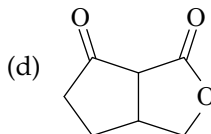
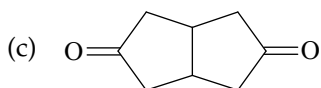
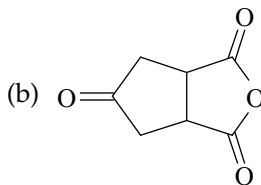
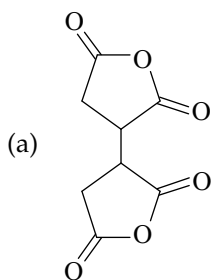


75. Identify major product for the following reaction



76. Identify S_N2 reaction which is favourable in a polar protic medium





Comprehension Type

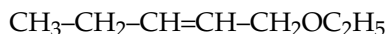
Passage 1

Suppose, a sample of an initially pure single enantiomer of tartaric acid $[\alpha]_D^{20} = +12.4^\circ$ (H_2O) underwent a chemical isomerization reaction which yielded 10 per cent of the other enantiomer and 20 per cent of the meso isomer, in addition to 70 per cent of the unchanged starting compound.

79. What is the enantiomeric excess (ee.) of the major enantiomer relative to the minor enantiomer?
 (a) 70 per cent (b) 700 per cent (c) 75 per cent (d) 87.5 per cent
80. If 1.0 g of the total mixture (including the meso isomer) was dissolved in 1 mL water and the rotation measured in a 1 dm cell, what value of α_D would one expect to observe?
 (a) +9.92 (b) +7.44 C (c) +12.4 (d) +8.68
81. From the total mixture, chromatographic separation on silica gel (a commonly used achiral solid adsorbent) might be expected to yield one of the three components in a pure state. Which one?
 (a) the minor enantiomer (b) the meso isomer
 (c) the major enantiomer (d) the 1:1 racemic form of tartaric acid

Passage 2

- (a) When 1-chloro-2-pentene is reacted with concentrated solution of sodium ethoxide, the rate of reaction depends on the concentration of both allylhalide and ethoxide ion. The product of reaction is exclusively



- (b) In dilute solution of sodium ethoxide, the reaction rate is dependent only on the concentration of allylhalide.
 (c) In the presence of traces of water, 1-chloro-2-pentene is slowly converted to a mixture of

1-chloro-2-pentene and 3-chloro-1-pentene

82. In case (a) the mechanism exclusively followed by reaction is

(a) $\text{S}_{\text{N}}1$ (b) $\text{S}_{\text{N}}2$ (c) E_1 (d) E_2

83. The intermediate formed in case (c) is

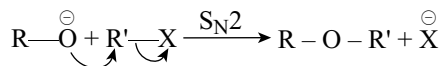
(a) $\text{CH}_3\text{--CH}_2\text{--}\overset{\oplus}{\text{CH}}\text{--CH=CH}_2$ (b) $\text{CH}_3\text{--CH}_2\text{--CH=CH--}\overset{\oplus}{\text{CH}}_2$
 (c) both (a) and (b) (d) none

84. Correct statement is

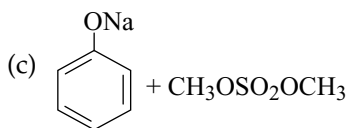
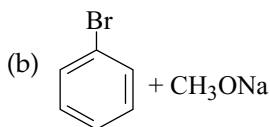
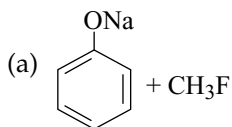
(a) The solvolysis of 1-chloro-2-pentene is greater than 1-chloropentane in water
 (b) The solvolysis of 1-chloro-2-pentene is lesser than 1-chloropentane in water
 (c) The solvolysis of 1-chloro-2-pentene and 1-chloropentane is equal in water
 (d) The solvolysis of 1-chloro-2-pentene and 1-chloropentane do not take place in water

Passage 3

Williamson's ether synthesis is one of the best method to prepare ether. In this method ethers can be prepared by the reaction of alkoxides with alkyl halide. The reaction takes place by $\text{S}_{\text{N}}2$ mechanism. The more basic alkoxide ion displaces the halide or other leaving group to produce ether. While going for Williamson's ether synthesis one should be aware of competitive reactions. The mechanism take palces as

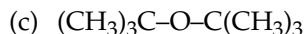
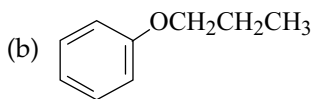
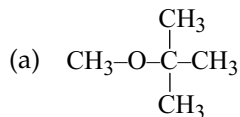


85. Which combination will be better to prepare anisole



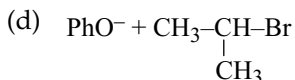
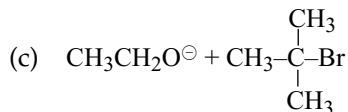
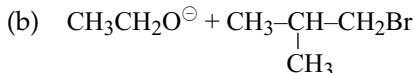
(d) Both (a) and (c) can be used

86. The ether that cannot be synthesized by the above method is



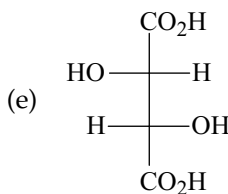
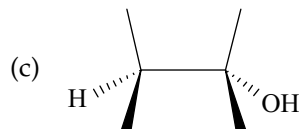
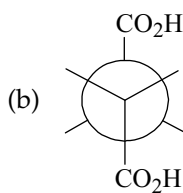
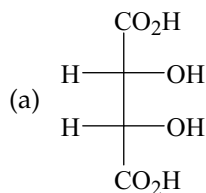
(d) All can be prepared

87. Which one of the following will give most effective yield of ether?



Passage 4

The specific rotation of (2R, 3R) - (+) - tartaric acid is $+12.4^\circ$ ($c = 2$, H_2O). The optical rotation of a solution made up from 10 g of a mixture B and C in 20 ml of water was measured in a 10 cm cell. The observed rotation was -3.1° .



88. What is the specific rotation of A?

(a) $+12.4^\circ$

(b) 0°

(c) -12.4°

(d) None of these

89. What is the specific rotation of the mixture?

(a) -3.1°

(b) -6.2°

(c) -0.62°

(d) $+12.4^\circ$

90. What is the relative ratio of B:C in the mixture?

(a) 0:100

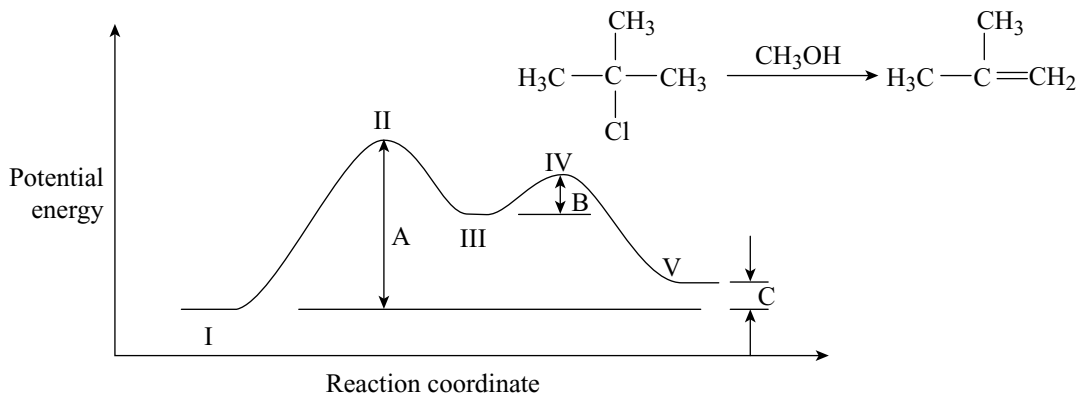
(b) 50:50

(c) 25:75

(d) 75:25

Passage 5

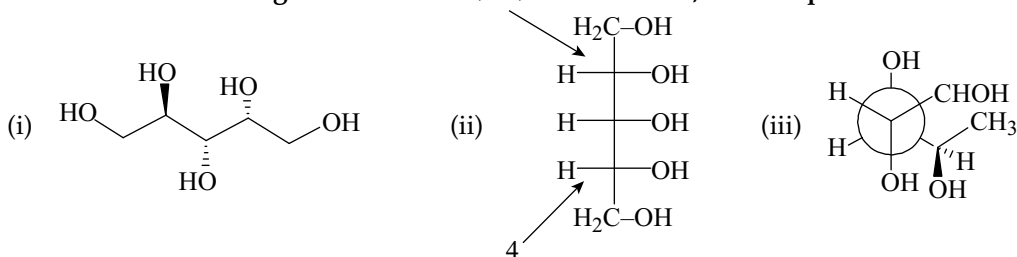
Based on the potential energy diagram for the following reaction

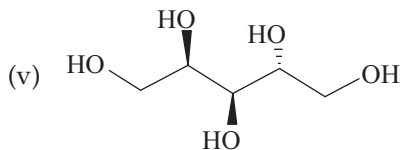
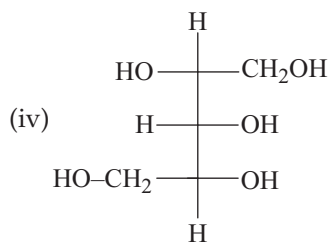


91. Is this an endothermic or exothermic reaction?
 (a) exothermic
 (b) There is not enough information to determine.
 (c) endothermic
 (d) This reaction can be either exothermic or endothermic.
92. What potential energy difference represents the heat of reaction?
 (a) A (b) B (c) C (d) Both A and B
93. Which of the following indicates the location of intermediate?
 (a) I (b) II (c) III (d) IV
94. According to the potential energy diagram, what is the possible mechanism for the reaction?
 (a) E_1 (b) E_2 (c) S_N1 (d) S_N2

Passage 6

Based on the following five structures (i–v) shown below, answer questions 99 to 106.



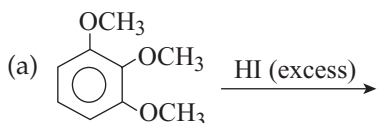


95. What type of isomers are (i) and (iv)?
 (a) diastereomers (b) conformational (c) identical (d) enantiomers
96. What type of isomers are (i) and (iii)?
 (a) diastereomers (b) conformational (c) identical (d) enantiomers
97. If the specific rotation of i is -60.0° , what is the specific rotation of iii, if 1.0 g of iii is dissolved in 10 mL of methanol and placed in a 1 dm tube?
 (a) $+60^\circ$ (b) 0° (c) 6° (d) -60°
98. If the specific rotation of i is -60.0° , what is the specific rotation of iv, if 1.0 g of iv is dissolved in 10 mL of methanol and placed in a 1 dm tube?
 (a) $+60^\circ$ (b) 0° (c) 6° (d) -60°
99. If the specific rotation of (i) is -60.0° , what is the observed rotation of a sample made up of a mixture of 1 g of (i) and 0.5 g of (v) when dissolved in water (10 mL) and placed in a 10 cm cell?
 (a) 60° (b) 0° (c) 6° (d) -60°
100. If the specific rotation of i is -60.0° , what is the optical purity of a sample of made up of a mixture of 1 g of (i) and 0.5 g of (v) when dissolved in water (10 mL)?
 (a) 50 per cent (b) 33 per cent (c) 100 per cent (d) 66 per cent
101. Assign the absolute configuration to carbon atoms 2 and 4 in compound (ii).
 (a) 2R, 4S (b) 2S, 4S (c) 2S, 4R (d) 2R, 4R
102. If the melting point of pure iv is 152°C , what is the melting point of pure (v)?
 (a) -152°C (b) 152°C
 (c) lower than 152°C (d) higher than 152°C

Matrix Type

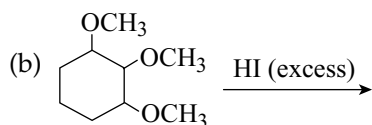
Match the reaction mechanism in Column I with their reaction conditions in Column II

103. Column I

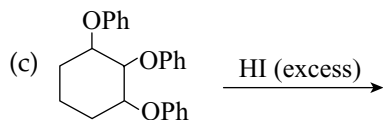


Column II

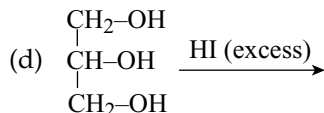
- (p) More than 4 moles of HI will be consumed



(q) Product will react with Na



(r) Product formed contains 2° halide



(s) CH_3I is one of the product

104. Column I

- (a) $\text{S}_{\text{N}}1$
 (b) $\text{S}_{\text{N}}2$
 (c) E_1
 (d) E_2

Column II

- (p) 3° alkyl halides $>$ 2° alkyl halides $>$ 1° alkyl halides
 (q) 1° alkyl halides $>$ 2° alkyl halides $>$ 3° alkyl halides
 (r) High concentration of strong base
 (s) Favourable medium is polar protic solvent

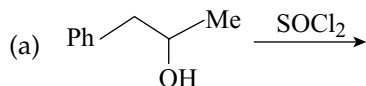
105. Column I

- (a) Walden inversion
 (b) Racemic mixture
 (c) Alkene $\xrightarrow[\text{reagent}]{\text{Bayer's}}$
 (d) Alkene $\xrightarrow{\text{Br}_2}$

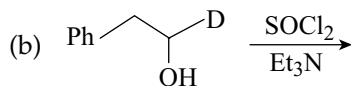
Column II

- (p) Cis-addition
 (q) Trans-addition
 (r) $\text{S}_{\text{N}}1$ reaction
 (s) $\text{S}_{\text{N}}2$ reaction

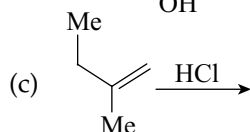
106. Column I (reaction)



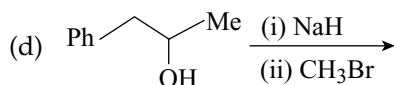
(p) Optically active compound



(q) Inversion of configuration



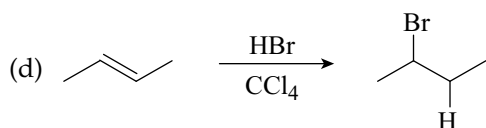
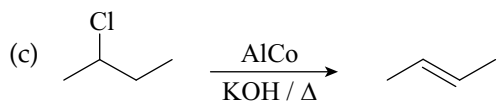
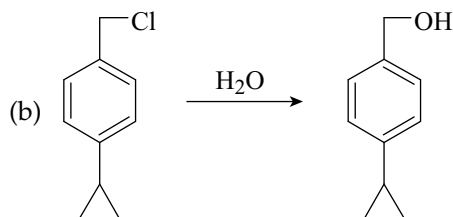
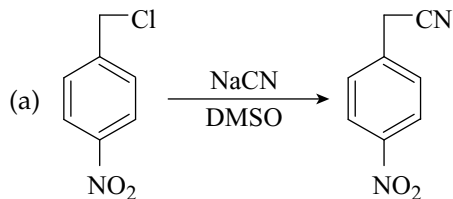
(r) Retention of configuration



(s) Optically inactive compound

(t) Carbocation intermediate

107. Column I



Column II

(p) Carbocation intermediate is formed

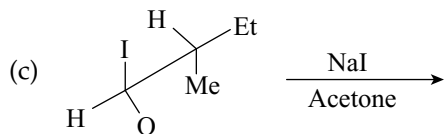
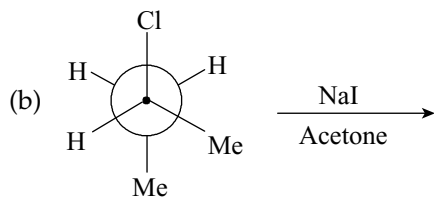
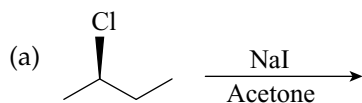
(q) S_N1

(r) S_N2

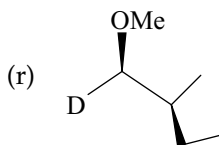
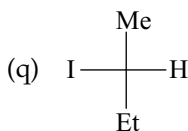
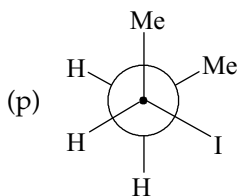
(s) Addition reaction

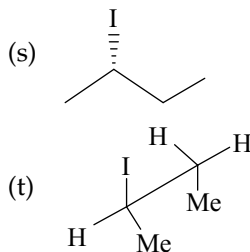
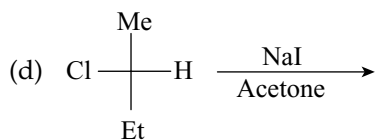
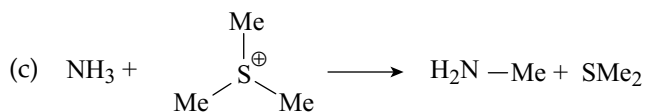
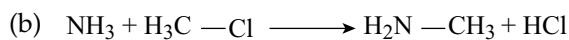
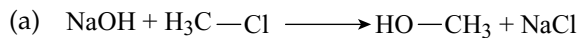
(t) Elimination reaction

108. Column I



Column II



**109. Column I****Column II**

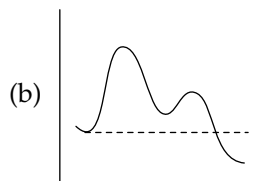
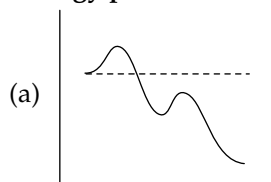
(p) Rate of reaction increases in polar protic solvent

(q) Rate of reaction increases in polar aprotic solvent

(r) $\text{S}_{\text{N}}2$ mechanism

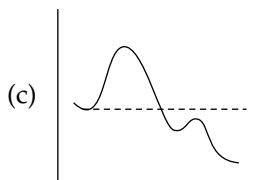
(s) $\text{S}_{\text{N}}1$ mechanism

(t) TS has positive charge [non-zwitter ion]

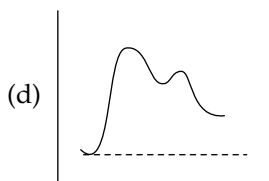
110. Column I
(energy profile)**Column II**
(chain propagation of halogenation)

(p) Iodination

(q) Fluorination

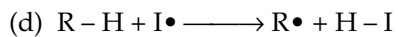
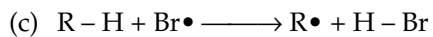
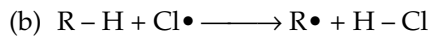
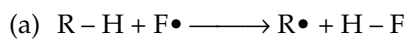


(r) Chlorination

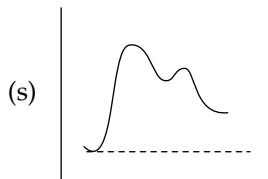
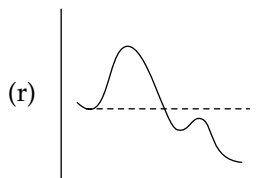
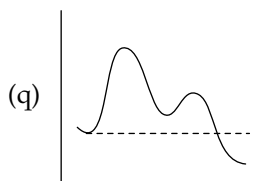
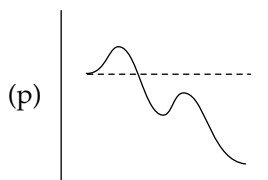


(s) Bromination

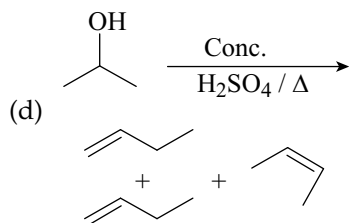
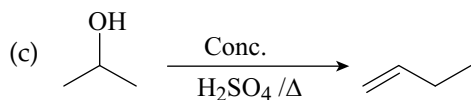
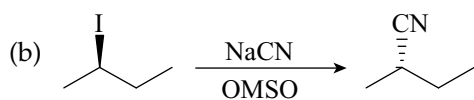
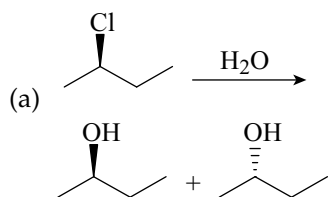
111. Column I
(RDS of halogenation)



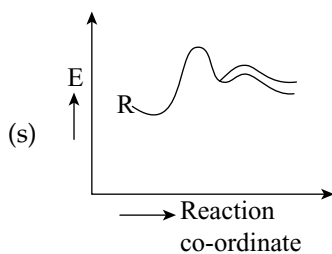
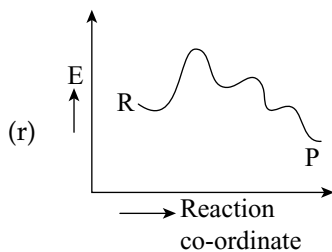
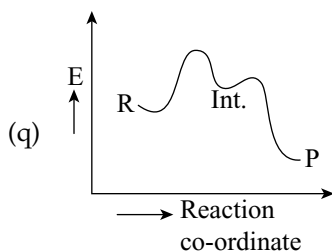
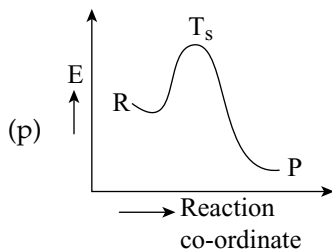
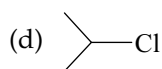
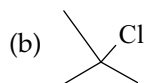
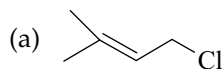
Column II
(energy profile)



112. Column I



Column II

113. Column I
(compounds)Column II
(rate of solvolysis)

(p) 0.07

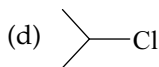
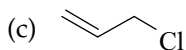
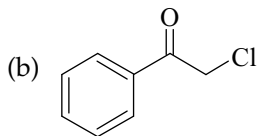
(q) 0.12

(r) 2100

(s) 130000

114. Column I
(compound)

(a) Me-Cl

Column II
(rate of reaction with NaI acetone)

(p) 0.02

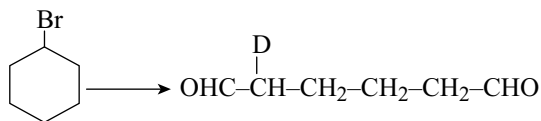
(q) 100000

(r) 200

(s) 79

Integer Type

115. Minimum number of steps involved in conversion of

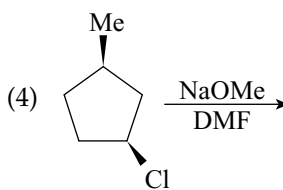
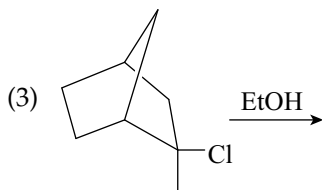
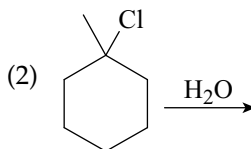
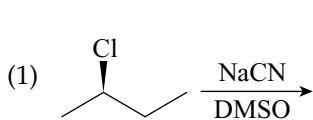


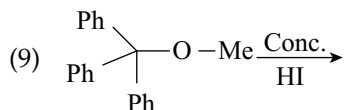
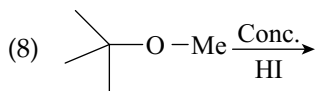
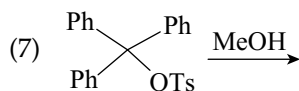
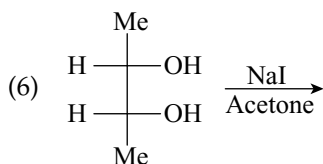
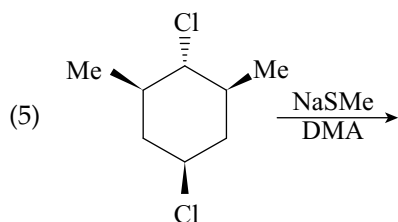
116. When (S)-2-chloropentane reacts with NaOH in 75 per cent EtOH and 25 per cent acetone follows rate law:

$$\text{Rate} = 2.35 \times 10^{-5} \left[\text{Structure of (S)-2-chloropentane} \right] \cdot [\text{OH}] + 5.30 \times 10^{-6} \left[\text{Structure of (R)-2-chloropentane} \right]$$

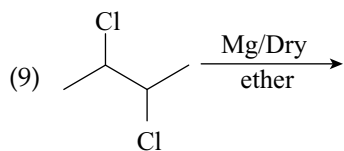
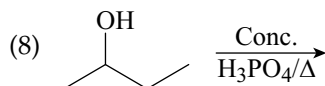
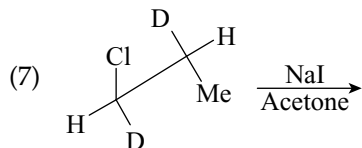
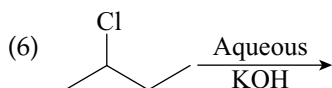
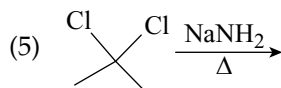
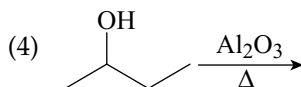
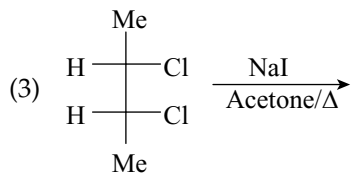
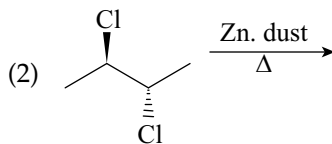
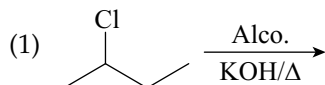
Then, calculate percentage of S_N1 product when concentration of $[\text{OH}] = 1.5$ molar.If percentage of S_N1 product = X, then calculate value of Y if:

$$Y = X - 5.$$

117. Identify total number of S_N1 reactions

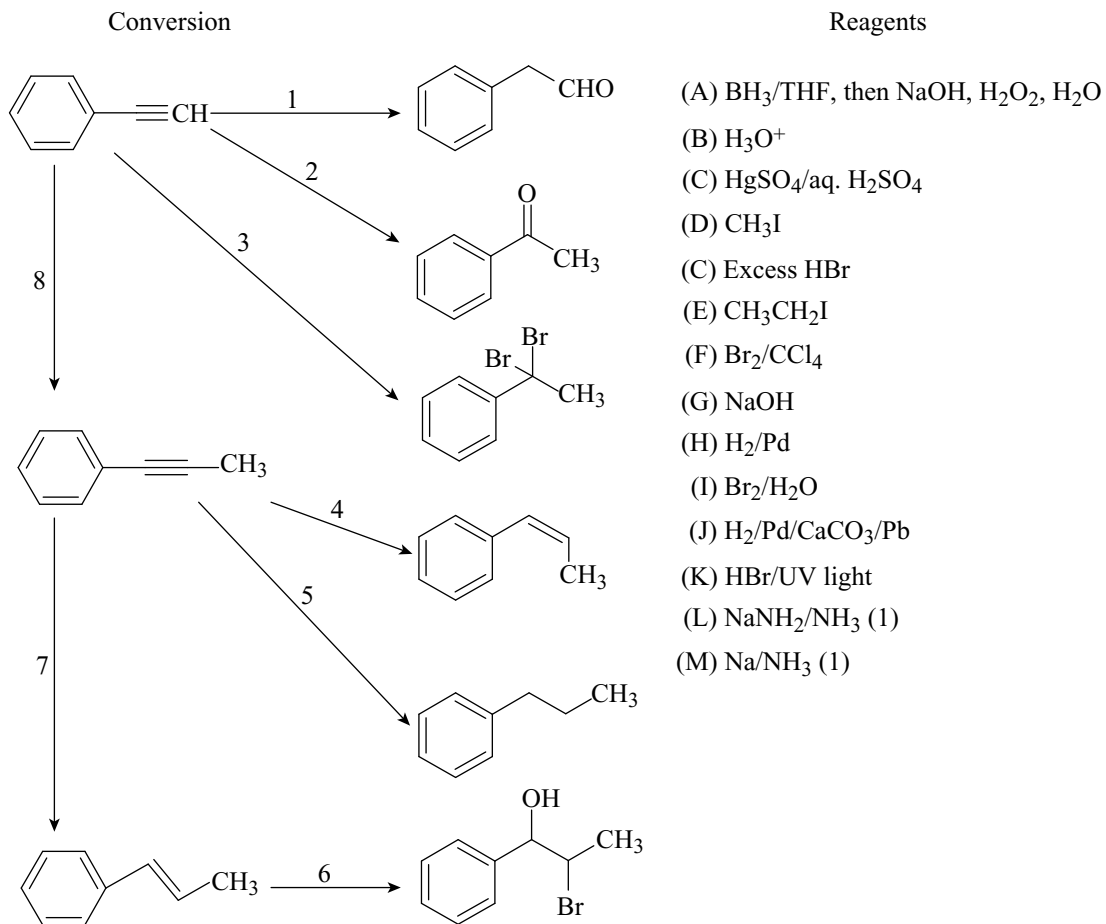


118. Identify total number of 'β'-elimination reactions



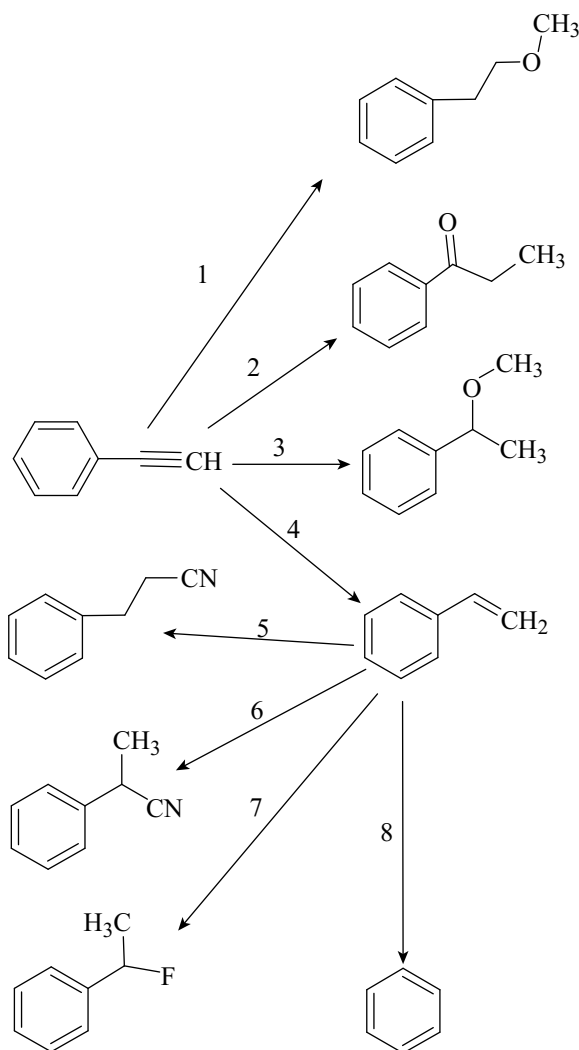
119. Calculate percentage of S_N1 product if (R) -2-chloro butane on reaction with NaOH/ H_2O and acetone gives 98 per cent inverted product.
120. Calculate total number of alkene products when 2-chloro-2-cyclobutyl hexane react with alcoholic KOH and heat.
121. Calculate total number of α -H present in alkene formed when 2, 3-dimethyl butanol react with concentrated H_2SO_4/Δ .

WORKBOOK EXERCISE 1



WORKBOOK EXERCISE 2

Conversion



Reagents

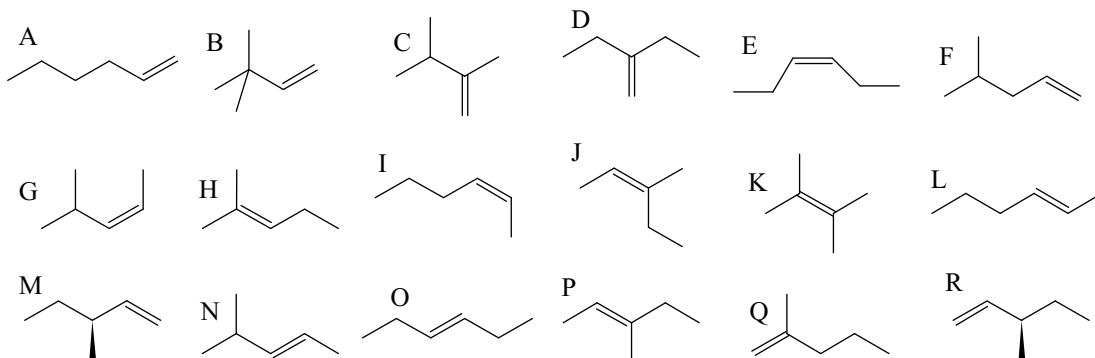
- (A) HgSO_4 , aq. H_2SO_4
- (B) 1. NaNH_2 then add MeI
2. HgSO_4 , aq. H_2SO_4
- (C) 1. NaNH_2 , then add MeI
2. BH_3 , then H_2O_2 , NaOH , H_2O
- (D) 1. BH_3 , then H_2O_2 , NaOH , H_2O
2. Na then MeI
- (E) 1. $\text{H}_2/\text{Lindlar's catalyst}$
2. BH_3 , then H_2O_2 , NaOH , H_2O
3. MeOH
- (F) 1. $\text{H}_2/\text{Lindlar's catalyst}$
2. BH_3 , then H_2O_2 , NaOH , H_2O
3. Na then MeI
- (G) 1. $\text{H}_2/\text{Lindlar's catalyst}$
2. aq. H_2SO_4
3. MeOH
- (H) 1. $\text{H}_2/\text{Lindlar's catalyst}$
2. aq. H_2SO_4
3. Na , then MeI
- (I) 1. aq. H_2SO_4 2. Na , then MeI
- (J) Na in NH_3 (1)
- (K) $\text{H}_2/\text{Lindlar's catalyst}$
- (L) H_2/Pd
- (M) HCN
- (N) 1. H_2/Pd 2. NaCH/DMSO
- (O) 1. aq. H_2SO_4 2. NaCN/DMSO
- (P) 1. HBr , H_2SO_2 2. NaCN/DMSO
- (Q) $\text{HF}/\text{Pyridine}$
- (R) 1. H_2/Pd or Ni 2. NaF/DMSO
- (S) 1. HBr , H_2O_2 2. NaF/DMSO
- (T) 1. HBr (dark, N_2) 2. NaF/EtOH
- (U) 1. HBr (dark, N_2) 2. NaF/DMSO
- (V) 1. HBr , H_2O_2 2. H_2S
- (W) 1. HBr , (dark, N_2) 2. H_2S
- (X) $\text{H}_2\text{S}/\text{EtOH}$
- (Y) 1. HBr (dark, N_2) 2. NaCN/DMSO

WORKBOOK EXERCISE 3

There are 18 alkene isomers with the molecular formula C_6H_{12} , and they are all drawn (A-R) in the box at the bottom of this page. You are presented with five flasks (1 to 5), each one containing a single pure sample of one of these alkenes. In a very long and excruciating lab class one afternoon, you run two separate reactions on each sample (i) ozonolysis (O_3 followed by Me_2S) and (ii) bromination (Br_2/CCl_4). For the ozonolysis reactions, you analyse what kind of stereoisomers (if any) are formed in each reaction. The results of the two reactions for each flask are summarized below that flask. Using your knowledge of alkene reactions, identify each sample (1 to 5) as ONE of the alkenes drawn below (A-R) by writing that letter in the small box below:

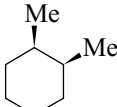
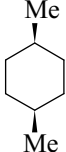
Key					
Racemic mixture (Rac) Diastereoisomers (Dia) Achiral meso compound Achiral, non-meso compound (Ach)	1	2	3	4	5
Number of unique ozonolysis (O_3) products	One	One	One	Two	Two
Reaction with Br_2 in CCl_4 gives	Ach	Rac	Meso	Ach	Dia
IDENTITY OF SAMPLE $\Rightarrow$					

Note: For one of the samples, there are two correct answers, but you only need to indicate one of them.



WORKBOOK EXERCISE 4

Match alkanes with their products.

Alkanes	Total Number of Mono-chloro Products	Total Number of Mono-chloro Fractions After Distillation
(1) 	(A) 6	(A) 2
(2) 	(B) 12	(B) 3
(3) 	(C) 14	(C) 4
(4) 	(D) 16	(D) 5
(5) 	(E) 8	(E) 6
(6) 	(F) 4	(F) 7
(7) 	(G) 5	(G) 8
(8) 	(H) 10	(H) 9
(9) $\begin{array}{c} \text{CH}_3 \\ \\ \text{H}-\text{C}-\text{Cl} \\ \\ \text{CH}_2 \\ \\ \text{CH}_3 \end{array}$	(I) 8	(I) 10
(10) $\begin{array}{c} \text{CH}_3 \\ \\ \text{H}-\text{C}-\text{OMe} \\ \\ \text{CH}_2 \\ \\ \text{CH}_3 \end{array}$	(J) 16	(J) 11
	(K) 7	(K) 12
	(L) 5	(L) 13
	(M) 13	(M) 14
	(N) 3	(N) 16

Answer Keys and Solutions to Workbook Exercises

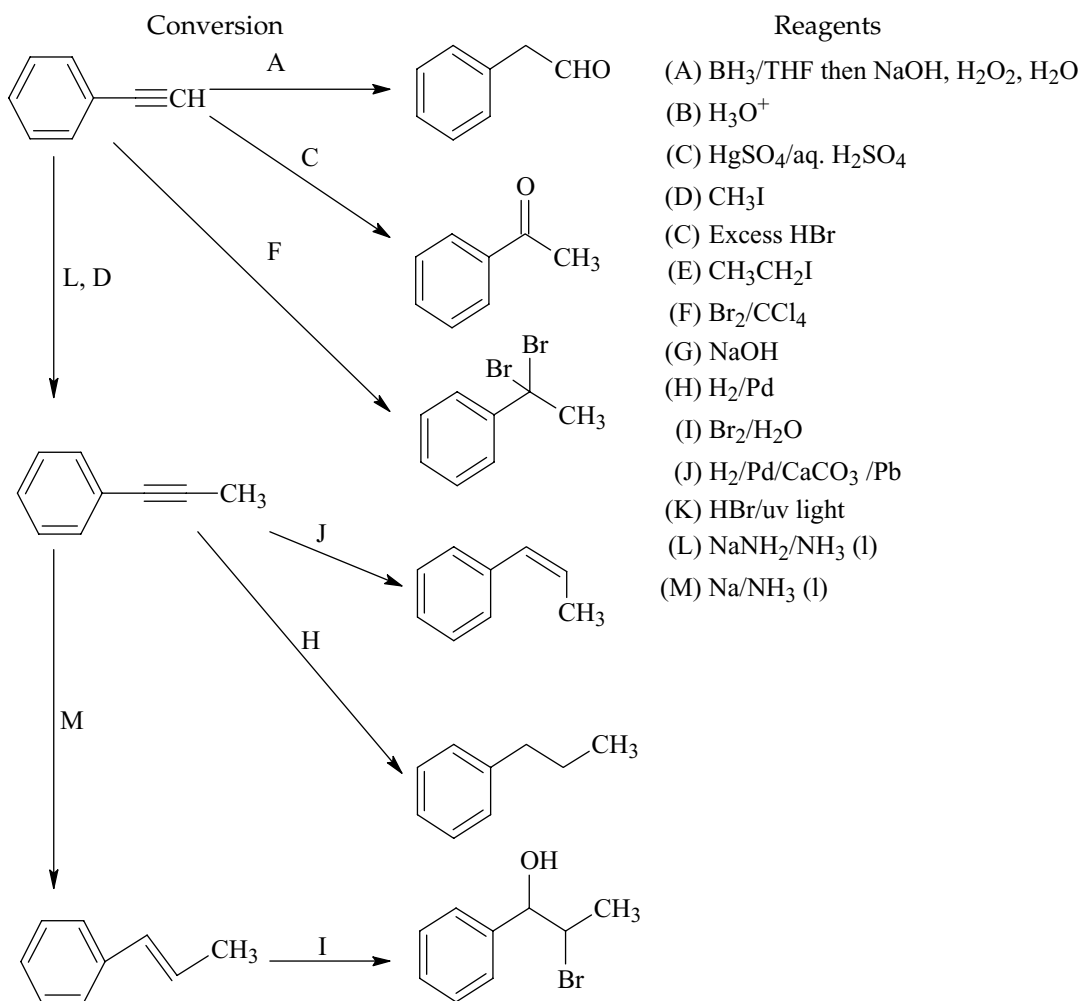
LEVEL 1

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
b	b	c	c	a	d	c	b	d	c	a	b	c	a	d
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
a	d	d	d	d	c	c	d	b	c	b	c	d	d	d
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
b	d	d	b	a	d	c	d	c	b	ab	b	a	c	c
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
b	a	a	d	a	d	d	b	d	a	c	a	b	a	d
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
d	a	d	b	a	a	b	b	d	c	b	b	b	a	a
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
c	b	a	c	d	d	d	a	a	a	b	a	a	a	b
91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
d	a	d	b	c	a	a	b	b	a	b	b	b	d	b
106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
b	d	a	c	d	c	d	c	a	d	c	cd	a	b	c
121	122	123	124	125	126	127	128	129	130					
d	b	d	a	b	b	c	b	bc	d					

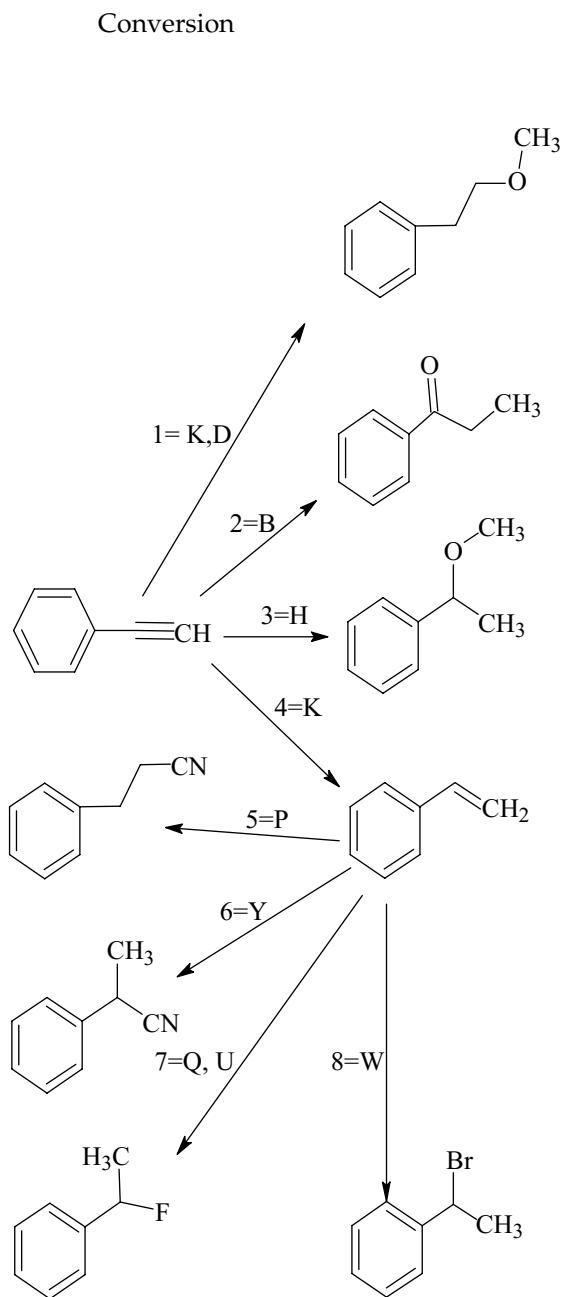
LEVEL 2

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
bd	a	a	a	d	ab	bcd	a	ac	acd	ad	abc	bcd	abc	bc
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
d	abc	ab	ac	a	abd	b	b	a	c	a	c	a	a	c
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
d	d	d	d	abd	cd	b	a	c	c	abd	b	c	c	d
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
d	b	a	c	c	d	d	a	c	a	b	d	d	a	b
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
a		a	bc	d	bc	acd	ab	abcd	abd	bc	abd	c	d	c
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
d	a	a	c	a	d	b	c	a	c	c	a	bcd	b	d
91	92	93	94	95	96	97	98	99	100	101	102	103(a)	103(b)	103(c)
c	c	c	a	a	d	a	b	d	c	ac	c	qs	pqrs	pqr
103(d)	104(a)	104(b)	104(c)	104(d)	105(a)	105(b)	105(c)	105(d)	106(a)	106(b)	106(c)	106(d)	107(a)	107(b)
pqr	ps	qr	ps	pr	s	r	q	p	pqr	pq	st	pr	r	pq
107(c)	107(d)	108(a)	108(b)	108(c)	108(d)	109(a)	109(b)	109(c)	109(d)	110(a)	110(b)	110(c)	110(d)	111(a)
t	ps	pst	pst	r	q	qr	pqr	qqr	pst	q	s	r	p	p
111(b)	111(c)	111(d)	112(a)	112(b)	112(c)	112(d)	113(a)	113(b)	113(c)	113(d)	114(a)	114(b)	114(c)	114(d)
r	q	s	r	p	q	s	s	r	p	q	r	q	s	p
115	116	117	118	119	120	121								
5	8.233	5	6	4	4	12								

WORKBOOK EXERCISE 1



WORKBOOK EXERCISE 2

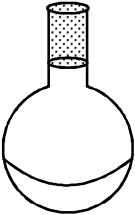
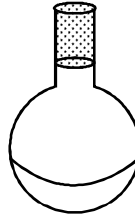


Reagents

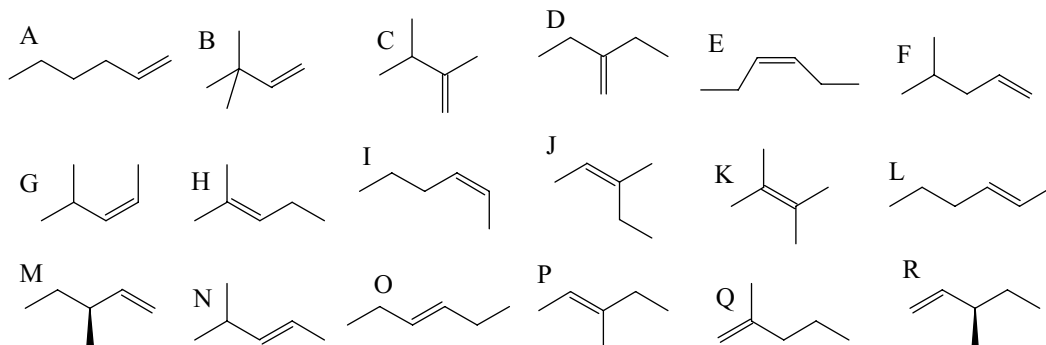
- (A) HgSO_4 , aq. H_2SO_4
- (B) 1. NaNH_2 then add MeI .
2. HgSO_4 , aq. H_2SO_4
- (C) 1. NaNH_2 then add MeI .
2. BH_3 , then H_2O_2 , NaOH , H_2O
- (D) 1. BH_3 , then H_2O_2 , NaOH , H_2O
2. Na then MeI
- (E) 1. H_2 / Lindlar catalyst.
2. BH_3 , then H_2O_2 , NaOH , H_2O
3. MeOH
- (F) 1. H_2 / Lindlar catalyst.
2. BH_3 , then H_2O_2 , NaOH , H_2O
3. Na then MeI
- (G) 1. H_2 / Lindlar catalyst.
2. aq. H_2SO_4
3. MeOH
- (H) 1. H_2 / Lindlar catalyst.
2. aq. H_2SO_4
3. Na then MeI
- (I) 1. aq. H_2SO_4
2. Na then MeI
- (J) Na in NH_3 (l)
- (K) H_2 / Lindlar catalyst.
- (L) H_2 / Pd
- (M) HCN
- (N) 1. H_2 / Pd .
2. NaCN / DMSO
- (O) 1. aq. H_2SO_4
2. NaCN / DMSO
- (P) 1. HBr , H_2O_2
2. NaCN / DMSO
- (Q) HF / Pyridine
- (R) 1. H_2 / Pd or Ni .
2. NaF / DMSO
- (S) 1. HBr , H_2O_2
2. NaF / DMSO
- (T) 1. HBr (dark, N_2)
2. NaF / EtOH
- (U) 1. HBr (dark, N_2)
2. NaF / DMSO
- (V) 1. HBr , H_2O_2
2. H_2S
- (W) 1. HBr (dark, N_2)
2. H_2S
- (X) H_2S / EtOH
- (Y) 1. HBr (dark, N_2)
2. NaCN / DMSO

WORKBOOK EXERCISE 3

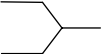
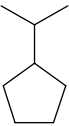
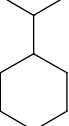
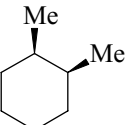
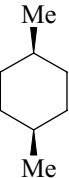
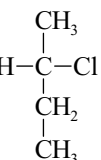
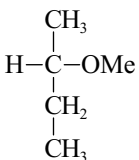
There are 18 alkene isomers with the molecular formula C_6H_{12} , and they are all drawn (A-R) in the box at the bottom of this page. You are presented with five flasks (1 to 5), each one containing a single pure sample of one of these alkenes. In a very long and excruciating lab class one afternoon, you run two separate reactions on each sample (i) ozonolysis (O_3 followed by Me_2S), and (ii) bromination (Br_2/CCl_4). For the ozonolysis reactions, you analyse what kind of stereoisomers (if any) are formed in each reaction. The results of the two reactions for each flask are summarized below that flask. Using your knowledge of alkene reactions, identify each sample (1 to 5) as as ONE of the alkenes drawn below (A-R) by writing that letter in the small box below:

Key					
Racemic mixture (Rac) Diastereoisomers (Dia) Achiral meso compound Achiral, non-meso compound (Ach)	1	2	3	4	5
					
	One	One	One	Two	Two
	Ach	Rac	Meso	Ach	Dia
IDENTITY OF SAMPLE	K	E	O	D	M, R

Note: For one of the samples, there are two correct answers, but you only need to indicate one of them.



WORKBOOK EXERCISE 4

Alkanes		Total Number of Mono-chloro Products	Total Number of Mono-chloro Fractions After Distillation
(1) 	--A,c	(A) 6	(a) 2
(2) 	--E,d	(B) 12	(b) 3
(3) 	--N,a		
(4) 	--H,e	(C) 14	(c) 4
(5) 	--B,f	(D) 16	(d) 5
(6) 	--C,g	(E) 8	(e) 6
(7) 	--C,f	(F) 4	(f) 7
(8) 	--K,d	(G) 5	(g) 8
(9) 	--L,d	(H) 10	(h) 9
(10) 	--K,e	(I) 8	(i) 10
		(J) 16	(j) 11
		(K) 7	(k) 12
		(L) 5	(l) 13
		(M) 13	(m) 14
		(N) 2	(n) 16

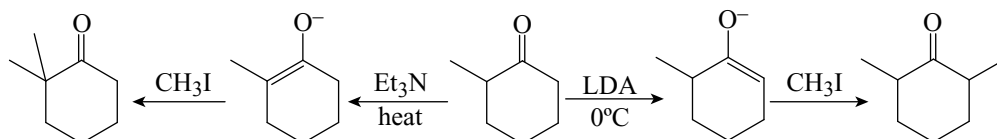
Carbonyl Compounds and Acid Derivatives

5

Question Bank

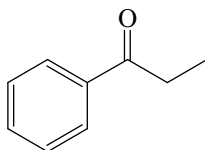
LEVEL 1

- When phenyl ethanoate is alkylated with chloroethane/ AlCl_3 , the para product is the major product. This is because
 - Alkylation usually occurs at the para position.
 - The $-\text{Cl}$ group is a weakly deactivating and an ortho/para director.
 - The $-\text{CO}_2\text{CH}_3$ group is an activator and an ortho/para director.
 - The $-\text{Et}$ group is weakly activating and an ortho/para director.
- Terminal alkynes react with 9-borabicyclononane (9-BBN) and then aq. alkaline hydrogen peroxide to give aldehydes because
 - Acid catalysed electrophilic addition of water follows Markovnikov's rule.
 - Acid catalysed electrophilic addition of water is anti-Markovnikov.
 - The radical addition is anti-Markovnikov.
 - Steric and electronic effects cause the boron to the terminal carbon.
- In general, ketones are more reactive towards nucleophiles than esters because
 - The α -protons of a ketone are more acidic than those of an ester.
 - The alkyl group in a ketone is an electron donating group due to hyperconjugation.
 - Alkoxy ($\text{RO}-$) groups are sterically larger than the related alkyl group.
 - Alkoxy ($\text{RO}-$) groups are stronger electron donating than alkyl groups via resonance.
- When 2-methylcyclohexanone reacts with a base and is then treated with an electrophile such as methyl iodide the product depends on the reaction conditions, see below. At high temperature, the major product is the geminal dimethyl compound. This is because

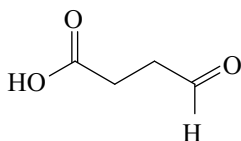


- Alkyl groups are weak electron donors due to inductive effects and hyperconjugation.
 - The enolate formed at high temperature is more stable.
 - diisopropyl amide (LDA) is a much stronger base than Et_3N .
 - Lithium diisopropyl amide (LDA) promotes an anti-Markovnikov type addition.
- The reaction of a Grignard reagent with a carboxylic acid does not give a secondary alcohol. This is because
 - Grignard reagents only react with the aldehydes, ketones, esters and epoxides.
 - The carboxylic acid is too sterically hindered to react.

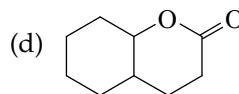
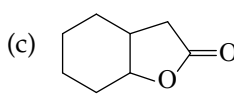
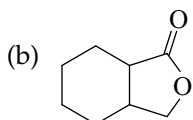
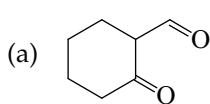
- (c) The carboxylic acid is not electrophilic enough to react.
 (d) The Grignard reagent is a base, so an acid–base reaction occurs.
6. When 1-phenylpropyne reacts with $\text{H}_2\text{O}/\text{HgSO}_4/\text{H}_2\text{SO}_4$, the major product is propiophenone shown below. This is because

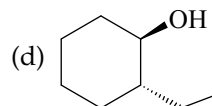
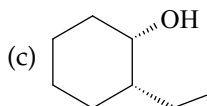
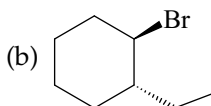
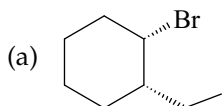
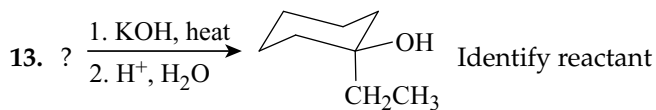
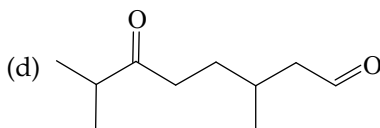
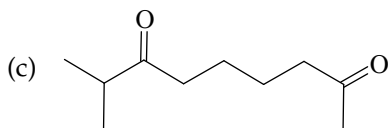
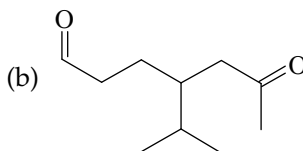
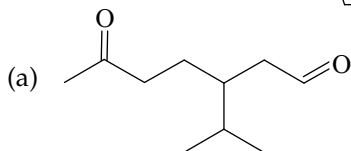
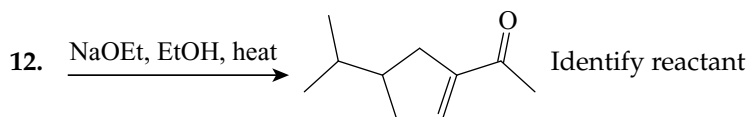
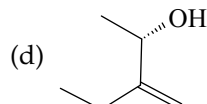
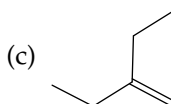
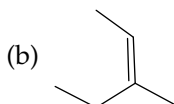
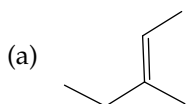
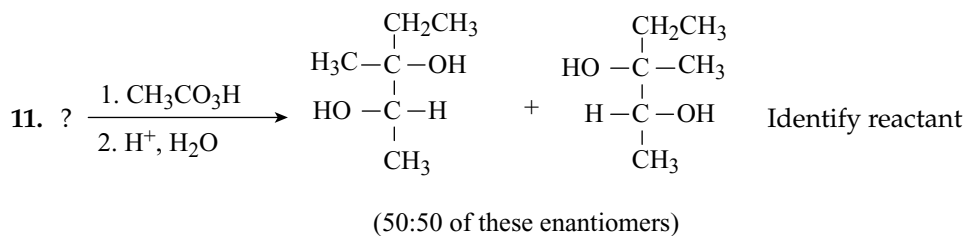
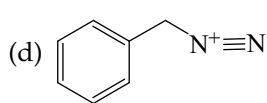
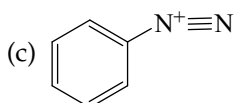
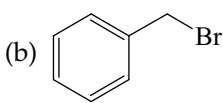
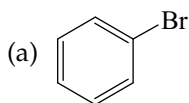
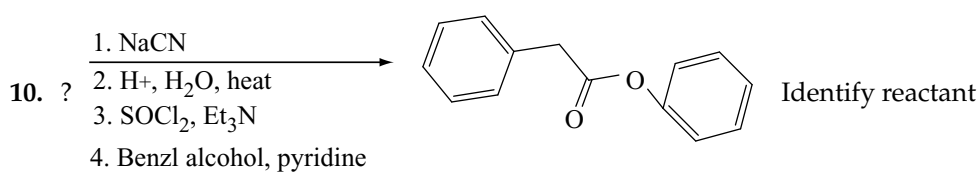


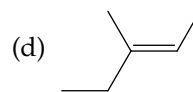
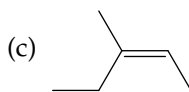
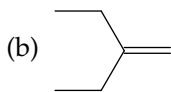
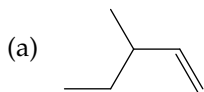
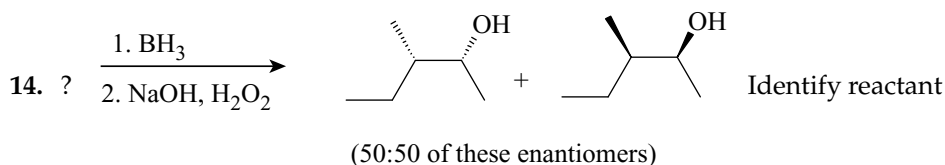
- (a) Alkyl groups are weak electron donors due to inductive effects and hyperconjugation.
 (b) Carbonyl groups are electron withdrawing groups due to resonance.
 (c) Phenyl groups can stabilise positive charge by resonance.
 (d) The reaction is controlled by steric factors.
7. When methyl benzoate is nitrated with $\text{HNO}_3/\text{H}_2\text{SO}_4$, the meta product is the major product. This is because
- (a) The $-\text{CO}_2\text{CH}_3$ group is a meta director.
 (b) The $-\text{OCH}_3$ is a meta director.
 (c) The $-\text{NO}_2$ group is deactivating and a meta director.
 (d) Nitration usually occurs at the meta position.
8. A chemist attempted to reduce the compound shown below with NaBH_4 , but could not isolate the desired product 4-hydroxybutanoic acid. This is because



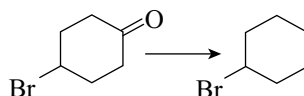
- (a) NaBH_4 reduced both the carboxylic acid and the aldehyde.
 (b) NaBH_4 is not reactive enough to reduce the carboxylic acid or the aldehyde.
 (c) An intermediate was formed that reacted to give a cyclic acetal.
 (d) An intermediate was formed that reacted to give a cyclic ester.
9. ? $\xrightarrow{\begin{array}{l} 1. \text{LDA then methyl iodide} \\ 2. \text{LiAlH}_4 \text{ then } \text{H}_3\text{O}^+ \\ 3. \text{TsOH, propanone, heat} \end{array}}$ Identify reactant







15. Appropriate reagent for the following conversion is



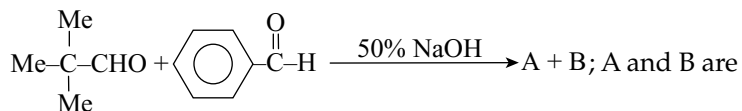
(a) NH_2NH_2 ; EtOH/OH^-

(b) Red P + I_2

(c) $\text{Zn-Hg}/\text{HCl}$

(d) All of them

16. In the given reaction



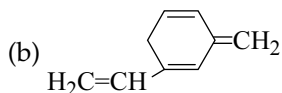
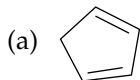
(a) $\text{Me}_3\text{C}-\text{CH}_2\text{OH} + \text{Ph}-\text{COO}^-$

(b) $\text{Me}_3\text{C}-\text{COO}^- + \text{Ph}-\text{CH}_2\text{OH}$

(c) $\text{Me}-\overset{\text{Me}}{\underset{\text{O}}{\parallel}{\text{C}}}-\text{Me} + \text{Ph}-\text{CH}_2\text{OH}$

(d) $\text{Me}-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{Me} + \text{Ph}-\overset{\text{CH}_3}{\underset{\text{OH}}{\text{CH}}}$

17. Which of the following gives glyoxal as one of the product on ozonolysis



(c) $\text{H}_3\text{C}=\text{CH}-\text{CH}=\text{CH}_2$

(d) Both (a) and (c)

18. The correct statement is

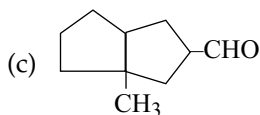
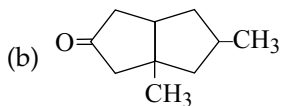
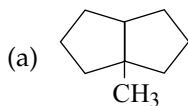
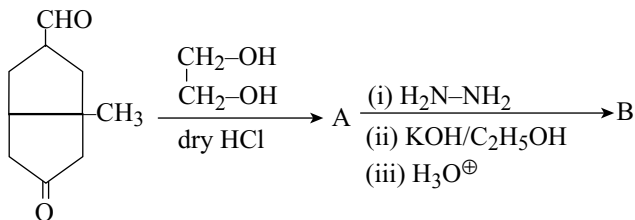
(a) chloral forms stable hydrate

(b) methanal is more reactive than ethanal towards nucleophilic addition

(c) protonated carbonyl group is more reactive towards KCN

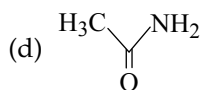
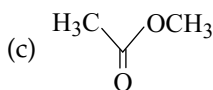
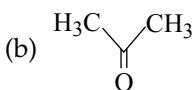
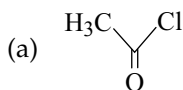
(d) all of them

19. In the given reaction sequence, the B is



(d) none of these

20. The maximum reactive towards the nucleophilic attack is



21. Match column I (reaction) with column II (name of the reaction) and select the correct answer using the codes given.

Column I

- (a) $2\text{C}_6\text{H}_5\text{CHO} \xrightarrow{20\% \text{ NaOH}}$
 (b) $\text{C}_6\text{H}_5\text{CHO} + \text{HCHO} \xrightarrow{20\% \text{ NaOH}}$
 (c) $2\text{C}_6\text{H}_5\text{CHO} \xrightarrow{\text{KCN}/(\text{alc.})}$
 (d) $\text{C}_6\text{H}_5\text{CHO} + (\text{CH}_3\text{CO})_2\text{O} \xrightarrow[\Delta]{\text{CH}_3\text{COONa}}$

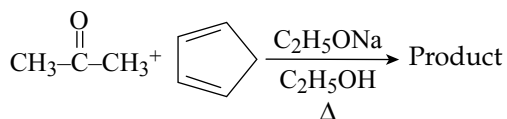
Column II

- (1) Benzoin condensation
 (2) Self-Cannizzaro reaction
 (3) Cross Cannizzaro reaction
 (4) Perkin reaction

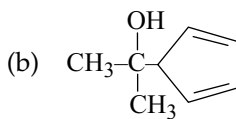
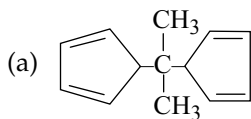
- (a) a - 1, b - 2, c - 3, d - 4
 (c) a - 1, b - 3, c - 4, d - 2

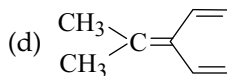
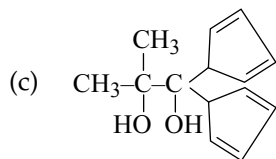
- (b) a - 2, b - 3, c - 1, d - 4
 (d) a - 2, b - 3, c - 4, d - 1

22. In the reaction

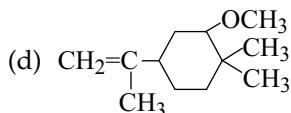
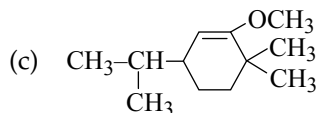
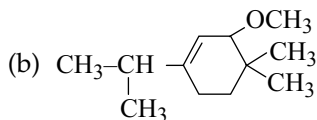
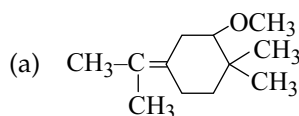
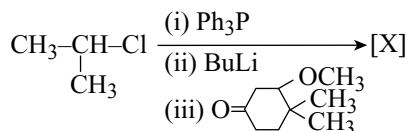


the product is





23. The product of the given reaction is



24. Which of the following gives positive Fehling's solution test

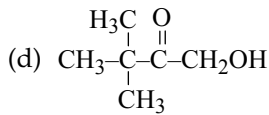
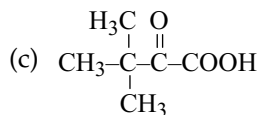
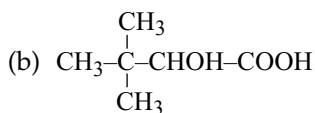
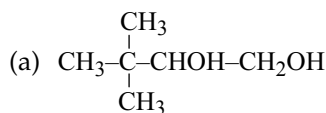
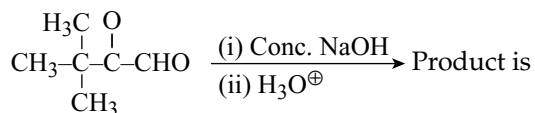
(a) Ph-CHO

(b) CH₃CHO

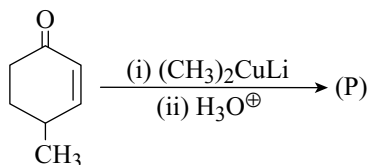
(c) CCl₃CHO

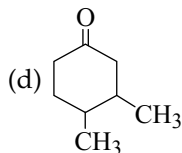
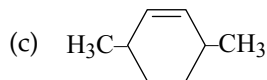
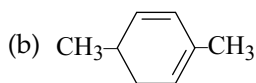
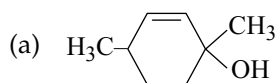
(d) All of them

25. In the given reaction

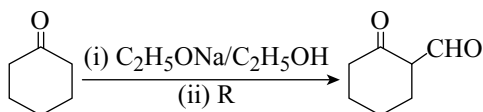


26. In the given reaction, (P) will be





27. In the given reaction



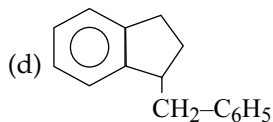
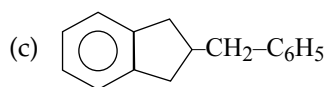
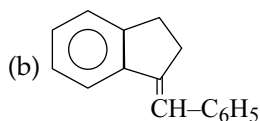
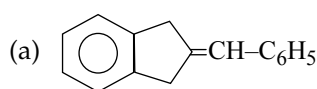
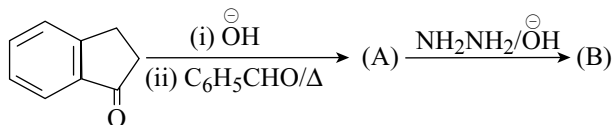
(R) would be

- (a) HCHO (b) CH₃CHO (c) HCOOC₂H₅ (d) (COOC₂H₅)₂

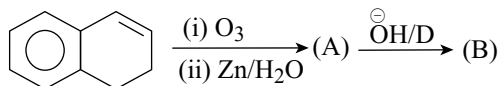
28. Aldehydes and ketones react with α -bromo esters and zinc in ether or benzene medium to yield

- (a) α -keto ester (b) β -hydroxy ester (c) γ -keto acids (d) β -keto ester

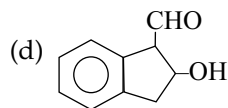
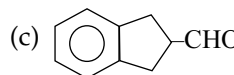
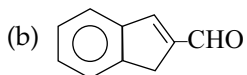
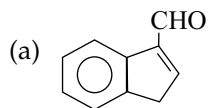
29. In the given reaction sequence, compound (B) is



30. In the given reaction sequence

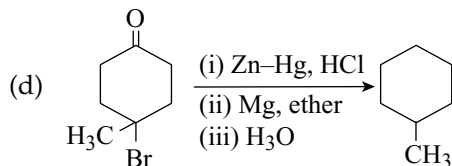
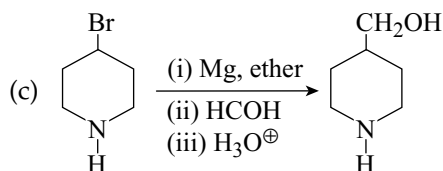
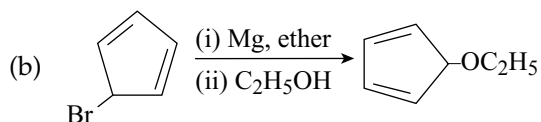
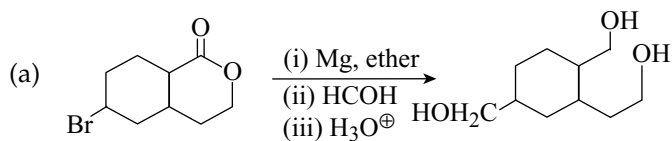


compound (B) is

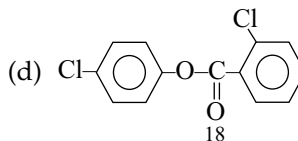
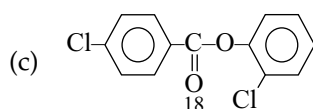
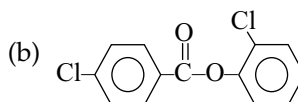
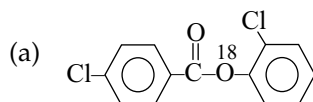
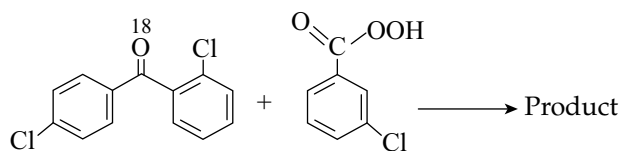


31. Two isomeric ketones, 3-hexanone and 2-pentanone can be distinguished by
 (a) I_2/NaOH (b) NaHSO_3 (c) NaCN/HCl (d) 2,4-DNP
32. Silver mirror test with Tollen's reagent is given by
 (a) $\text{C}_6\text{H}_5\text{CHO}$ (b) Ph-OH
 (c) $\text{C}_6\text{H}_5\text{-CH}_2\text{-}\overset{\text{O}}{\parallel}\text{C-CH}_2\text{OH}$ (d) CH_3CHO

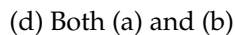
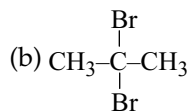
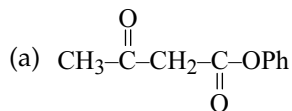
33. Which of the following is a correct conversion?



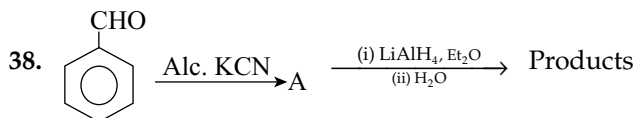
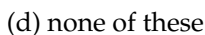
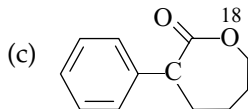
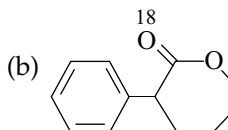
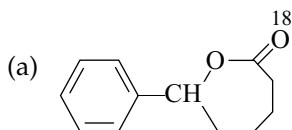
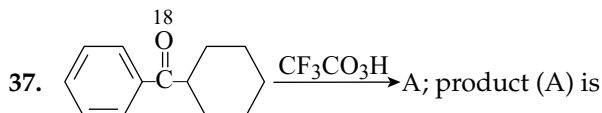
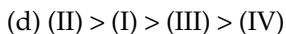
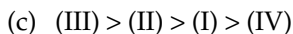
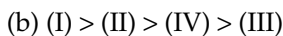
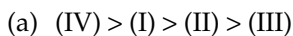
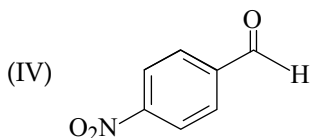
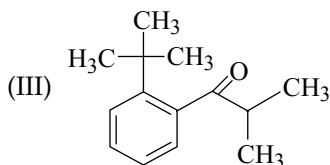
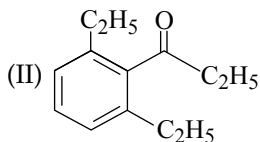
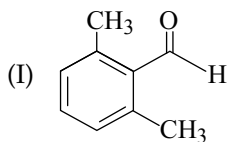
34. For the given reaction, the product is



35. Which one of the following does not respond positive to iodoform test even on heating?

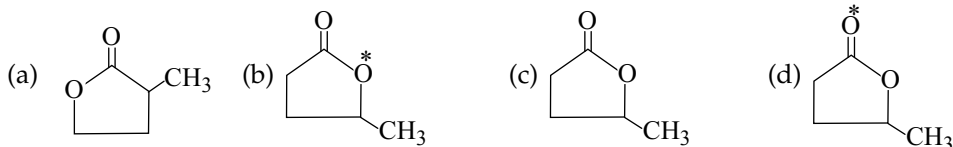
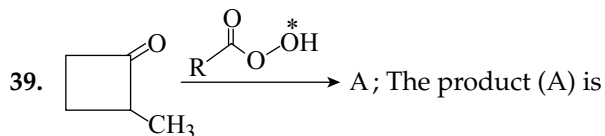


36. The rate of reduction of the following carbonyl compounds by NaBH_4



The number of possible stereoisomers in the product mixture is

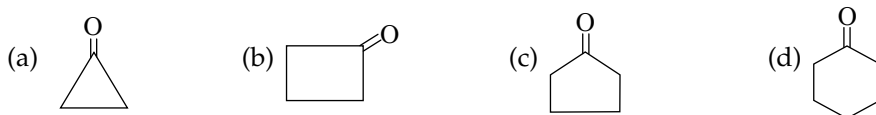




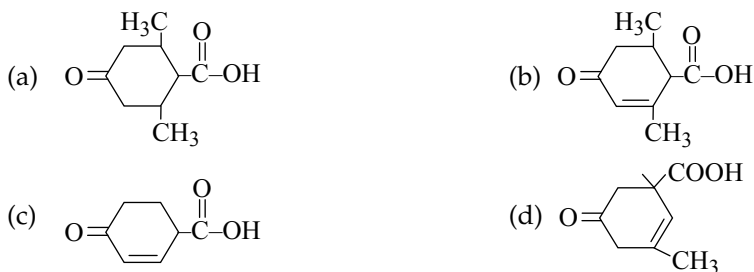
40. The incorrect statement amongst the following is

- The order of reduction reaction by LiAlH_4 and that of the reduction by NaBH_4 each is two
- Al-H bond is more ionic than B-H bond and hence LiAlH_4 can produce larger concentration of hydride ion than that of NaBH_4
- LiAlH_4 cannot reduce $-\text{NO}_2$ group
- In spite of very small rate constant of reduction of any carbonyl function other than aldehydes and ketones, the rate of reduction with LiAlH_4 becomes appreciable due to large concentration of hydride ion

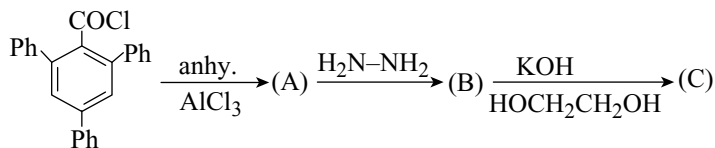
41. Which of the following ketones has the largest equilibrium constant for addition of water?

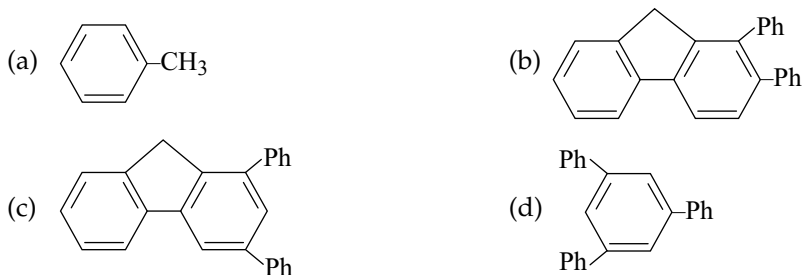


42. Ethyl acetoacetate react with $\text{CH}_3\text{C}(=\text{O})\text{H}$ in the presence of base/ Δ followed by acidic hydrolysis and subjected to heat, product formed is



43. Identify the product (C) of the given sequence of reaction





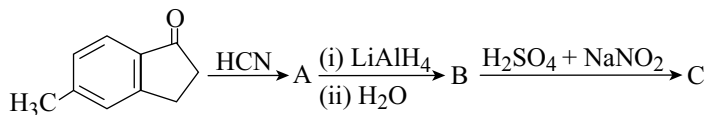
44. Which of the following gives positive Fehling's solution test



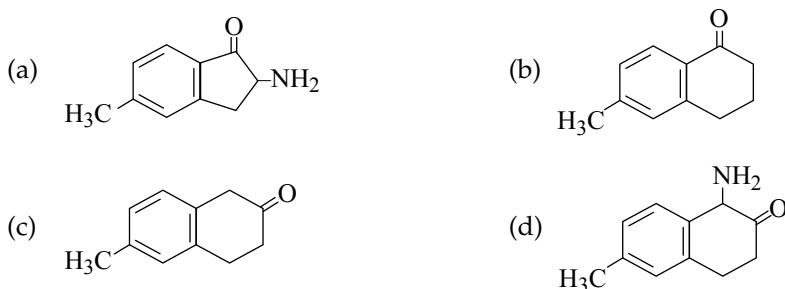
45. Which of the following geminal diol is most unstable?



46. In the given reactions, sequence



The compound C is



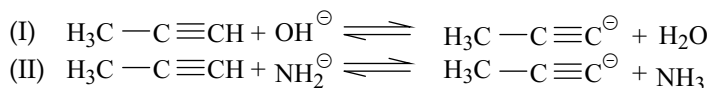
47. What will be the pH of an acetate-acetic acid solution when the ratio of $[\text{CH}_3\text{CO}_2^-]/[\text{CH}_3\text{CO}_2\text{H}]$ is 10? (A table of pK_a data is given below.)

- (a) 5.76
(b) 4.76
(c) 3.76
(d) 1.76

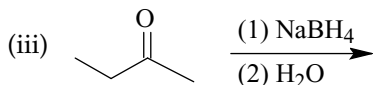
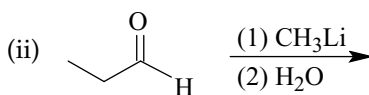
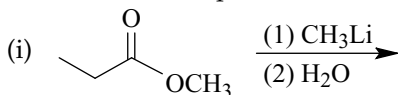
Some useful	pK_a values
$\text{CH}_3\text{CO}_2\text{H}$	4.76
H_3PO_4	2.2
H_2OPO_4	7.2
HPO_4^{2-}	12.4

48. If the internal pH of a muscle cell is 6.8. Which one of the following statements about the concentrations of PO_4^{3-} , HPO_4^{2-} , $\text{H}_2\text{PO}_4^{1-}$, and H_3PO_4 is true?
(Hint: use the table of pK_a s given in the previous question.)
- The species with the highest concentration is H_3PO_4 .
 - The species with the highest concentration is $\text{H}_2\text{PO}_4^{1-}$.
 - The species with the highest concentration is PO_4^{3-} .
 - The species with the highest concentration is HPO_4^{2-} .
49. Choose the incorrect statement about the following acid/base reactions involving propyne and its anion propynide.

	pK_a
H_2O	16
$\text{H}_3\text{C}-\text{C}\equiv\text{CH}$	25
NH_3	34

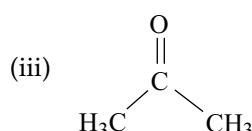
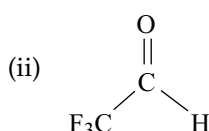
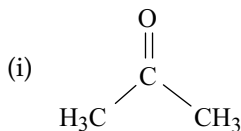


- The equilibrium in equation (i) lies to the left.
 - The equilibrium in equation (ii) lies to the right.
 - In reaction (ii) ammonia, NH_3 , acts as a base.
 - Propyne, $\text{CH}_3\text{C}\equiv\text{CH}$, is a stronger acid than ammonia, NH_3 .
50. Choose among the following reactions those that will give a racemic mixture of two enantiomers as the products.



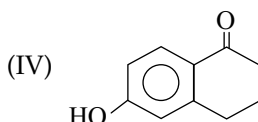
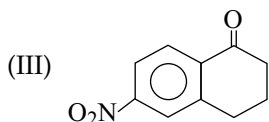
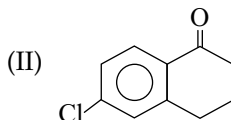
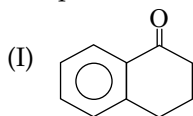
- i
- ii
- iii
- ii + iii

51. Choose the order that has the following carbonyl groups correctly arranged with respect to increasing reactivity with NaBH_4 .



- $\text{i} < \text{iii} < \text{ii}$
 - $\text{i} < \text{ii} < \text{iii}$
 - $\text{ii} < \text{i} < \text{iii}$
 - $\text{iii} < \text{i} < \text{ii}$
- increasing acidity $\rightarrow$ increasing acidity $\rightarrow$ increasing acidity $\rightarrow$ increasing acidity $\rightarrow$

52. Which of the following is the correct order for hydrate formation of the given compounds?

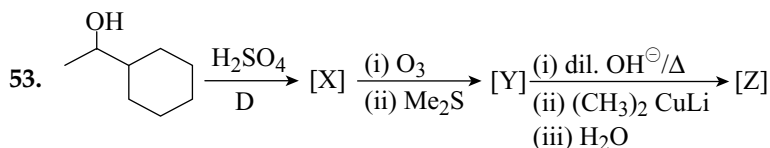


(a) (I) > (II) > (III) > (IV)

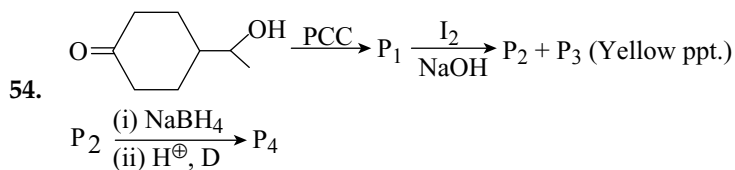
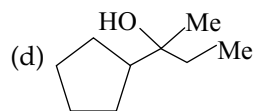
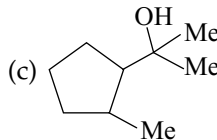
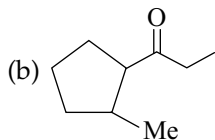
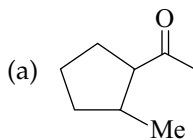
(b) (IV) > (I) > (II) > (III)

(c) (III) > (II) > (I) > (IV)

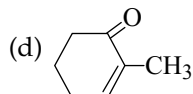
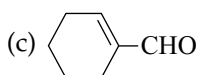
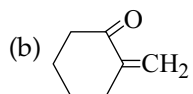
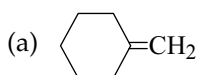
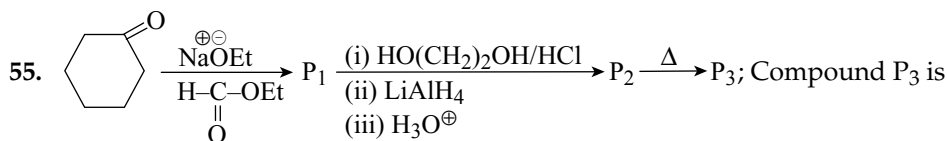
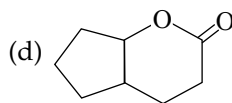
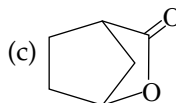
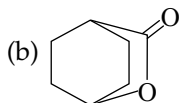
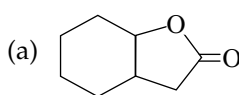
(d) (III) > (I) > (IV) > (II)



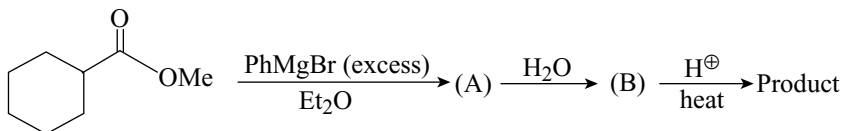
In the above sequence, compound [Z] is

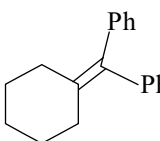
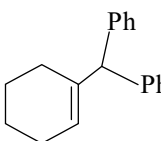
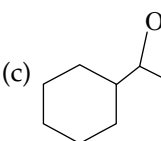
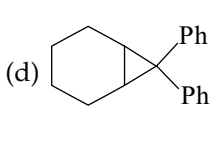


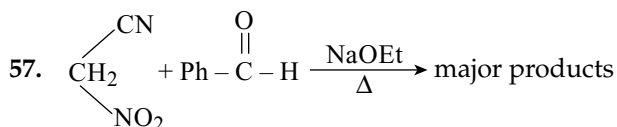
Final compound P₄ is

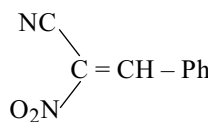
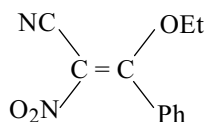
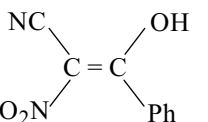


56. Product "P" obtained by the reaction

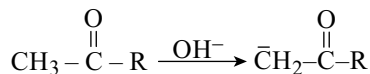


- (a)  (b)  (c)  (d) 



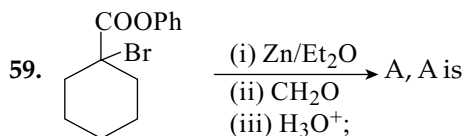
- (a)  (b)  (c)  (d) None of these

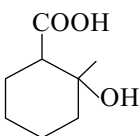
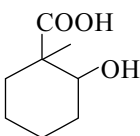
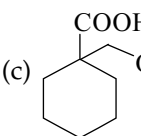
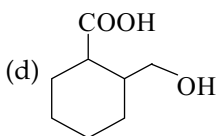
58. Aldehydes and ketones having α -hydrogen convert into carbanion in the presence of base as follows



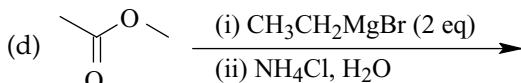
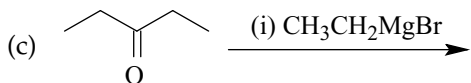
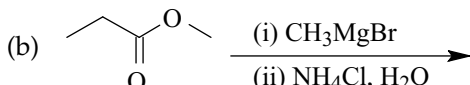
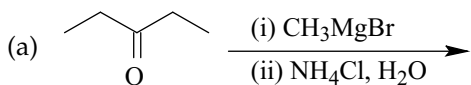
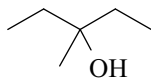
This carbanion is stable because of

- (a) Presence of -I group ($-\text{CO}-$)
 (b) Delocalisation of negative charge by pi bond
 (c) Presence of +I group (i.e., R) on $-\text{CO}-$
 (d) Both (a) and (b)

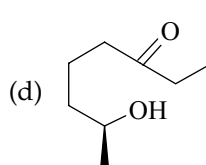
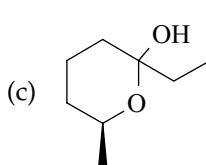
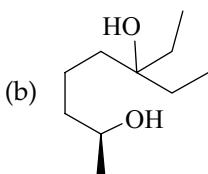
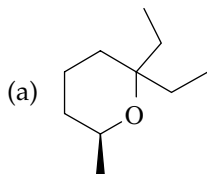
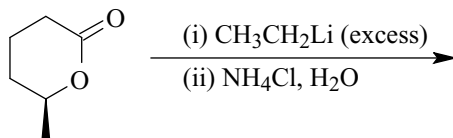


- (a)  (b)  (c)  (d) 

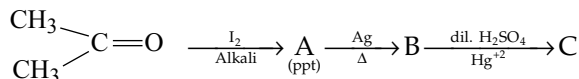
60. Choose the reaction that would not be successful for the preparation of the following alcohol.



61. Choose the major product of the following reaction sequence.



62. The compounds A, B and C in the reaction sequence

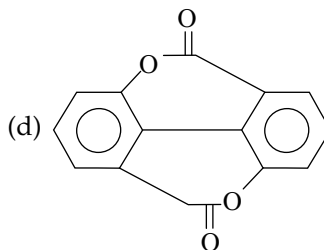
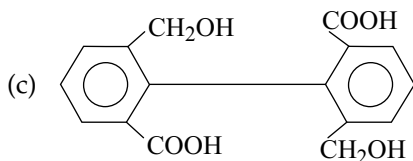
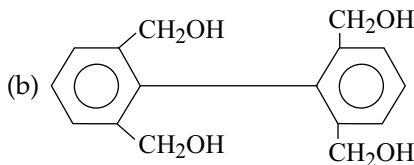
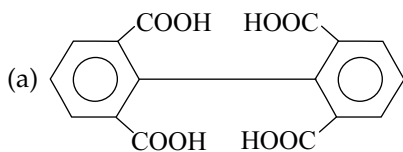


are given by the set

- (a) Iodoform, ethylene, ethyl alcohol
(c) Iodoform, propyne, acetone

- (b) Iodoform, acetylene, acetaldehyde
(d) Iodoform, 2-propanol, propanone

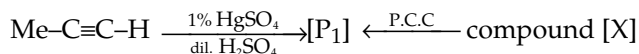
63. Any one of the product formed is

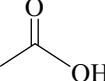
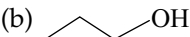
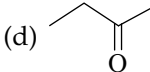


64. A compound "A" formula of $C_3H_6Cl_2$ on reaction with alkali can give "B" of formula C_3H_6O or "C" of formula C_3H_4 . "B" on oxidation gave a compound of the formula $C_3H_6O_2$. "C" with dilute H_2SO_4 containing Hg^{2+} ion gave "D" of formula C_3H_6O , which with bromine and alkali gave the sodium salt of $C_2H_4O_2$. Then "A" is

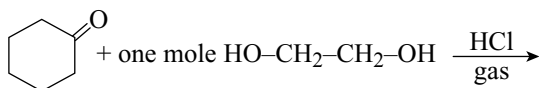
- (a) $CH_3CH_2CHCl_2$ (b) $CH_3CCl_2CH_3$
(c) $CH_2ClCH_2CH_2Cl$ (d) $CH_3CHClCH_2Cl$

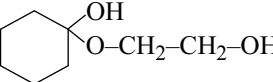
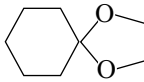
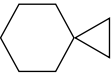
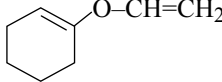
65. Identify compound [X] in the given reaction



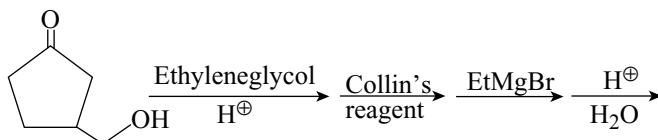
- (a)  (b)  (c)  (d) 

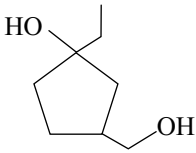
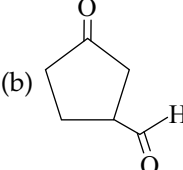
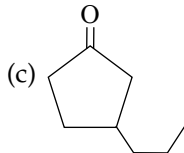
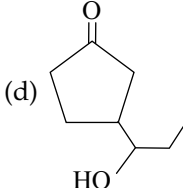
66. What is the major product obtained from the following reaction



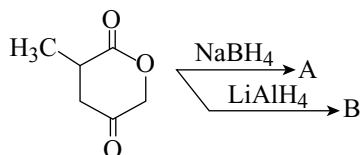
- (a)  (b) 
(c)  (d) 

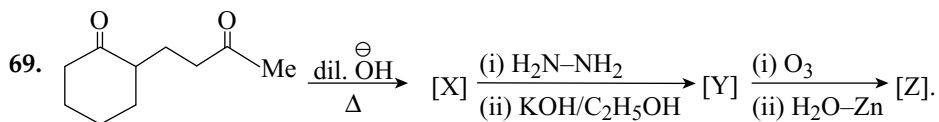
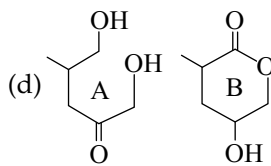
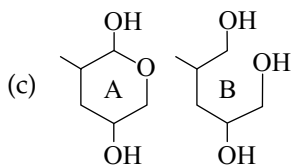
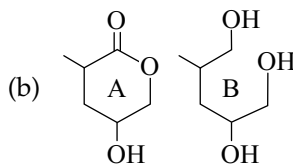
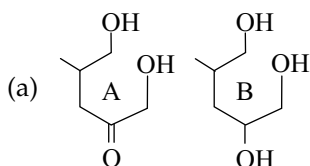
67. Give the product from the following reaction sequence



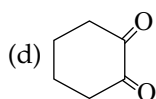
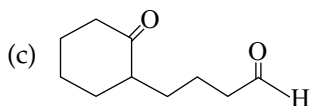
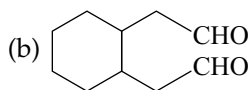
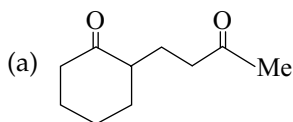
- (a)  (b)  (c)  (d) 

68. Predict the major organic product from the following reaction

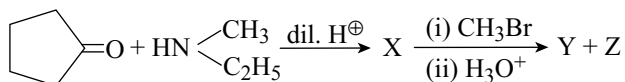




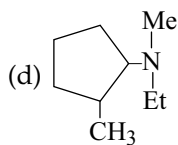
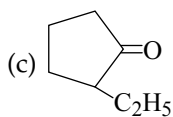
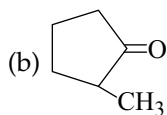
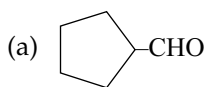
The product (Z) is



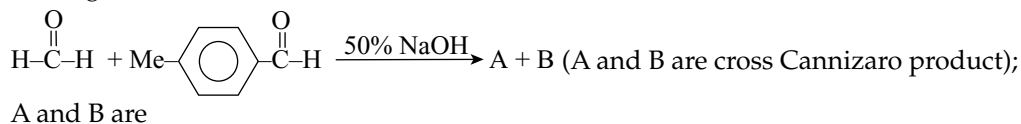
70. The final product (Y) of the following reaction sequence is



Z + Hinsberg reagent $\xrightarrow{\text{KOH}}$ (No dissolution/solid residue)

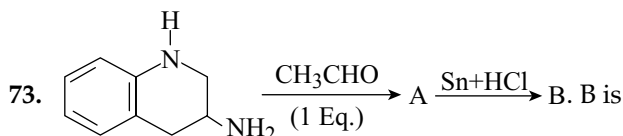
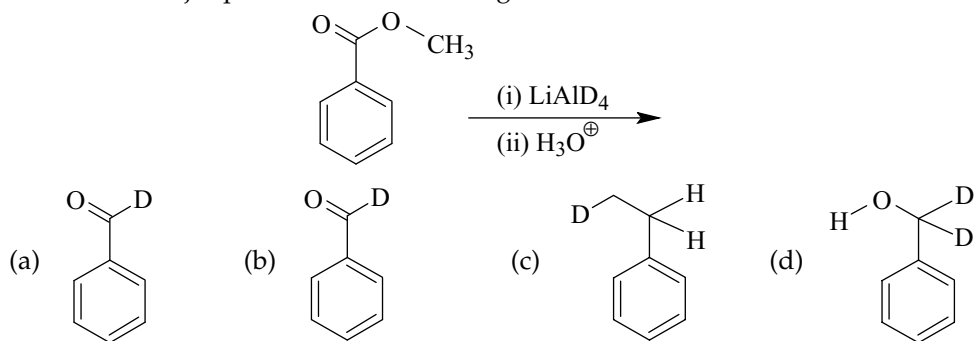


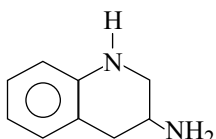
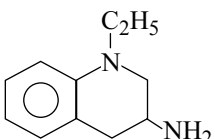
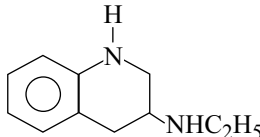
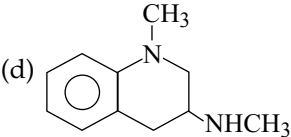
71. In the given reaction



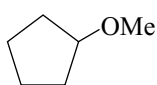
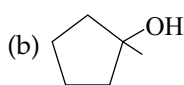
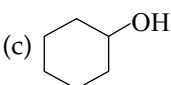
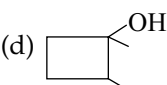
- (a) $\text{HCOO}^- + \text{Me}-\text{C}_6\text{H}_4-\text{CH}_2\text{OH}$ (b) $\text{CH}_3\text{OH} + \text{Me}-\text{C}_6\text{H}_4-\text{COO}^-$
 (c) $\text{HCOO}^- + \text{Me}-\text{C}_6\text{H}_4-\text{COO}^-$ (d) $\text{CH}_3\text{OH} + \text{Me}-\text{C}_6\text{H}_4-\text{CH}_2\text{OH}$

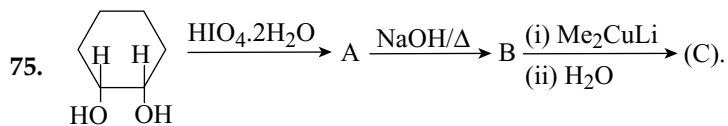
72. Choose the major product of the following reaction.



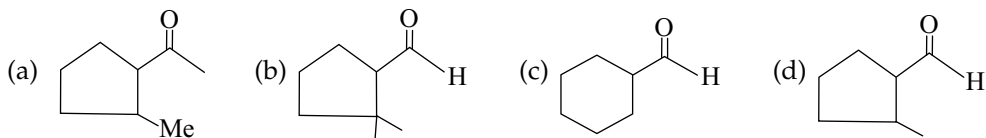
- (a)  (b) 
 (c)  (d) 

74. An organic compound A ($\text{C}_6\text{H}_{12}\text{O}$) neither decolourise bromine water nor changes the colour of acidic dichromate solution. A on heating with H_2SO_4 produces an alkene which on oxidative ozonolysis gives B ($\text{C}_6\text{H}_{10}\text{O}_3$), which gives a yellow precipitate with NaOH/I_2 . The most probable structure of A is

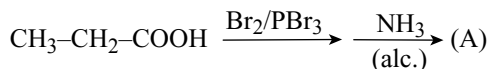
- (a)  (b)  (c)  (d) 



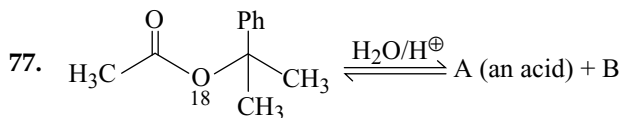
Identify "(C)" product



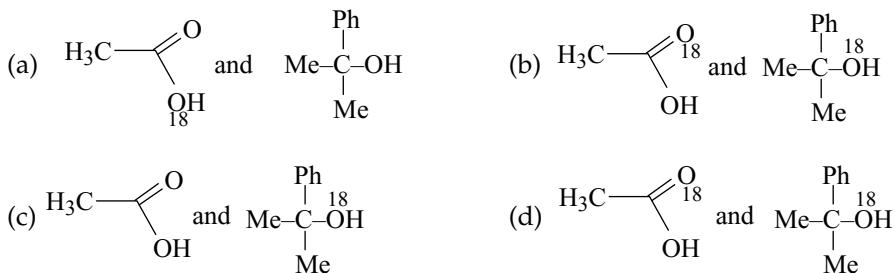
76. In the given reaction



(a) would be

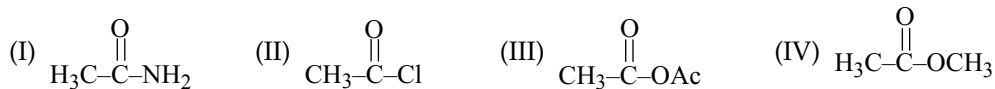


where A and B are, respectively



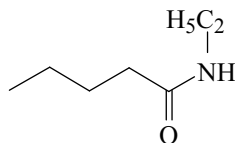
(a) H_2/Pd (b) LiAlH_4 (c) NaBH_4 (d) CH_3MgBr

79. Arrange in increasing order of reactivity towards nucleophile



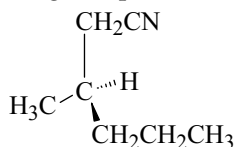
- (a) (II) > (IV) > (I) > (III) (b) (II) > (III) > (IV) > (I)
 (c) (III) > (II) > (I) > (IV) (d) (III) > (IV) > (II) > (I)

80. What is the name of the following compound?



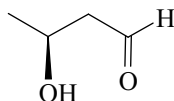
- (a) trans-N-ethyl-3-pentenamide (b) trans-1-ethyl-3-pentenamide
 (c) trans-N-ethyl-4-pentenamide (d) trans-1-ethyl-4-pentenamide

81. What is the name of the following compound?



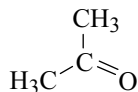
- (a) (R)-3-methylhexaneamine (b) (S)-3-methylhexaneamine
 (c) (R)-3-methylhexanenitrile (d) (S)-3-methylhexanenitrile

82. What is the name of the following compound?



- (a) (R)-3-hydroxybutanal (b) (S)-3-hydroxybutanal
 (c) (R)-3-hydroxybutanol (d) (S)-3-hydroxybutanol

83. What is not the name of the following compound?

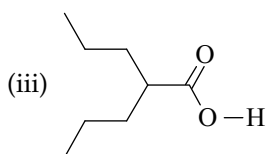
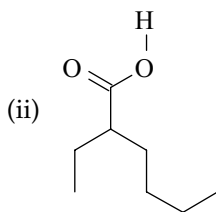
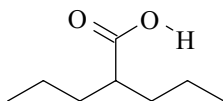
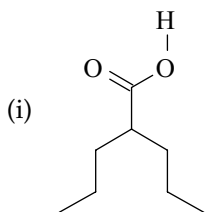


- (a) acetone (b) dimethyl ketone (c) 2-oxoketone (d) propanone

84. Which of the following compound has the highest boiling point?

- (a) $\text{CH}_3\text{CH}_2\text{CH}_3$ (b) $\text{CH}_3\text{CO}_2\text{H}$ (c) CH_3COCH_3 (d) $\text{CH}_3\text{CH}_2\text{OH}$

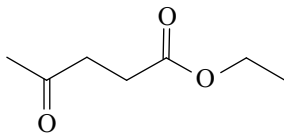
85. Valproic acid found clinical use as an anticonvulsant and mood-stabilizing drug that has proven useful in the treatment of epilepsy, bipolar disorder, and other problems. Choose the constitutional isomer(s) of valproic acid.



valproic acid

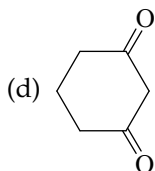
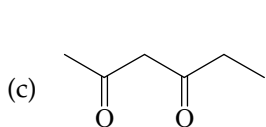
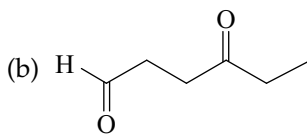
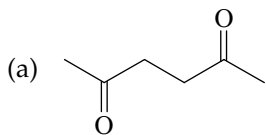
- (a) i (b) ii (c) iii (d) i and ii

86. What is the name of the following compound?

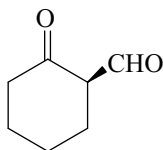


- (a) ethyl 4-oxopentanoate (b) ethyl ester methyl ketone
(c) ethyl 4-ketonepentanoate (d) ethyl 4-ketone pentyl ester

87. What is the structure of 2,4-hexanedione?



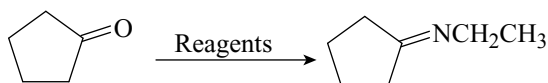
88. What is the name of the following compound?



- (a) (R)-2-formylcyclohexanone
- (b) (R)-2-aldehydecyclohexanone
- (c) (S)-2-formylcyclohexanone
- (d) (S)-2-aldehydecyclohexanone

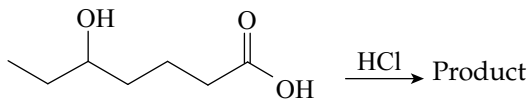
89. If only one equivalent of NH_3 (one mole of NH_3 reacts with one mole of cyclohexyl chloride) was used, the reaction may not go to completion. Why?
- (a) NH_3 is not a strong nucleophile. Therefore, more NH_3 is needed.
 - (b) Cl is not a good leaving. Therefore, more NH_3 is needed.
 - (c) Cyclohexyl group presents large steric hindrance. Therefore, more NH_3 is needed.
 - (d) NH_3 will be protonated by HCl as reaction proceeds. Therefore, the reaction will be incomplete.

90. What could be the best reagents for the following reaction?



- (a) H_2NEt and $\text{CH}_3\text{CO}_2\text{H}$
- (b) H_2NEt and HCl
- (c) H_2NEt and NaOH
- (d) H_2NEt and $\text{Na}^+ - \text{OCH}_3$

91. What could be the product for the following reaction?

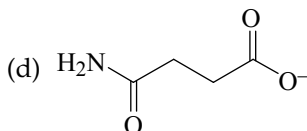
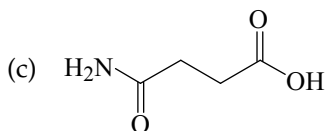
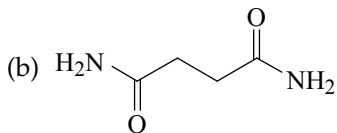
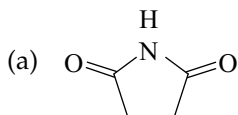
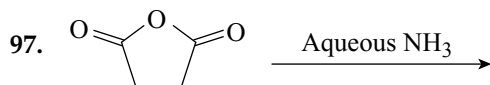
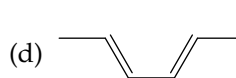
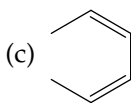
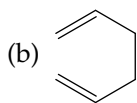
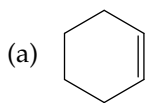
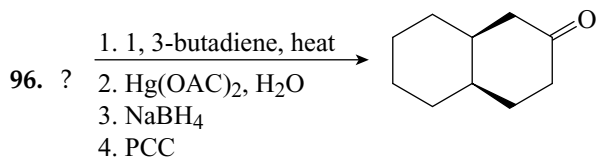
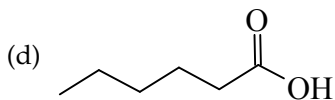
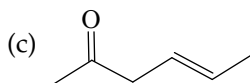
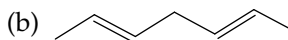
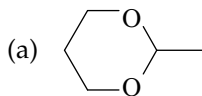
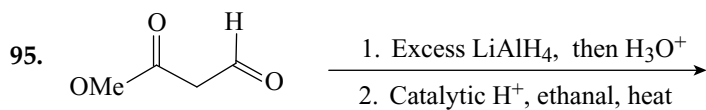
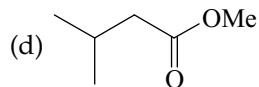
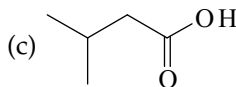
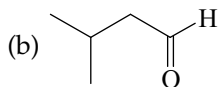
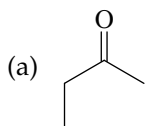
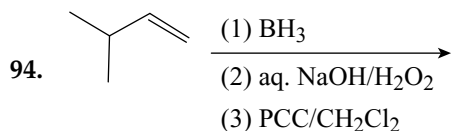
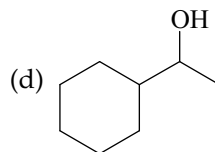
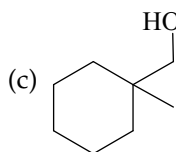
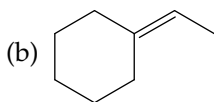
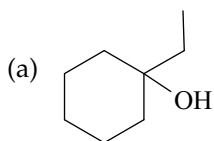


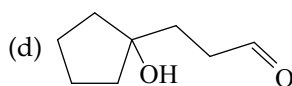
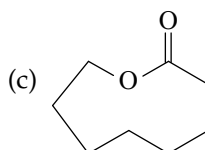
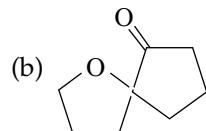
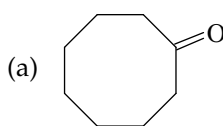
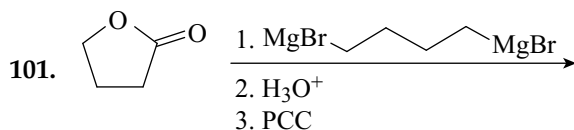
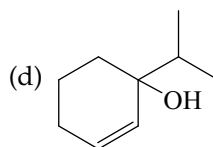
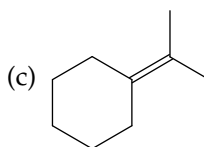
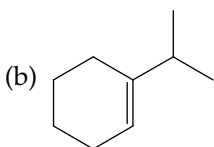
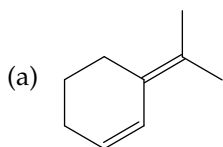
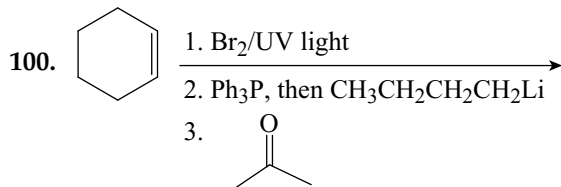
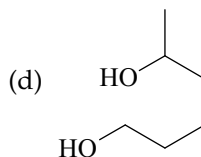
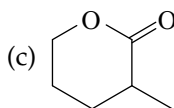
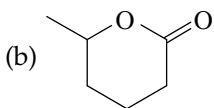
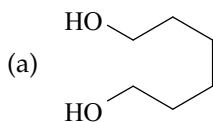
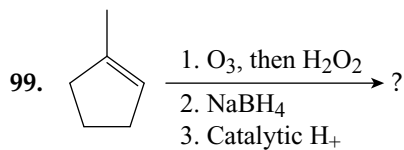
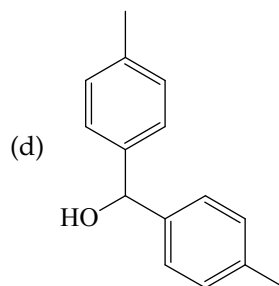
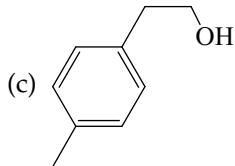
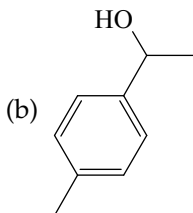
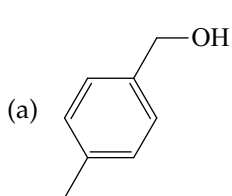
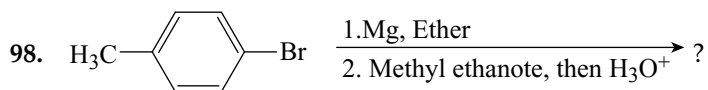
- (a) CCC(Cl)CCC(=O)O
- (b) CCC(O)CCC(=O)Cl
- (c) CCC1OCCCC1Cl
- (d) CCC1OCCCC1=O

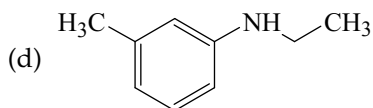
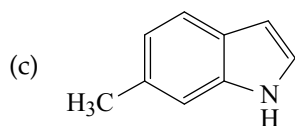
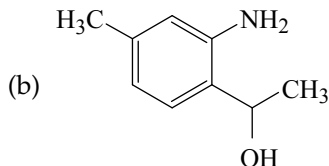
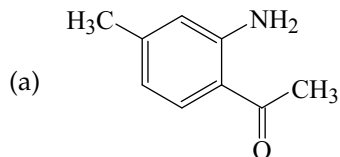
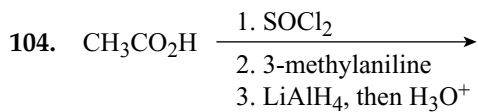
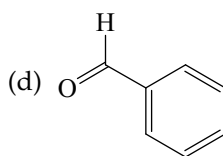
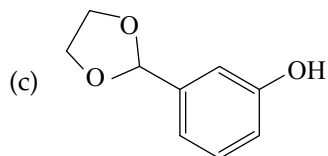
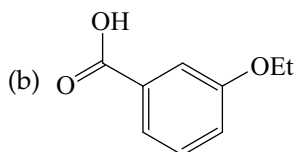
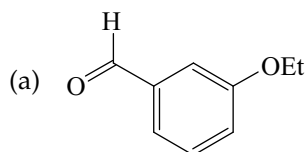
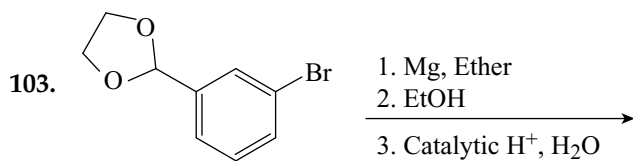
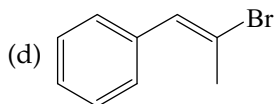
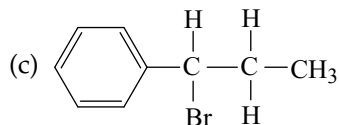
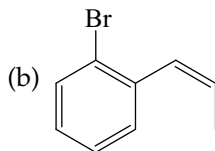
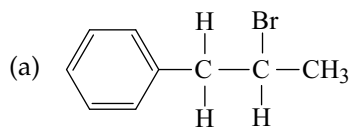
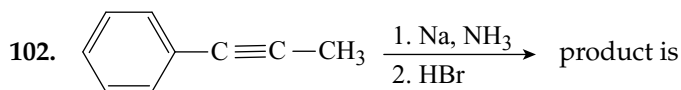
92. What is the order of increasing boiling points (from the lowest to highest) for the following compounds?

- (I) CCC(N)=O (II) CCC(=O)Cl (III) CCC(=O)O
- (a) II, I, III
 - (b) II, III, I
 - (c) I, III, II
 - (d) III, I, II

93. C1CCCCC1=C $\xrightarrow{\begin{matrix} (1) \text{CH}_3\text{CO}_3\text{H} \\ (2) \text{CH}_3\text{MgBr then H}_3\text{O}^+ \end{matrix}}$ product is



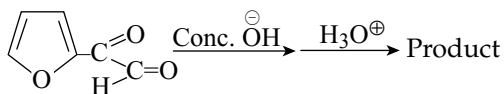




LEVEL 2

Single and Multiple-choice Type

1. The product is



- (a) (b) (c) (d)

2. $\xrightarrow[\text{(ii) Me}_2\text{S}]{\text{(i) O}_3}$ (A) $\xrightarrow[\Delta]{\text{dil. } \text{OH}^-}$ Products

In the above sequence the final products of the reaction is/are

- (a) (b) (c) (d)

3. $\xrightarrow[\text{(ii) } ^-\text{OH}/\text{ROH}]{\text{(i) H}_2\text{N}-\text{NH}_2}$ Product (A)
 $\xrightarrow{\text{Zn-Hg}}$ Product (B)
 $\xrightarrow{\text{HCl}}$ Product (C)
 $\xrightarrow{\text{NaBH}_4}$ Product (C)

Correct option is

- (a) $\text{A} = \text{B} = \text{H}_3\text{C}-\text{C}_6\text{H}_4-\text{CH}_2\text{CH}_2\text{F}$; $\text{C} = \text{H}_3\text{C}-\text{C}_6\text{H}_4-\text{CH}_2\text{CH}_2\text{OH}$
 (b) $\text{A} = \text{H}_3\text{C}-\text{C}_6\text{H}_4-\text{CH}=\text{CH}_2$; $\text{B} = \text{H}_3\text{C}-\text{C}_6\text{H}_4-\text{CH}_2\text{CH}_2\text{F}$
 (c) $\text{C} = \text{HOCH}_2-\text{C}_6\text{H}_4-\text{CH}_2\text{CH}_2\text{F}$
 (d) $\text{B} = \text{H}_3\text{C}-\text{C}_6\text{H}_4-\text{CH}_2\text{CH}_2\text{F}$; $\text{C} = \text{HOCH}_2-\text{C}_6\text{H}_4-\text{CH}_2\text{CH}_3$

4. (A) $\xrightarrow[\text{(ii) BuLi}]{\text{(i) Ph}_3\text{P}}$  $\xrightarrow{\text{(iii) (B)}}$

In the above reaction, (A) and (B) will respectively be

- (a)  and CH_3CHO
- (b) $\text{CH}_3\text{CH}_2\text{Cl}$ and 
- (c)  and HCHO
- (d)  and CH_3CHO

5. Which of the following pairs will form the same product with phenyl hydrazine/ H^+ ?

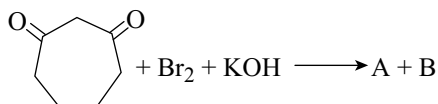
- (a) $\text{CH}_2\text{OH}-\text{CO}-\text{CH}_3$ and $\text{CHO}-\text{CHOH}-\text{CH}_3$
 (b) Glucose and mannose
 (c) Glucose and fructose
 (d) Glucose and sucrose

6. Which is/are untrue statement(s)?

- (a) Protonation increases electrophilic nature of carbonyl group
(b) CF_3SO_3^- is a better leaving group than CH_3SO_3^-
(c) Benzyl carbonium ion is stabilised by resonance

- (d) $\text{CCl}_3\text{CH} \begin{matrix} \diagup \text{OH} \\ \diagdown \text{OH} \end{matrix}$ is unstable

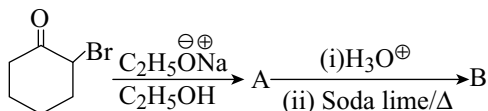
7. Identify the product in the following reaction



A and B are

- (a) CHBr_3 (b)  (c)  (d) 

8. In the following reaction

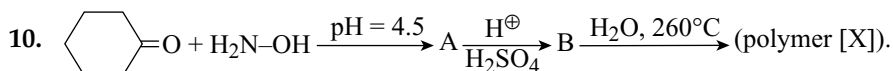
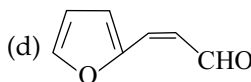
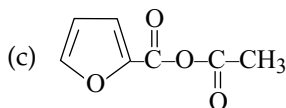
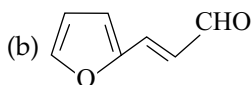
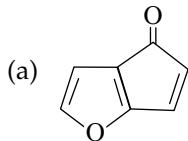


Product A and B are

- (a)  (b)  (c)  (d) 

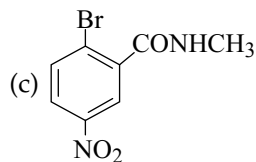
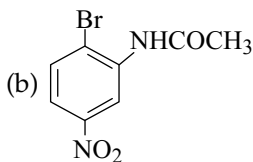
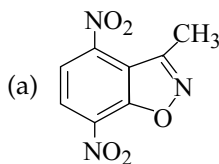
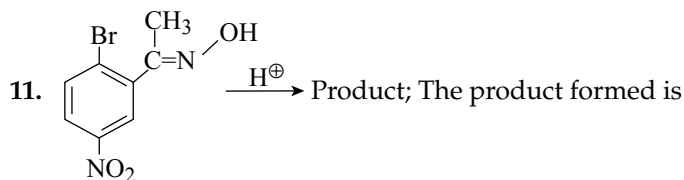


Product (C) is/are



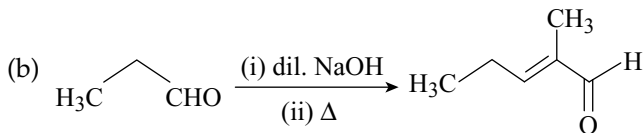
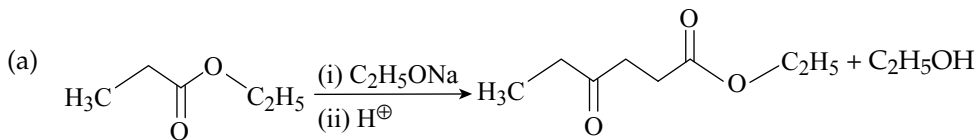
Correct statements about [X] are

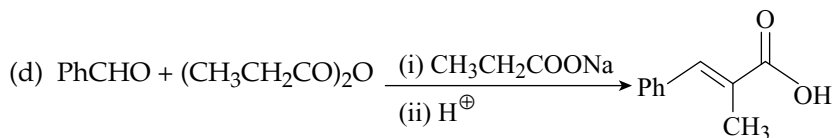
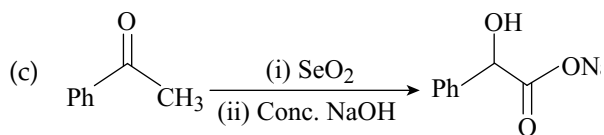
- (a) the polymer is bakelite
- (b) the polymer is Nylon-6
- (c) the polymer contains six carbon atoms in the repeating units
- (d) it is condensation polymer(s)



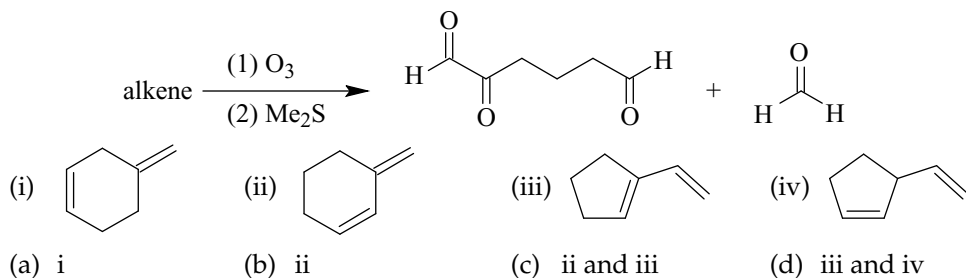
(d) None of these

12. Select the correct reactions

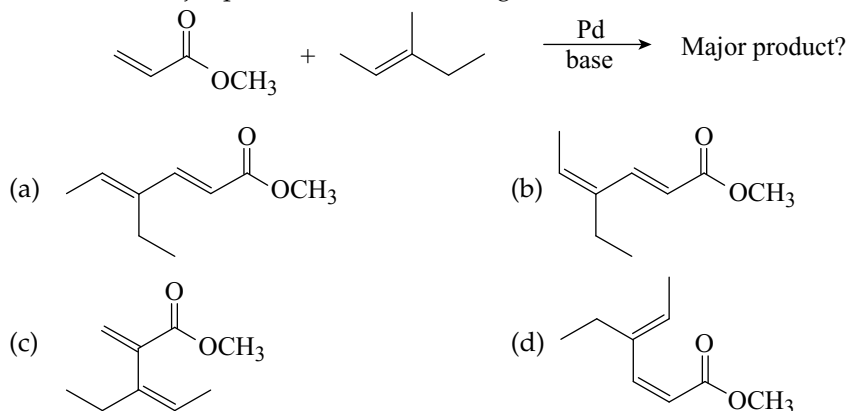




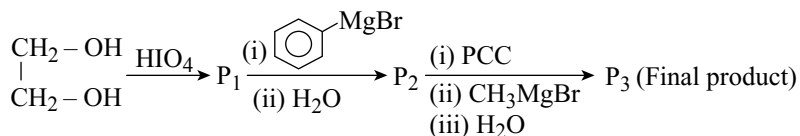
13. Choose the alkene(s) that would give the two products of the following ozonolysis reaction.



14. Choose the major product of the following reaction



15. Which of the following is correct for the final product of the given sequence of reaction



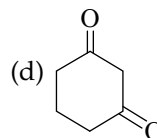
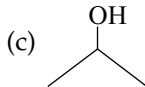
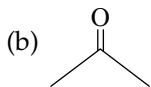
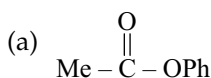
- (a) Compound P₃ on oxidation with (CrO₃ + H₂SO₄) gives a compound which gives 2, 4 DNP test
 (b) Compound P₃ on reaction with I₂ + NaOH gives yellow ppt
 (c) Compound P₃ on reaction with ceric ammonium nitrate gives red colouration

(d) Compound P₃ on reaction with MnO₂ gives carboxylic acid

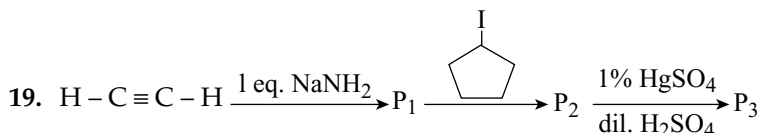
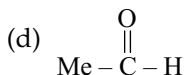
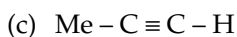
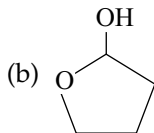
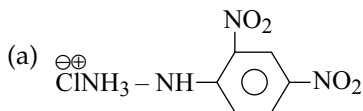
16. Identify correct method of preparation of acetaldehyde from reaction of cyanide

- (a) $\text{Me} - \text{C} \equiv \text{N} \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) DIBAL}}$ (b) $\text{Me} - \text{C} \equiv \text{N} \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) SnCl}_2 + \text{HCl}}$
- (c) $\text{Me} - \text{C} \equiv \text{N} \xrightarrow[\text{(ii) dill. NaOH}]{\text{(i) Conc. H}_2\text{SO}_4}$ (d) $\text{Me} - \text{C} \equiv \text{N} \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) Pd/BaSO}_4 / \text{H}_2}$

17. Identify compounds that give iodoform test



18. Identify compounds which give PPT with Tollen's reagent



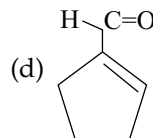
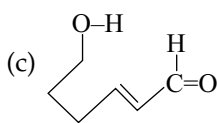
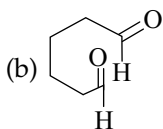
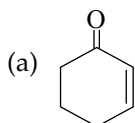
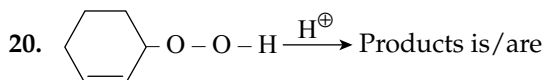
Identify the correct statement for the above reaction sequence?

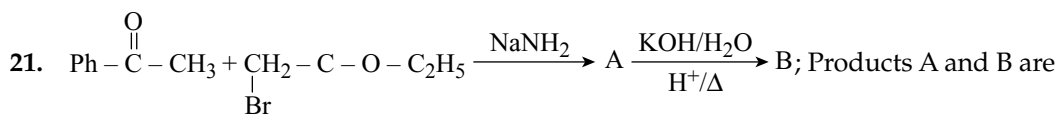
(a) P₃ – Product gives +ve iodoform test

(b) P₂ – Product on reaction with HB(Sia)_2 and $\text{H}_2\text{O}_2/\text{OH}^-$ gives

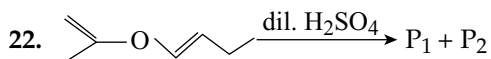
(c) P₃ – Product $\xrightarrow{\text{CH}_2=\text{PPh}_3} \text{X} \xrightarrow[\text{(ii) NaOH/H}_2\text{O}_2]{\text{(i) B}_2\text{H}_6 \cdot \text{THF}} \text{cyclopentyl-2-methylpropan-1-ol}$

(d) P₃ – Product gives Tollen's test



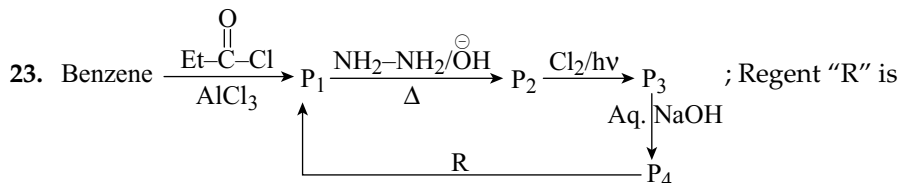


- (a) $\text{Ph}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2$ (b) $\text{Ph}-\overset{\text{O}}{\text{C}}(\text{CH}_3)-\text{CH}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OC}_2\text{H}_5$
- (c) $\text{Ph}-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$ (d) $\text{Ph}-\underset{\text{CH}_3}{\text{CH}}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$



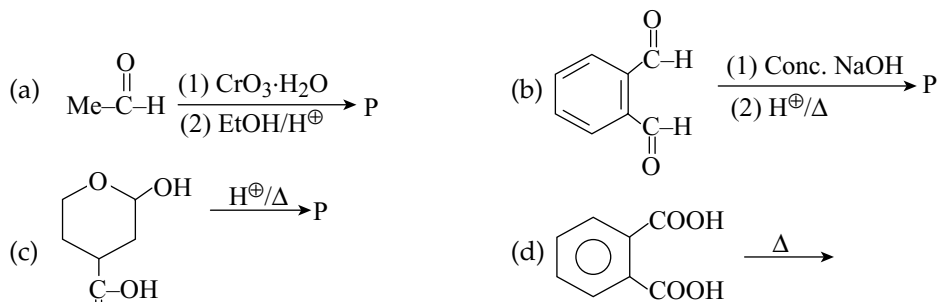
P_1 and P_2 products are identified by

- (a) Tollen's reagent (b) Iodoform test
(c) $\text{Br}_2 + \text{H}_2\text{O}$ test (d) 1% alkaline KMnO_4

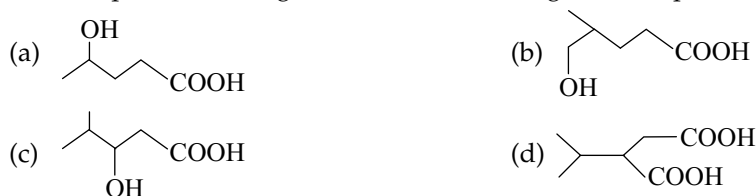


- (a) MnO_2/Δ (b) P.C.C. (c) $\text{Br}_2 + \text{H}_2\text{O}$ (d) LAH

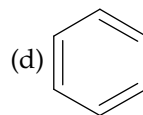
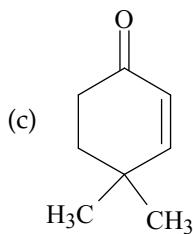
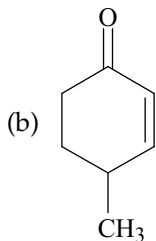
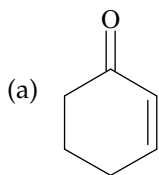
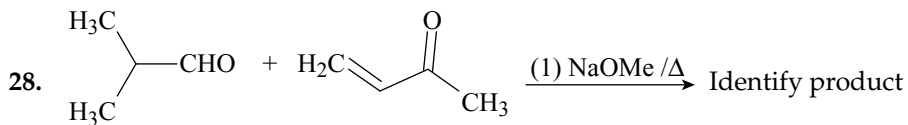
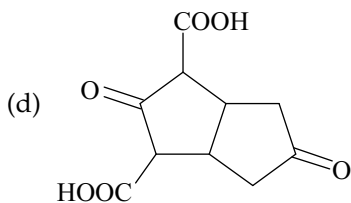
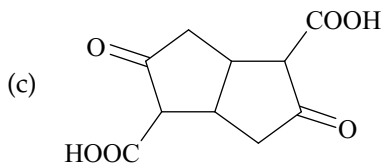
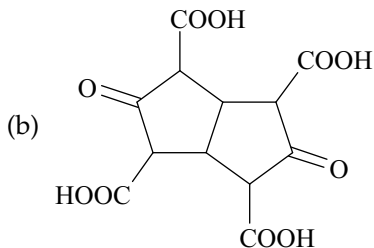
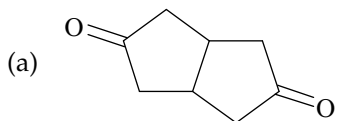
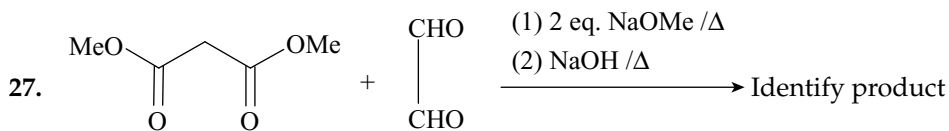
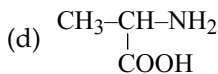
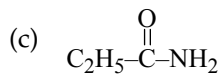
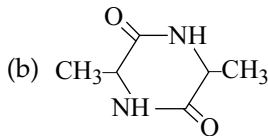
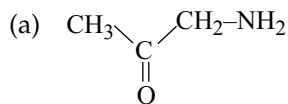
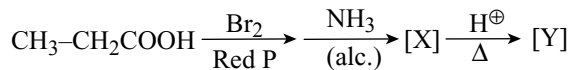
24. Among the following reactions identify lactonisation reactions

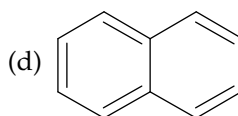
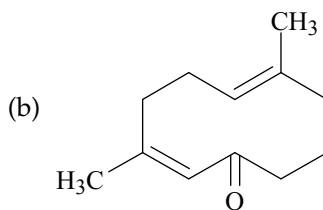
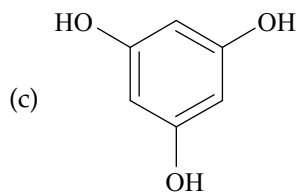
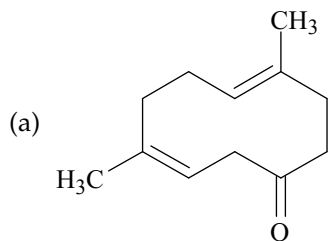
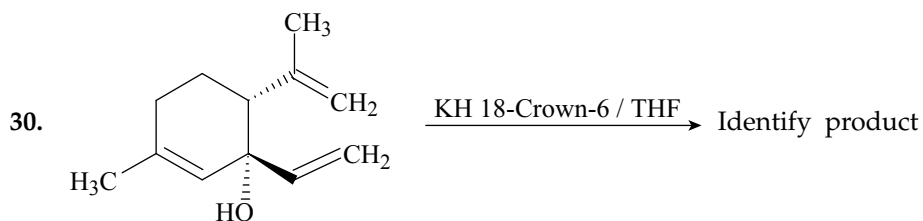
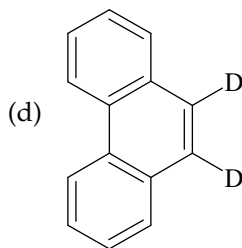
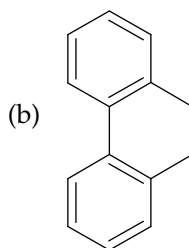
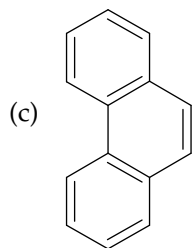
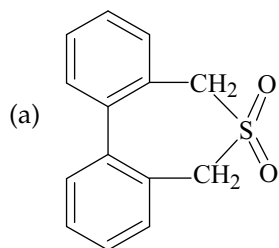
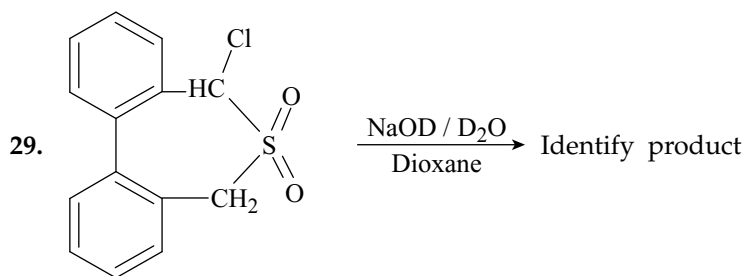


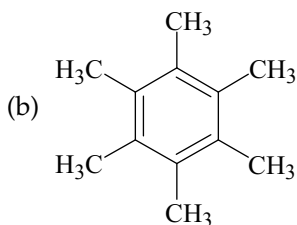
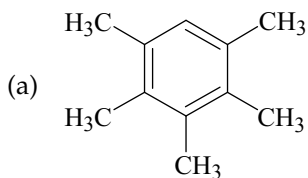
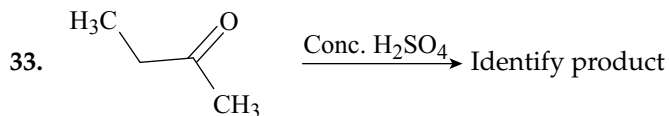
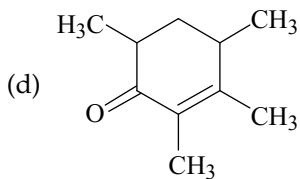
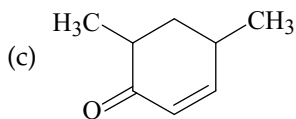
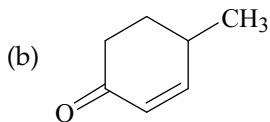
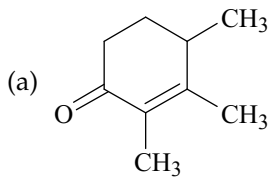
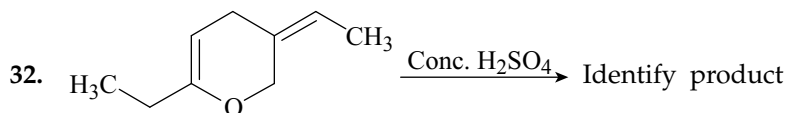
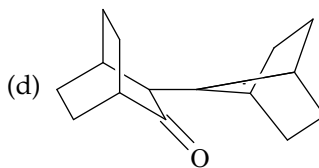
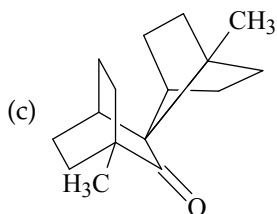
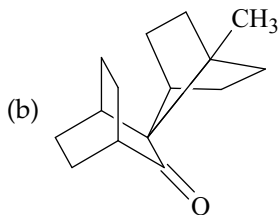
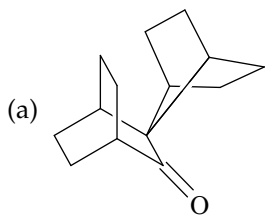
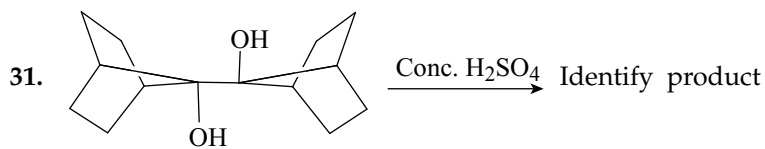
25. The compound which gives lactone on heating or in the presence of Conc. H^+ medium

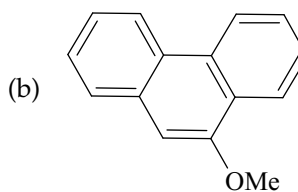
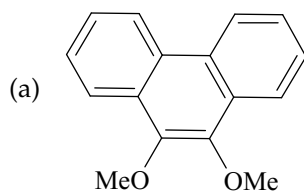
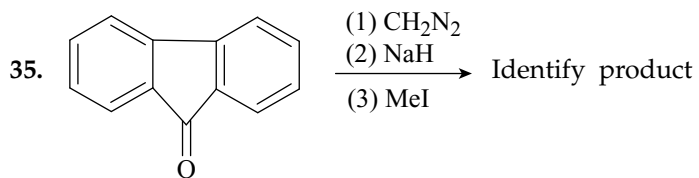
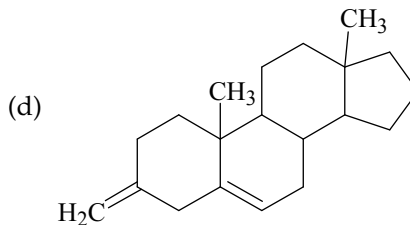
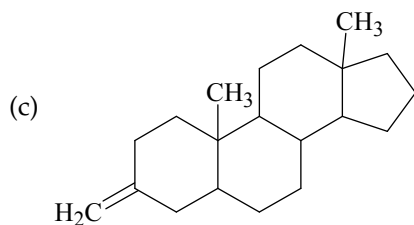
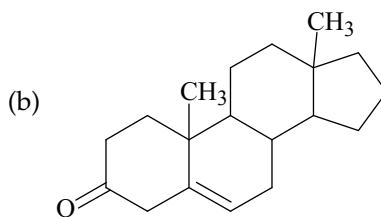
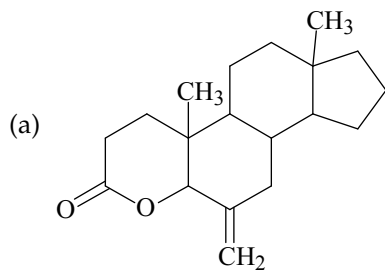
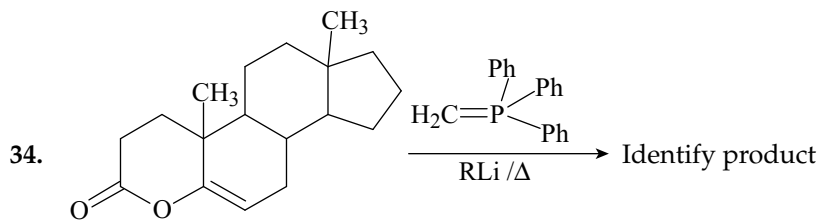
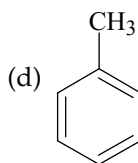
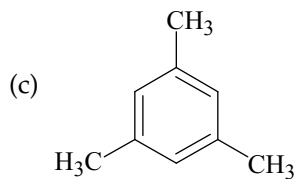


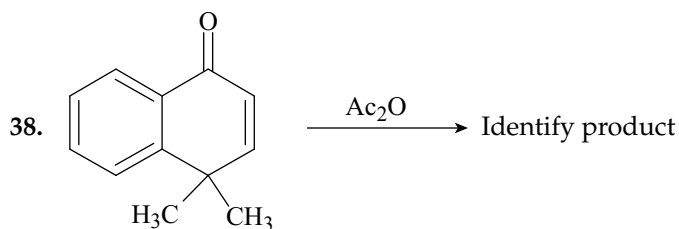
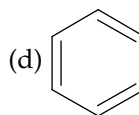
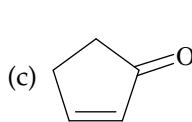
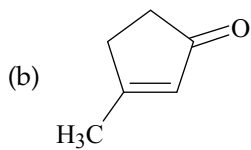
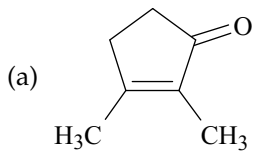
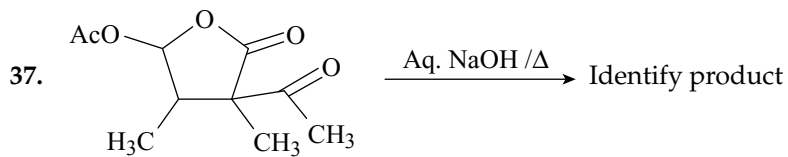
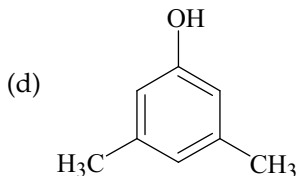
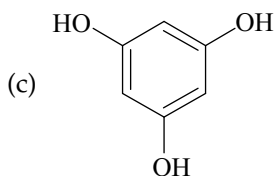
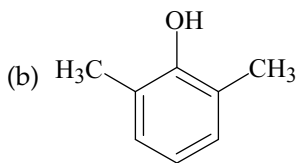
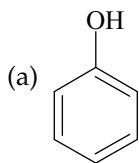
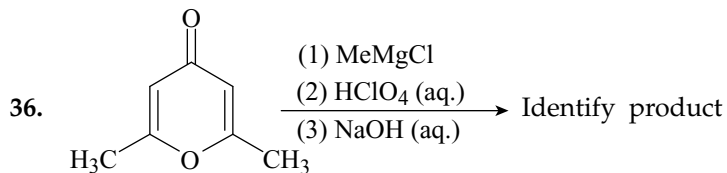
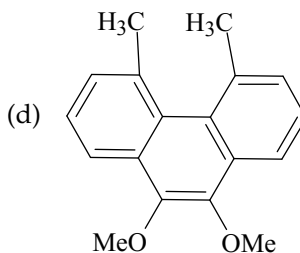
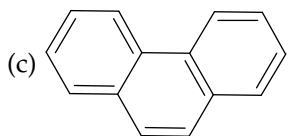
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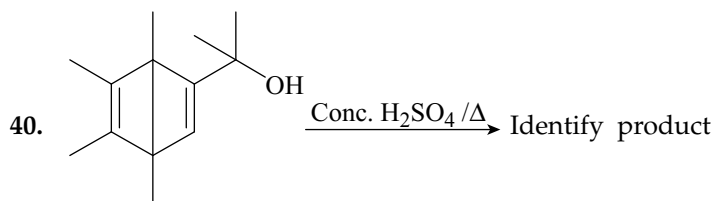
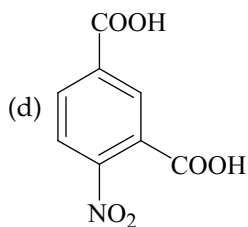
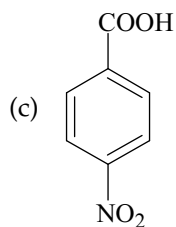
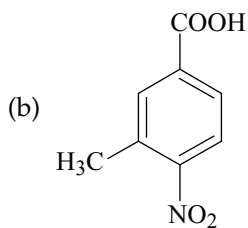
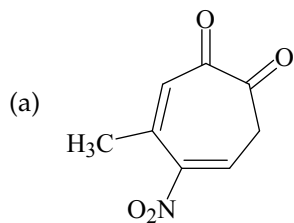
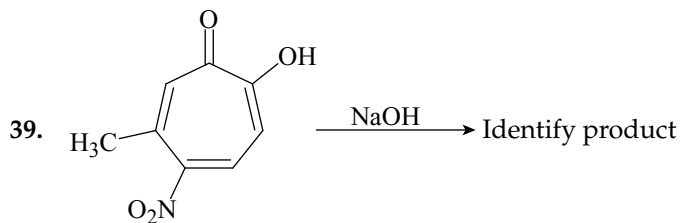
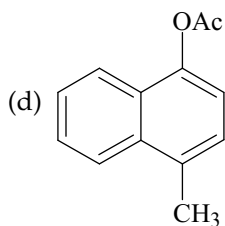
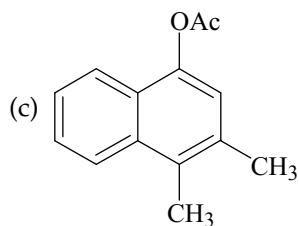
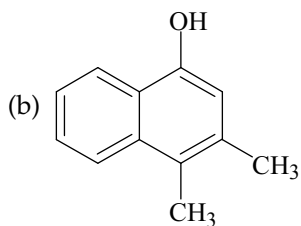
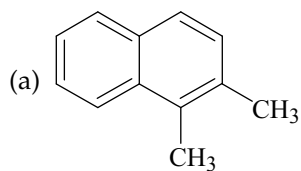


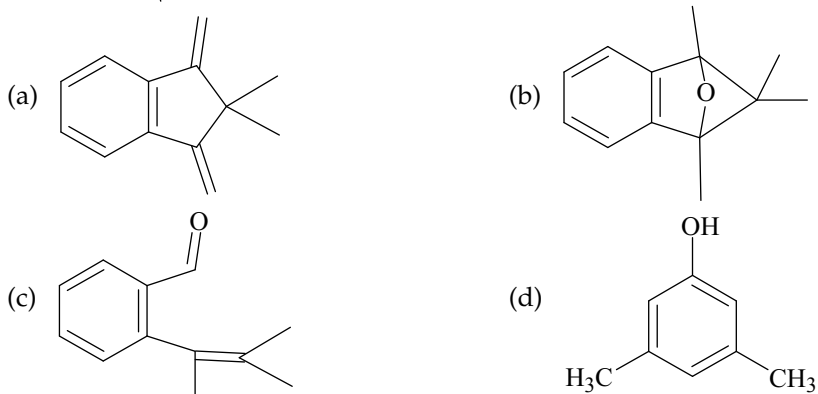
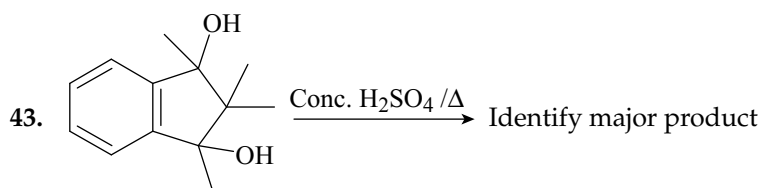
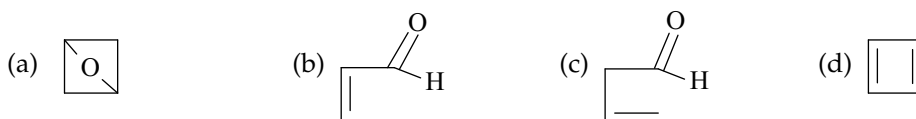
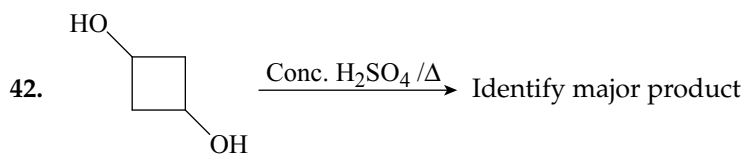
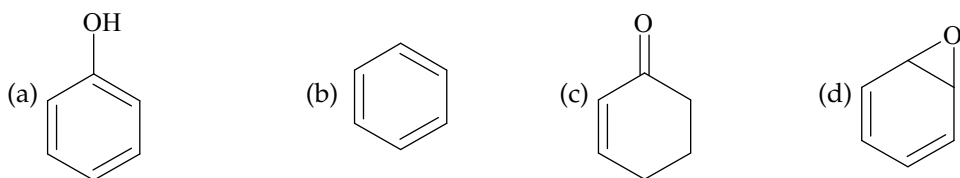
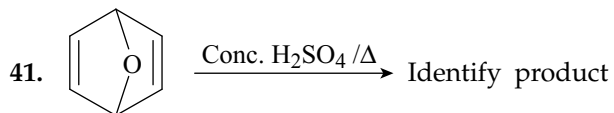
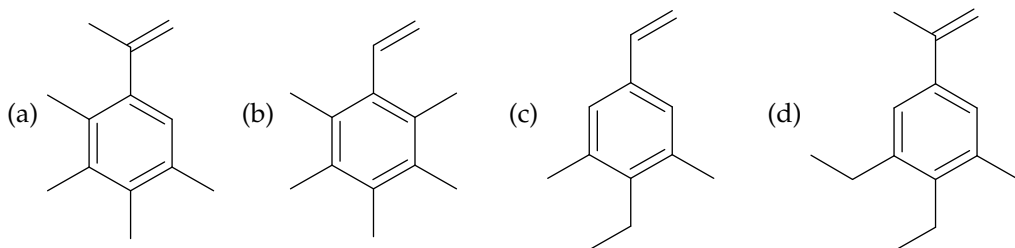


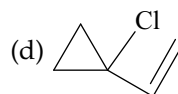
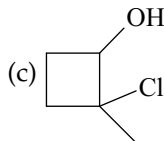
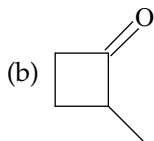
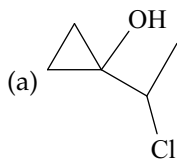
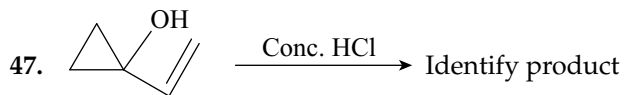
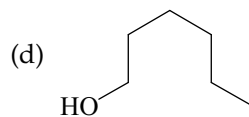
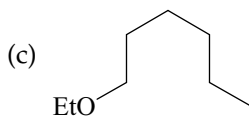
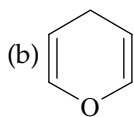
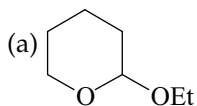
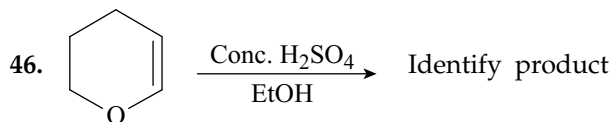
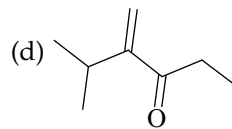
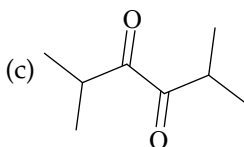
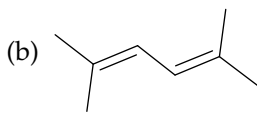
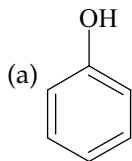
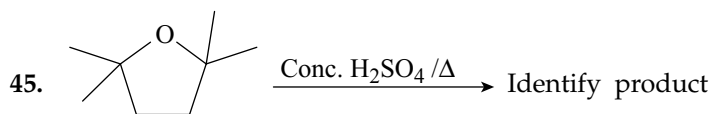
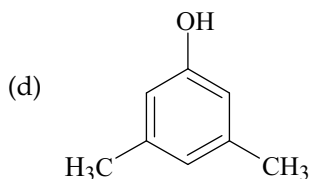
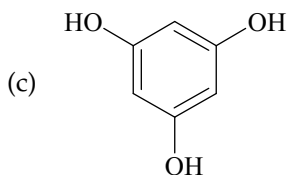
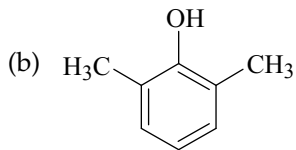
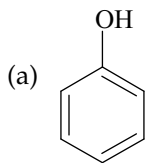
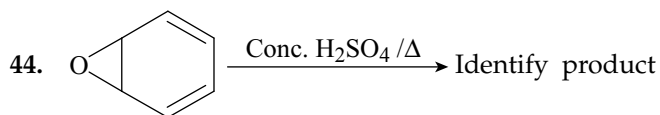


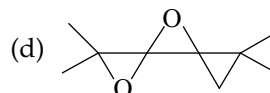
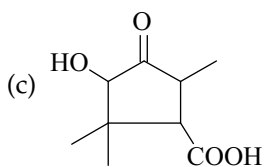
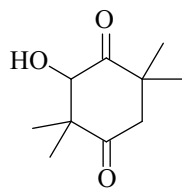
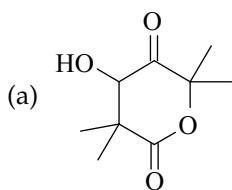
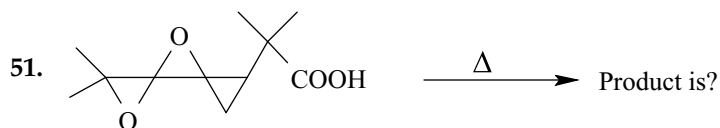
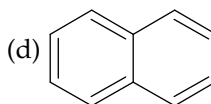
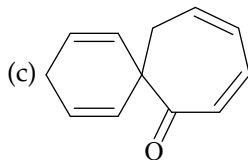
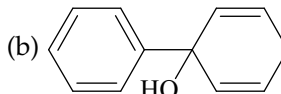
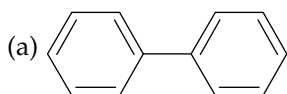
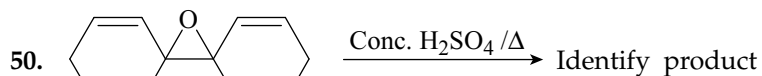
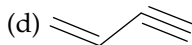
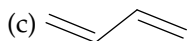
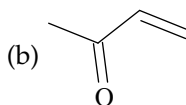
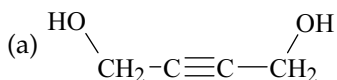
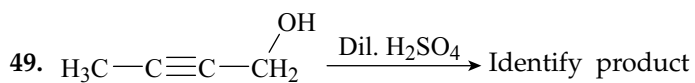
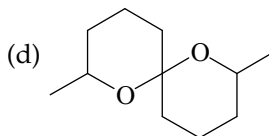
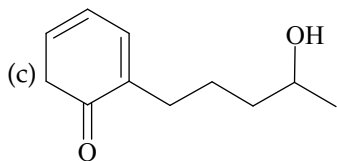
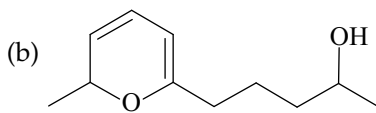
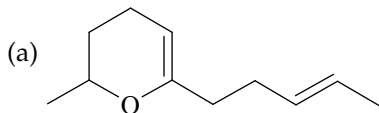
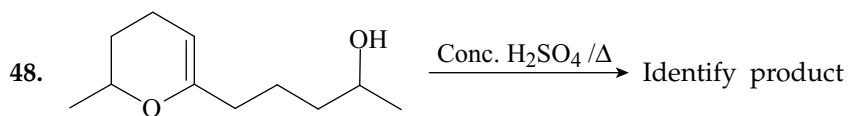








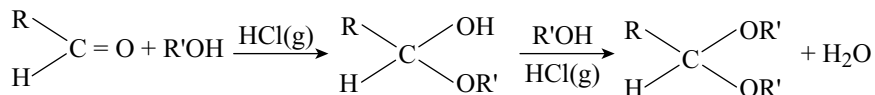




Comprehension Type

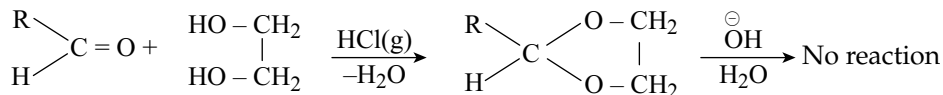
Passage 1

If we take an alcohol solution of an aldehyde (or ketone) and pass into it a small amount of gaseous HCl the hemiacetal forms, and hemiacetal reacts with a second molar equivalent of the alcohol to produce an acetal (or ketal).

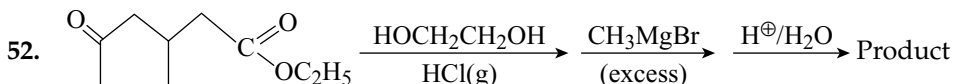
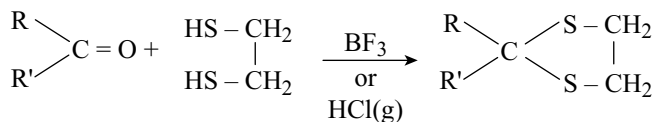


All the steps in the formation of an acetal from an aldehyde (or ketone) are reversible. With excess of alcohol, equilibrium favours forward (acetal formation) direction.

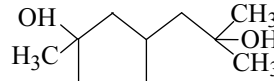
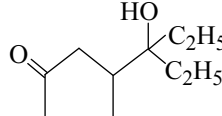
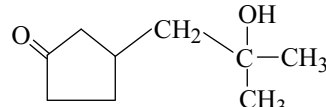
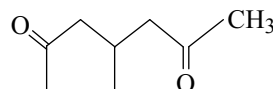
Although acetals are hydrolysed to aldehydes and ketones in aqueous acid, they are stable in basic solution. Because of this property, it is a convenient method for protecting aldehyde and ketone from a series reaching in basic solution.

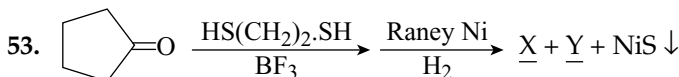


Similarly, thioacetal (ketal) formation takes place.



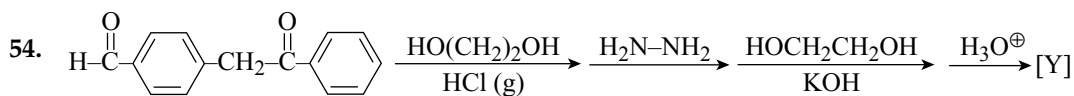
The final product of the reaction is

- (a)  (b) 
- (c)  (d) 

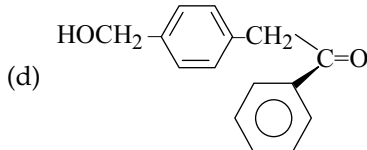
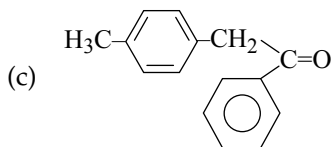
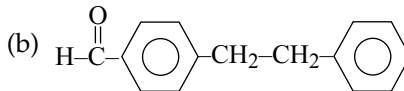
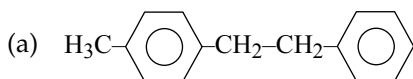


Correct statement for the given reaction is

- (a) X and Y are structural isomers
 (b) both X and Y gives only one monochlorinated product
 (c) X on oxidative ozonolysis gives gluteric acid
 (d) X and Y on reaction with $\text{H}_2\text{N}-\text{OH}$ gives oxime

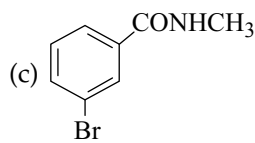
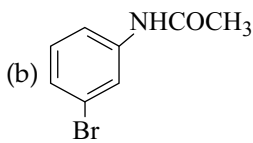
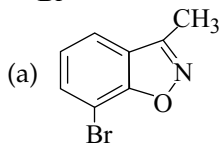
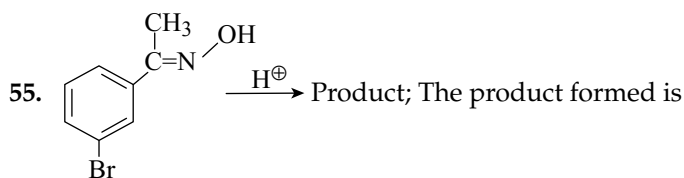
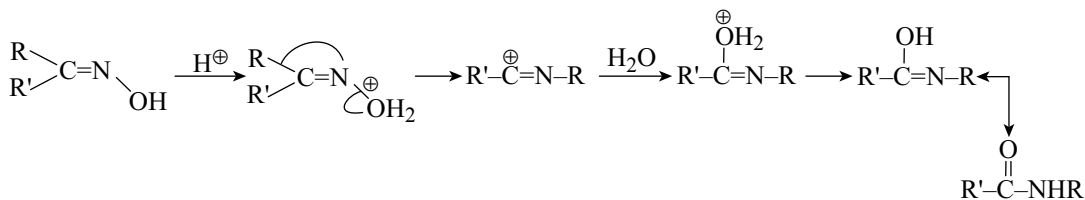


The product (Y) is

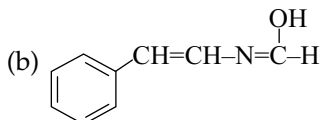
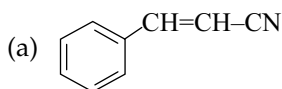
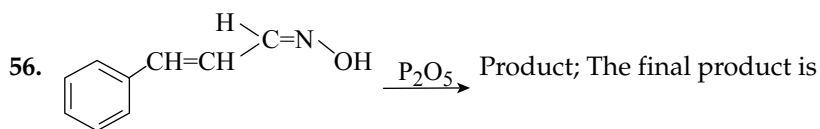


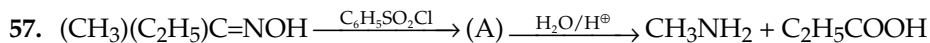
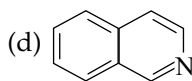
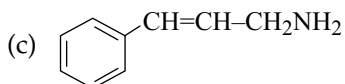
Passage 2

In organic chemistry various reactions take place by rearrangements. These rearrangements can be classified on the basis of migratory group and its nature. One of the well-known rearrangement is the formation of N-substituted amides by rearrangement of aldoximes or ketoximes. This is known as Beckmann rearrangement. It is catalysed by various acidic reagents. The mechanism of this reaction is given as



(d) none of these





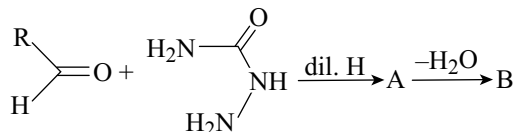
The formation of final product suggest the migration of

- (a) $-\text{CH}_3$ group (b) $-\text{C}_2\text{H}_5$ group
(c) $\text{C}_6\text{H}_5-\text{SO}_2$ -group (d) cannot predicated without structure

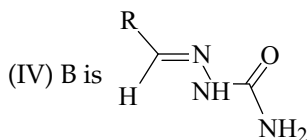
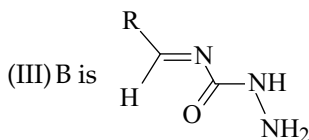
Passage 3

Aldehydes and ketones show nucleophilic addition reactions. The α -hydrogens in aldehydes and ketones are acidic. Therefore, aldehydes and ketones having at least one α -H undergo aldol condensation in the presence of a base such as NaOH. Aldehydes having no α -hydrogen undergo Cannizzaro reaction in the presence of concentrated alkali. The carbonyl group of aldehydes and ketones can be reduced to methylene group by Clemmensen or Wolf Kishner reduction. Aldehydes are easily oxidised to carboxylic acids by Tollen's reagent and Fehling's solution. Functional derivatives of carboxylic acids undergo nucleophilic acyl substitution with nucleophiles.

58. In the reaction given hereunder, correct statement (s) about A and B are



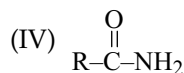
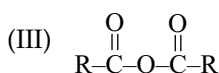
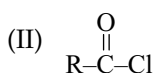
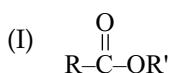
- (I) A is optically active and dextrorotatory
(II) A is racemic mixture and optically inactive



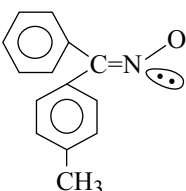
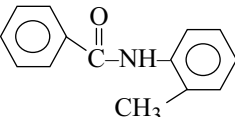
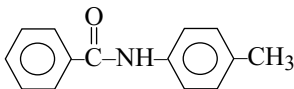
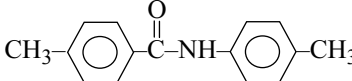
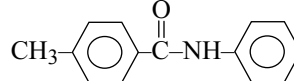
- (a) (I) and (III) (b) (II) and (IV) (c) (II) and (III) (d) (I) and (IV)

59. To convert $\text{CH}_2=\text{CH}-\text{CH}_2\text{CHO}$ to $\text{CH}_2=\text{CH}-\text{CH}_2\text{COOH}$, the best reagent will be
(a) KMnO_4 (b) P.C.C. (c) $[\text{Ag}(\text{NH}_3)_2]\text{OH}$ (d) HIO_4

60. The reactivity order of the following toward nucleophilic substitution will be



- (a) (I) > (III) > (IV) > (II) (b) (II) > (III) > (I) > (IV)
(c) (III) > (IV) > (I) > (II) (d) (III) > (I) > (IV) > (II)

66.  $\xrightarrow[\Delta]{\text{PCl}_5}$ (A); Product (A) of the reaction is
- (a) 
- (b) 
- (c) 
- (d) 

Passage 6

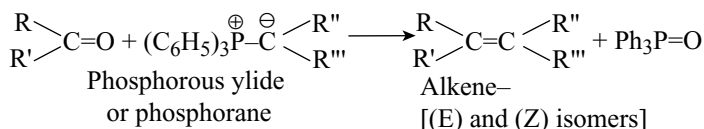
Ethanal is heated with a base in two separate testtubes. In one testtube nothing except the two mentioned reactants were present. In another testtube, malonic acid, $\text{CH}_2(\text{COOH})_2$ was also added. The final products of both testtubes add easily 1, 3-butadiene molecule. The product obtained in the first test tube can be converted into second product identical in all respects with that obtained in the second testtube.

67. The respective reaction involved in the two testtubes are
- Aldol condensation and Cannizaro reaction
 - Aldol condensation and Perkin reaction
 - Aldol condensation and Knoevenagel reaction
 - Aldol reaction and Claisen condensation
68. Which feature is common in the products of both reactions?
- presence of an acidic group
 - presence of an aldehydic group
 - presence of α, β unsaturation
 - presence of a ketonic group
69. What would have been the final product when ethanol of the first testtube is replaced by propanal?
- $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_2\text{CHO}$
 - $\text{CH}_3\text{CH}_2\text{CH}=\overset{\text{CH}_3}{\text{C}}\text{CHO}$
 - $\text{CH}_3\text{CH}_2\overset{\text{CH}_3}{\text{C}}=\text{CH}-\text{CHO}$
 - $\text{CH}_3\text{CH}=\overset{\text{CH}_3}{\text{CH}}-\text{CHCHO}$

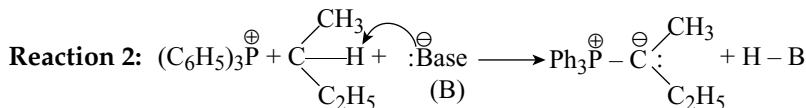
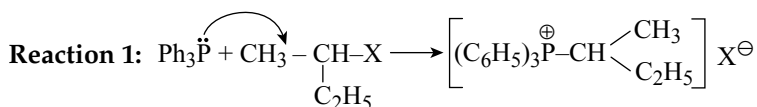
Passage 7

Aldehydes and ketones react with phosphorous ylides to yield alkenes and triphenylphosphine oxide. (An ylide is a neutral molecule having a negative carbon adjacent to appositive heteroatom).

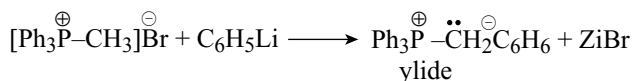
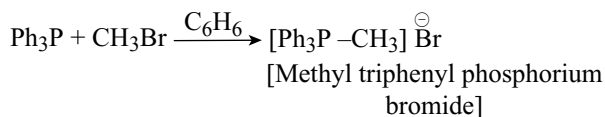
Phosphorous ylides are also called phosphoranes.



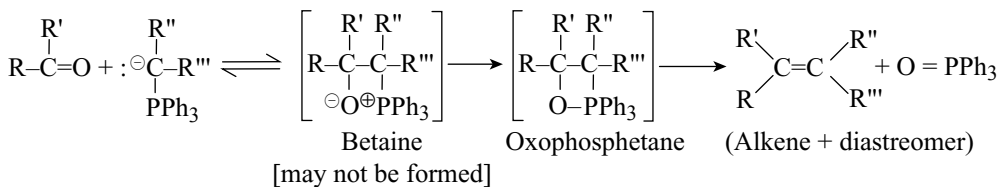
Through reaction, known as the Wittig reaction, phosphorus ylides are easily prepared from triphenylphosphine and alkyl halides. Their preparation involves two reactions:



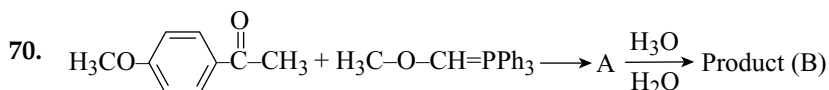
Specific example:



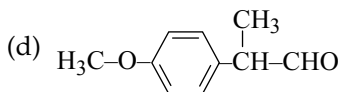
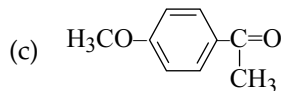
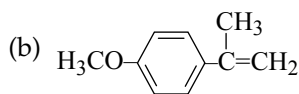
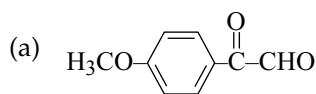
Mechanism of Wittig reaction



The mechanism suggested that the ylide, acting as a carbonion, attacks the carbonyl carbons of the aldehydes or ketone to form an unstable intermediate with separated charge called a betaine. In the next step, betaine forms oxaphosphetane, which spontaneously loses triphenylphosphine oxide to become an alkene.

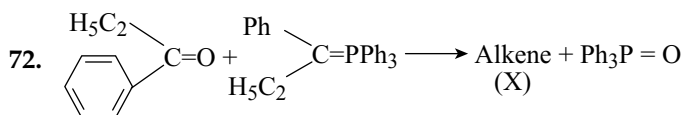


In the given sequence of reaction, the product (B) is



71. Which of the following cannot be used in the preparation of ylide?

- (a) $\text{CH}_3\text{CH}_2\text{Br}$ (b) $(\text{CH}_3)_3\text{C-I}$ (c) (d) Both (b) and (c)

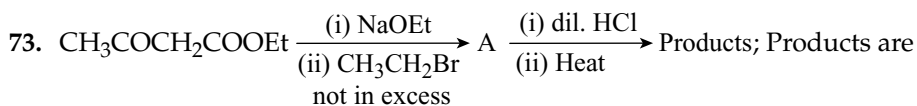
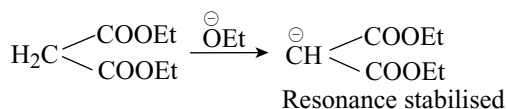


In the given reaction, the correct statement for alkene (X) is

- (a) X on reaction with Br_2/CCl_4 forms an optically inactive mixture
 (b) X on reaction with alk. KMnO_4 forms an optically inactive mixture
 (c) X on reaction with alk. KMnO_4 forms an optically active mixture
 (d) Both (a) and (b)

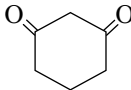
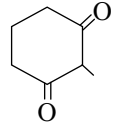
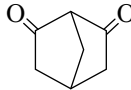
Passage 8

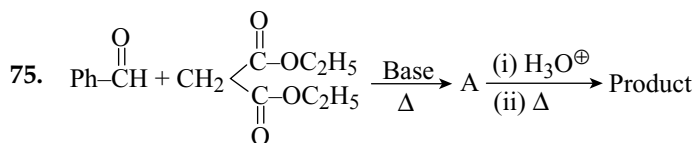
Active methylene compounds are of great importance in synthetic chemistry. When $-\text{CH}_2$ (methylene) group is flanked between two electron withdrawing groups, then its hydrogen becomes acidic and can be replaced easily. The acidity of methylene hydrogen depends upon the strength of electron withdrawing group. Acetoacetic ester, malonic ester, etc. are the examples of active methylene compound. In synthetic application acidic hydrogen is trapped by a strong base to get resonance stabilised anion which reacts accordingly to give the desired product.



- (a) $\text{CH}_3\text{COCH}_3 + \text{CO}_2 + \text{EtOH}$
 (b) 2 mol of $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3 + \text{CO}_2$
 (c) $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3 + 2\text{EtOH}$
 (d) $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3 + \text{EtOH} + \text{CO}_2$

74. Which one of the following is most effective as an active methylene compound?

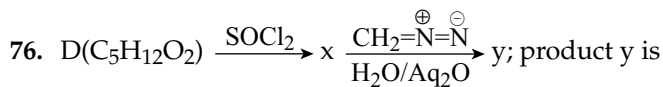
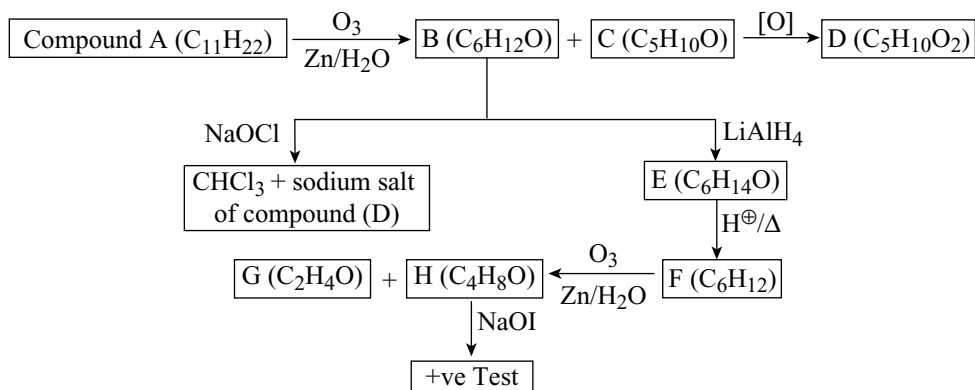
- (a)  (b) $\text{CH}_3\text{COCH}_2\text{COOEt}$ (c)  (d) 

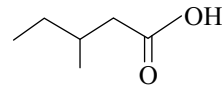
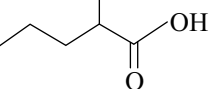
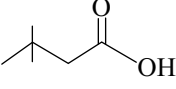
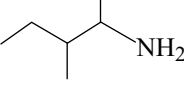


The final product is

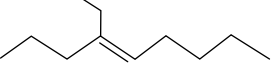
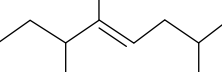
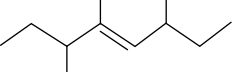
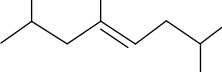
- (a) $\text{Ph}-\text{COOH}$ (b) $\text{Ph}-\text{Ph}-\overset{\text{O}}{\parallel}\text{C}-\text{CH}=\text{CH}-\text{COOH}$
 (c) $\text{Ph}-\text{CH}=\text{CH}-\text{COOH}$ (d) $\text{Ph}-\text{CH}=\text{CH}-\text{CHO}$

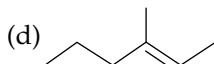
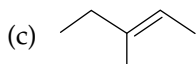
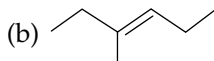
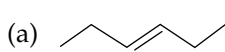
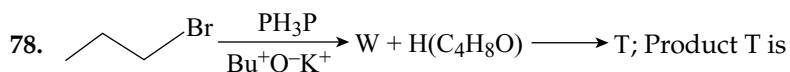
Passage 9



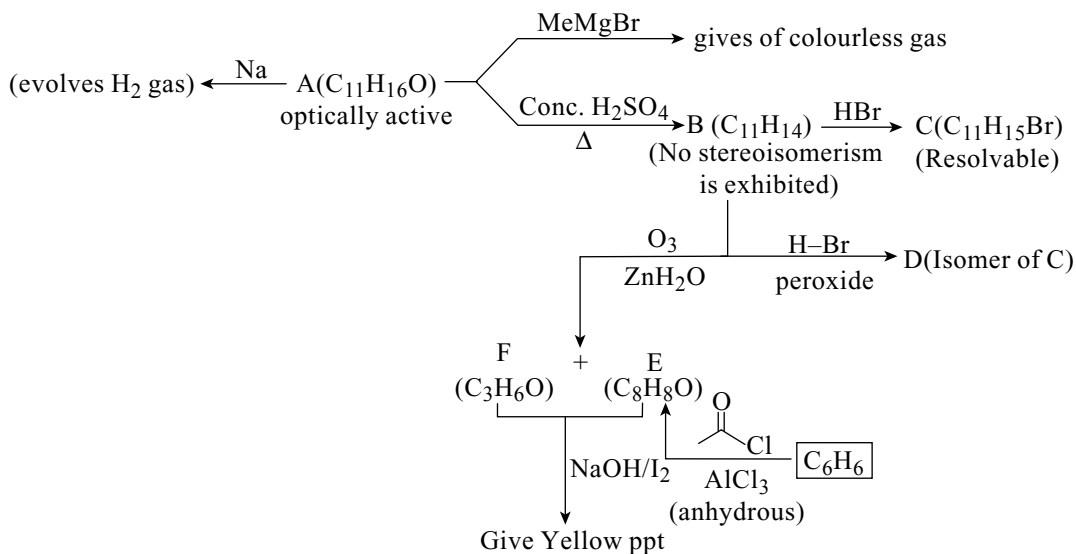
- (a)  (b) 
 (c)  (d) 

77. Compound A is

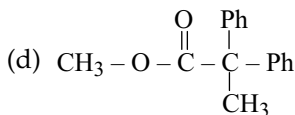
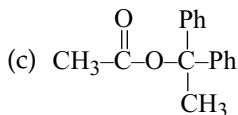
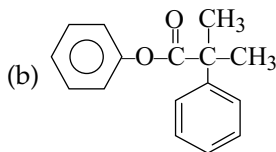
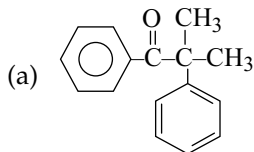
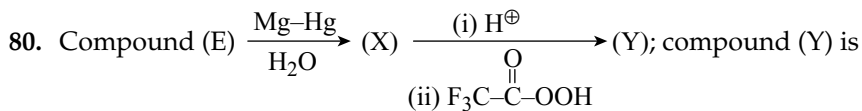
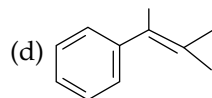
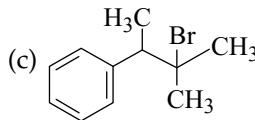
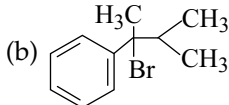
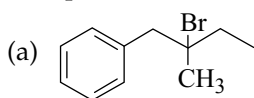
- (a)  (b) 
 (c)  (d) 



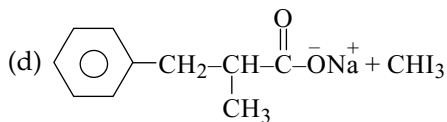
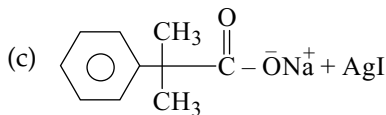
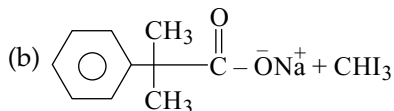
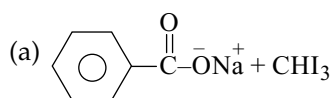
Passage 10



79. Compound (C) is



81. Compound (A) $\xrightarrow[\text{Acetone (aq.)}]{\text{H}_2\text{CrO}_4}$ (R) $\xrightarrow{\text{NaOI}}$ (S) + (T). Compound (S) and (T) respectively are
Yellow ppt.



Matrix Type

82. Match the reactions in column I with their reagents in column II.

Column I

- (a) $\text{CH}_3\text{-CH}_2\text{=CH-CHO} \rightarrow \text{CH}_3\text{CH=CHCOOH}$
 (b) $\text{CH}_3\text{-CH=CH-CHO} \rightarrow \text{CH}_3\text{CH=CH-CH}_2\text{OH}$
 (c) $\text{Ph-CH=CH-CHO} \rightarrow \text{Ph-CH}_2\text{-CH}_2\text{-CHO}$
 (d) $\text{CH}_2\text{=CH-CH}_2\text{CHO} \rightarrow \text{CH}_2\text{=CH-CH}_2\text{CH}_2\text{OH}$

Column II

- (p) LiAlH_4
 (q) NaBH_4
 (r) Pd-C/H_2
 (s) $\text{Ag(NH}_3)_2^+$

83. Match the columns.

Column I

- (a)
 (b)
 (c) $\text{CH}_3\text{-C(=O)-CH}_2\text{-CH}_2\text{-CH}_2\text{-C(=O)-H} \xrightarrow{\text{HO}} \text{(A)}$
 (d)

Column II (In the given sequence)

- (p) Formation of six member ring takes place
 (q) Final product is ketone
 (r) Final product formed will give positive Tollen's test
 (s) Final product formed will react with 2,4-DNP. (2, 4-di-nitro-phenyl hydrazine)

84. Match the chemical compounds in Column I with the reagents used to test them in Column II.

Column I

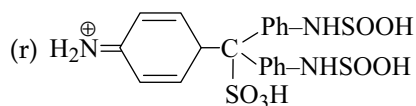
- (a) Ethanol
 (b) Glucose

Column II

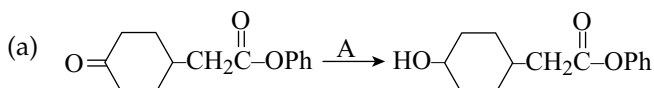
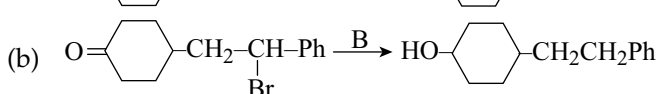
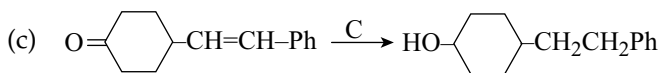
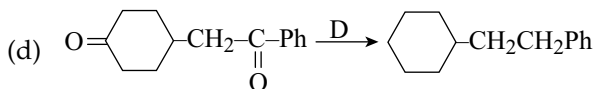
- (p) $[\text{Ag(NH}_3)_2]\text{OH}$
 (q) Fehling's solution

(c) Glyoxal

(d) Acetone


 (s) I_2/NaOH

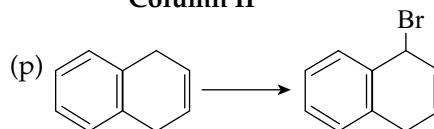
85. Match the columns.

Column I (Reaction)
**Column II
(Reducing reagent)**

 (p) $\text{Zn} + \text{Hg}/\text{HCl}$

 (q) LiAlH_4

 (r) NaBH_4

 (s) $\text{Red P} + \text{HI}$

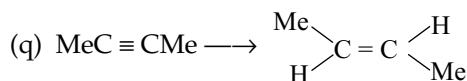
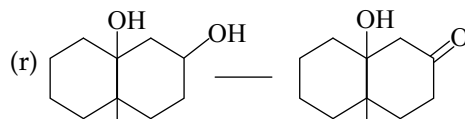
86. Match the columns.

Column I
Column II

(a) PCC (Pyridiniumchlorochromate)

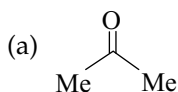


(b) NBS (N-Bromosuccinimide)

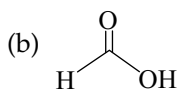

 (c) AlPO_4

 (d) $\text{Li}/\text{Liquid NH}_3$

 (s) $\text{CH}_3\text{COOH} \longrightarrow \text{CH}_2 = \text{C} = \text{O}$

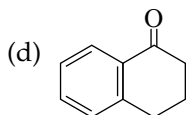
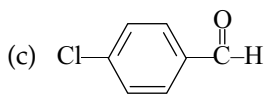
87. Match column I with column II.

Column I
Column II


(p) 2, 4-DNP test



(q) Tollen's reagent test

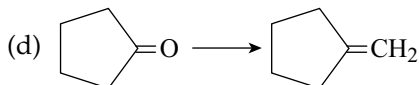
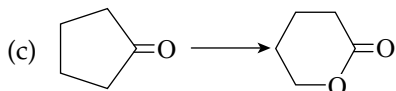
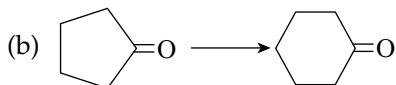
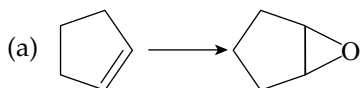


(r) $I_2 + NaOH$ (Iodoform test)

(s) Fehling's solution test

88. Match column I with column II.

Column I (conversion)

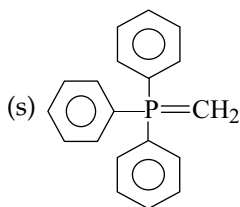


Column II (reagents)

(p) HCN ; H_2/Ni ; HNO_2

(q) CH_2N_2

(r) m-CPBA (Meta-chloroperbenzoic acid)



89. Match Column I with Column II.

Column I

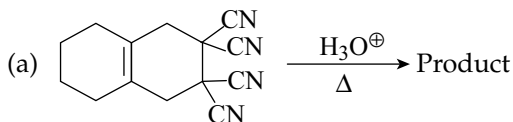
- (a) Aldol condensation
- (b) Cannizzaro reaction
- (c) Reformatsky reaction
- (d) Benzoin condensation

Column II

- (p) Hydride ion transfer
- (q) Zinc enolate
- (r) $\bar{C}N$ as catalyst
- (s) Enolate ion

90. Match the columns.

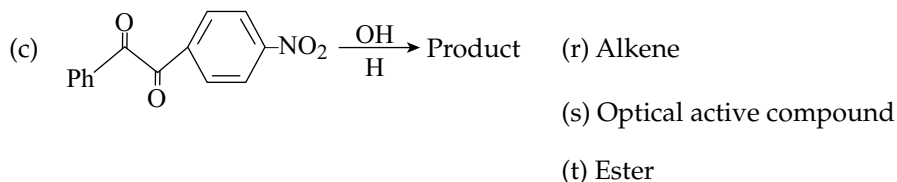
Column I



Column II

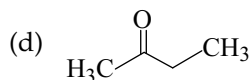
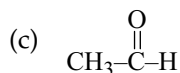
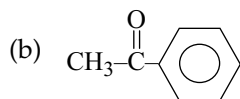
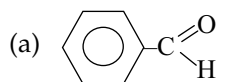
(p) Anhydride

(q) Carboxylic acid



91. Match the columns.

Column I (compound)



Column II (Tests)

(p) 2,4-DNP test

(q) Yellow ppt. with NaOH + I₂

(r) Red ppt. with Fehling's solution

(s) Silver mirror with Tollen's reagent

92. Match the columns.

Column I (Reagents)

(a) Tollen's reagent give white ppt. with

(b) Br₂ + H₂O test given by

(c) Product of reaction of acetylene with 1% HgSO₄ and dil. H₂SO₄

(d) Pd/H₂ reacts with

Column II

(p) Me-CH=CH-Me

(q) Me-C≡C-H

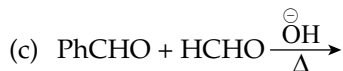
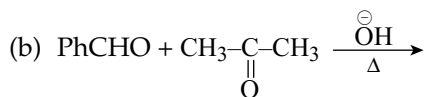
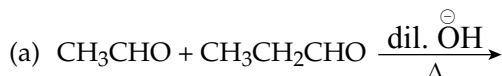
(r) $\text{Cl}^+\text{N}(\text{H}_3) - \text{NH}_2$

(s) 

(r) 

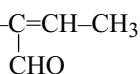
93. Match column I with column II.

Column I

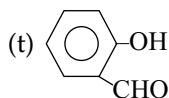
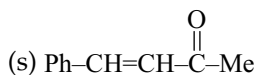
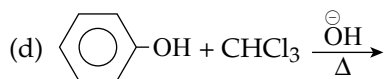


Column II

(p) $\text{CH}_3\text{-CH}_2\text{CH=CH-CHO}$

(q) 

(r) HCOO^-



94. Match the column I with column II.

Column I

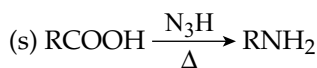
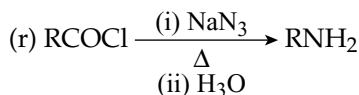
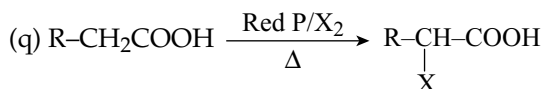
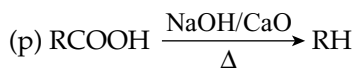
(a) Schimdt reaction

(b) Curtius reaction

(c) Decarboxylation

(d) HVZ reaction

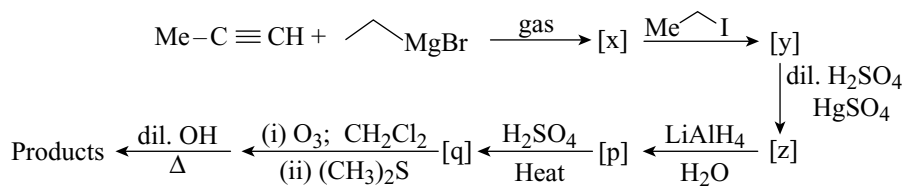
Column II



Integer Type

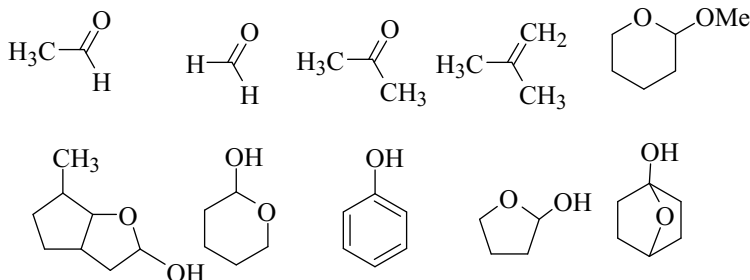
95. How many products are possible when ethanal and phenyl ethanal (mixture) is treated with dil. NaOH at about 10°C.

96. For the given sequence of reaction

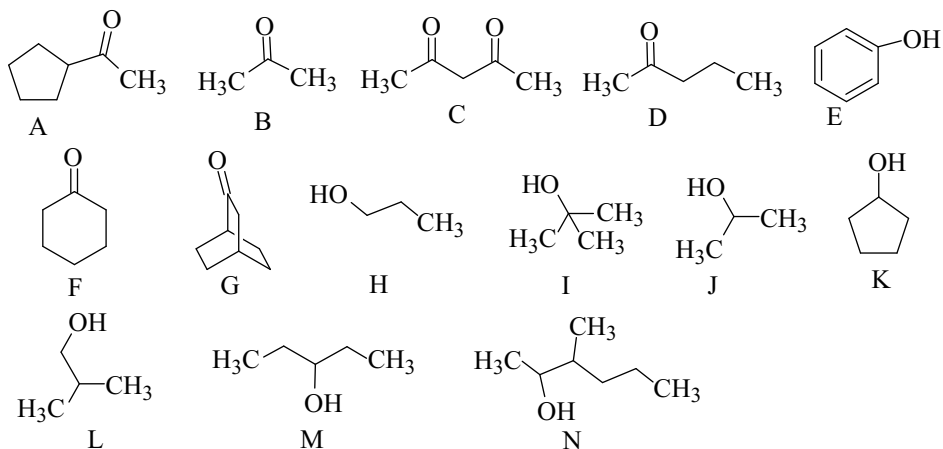


How many products are obtained finally?

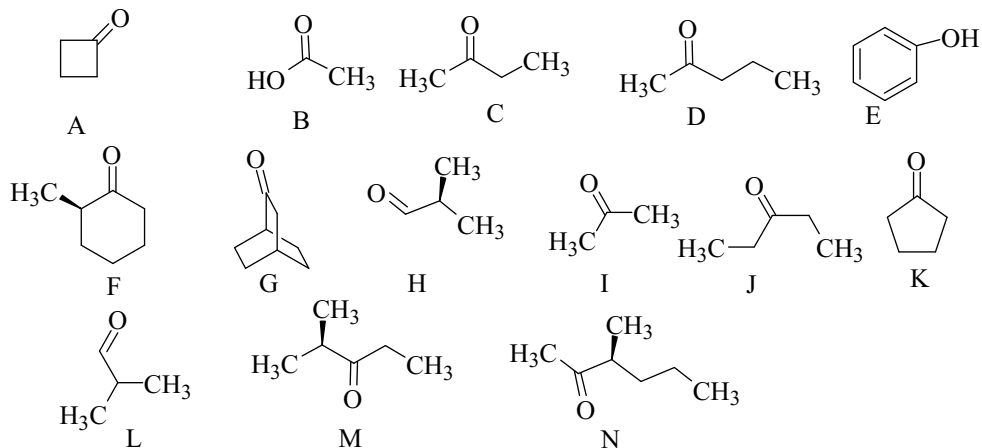
97. Identify the total number of compounds that give positive test with Tollen's reagent



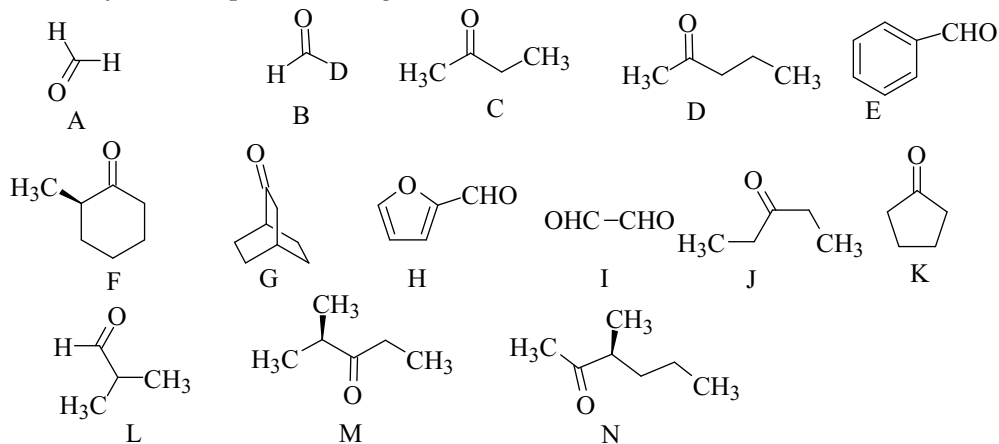
98. Identify the total number of compounds that give positive iodoform test



99. Identify the total number of compounds that give diastereomeric product on reaction with MeMgCl



100. Identify that compounds that give Cannizaro reaction

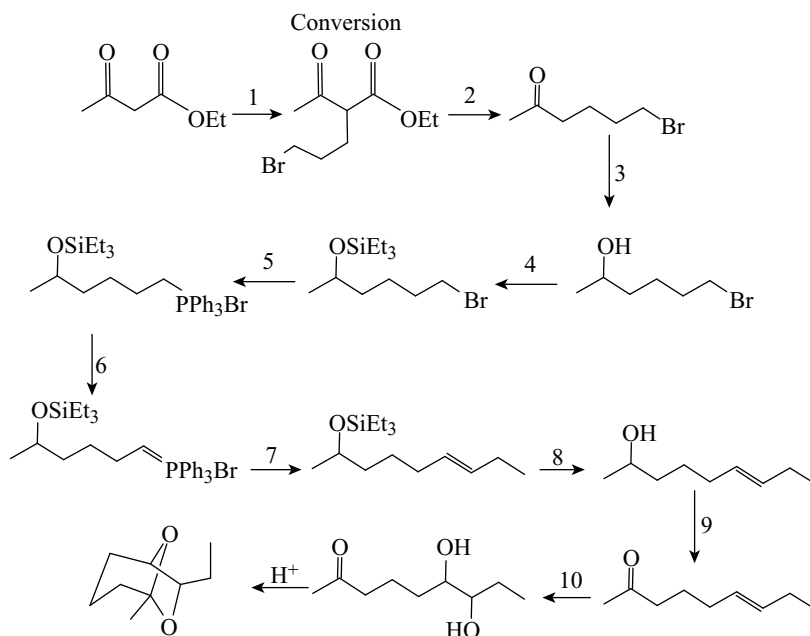


WORKBOOK EXERCISE 1

Identify reagents (1 to 10) used in the following conversion from reagent present in second Column II (A to P).

Column I (Conversion)

Column II (Reagent)



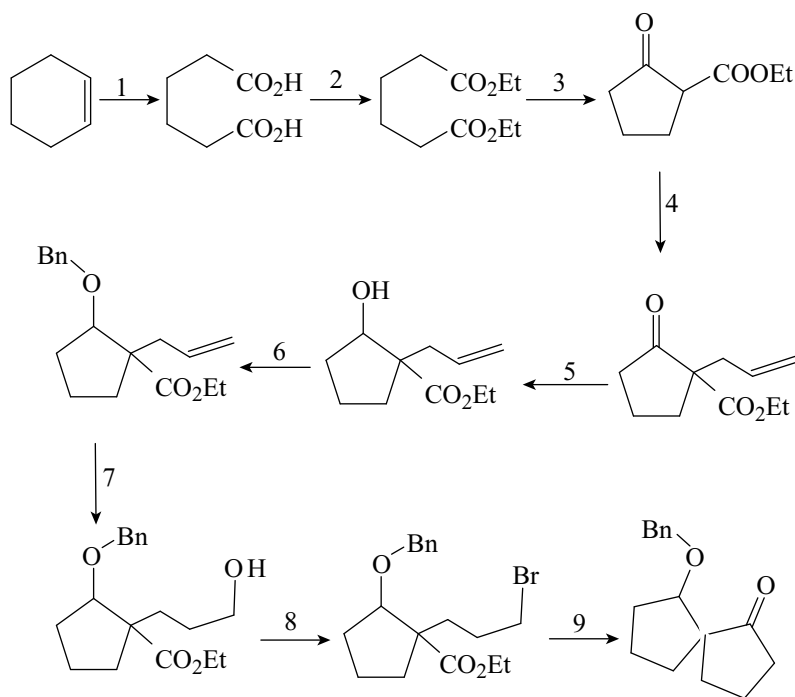
- Reagent
- A. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Li} / \text{THF}$
 - B. $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$
 - C. $\text{Et}_3\text{SiCl} / \text{Et}_3\text{N}$
 - D. $\text{Et}_3\text{SiH} / \text{pyridine}$
 - E. CH_3CHO
 - F. $\text{CH}_3\text{CH}_2\text{CHO}$
 - G. PDC
 - H. PPh_3
 - I. $\text{CH}_2 = \text{CHCH}_2\text{Br}$
 - J. $\text{H}_3\text{O}^+ / \Delta$
 - K. $\text{Conc. H}_2\text{SO}_4 / \Delta$
 - L. NaOEt then $\text{Br}(\text{CH}_2)_3\text{Br}$
 - M. Et_3N
 - N. NaBH_4 then H_3O^+
 - O. Bu_4NF then H_2O
 - P. $\text{KMnO}_4 / \text{aq. NaOH} / 0^\circ\text{C}$

WORKBOOK EXERCISE 2

Identify reagents (1 to 10) used in the following conversion from reagent present in second Column II (A to T).

Column I (Conversion)

Column II (Reagent)



NOTE: Bn = Benzyl group, PhCH_2^-

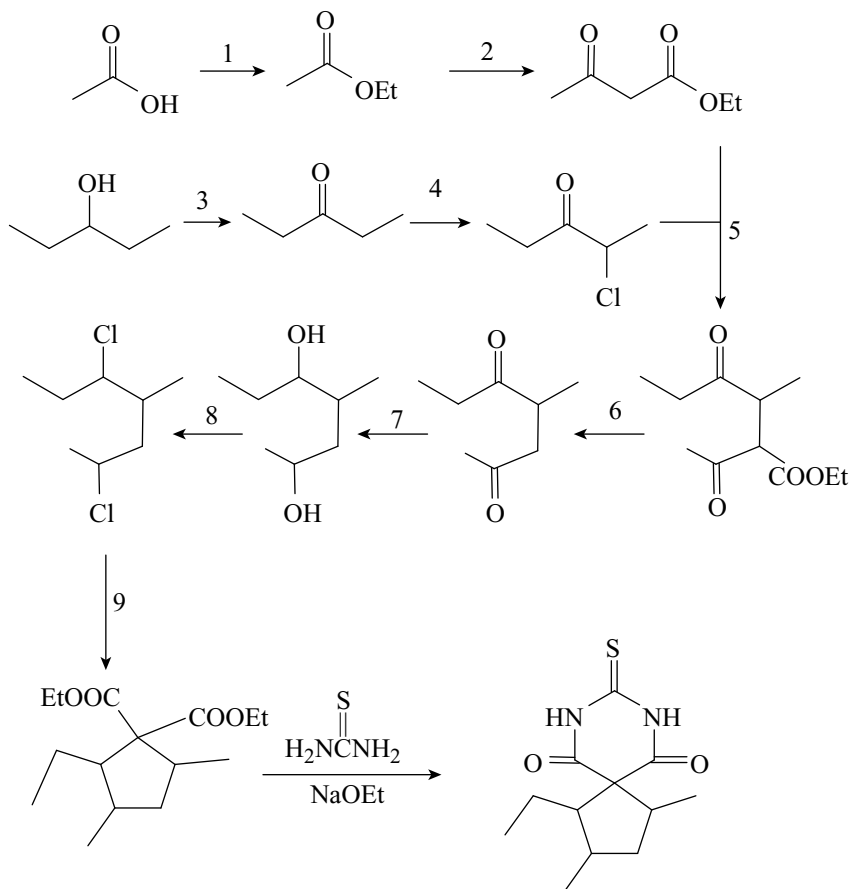
- (A) NaBH_4 then H_3O^+
- (B) LiAlH_4 then H_3O^+
- (C) O_3 then $\text{Zn} / \text{CH}_3\text{CO}_2\text{H}$
- (D) O_3 then H_2O_2
- (E) NaOH
- (F) Na dissolved in EtOH
- (G) $\text{AC. H}_3\text{O}^+$
- (H) BH_3 then $\text{NaOH} / \text{H}_2\text{O}_2$
- (I) Na then add PhBr
- (J) Na then add PhCH_2Br
- (K) Pyridinium chlorochromate
- (L) $\text{Na}_2\text{Cr}_2\text{O}_7 / \text{H}_3\text{O}^+$
- (M) $\text{NaBr} / \text{acetone}$
- (N) PBr_3
- (O) $\text{Br}_2 / \text{h}\nu$
- (P) $\text{LDA} / \text{THF} / -78^\circ\text{C}$
- (Q) $\text{Mg} / \text{Et}_2\text{O}$
- (R) $\text{EtOH} / \text{H}^+ / \text{heat}$
- (S) $\text{CH}_2=\text{CH}-\text{CH}_2\text{MgBr}$
- (T) NaOEt then $\text{CH}_2=\text{CH}-\text{CH}_2\text{Br}$

WORKBOOK EXERCISE 3

Identify reagents (1 to 10) used in the following conversion from reagent present in second Column II (A to P).

Column I (Conversion)

Column II (Reagent)



- (A) NaH
- (B) Na / THF then MeOH
- (C) H_2 / Pd
- (D) NaOH / Cl_2
- (E) HCl / Cl_2
- (F) PCl_3
- (G) 1 mol. eq. LiAlH_4 then H_3O^+
- (H) Pyridinium chlorochromate
- (I) H_2O_2
- (J) $\text{H}_3\text{O}^+ / \Delta$
- (K) Conc. $\text{H}_2\text{SO}_4 / \Delta$
- (L) EtOH / H^+ / Δ
- (M) NaOEt / EtOH
- (N) LiNiPr_2 / THF
- (O) Aq. NaOH / Δ
- (P) NaOEt / EtOH / $\text{CH}_2(\text{CO}_2\text{Et})_2$

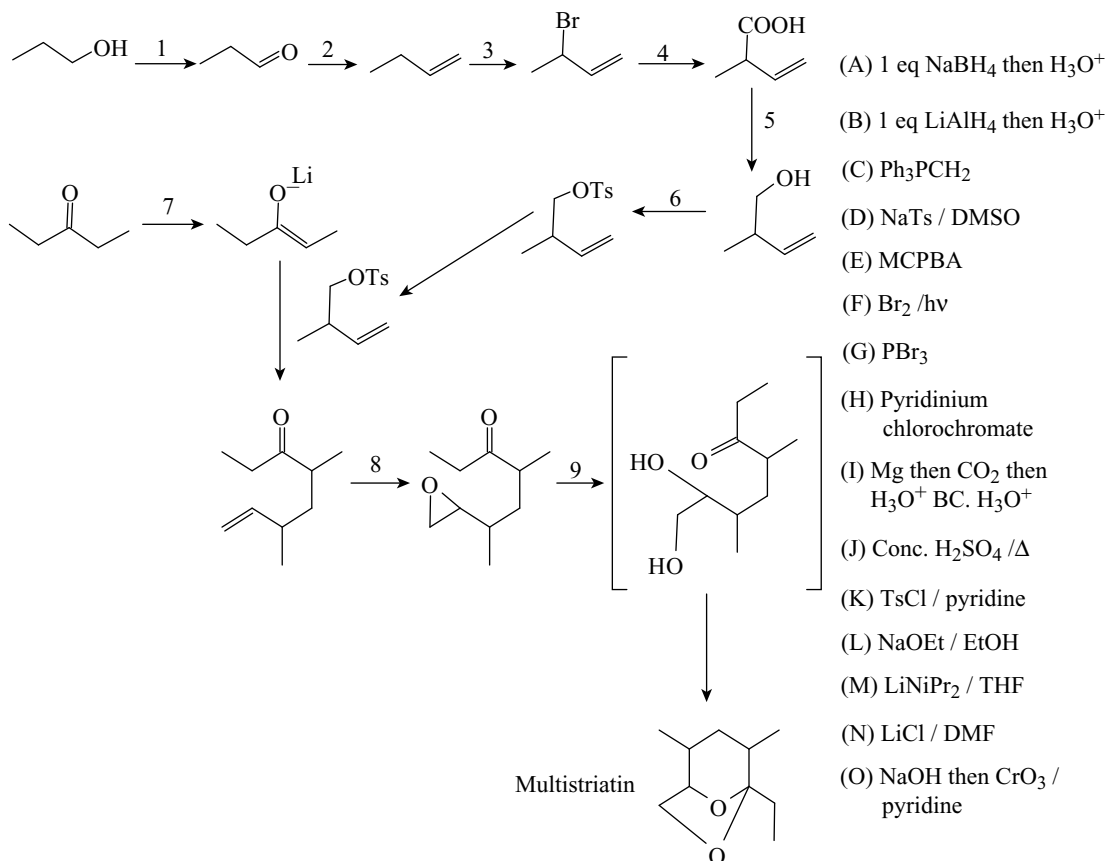
Spirothiobarbital

WORKBOOK EXERCISE 4

Identify reagents (1 to 10) used in the following conversion from reagent present in second Column II (A to O).

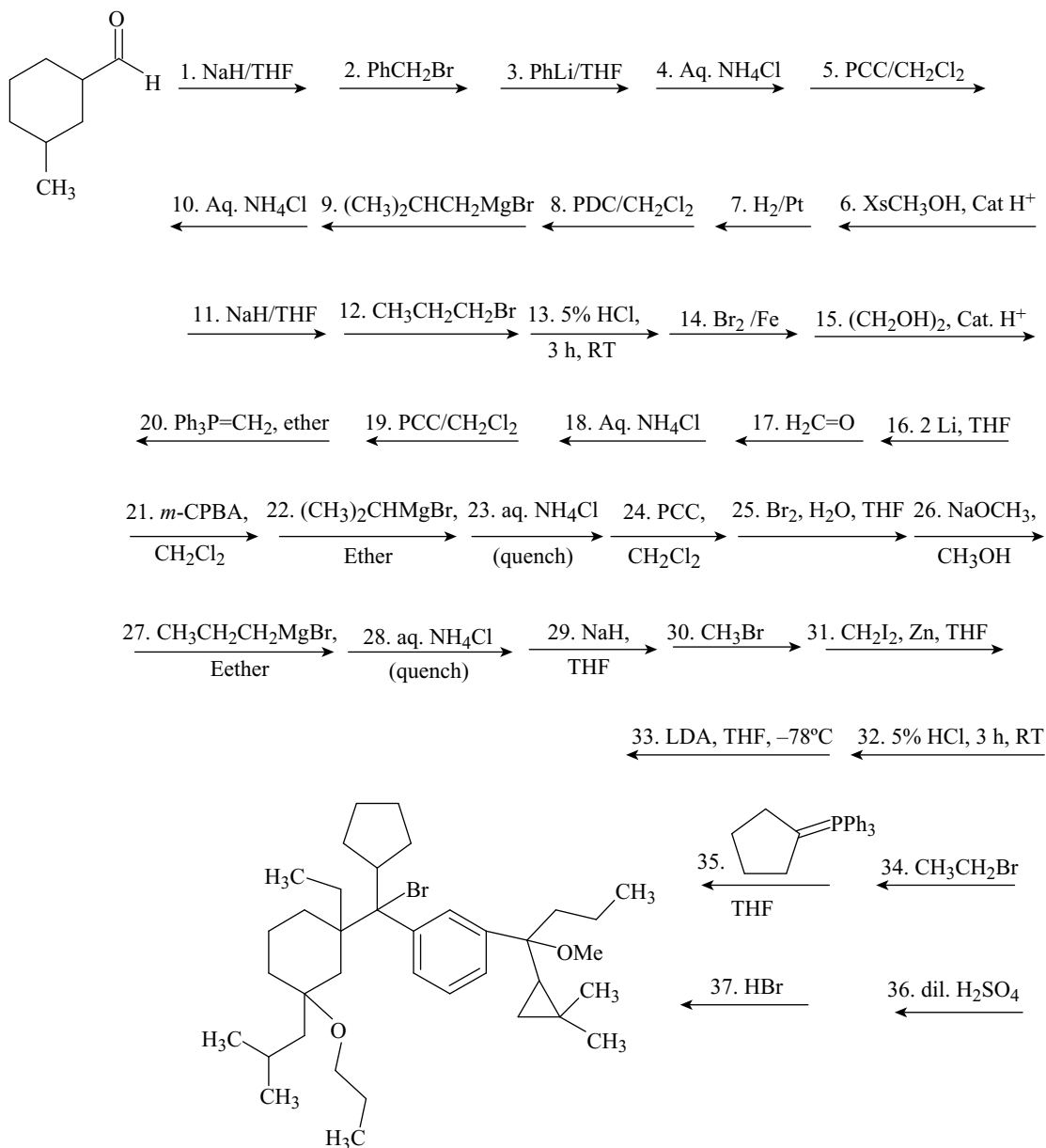
Column I (Conversion)

Column II (Reagent)



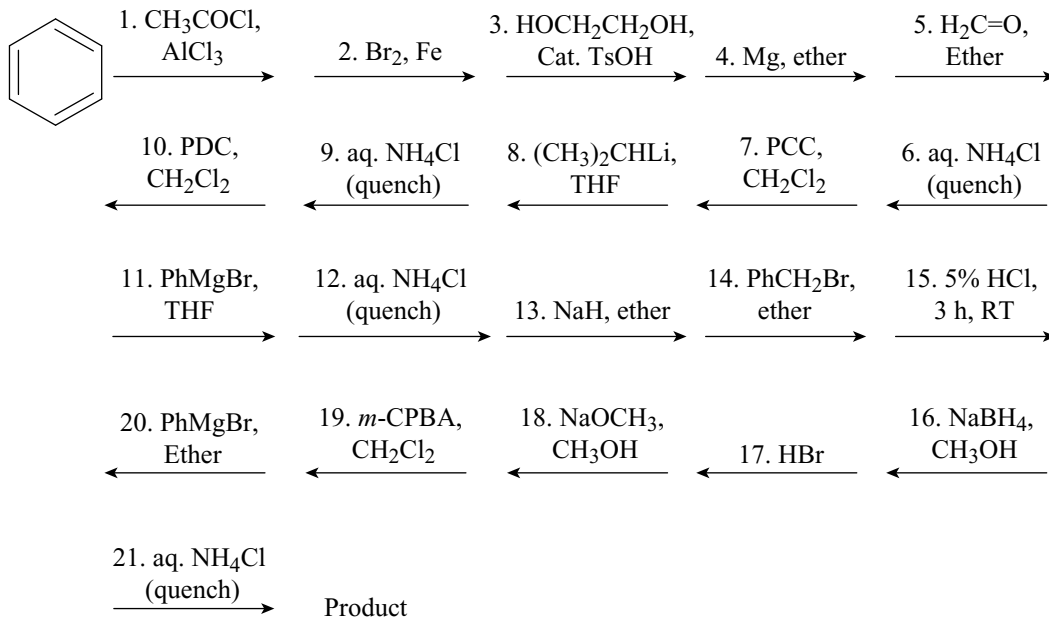
WORKBOOK EXERCISE 5

Identify the intermediate product in the following conversion



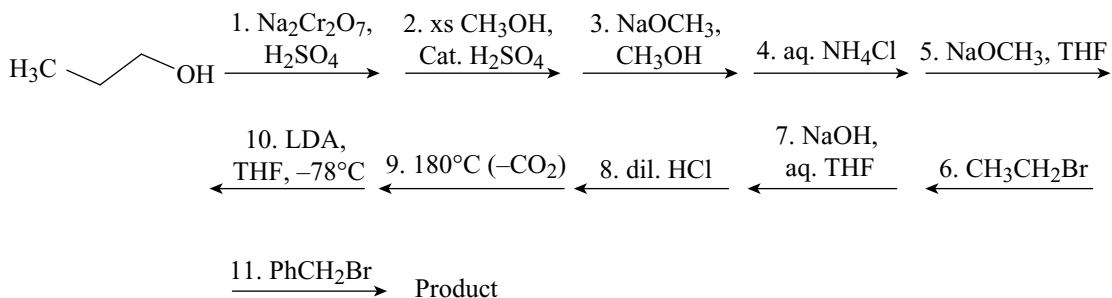
WORKBOOK EXERCISE 6

Identify the intermediate product in the following conversion



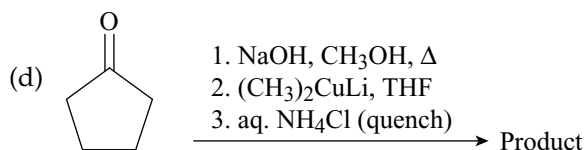
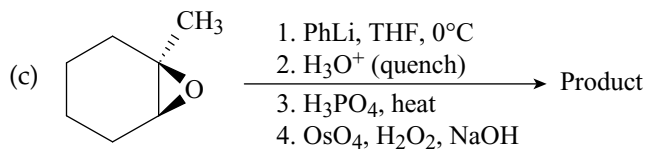
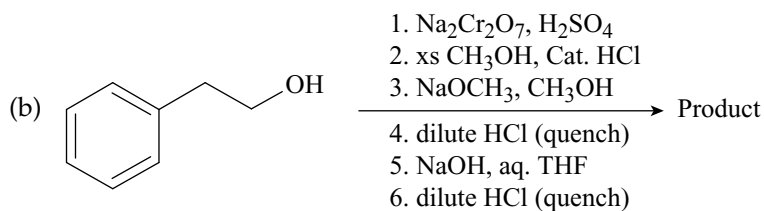
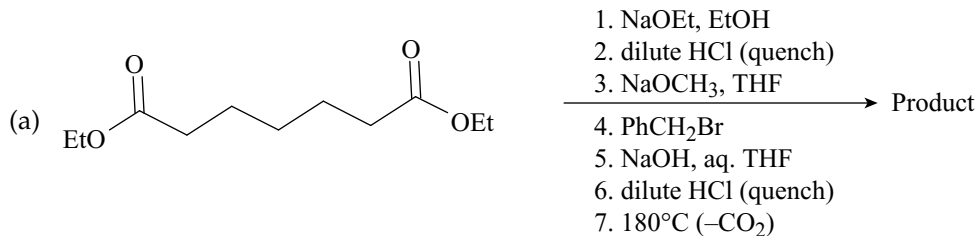
WORKBOOK EXERCISE 7

Identify the intermediate product in the following conversion

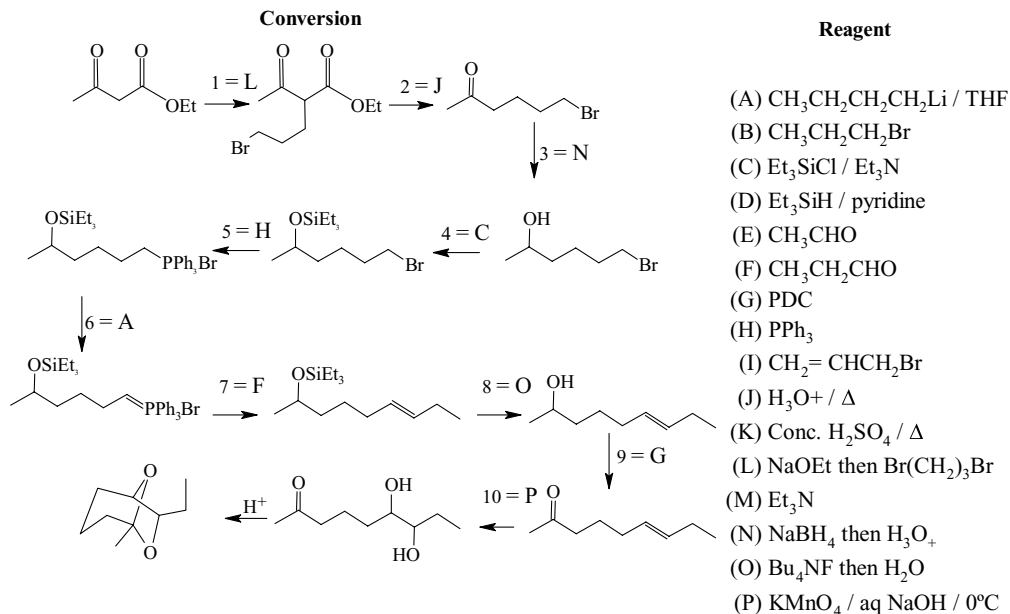


WORKBOOK EXERCISE 8

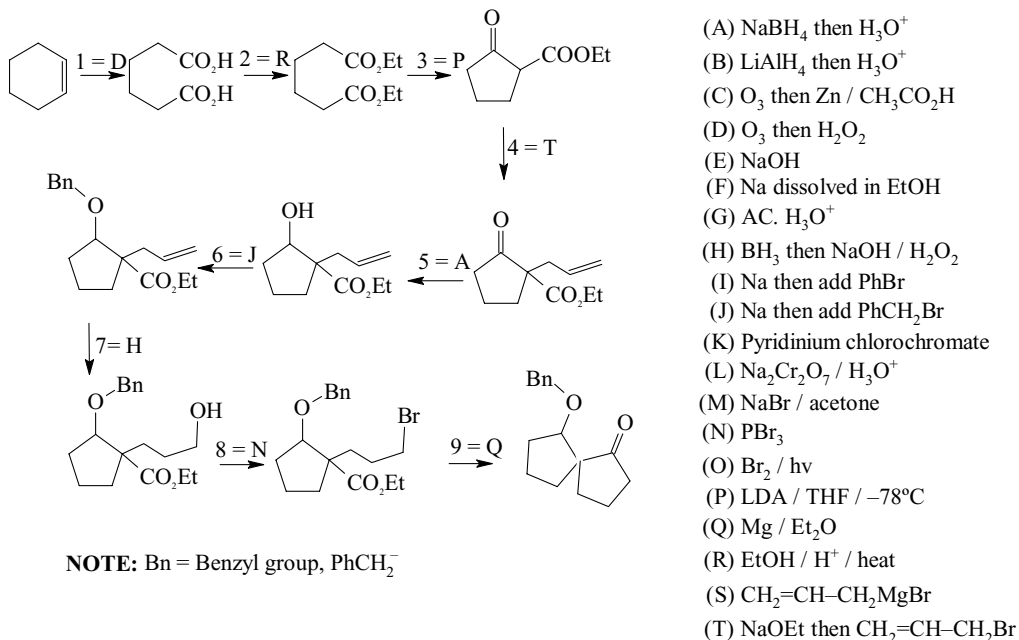
Identify the final product in the following reaction sequence



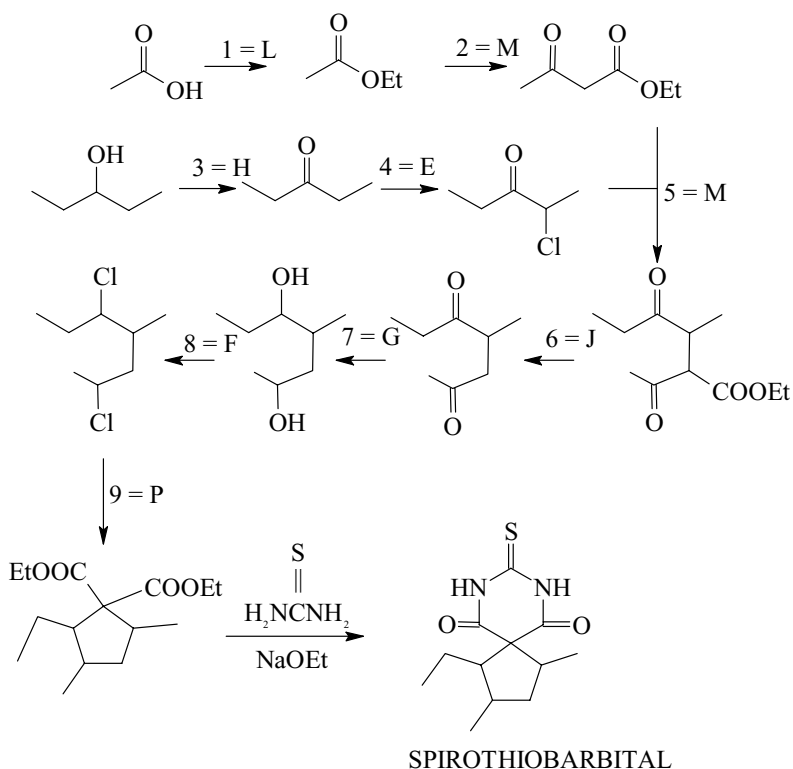
WORKBOOK EXERCISE 1



WORKBOOK EXERCISE 2



WORKBOOK EXERCISE 3



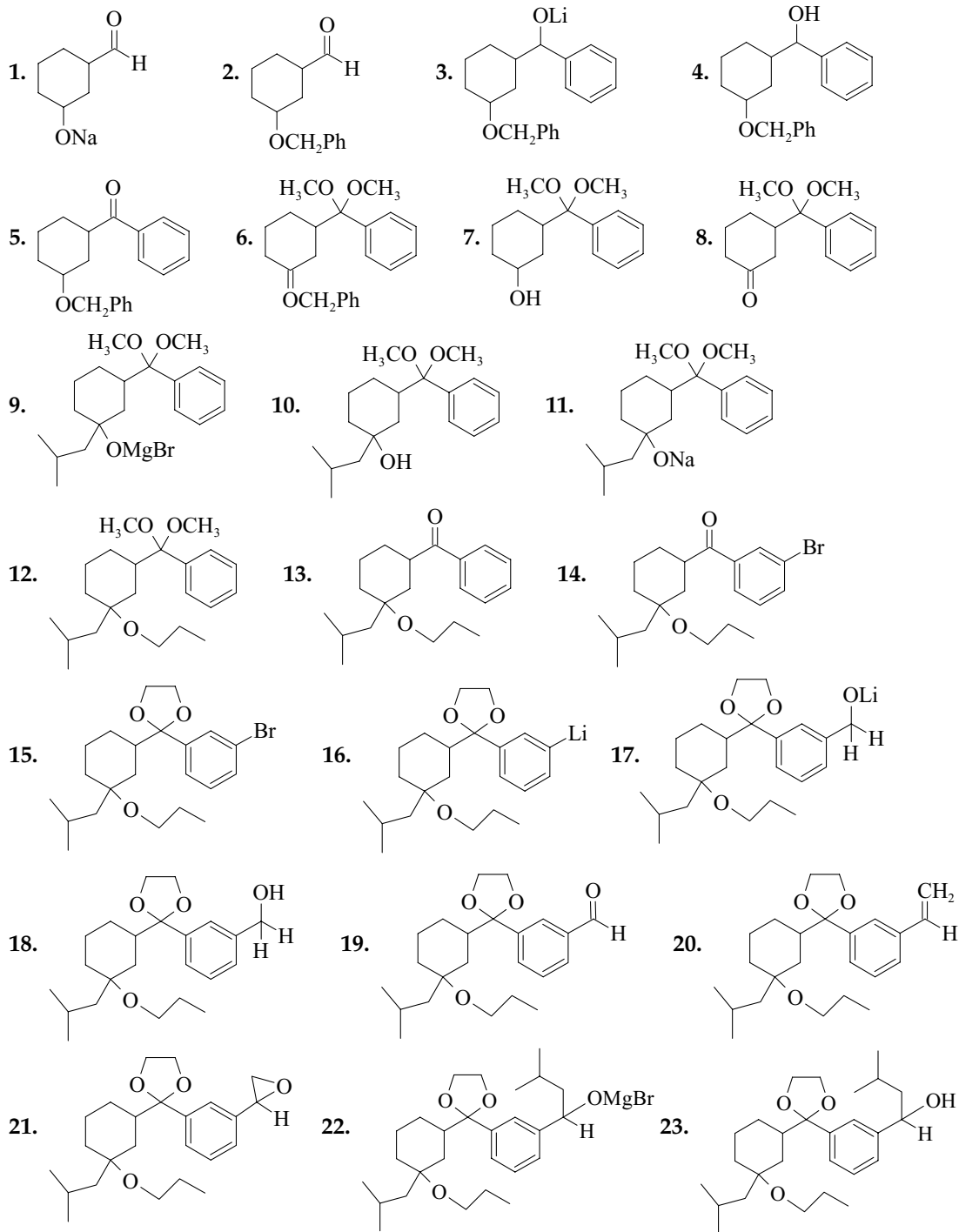
- (A) NaH
- (B) Na / THF then MeOH
- (C) H_2 / Pd
- (D) $\text{NaOH} / \text{Cl}_2$
- (E) HCl / Cl_2
- (F) PCl_3
- (G) 1 mol. eq. LiAlH_4 then H_3O^+
- (H) Pyridinium chlorochromate
- (I) H_2O_2
- (J) $\text{H}_3\text{O}^+ / \Delta$
- (K) Conc. $\text{H}_2\text{SO}_4 / \Delta$
- (L) $\text{EtOH} / \text{H}^+ / \Delta$
- (M) $\text{NaOEt} / \text{EtOH}$
- (N) $\text{Li NiPr}_2 / \text{THF}$
- (O) Aq. NaOH / Δ
- (P) $\text{NaOEt} / \text{EtOH} / \text{CH}_2(\text{CO}_2\text{Et})_2$

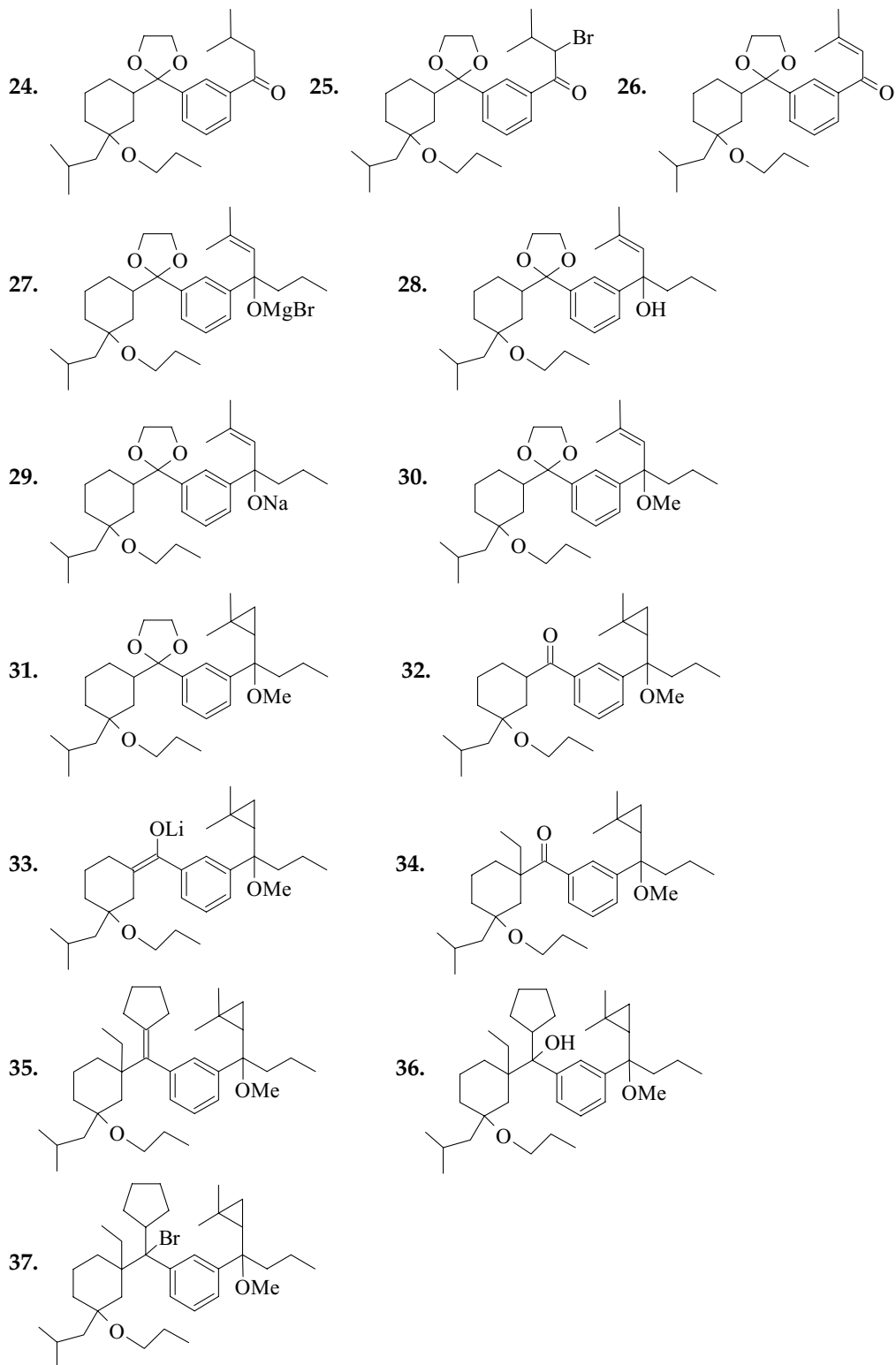
WORKBOOK EXERCISE 4



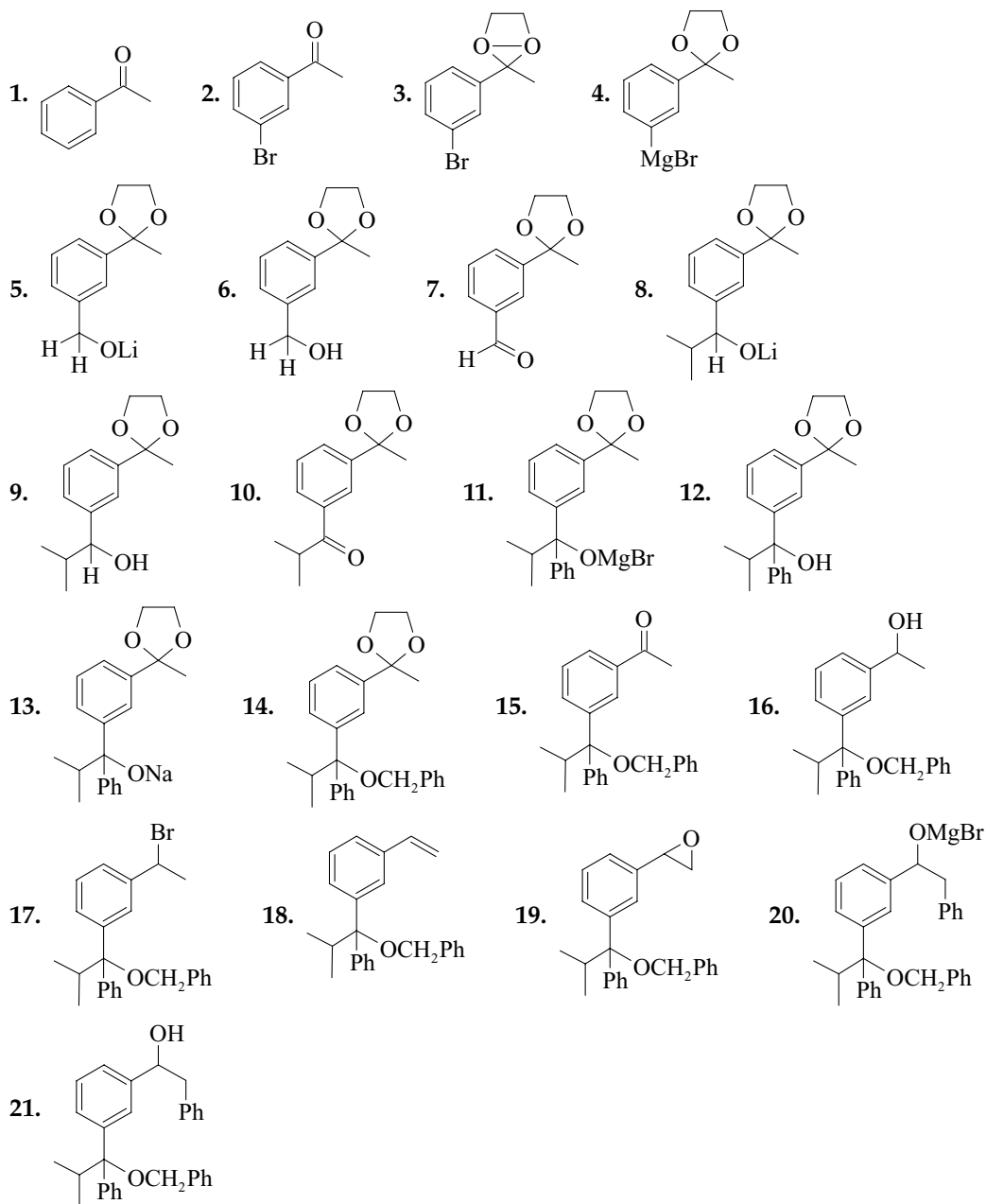
- (A) 1 eq NaBH_4 then H_3O^+
(B) 1 eq LiAlH_4 then H_3O^+
(C) Ph_3PCH_2
(D) $\text{NaOH} / \text{DMSO} / \text{H}_2\text{O}$
(E) MCPBA
(F) $\text{Br}_2 / \text{h}\nu$
(G) PBr_3
(H) Pyridinium chlorochromate
(I) Mg then CO_2 then H_3O^+ BC. H_3O^+
(J) Conc. $\text{H}_2\text{SO}_4 / \Delta$
(K) $\text{TsCl} / \text{pyridine}$
(L) $\text{NaOEt} / \text{EtOH}$
(M) $\text{Li NiPr}_2 / \text{THF}$
(N) LiCl / DMF
(O) NaOH then $\text{CrO}_3 / \text{pyridine}$

WORKBOOK EXERCISE 5

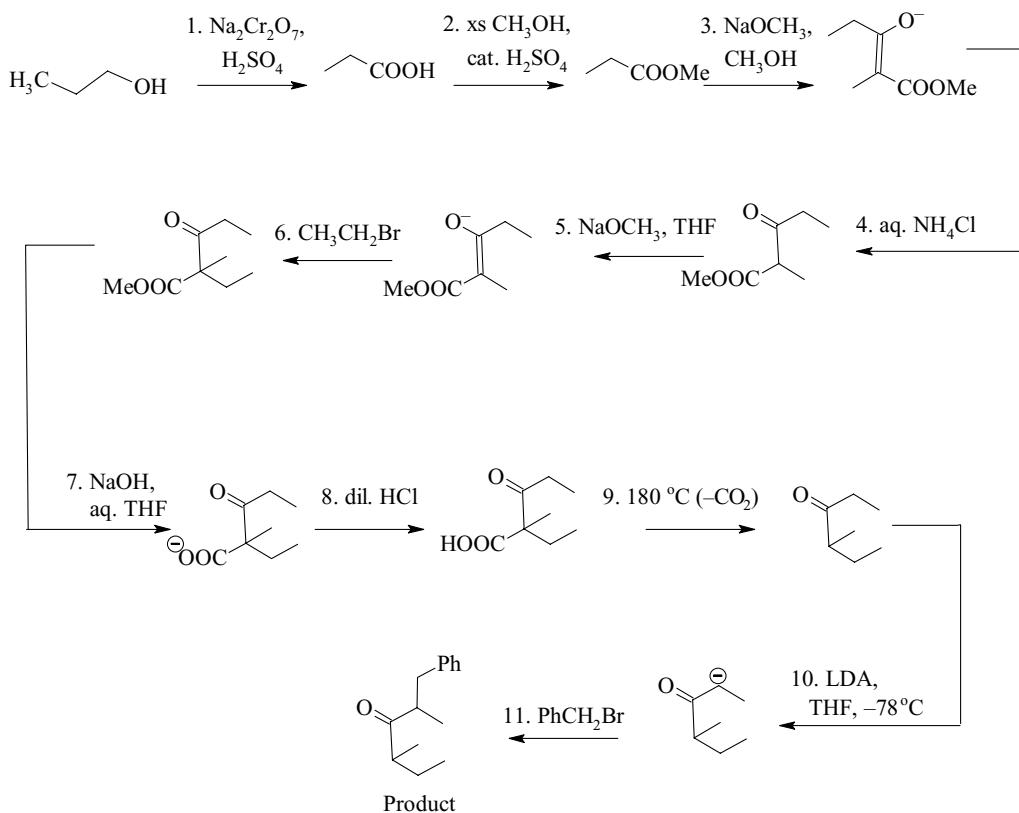




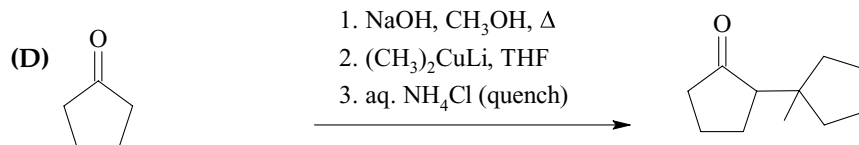
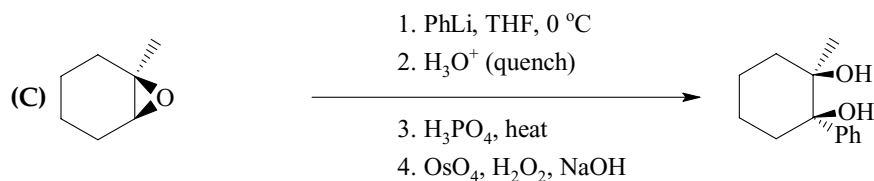
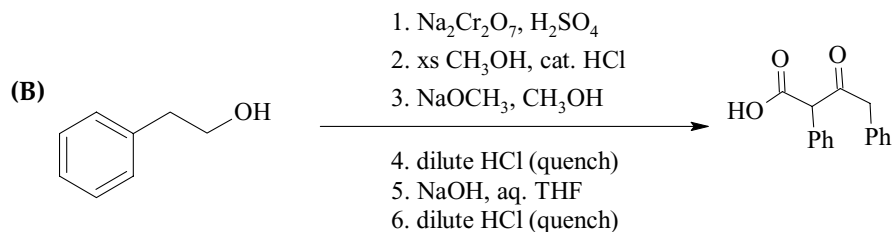
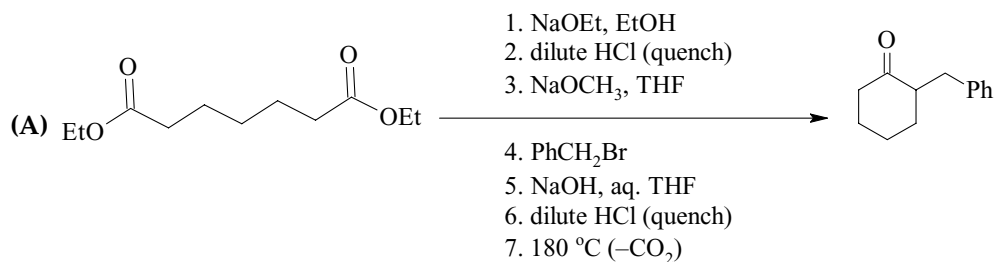
WORKBOOK EXERCISE 6



WORKBOOK EXERCISE 7



WORKBOOK EXERCISE 8



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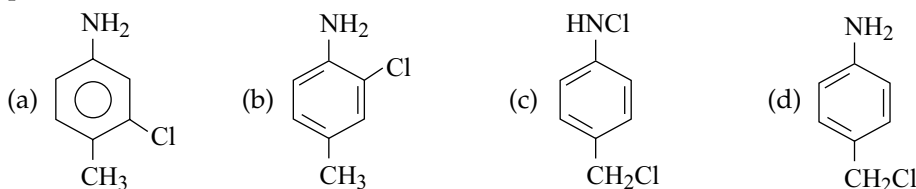
ESR Amines and Phenols

6

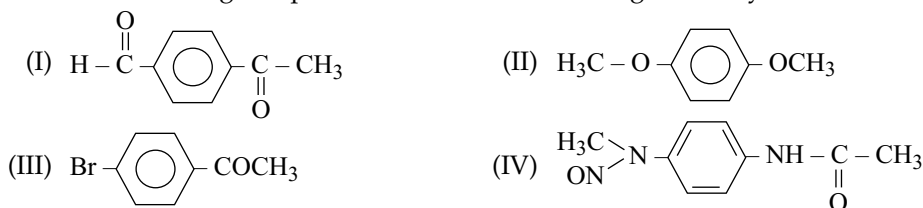
Question Bank

LEVEL 1

1. In the given reaction $\text{H}_2\text{N}-\text{C}_6\text{H}_4-\text{CH}_3 \xrightarrow[\text{Pyridine}]{(\text{CH}_3\text{CO})_2\text{O}} \text{A} \xrightarrow[\text{AlCl}_3]{\text{Cl}_2} \text{B} \xrightarrow{\text{H}_3\text{O}^+} \text{C}$ the product (C) will be

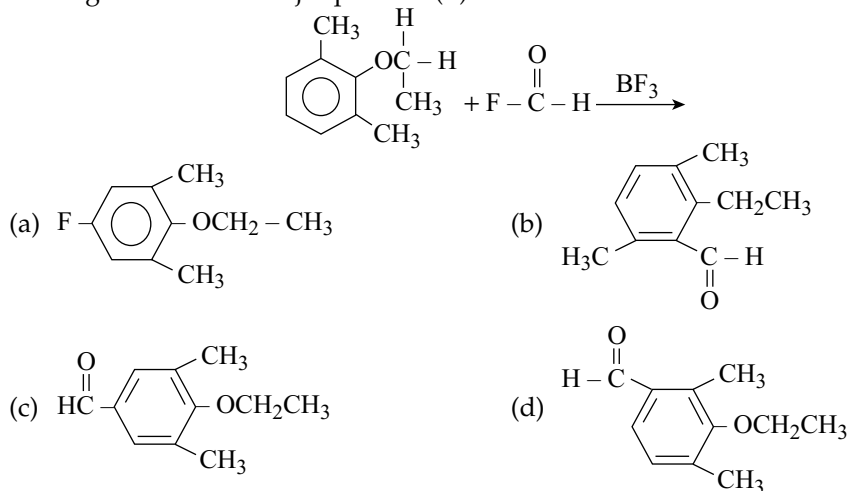


2. Rank the following compounds in order of decreasing reactivity for nitration.

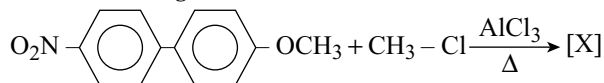


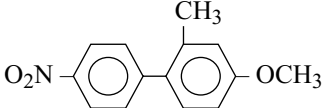
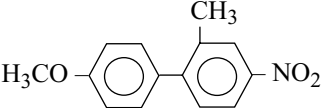
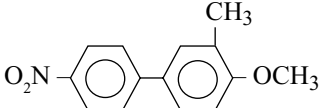
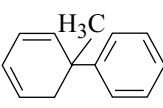
Select the correct answer from the following:

- (a) $\text{IV} > \text{II} > \text{I} > \text{III}$ (b) $\text{II} > \text{IV} > \text{III} > \text{I}$
(c) $\text{II} > \text{IV} > \text{I} > \text{III}$ (d) $\text{I} > \text{II} > \text{III} > \text{IV}$
3. In the given reaction major product (P) will be

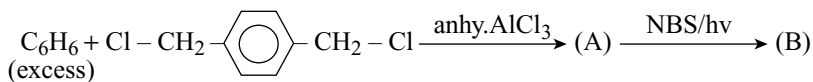


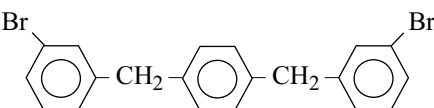
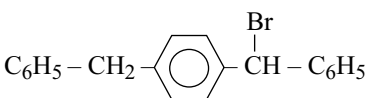
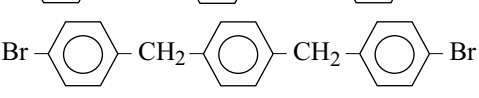
4. The major product [X] of the given reaction is



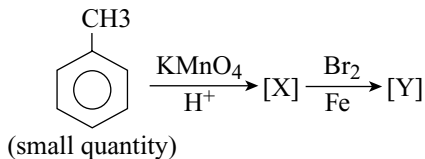
- (a)  (b) 
- (c)  (d) 

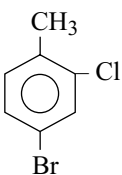
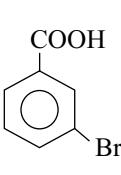
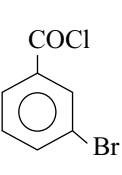
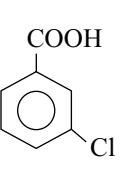
5. In the given reaction sequence, identify (B)

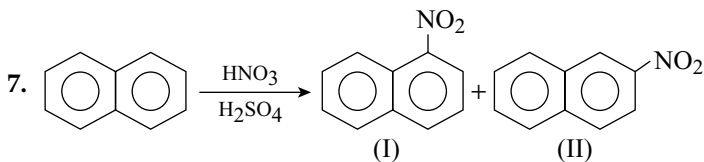


- (a)  (b) 
- (c)  (d) 

6. The final product of the given reaction is

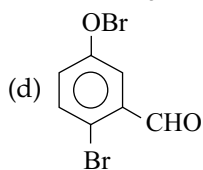
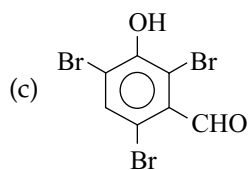
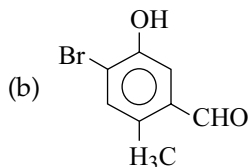
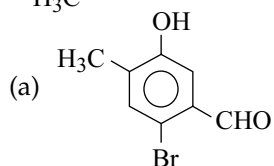
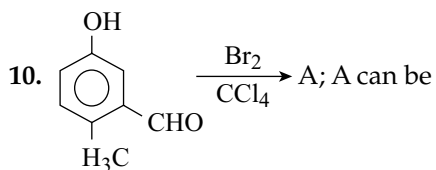
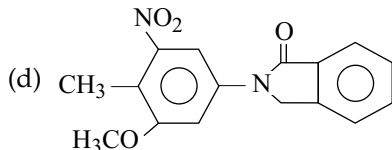
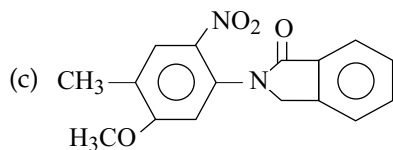
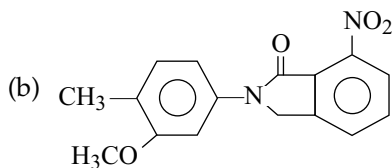
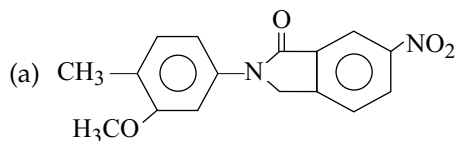
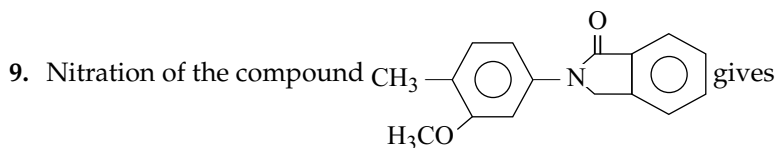
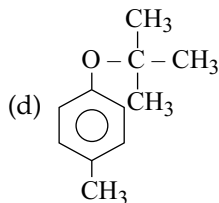
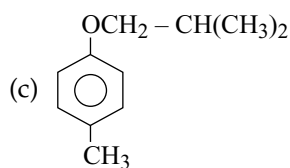
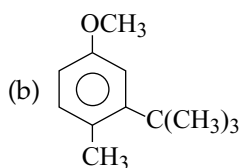
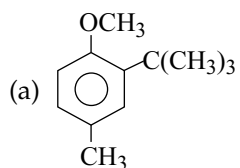
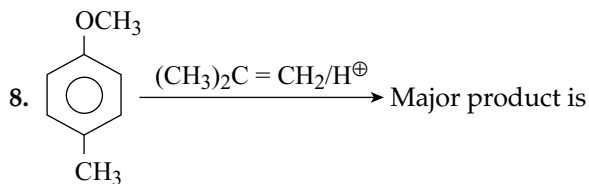


- (a)  (b) 
- (c)  (d) 

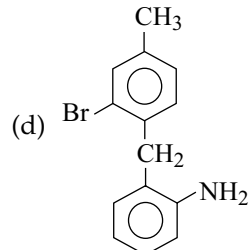
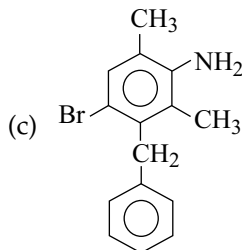
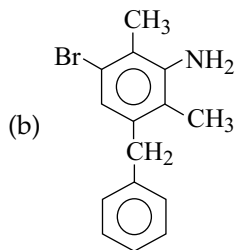
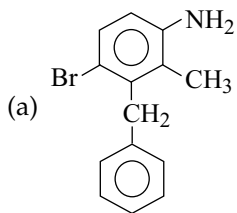
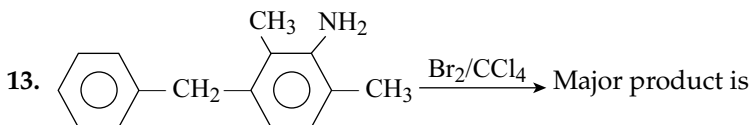
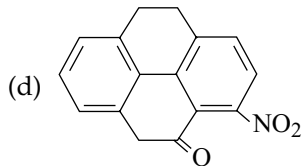
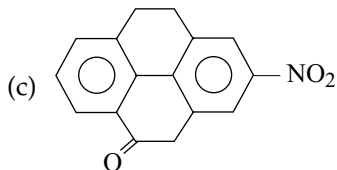
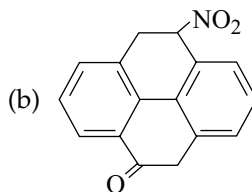
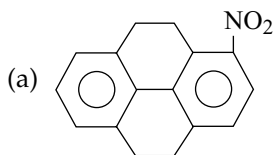
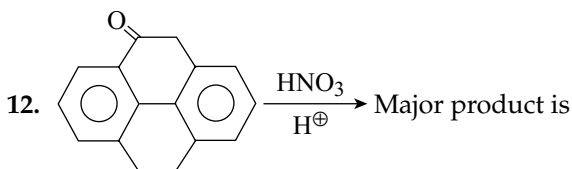
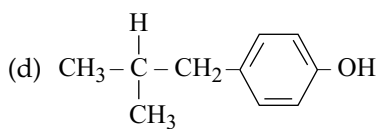
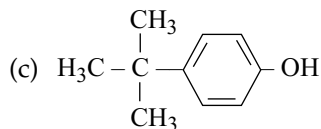
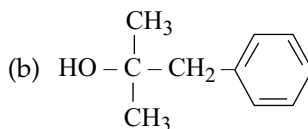
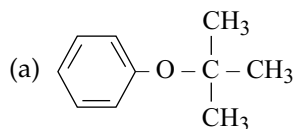
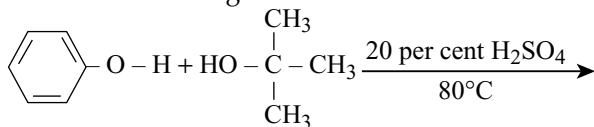


Which statement is correct?

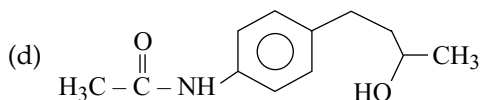
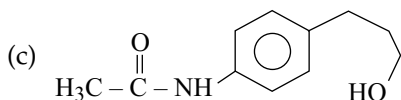
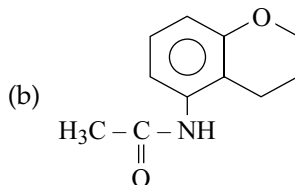
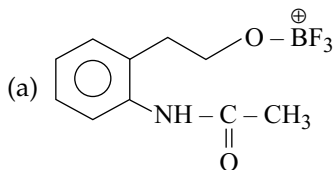
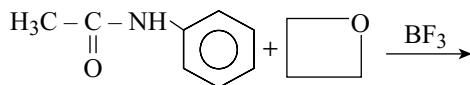
- (a) (I) form in large quantity (b) (II) form in large quantity
(c) (I), (II) form in equal quantity (d) naphthalene does not show nitration



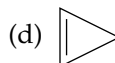
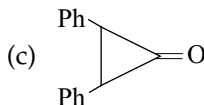
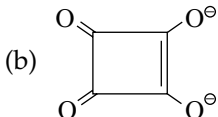
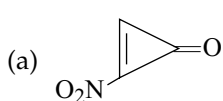
11. The major product of the following reaction is



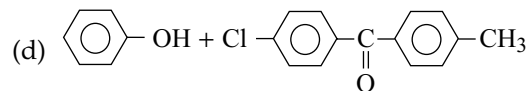
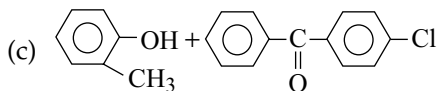
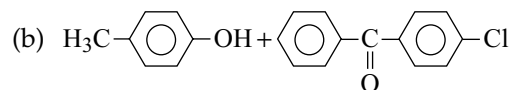
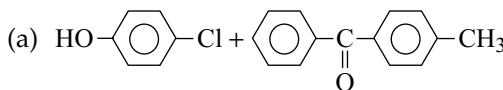
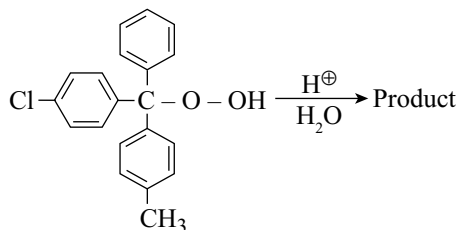
14. The product of the following reaction is



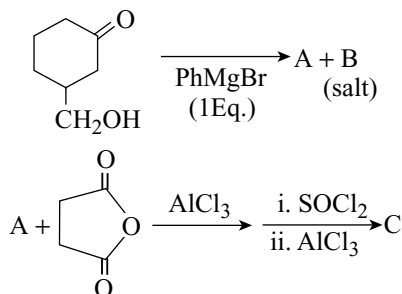
15. Which has equal carbon–carbon bond length?

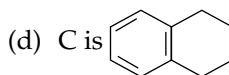
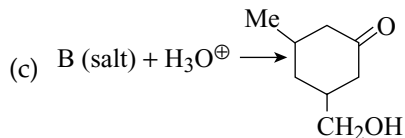
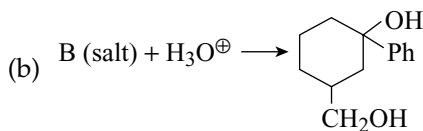
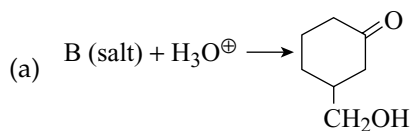


16. Product of the reaction is

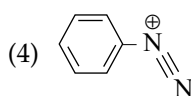
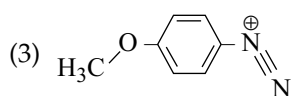
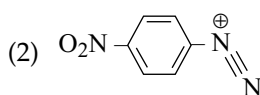
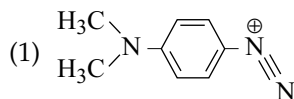


17. For the following reaction, correct options are



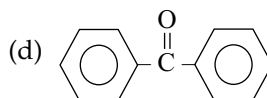
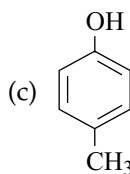
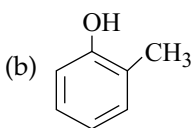
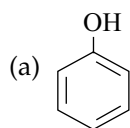
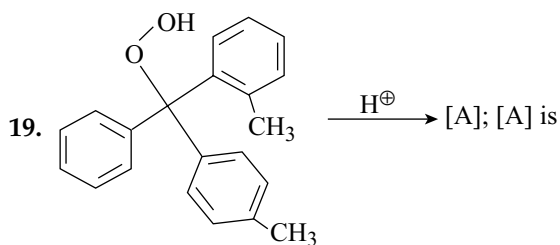


18. Consider the following ions

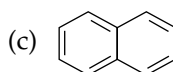
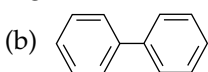
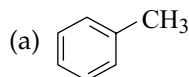


The reactivities of these ions in azo-coupling reactions (under similar conditions) will be such that

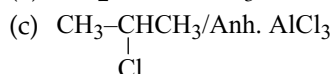
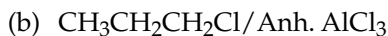
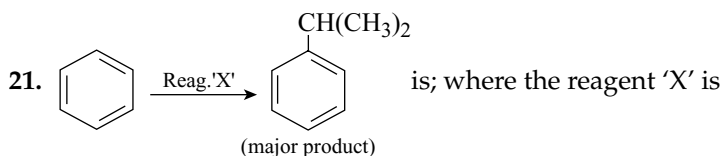
- (a) $1 < 4 < 2 < 3$ (b) $1 < 3 < 4 < 2$ (c) $3 < 1 < 2 < 4$ (d) $3 < 1 < 4 < 2$



20. Which of the following is/are more reactive towards nitration than benzene?

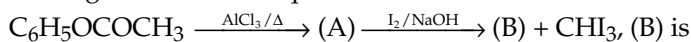


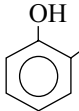
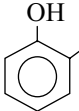
(d) All of these



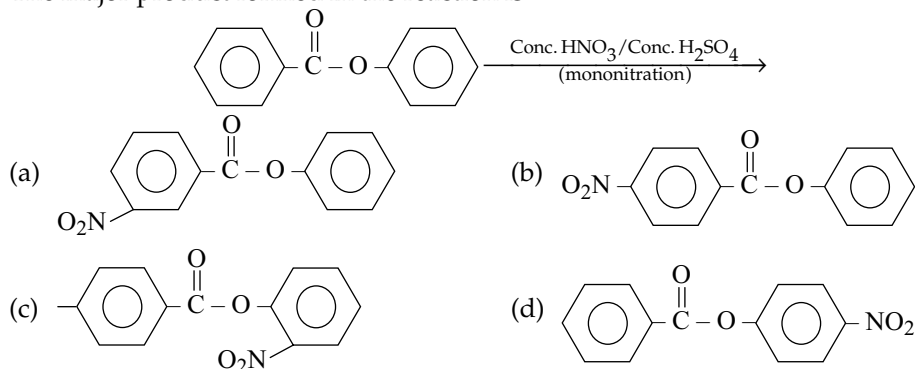
(d) All of these

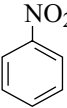
22. In the given reaction sequence

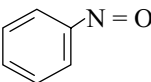
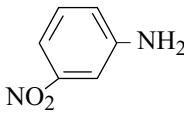
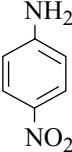
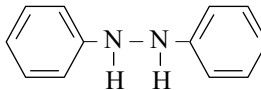


- (a) $\text{C}_6\text{H}_5\text{COONa}$ (b) $\text{C}_6\text{H}_5\text{COOH}$ (c)  (d) 

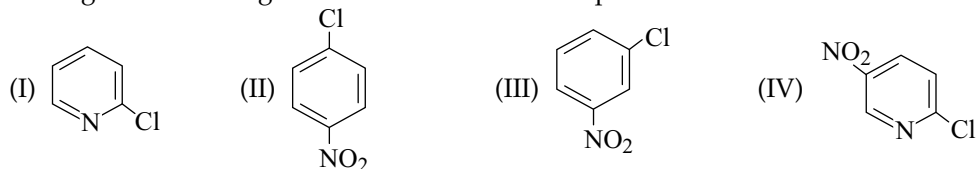
23. The major product formed in the reaction is



24.  $\xrightarrow{\text{Fe/HCl}} (\text{A}) \xrightarrow[\text{H}_2\text{SO}_4]{\text{HNO}_3} (\text{B})$. The major product (b) is

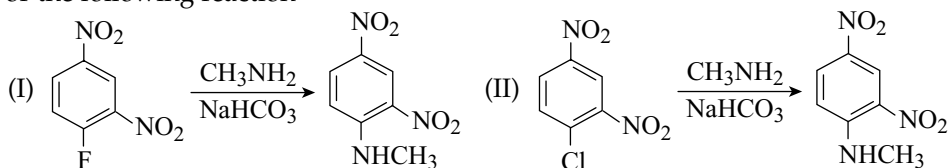
- (a)  (b) 
 (c)  (d) 

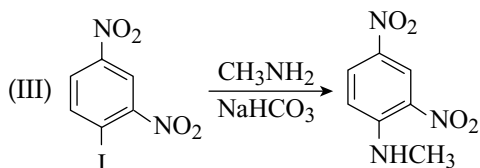
25. Arrange the following in the order of their nucleophilic substitution reaction



- (a) $\text{II} > \text{IV} > \text{I} > \text{III}$ (b) $\text{I} > \text{IV} > \text{II} > \text{III}$ (c) $\text{IV} > \text{I} > \text{II} > \text{III}$ (d) $\text{IV} > \text{II} > \text{I} > \text{III}$

26. For the following reaction



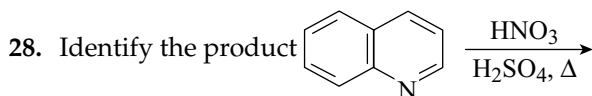


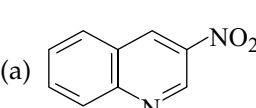
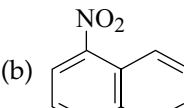
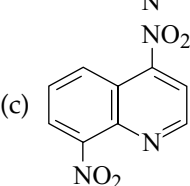
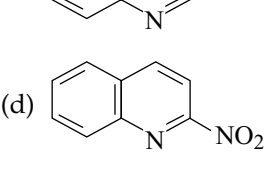
Correct order of reactivity

- (a) I = II = III (b) I > II > III (c) I < II < III (d) II > III > I

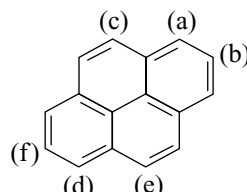
27. What is correct order of rate of nitration of the following compounds?

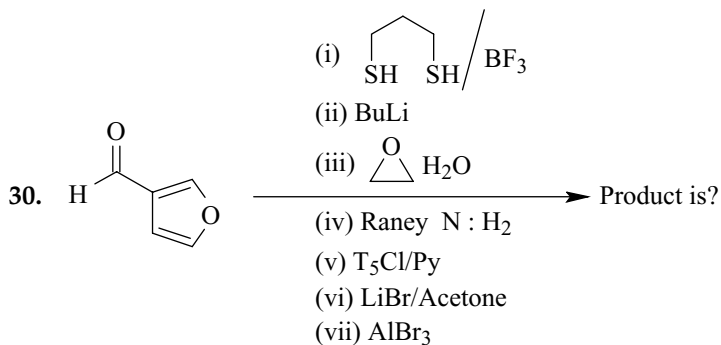
- (I) $\text{C}_6\text{H}_5\text{CH}_3$ (II) C_6H_6 (III) $\text{C}_6\text{H}_5\text{Br}$
 (IV) $\text{C}_6\text{H}_5\text{NR}_3^+$ (V) $\text{C}_6\text{H}_5\text{NMe}_2$
 (a) (IV) > (III) > (II) > (I) > (V) (b) (V) > (III) > (II) > (I) > (IV)
 (c) (V) > (I) > (II) > (III) > (IV) (d) (V) > (IV) > (II) > (I) > (III)

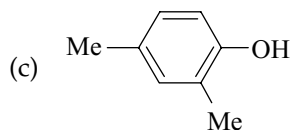
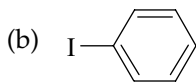
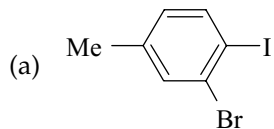
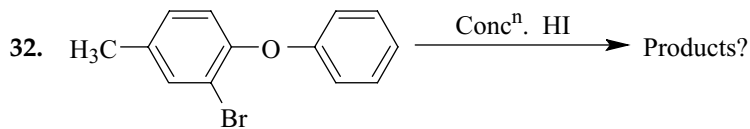
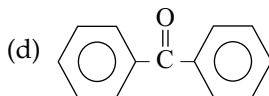
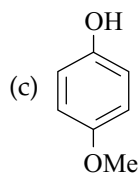
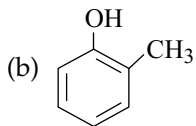
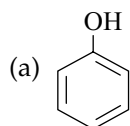
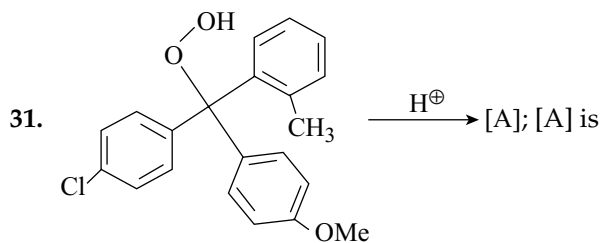
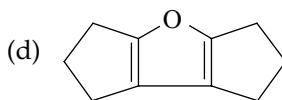
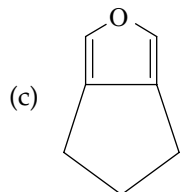
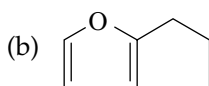
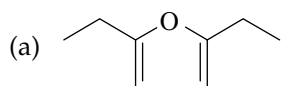


- (a)  (b) 
 (c)  (d) 

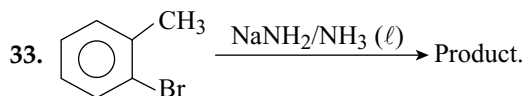
29. Which centre (s) more reactive towards EAS in the given molecule?

- 
 (a) a (b) d (c) c (d) e

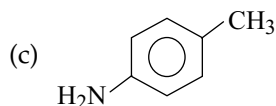
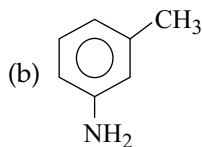
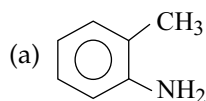




(d) No Reaction

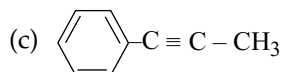
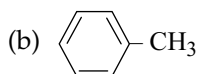
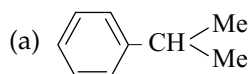


Product is



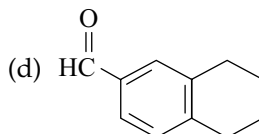
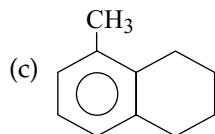
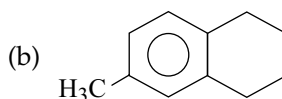
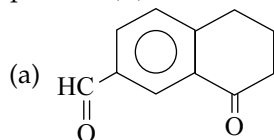
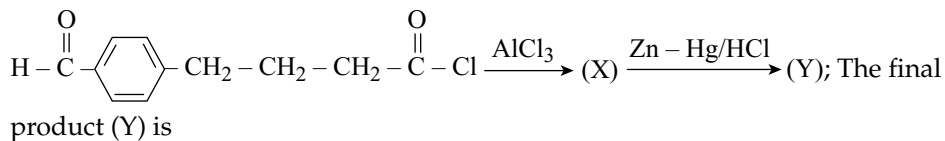
(d) Mixture of (a) and (b)

34. Which compound on oxidation with acidified KMnO_4 solution gives benzoic acid?

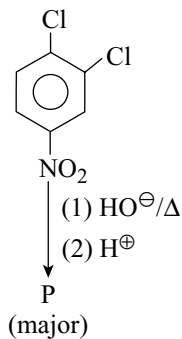


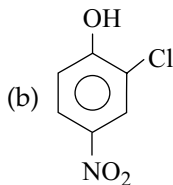
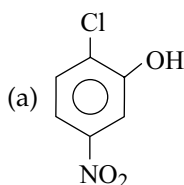
(d) All of these

35. In the given reaction



36. Based on the following reaction, the major product would be

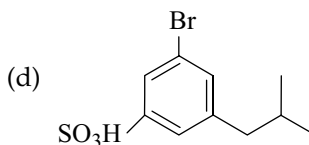
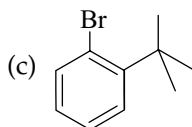
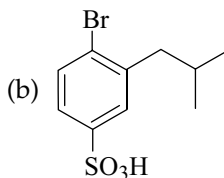
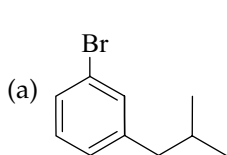
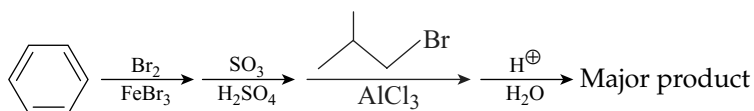




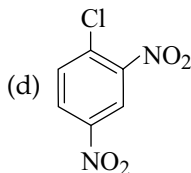
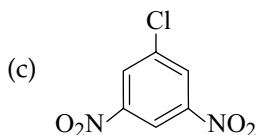
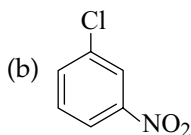
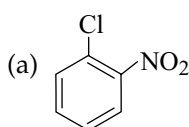
(c) Both in equal proportions

(d) None of these

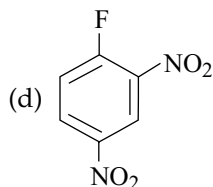
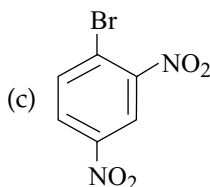
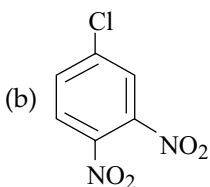
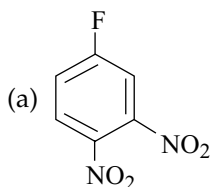
37. Give the major product from the following reaction sequence



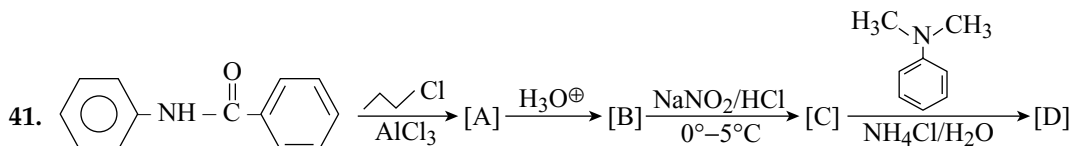
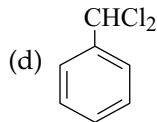
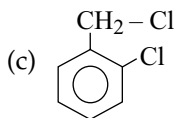
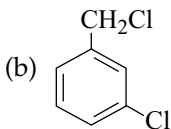
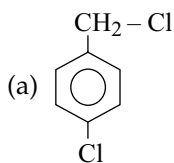
38. Which one of the following compounds will be most readily hydrolyzed in aqueous alkali?



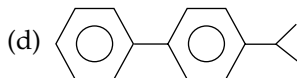
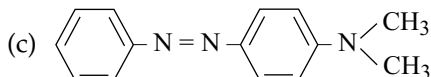
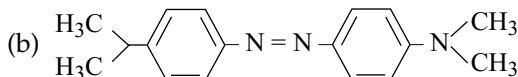
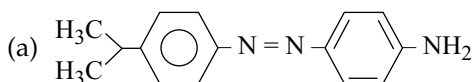
39. Which of the following is most reactive towards the reaction with NaOMe?



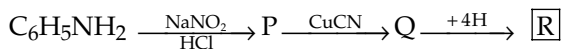
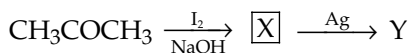
40. An aromatic compound 'A' $C_7H_6Cl_2$, gives $AgCl$ on boiling with alcoholic $AgNO_3$ solution and yields C_7H_7OCl on treatment with $NaOH$. 'A' on oxidation gives monochloro benzoic acid which affords only one mononitro derivative. The compound 'A' is



Product [D] is

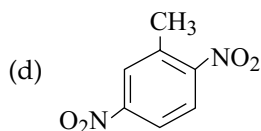
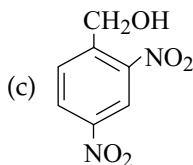
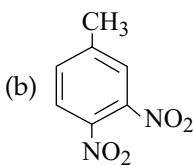
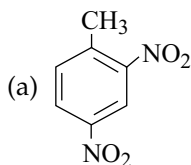


42. Identify B, X and R respectively in the following sequence of reactions

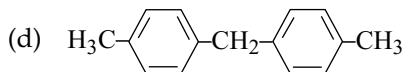
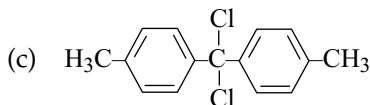
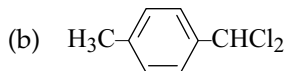
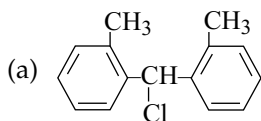


- (a) C_2H_5COOH , CHI_3 , $C_6H_5CH_2NH_2$
 (b) C_2H_5COOH , CH_3I , C_6H_5COOH
 (c) $C_2H_5CH_2NH_2$, CH_3I , C_6H_5COOH
 (d) C_2H_5COOH , C_2H_5I , $C_6H_5CONH_2$

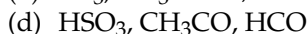
43. p-nitrotoluene on further nitration gives



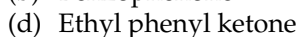
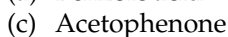
44. Which of the following structures correspond to the product expected, when excess of toluene reacts with CH_2Cl_2 in presence of anhydrous AlCl_3 ?



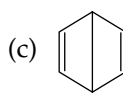
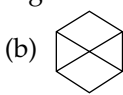
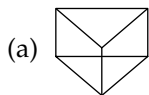
45. In the sulphonation, acetylation and formylation of benzene the group of effective electrophiles would be



46. When benzene is heated with acetic anhydride in the presence of anhydrous aluminium chloride at 80°C , the product formed is?

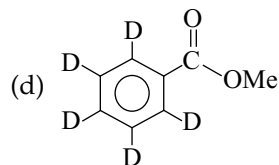
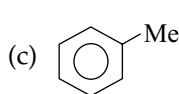
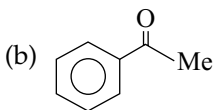
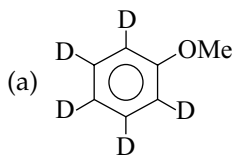


47. Which of the following is not the structural formulae of benzene?

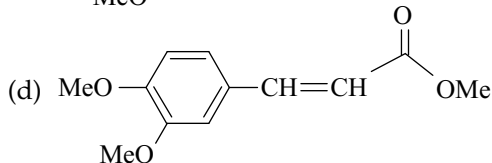
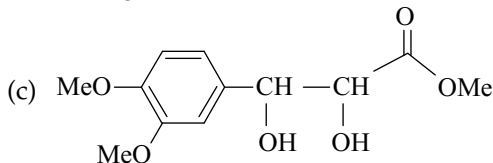
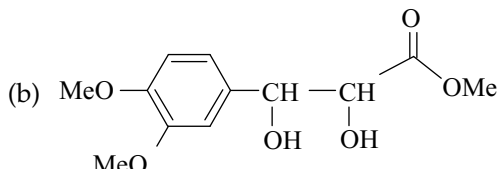
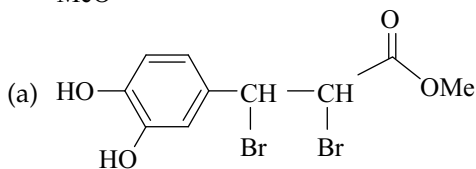


(d) None of these

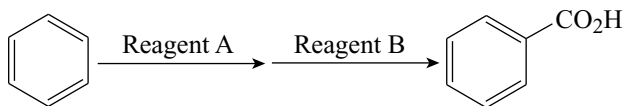
48. Most reactive towards nitration is



49. Major Products is?

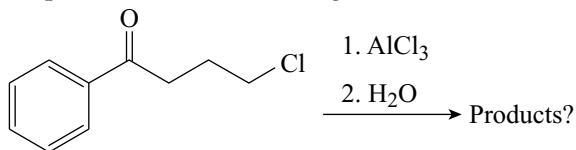


50. What reagents (conditions) are needed for the following reaction?



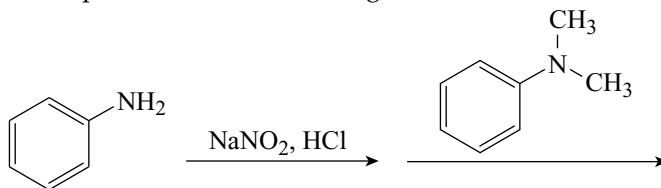
- (a) reagent A: $\text{CH}_3\text{COCl}/\text{AlCl}_3$; reagent B: $\text{Sn}, \text{HCl}, \text{heat}$
 (b) reagent A: $\text{CH}_3\text{CH}_2\text{Cl}/\text{AlCl}_3$; reagent B: $\text{KMnO}_4, \text{heat}$
 (c) reagent A: $\text{CH}_3\text{COCl}/\text{AlCl}_3$; reagent B: $\text{H}_2\text{NNH}_2, \text{KOH}, \text{H}_2\text{O}, \text{heat}$
 (d) reagent A: $\text{HNO}_3, \text{H}_2\text{SO}_4$; reagent B: $\text{Sn}, \text{HCl}, \text{heat}$

51. What could be the product for the following reaction?



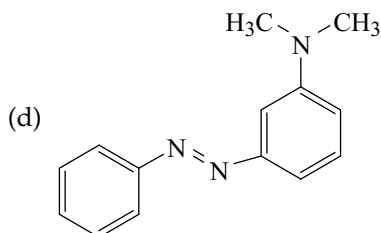
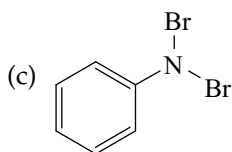
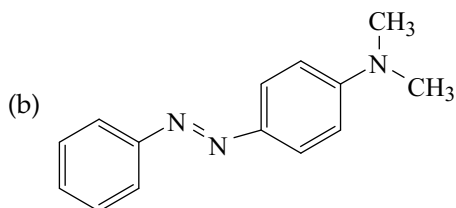
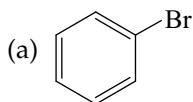
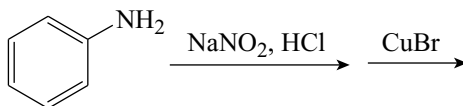
- (a)
- (b)
- (c)
- (d)

52. What could be the product for the following reaction?

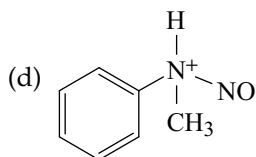
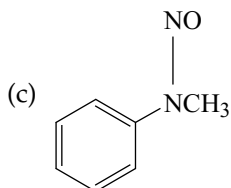
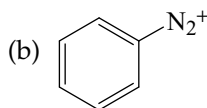
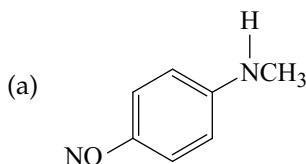


- (a)
- (b)
- (c)
- (d)

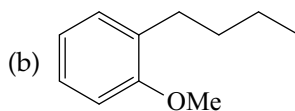
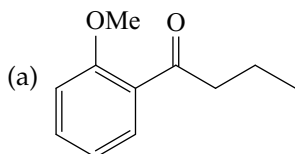
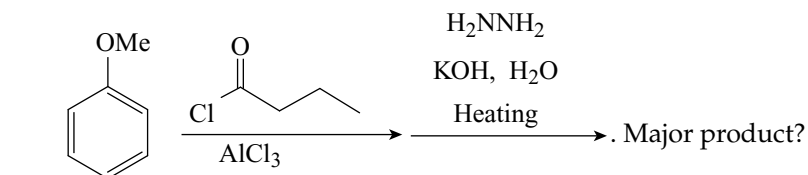
53. What could be the product for the following reaction?

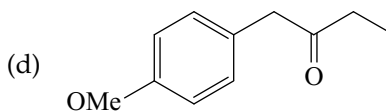
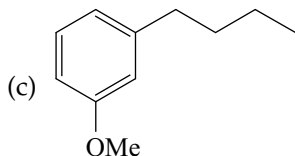


54. What could be the product for the following reaction $\text{Ph}-\text{NH}-\text{Me} \xrightarrow[\text{HCl}]{\text{NaNO}_2} ?$

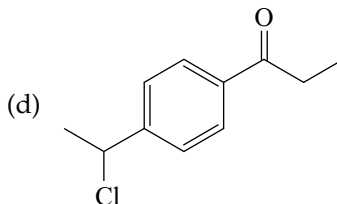
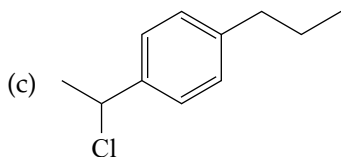
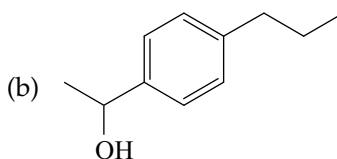
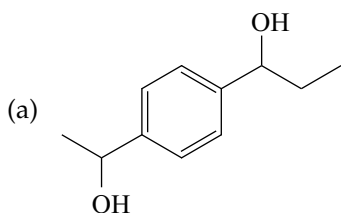
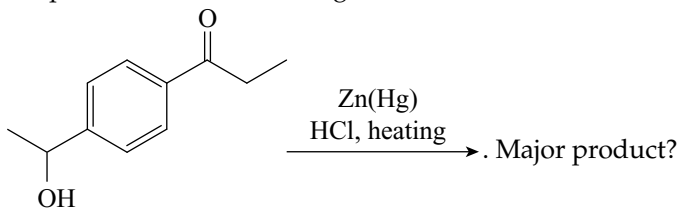


55. What could be the major product for the following reaction?

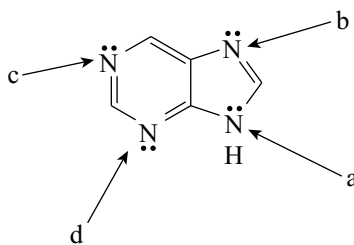




56. What could be the product for the following reaction?



57. For the following compound, which nitrogen is most apt to be protonated?



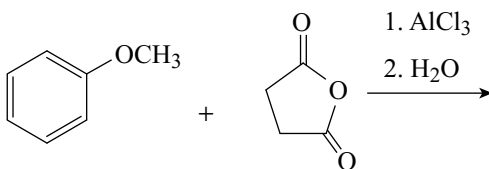
(a) Nitrogen indicated by arrow 'a'

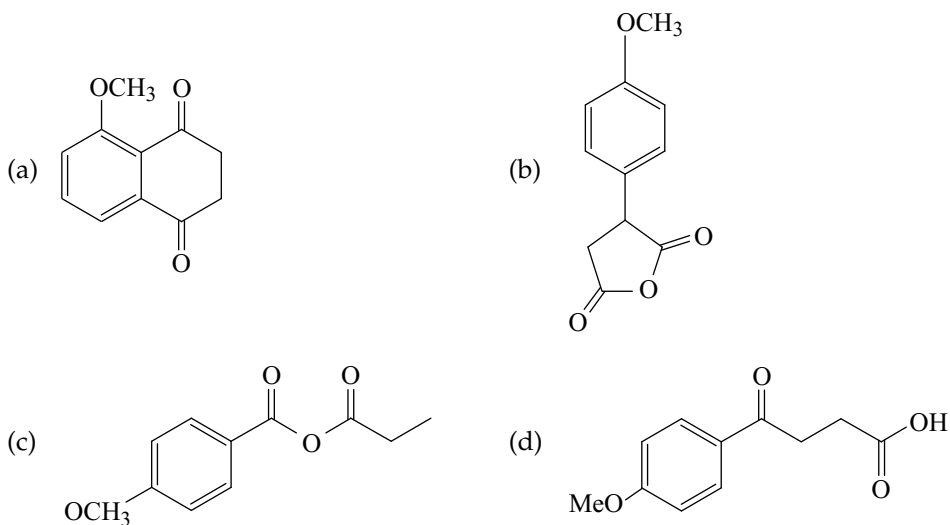
(b) Nitrogen indicated by arrow 'b'

(c) Nitrogen indicated by arrow 'c'

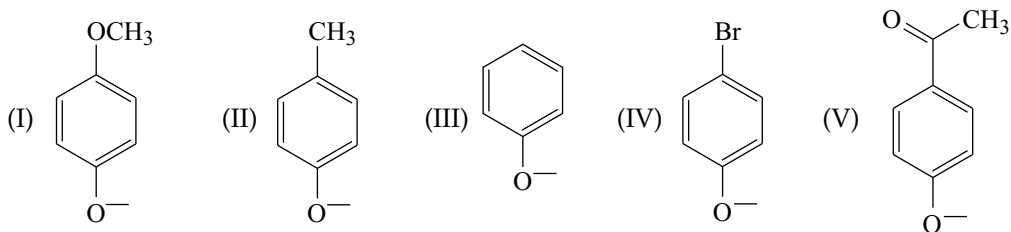
(d) Nitrogen indicated by arrow 'd'

58. What could be the product for the following reaction?





59. What is the correct order of decreasing basicity for the following anions (from the most to the least)?



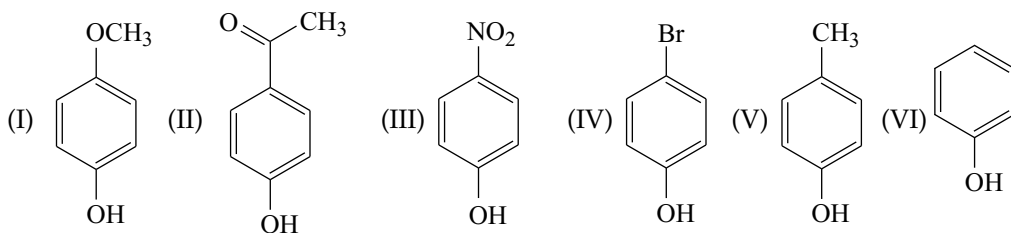
(a) I > II > III > IV > V

(b) I > II > III > V > IV

(c) I > II > IV > III > V

(d) V > IV > III > II > I

60. What is the correct order of decreasing acidity for the following phenol and phenol derivatives (from the most to the least)?



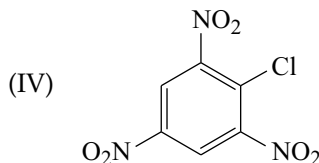
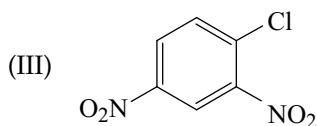
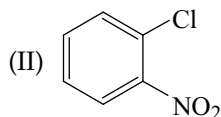
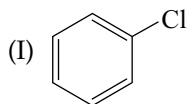
(a) I > II > III > IV > V > VI

(b) II > III > IV > VI > V > I

(c) III > II > IV > VI > V > I

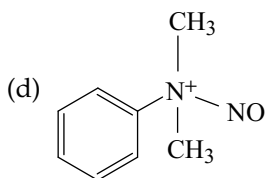
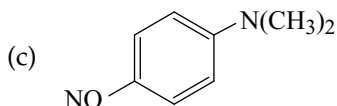
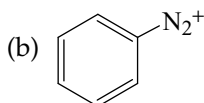
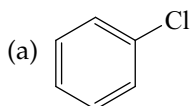
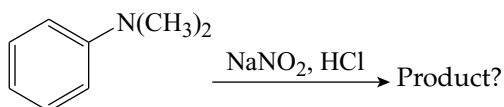
(d) II > III > IV > VI > I > V

61. What is the correct order of decreasing reactivity (fastest to slowest) toward nucleophilic aromatic substitution for the following compounds?

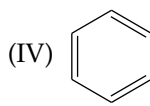
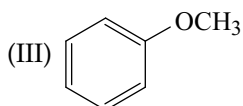
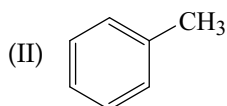
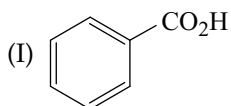


- (a) I > II > III > IV (b) II > III > IV > I (c) III > II > IV > I (d) IV > III > II > I

62. What could be the product for the following reaction?

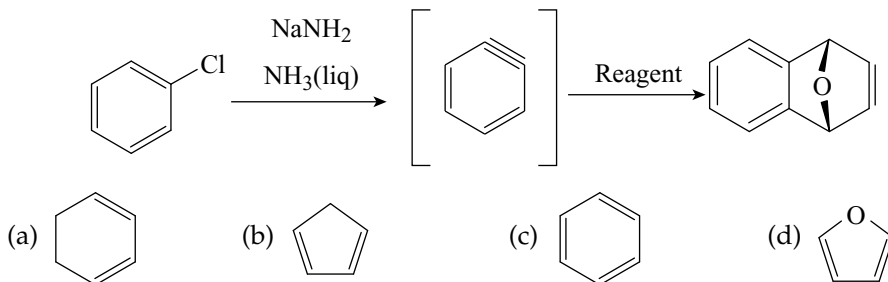


63. What is the correct order of decreasing reactivity (fastest to slowest) toward electrophilic aromatic substitution for the following compounds?

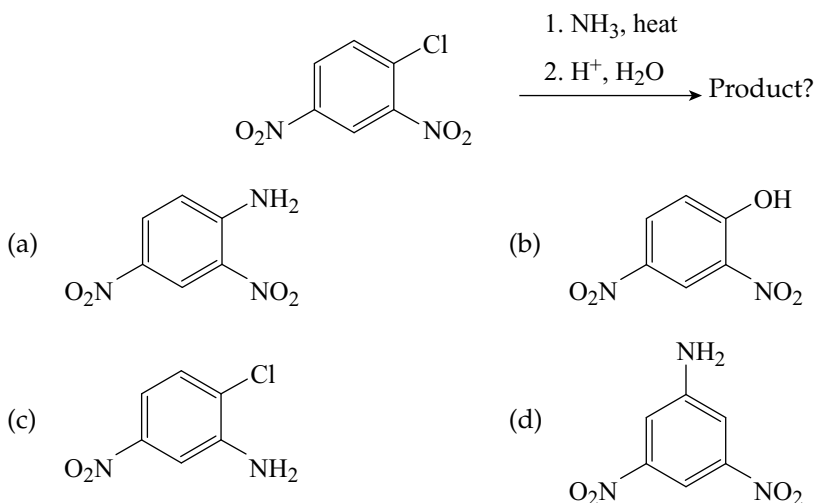


- (a) I > II > III > IV (b) II > I > IV > III
(c) III > II > IV > I (d) IV > III > II > I

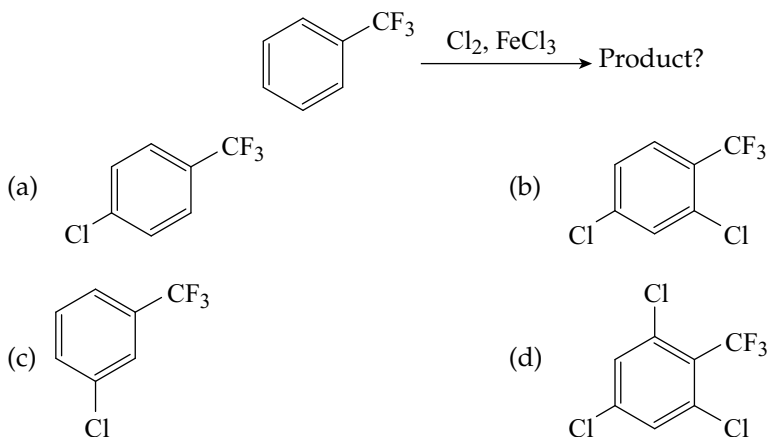
64. What could be the product for the following reaction?



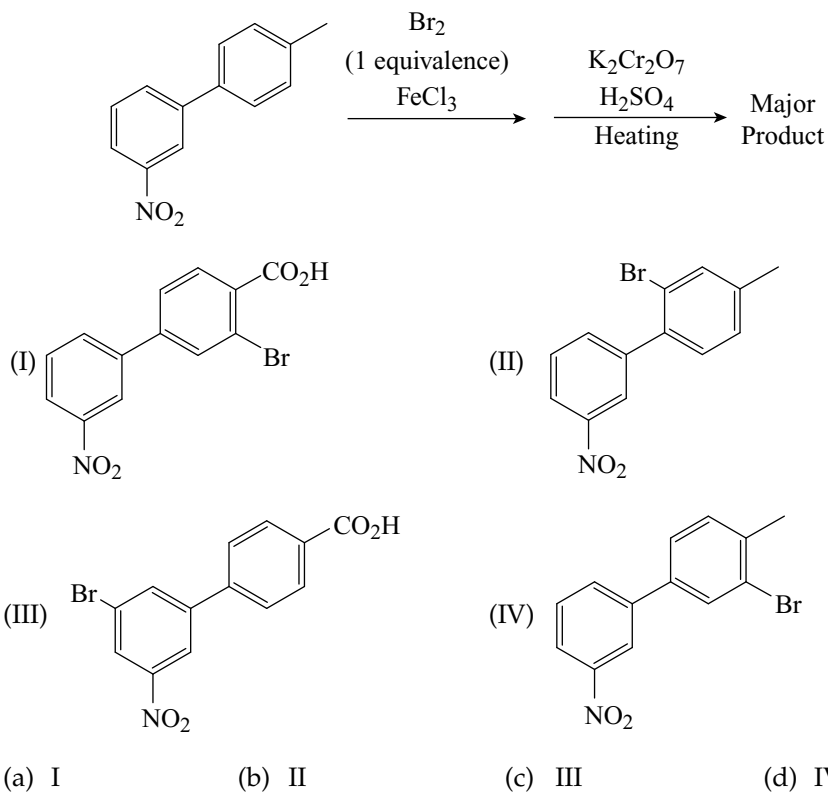
65. What could be the product for the following reaction?



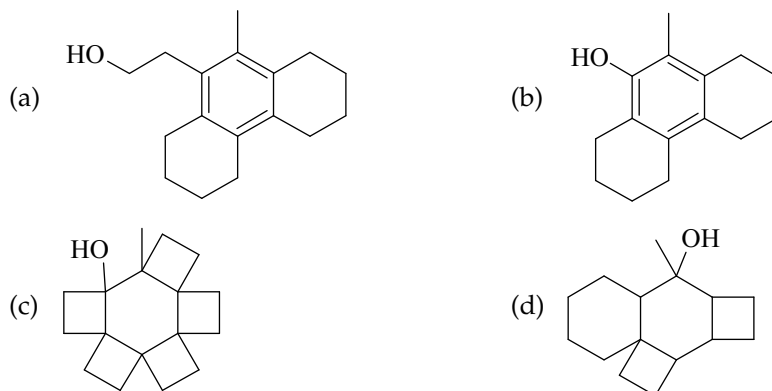
66. Which could be the major product of the following reaction?



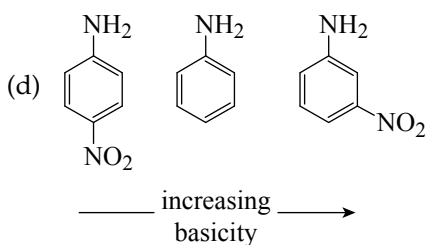
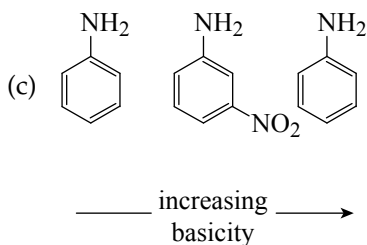
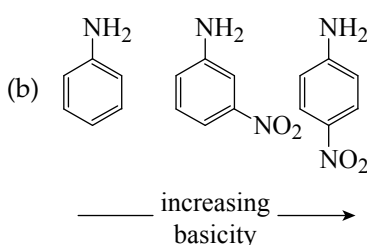
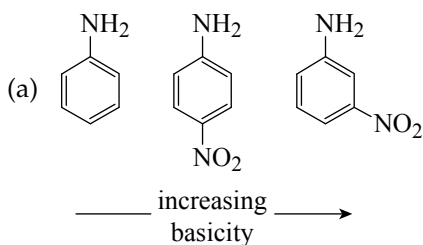
67. Which could be the major product of the following reaction?



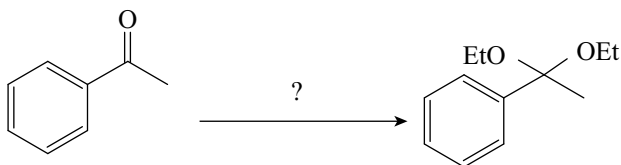
68.  $\xrightarrow[\Delta]{\text{dil H}_2\text{SO}_4}$ Products are?



69. Choose the order that has the following compounds correctly arranged with respect to increasing basicity.



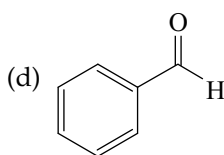
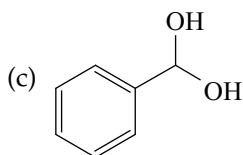
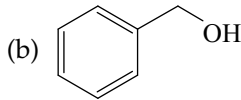
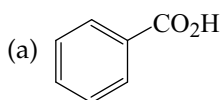
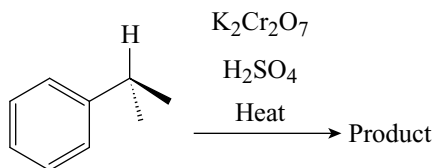
70. What could be the reagent and reaction condition for the following transformation?



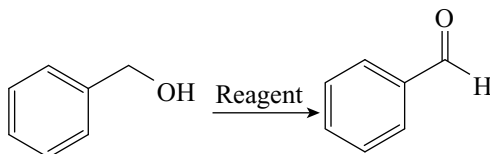
- (a) ethanol, NaOH
(c) methanol, NaOH

- (b) ethanol, H^+
(d) methanol, H^+

71. What could be the product for the following reaction?

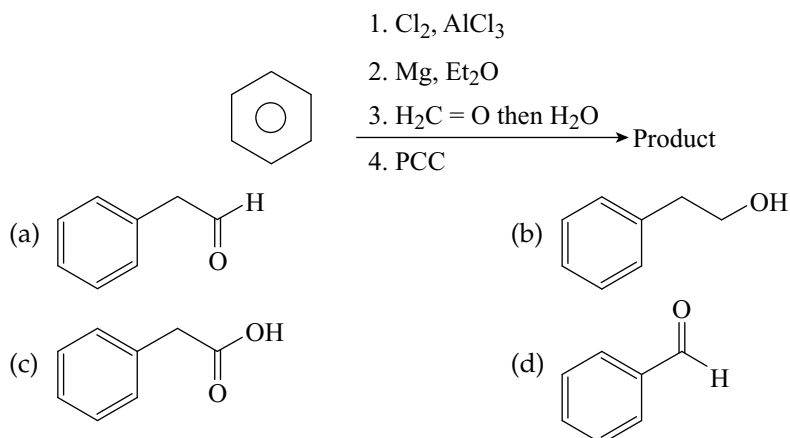


72. What could be the reagent to complete the following reaction?

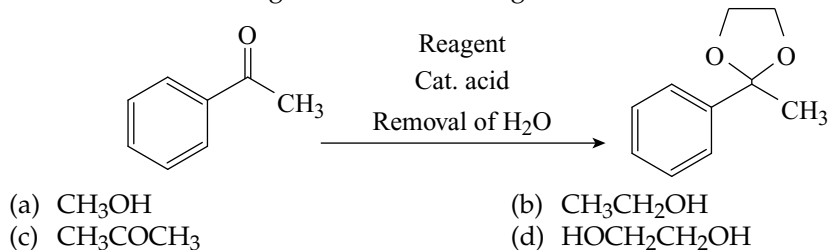


- (a) HCrO_4 (b) $\text{K}_2\text{Cr}_2\text{O}_7$ (c) PCC in dried CH_2Cl_2 (d) OsO_4

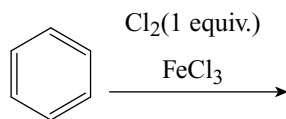
73. What could be the product for the following reaction?



74. What could be the reagent for the following reaction?

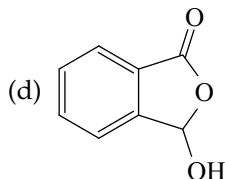
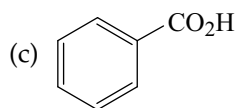
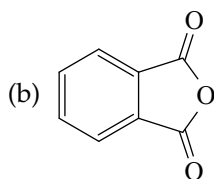
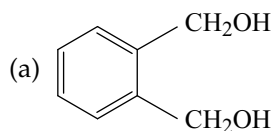
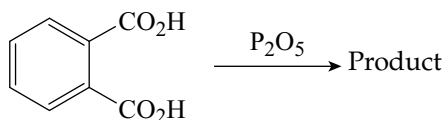


75. What is the expected product for the following reaction?

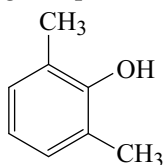


- (a) (b)
 (c) (d)

76. What could be the product for the following reaction?



77. What is the name of the following compound?



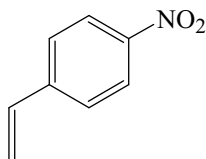
(a) 2,6-Dimethylphenol

(b) 1,5-Dimethylphenol

(c) 2,6-Dimethylanisole

(d) 1,5-Dimethylanisole

78. What is the name of the following compound?



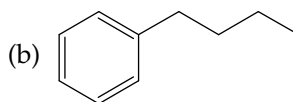
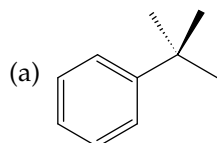
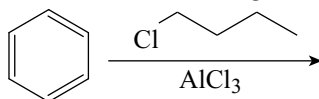
(a) p-aminotoluene

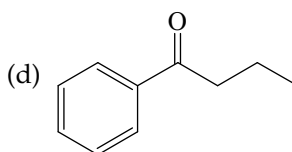
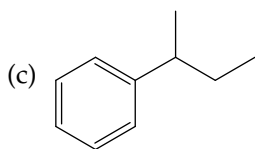
(b) p-nitrotoluene

(c) p-nitrostyrene

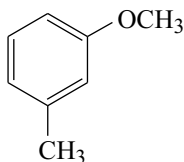
(d) 3-aminostyrene

79. What should be the major product for the following reaction?





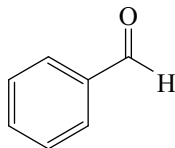
80. What is the name of the following compound?



- (a) p-methylphenol
(c) o-methylanisole

- (b) m-methylphenol
(d) m-methylanisole

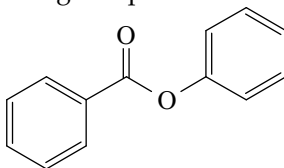
81. What is the name of the following compound?



- (a) benzylcarbonyl
(c) phenylaldehyde

- (b) benzaldehyde
(d) phenylketone

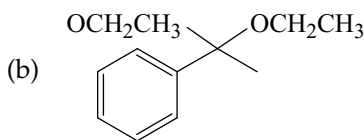
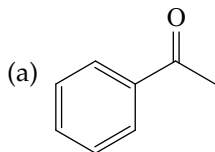
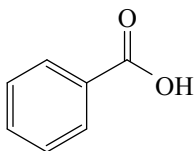
82. What is the name of the following compound?

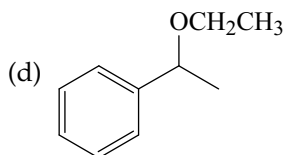
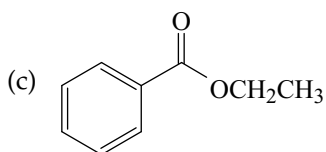


- (a) benzyl phenoate
(c) benzyl benzoate

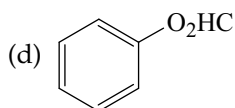
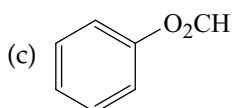
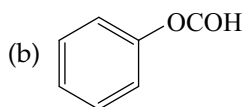
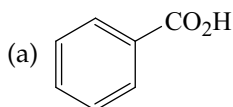
- (b) phenyl benzoate
(d) phenyl phenoate

83. What could be the product for the following reaction?

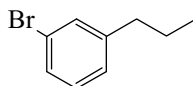




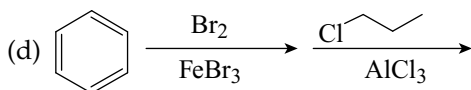
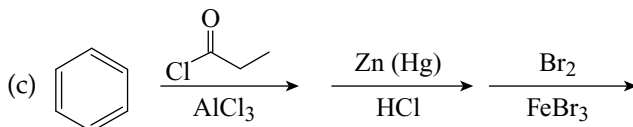
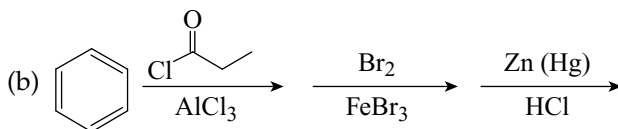
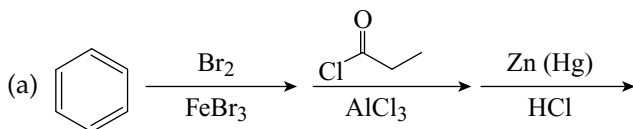
84. Which of the following structures is benzoic acid?



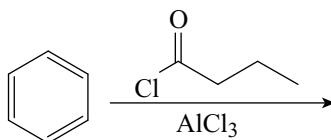
85. Which of the following reaction sequences would be the best for synthesizing the compound, 1-bromo-3-propylbenzene?

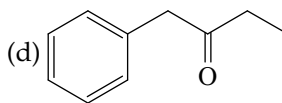
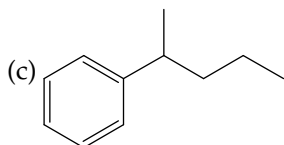
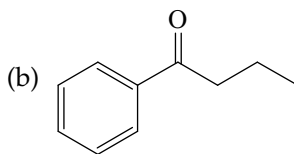
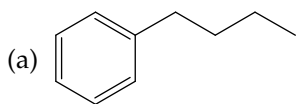


1-Bromo-3-propylbenzene

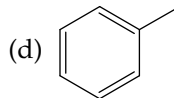
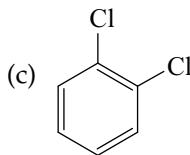
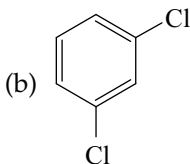
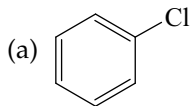
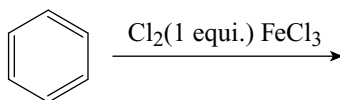


86. What is the expected product for the following reaction?

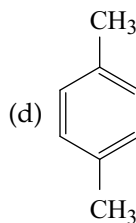
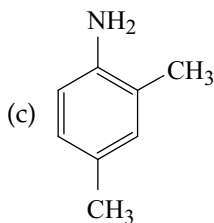
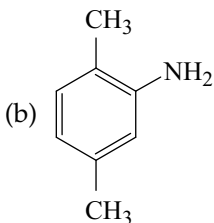
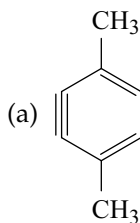
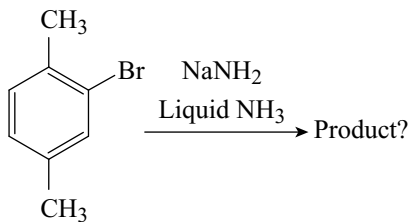




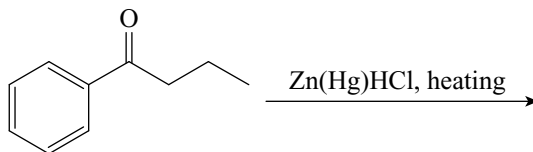
87. What is the expected product for the following reaction?

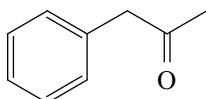
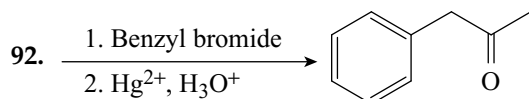
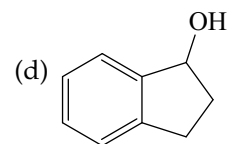
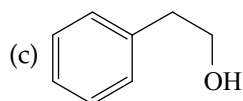
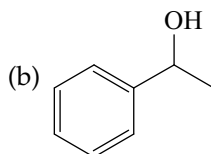
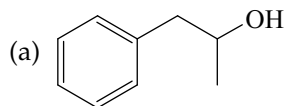
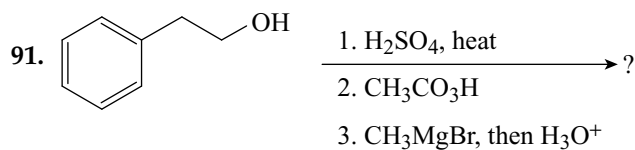
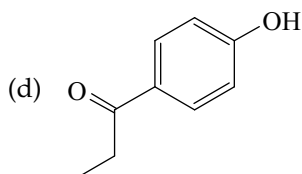
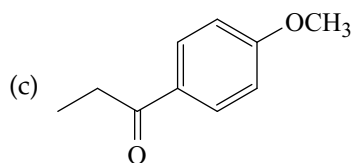
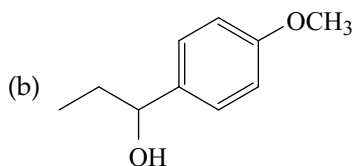
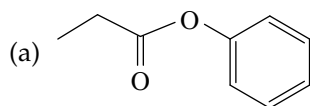
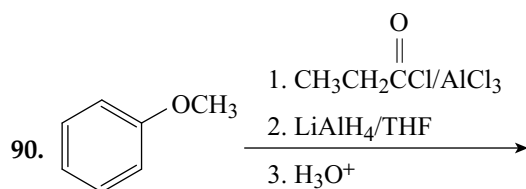
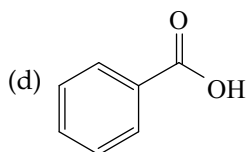
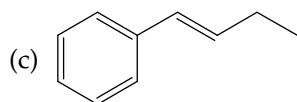
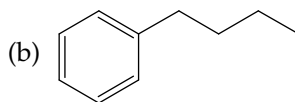
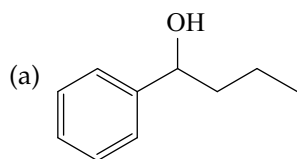


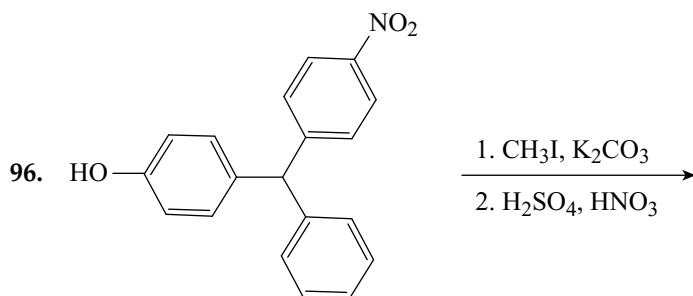
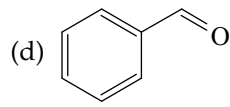
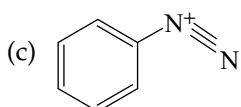
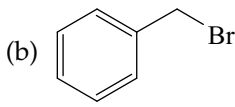
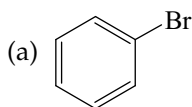
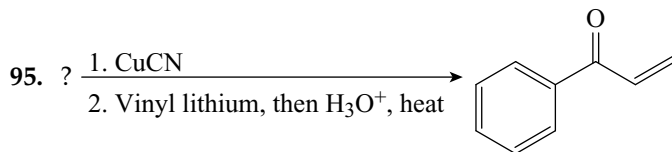
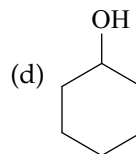
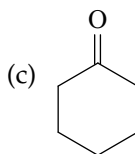
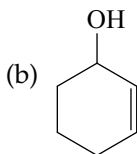
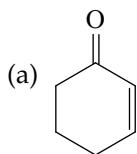
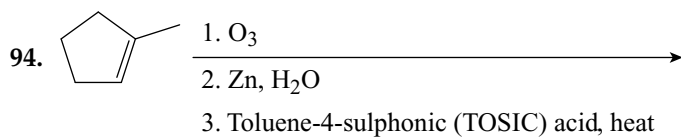
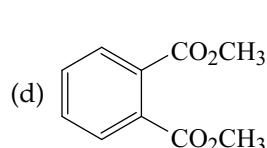
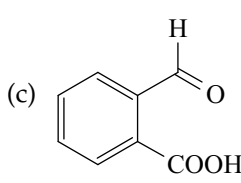
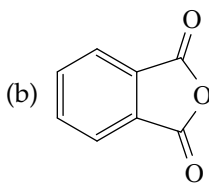
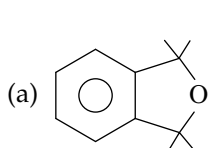
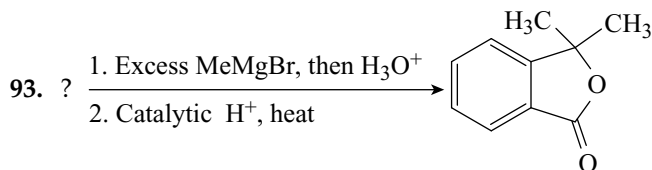
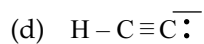
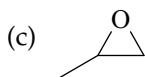
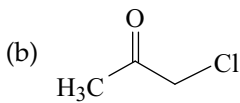
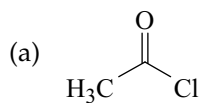
88. What could be the product for the following reaction?

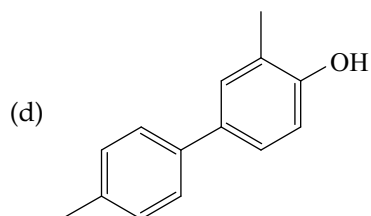
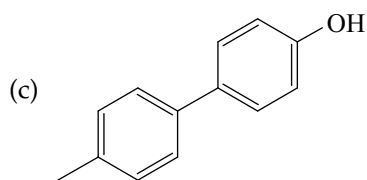
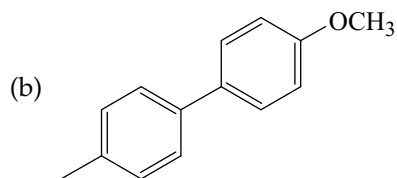
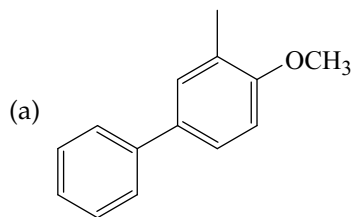
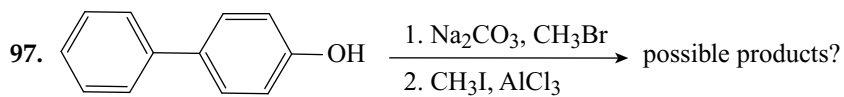
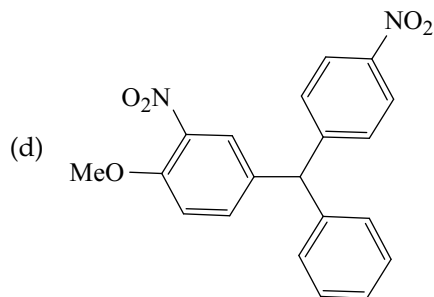
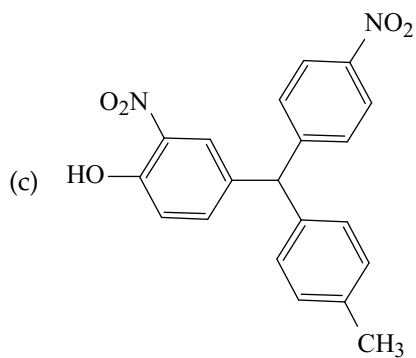
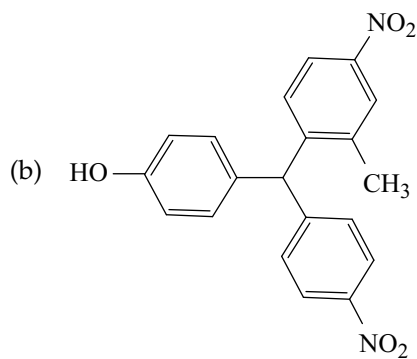
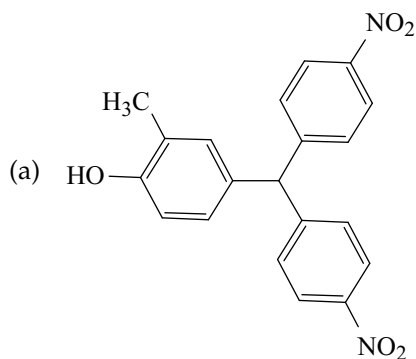


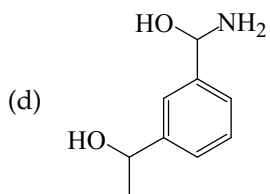
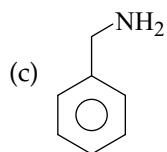
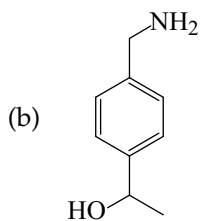
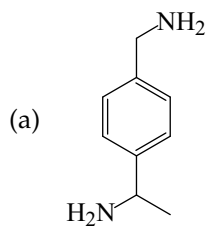
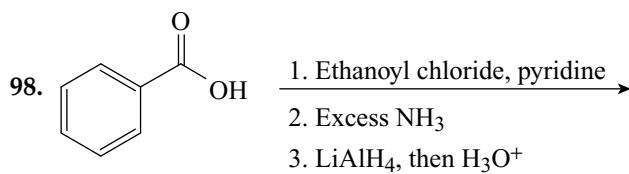
89. What could be the product for the following reaction?







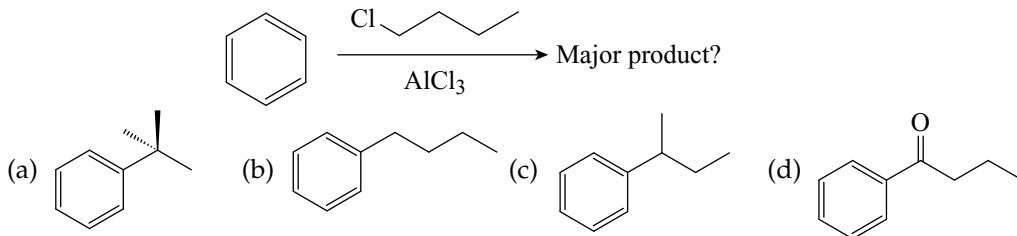




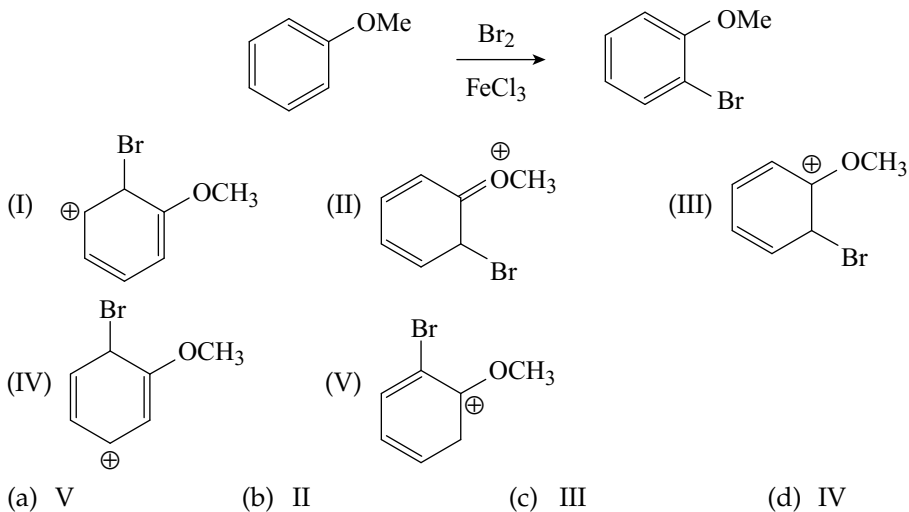
LEVEL 2

Single and Multiple-choice Type

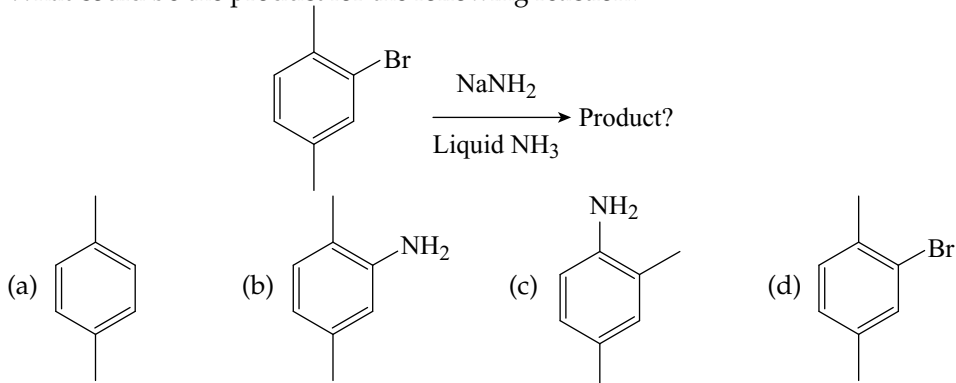
1. What should be the major product for the following reaction?



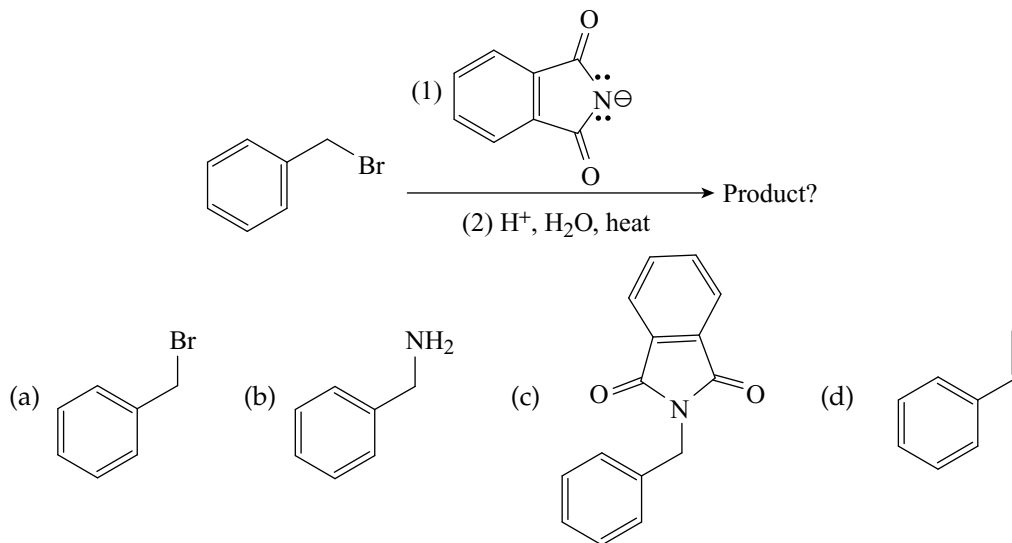
2. Which of the following is not the resonance structure of intermediate from the listed electrophilic aromatic substitution?



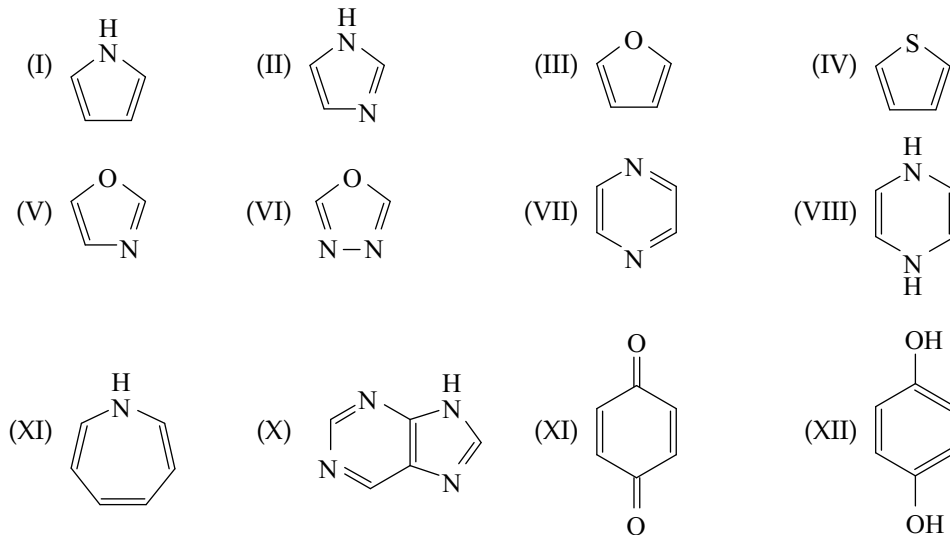
3. What could be the product for the following reaction?



4. What could be the product for the following reaction?



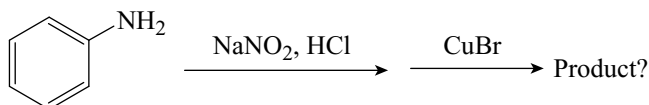
5. Which of the following compounds are aromatic compounds?

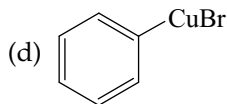
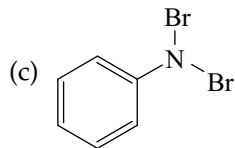
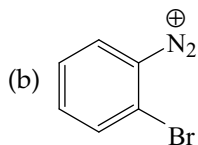
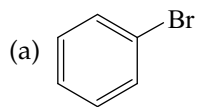


- (a) I, II, III, IV, V, VI, VIII, X, XII
(c) I, II, III, VI, VIII, X, XI, XII

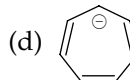
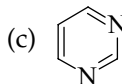
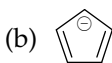
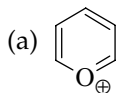
- (b) I, II, III, V, VII, VIII, IX, X, XII
(d) I, II, III, IV, V, VI, VII, X, XII

6. What could be the product for the following reaction?

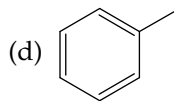
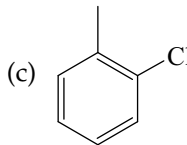
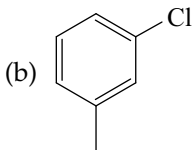
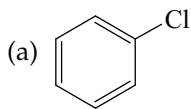
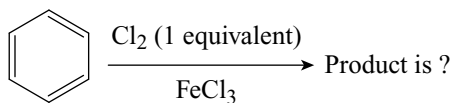




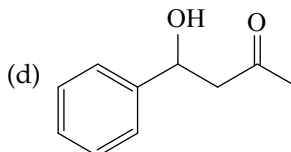
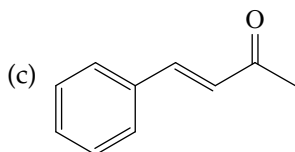
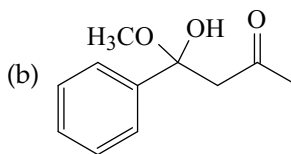
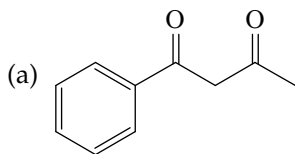
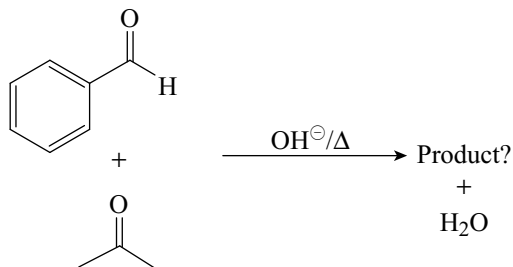
7. Which one of the following compounds is antiaromatic?



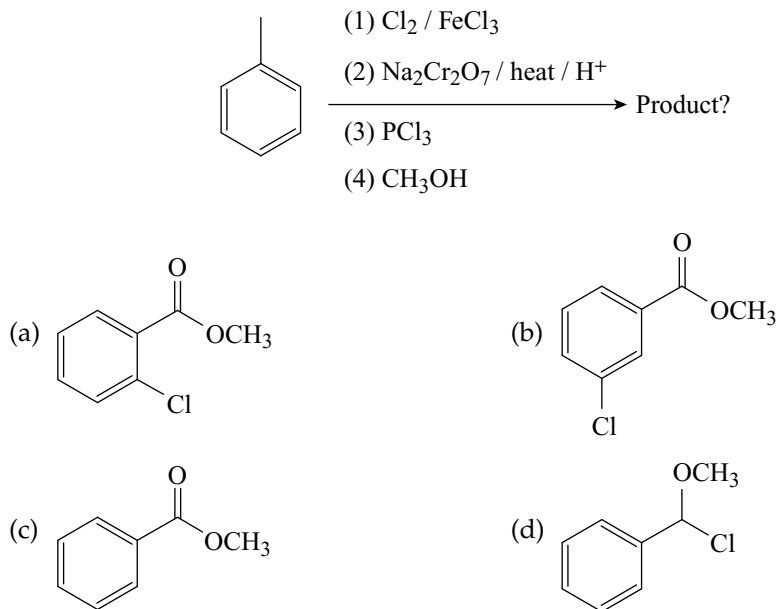
8. What is the expected product for the following reaction?



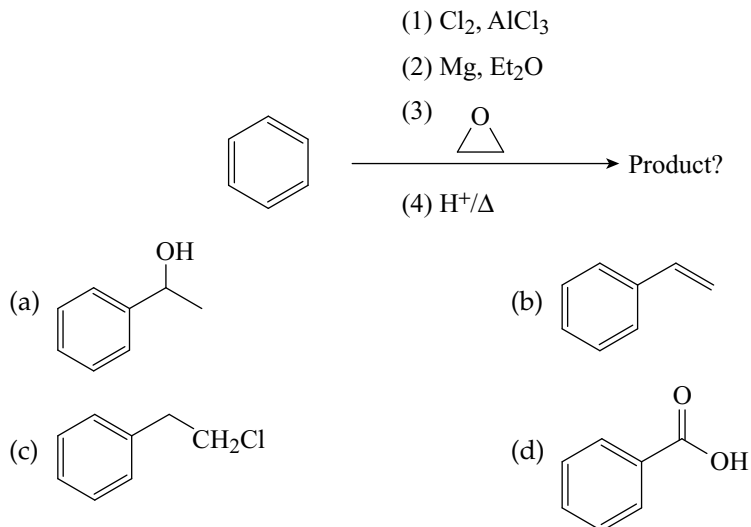
9. What could be the product for the following reaction?



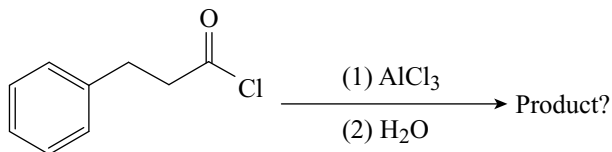
10. What could be the product for the following reaction?

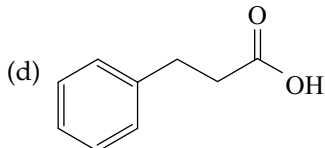
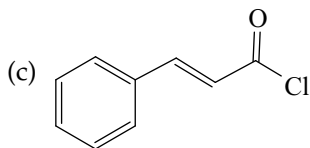
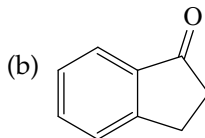
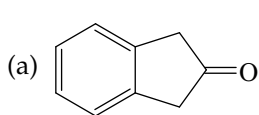


11. What could be the product for the following reaction?



12. What could be the product for the following reaction?



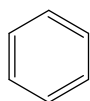


13. What could be the product for the following reaction?

(1) $\text{Cl}_2, \text{AlCl}_3$

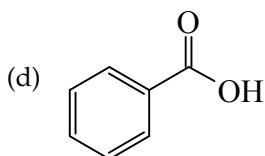
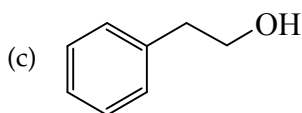
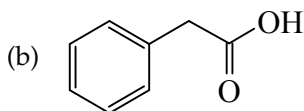
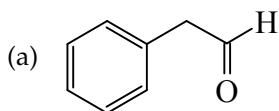
(2) $\text{Mg}, \text{Et}_2\text{O}$

(3) then H_2O

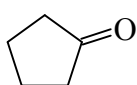


(4) $\text{K}_2\text{Cr}_2\text{O}_7, \text{H}^+$
room temp

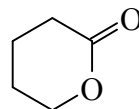
Product?



14. What could be the reagent for the following reaction?



Reagent $\longrightarrow$



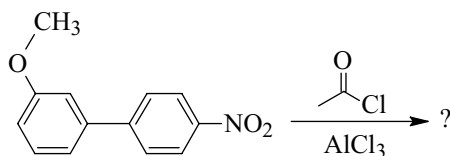
(a) PhCO_2H

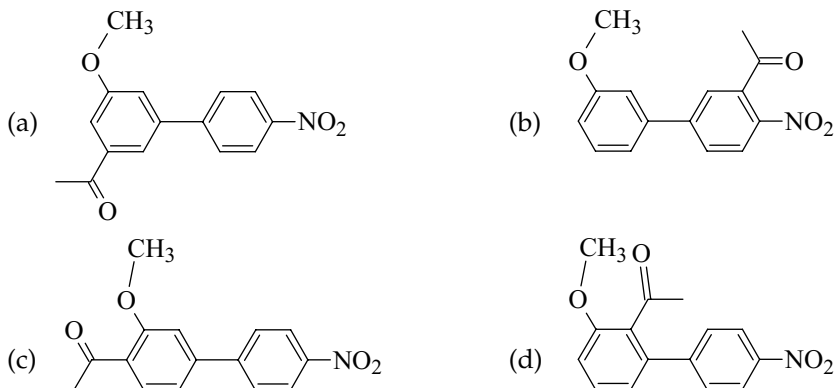
(b) PhCO_3H

(c) OsO_4

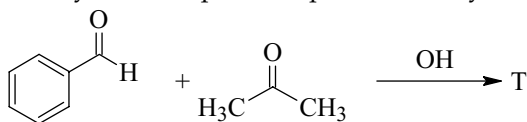
(d) $\text{Na}_2\text{Cr}_2\text{O}_7$

15. Which of the indicated compounds would be the major product in the following Friedel Crafts reaction?

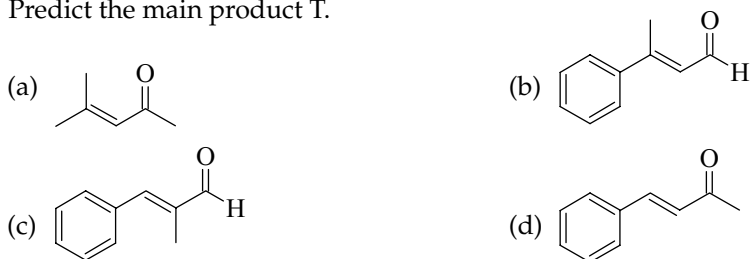




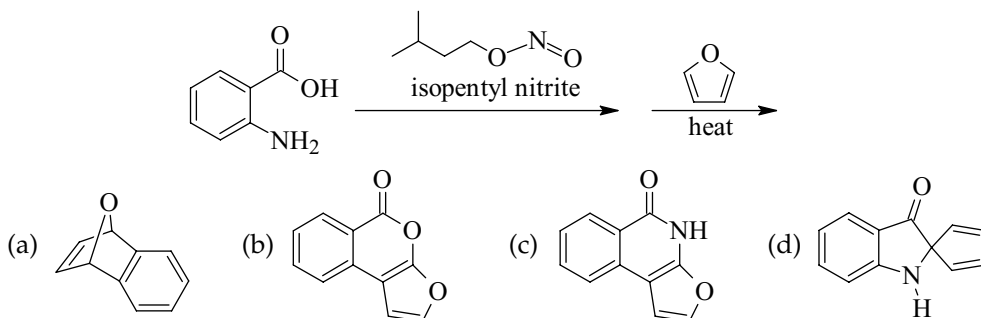
16. The following reaction yields compound T predominately.



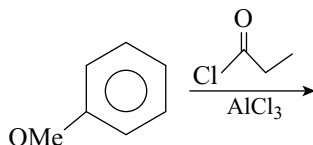
Predict the main product T.

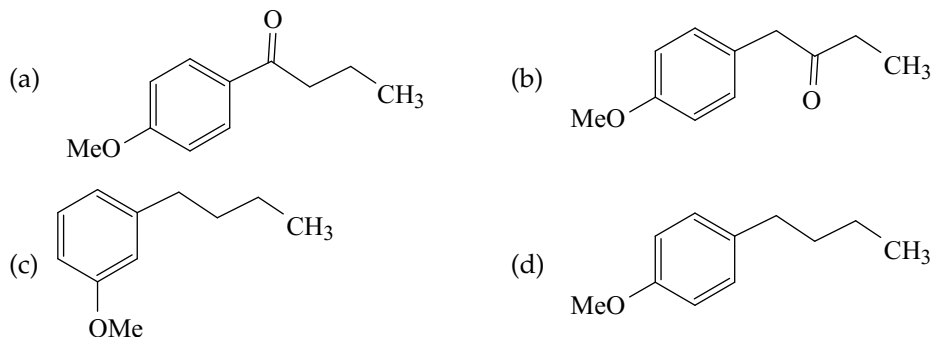


17. The compound isopentyl nitrite is a source of NO^+ ions and will react with an amine to generate a diazonium cation. Predict the product of the following reaction sequence.

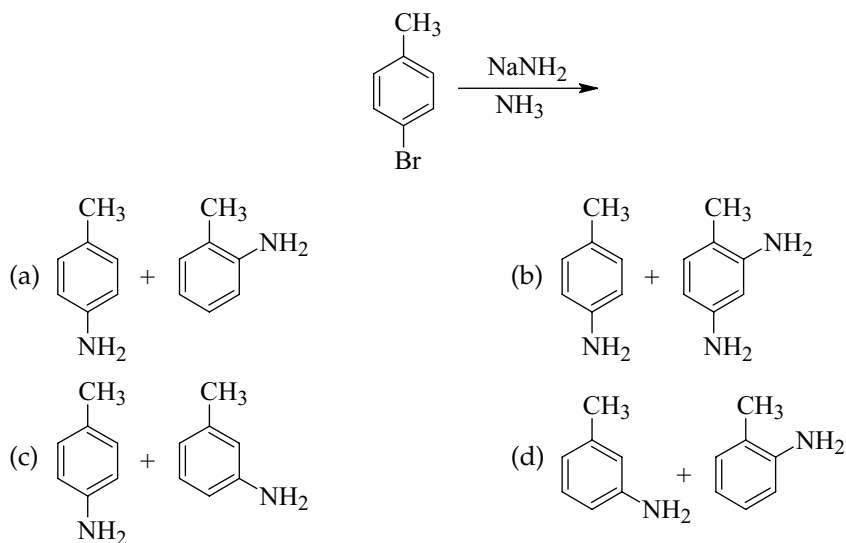


18. What could be the major product for the following Reaction?

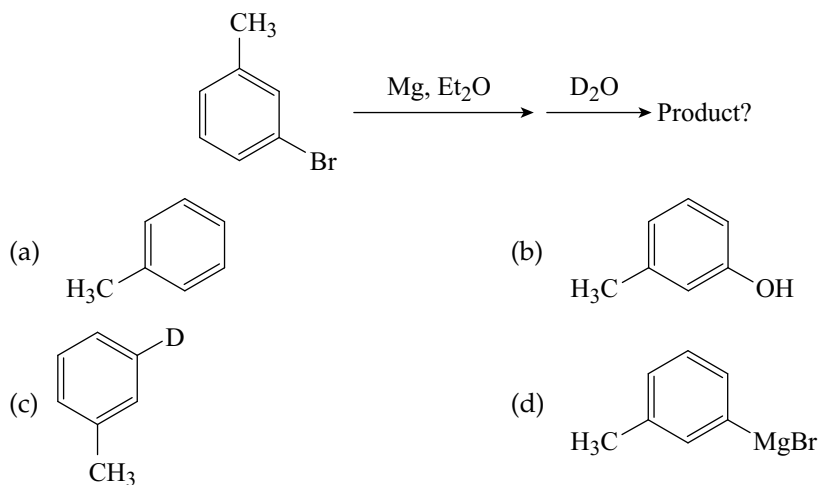




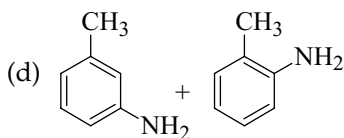
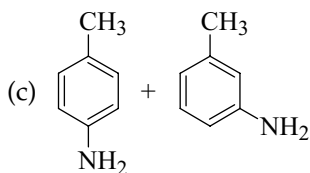
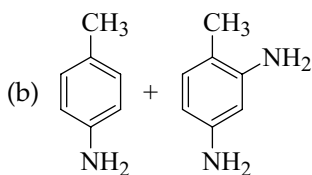
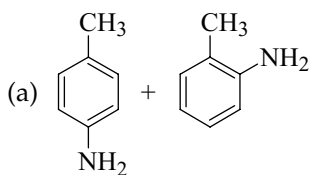
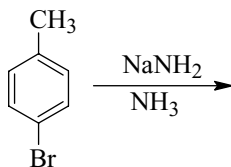
19. The following reaction gives two main products. Identify the products.



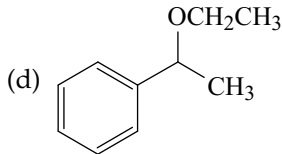
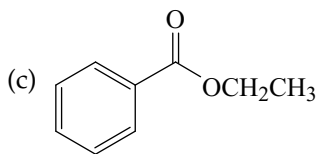
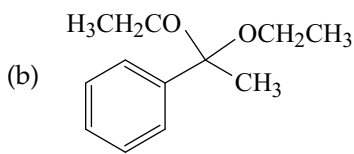
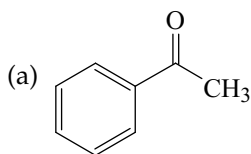
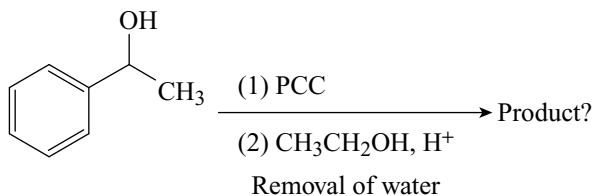
20. What could be the product for the following reaction?



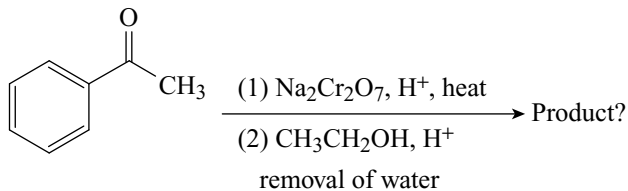
21. The following reaction gives two main products. Identify the products.

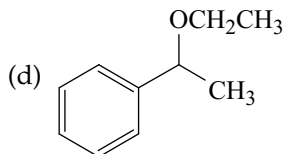
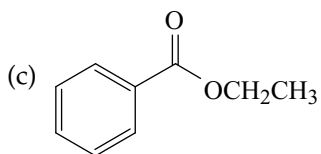
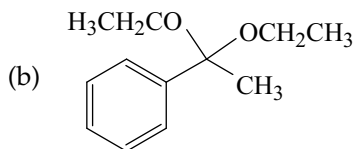
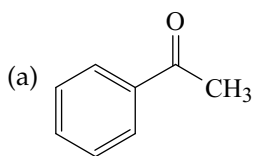


22. What could be the product for the following reaction?

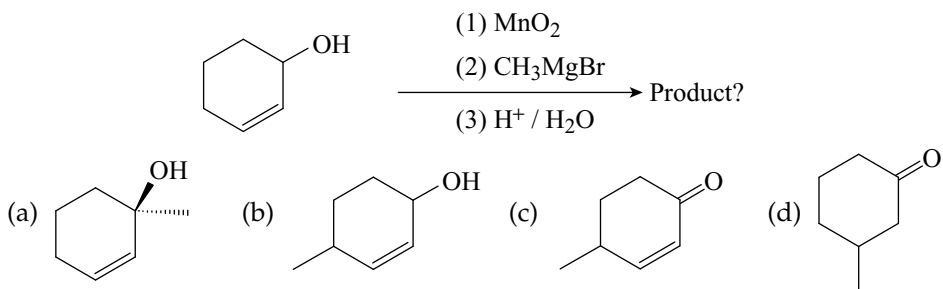


23. What could be the product for the following reaction?

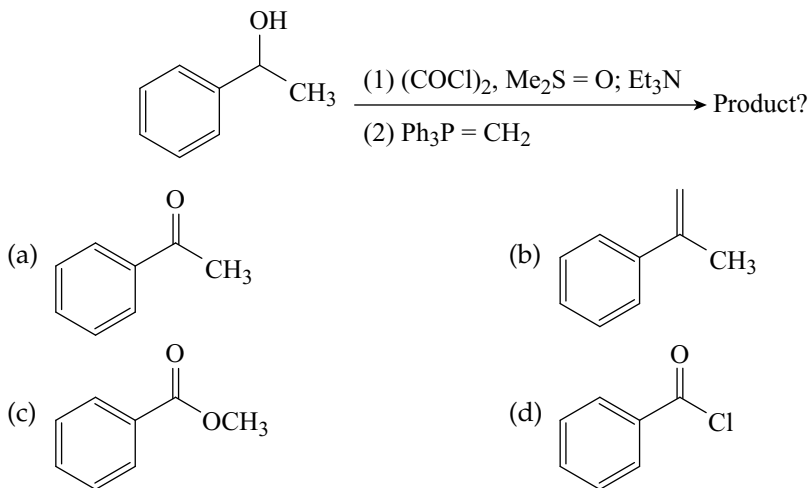




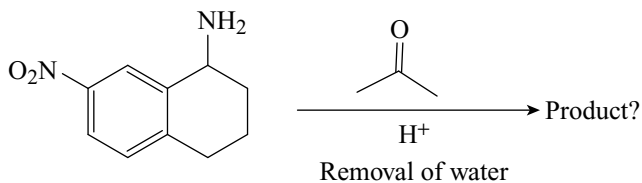
24. What could be the product for the following reaction?

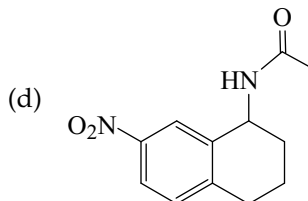
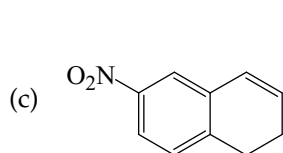
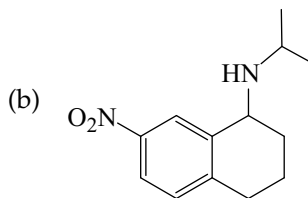
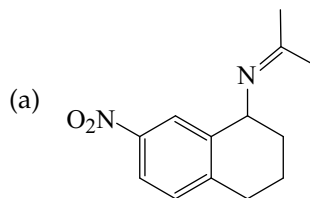


25. What could be the product for the following reaction?

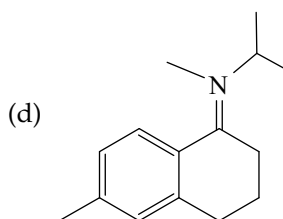
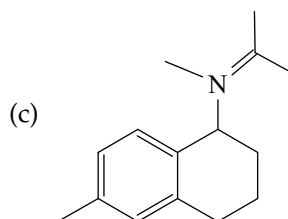
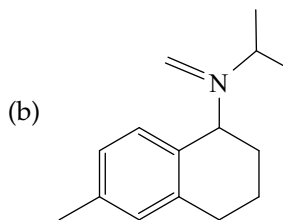
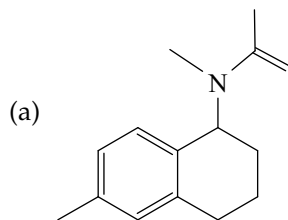
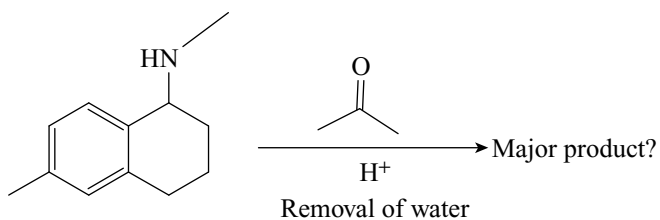


26. What could be the product for the following reaction?

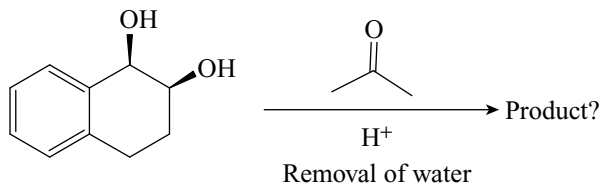


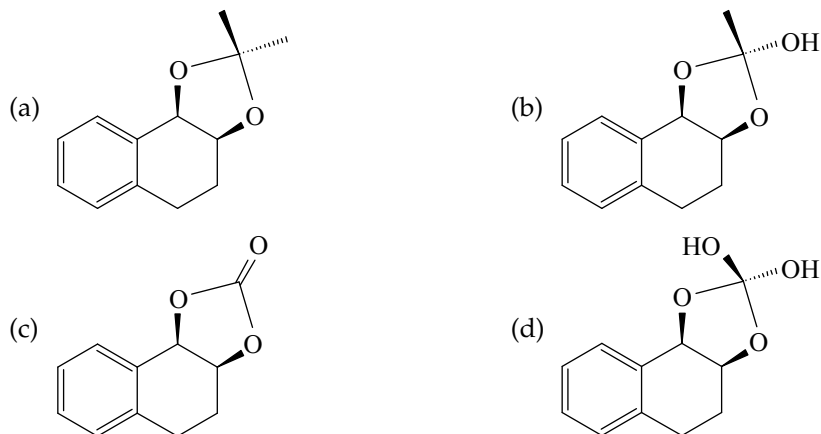


27. What could be the major product for the following reaction?

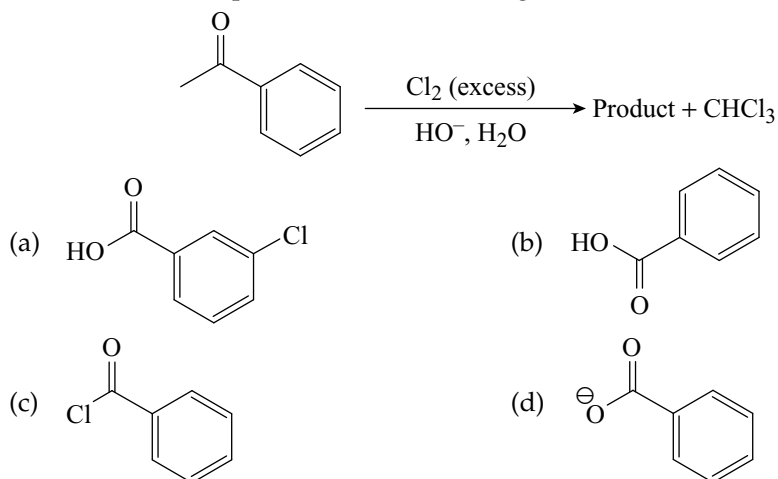


28. What could be the product for the following reaction?

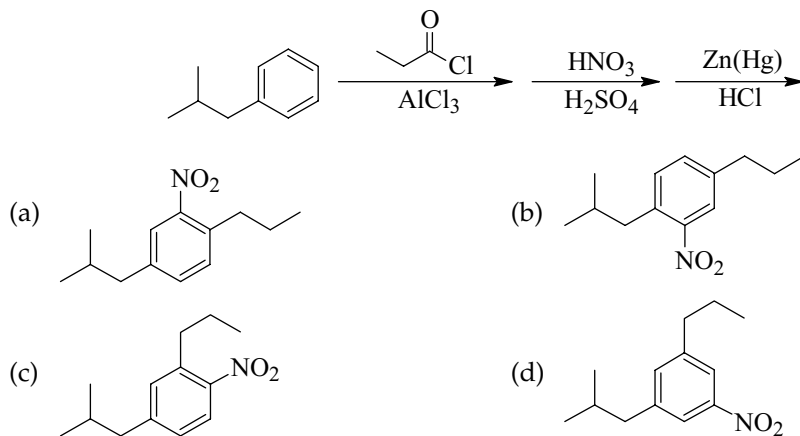




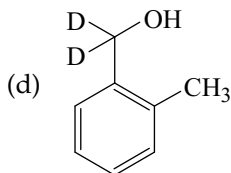
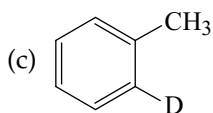
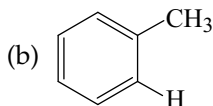
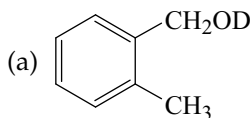
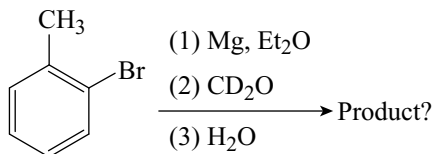
29. What could be the product for the following reaction?



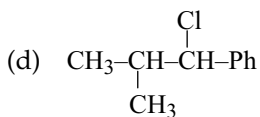
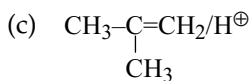
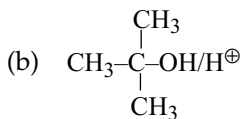
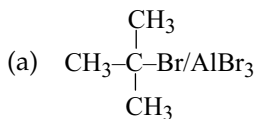
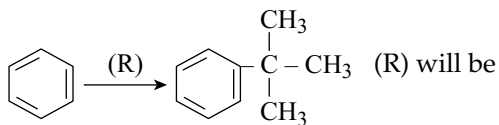
30. Predict the product of the following reaction sequence.



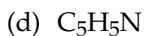
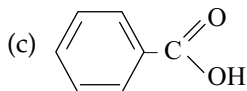
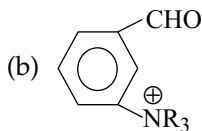
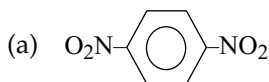
31. What could be the product for the following reaction?



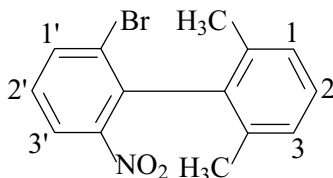
32. For the given reaction:



33. Which of the following compound will not give Friedel-Crafts reaction?

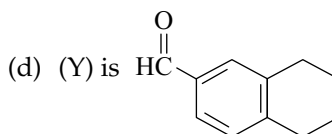
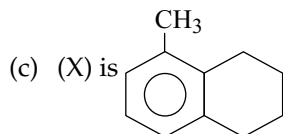
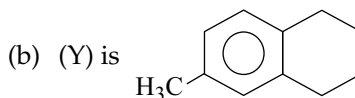
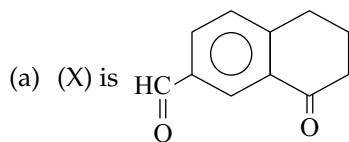
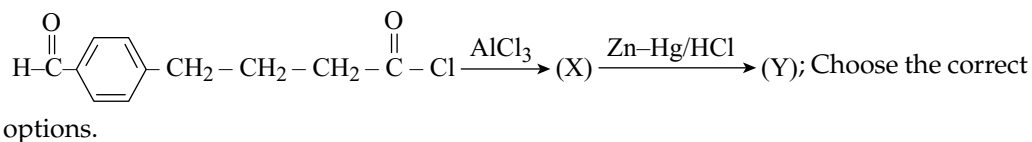


34. In the given reaction, electrophilic substitution will take place readily at the carbon?



- (a) 1' (b) 1 (c) 3 (d) 3'

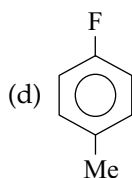
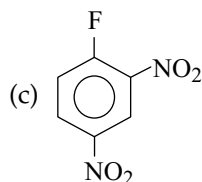
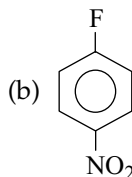
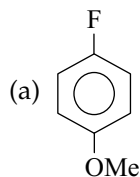
35. In the given reaction



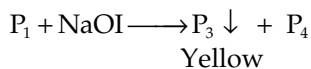
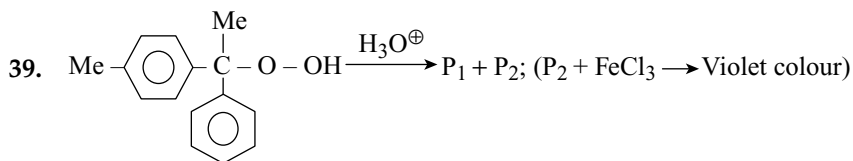
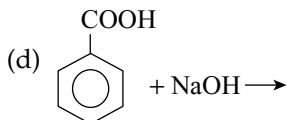
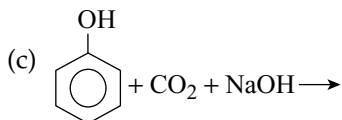
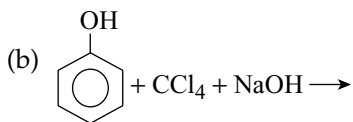
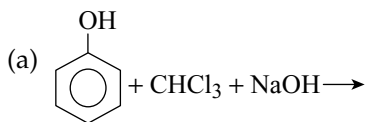
36. The type of substitution reactions of benzenoid hydrocarbons are

- (a) elimination (b) electrophilic
(c) nucleophilic (d) free radical

37. Among the following compounds, which liberates F^- on reaction with MeO^- ?



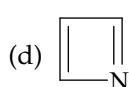
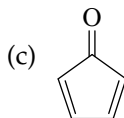
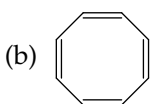
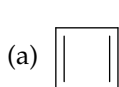
38. Among the following reactions, which form salicylic acid (after acidification)?



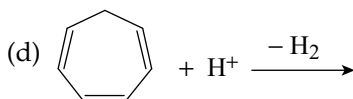
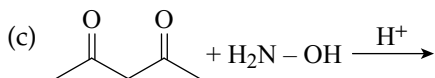
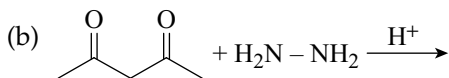
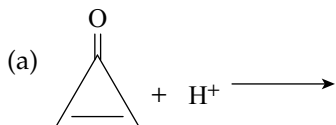
Correct statement for the above sequence is

- (a) P_3 on reaction with Ag gives acetylene
- (b) P_4 on reaction with sodalime gives toluene
- (c) P_4 on reaction with sodalime gives benzene
- (d) P_1 on reaction with 2,4-DNP gives yellow compound

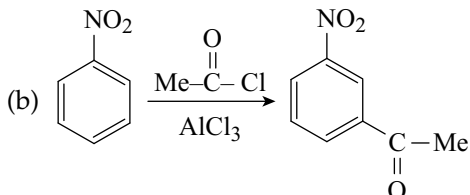
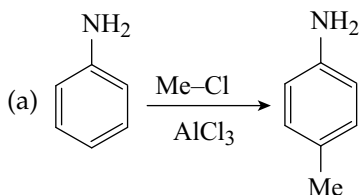
40. Identify compounds which are unstable at room temperature.

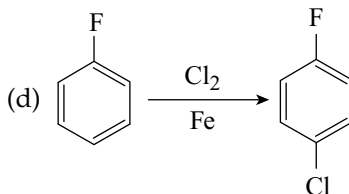
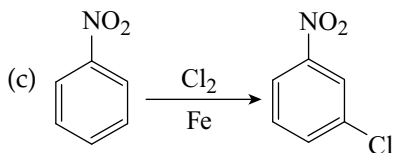


41. Identify reactions which give aromatic product.

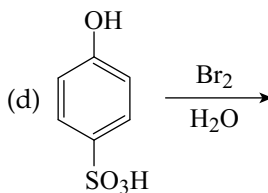
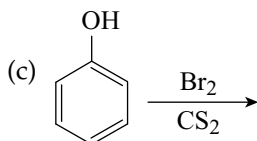
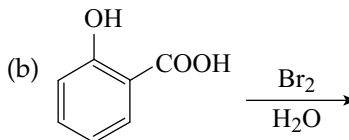
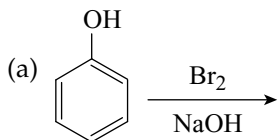


42. Identify reactions which are not feasible.

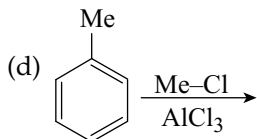
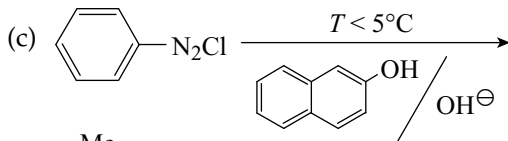
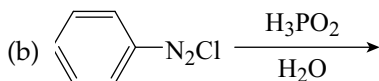
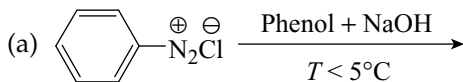




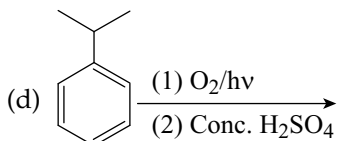
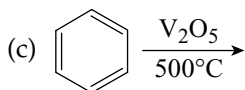
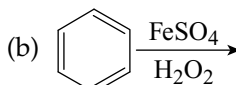
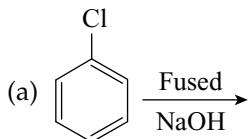
43. Identify reactions that give tribromo substituted product.



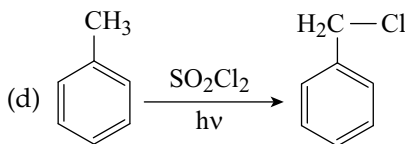
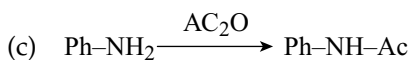
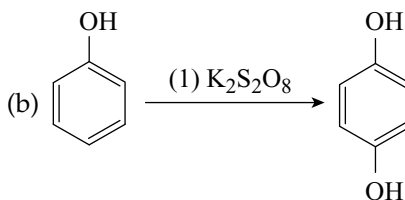
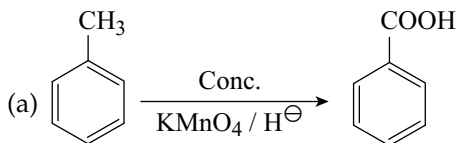
44. Identify coupling reactions.



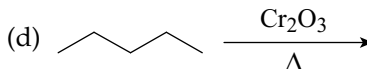
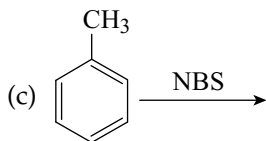
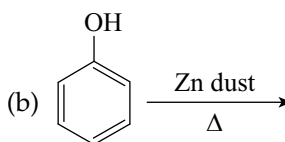
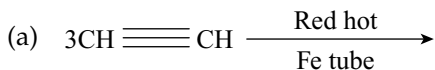
45. Identify reactions which give phenol product.



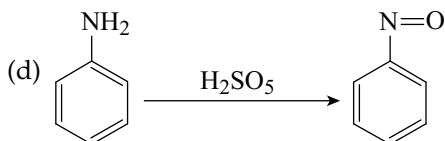
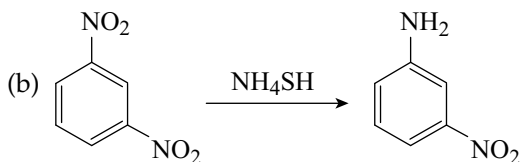
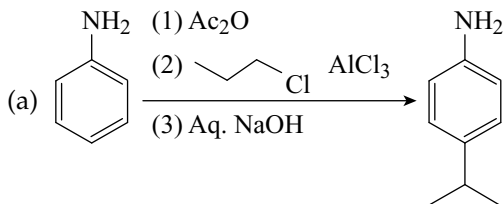
46. Identify correctly matched reactions with their products.



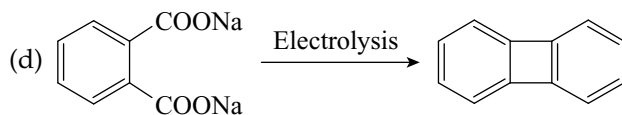
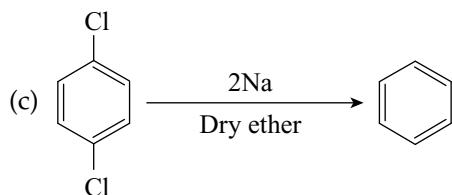
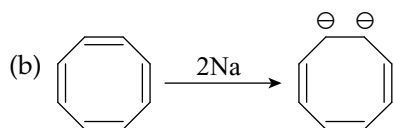
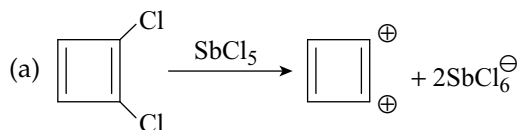
47. Identify method of preparation of benzene.



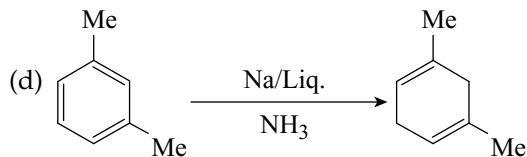
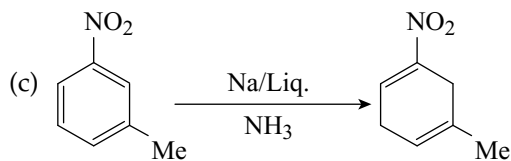
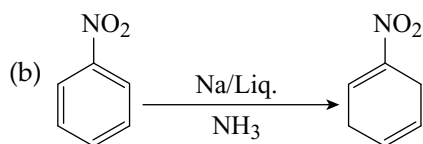
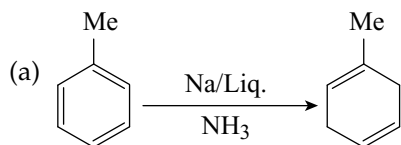
48. Identify correctly matched reactions with their products.



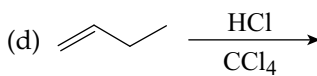
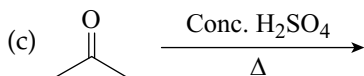
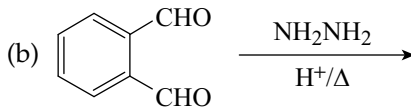
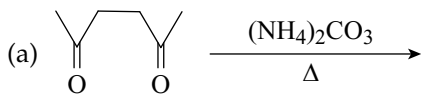
49. Identify correctly matched reactions with their products.



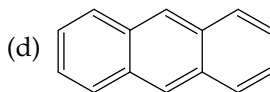
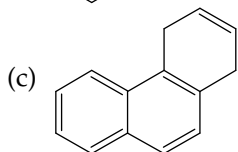
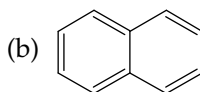
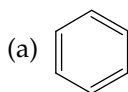
50. Identify correctly matched reaction with their products.



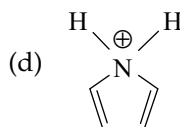
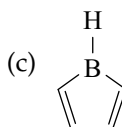
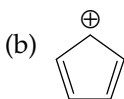
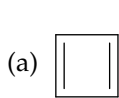
51. Identify reactions that give aromatic products.



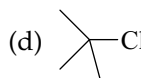
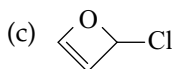
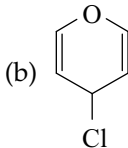
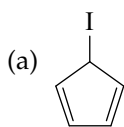
52. Annealation takes place in which of the following compounds?



53. Which of the following is an anti-aromatic compound?



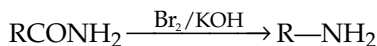
54. Which compound does not give SN' /solvolysis reaction?



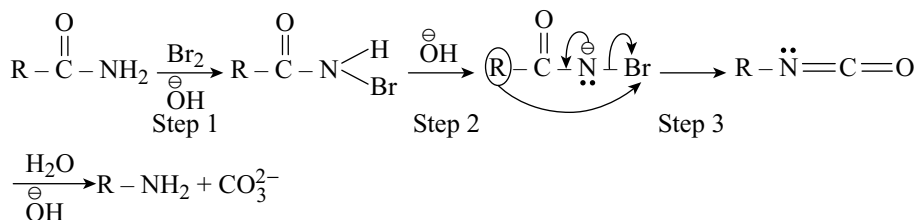
Comprehension Type

Passage 1

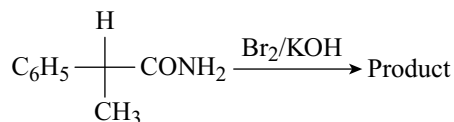
The conversion of an amide into an amine with one carbon atom less by the action of alkaline hydrohalite is known as Hofmann bromamide rearrangement:



The most important feature of the reaction is the rearrangement of N-bromamide anion to isocyanate:



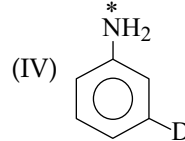
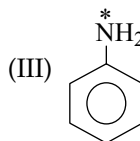
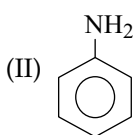
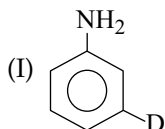
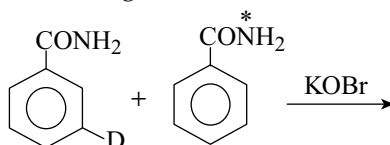
55. The product of the reaction



Product amine will be

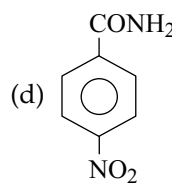
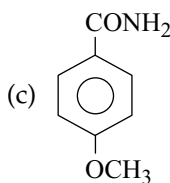
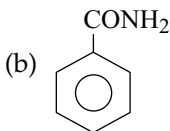
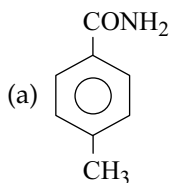
- (a) S-amine (b) R-amine
(c) 50:50 mixture of (+) and (–) amine (d) 30:70 mixture of (+) and (–) amine

56. Predict the product in the following reaction



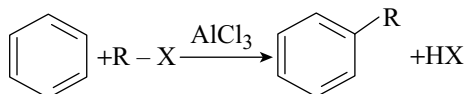
- (a) I and II (b) I and III (c) II and III (d) I and IV

57. Which of the following can undergo Hofmann bromamide reaction most easily?



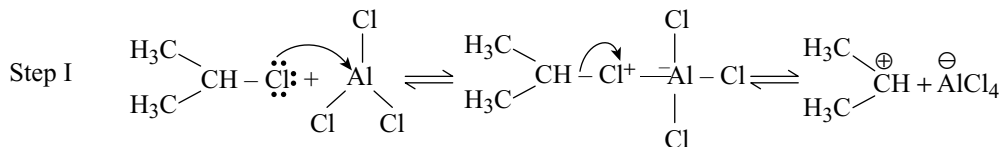
Passage 2

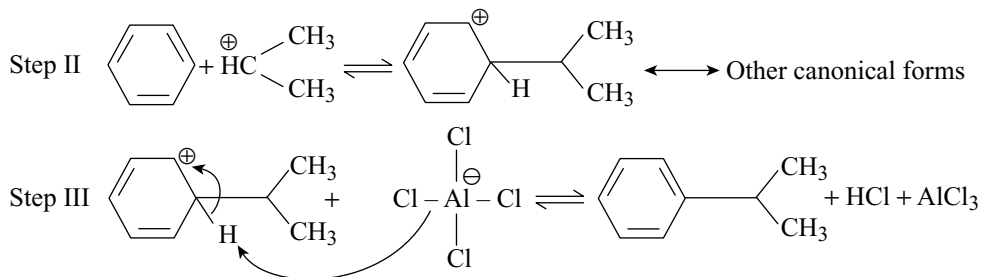
A general equation for a Friedel-Crafts alkylation reaction is the following



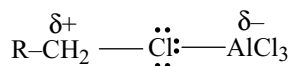
Alkyl halides by themselves are insufficiently electrophilic to react with benzene. Further, AlCl_3 serves as a Lewis acid catalyst to enhance the electrophilicity of the alkylating agent. The mechanism for the reaction is shown in the following steps

Mechanism for the reaction:

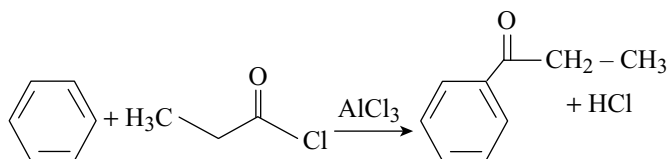




With R-X is a primary halide, the carbon halogen bond is nearly broken and one in which the carbon atom has a considerable +ve charge.



this complex acts as the electrophile. The Friedel-Crafts acylation reaction is an effective means of introducing an acyl group into an aromatic ring. The reaction is often carried out by treating the aromatic compound with an acyl halide.

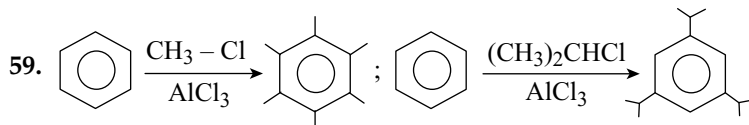


58. Consider the following statements for the given reaction

- (1) The Friedel-Crafts reaction is an electrophilic aromatic substitution.
- (2) First step of the reaction is the rate-determining step.
- (3) Second step of the reaction is the rate-determining step.
- (4) Third step is an acid-base reaction.

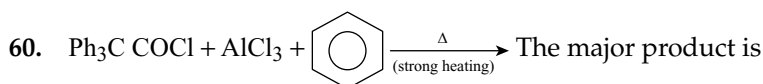
Now, of these statements

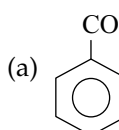
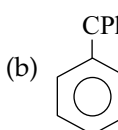
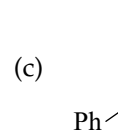
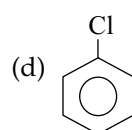
- | | |
|----------------------------------|---------------------------------------|
| (a) (1) and (2) are correct | (b) (1), (2) and (3) are correct |
| (c) (1), (3) and (4) are correct | (d) (1), (2), (3) and (4) are correct |



What is the reason for trisubstituted product in the second case?

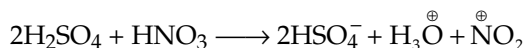
- | | |
|------------------------------------|---------------------------|
| (a) less positive inductive effect | (b) more steric effect |
| (c) less hyperconjugation | (d) more mesomeric effect |



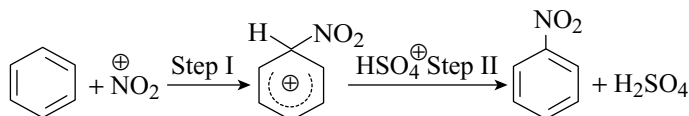
- | | | | |
|---|---|---|--|
| (a)  | (b)  | (c)  | (d)  |
|---|---|---|--|

Passage 3

Nitration of benzene or any aromatic system is done by treating it with a mixture of Conc. HNO_3 and Conc. H_2SO_4 (called nitrating mixture). These two acids react together according to the equation given below, to form nitronium ion which act as an electrophile:



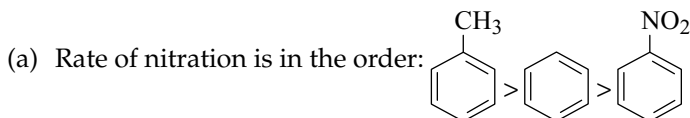
NO_2^+ ion, an electrophile then attacks benzene or any aromatic system to accomplish nitration in the following two steps with arenium ion as the reaction intermediate:



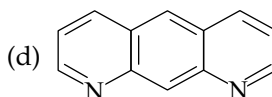
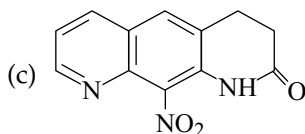
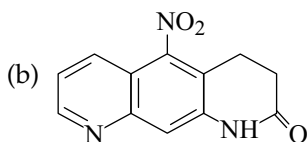
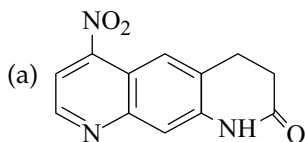
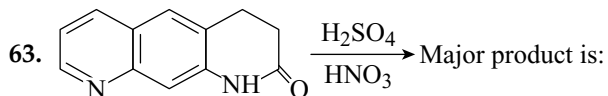
The direct nitration of aniline or phenol with nitrating mixture gives very poor yield of nitro product because benzene ring attached to $-\text{NH}_2$ or $-\text{OH}$ group is very sensitive to oxidation and major part of aniline or phenol is oxidized to give a black tar mass that

contains mainly benzoquinone, $\text{O}=\text{C}_6\text{H}_4=\text{O}$. Although $-\text{NH}_2$ group is o/p-orienting, m-nitro derivative is also formed. The ring is protected from oxidation by acetylating $-\text{NH}_2$ group before subjecting it to nitration. After nitration, product is hydrolyzed to get deacetylated product (o- and p-nitro derivatives). Nitration of phenol is carried out by Dil. HNO_3 . It is believed that nitrous acid (present as an impurity) interacts with HNO_3 to give nitrosonium ion, an electrophile which reacts with phenol to give o- and p-nitrosophenol, the latter being the principal product, according to the same mechanism as that of nitration mentioned above. Nitrosophenol is then oxidized by HNO_3 to nitrophenol while HNO_3 is itself reduced to HNO_2 .

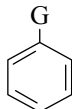
61. Regarding the nitration of phenol with Dil. HNO_3 as described above, the incorrect statement is
- HNO_3 acts as an acid and also as an oxidant.
 - Nitrous acid acts as a base.
 - The reaction intermediate is an arenium ion.
 - Amount of HNO_2 goes on decreasing with the progress of nitration.
62. Correct statement is/are

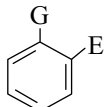
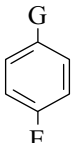


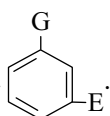
- Rate of nitration of benzene and that of hexa deuterated benzene occur almost at the same rate
- By increasing the concentration of acids ($\text{HNO}_3 + \text{H}_2\text{SO}_4$), the rate of nitration increases
- All of the above



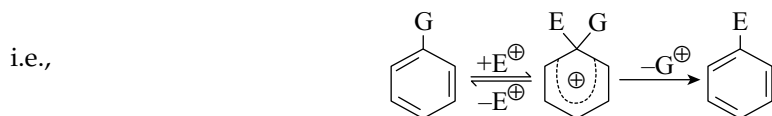
Passage 4

It is very well known that when a compound  reacts with an electrophile then we

either get ortho- or para-substituted product, i.e.,  +  or we get a meta-

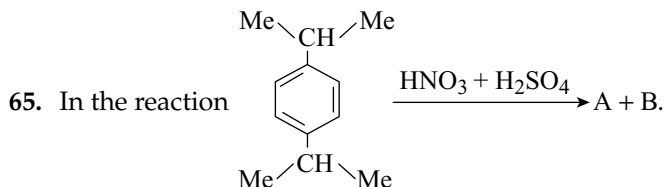
substituted product, i.e., .

This depends on the electron-releasing or electron-withdrawing power of the group, i.e., +R/-R or +M/-M effect of group G. There is another theory that is the electrophile attacks the ring carbon where the substituent G is already attached.

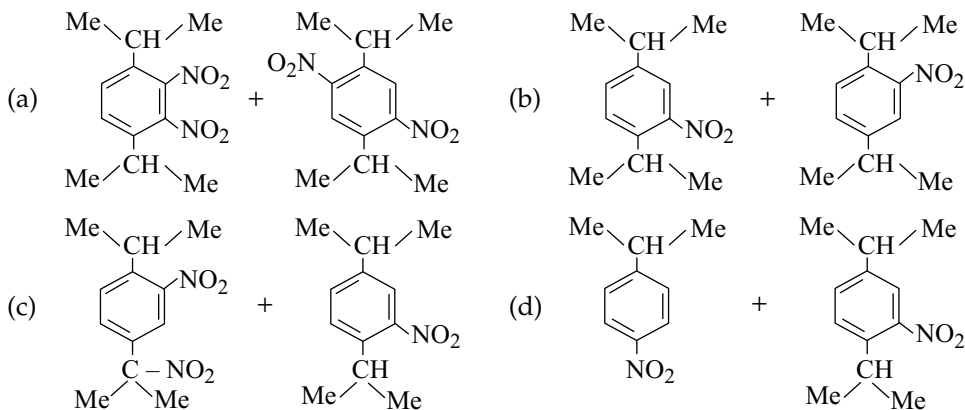


This is called ipso substitution reaction.

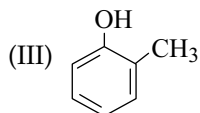
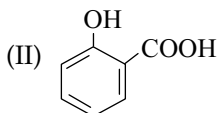
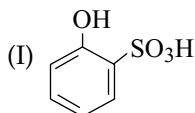
64. The factor which is expected to promote ipso substitution is
- The group G should be a strong electron withdrawing group
 - A group which is highly electron withdrawing should be attached at ortho or para position with respect to the group G
 - The group G should leave as G^+ easily, i.e., G^+ should be highly stable
 - The group G should leave easily, i.e., G^+ should be highly unstable



A and B are respectively



66. In which of the following can you expect ipso addition?



(a) (I) only

(b) (II) only

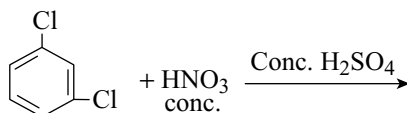
(c) (I) and (II) only

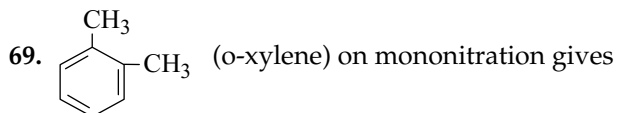
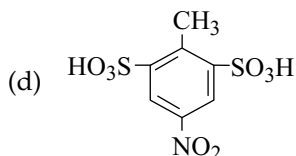
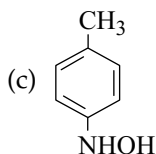
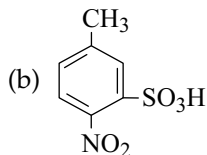
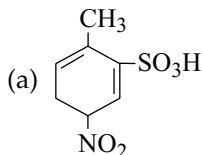
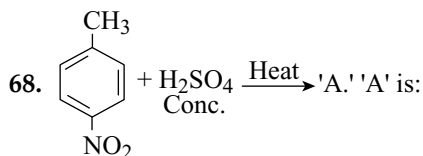
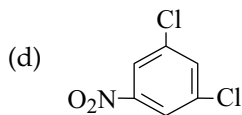
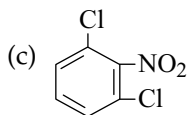
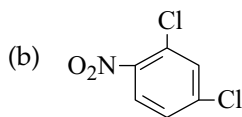
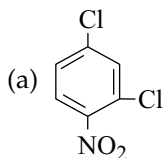
(d) (I), (II), and (III)

Passage 5

When a second substituent is introduced in benzene ring, it is directed by group already present on benzene ring. Electron-releasing groups are activating, therefore o and p-directing, whereas electron-withdrawing groups are deactivating, therefore m-directing. Halogens, although they are electron withdrawing due to $-I$ effect but still o- and p-directing due to $+R$ effect. $-N=O$ group is also deactivating but o- and p-directing due to presence of long pair of electrons like halogens it shows $+R$ effect. When a third substituent is introduced into a disubstituted product, the o-isomer gives two, the m-isomer gives three while p-isomer gives only one product. This method is called Korner's method. The major product is formed such that it has minimum steric hindrance.

67. Which of the following is not formed at all?



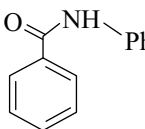


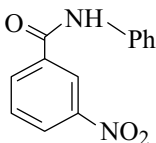
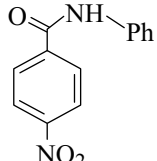
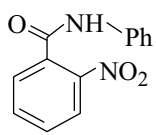
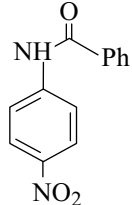
- (a) two products (b) three products (c) one product (d) four products

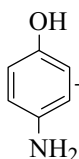
Passage 6

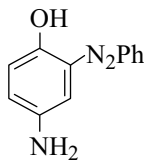
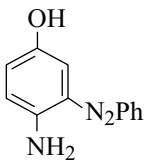
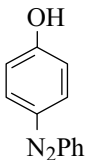
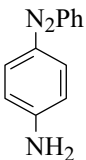
When a mono substituted benzene derivative, $\text{C}_6\text{H}_5\text{Y}$, undergoes further electrophilic substitution, e.g., nitration, the incoming substituent may be incorporated at the o-, m- or p- position and the overall rate at which substitution takes place may be faster or slower than with benzene itself. It has been observed that substitution occurs so as to yield either predominantly the m-isomer or a mixture of o- and p-isomers, in the former case the overall rate of attack is always slower than on benzene itself, in the latter case the overall rate of attack is usually faster than on benzene itself. This is due to electronic effects that Y can exert.

Substituent, Y is thus classified as m- or o-/p-directing; if the substituent induces faster overall attack than on benzene itself then the substituent is said to be activating, if slower then deactivating.

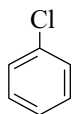
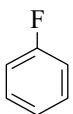
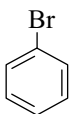
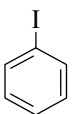
70.  on mononitration gives the major product

- (a)  (b)  (c)  (d) 

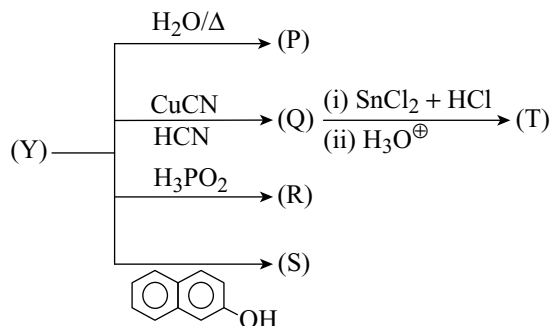
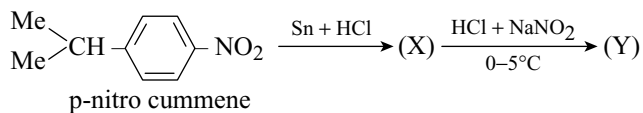
71.  + PhN_2^+ $\xrightarrow{\text{pH} = 9-10}$ X (major product). X will be

- (a)  (b)  (c)  (d) 

72. Which among the following will give highest yield of p-isomer during nitration?

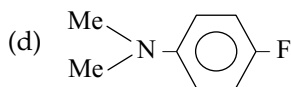
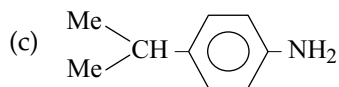
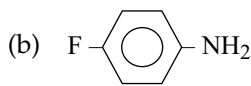
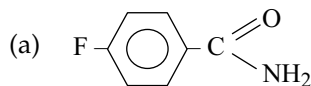
- (a)  (b)  (c)  (d) 

Passage 7



For the given reaction sequence, answer the following

73. (Y) $\xrightarrow[\Delta]{\text{HBF}_4}$ $\xrightarrow[\Delta]{\text{KMnO}_4/\text{H}^+}$ $\xrightarrow[\text{H}_2\text{SO}_4]{\text{N}_2\text{H}}$ Final product; final product is



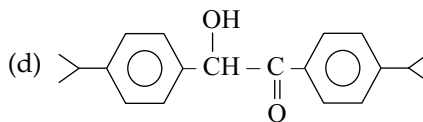
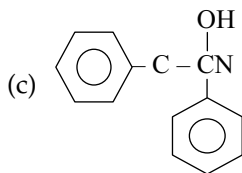
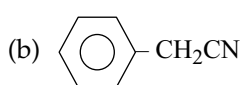
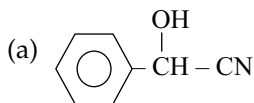
74. (R) $\xrightarrow[\text{(ii) NaOH + I}_2]{\text{(i) CH}_3\text{COCl/AlCl}_3}$ (A) + (B) $\downarrow$
Yellow ppt.

(A) $\xrightarrow[\text{(iii) (R)/AlCl}_3]{\text{(i) H}_3\text{O}^+ \text{ (ii) SOCl}_2}$ Product (L)

The correct statement about the product (L) is

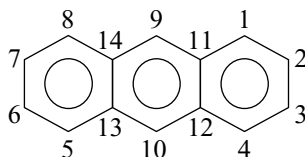
- (a) Product (L) on reaction with Tollen's reagent gives silver mirror
- (b) Product (L) on reaction with $\text{H}_2\text{N-NH}_2/\text{OH}^-$, Δ gives diphenyl methane
- (c) Product (L) on reaction with HCN gives a compound which contains a chiral centre
- (d) All of these

75. Compound (T) $\xrightarrow[\text{KCN}]{\text{Alc.}}$ Product is



Passage 8

Consider the aromatic anthracene molecule, $\text{C}_{14}\text{H}_{10}$, shown in the figure:



Approximate calculation of the π -bond order for C–C bonds yield the following results

Bond	π -bond order
1–2	0.738
1–11	0.535
2–3	0.586
9–11	0.606
11–12	0.485

Also, considering the electron displacement effect in combination with bond order data, answer the following questions.

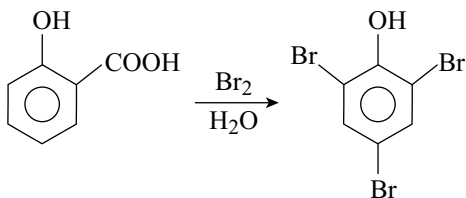
76. Which of the following carbon is most likely to be attacked by an electrophile (NO_2^+ in acetic anhydride at 15–20°C)?
 (a) C–1 (b) C–9
 (c) C–2 (d) C–11
77. Which of the following C–C bond has least percentage of σ -bond character?
 (a) $\text{C}_1\text{--C}_2$ (b) $\text{C}_1\text{--C}_{11}$
 (c) $\text{C}_{11}\text{--C}_{12}$ (d) $\text{C}_2\text{--C}_3$
78. Which of the following (C–C) bonds are shortest and longest respectively?
 (a) 11–12 and 1–2 (b) 1–2 and 11–12
 (c) 9–11 and 1–2 (d) 2–3 and 9–11

Passage 9

Benzene gives electrophilic substitution reaction with strong electrophilic. The leaving group of this reaction is H in the form of H^+ , $-\text{COOH}$ in the form of CO_2 and $-\text{SO}_3\text{H}$ in the form of SO_2 . The reaction is known as aromatic electrophilic substitution (ArSE) reaction.

79. Benzene can be obtained by
 (a) Conc. HNO_3 /Conc. H_2SO_4 /Benzene
 (b) NO_2BF_4 /Benzene
 (c) NO_2ClO_4 /Benzene
 (d) Phenol/Zn
80. Consider the following statements
 (1) Benzene reacts with electrophile to form reaction intermediate known as cyclohexadienyl cation.
 (2) Formation of carbon-electrophilic bond is rate-determining step in ArSE reaction.
 (3) Breaking of C – H bond is rate-determining step
 (4) Sulphonation of benzene is a reversible reaction
- The correct statements from the above are
 (a) (1), (2), (3) and (4) (b) (1), (2) and (4)
 (c) (1), (3) and (4) (d) (2), (3) and (4)

81. In the given reaction

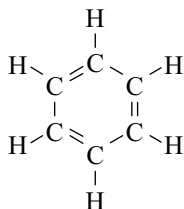


The leaving group(s) in the form of electrophile is

- (a) $2\text{H}^{\oplus}$ only (b) One $\text{H}^{\oplus}$ and one CO_2
 (c) $2\text{H}^{\oplus}$ and one CO_2 (d) One CO_2 only

Passage 10

Based on the Lewis structure of benzene



Benzene

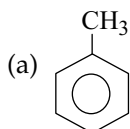
82. What is the bond angle for each H-C-C and each C-C-C on benzene?
 (a) 120° and 120° (b) 109.5° and 120° (c) 120° and 109.5°
 (d) 180° and 120° (e) None of these
83. What one of the following is the best description for the overall shape of benzene molecule?
 (a) Linear (b) Triangular (c) Tetrahedron
 (d) Planar (e) Goofy

Matrix Type

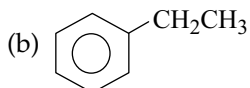
84. Match the columns.

Column I (alkene)

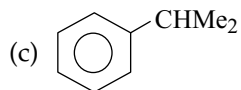
Column II (o/p ratio in nitration)



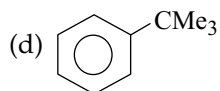
(p) 0.22



(q) 0.92



(r) 1.57



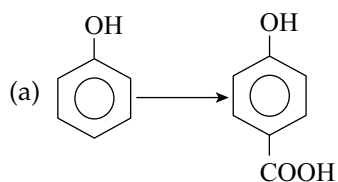
(s) 0.48

85. Match the columns.

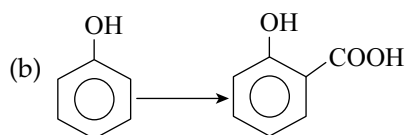
Column I

Column II

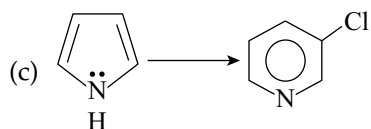
(reagents used for the desired product in major amount)



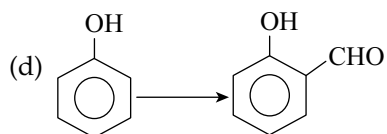
(p) $\text{CHCl}_3/\text{NaOH}$



(q) CCl_4/NaOH



(r) NaOH followed by CO_2

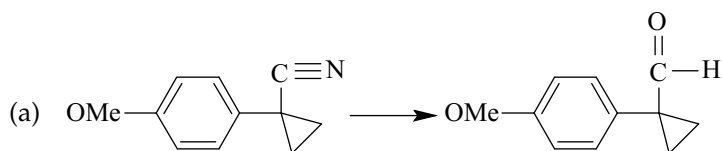


(s) KOH followed by CO_2

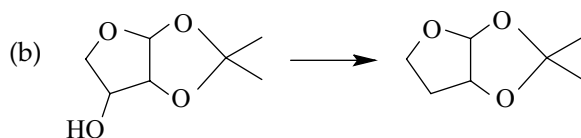
86. Identify reaction correct match with its reagent

Reaction

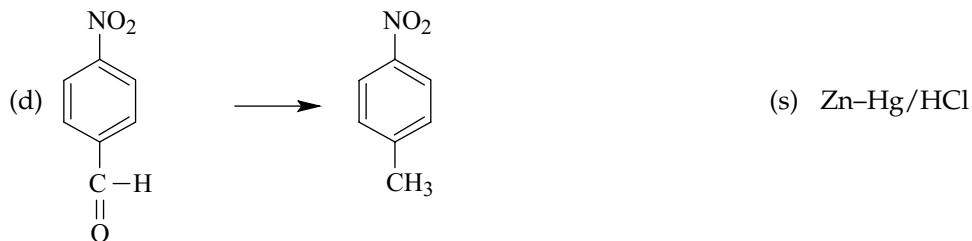
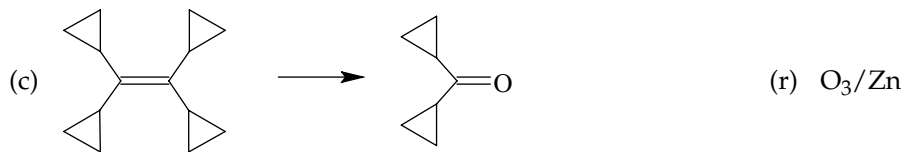
Reagent



(p) $\text{DIBAL-H}/\text{H}_3\text{O}^+$

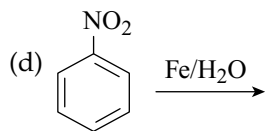
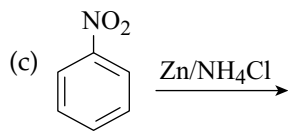
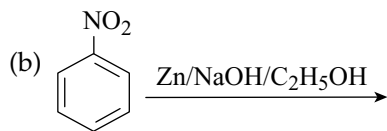
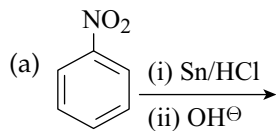


(q) $\text{T}_5\text{Cl}/\text{LAH}$

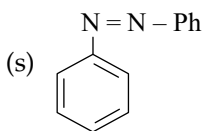
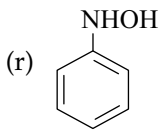
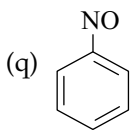
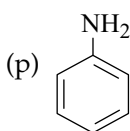


87. Match the reduction in **Column I** with their products listed in **Column II**.

Column I



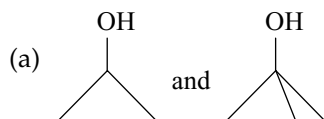
Column II



88. Match the processes in **Column I** with properties in **Column II**.

Column I

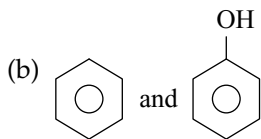
(pair of compounds)



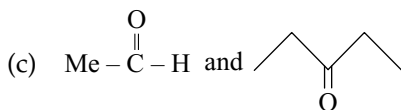
Column II

(reagent used to distinguish pair of compounds)

(p) $\text{Br}_2/\text{H}_2\text{O}$ test



(q) Tollen's reagent



(r) Iodoform test



(s) Lucas reagent

(t) Ammonical cuprous chloride

89. Match the **Column I** with **Column II**.

Column I

Column II

(a) Toluene

(p) On oxidation by KMnO_4 in acid medium gives benzoic acid

(b) Cumene

(q) Used in the manufacture of phenol

(c) Benzene

(r) Symmetrical trimethyl benzene

(d) Mesitylene

(s) On oxidation ($\text{V}_2\text{O}_5/\text{O}_2$) at $250-450^\circ\text{C}$ gives maleic anhydride

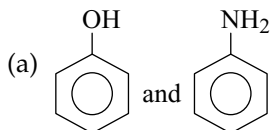
90. Match the columns.

Column I

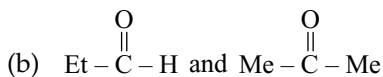
Column II

(pair of compounds)

(reagents used for identification)



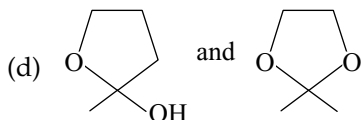
(p) $\text{Br}_2 + \text{H}_2\text{O}$ test



(q) $\text{CHCl}_3 + \text{NaOH}$ test



(r) Iodoform test

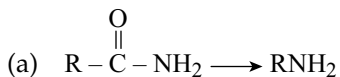


(s) Tollen's reagent

91. Match the columns.

Column I (reaction)

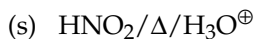
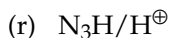
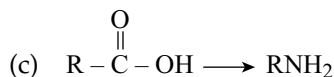
Column II (reagents)



(p) KOBBr

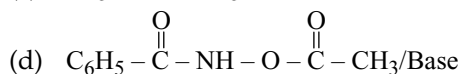
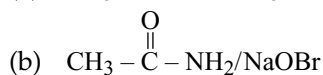
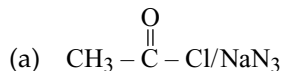


(q) $\text{OH}^-/\text{H}_2\text{O}$



92. Match the columns.

Column I (pair)



Column II

(p) Lossen rearrangement

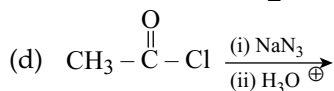
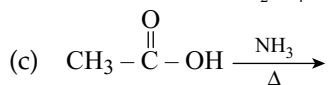
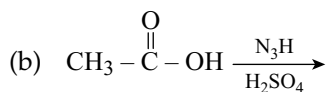
(q) Schmidt rearrangement

(r) Hofmann rearrangement

(s) Curtius rearrangement

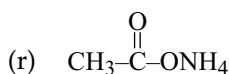
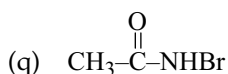
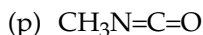
93. Match the **Column I** with **Column II**.

Column I



Column II

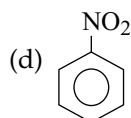
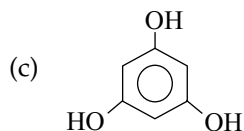
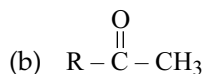
(Intermediate/Product)



94. Match **Column I** with **Column II**.

Column I (elements/compounds)

(a) Halogens



Column II (tests)

(p) $\text{Br}_2/\text{H}_2\text{O}$

(q) Baker-Mulliken test

(r) Iodoform test

(s) Beilstein test

95. Match Column I with Column II.

Column I [name of reaction]

- (a) Haloform reaction
- (b) Aldol addition
- (c) Witting reaction
- (d) Hofmann bromamide reaction

Column II [substrate(s) of reaction]

- (p) α -methyl carbonyl compound
- (q) Acid amide
- (r) Aldehyde
- (s) Halide and carbonyl

96. Column I (reaction)

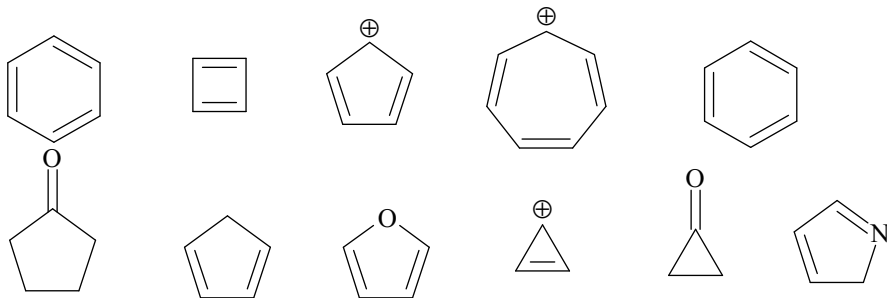
- (a) Wurtz reaction
- (b) Dehydration of alcohol with Conc. H_3PO_4
- (c) Kolbe's electrolysis
- (d) Diazotisation

Column II (intermediate)

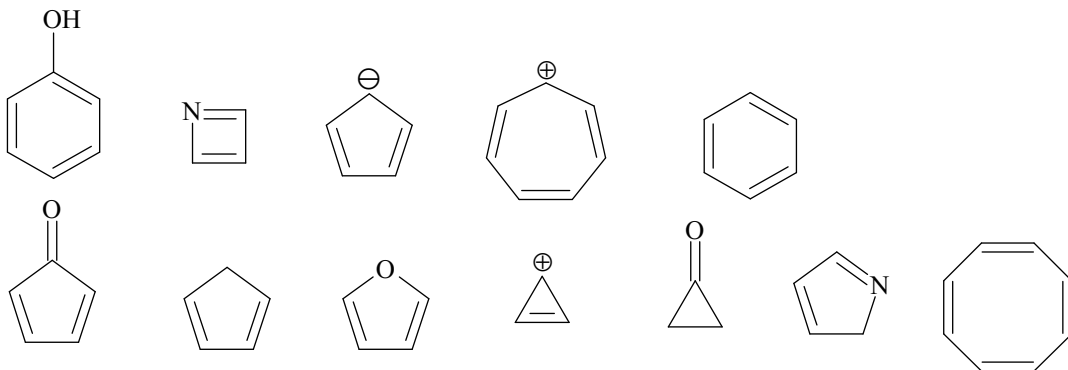
- (p) Carbocation
- (q) Electrophile (excluding free radicals)
- (r) Free radical
- (s) Carbanion
- (t) N_2 gas liberated in intermediate step

Integer Type

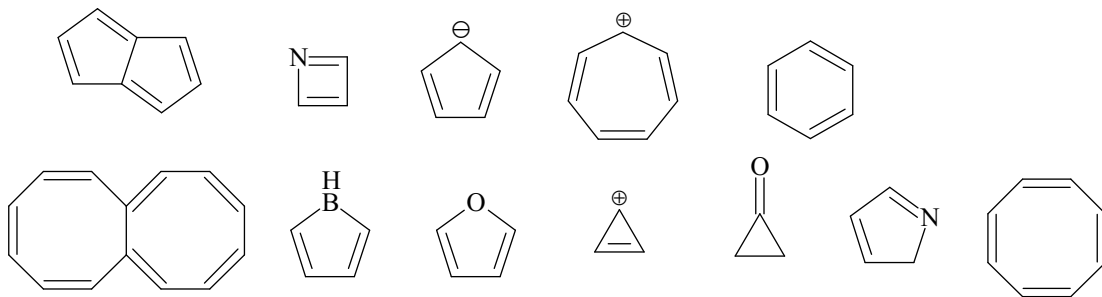
97. Identify compounds which are aromatic.



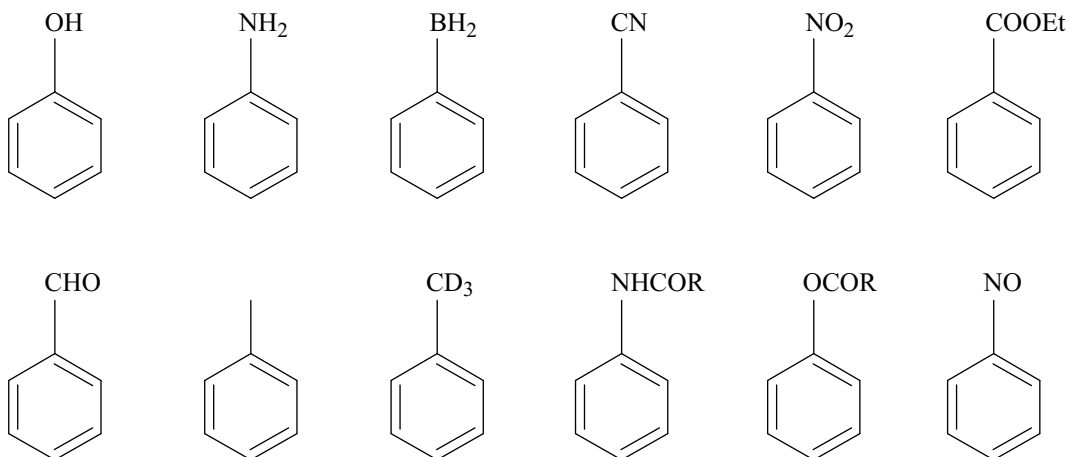
98. Identify compounds which are nonaromatic.



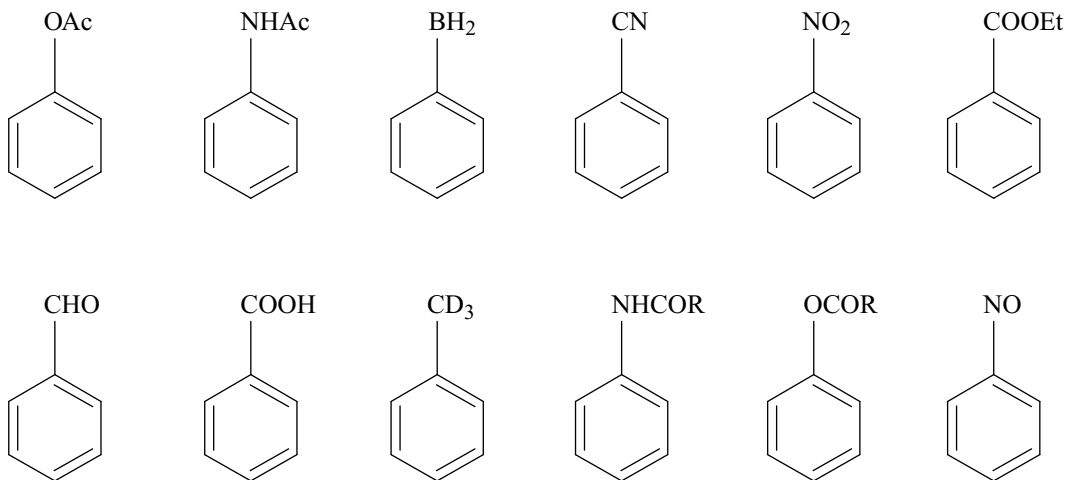
99. Identify compounds which are antiaromatic.



100. Identify compounds which react faster than benzene in ArSE reaction?

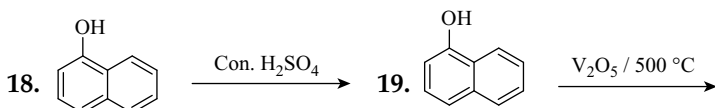
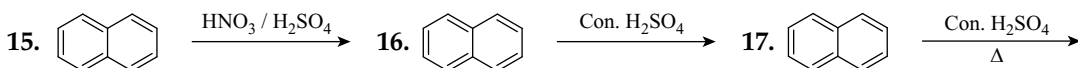
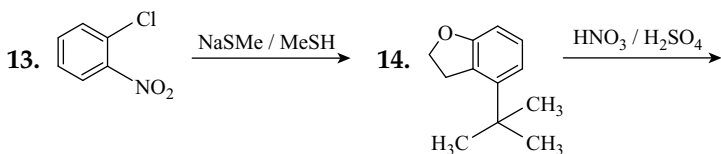
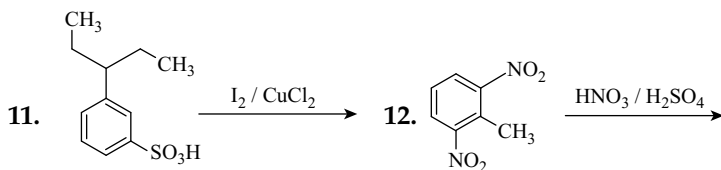
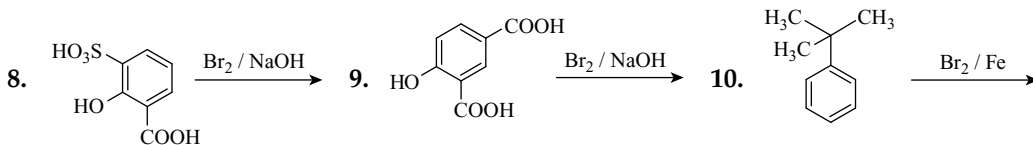
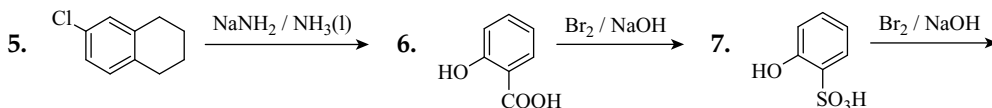
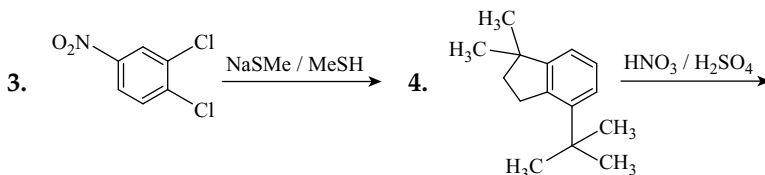
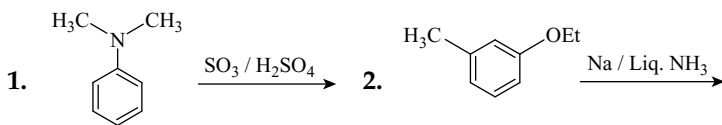


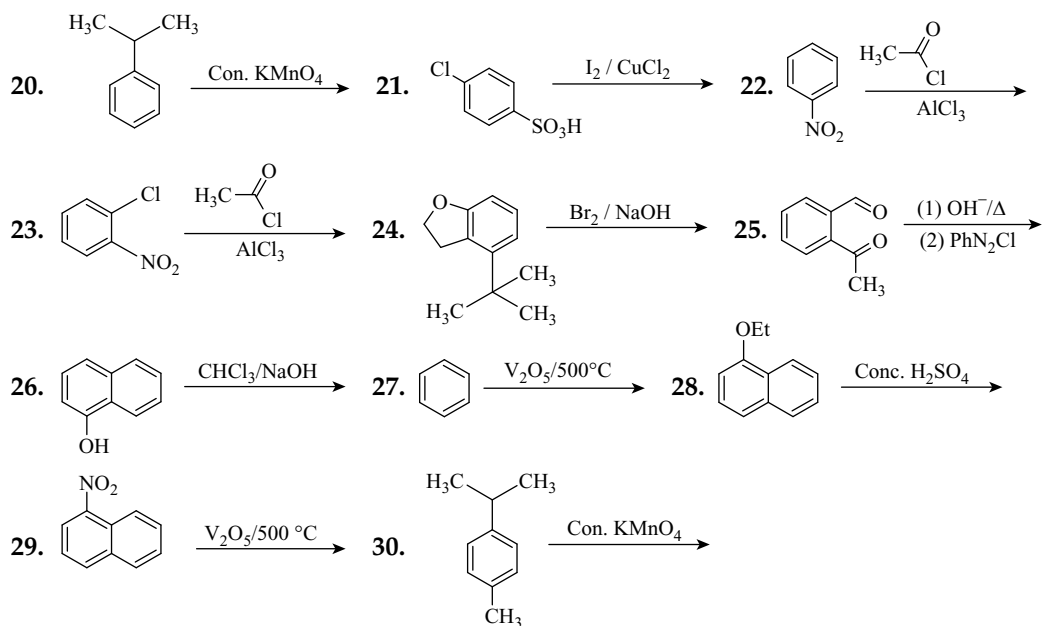
101. Identify compounds which react slower than benzene in ArSE reaction?



WORKBOOK EXERCISE 1

Identify complete reaction sequence

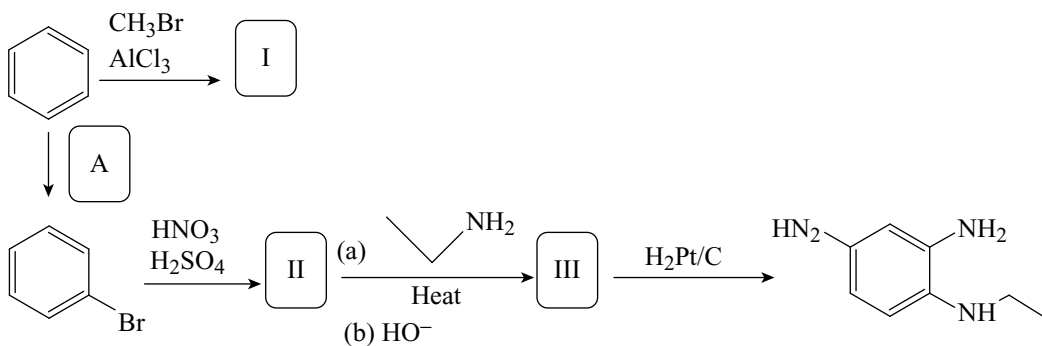




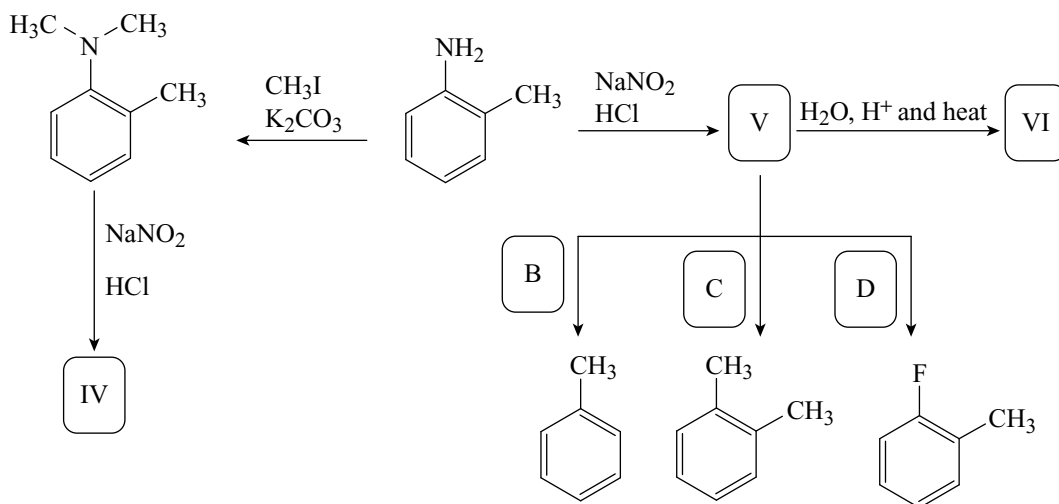
WORKBOOK EXERCISE 2

Identify reagent used and intermediate products in the following conversion.

A. Multiple-step synthesis

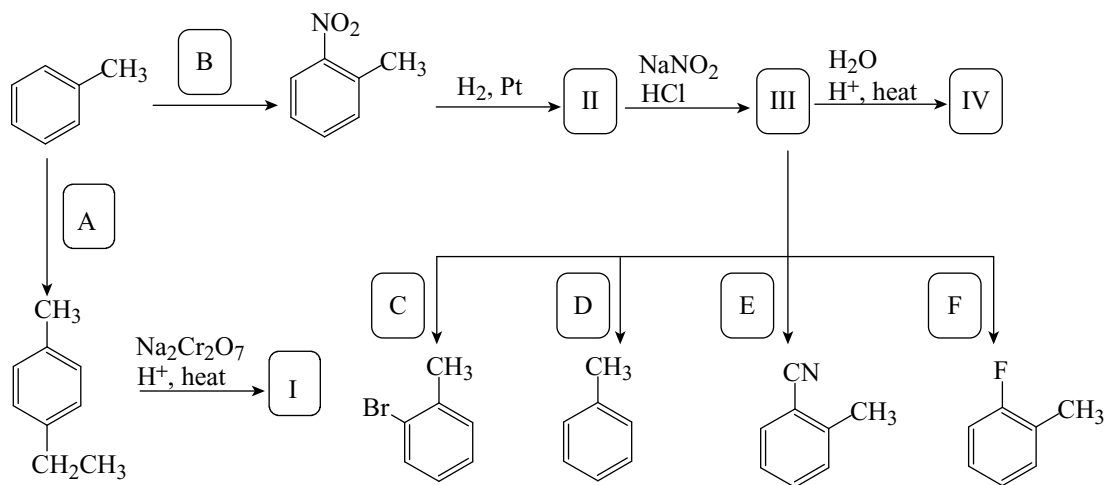


B. Multiple-step synthesis



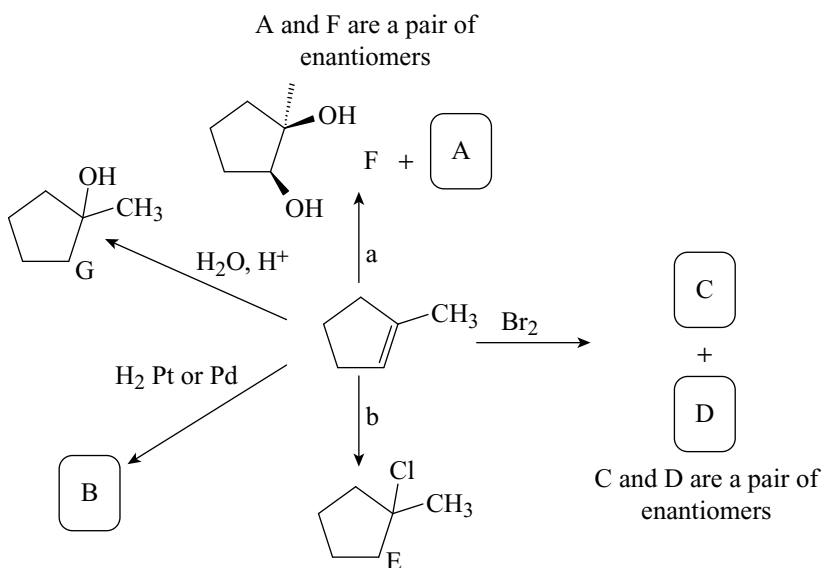
WORKBOOK EXERCISE 3

Identify reagent used and intermediate products in the following conversion.



WORKBOOK EXERCISE 4

Identify reagent used and intermediate products in the following conversion.

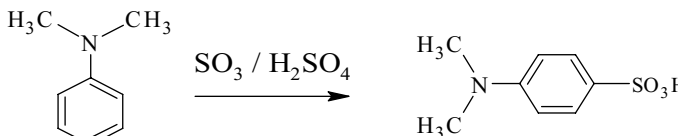
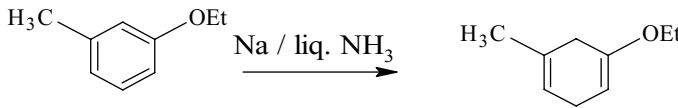
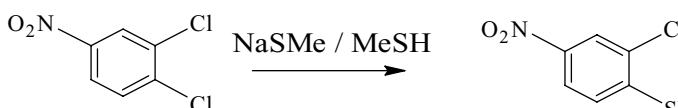
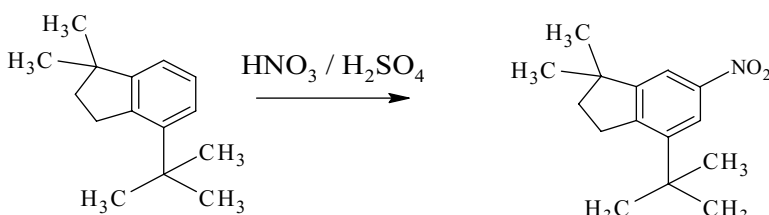
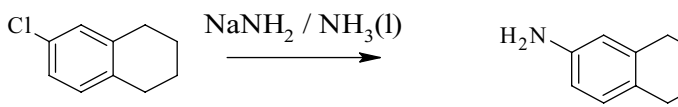
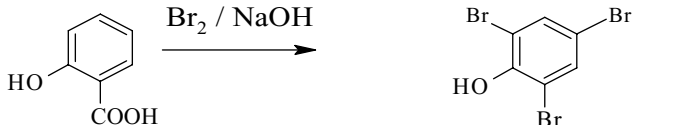
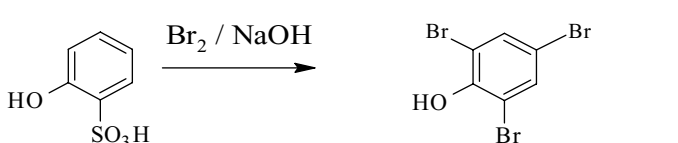
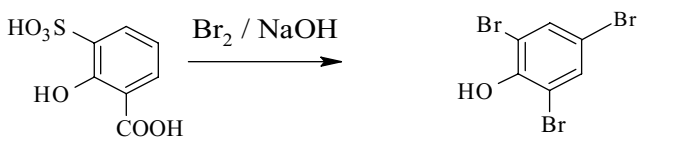
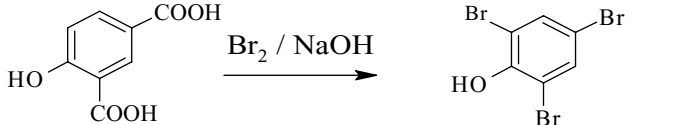


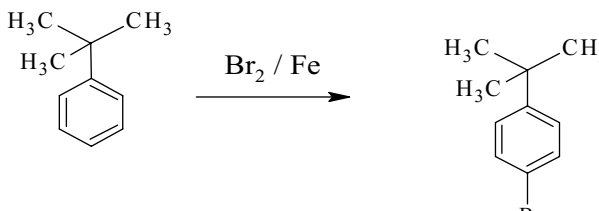
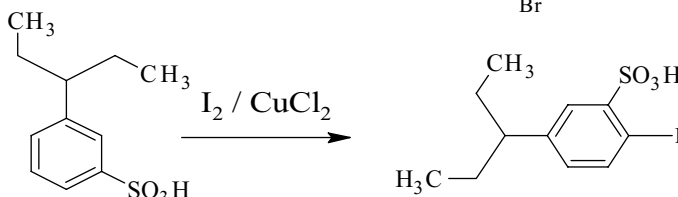
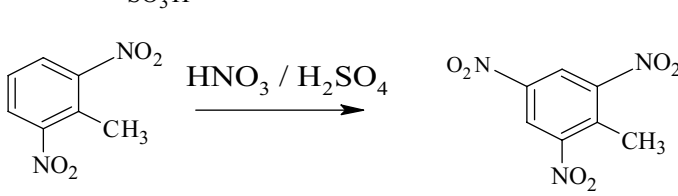
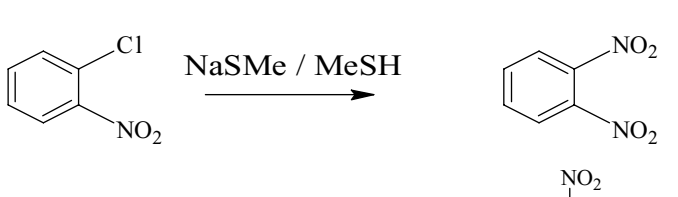
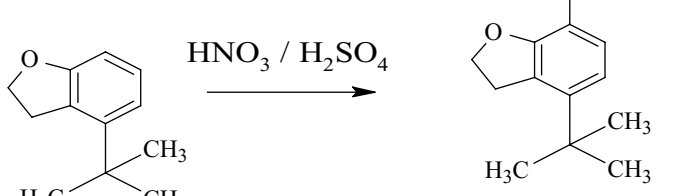
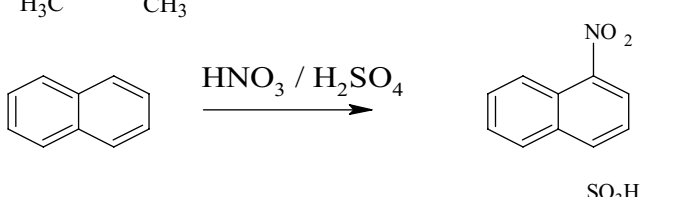
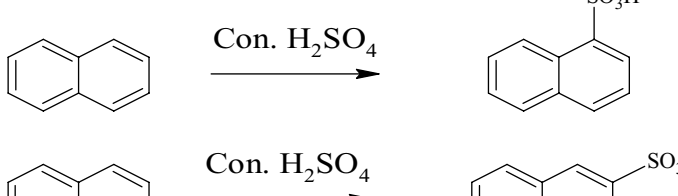
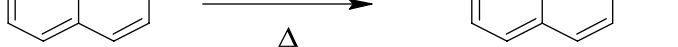
LEVEL 2

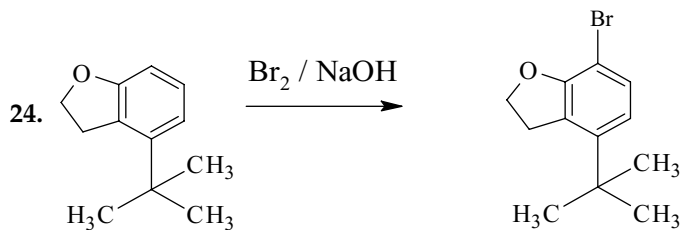
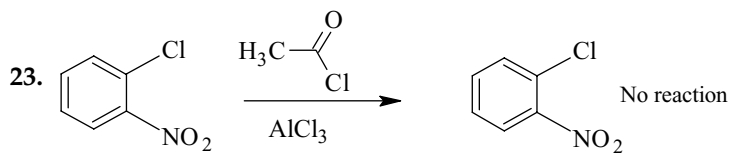
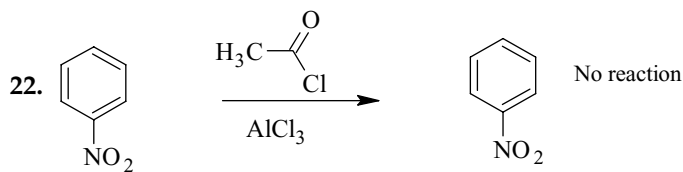
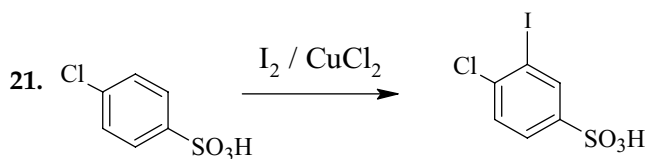
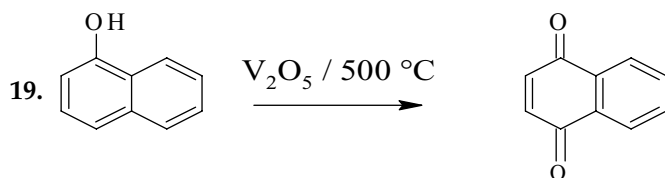
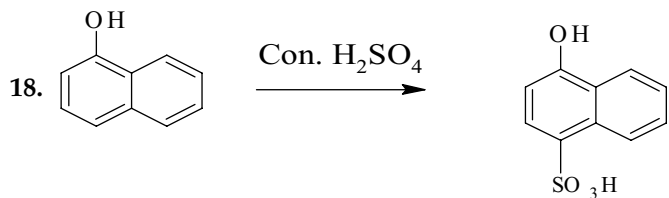
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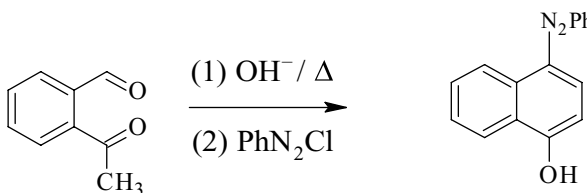
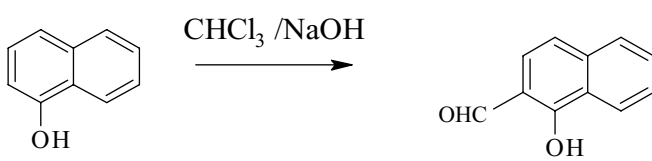
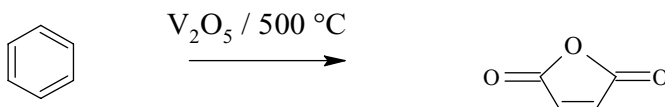
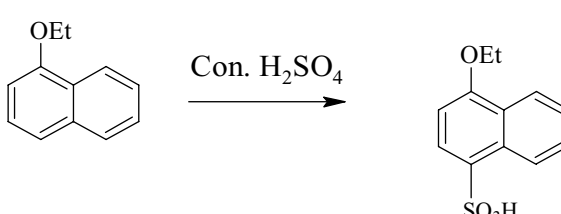
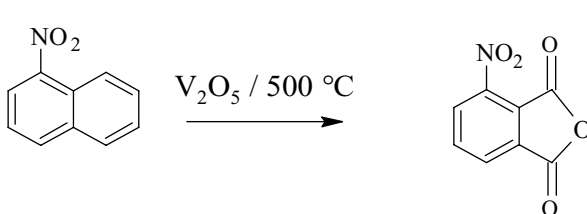
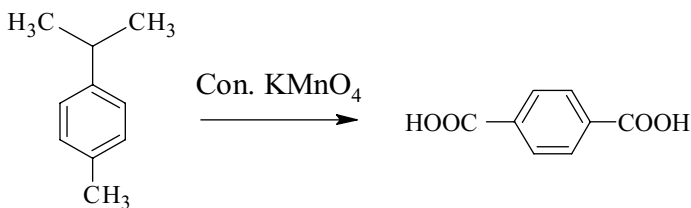
WORKBOOK EXERCISE 1

Identify complete reaction sequence.

1. 
CN(C)c1ccccc1.O=S(=O)(O)O>>CN(C)c1ccc(S(=O)(=O)O)cc1
2. 
CC1=CC=C(C)C=C1OCC.N>>CC1=CCCCC1OCC
3. 
Clc1cc(Cl)c(Cl)cc1.CS>>Clc1cc(Cl)c(SC)cc1
4. 
CC1(C)C2=CC=CC=C2C1(C)C(C)(C)C3=CC=CC=C3C2(C)C1(C)C2.C=O>>CC1(C)C2=CC(=C(C=C2)C1(C)C(C)(C)C3=CC=CC=C3C2(C)C1(C)C2)[N+](=O)[O-]
5. 
ClC1=CC=CC2=CC=CC=C12.N>>NC1=CC=CC2=CC=CC=C12
6. 
OC(=O)c1ccccc1O.Br>>OC(=O)c1c(Br)cc(Br)cc1Br
7. 
OS(=O)(=O)c1ccccc1O.Br>>OC(=O)c1c(Br)cc(Br)cc1Br
8. 
OS(=O)(=O)c1ccc(O)c(C(=O)O)c1.Br>>OC(=O)c1c(Br)cc(Br)cc1Br
9. 
OC(=O)c1cc(O)cc(C(=O)O)cc1.Br>>OC(=O)c1c(Br)cc(Br)cc1Br

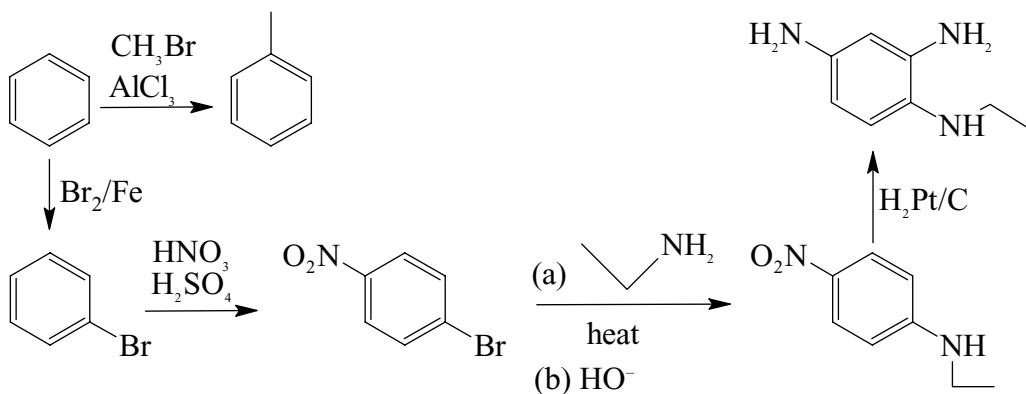
10.  CC(C)(C)c1ccccc1.BrBr>[Fe]>CC(C)(C)c1ccc(Br)cc1
11.  CC(C)Cc1cccc(S(=O)(=O)O)c1.II>ClCu>CC(C)Cc1ccc(I)c(S(=O)(=O)O)c1
12.  Cc1cc([N+](=O)[O-])cc([N+](=O)[O-])cc1>O=[N+]([O-])O>OS(=O)(=O)O>Cc1c([N+](=O)[O-])cc([N+](=O)[O-])cc1[N+](=O)[O-]
13.  Clc1ccccc1[N+](=O)[O-]>CS[Na]>CS>O=[N+]([O-])c1ccccc1[N+](=O)[O-]
14.  CC(C)(C)C1C(C)OC2=CC=CC=C2O1>O=[N+]([O-])O>OS(=O)(=O)O>CC(C)(C)C1C(C)OC2=CC(=CC=C2C2=CC=CC=C2O1)[N+](=O)[O-]
15.  c1ccc2ccccc2c1>O=[N+]([O-])O>OS(=O)(=O)O>O=[N+]([O-])c1cccc2ccccc12
16.  c1ccc2ccccc2c1>OS(=O)(=O)O>>OS(=O)(=O)c1cccc2ccccc12
17.  c1ccc2ccccc2c1>OS(=O)(=O)O>>OS(=O)(=O)c1ccc2ccccc2c1



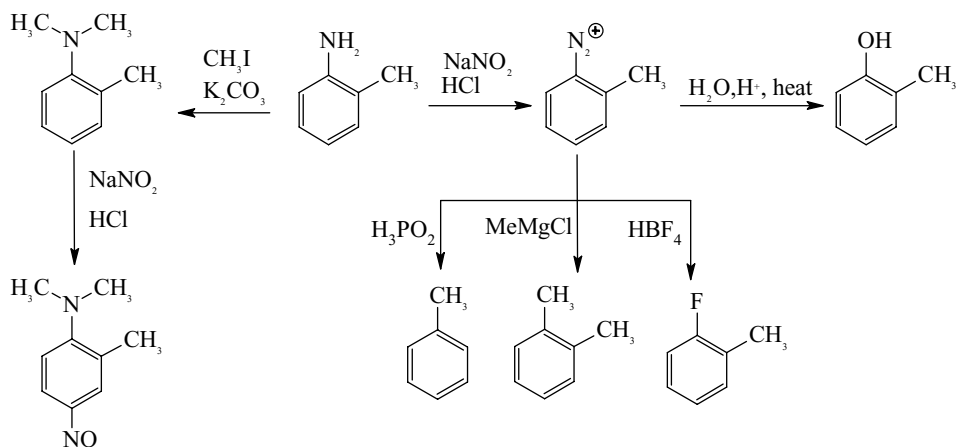
25.  CC(=O)c1ccccc1O>>O=C1C(=O)C(=O)c2ccccc2O1
26.  Oc1ccccc2ccccc12>>O=Cc1c(O)cccc2ccccc12
27.  c1ccccc1>>O=C1OC(=O)C=C1
28.  CCOC1=CC=CC2=CC=CC=C12>>CCOC1=CC=CC(=C1)S(=O)(=O)C2=CC=CC=C2
29.  [O-][N+](=O)c1cccc2ccccc12>>O=C1OC(=O)C2=CC(=CC=C2)[N+](=O)[O-]C3=CC=CC=C13
30.  CC(C)c1ccc(C)cc1>>OC(=O)c1ccc(cc1)C(=O)O

WORKBOOK EXERCISE 2

A. Multiple-step Synthesis

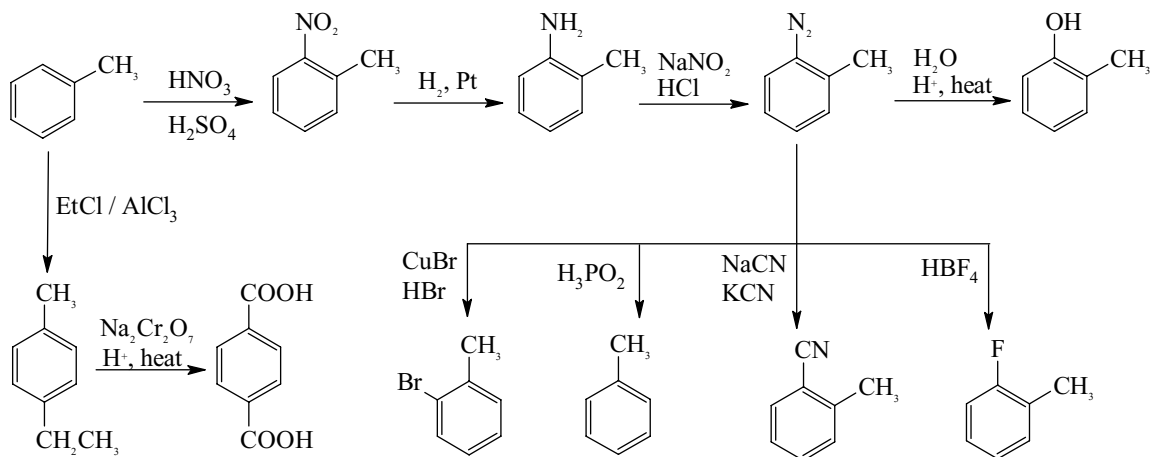


B. Multiple-step Synthesis



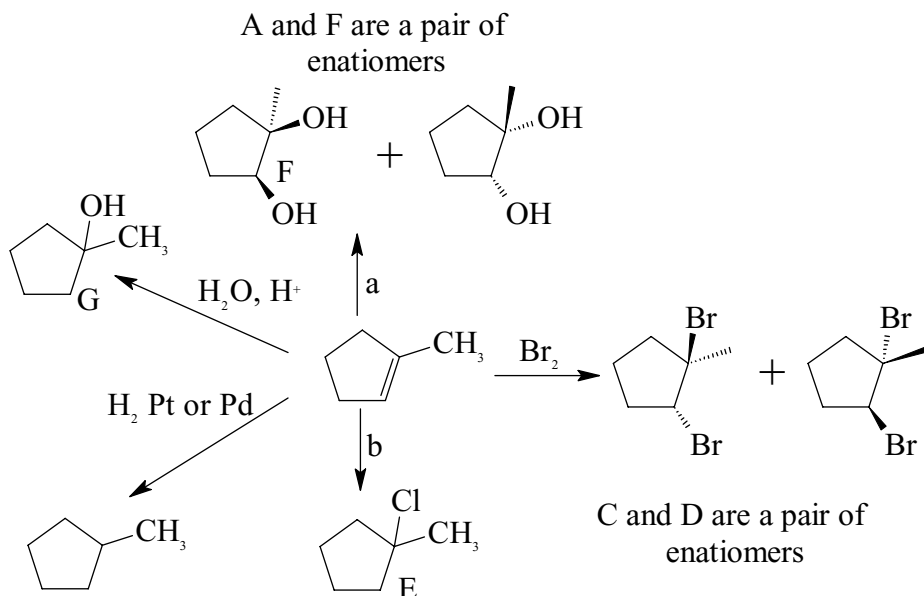
WORKBOOK EXERCISE 3

Identify reagent used and intermediate products in following conversion.

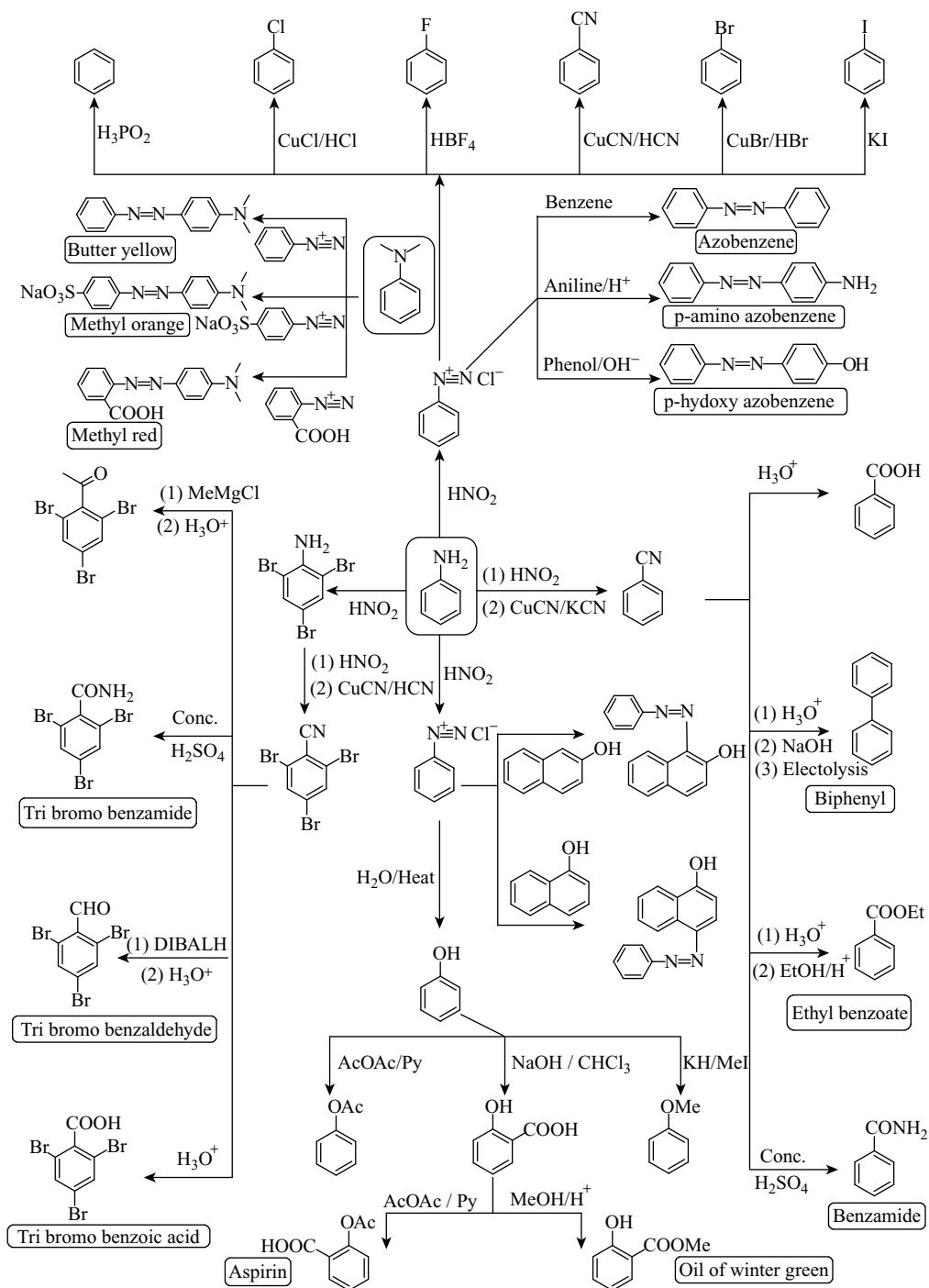


WORKBOOK EXERCISE 4

Identify reagent used and intermediate products in following conversion.



Reaction Mechanism Chart



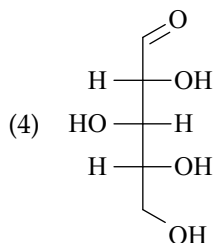
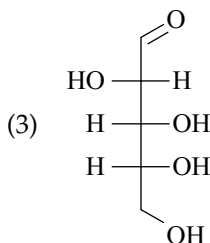
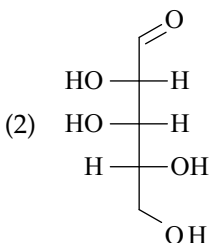
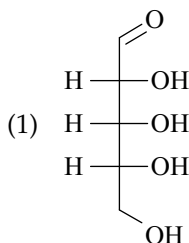
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Biomolecules

Question Bank

LEVEL 1

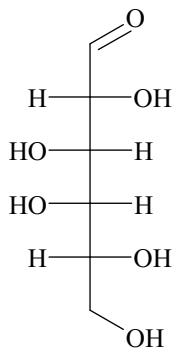
- Which of the following statements is correct?
 - The Ruff procedure lengthens an aldose chain and gives a single product.
 - The Ruff procedure shortens an aldose chain and gives two epimers.
 - The Kiliani-Fischer procedure shortens an aldose chain and gives a single product.
 - The Kiliani-Fischer procedure lengthens an aldose chain and gives two epimers.
- Which of the following is not a disaccharide?
 - sucrose
 - mannose
 - lactose
 - maltose
- Which statement about the pyranose form of mannose is not correct?
 - it exists as two anomeric stereoisomers.
 - it reacts with Tollen's reagent to give a silver mirror (i.e., it is a reducing sugars).
 - reaction with excess CH_3I and AgOH gives a non-reducing penta-O-methyl derivative.
 - it resists reduction with aqueous sodium borohydride.
- Two aldopentoses X and Y give the same osazone derivative. X is oxidised to an optically active aldaric acid by dilute nitric acid. Ruff degradation of Y gave a tetrose which was similarly oxidised to an optically active aldaric acid. Assign the structures of X and Y from the following list.



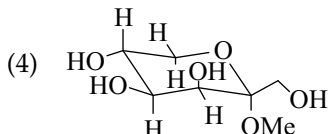
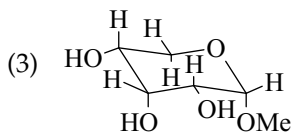
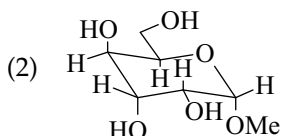
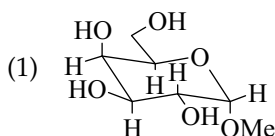
- X = 1 and Y = 4
- X = 2 and Y = 3

- X = 4 and Y = 1
- X = 3 and Y = 2

5. Which of the structures 1 through 4 is methyl-D-galactopyranoside?



D-Galactose



(a) 1

(b) 2

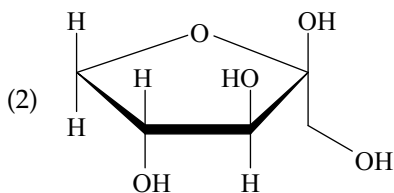
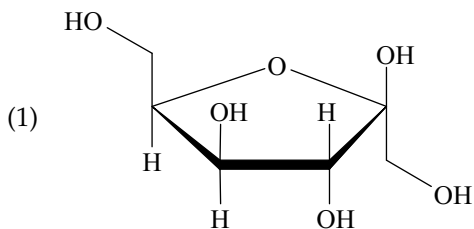
(c) 3

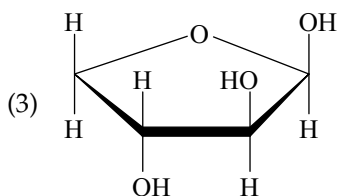
(d) 4

6. What is invert sugar, and why is it so named?

- the sugar mixture from hydrolysis of sucrose; fructose is isomerised to glucose.
- the sugar mixture from hydrolysis of sucrose; the optical rotation changes from (+) to (-).
- the sugar mixture from hydrolysis of starch; α -glycosidic bonds are changed to β -glycosidic bonds.
- the sugar mixture from hydrolysis of starch; glucose is isomerised to fructose.

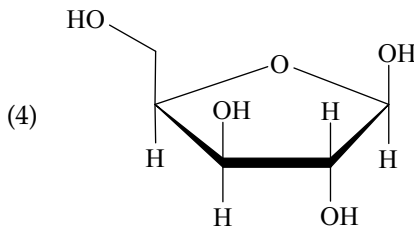
7. Which of the following compounds is a β -aldopentafuranose?





(a) 1

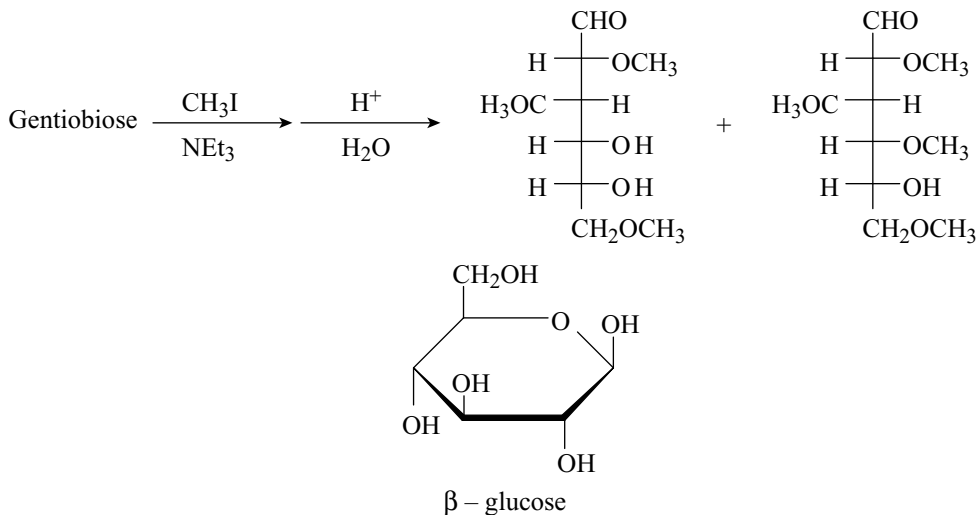
(b) 2



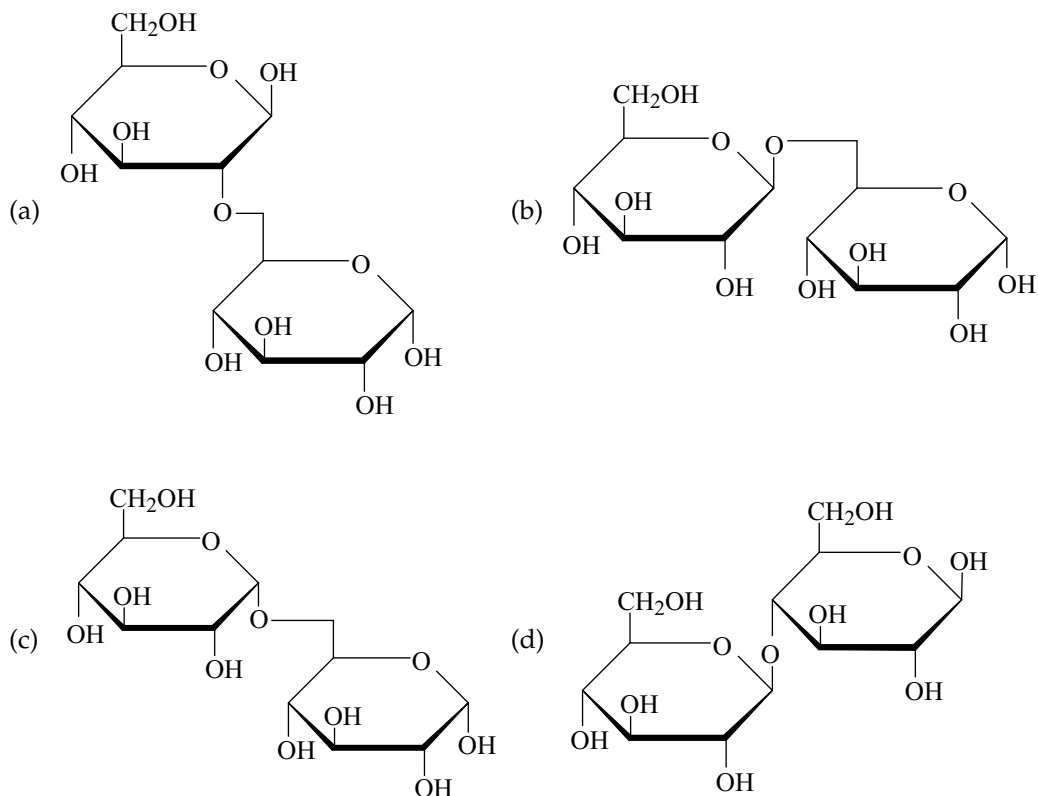
(c) 3

(d) 4

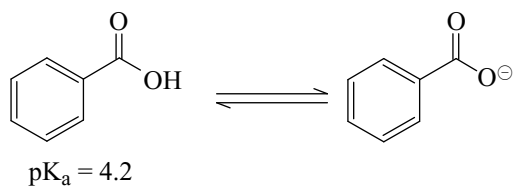
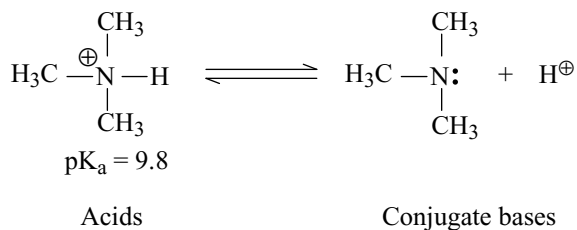
8. When octa-O-methyl D-cellobiose is hydrolysed by an aqueous acid, two O-methylated glucose derivatives are formed. One is a tetramethyl derivative, and the other is a trimethyl derivative. Why is a single methyl substituent lost in this process?
- one methoxy group is lost by β -elimination.
 - one methoxy group is an ester and the others are all ethers.
 - one methoxy group is part of an acetal, the others are all ethers.
 - one glucose is an α -methyl glycoside; the other is a β -methyl glycoside.
9. Acid-catalysed reaction of D-glucose with benzaldehyde produces the 4,6-O-benzylidene derivative. Reduction with NaBH_4 , followed by excess HIO_4 cleavage and acid hydrolysis yields a $\text{C}_4\text{H}_8\text{O}_4$ tetrose and benzaldehyde. What is the configuration of this tetrose?
- 2S, 3S
 - 2R, 3S
 - 2R, 3R
 - 2S, 3R
10. Gentiobiose is a disaccharide incorporated into the chemical structure of crocin, the chemical compound that gives saffron its colour. Deduce the structure of gentiobiose from the following information.
- Acid hydrolysis of gentiobiose yields only D-glucose.
 - Gentiobiose is hydrolysed by β -glucosidase enzymes, but not by α -glucosidase enzymes.
 - Gentiobiose is a reducing sugar and reacts with sodium borohydride.
 - Methylation of gentiobiose followed by hydrolysis of the glycosidic bonds yields the two D-glucose derivatives shown below.

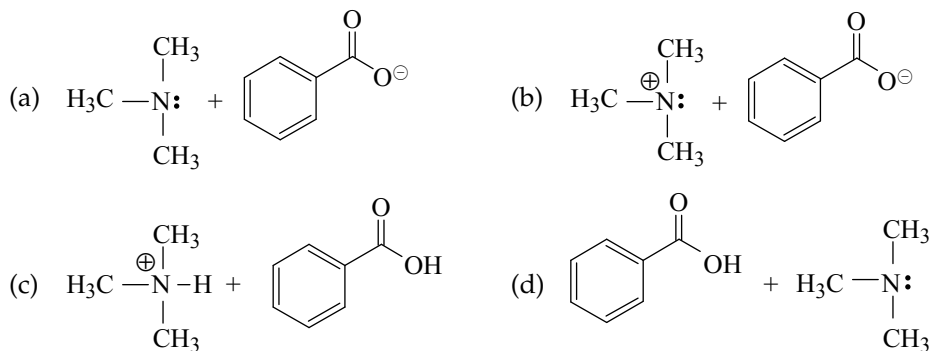


What is the structure of gentiobiose?



11. Choose the answer that has correctly selected the major species of the following two acids to be present in blood naturally buffered at pH = 7.2.





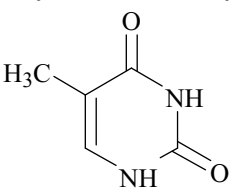
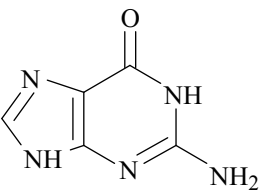
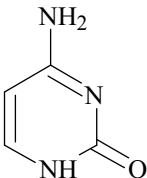
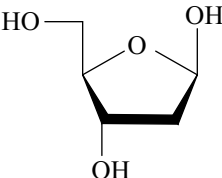
[Possible species in blood buffered at pH = 7.2]

12. Which one of the following is a non-reducing sugar?
 (a) Glucose (b) Mannose (c) Fructose (d) Sucrose
13. Which of the following compounds will not show mutarotation?
 (a) Methyl- α -D-glucopyranoside (b) α -D (+) glucopyranose
 (c) β -D (+) glucopyranose (d) β -D (+) galactopyranose
14. α -D-glucose and β -D-glucose are
 (a) anomers (b) C₂-epimers (c) C₃-epimers (d) enantiomers
15. At isoelectric point, the amino acid has
 (a) Least viscosity (b) Maximum surface tension
 (c) Maximum solubility (d) All of these
16. Which one of the following is not correct for monosaccharides?
 (a) They are optically active polyhydroxy carbonyl compounds
 (b) Fructose is ketose sugar and hence it does not give red precipitate with Fehling's solution
 (c) α -D (+) glucose and β -D (+) glucose are anomers
 (d) Glucose and mannose are epimers
17. Which of the following is not a surfactant?
 (a) $\text{CH}_3 - (\text{CH}_2)_{15} - \overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{N}^+}} - \text{CH}_3\text{Br}^-$ (b) $\text{CH}_3 - (\text{CH}_2)_{14} - \text{CH}_2 - \text{NH}_2$
 (c) $\text{CH}_3 - (\text{CH}_2)_{16} - \text{CH}_2\text{OSO}_2^- \text{Na}^+$ (d) $\text{OHC} - (\text{CH}_2)_{14} - \text{CH}_2 - \text{COO}^- \text{Na}^+$
18. Tertiary structure of protein contains which type of forces?
 (a) Electrostatic forces only
 (b) Electrostatic as well as hydrogen bonds
 (c) Electrostatic, hydrogen bonds and Vander waal forces
 (d) Electrostatic, hydrogen bonds, Vander waal forces, dipole-dipole attractions and disulphide crosslinkage.

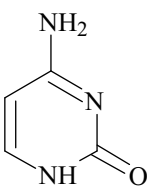
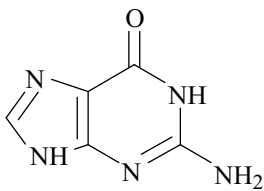
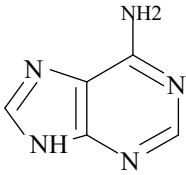
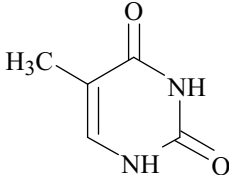
19. Which of the following is an addition and homopolymer?

- (I) Neoprene (II) Polystyrene (III) Nylon-6,6 (IV) Nylon-6
(a) only (I) (b) (I), (II) and (IV)
(c) (I) and (II) (d) (I), (II), (III) and (IV)

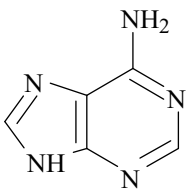
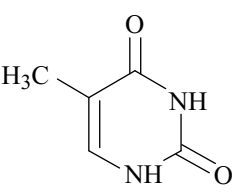
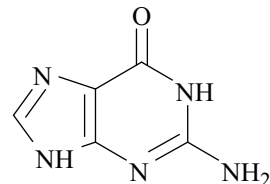
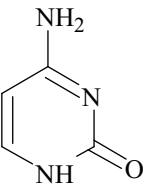
20. Identify structure of thymine

- (a) 
- (b) 
- (c) 
- (d) 

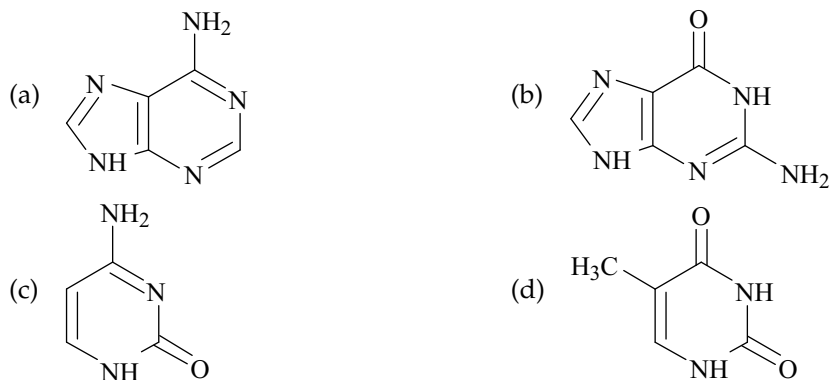
21. Identify structure of adenine

- (a) 
- (b) 
- (c) 
- (d) 

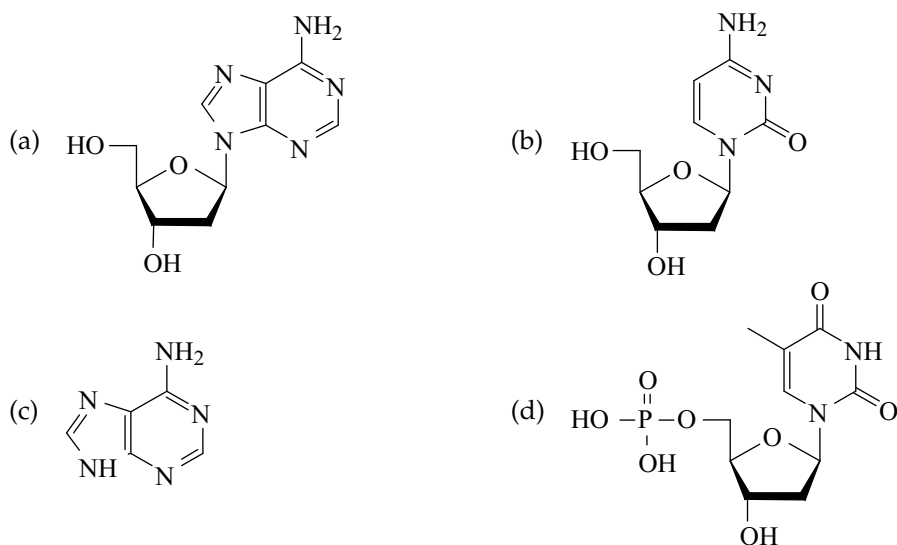
22. Identify structure of cytosine

- (a) 
- (b) 
- (c) 
- (d) 

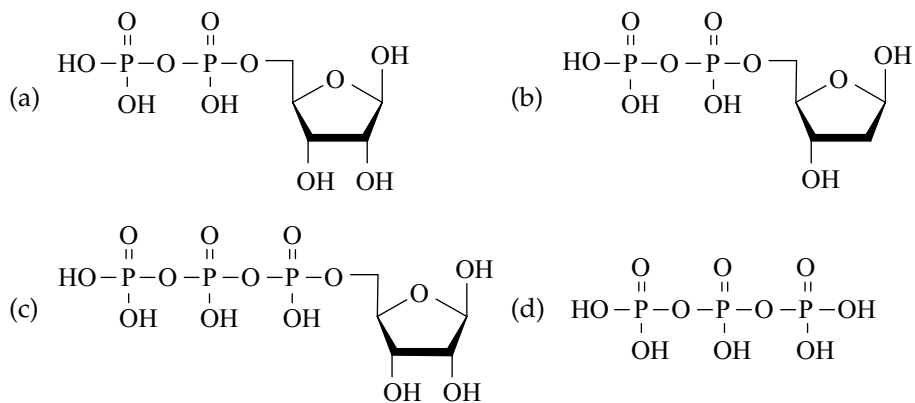
23. Identify structure of guanine



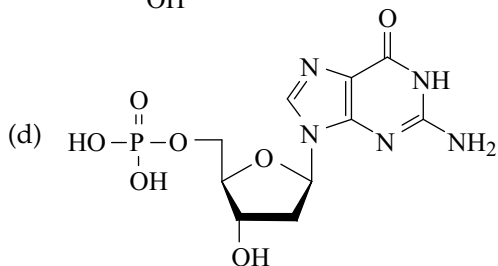
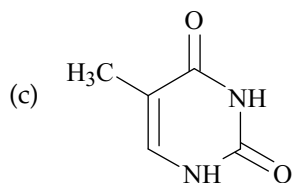
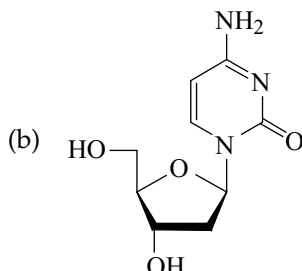
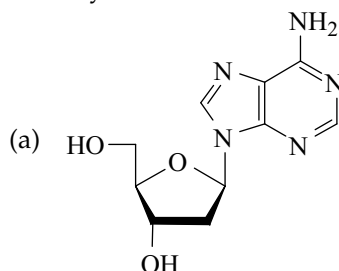
24. Identify structure of adenine deoxyribo-nucleoside



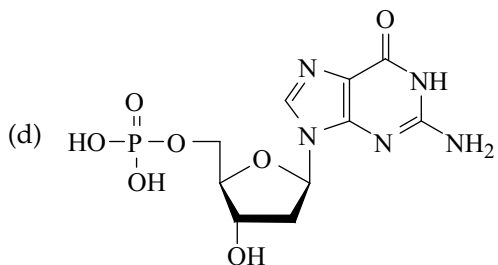
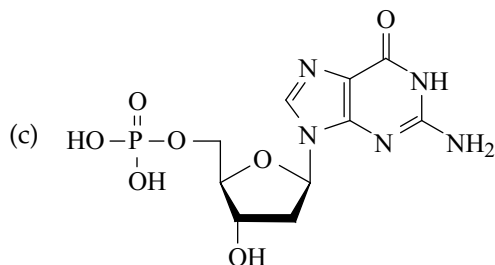
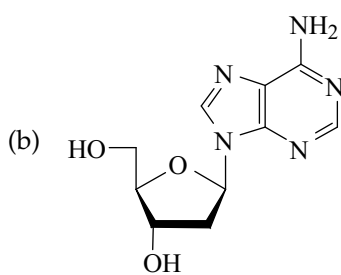
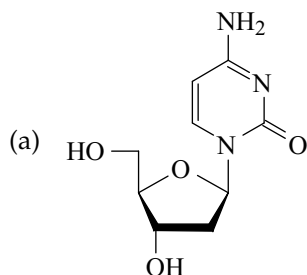
25. Identify structure of Riboso-5-triphosphate



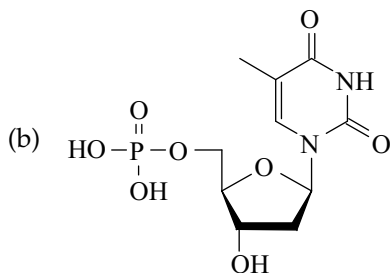
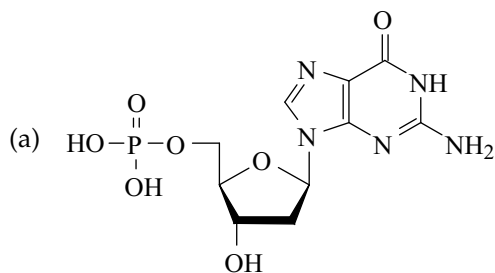
26. Identify structure of Guanine deoxyribo-nucleotide

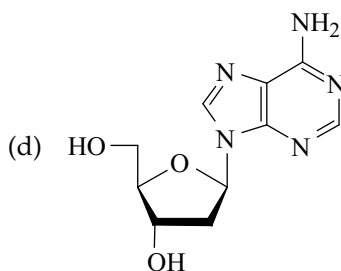
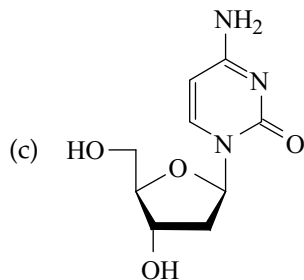


27. Identify structure of Cytosine deoxyribo-nucleoside

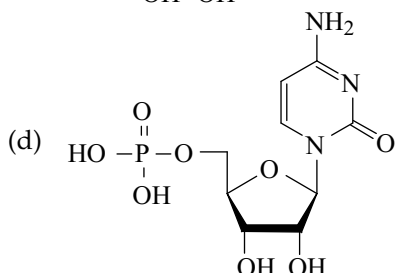
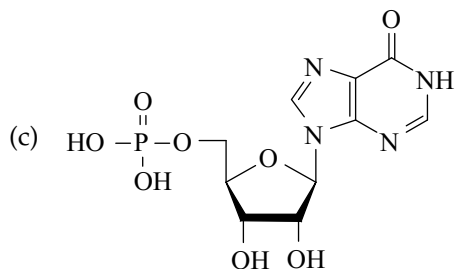
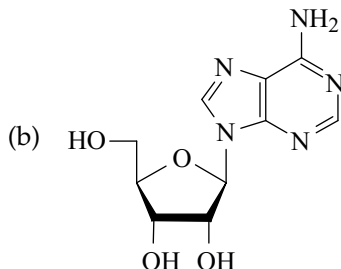
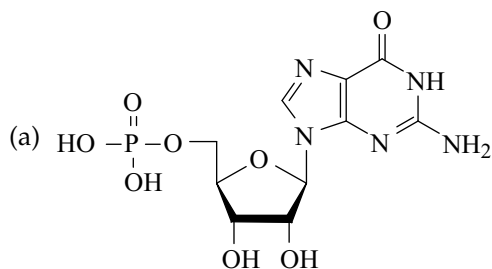


28. Identify structure of Thymine deoxyribo-nucleotide

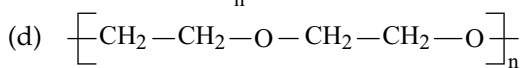
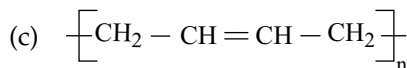
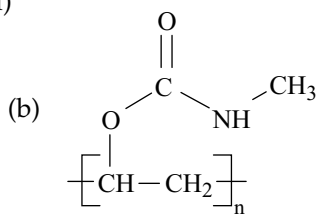
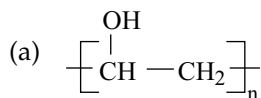




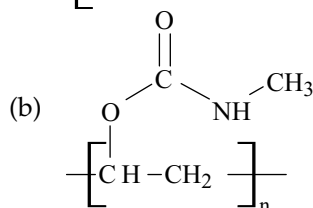
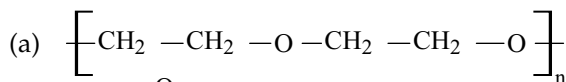
29. Identify structure of Guanine ribonucleotide

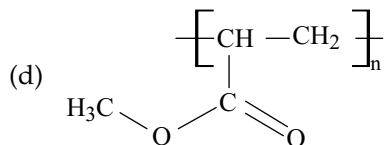
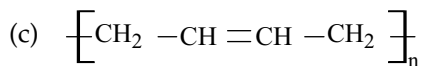


30. Identify structure of poly(dithylene glycol)

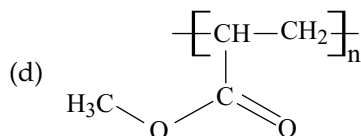
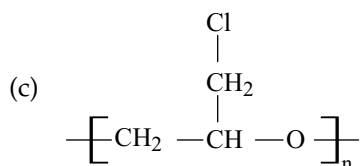
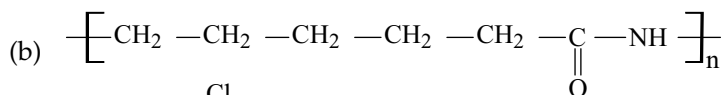
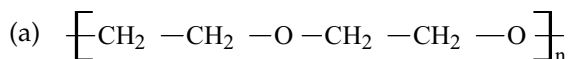


31. Identify structure of poly(methyl acrylate)

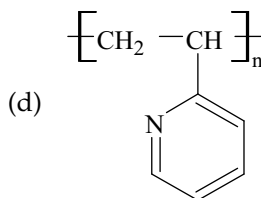
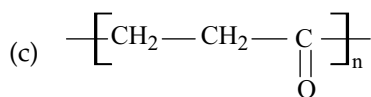
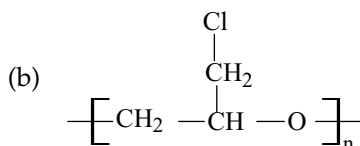
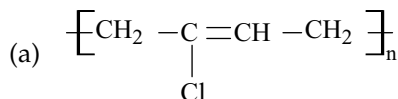




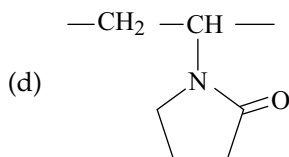
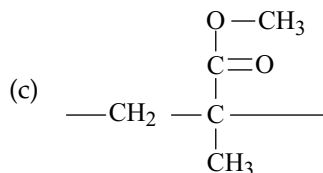
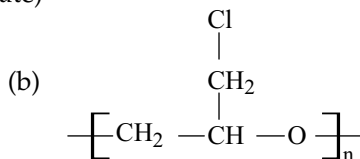
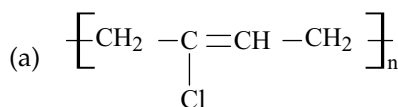
32. Identify structure of Nylon-6



33. Identify structure of neoprene

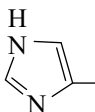


34. Identify structure of poly(methylmetacrylate)



35. Among the following polymer identify homopolymer
 (a) Nylon-6 (b) Buna-N (c) Buna-S (d) Polyethylene
36. Among the following polymers, identify the copolymer
 (a) Nylon-6 (b) Starch (c) PVC (d) Protein
37. Which of these is a hypnotic?
 (a) Metaldehyde (b) Acetaldehyde (c) Paraldehyde (d) None of these
38. An ester used as medicine is
 (a) ethyl acetate (b) methyl acetate (c) methyl salicylate (d) ethyl benzoate
39. Nylon threads are made of
 (a) polyvinyl polymer (b) polyester polymer
 (c) polyamide polymer (d) polyethylene polymer
40. Paracetamol is an
 (a) antibiotic (b) antipyretic (c) antimalarial (d) analgesic
41. Antiseptic, chlorozyleneol is
 (a) 4-chloro-3,5-dimethyl phenol (b) 3-chloro-4,5-dimethyl phenol
 (c) 4-chloro-2,5-dimethyl phenol (d) 5-chloro-3,4-dimethyl phenol
42. Phenacetin is used as on
 (a) antipyretic (b) antiseptic (c) analgesic (d) antimalarial
43. Which of the following colours is imparted by alizarin dye in the presence of Cr^{3+} ion?
 (a) Violet (b) Brown-red (c) Pink (d) Red
44. Which of the following polymers can be used for lubrication and as an insulator?
 (a) SBR (b) PVC (c) PTFE (d) PAN
45. A hybrid rocket propellant uses
 (a) a liquid oxidiser and a solid fuel
 (b) a composite solid propellant
 (c) a biliquid propellant
 (d) a solid liquid and gas as a propellant
46. Buna-N synthetic rubber is a copolymer of
 (a) $\text{H}_2\text{C}=\text{CH}-\underset{\text{C}}{\overset{\text{Cl}}{|}}=\text{CH}_2$ and $\text{H}_2\text{C}=\text{CH}-\text{CH}=\text{CH}_2$
 (b) $\text{H}_2\text{C}=\text{CH}-\text{CH}=\text{CH}_2$ and $\text{H}_5\text{C}_6-\text{CH}=\text{CH}_2$
 (c) $\text{H}_2\text{C}=\text{CH}-\text{CN}$ and $\text{H}_2\text{C}=\text{CH}-\text{CH}=\text{CH}_2$
 (d) $\text{H}_2\text{C}=\text{CH}-\text{CN}$ and $\text{H}_2\text{C}=\text{CH}-\underset{\text{C}}{\overset{\text{Cl}}{|}}=\text{CH}_2$

47. Among cellulose, polyvinyl chloride, nylon and natural rubber, the polymer in which the intermolecular force of attraction is weakest is
 (a) nylon (b) polyvinyl chloride
 (c) cellulose (d) natural rubber

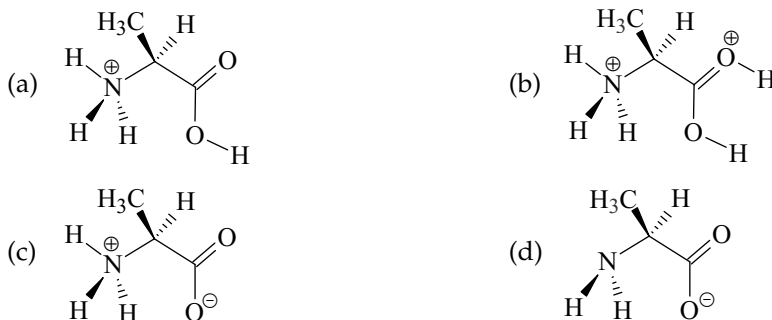
48. The drug  is used as an
 (a) antacid (b) analgesic (c) antimicrobial (d) antiseptic

49. What is the complementary RNA sequence for the DNA segment AATCAGTT?
 (a) AAUCAGUU (b) CCAUCGAA (c) AACUGAUU (d) UUAGUCAA

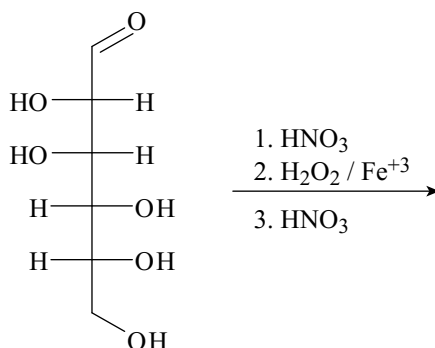
50. In DNA replication the complementary nucleotide sequence for 5'-ACGT-3' is which of the following?
 (a) 3'-ACGT-5' (b) 3'-TGCA-5' (c) 5'-AGCT-3' (d) 5'-TCGA-3'

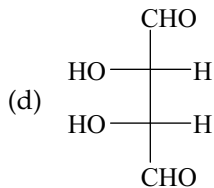
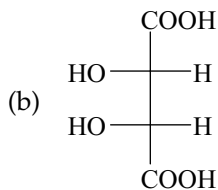
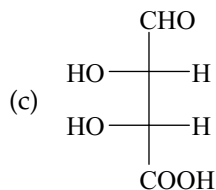
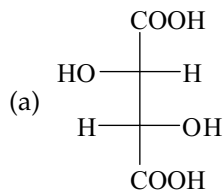
51. Which of the following is not a disaccharide?
 (a) sucrose (b) mannose (c) lactose (d) maltose

52. Shown below are five representations of the molecule alanine. Which one best represents the structure of alanine in the blood stream (a buffered solution of about pH 7.4)? The pK_a values for alanine are 2.3 ($-\text{CO}_2\text{H}$) and 9.7 ($-\text{NH}_3^+$).

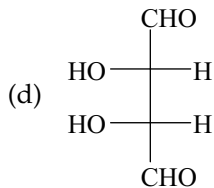
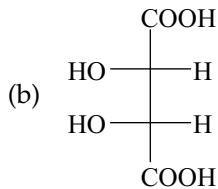
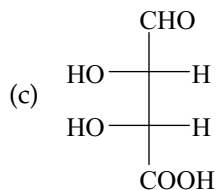
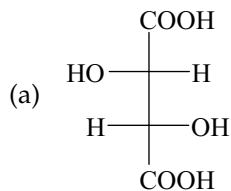
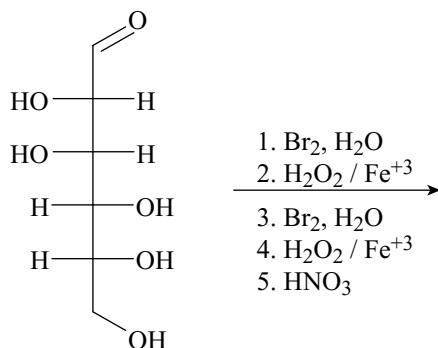


53. Identify the final product of the reaction

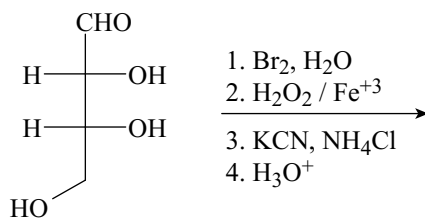


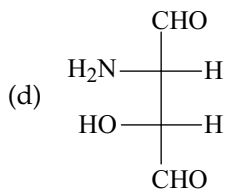
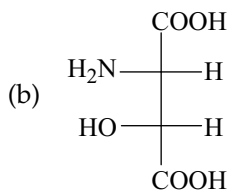
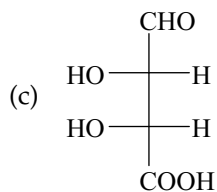
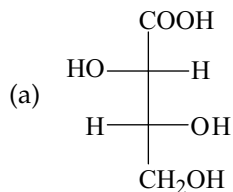


54. Identify the final product of the reaction

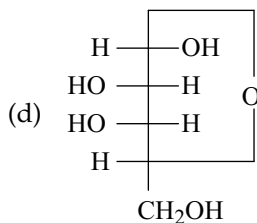
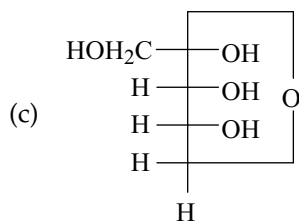
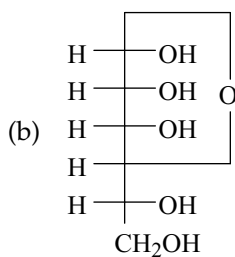
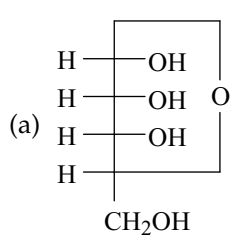


55. Identify the final product of the reaction

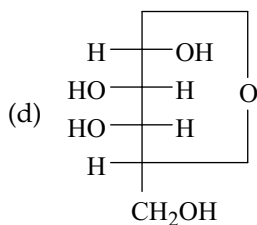
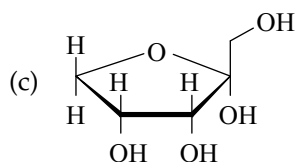
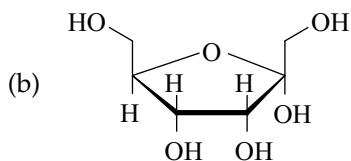
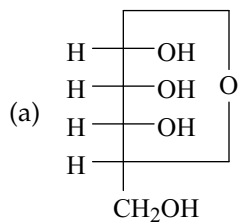




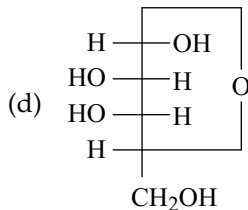
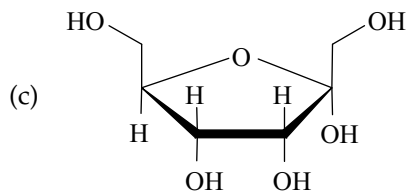
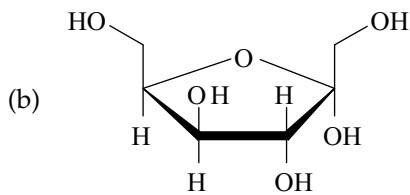
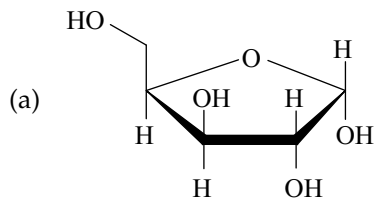
56. Identify structure of α -D-Lyxofuranose



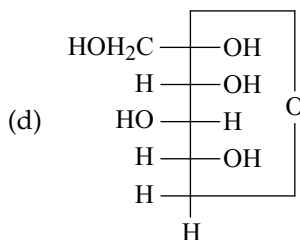
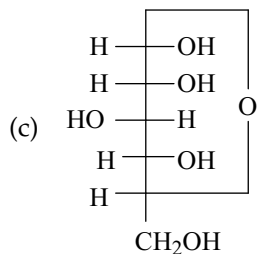
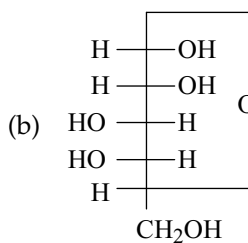
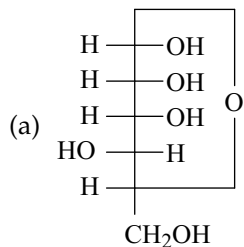
57. Identify structure of α -D-Ribulofuranose



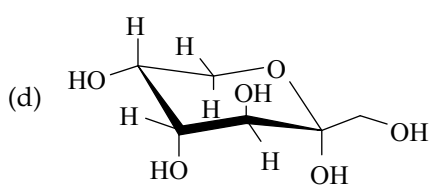
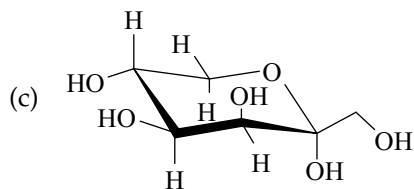
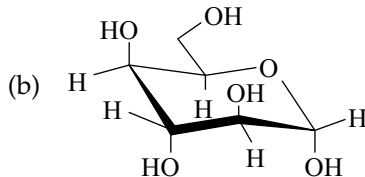
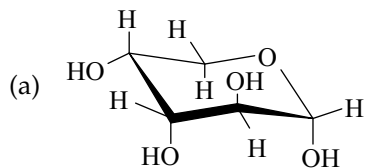
58. Identify structure of α -D-Xylofuranose



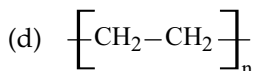
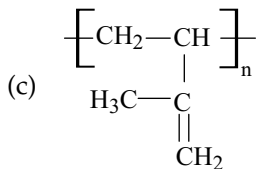
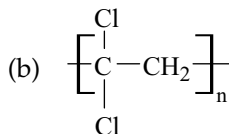
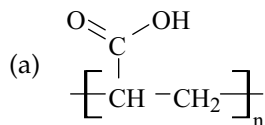
59. Identify structure of α -D-Glucopyranose



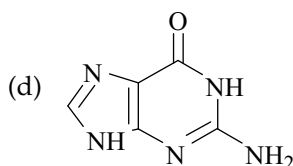
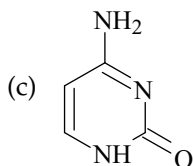
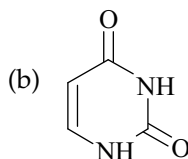
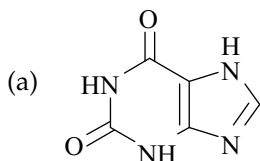
60. Identify structure of α -D-Arabinopyranose



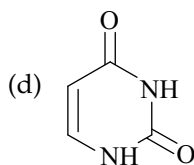
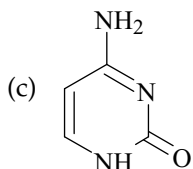
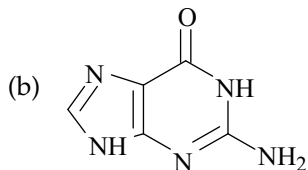
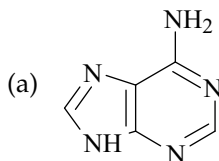
61. Identify structure of polyethene



62. Identify structure of guanine



63. Identify structure of adenine



64. The pupils of eyes are dilated with a very dilute solution of an alkaloid which is

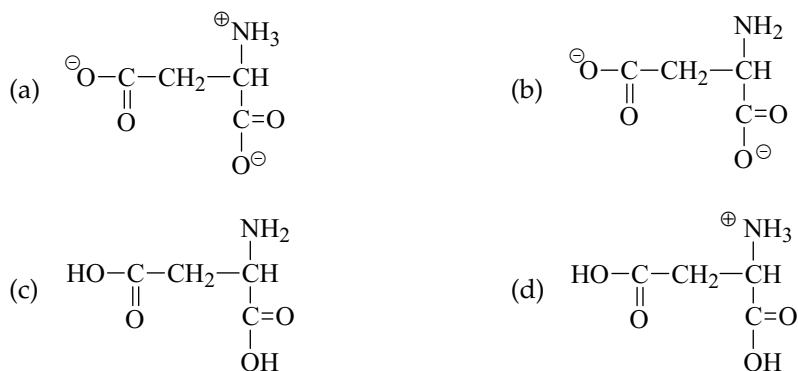
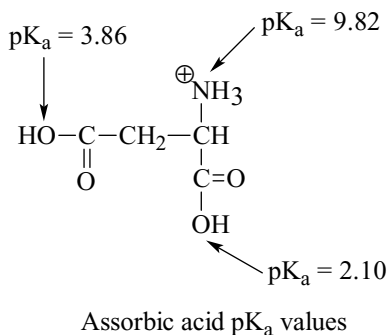
(a) adrenaline

(b) atropine

(c) equanil

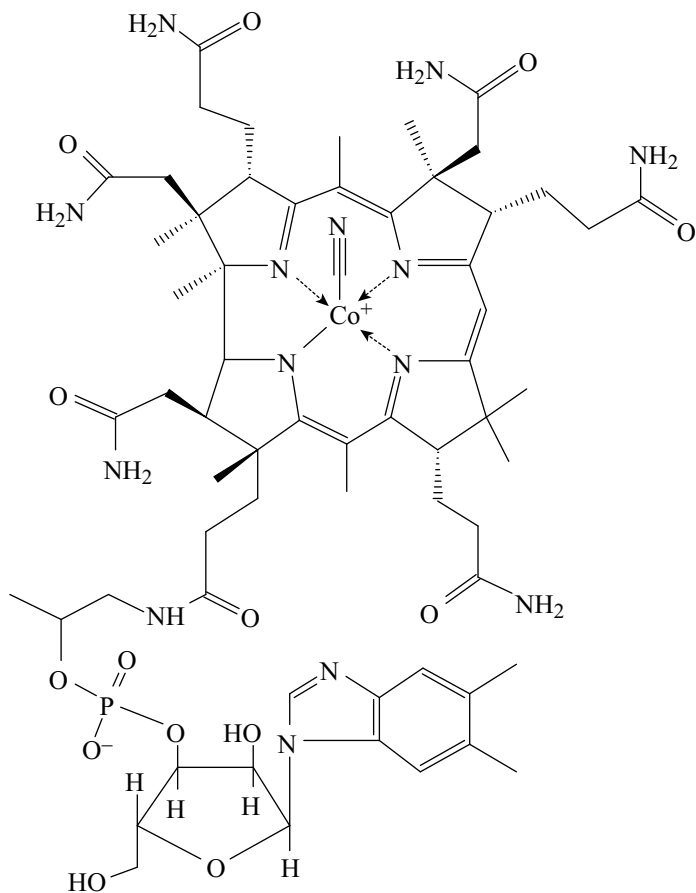
(d) ephedrine

65. The pK_a values of aspartic acid, a naturally occurring amino acid are shown to the right. Choose from below, the structure of the major species present at a pH of 7.2.



66. Penicillin was first discovered by
 (a) A. Fleming (b) L. Pasteur
 (c) G. Thompson (d) A. Noble
67. The poisonous gas evolved in Bhopal tragedy was
 (a) COCl_2 (b) CH_3NCO
 (c) CH_3CN (d) CO
68. In making lemon pickle
 (a) chillies are used to kill all germs and bacteria
 (b) lemon oil acts as a preservative
 (c) salt plays the same role as sugar in jam
 (d) citric acid acts as a preservative

69. Identify the sugar present in vitamin B₁₂



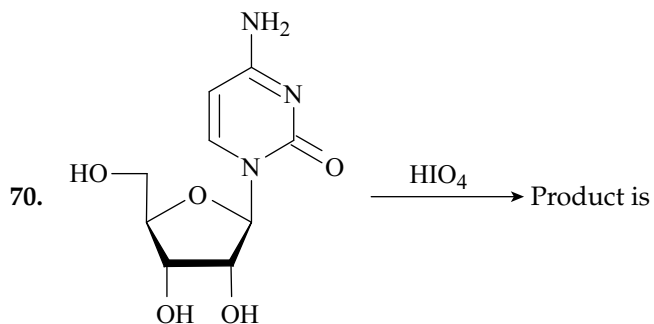
Vitamin B₁₂

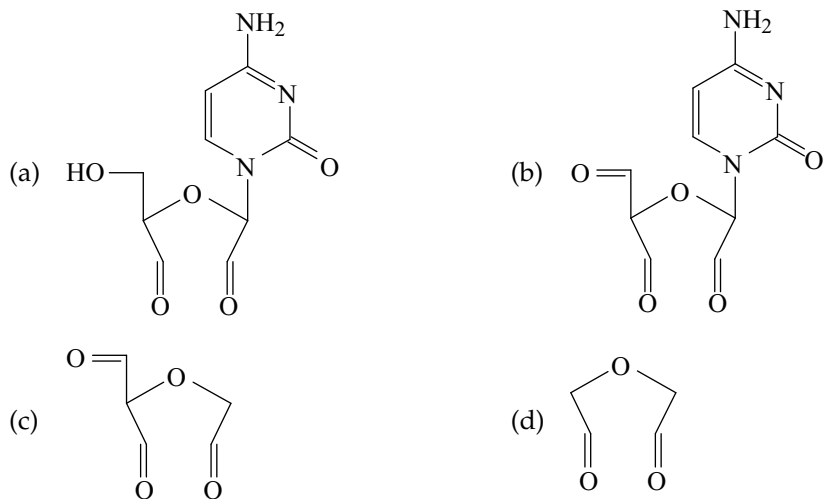
(a) Triose

(b) Pentose

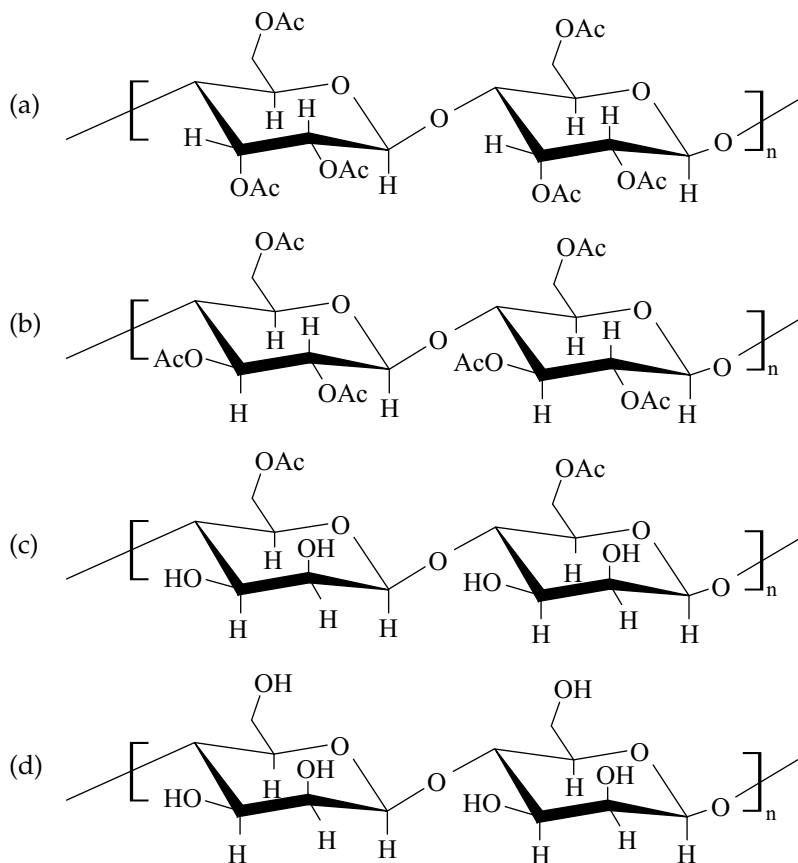
(c) Hexose

(d) Tetrose

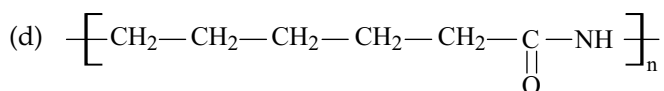
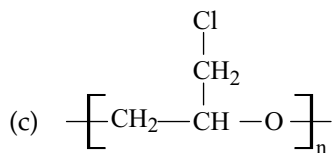
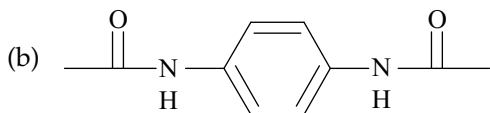
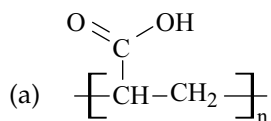




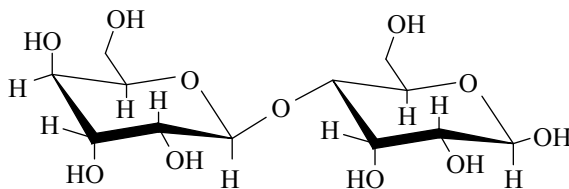
71. Identify structure of cellulose triacetate is



72. Identify the structure which gives Biurate test



73. Identify the correct statement about the following disaccharide



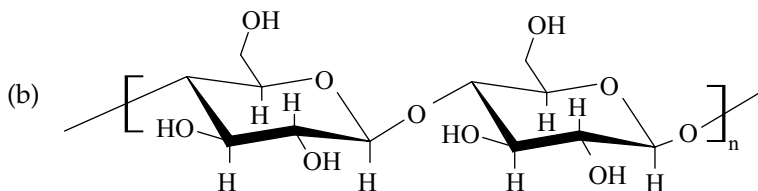
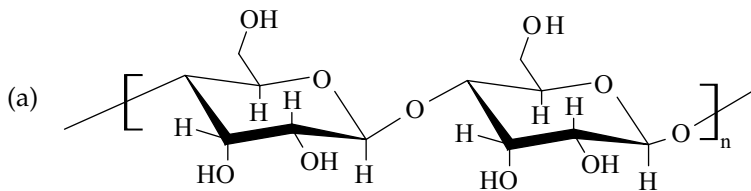
(a) Both rings are pyranose

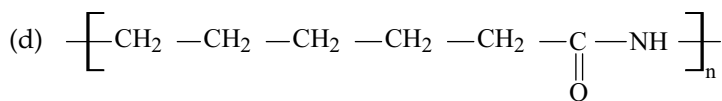
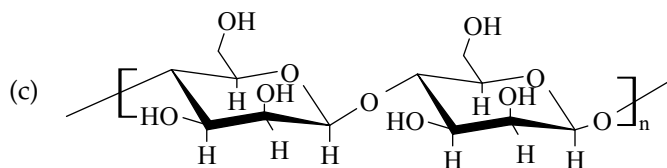
(b) Reducing sugar

(c) Non-reducing sugar

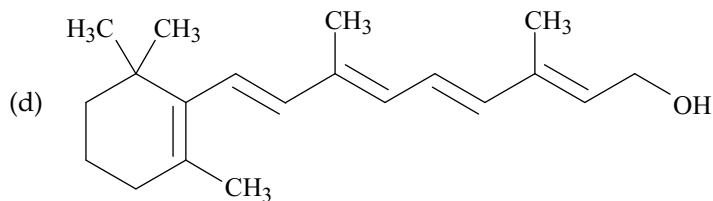
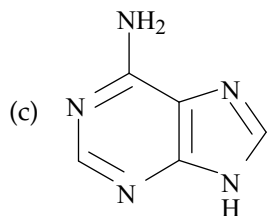
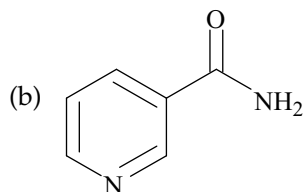
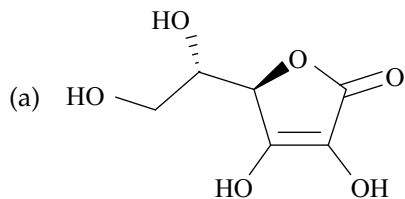
(d) (a) and (b) both are correct

74. Identify the structure of cellulose

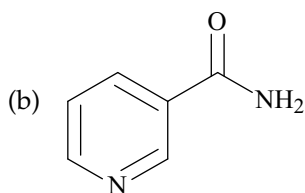
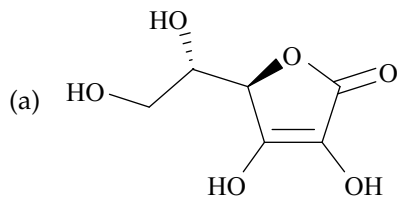


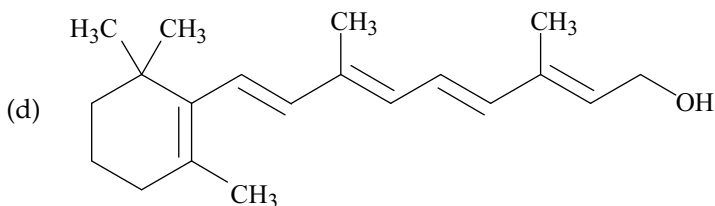
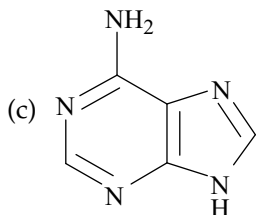


75. Identify the structure of Vitamin C

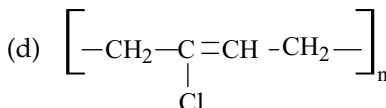
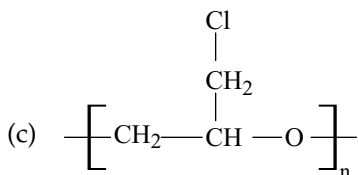
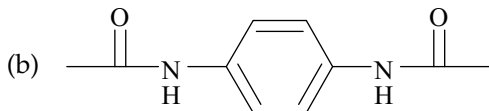
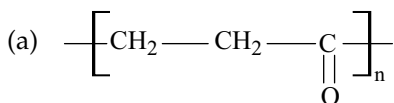


76. Identify the structure of Vitamin A





77. Identify the structure of neoprene



78. Synthetic polymer which resembles natural rubber is

- (a) chloroprene (b) glyptal (c) nylon (d) neoprene

79. D-(+)-Glucose shows mutarotation because

- (a) it undergoes interconversion with D-(-)-fructose
 (b) it undergoes interconversion between its pyranose structure and furanose structure
 (c) it undergoes interconversion between its α -D-(+)-glucopyranose and β -D-(+)-glucopyranose structures via the open-chain structure
 (d) it is dextrorotatory

80. Which of the following is not a biopolymer?

- (a) Nucleic acid (b) Rubber (c) Protein (d) Starch

81. Lysergic acid diethylamide (LSD) is

- (a) antibiotic (b) synthetic fibre
 (c) psychedelic drug (d) sweetening agent

82. PVC polymer can be prepared by which of the monomers?

- (a) $\text{H}_2\text{C}=\text{CHCl}$ (b) $\text{C}_6\text{H}_5\text{CH}=\text{CH}_2$ (c) $\text{CH}_3\text{CH}=\text{CH}_2$ (d) $\text{H}_2\text{C}=\text{CH}_2$

83. Diabetes is detected by testing the urine of a patient usually with

- (a) Benedict's reagent (b) Nessler's reagent
 (c) Fenton's reagent (d) Tollen's reagent

84. Which of the following sets contains only addition polymers?

- (a) Bakelite, PVC, polyethylene (b) Polyethylene, PVC, acrilon
 (c) Buna-S, nylon, polybutadiene (d) Polyethylene, polypropylene, terylene

85. Which of the following is an example of azo dye?
(a) Malachite green (b) Congo red
(c) Martius yellow (d) Indigo
86. Which of the following is not an example of natural polymer?
(a) Leather (b) Silk (c) Wool (d) Nylon
87. Heroin is a derivative of
(a) nicotine (b) morphine (c) cocaine (d) caffeine
88. Which of the following fibres is not made up of polyamides?
(a) Natural silk (b) Artificial silk (c) Wool (d) Nylon
89. An aldose is converted into its next higher homologue by
(a) Amadori rearrangement (b) Wohl's method
(c) Killiani's reaction (d) Mutarotation
90. Nylon is
(a) polyethylene methyl acrylate fibre (b) polyamide fibre
(c) polythene derivative (d) polyester fibre
91. Polythene is a resin obtained by polymerisation of
(a) butadiene (b) ethylene (c) styrene (d) isoprene
92. Which of the following is used as an antipyretic?
(a) Chloroquine (b) Paracetamol
(c) Chloramphenicol (d) LSD
93. Point out the wrong statement
(a) Phenacetin is a very important antibiotic
(b) Penicillin was discovered by A. Flemming
(c) Chloroquine is an antimalarial drug
(d) Ether is an anaesthetic
94. The aqueous solution of carbohydrate gave a dark blue colour with iodine solution. The carbohydrate is
(a) glucose (b) sucrose (c) fructose (d) starch
95. Aspirin is a/an
(a) anaesthetic (b) antipyretic (c) tranquillizer (d) narcotic
96. Which one of the following is a protein fibre?
(a) Rayon (b) Cotton (c) Silk (d) Polyester
97. The monomers used in the manufacture of nylon-6, 6 are
(a) adipic acid and butadiene (b) sebacic acid and hexamethylene diamine
(c) sebacic acid and butadiene (d) adipic acid and hexamethylene diamine

98. The antibiotic used for the treatment of typhoid is
(a) penicillin (b) terramycin
(c) chloramphenicol (d) sulphadiazine
99. A polyamide synthetic polymer prepared by prolonged heating of caprolactum is
(a) glyptal (b) nylon-6 (c) nylon-6, 10 (d) nylon-6, 6
100. Reserine is
(a) hormone (b) antibiotic (c) vitamin (d) tranquillizer
101. Which one of the following is a chromophore group?
(a) $-\text{SO}_3\text{H}$ (b) $-\text{OH}$ (c) $-\text{N}=\text{N}-$ (d) $-\text{NH}_2$
102. The specific rotation of a freshly prepared solution of $\alpha\text{-D-glucose}$ changes from a value of x° to a constant value of y° . The value of x and y are respectively
(a) 112° and 52.5° (b) 19° and 52.5° (c) 52.5° and 19° (d) 52.5° and 112°
103. The basic dye among the following is
(a) congo red (b) antiline yellow (c) alizarin (d) indigo
104. Which of the following is an chain growth polymer?
(a) Glyptal (b) Polypropylene (c) Nylon-6 (d) Nylon-6, 6
105. Starch is a polymer of
(a) ribose (b) glucose (c) lactose (d) fructose
106. The presence of carbohydrate in a sample can be detected by
(a) Benedict's test (b) Tollen's test (c) Biuret test (d) Molish test
107. Which of the following is an example of condensasion polymer?
(a) Nylon-6, 6 (b) Bakelite (c) Buna-S rubber (d) All of these
108. Molisch test is used for the detection of
(a) alkaloid (b) carbohydrates (c) alkyl halide (d) fats
109. In making lemon pickle
(a) citric acid acts as a preservative
(b) lemon oil acts as a preservative
(c) salt plays the same role as sugar in jam
(d) chillies are used to kill all germs and bacteria
110. Neoprene is a polymer of
(a) butadiene (b) isoprene (c) styrene (d) chloroprene
111. Which of the following is an antibiotic?
(a) Paracetamol (b) Aspirin (c) Terramycin (d) Chloroquine

112. Hydrazine as a drug is used in the treatment of
(a) typhoid (b) malaria (c) cholera (d) tuberculosis
113. Arsenic drugs are mainly used in the treatment of
(a) jaundice (b) syphilis (c) typhoid (d) cholera
114. Sorbitol can be obtained by the reduction of
(a) fructose (b) glucose (c) sarbose (d) all of these
115. Bakelite is obtained from phenol by reacting with
(a) chlorobenzene (b) acetaldehyde
(c) formaldehyde (d) acetal
116. Which of the following is an alkaloid?
(a) Cocaine (b) Atropine (c) Nicotine (d) All of these
117. A raw material used in making nylon is
(a) isoprene (b) butadiene (c) adipic acid (d) ethylene
118. Polypropylene is not used in
(a) clothes (b) parachute ropes
(c) ropes (d) heat-resistant plastics
119. The drug which is effective in curing malaria is
(a) analgin (b) aspirin (c) salol (d) quinine
120. A substance which can act both as an antiseptic and disinfectant is
(a) phenol (b) aspirin (c) analgin (d) sodium pentothal
121. Dacron is an example of
(a) polypropylene (b) polyamide (c) polyurethane (d) polyester
122. Which of the following is used as an anaesthetic?
(a) CO_2 (b) N_2O (c) CH_4 (d) N_2
123. Dettol consists of
(a) cresol + ethanol (b) chloroxylenol + terpeneol
(c) xylenol + terpeneol (d) none of the above
124. The pupils of eyes are dilated with a very dilute solution of an alkaloid which is
(a) atropine (b) adrenaline (c) equanil (d) ephedrine
125. Orlon is a unit of
(a) vinylcyanide (b) acrolein (c) glycol (d) isoprene
126. Penicillin was first discovered by
(a) A. Noble (b) L. Pasteur (c) G. Thompson (d) A. Fleming

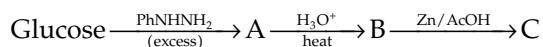
127. The substances which relieve anxiety, reduce mental tension and induce sleep are called
(a) tranquillizers (b) antipyretics (c) analgesics (d) anaesthetics
128. To which class of dyes does phenolphthalein belong?
(a) Phthalein dyes (b) Nitro dyes
(c) Triphenylmethane dyes (d) Azo dyes
129. The poisonous gas evolved in Bhopal tragedy was
(a) CH_3CN (b) CH_3NCO (c) COCl_2 (d) CO
130. (+)-Sucrose is made up of
(a) L-fructose and L-glucose (b) D-glucose and L-fructose
(c) D-fructose and L-glucose (d) D-glucose and L-fructose
131. Which of the following sugars is obtained from nature, usually in a levorotatory form?
(a) Maltose (b) Fructose (c) Sucrose (d) Glucose
132. An example of natural biopolymer is
(a) rubber (b) nylon (c) teflon (d) DNA
133. In the ring structure of fructose, the anomeric carbon is
(a) C-6 (b) C-5 (c) C-2 (d) C-1
134. Monosaccharides are assigned D- or L-configuration depending upon whether its standard Fischer projection formula has a right or left orientation respectively of the hydroxy group at the
(a) anomeric carbon (b) bottom most asymmetric carbon
(c) carbon number 2 (d) topmost asymmetric carbon
135. Teflon is a polymer of the monomer
(a) $\text{CHCl} = \text{CHCl}$ (b) $\text{CHF} = \text{CHCl}$
(c) $\text{CHF} = \text{CH}_2$ (d) $\text{F}_2\text{C} = \text{CF}_2$
136. Cane sugar and invert sugar are
(a) levorotatory and dextrorotatory respectively
(b) dextrorotatory and levorotatory respectively
(c) both levorotatory
(d) both dextrorotatory
137. The major compound of sugar present in honey is
(a) lactose (b) glucose (c) invert sugar (d) sucrose
138. Which one of the following statements is wrong?
(a) PVC stands for polyvinyl chloride
(b) Buna-S stands for natural rubber
(c) PTFE stands for teflon
(d) PMMA stands for polymethyl methyl acrylate

139. Cellulose is a linear polymer of
(a) α -D-fructose (b) β -D-glucose (c) α -D-glucose (d) β -D-fructose
140. Aldoses can be differentiated from ketoses by
(a) periodic acid (b) Tollen's reagents (c) bromine water (d) both (a) and (c)
141. Which of the following sugars exhibits mutarotation?
(a) Fructose (b) Glucose (c) Lactose (d) All of these
142. Heating of rubber with sulphur is known as
(a) vulcanisation (b) bessemerisation (c) galvanisation (d) sulphonation
143. Which of the following is the C-2 epimer of D-glucose?
(a) D-Fructose (b) L-Glucose (c) D-Galactose (d) D-Mannose
144. Glucose reacts with an excess of phenylhydrazine to form
(a) glucosone (b) glucose phenylhydrazone
(c) glucosazone (d) fructose phenylhydrazone
145. Which of the following is a natural polymer?
(a) Bakelite (b) Polythene (c) Buna-S (d) Protein
146. Which of the following pairs of sugars form identical osazone derivative?
(a) D-(+)-Glucose and D-(-)-arabinose (b) D-(+)-Glucose and D-(-)-fructose
(c) D-(+)-Glucose and (+)-maltose (d) D-(+)-Glucose and (+)-lactose
147. Which of the following sugars forms an osazone derivative which is identical with that of D-glucose?
(a) D-Mannose (b) D-Galactose (c) D-Fructose (d) Both (a) and (c)
148. Upon hydrolysis lactose breaks down into
(a) glucose and arabinose (b) glucose and fructose
(c) glucose and galactose (d) glucose and mannose
149. Cellulose is a linear polymer of
(a) β -glucose (b) α -fructose (c) α -glucose (d) amylose
150. On hydrolysis of starch, we finally get
(a) sucrose (b) glucose (c) fructose (d) both (b) and (c)
151. D-glucose and D-fructose all form the same osazone derivative because all of them have
(a) the same configuration at C-5
(b) the same constitution at C-1 and C-2
(c) the same constitution
(d) the same constitution and configuration at C-3, C-4, C-5 and C-6 but different constitution and configuration at C-1 and C-2 which becomes identical by osazone formation

152. The fibre obtained by the condensation of hexamethylene diamine and adipic acid is
(a) nylon-6, 6 (b) dacron (c) rayon (d) teflon
153. In the formation of osazone derivatives of aldohexoses and ketohexoses, the carbon atom(s) that participate(s) in the reactions is
(a) C-1 and C-2 (b) C-2 and C-3 (c) C-1 (d) C-2
154. In an aqueous solution of D-glucose the percentages of α - and β -anomer at the equilibrium condition are respectively
(a) 20% and 80% (b) 80% and 20% (c) 36% and 64% (d) 64% and 36%
155. Which of the following is a polysaccharide?
(a) Cellobiose (b) Polyethene (c) Cellulose (d) Nylon
156. Which one of the following is a thermosetting polymer?
(a) SBR (b) Nylon-6, 6 (c) Bakelite (d) Nylon-6
157. Glucose gives positive silver mirror test with ammoniacal silver nitrate because it contains
(a) aldehyde group (b) hydroxy group
(c) ketone group (d) vicinal diol group
158. Which functional groups of glucose interact to form cyclic hemiacetal leading to pyranose structure?
(a) Ketone group at C-2 and hydroxy group at C-5
(b) Aldehyde group and hydroxy group at C-5
(c) Aldehyde group and hydroxy group at C-4
(d) Aldehyde group and hydroxy group at C-6
159. Which of the following reagents may be used to identify glucose?
(a) NaHSO_3 (b) Ammoniacal AgNO_3 solution
(c) CHCl_3 and alcoholic KOH (d) Neutral FeCl_3 solution
160. Fructose reduces Fehling's solution due to the presence of
(a) hydroxy group (b) ketone group
(c) aldehyde group (d) α -hydroxyketone group
161. Natural rubber is a polymer of
(a) butadiene (b) isoprene (c) chloroprene (d) neoprene
162. The formation of furanose structure of fructose involves the interaction of functional groups present at
(a) C-1 and C-4 (b) C-1 and C-5 (c) C-2 and C-5 (d) C-2 and C-6
163. The widely used plastic PVC is a polymerisation product of
(a) $\text{CHCl} = \text{CHCl}$ (b) $\text{H}_2\text{C} = \text{CCl}_2$ (c) $\text{H}_2\text{C} = \text{CH}_2$ (d) $\text{H}_2\text{C} = \text{CHCl}$

164. D-Glucose reacts with anhydrous methyl alcohol in the presence of dry HCl gas to form
- β -methyl D-glucopyranoside
 - α -methyl D-glucopyranoside
 - 2, 3, 4, 5, 6-penta-O-methyl D-glucose
 - both (a) and (b)

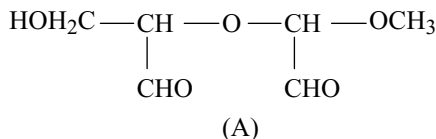
165. Consider the following sequence of reactions



The product (C) is expected to be

- mannose
 - fructose
 - sorbitol
 - arabinose
166. Which of the following fibres is made of polyamides?
- Raon
 - Orlon
 - Nylon
 - Dacron
167. In the multi-step conversion of an aldose into next higher aldose by Kiliani-Fischer synthesis, the reagent employed in the first step is
- HCN
 - NH₂OH
 - Br₂/H₂O
 - C₆H₅NHNH₂
168. Salol can be used as
- antiseptic
 - antipyretic
 - both (a) and (b)
 - none of these
169. When methyl D-glucopyranoside is oxidised with periodic acid, how many moles of the oxidising agent are consumed per mole of the sugar?
- 2
 - 3
 - 4
 - 5
170. The urine of a diabetic patient contains
- glucose
 - fructose
 - sucrose
 - galactose
171. D-Glucose and D-mannose are
- epimers
 - anomers
 - diastereomers
 - both (a) and (c)
172. In which of the following sets do the carbohydrates contain disaccharides only?
- Maltose, lactose, starch
 - Sucrose, lactose, cellulose
 - Maltose, sucrose, lactose
 - Maltose, lactose, mannose
173. When glucose is treated with an excess of HIO₄, the products formed are
- one mole of oxalic acid (COOHCOOH), three moles of HCOOH and one mole of HCHO
 - five moles of HCOOH and one mole of HCHO
 - one mole of glyoxal (CHOCHO), one mole of HCHO and three moles of HCOOH
 - six moles of HCOOH
174. Malisch test is performed for the detection of
- carbohydrates
 - fats
 - alkaloids
 - alkyl halides

175. Methyl D-glucoside on reaction with HIO_4 consumes two moles of the reagent and produces the dialdehyde (A) and one mole of HCOOH .



The result of this reaction proves that glucose has

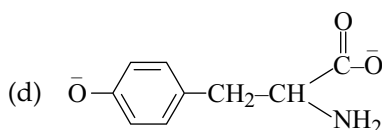
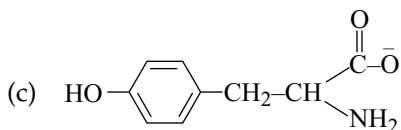
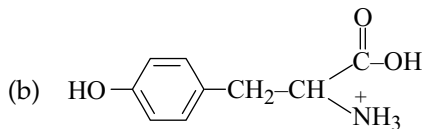
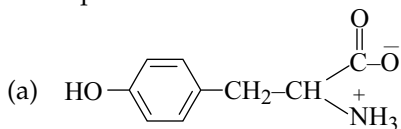
- (a) a four-membered ring structure (b) a furanose structure
(c) an open-chain structure (d) a pyranose structure
176. Plants store their reserve carbohydrates in the form of
(a) starch (b) cellulose (c) glycogen (d) cellobiose
177. Which one is an example of vat dye?
(a) Indigo (b) Alizarin (c) Malachite green (d) Congo red
178. Starch consists of two polymeric units, namely
(a) glycogen and collagen (b) cellulose and cellobiose
(c) amylose and amylopectin (d) inulin and pectin
179. Ampicillin is
(a) an antibiotic (b) an analgesic (c) an antimalarial (d) an antipyretic
180. Charring of sugar is due to
(a) reduction and hydration (b) reduction
(c) dehydration (d) oxidation
181. Isoprene substances are used for making
(a) natural rubber (b) propane (c) petrol (d) liquid fuel
182. Glucose gives silver mirror with ammoniacal silver nitrate because it has
(a) ketone (b) aldehyde (c) ester (d) alcoholic AgNO_3
183. Lactose on hydrolysis gives
(a) glucose + galactose (b) fructose
(c) fructose + glucose (d) glucose
184. Chloroquine is
(a) an antipyretic (b) an antibiotic (c) an antimalarial (d) an analgesic
185. Terylene is a condensation polymer of ethylene glycol and
(a) salicylic acid (b) terephthalic acid
(c) benzoic acid (d) phthalic acid

186. Which of the following groups would you introduce into a dye to make it water soluble?
(a) $-\text{NO}_2$ (b) $-\text{SO}_3\text{H}$ (c) $-\text{Cl}$ (d) $-\text{OH}$
187. Polymer obtained by condensation polymerisation is
(a) PVC (b) teflon
(c) polythene (d) phenol-formaldehyde resin
188. A dye absorbs the wavelength corresponding to blue colour of light. The observed colour will be
(a) blue (b) orange (c) green (d) red
189. The product of addition polymerisation reaction is
(a) polythene (b) glyptal (c) nylon (d) terylene
190. With which of the following cations, alizarin will impart a violet colour on the fabrics?
(a) Cr^{3+} (b) Fe^{3+} (c) Ba^{2+} (d) Al^{3+}
191. Glucose cannot be classified as
(a) a hexose (b) a carbohydrate
(c) an aldose (d) an oligosaccharide
192. In the molecule of D-fructofuranose, the anomeric carbon is
(a) C-5 (b) C-2 (c) C-1 (d) C-6
193. Glucose is treated with excess of HIO_4 . How many moles of HIO_4 will be consumed per mole of the sugar?
(a) 5 (b) 4 (c) 3 (d) 2

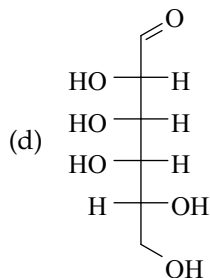
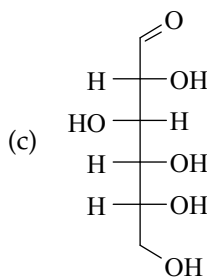
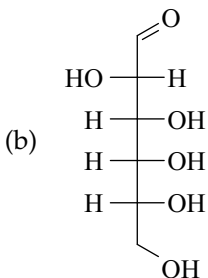
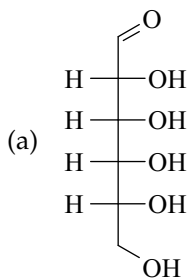
LEVEL 2

Single and Multiple-choice Type

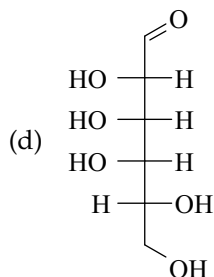
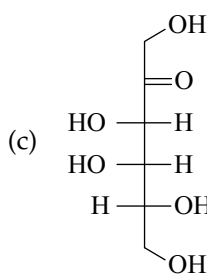
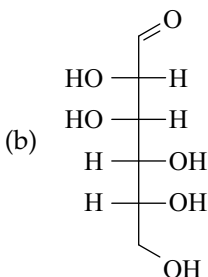
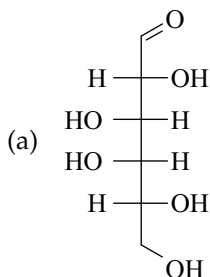
1. Which of the following represents (tyrosine) $\text{HO}-\text{C}_6\text{H}_4-\text{CH}_2-\text{CH}(\text{NH}_2)\text{COOH}$ at a pH = 2 and pH = 12?



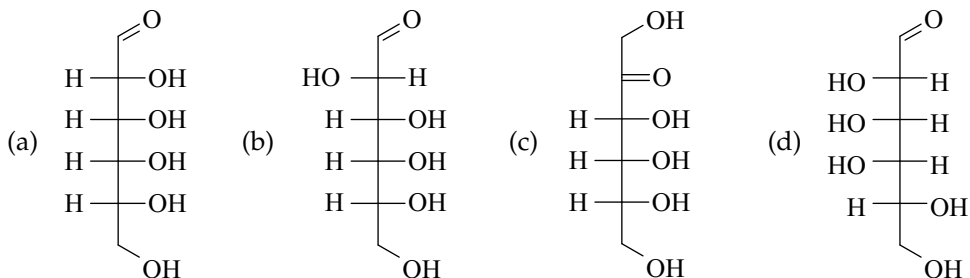
2. Which of the following is/are co-polymer (s)?
 (a) Nylon-66 (b) Bakelite (c) Teflon (d) Polystyrene
3. Which of the following do not undergo hydrolysis?
 (a) Glucose (b) Fructose (c) Cane sugar (d) Maltose
4. Among the following polymers, peptide linkage is present in
 (a) Nylon-66 (b) Protein (c) Nylon-6 (d) Tafflon
5. Identify the compound that gives same Osazone



6. Identify the compound that gives same Osazone



7. Identify the compounds which are inter-convertible in basic medium



8. Among the following polymers, identify the co-polymer

- (a) Nylon-6 (b) Buna-N (c) Buna-S (d) Protein

9. Among the following polymers, identify the co-polymer

- (a) Terylene (b) Glyptal (c) PVC (d) Protein

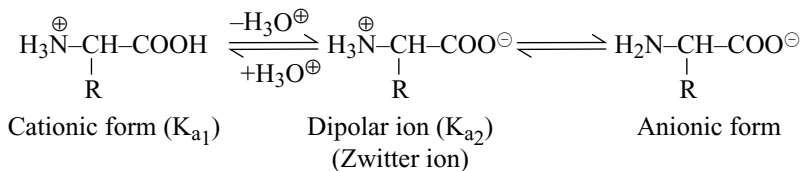
10. Among the following polymers, identify the homopolymer

- (a) Bakelite (b) Dacron (c) Natural rubber (d) Orlon

Comprehension Type

Passage 1

Amino acids contain both a basic group ($-\text{NH}_2$) and an acidic group ($-\text{CO}_2\text{H}$). In the dry solid state, amino acids exist as dipolar ions, a form in which the carboxylic group is present as a carboxylate ion, $-\text{CO}_2^-$ and the amino group is present as an ammonium ion, $-\text{NH}_3^+$. In aqueous solution, an equilibrium exists between the dipolar ion (zwitter ions) and the anionic and cationic forms of an amino acid.



In strongly acidic solutions ($\text{pH} \approx 0$) all amino acids are present primarily as cations, in strongly basic solution they are present as anions. At some intermediate pH, called the isoelectric point (pI), the concentration of the anions and cations is equal. Each amino acid has particular isoelectric point.

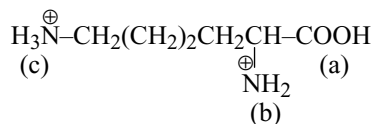
11. For the phenyl alanine, the first and second dissociation constants are given as $\text{pK}_{a1} = 1.8$ and $\text{pK}_{a2} = 9.1$. What will be the pH at isoelectric point?

- (a) 4.5 (b) 5.45 (c) 7 (d) 4.2

12. The formula of lysine is $\text{H}_2\text{N}(\text{CH}_2)_3 \cdot \text{CH}_2\text{---CH---COOH}$ for lysine the value of pK_{a1} , pK_{a2} and pK_{a3} are 2.2, 9.0 and 10.5 respectively. The pH at isoelectric point of lysine is

- (a) 5.6 (b) 7 (c) 6.35 (d) 9.75

13. The dicationic form of the lysine is shown as



Arrange the shown hydrogen in the order of their acidic strength

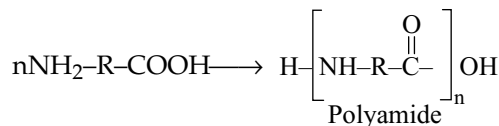
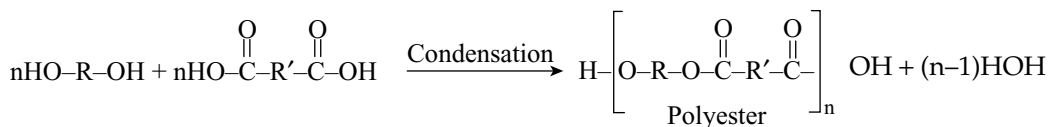
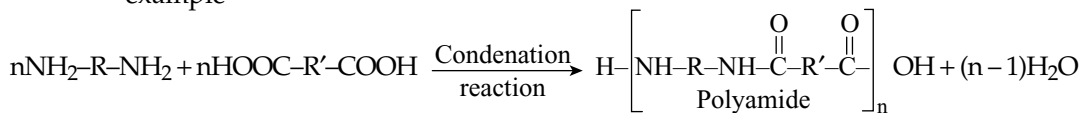
- (a) $a > b > c$ (b) $c > b > a$ (c) $a > b = c$ (d) $b = c > a$

Passage 2

Classification of Polymers

(A) Condensation polymers and condensation polymerisation:

- Condensation polymerisation is also known as step polymerisation.
- For condensation polymerisation, monomers should have at least two functional groups. Both functional groups may be same or different.
- Monomers having only two functional groups always give a linear polymer. For example

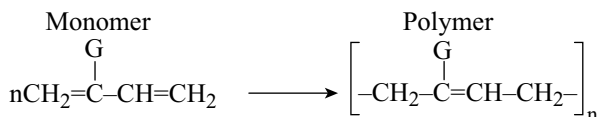


- Condensation polymers do not contain all the atoms initially present in the monomers. Some atoms are lost in the small molecule that is eliminated.
- Monomer having three functional groups always gives cross-linked polymer. Examples are: urea formaldehyde resin, phenol-formaldehyde resin, alkyde resin.

(B) Addition polymer:

- Polymers which are formed by addition reaction are known as addition polymers.
- If monomer is ethylene or its derivative then addition polymer is either linear polymer or branch-chain polymer. Examples are: Polyethylene, polyvinylchloride, polystyrene, polytetrafluoroethylene, polyvinyl cyanide, etc.
- If monomer is 1,3-butadiene or 2-substituted-1,3 butadiene $\left(\text{CH}_2=\underset{\text{G}}{\text{C}}-\text{CH}=\text{CH}_2 \right)$ then

polymer is always branch chain polymer which is always an elastomer. Example are:



- (i) $G = H$, 1,3-butadiene (i) Polybutadiene
 (ii) $G = CH_3$, 2-methyl-1,3-butadiene (ii) Polyisoprene
 (iii) $G = Cl$, 2-chloro-1,3-butadiene or chloroprene (iii) Polychloroisoprene (Neoprene)
 (iv) Addition polymers retain all the atoms of the monomer units in the polymer.
 (v) Addition polymerisation takes place in three steps: Initiation, chain propagation and chain termination.
 (vi) Addition polymers are also known as chain growth polymers.
14. Which one of the following is a monomer of condensation polymer?
 (a) $CH_3(CH_2)_5COOH$ (b) $HO-CH_2-CH_2-OH$
 (c) $CH_3-CH=CH-NH_2$ (d) $CH_3-CH=CH-CH_2-CONH_2$
15. Natural rubber is which type of polymer?
 (a) Condensation (b) Addition (c) Coordinate (d) None of these
16. Among the following, identify the condensation polymer
 (a) Nylon-6 (b) Buna-N (c) Polystyrene (d) Nylon-6-6

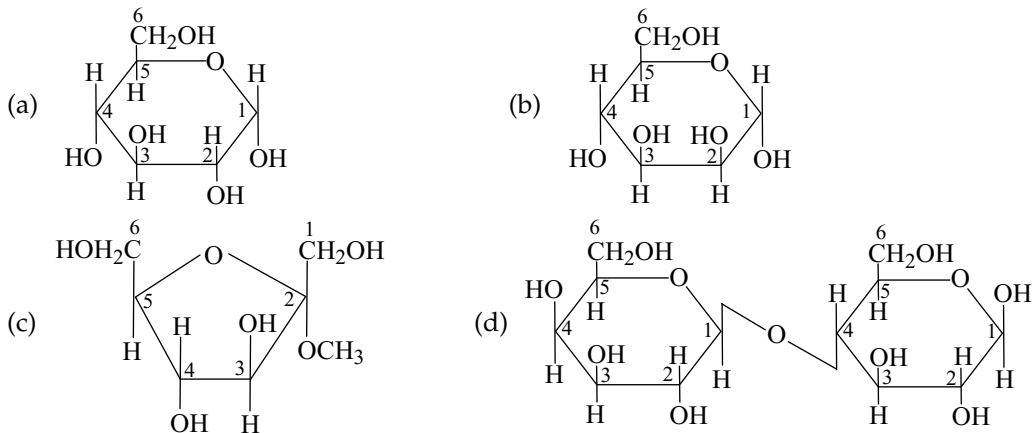
Passage 3

The carbohydrates in which the following properties are found are called reducing sugars.

The reduction of Tollen's reagent, Fehling's solution, Benedict's solution.

All monosaccharides are reducing sugars. Contrary to this the carbohydrates in which the above properties are not found are called non-reducing sugars. For example, sucrose, starch, cellulose, etc. The disaccharides, in which carbonyl group of any one monosaccharide from both the monosaccharides, is present as hemiacetal or ketal and does not involve in glycosidic linkage formation, are reducing. For example, maltose and lactose. Two diastereomeric monosaccharides are called epimers when their configuration differs only at stereogenic centre. Monosaccharides form glycoside with alcohol in the presence of an acid. All monosaccharides show mutarotation.

17. Which of the following is a non-reducing sugar?



18. Which one of the following is C-2 epimers?
 (a) Glucose and galactose (b) Glucose and mannose
 (c) Glucose and fructose (d) Galactose and fructose
19. Which of the following statements about lactose is not correct?
 (a) It does not reduce Tollen's reagent
 (b) It is a disaccharide. On hydrolysis, it produces two monosaccharides
 (c) It possesses glycosidic linkage
 (d) All of them

Passage 4

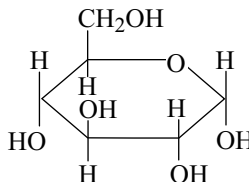
Monosaccharides have —CHO (or C=O) and —OH groups, so they undergo usual oxidation and reduction. Further, monosaccharides form osazone when treated with excess of phenylhydrazine (3 equivalents). In osazone formation only the first two carbon atoms are involved. Thus monosaccharides having identical configuration on rest of C atoms except first two will form same osazone. The same is the case with glucose and fructose.

A, B and C are three hexoses and form same osazone D. Compounds A to D behave as shown below:

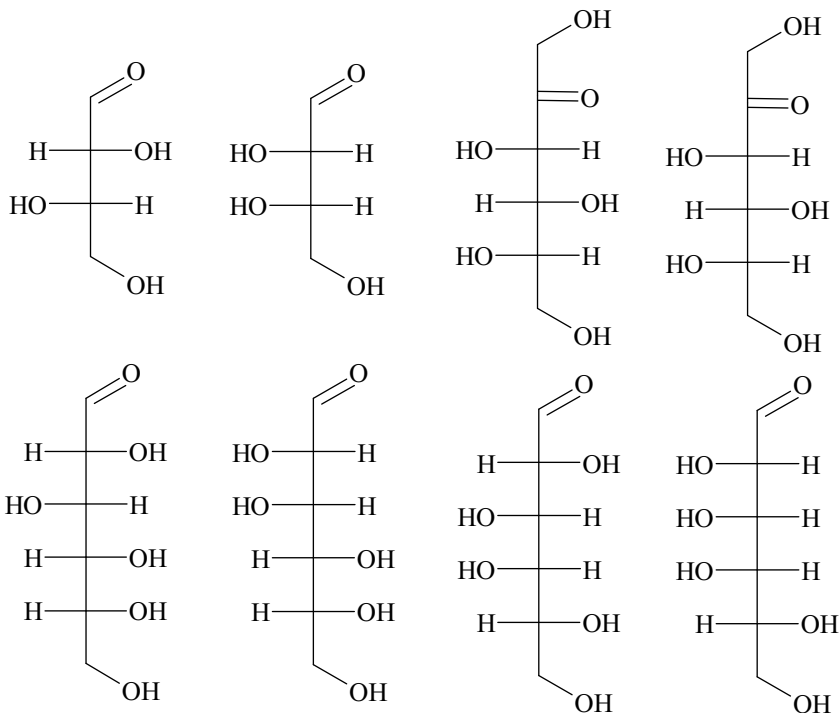
- (i) $\text{D} \xrightarrow{\text{HCl}} \xrightarrow[\text{CH}_3\text{COOH}]{\text{Zn}} \text{D-Fructose}$
 (ii) $\text{A} \xrightarrow{\text{Ni, H}_2} \xrightarrow{\text{HNO}_3} \xrightarrow[\text{H}_3\text{O}^+]{\text{Na-Hg}} \text{B} + \text{C}$
 (iii) $\text{B} \xrightarrow{\text{HNO}_3} \text{Optically active glyceric acid}$
 (iv) $\text{C} \xrightarrow{\text{HNO}_3} \text{Optically inactive glyceric acid}$

20. Compound D is an osazone which can be obtained from
 (a) Only one compound (b) Two compounds
 (c) Three compounds (d) Four compounds
21. Compound A should be
 (a) D-glucose (b) D-fructose (c) L-glucose (d) L-fructose
22. Compound B and C, respectively, are
 (a) D-glucose and D-mannose (b) D-mannose and D-glucose
 (c) D-glucose and L-glucose (d) D-glucose and L-mannose

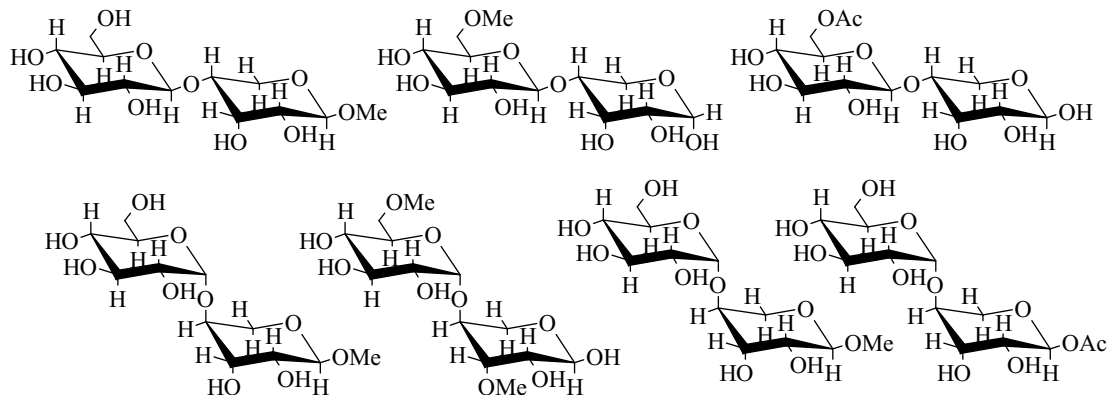
Integer Type

23.  $\xrightarrow[\text{H}_2\text{O}]{\text{Br}_2}$ Compound (A) ; Number of chiral centre in compound A is

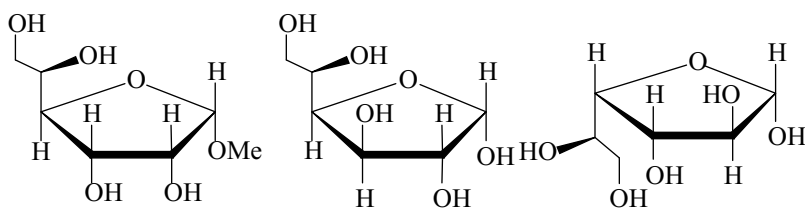
24. Identify the total number of carbohydrates which have D-configuration

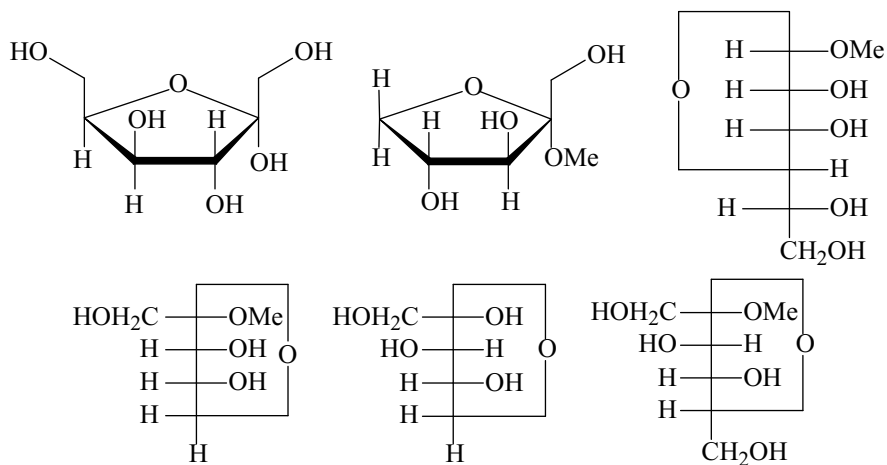


25. Identify the total number of carbohydrates which give mutarotation

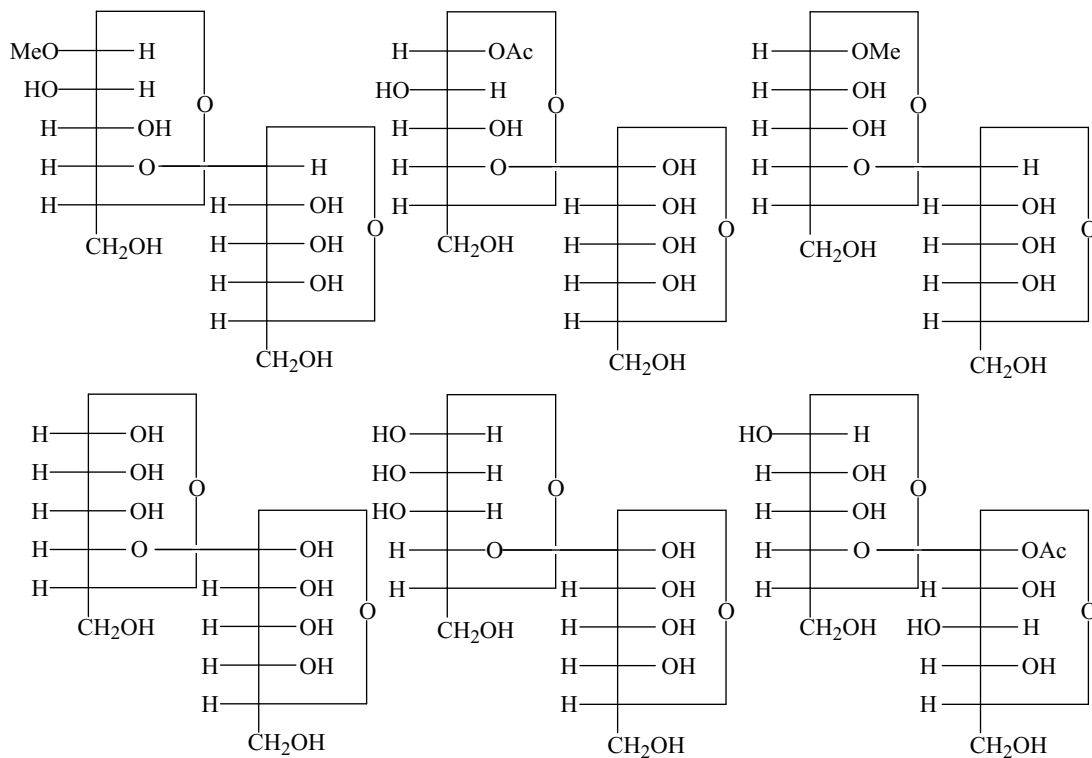


26. Identify the total number of carbohydrates which give mutarotation





27. Identify total number of carbohydrates which give $\oplus$ ve tollen test.



Answer Keys

LEVEL 1

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
c	b	d	c	a	b	d	c	c	d	b	d	a	a	c
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
ab	b	d	c	a	c	d	b	a	c	d	a	b	a	d
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
d	b	a	c	ad	d	c	c	c	ab	a	a	a	c	a
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
c	d	a	d	b	b	c	a	b	a	d	c	a	c	a
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
d	d	a	b	d	a	b	c	b	a	b	b, d	d	b	a
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
d	d	d	c	b	c	a	a	b	b	d	b	c	c	b
91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
b	b	a	d	b	c	d	c	b	d	c	a	b	b	b
106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
d	ab	b	c	d	ab	d	b	b	c	d	c	b	d	a
121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
d	b	b	a	a	d	a	a	b	bd	c	d	c	b	d
136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
a	d	b	c	d	d	a	d	c	d	b	d	c	a	b
151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
d	a	a	c	c	c	a	b	b	d	b	c	d	d	b
166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
c	a	a	a	a	a	c	b	a	d	a	a	c	a	c
181	182	183	184	185	186	187	188	189	190	191	192	193		
a	b	a	c	b	abd	d	b	a	a	d	b	a		

LEVEL 2

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
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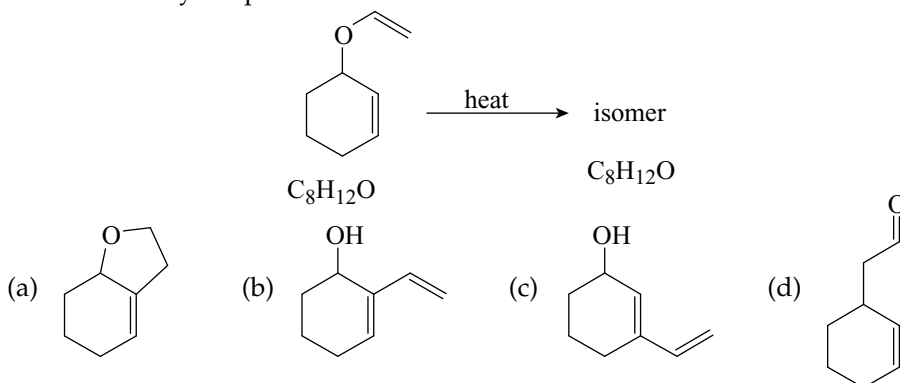
Organic Reaction Mechanisms and Reagents

8

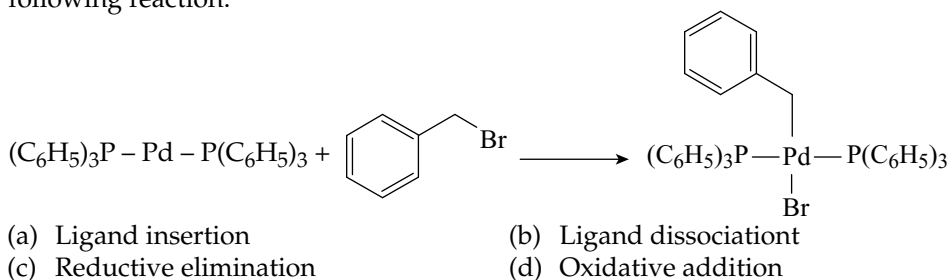
Question Bank

LEVEL 1

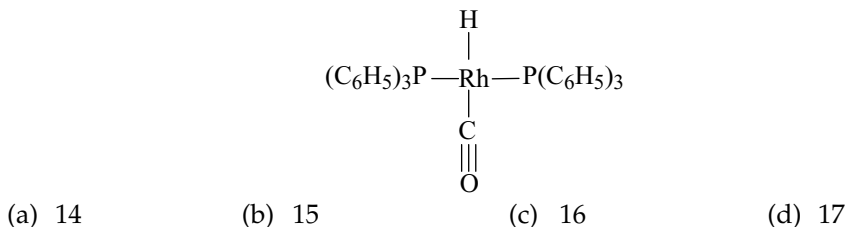
1. When the compound shown below is heated it undergoes a rearrangement to form an isomer. Identify the product.



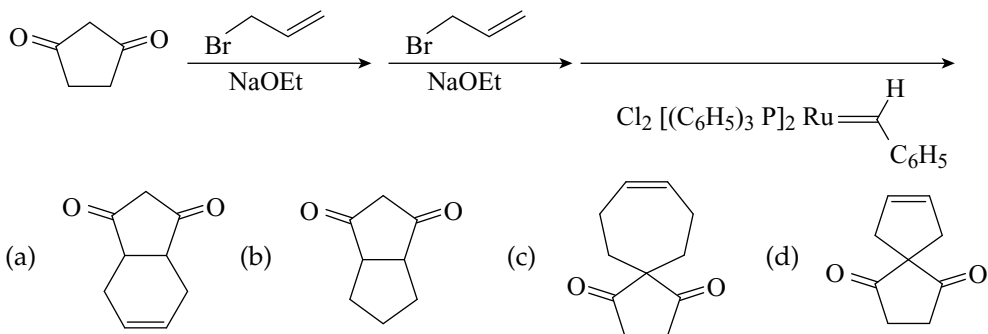
2. Organometallic reactions can be classified into fundamental reaction types. Classify the following reaction.



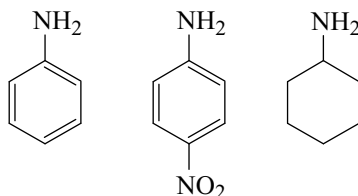
3. What is the electron count for the following transition metal complex?



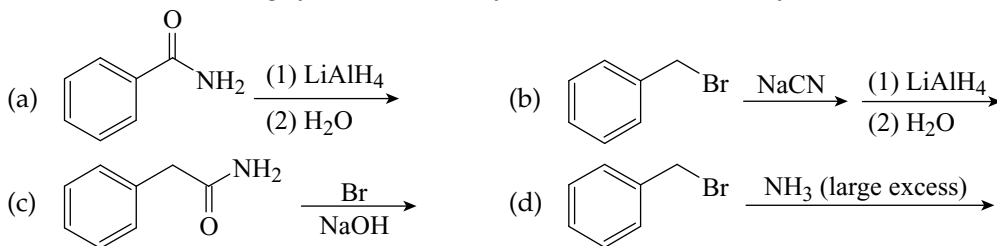
4. Predict the product of the following reaction sequence.



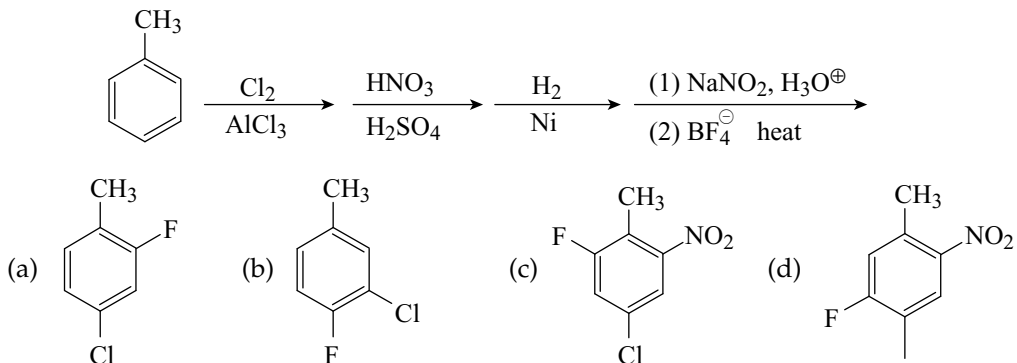
5. Consider the relative basicity of these three amines. Which statement is true?



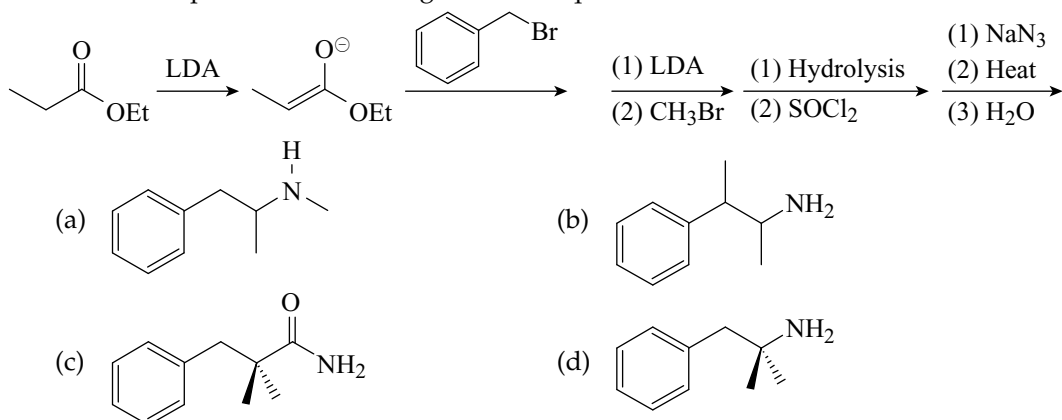
- (a) Cyclohexylamine is the strongest base and aniline is the weakest base
 (b) Cyclohexylamine is the strongest base and 4-nitroaniline is the weakest base
 (c) Aniline is the strongest base and cyclohexylamine is the weakest base
 (d) Nitroaniline is the strongest base and cyclohexylamine is the weakest base
6. Which of the following syntheses of benzylamine is the least likely to work?



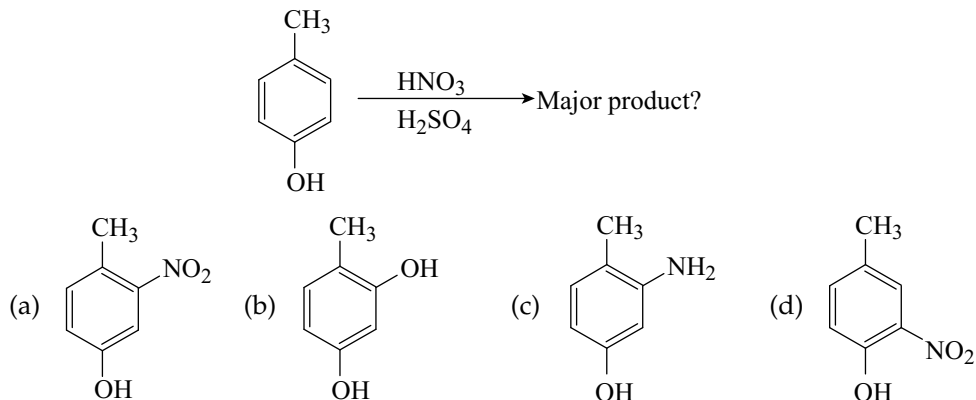
7. Predict the product of the following reaction sequence.



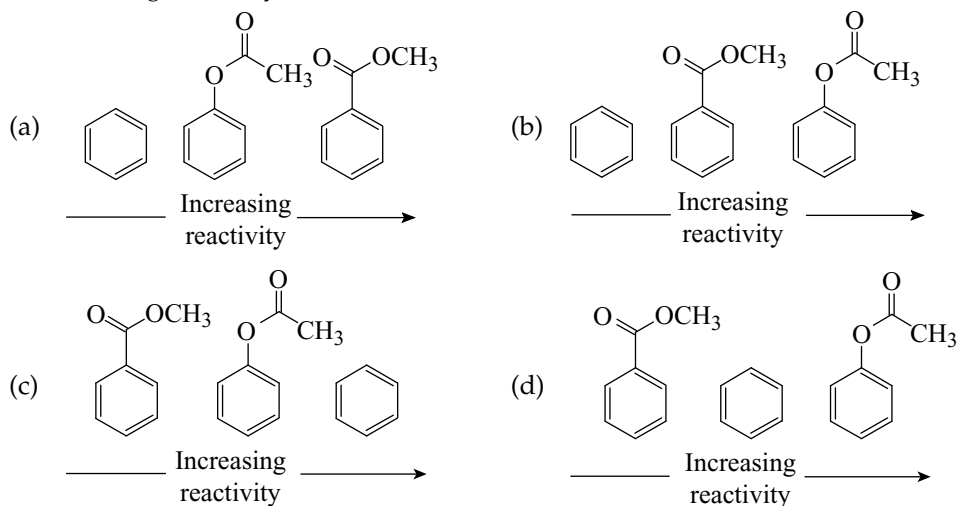
8. Predict the product of following reaction sequence.



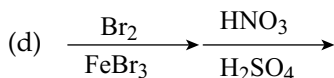
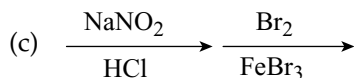
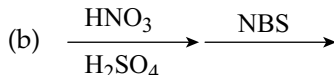
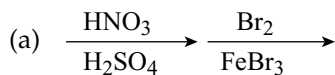
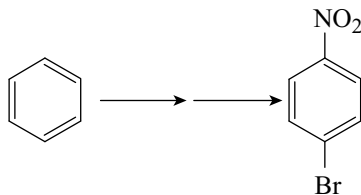
9. Choose the major product of the following reaction.



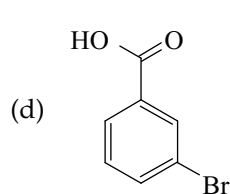
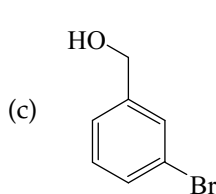
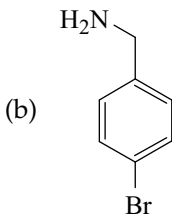
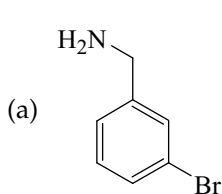
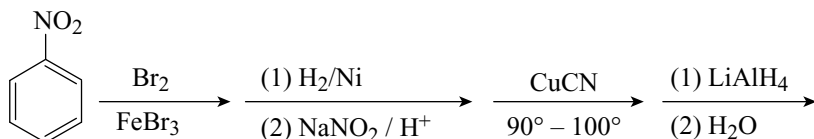
10. Choose the answer that has the following compounds arranged correctly with respect to increasing reactivity with $\text{Br}_2/\text{FeBr}_3$.



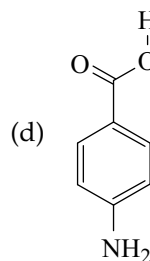
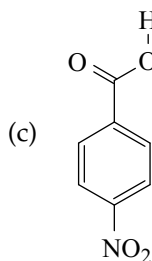
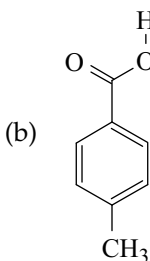
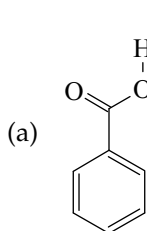
11. Choose the reaction sequence that could be used to perform the following transformation.



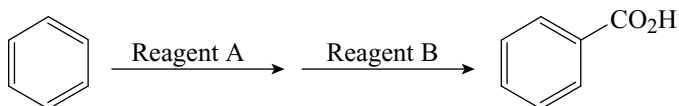
12. Predict the major product of the following reaction



13. Which of the following compounds is the most acidic? (lowest pK_a)

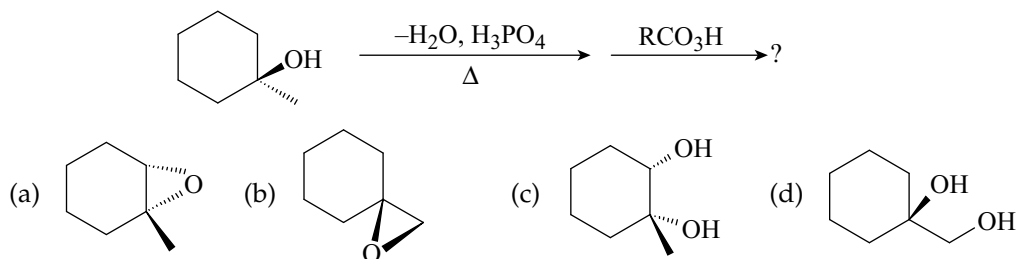


14. What could be reagents A and B for the following reactions?

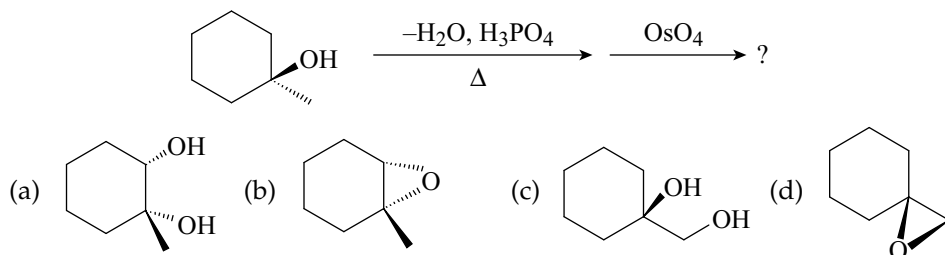


- (a) Reagent A: CH₃CH₂Cl / AlCl₃; reagent B: Na₂Cr₂O₇, heat
 (b) Reagent A: CH₃COCl / AlCl₃; reagent B: Na₂Cr₂O₇, heat
 (c) Reagent A: HNO₃, H₂SO₄; reagent B: RCO₃H, heat
 (d) Reagent A: CH₃COCl / AlCl₃; reagent B: H₂, Ni, heat

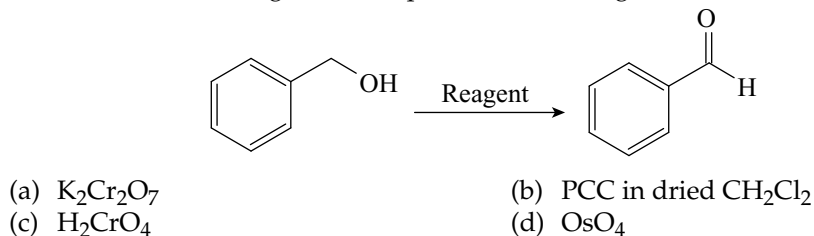
15. What could be the major product from the following reactions?



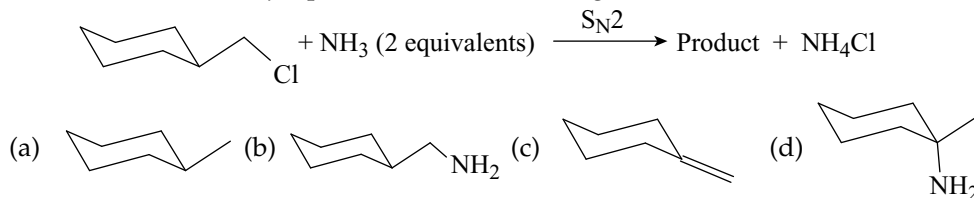
16. What could be the major product from the following reactions?



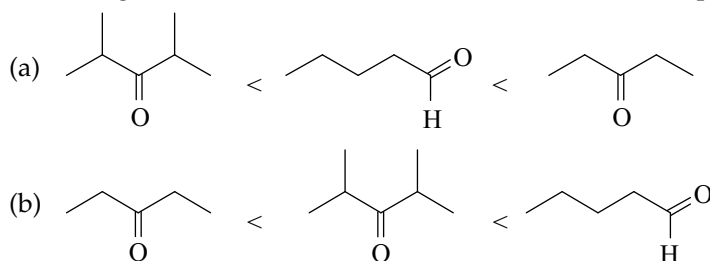
17. What could be the reagent to complete the following reaction?

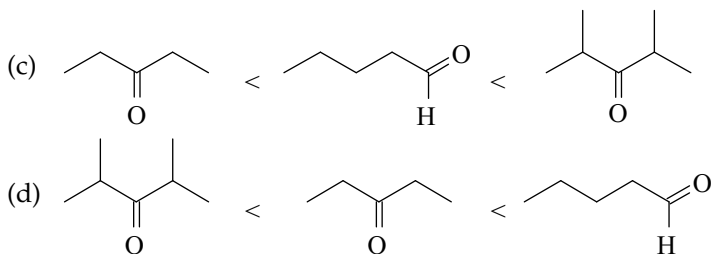


18. What could be the major product for the following reaction?

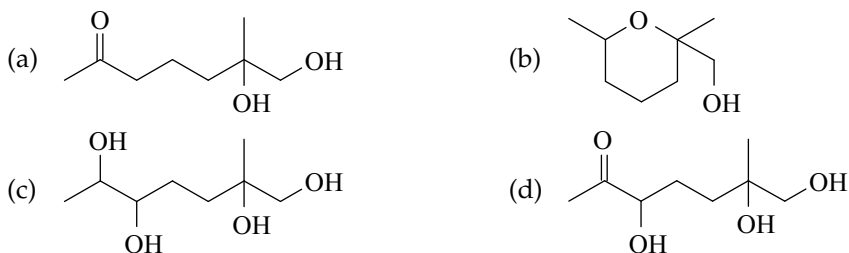
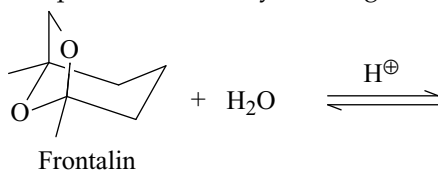


19. Choose order that has the following compounds correctly arranged with respect to increasing rate of reaction with LiAlH_4 (most reactive compound on the right).

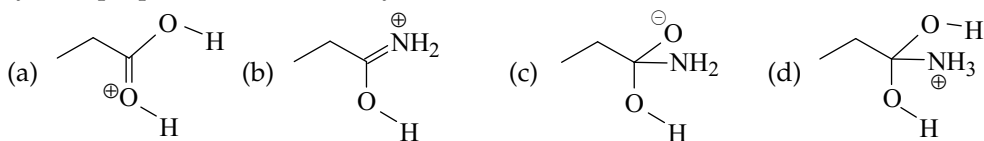




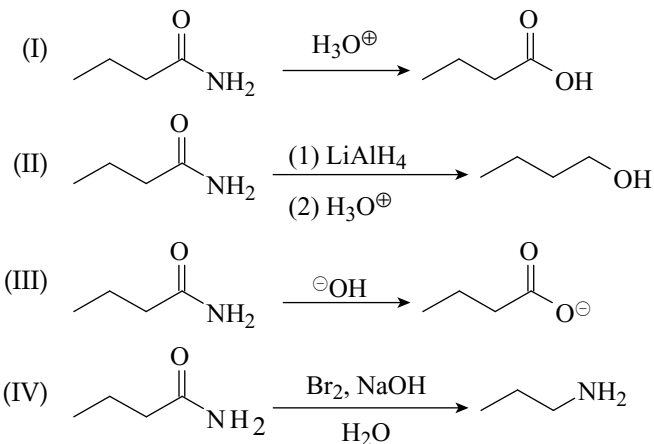
20. Choose the following compound that will cyclize to give the pheromone frontalin.



21. Choose the species that does not represent an intermediate in the acid-catalyzed hydrolysis of propionamide to carboxylic acid.

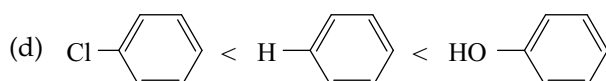
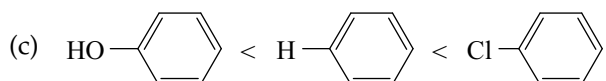
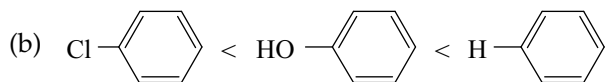
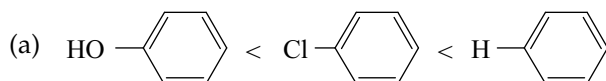


22. Choose the reaction(s) that will not proceed as shown hereunder.

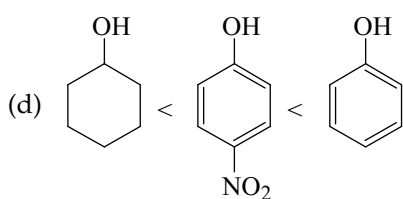
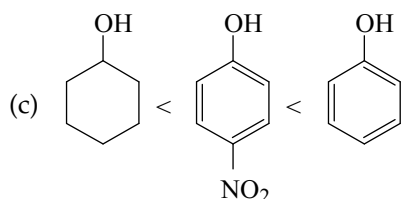
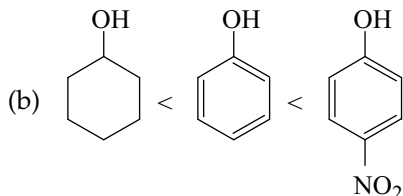
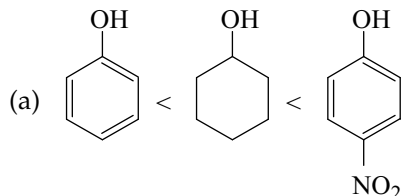


- (a) Only (I) (b) Only (II) (c) Only (III) (d) Only (IV)

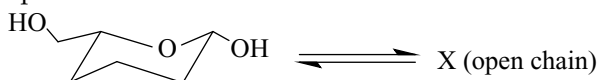
23. Choose the order that has the following aromatic compounds correctly arranged with respect to increasing reactivity towards $\text{Br}_2/\text{FeBr}_3$



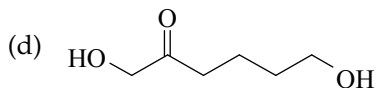
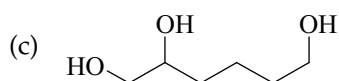
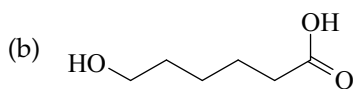
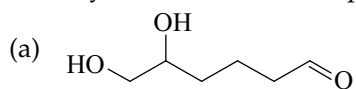
24. Choose the order that has the following alcohols correctly arranged with respect to increasing acidity.



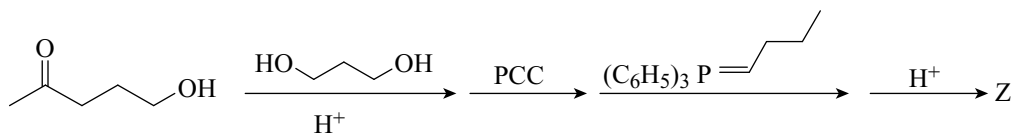
25. The compound shown below is a cyclic hemiacetal. It is in equilibrium with an acyclic open chain compound X.



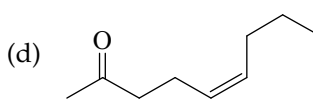
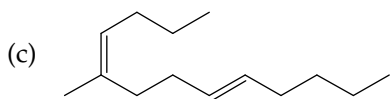
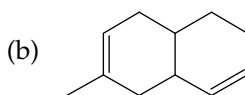
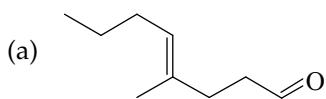
Identify the structure of compound X.



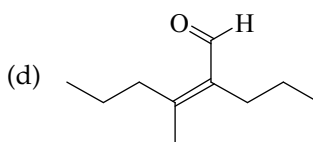
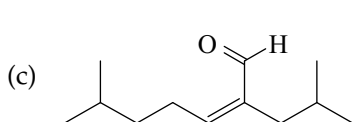
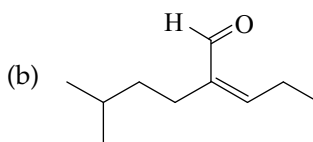
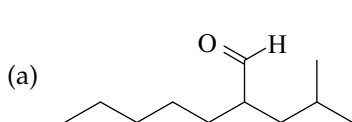
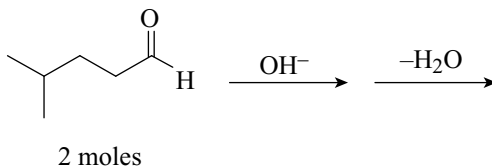
26. The reaction sequence shown below gives compound Z as the main product.



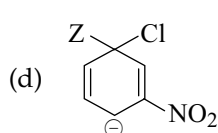
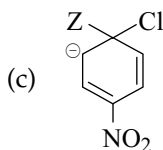
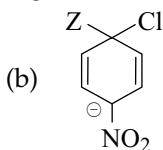
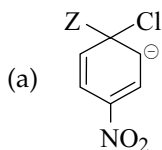
Identify the structure of compound Z.



27. Predict the product of the following aldol condensation.

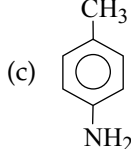
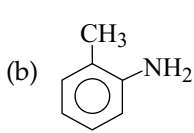
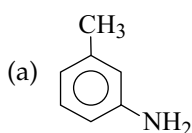


28. Which of the following structure is more stable?



29. Predict the nature of P in the following reaction

P (Major product)



(d) All the three

30. Which of the following is most reactive towards aqueous HBr?

(a) 1-Phenyl-2-propanol
(c) 3-Phenyl-1-propanol

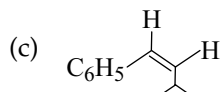
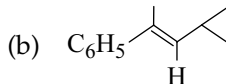
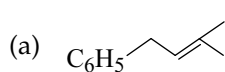
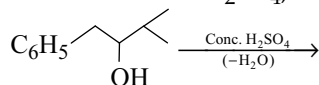
(b) 1-Phenyl-1-propanol
(d) 2-Phenyl-1-propanol

31. Ethylbenzene when treated with chlorine in presence of light mainly gives

(a) β -phenylethyl chloride
(c) o-chloroethyl benzene

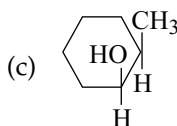
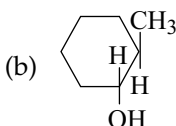
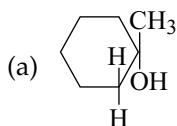
(b) α -phenylethyl chloride
(d) o-and p-chloroethylbenzene

32. The following alcohol is treated with Conc. H_2SO_4 , the major product obtained is



(d) All the three will be formed in equal amounts

33.  $\xrightarrow[\text{(ii) H}_2\text{O}_2, \text{OH}^-]{\text{(i) B}_2\text{H}_6}$ X. The compound X is



(d) Both (b) and (c)

34. Give the nature of A and B in the given reaction



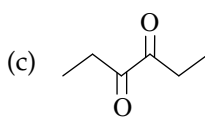
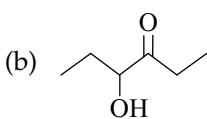
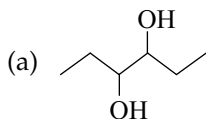
(a) A and B both are $(\text{CH}_3)_2\text{C}=\text{CH}_2$

(b) A and B, both are $(\text{CH}_3)_2\text{CO} + \text{CH}_2\text{O}$

(c) A is $(\text{CH}_3)_3\text{COH}$, while B is $(\text{CH}_3)_2\text{C}=\text{CH}_2$ or $(\text{CH}_3)_2\text{CO}$

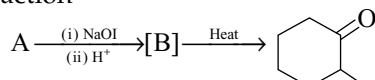
(d) A and B, both are $(\text{CH}_3)_3\text{COH}$, i.e., there is no reaction

35. Which of the following is liable to be oxidized by periodic acid?

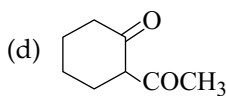
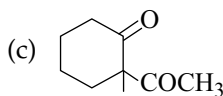
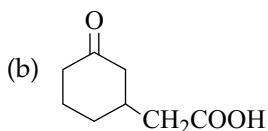
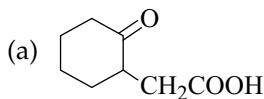


(d) All of these

36. From the given set of reaction



starting compound A corresponds to



37. Methanoic acid is heated with Conc. H_2SO_4 , to form

(a) CO

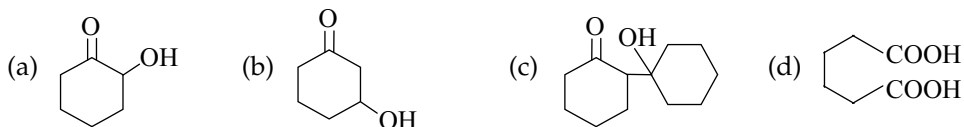
(b) CO_2

(c) CH_4

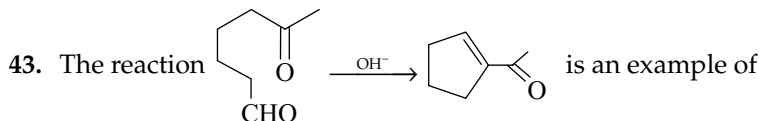
(d) $(\text{COOH})_2$

38. When ethane-1,2-dioic acid is heated with Conc. H_2SO_4 , it gives
 (a) $\text{CO} + \text{HCOOH}$ (b) $\text{CO}_2 + \text{HCOOH}$
 (c) $\text{CO} + \text{CO}_2 + \text{HCOOH}$ (d) $\text{CO} + \text{CO}_2 + \text{H}_2\text{O}$
39. When sodium formate is heated with soda lime, we get
 (a) CH_4 (b) H_2 (c) sodium oxalate (d) no action
40. Sodium formate is heated at 360°C to give
 (a) CO (b) CO_2 (c) sodium oxalate (d) no action

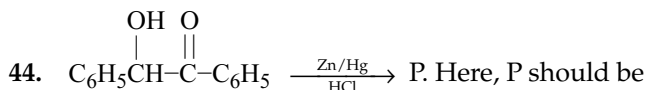
41. When cyclohexanone is treated with Na_2CO_3 solution, we get

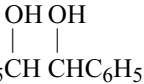
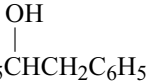


42. Which of the following reagent reacts in different ways with CH_3CHO , HCHO and $\text{C}_6\text{H}_5\text{CHO}$?
 (a) Fehling's solution (b) $\text{C}_6\text{H}_5\text{NHNH}_2$
 (c) Ammonia (d) HCl



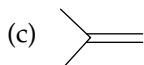
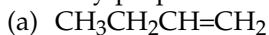
- (a) Oxidation reaction (b) Reduction
 (c) Both (d) Aldol condensation



- (a)  (b) 
 (c) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{C}_6\text{H}_5$ (d) $\text{C}_6\text{H}_5\text{CH}=\text{CHC}_6\text{H}_5$

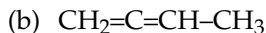
45. Nitrobenzene can be reduced to aniline by
 (I) H_2/Ni (II) Sn/HCl (III) Zn/NaOH (IV) LiAlH_4
 (a) (I), (II) and (III) (b) (I) and (II) (c) (I), (II) and (IV) (d) only (II)
46. 1-Methylcyclopentene can be converted into 2-methylcyclopentanol by
 (a) acid-catalyzed hydration
 (b) hydroboration oxidation
 (c) epoxide formation followed by reduction with LiAlH_4
 (d) oxymercuration-demercuration

47. 2-Methylpropanol-2 can be obtained by the acid-catalyzed hydration of



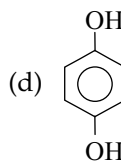
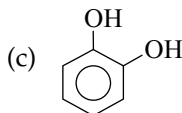
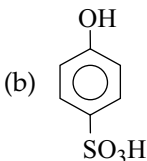
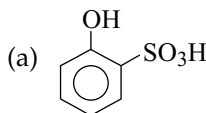
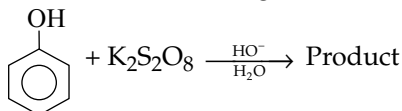
(d) Either of the three

48. Predict the nature of P in the following reaction $\text{CH}_3\text{C}\equiv\text{CCH}_3 \xrightarrow[\text{heat}]{\text{NaNH}_2/\text{inert solvent}} \text{P}$

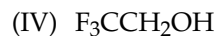
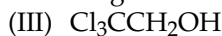
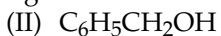
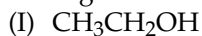


(d) No reaction

49. Identify the nature of product in the following reaction



50. Arrange the following alcohols in order of increasing ease of dehydration



(a) (II) < (I) < (IV) < (III)

(b) (IV) < (III) < (II) < (I)

(c) (IV) < (III) < (I) < (II)

(d) (II) < (I) < (III) < (IV)

51. 1,2-Diethylbenzene on ozonolysis gives.....different products

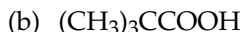
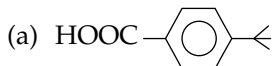
(a) 1

(b) 2

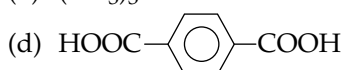
(c) 3

(d) 4

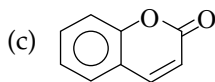
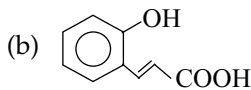
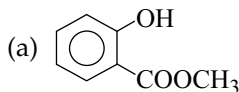
52. $\xrightarrow{\text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4}$ Z. Here, Z is



(c) Both (a) and (b)



53. When o-hydroxybenzaldehyde is heated with ethanoic anhydride in the presence of sodium ethanoate, compound formed during the reaction is



(d) Both (b) and (c)

54. A new carbon-carbon bond is formed in

(I) Aldol condensation

(II) Kolbe's reaction

(III) Reimer-Tiemann reaction

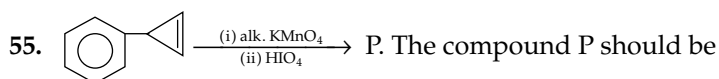
(IV) Wurtz Fittig reaction

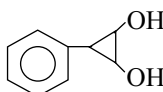
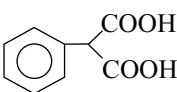
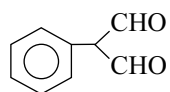
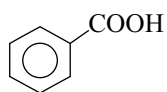
(a) (I), (III)

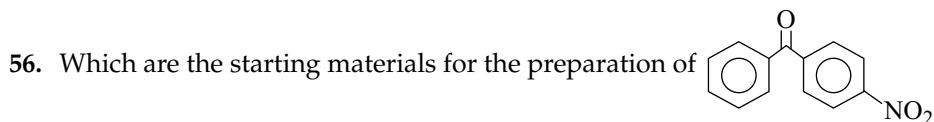
(b) (II), (III)

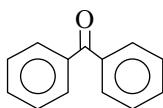
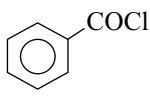
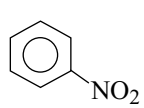
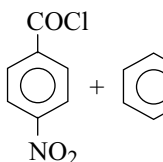
(c) (I), (III), (IV)

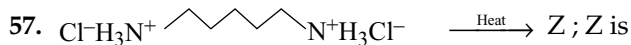
(d) All of these

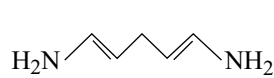
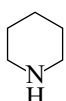
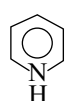


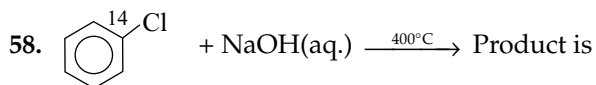
- (a)  (b)  (c)  (d) 

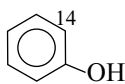
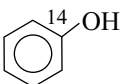


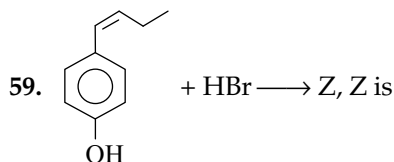
- (a)  $\xrightarrow[\text{Conc. H}_2\text{SO}_4]{\text{Conc. HNO}_3}$ (b)  +  $\xrightarrow[\text{AlCl}_3]{\text{Anhydrous}}$
- (c)  +  $\xrightarrow[\text{AlCl}_3]{\text{Anhydrous}}$ (d) Any of the three

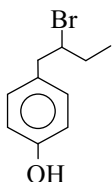
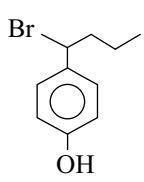
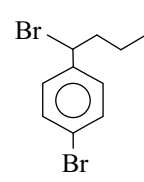
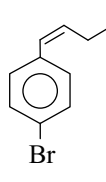


- (a)  (b) 
- (c)  (d) 



- (a)  (b)  (c) Both (a) and (b) (d) No reaction



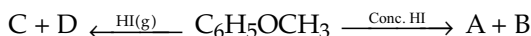
- (a)  (b) 
- (c)  (d) 



Predict the nature of product and the type of reaction involved in their formation.

- (a) Me_2CHI and Me_3COH , formed by $\text{S}_{\text{N}}1$ reaction
- (b) Me_2CHI and Me_3Cl , formed by $\text{S}_{\text{N}}1$ reaction
- (c) Me_2CHI and Me_3COH , formed by $\text{S}_{\text{N}}2$ reaction
- (d) Me_2CHOH and Me_3Cl , formed by $\text{S}_{\text{N}}2$ reaction

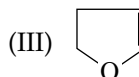
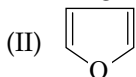
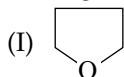
61. Anisole is treated with HI under two different conditions



The nature of A to D will be

- (a) A and B are CH_3I and $\text{C}_6\text{H}_5\text{OH}$, while C and D are CH_3OH and $\text{C}_6\text{H}_5\text{I}$
- (b) A and B are CH_3OH and $\text{C}_6\text{H}_5\text{I}$, while C and D are CH_3I and $\text{C}_6\text{H}_5\text{OH}$
- (c) Both A and B as well as both C and D are CH_3I and $\text{C}_6\text{H}_5\text{OH}$
- (d) A and B are CH_3I and $\text{C}_6\text{H}_5\text{OH}$, while there is no reaction in the second case.

62. Arrange the following in decreasing order of solubility in water

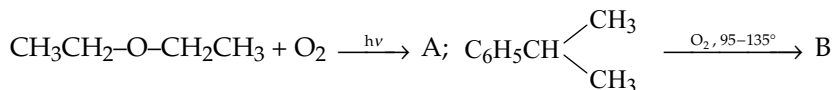


- (a) (I) > (III) > (II) (b) (III) > (II) > (I) (c) (II) > (III) > (I) (d) All are equally soluble

63. The ethereal linkage ($-\text{C}-\text{O}-\text{C}-$) is cleaved by

- (a) HBr (b) HNO_3 (c) Both (a) and (b) (d) None

64. Predict the compounds A and B in the following reactions



- (a) $\text{CH}_3\text{CH}_2-\text{O}-\text{O}-\text{CH}_2\text{CH}_3$ and $\text{C}_6\text{H}_5-\overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}}-\text{O}-\text{OH}$ respectively
- (b) $\text{CH}_3\text{CH}_2-\text{O}-\text{O}-\text{CH}_2\text{CH}_3$ and $\text{C}_6\text{H}_5-\overset{\text{CH}_3}{\underset{\text{CH}_2-\text{O}-\text{OH}}{\text{CH}}}$ respectively
- (c) $\text{CH}_3-\overset{\text{OOH}}{\text{CH}}-\text{O}-\text{CH}_2\text{CH}_3$ and $\text{C}_6\text{H}_5-\overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}}-\text{O}-\text{OH}$ respectively
- (d) No reaction and $\text{C}_6\text{H}_5-\overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}}-\text{O}-\text{OH}$ respectively

65. Products (P_2) $\xleftarrow{\text{anhy. HI}}$ $(\text{CH}_3)_3\text{C}-\text{O}-\text{CH}_3 \xrightarrow{\text{Conc. HI}}$ Products (P_1)

The products P_1 and P_2 respectively are

- (a) $(\text{CH}_3)_3\text{COH} + \text{CH}_3\text{I}$ and $(\text{CH}_3)_3\text{Cl} + \text{CH}_3\text{OH}$
 (b) $(\text{CH}_3)_3\text{Cl} + \text{CH}_3\text{OH}$ and $(\text{CH}_3)_3\text{COH} + \text{CH}_3\text{I}$
 (c) $(\text{CH}_3)_3\text{Cl} + \text{CH}_3\text{OH}$ in both cases
 (d) CH_3I and $(\text{CH}_3)_3\text{COH}$ in both cases

66. Cyclobutylethene is treated with dil. H_2SO_4 to form

- (a) 2-cyclobutylethanol (b) 1-cyclobutyl-2-ethanol
 (c) 2-methylcyclopentanol (d) 1-methylcyclopentanol

67. The decreasing order of acidic character of the compounds is

$\text{CH}_3\text{C} \equiv \text{CH}$, MeOH , Me_2CHOH , Me_3COH , H_2O

- (a) $\text{CH}_3\text{C} \equiv \text{CH} > \text{Me}_3\text{COH} > \text{Me}_2\text{CHOH} > \text{MeOH} > \text{H}_2\text{O}$
 (b) $\text{MeOH} > \text{Me}_2\text{CHOH} > \text{Me}_3\text{COH} > \text{H}_2\text{O} > \text{CH}_3\text{C} \equiv \text{CH}$
 (c) $\text{Me}_3\text{COH} > \text{Me}_2\text{CHOH} > \text{MeOH} > \text{H}_2\text{O} > \text{CH}_3\text{C} \equiv \text{CH}$
 (d) $\text{H}_2\text{O} > \text{MeOH} > \text{Me}_2\text{CHOH} > \text{Me}_3\text{COH} > \text{CH}_3\text{C} \equiv \text{CH}$

68. Arrange the following in the decreasing order of acidic strength

(I) Phenol (II) p-nitrophenol (III) m-cresol (IV) p-cresol

- (a) (II) > (III) > (IV) > (I) (b) (II) > (I) > (III) > (IV)
 (c) (II) > (I) > (IV) > (III) (d) (III) > (IV) > (II) > (I)

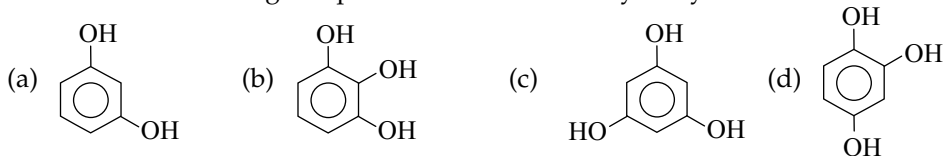
69. Which of the following will be most acidic

- (a) o-Aminophenol (b) p-Aminophenol
 (c) m-Aminophenol (d) None of these

70. Arrange the following in increasing acidic character

- (I) Phenol (II) m-nitrophenol
 (III) m-chlorophenol (IV) m-cresol
 (a) (IV) < (I) < (III) < (II) (b) (IV) < (I) < (II) < (III)
 (c) (I) < (IV) < (III) < (II) (d) (III) < (II) < (IV) < (I)

71. Which of the following compounds can react with hydroxylamine?



72. Which of the following statement is true regarding amount of AlCl_3 required during Friedel-Craft acetylation using acetyl chloride or acetic anhydride?

- (a) Both require same amount
 (b) Acetylation with acetyl chloride requires more amount
 (c) Acetylation with acetic anhydride requires more amount
 (d) Nothing is definite

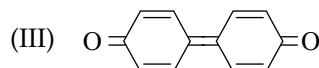
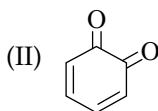
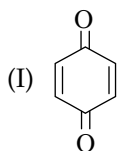
73. Which of the following gives effervescence of CO_2 with NaHCO_3 solution?

- (a) HCOOH (b) 2,4,6-trinitrophenol
(c) Both (a) and (b) (d) None of these

74. 2,4,6-Trinitrophenol can be prepared in good yield

- (a) by the nitration of 2,4-dinitrochlorobenzene
(b) by the nitration of 2,4-dinitrophenol
(c) by both (a) and (b)
(d) neither by (a) nor by (b)

75. The correct stability order of the following three quinones is



- (a) (I) > (III) > (II) (b) (I) = (III) > (II)
(c) (I) = (II) > (III) (d) (III) > (I) > (II)

76. Which of the following is most stable, and which one is least stable?

- (I) HCHO (II) CH_3CHO (III) CH_3COCH_3 (IV) Cl_3CCHO

- (a) (III) is most stable and (I) is least (b) (IV) is most stable and (I) is least
(c) (III) is most stable and (IV) is least (d) All the four are equally stable

77. Which of the following has highest and lowest hydration equilibrium constant?

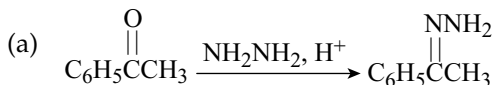
HCHO , CH_3CHO , CH_3COCH_3

- (a) HCHO -Highest, CH_3CHO -Lowest
(b) CH_3CHO -Highest, HCHO -Lowest
(c) HCHO -Highest, CH_3COCH_3 -Lowest
(d) CH_3COCH_3 -Highest, HCHO -Lowest

78. $\text{CH}_3\text{COCH}_2\text{Cl} \xrightarrow{\text{OH}^-, \text{Cl}_2}$ Product P is

- (a) $\text{ClCH}_2\text{COCH}_2\text{Cl}$ (b) $\text{CH}_3\text{COCHCl}_2$
(c) Both (a) and (b) (d) $\text{ClCH}_2\text{COOH} + \text{CH}_3\text{Cl}$

79. Which of the following is an example of nucleophilic addition?



- (c) Both (a) and (b)
(d) None of the two

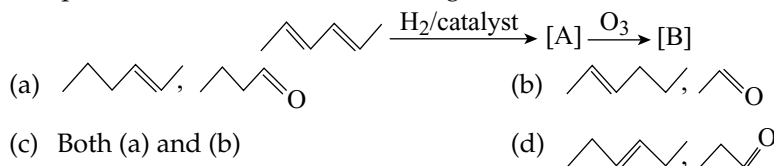
80. Propanal and propanone, both have same molecular formula (C_3H_6O), what do you expect about their boiling points?
- Both have same boiling point.
 - Boiling point of propanal is higher than the boiling point of propanone.
 - Boiling point of propanal is lower than the boiling point of propanone.
 - Nothing can be predicted.

81. Which of the following statement is false about 1,3-dithiane,  ?
- 1,3-Dithiane can react with RLi
 - It can be alkylated by $CH_3CH_2CH_2Br$
 - It can be alkylated by Me_2CHX
 - 1,3-Dithiane can be used for preparing aldehydes and ketones

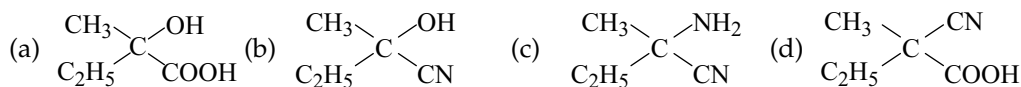
82. In dilute aqueous solution, formaldehyde exists as
- Formaldehyde
 - Paraldehyde
 - Trioxymethylene
 - Methyleneglycol

83. The major driving force for the hydration of chloral is
- less steric hinderance in the product
 - less force of repulsion in the product
 - hydrogen bonding in the product
 - electronegativity of the three chlorine atoms

84. The products A and B in the following reactions are



85. $CH_3\overset{\overset{O}{||}}{C}C_2H_5 + NaCN + NH_4Cl \xrightarrow{NH_4OH} Z$; Z is

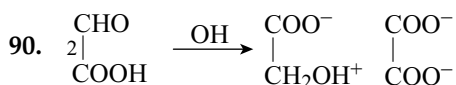


86. The compound having the formula, $CH_3-\overset{\overset{O}{||}}{C}-CH_2-\overset{\overset{O}{||}}{C}-OC_2H_5$
- forms dioxime
 - undergoes iodoform test
 - both (a) and (b)
 - neither of the two

87. Fehling's solution can be used for distinguishing between
- CH_3CHO and C_6H_5CHO
 - CH_3CHO and CH_3COCH_2OH
 - Both (a) and (b)
 - None of these

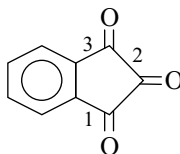
88. Which of the following is least reactive with a nucleophile?
- Methanal
 - Propanone
 - 3-Pentanone
 - 2-Pentanone

89. Aldehydic group can be protected
 (a) by acetal formation against alkaline oxidizing agents.
 (b) by mercaptal formation against acidic oxidizing agents.
 (c) both (a) and (b)
 (d) none of the above

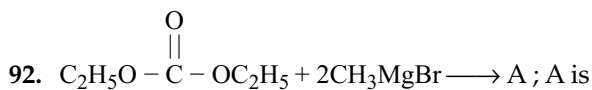


The above reaction can said to be an example of

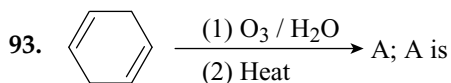
- (a) Intramolecular Cannizzaro reaction (b) Intermolecular Cannizzaro reaction
 (c) Crossed Cannizzaro reaction (d) Tischenko reaction
91. Ninhydrin has three keto groups, which of the keto group is expected to be hydrated most easily?



- (a) 2 (b) 3 (c) 1 (d) All are equally hydrated



- (a)
$$\begin{array}{c} \text{OH} \\ | \\ (\text{CH}_3)_2\text{C}-\text{C}_2\text{H}_5 \end{array}$$
 (b)
$$\begin{array}{c} \text{OH} \\ | \\ \text{C}_2\text{H}_5\text{C}-\text{CH}_3 \\ | \\ \text{C}_2\text{H}_5 \end{array}$$
 (c) CH_3COOH (d) CH_3COCH_3



- (a) $\text{CH}_2(\text{COOH})_2$ (b) CH_3COOH (c) HCOOH (d) HCHO

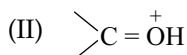
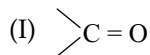
94. Which is least reactive towards addition of HCl?

- (a) $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$ (b) $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$
 (c) $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CCH}_3$ (d) $\text{CH}_3\text{CH}=\text{CHCHO}$

95. Which of the following is true regarding preparation of aldehydes and ketones?

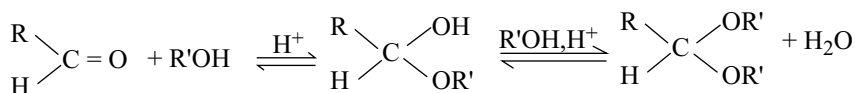
- (I) Both can be prepared by the oxidation of the concerned alcohol with copper at about 250°C .
 (II) Both can be prepared by the oxidation of the concerned alcohol by Oppenauer oxidation.
 (III) Both can be prepared by the oxidation of respective alcohol with acidic dichromate.
 (a) (I) (b) (II) and (III) (c) (I) and (III) (d) All the three

96. Observe the following structures and pick up the correct statement.



- (a) Carbonyl carbon of (I) is more electrophilic than that of (II).
 (b) Carbonyl carbon of (I) is less electrophilic than that of (II).
 (c) Carbonyl carbon of both structures has equal electrophilic character.
 (d) It depends upon the complete structure of the compound

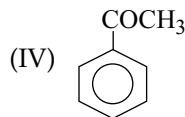
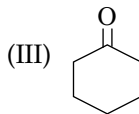
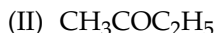
97. Acetal formation is a reversible reaction



Under what conditions, the reaction can be forced to proceed only in right (forward) direction?

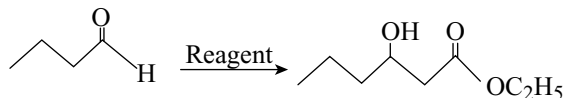
- (a) Using excess of alcohol
 (b) Using high temperature
 (c) Using dilute acid and excess of alcohol
 (d) Using dry acid and excess of alcohol

98. Which of the following does not react with sodium bisulphite?



- (a) (III) and (IV) (b) (II) and (IV) (c) (I) (d) All reacts

99. Which of the following reagent can be used for carrying out the reaction outlined below?



- (a) $\text{BrMgCH}_2\text{COOC}_2\text{H}_5$ (b) $\text{BrZnCH}_2\text{COOC}_2\text{H}_5$
 (c) $\text{LiCH}_2\text{COOC}_2\text{H}_5$ (d) Any of the three

100. Which of the following statement is false?

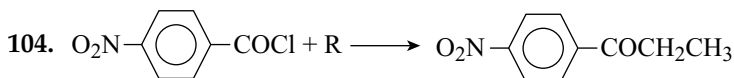
- (a) Cannizzaro reaction is given by aldehydes in presence of alkali
 (b) Aldol condensation is given by aldehydes in presence of alkali
 (c) Aldol condensation is given by aldehydes and ketones in presence of acids
 (d) None of these

101. Carbonyl compounds, sensitive to both acids as well as bases, can be reduced to hydrocarbons by

- (a) Clemmensen reduction (b) Wolf-Kishner reduction
 (c) Thioacetal reduction (d) All of the three

102. What should be the product when ethylmethyl ketone is treated with peracetic acid
 (a) Ethyl acetate (b) Methyl propanoate
 (c) Both (a) and (b) (d) Only acetic acid

103. Which of the following complex hydride is a stronger reducing agent?
 (a) $\text{Li}^+[\text{AlH}_4]^-$ (b) $\text{Li}^+[\text{Al}(\text{OCMe}_3)_3\text{H}]^-$
 (c) $\text{Al}(\text{CH}_2\text{CHMe}_2)_2\text{H}$ (d) All are strong reducing agents



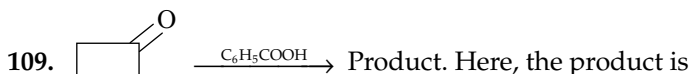
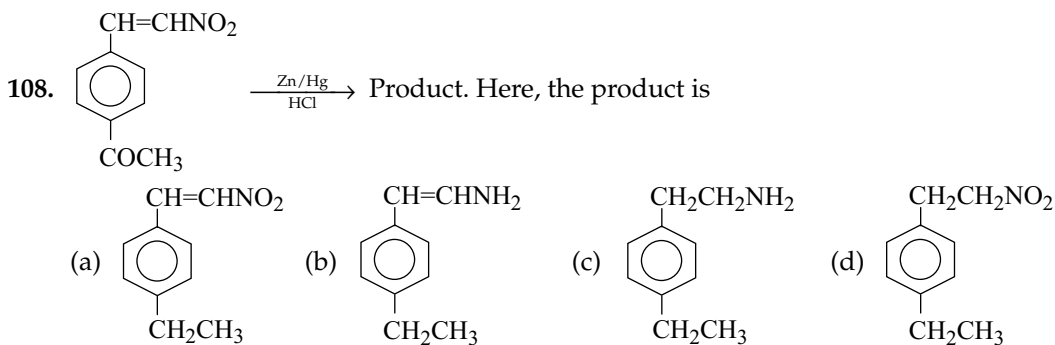
The reagent R may be

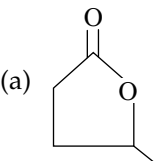
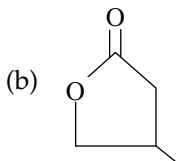
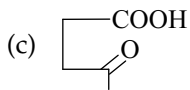
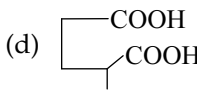
- (I) $\text{CH}_3\text{CH}_2\text{MgBr}$ (II) $\text{CH}_3\text{CH}_2\text{Li}$
 (III) $(\text{CH}_3\text{CH}_2)_2\text{Cd}$ (IV) $(\text{CH}_3\text{CH}_2)_2\text{CuLi}$
 (a) (I) or (II) (b) (I) or (II) or (III) (c) (III) or (IV) (d) Any of the four

105. Which one does not belong to the same compound?
 (a) Paraformaldehyde (b) Paraldehyde
 (c) Trioxane (d) Formalin

106. Which of the following is not a good reagent in Wittig reaction?
 (a) $\text{Ph}_3\text{P}=\text{CH}_2$ (b) $\text{Ph}_3\text{P}^+\text{CH}^-\text{CH}_2\text{CH}_3$
 (c) $(\text{CH}_3)_3\text{CCH}=\text{PPh}_3$ (d) $\text{C}_6\text{H}_5\text{COCH}=\text{PPh}_3$

107. Which of the following is not formed in iodoform reaction?
 (a) $\text{CH}_3\text{COCH}_2\text{I}$ (b) $\text{ICH}_2\text{COCH}_2\text{I}$ (c) $\text{CH}_3\text{COCHI}_2$ (d) CH_3COCl_3



- (a)  (b)  (c)  (d) 

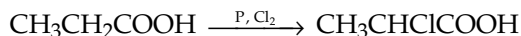
110. Acetyl chloride does not react with
 (a) Water (b) Sodium acetate
 (c) 2-methylpropene (d) It reacts with all the three
111. Which of the following statement is not true?
 (a) At room temperature, formyl chloride is present in the form of CO and HCl.
 (b) Acetamide behaves as a weak base as well as a weak acid.
 (c) $\text{CH}_3\text{CONH}_2 \xrightarrow{\text{LiAlH}_4} \text{CH}_3\text{CH}_2\text{NH}_2$.
 (d) None of the three.
112. Which of the following reaction is possible?
 (I) $\text{CH}_3\text{COCl} + \text{H}_2\text{O} \longrightarrow \text{CH}_3\text{COOH} + \text{HCl}$
 (II) $\text{CH}_3\text{COOCH}_3 + \text{HBr} \longrightarrow \text{CH}_3\text{COBr} + \text{CH}_3\text{OH}$
 (III) $\text{CH}_3\text{CONH}_2 + \text{HBr} \longrightarrow \text{CH}_3\text{COBr} + \text{NH}_3$
 (IV) $\text{CH}_3\text{COOCOCH}_3 + \text{H}_2\text{O} \longrightarrow 2\text{CH}_3\text{COOH}$
 (a) (I) and (IV) (b) (I), (III) and (IV) (c) (I), (II) and (IV) (d) All the four
113. Which of the following is not possible?
 (a) $\text{ICH}_2\text{COOH} + \text{NaCl} \xrightarrow{\text{Acetone}} \text{ClCH}_2\text{COOH} + \text{NaI}$
 (b) $\text{ClCH}_2\text{COOH} + \text{NaI} \xrightarrow{\text{Acetone}} \text{ICH}_2\text{COOH} + \text{NaCl}$
 (c) Both (a) and (b)
 (d) None of these
114. Acid amide do not undergo the usual properties of carbonyl, C=O group because
 (a) it is a weak base (b) it is a weak acid
 (c) it is amphoteric (d) its carbonyl carbon is not electron deficient
115. Which of the following statement is true regarding aspirin, a commonly used antipyretic and analgesic? Given pK_a for aspirin = 3.5; pH in stomach and small intestine is 2.5 and 8, respectively.
 (a) It is completely ionized in the stomach and almost un-ionized in the small intestine.
 (b) It is ionized in the small intestine and almost un-ionized in the stomach.
 (c) It is ionized in the stomach and almost un-ionized in the small intestine.
 (d) It is neither ionized in stomach nor in intestine.
116. Which of the following will undergo alkaline hydrolysis most rapidly?
- (a)

(b)

(c)

(d)

117. HVZ reaction involves the use of P and Cl_2

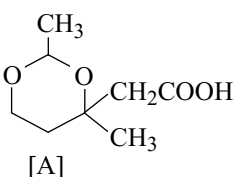


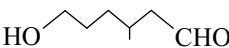
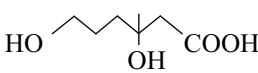
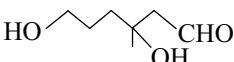
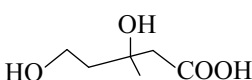
The function of phosphorus is

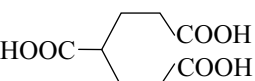
- (a) as a catalyst
- (b) in the formation of PCl_3 which carries out halogenation at the α -carbon atom
- (c) in the formation of PCl_3 which converts $-\text{COOH}$ into $-\text{COCl}$
- (d) none of the these

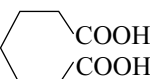
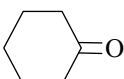
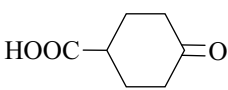
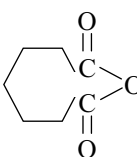
118. Hydrolysis of esters in presence of an acid is a reversible reaction. What is true about ester hydrolysis in presence of a base?

- (a) It is irreversible because salts of carboxylic acids are insoluble.
- (b) It is irreversible because salts of carboxylic acids have high melting points.
- (c) It is irreversible because carboxylate ion is resonance stabilized.
- (d) It is a reversible reaction.

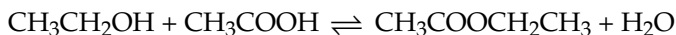
119. Compound A is formed by the interaction of 

- (a) CH_3COOH and 
- (b) CH_3CHO and 
- (c) $\text{CH}_3\text{COCH}_2\text{COOH}$ and 
- (d) CH_3CHO and 

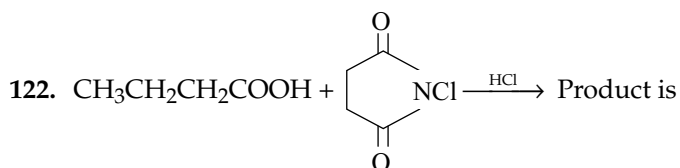
120. What is the main product when  is heated?

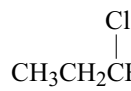
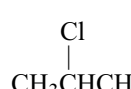
- (a) 
- (b) 
- (c) 
- (d) 

121. The yield of ester in esterification can be increased by

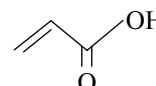
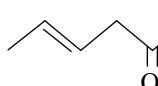
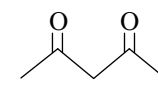


- (a) removing water
- (b) taking ethanol in excess
- (c) taking acetic acid in excess
- (d) all the above factors

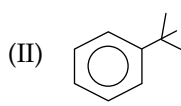
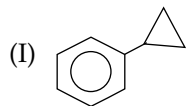


- (a)  (b) $\text{ClCH}_2\text{CH}_2\text{CH}_2\text{COOH}$
 (c)  (d) All the three

123. The correct order of decarboxylation of the three acids is

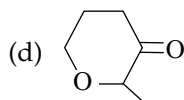
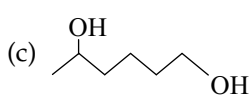
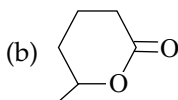
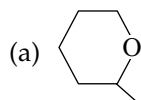
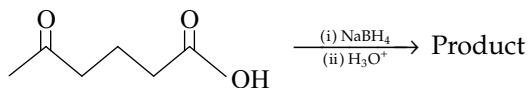
- (I)  (II)  (III) 
 (a) (III) > (II) > (I) (b) (III) = (II) > (I) (c) (III) > (II) = (I) (d) (III) = (II) = (I)

124. Which statement is true regarding oxidation of the following two compounds?

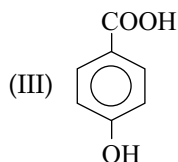
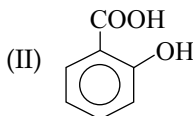
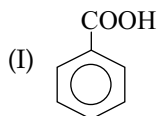


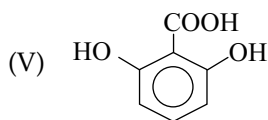
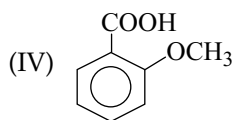
- (a) Both are oxidizable to benzoic acid under similar conditions
 (b) It is very difficult to oxidize either of the two
 (c) Compound (I) is oxidizable to benzoic acid easily while compound (II) is oxidizable only under vigorous conditions to benzoic acid
 (d) Compound (I) is oxidizable to benzoic acid, while (II) is oxidizable only under vigorous conditions to 2,2-dimethylpropanoic acid

125. Predict the nature of end product in the following reaction



126. The correct order for the acidic character of the following carboxylic acids is





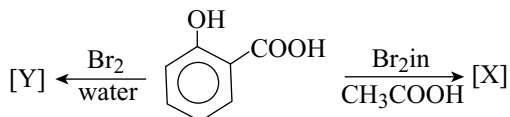
(a) (IV) > (I) > (II) > (III) > (V)

(b) (V) > (II) > (III) > (I) > (IV)

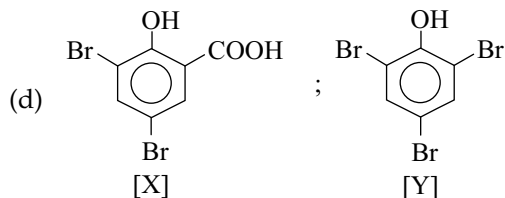
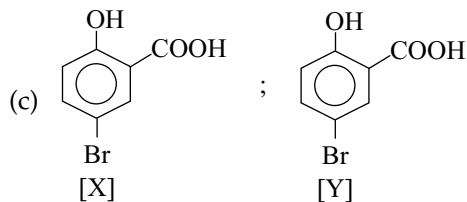
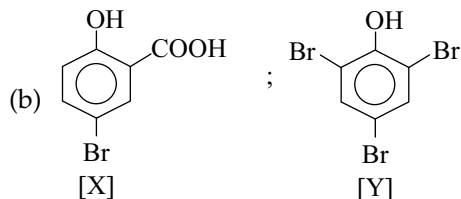
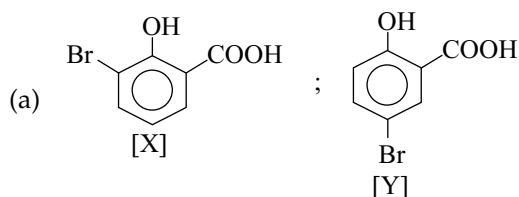
(c) (V) > (II) > (IV) > (III) > (I)

(d) (V) > (II) > (IV) > (I) > (III)

127. Salicylic acid is treated with bromine under two different conditions.



Predict the nature of [X] and [Y] in the following reactions



128. Which of the following statements is true?

- (a) Hydrogen bonding always increases the acidic character of a species.
- (b) Hydrogen bonding always decreases the acidic character of a species.
- (c) Hydrogen bonding may increase or decrease the acidic character of a species.
- (d) Hydrogen bonding play no role in determining the acidity of a species.

129. Choose the correct statement regarding acidic character of acetic acid, CH_3COOH and peroxyacetic acid, CH_3COOOH .
- Peroxyacetic acid is stronger acid than acetic acid since the former has one extra oxygen, an electronegative element.
 - Peroxyacetic acid is stronger than acetic acid because its conjugate base is a weaker base than acetate.
 - Peroxyacetic acid is weaker than acetic acid because its conjugate base is less stable than that of acetate ion.
 - Both are equally strong.

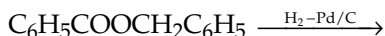
130. A carboxylic acid can best be converted into acid chloride by using
- PCl_5
 - SOCl_2
 - HCl
 - ClCOCOCI

131. The yield of acid amide in the reaction, $\text{RCOCl} + \text{NH}_3 \longrightarrow \text{RCONH}_2$, is maximum when
- acid chloride and ammonia are treated in equimolar ratio
 - acid chloride and ammonia are treated in 1:2 molar ratio
 - acid chloride and ammonia are treated in 2:1 molar ratio
 - All the three give nearly similar result

132. Which of the following statement is not upto the mark?

- $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OR}' \xrightarrow{\text{OH}^-} \text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}^- + \text{R}'\text{OH}$ (a base-catalyzed reaction)
- $\text{CH}_3\text{COOC}_2\text{H}_5 \xrightarrow{\text{OH}^-} \text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH}$ (reaction involves acyl-oxygen fission)
- $\text{C}_6\text{H}_5\text{COOH} + \text{CH}_3\text{OH} \xrightarrow{\text{H}^+} \text{C}_6\text{H}_5\text{COOCH}_3 + \text{H}_2\text{O}$ (reaction involves acyl-oxygen fission)
- $\text{CH}_3\text{COOCH}(\text{CH}_3)\text{C}_2\text{H}_5 \xrightarrow{\text{OH}^-} \text{CH}_3\text{COO}^- + \text{HOCH}(\text{CH}_3)\text{C}_2\text{H}_5$ (configuration of the reactant is retained)

133. The products in the following reaction are



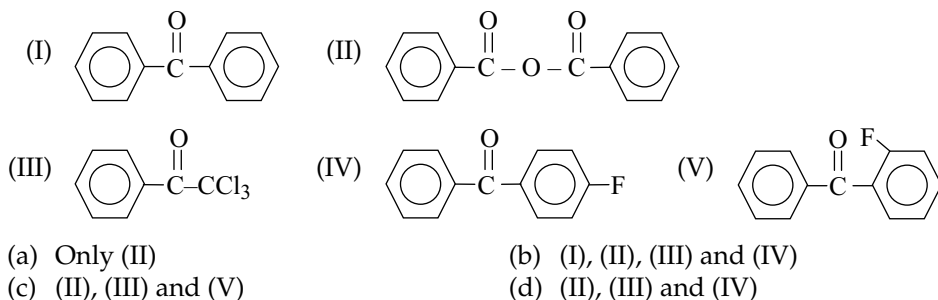
- $\text{C}_6\text{H}_5\text{CH}_2\text{OH} + \text{C}_6\text{H}_5\text{CH}_2\text{OH}$
- $\text{C}_6\text{H}_5\text{CH}_3 + \text{C}_6\text{H}_5\text{CH}_3$
- $\text{C}_6\text{H}_5\text{CH}_2\text{OH} + \text{C}_6\text{H}_5\text{CH}_3$
- $\text{C}_6\text{H}_5\text{COOH} + \text{C}_6\text{H}_5\text{CH}_3$

134. Pyrolysis of $\text{CH}_3\overset{\text{CH}_3}{\underset{|}{\text{COOCH}}}\text{CH}_2\text{CH}_3$ gives

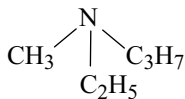
- 1-butene and 2-butene in equimolar ratio
- 1-butene and 2-butene in 1:2 molar ratio
- 1-butene and 2-butene in 3:2 molar ratio
- 1-butene and 2-butene in 2:3 molar ratio

135. The relative stability of the four acid derivatives towards nucleophiles is
 (a) Amide > Ester > Acid anhydride > Acid chloride
 (b) Amide > Acid anhydride > Ester > Acid chloride
 (c) Acid chloride > Acid anhydride > Ester > Amide
 (d) Acid chloride > Ester > Acid anhydride > Amide
136. Which of the following statement is true about the hydrolysis of acetic anhydride?
 (I) It is more easily hydrolyzed in acidic medium than in neutral.
 (II) It is more easily hydrolyzed in alkaline medium than in neutral.
 (III) It is equally hydrolyzed in all the three media.
 (IV) It is more easily hydrolyzed in neutral than in acidic media.
 (V) It is more easily hydrolyzed in neutral than in alkaline medium.
 (a) (I) and (II) (b) (III) (c) (IV) and (V) (d) (I) and (V)

137. Which of the following compounds can undergo nucleophilic substitution easily?

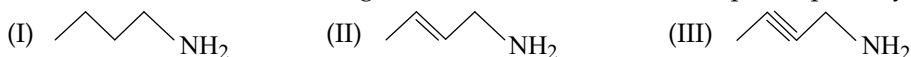


138. Which statement is true regarding the following structure?



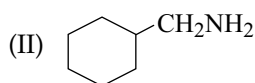
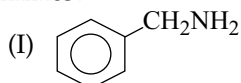
- (a) It is a chiral molecule (b) It exists in two resolvable optically active forms
 (c) Both (a) and (b) (d) Neither (a) nor (b)
139. Introduction of a methyl group in ammonia markedly increases the basic strength of ammonia in aq. solution, but introduction of the second methyl group increases only marginally the basic strength of methyl amine in water. This is due to
 (a) different type of hybridization in the two amines.
 (b) protonated dimethyl amines are more solvated than methyl amine.
 (c) protonated dimethyl amine is more solvated than the protonated methyl amine.
 (d) protonated dimethyl amine is less stable than the protonated methyl amine.
140. The basic character of ethyl amine, diethyl amine and triethyl amine in chlorobenzene is
 (a) $\text{C}_2\text{H}_5\text{NH}_2 < (\text{C}_2\text{H}_5)_2\text{NH} < (\text{C}_2\text{H}_5)_3\text{N}$ (b) $\text{C}_2\text{H}_5\text{NH}_2 < (\text{C}_2\text{H}_5)_3\text{N} < (\text{C}_2\text{H}_5)_2\text{NH}$
 (c) $(\text{C}_2\text{H}_5)_3\text{N} < (\text{C}_2\text{H}_5)_2\text{NH} < \text{C}_2\text{H}_5\text{NH}_2$ (d) $(\text{C}_2\text{H}_5)_3\text{N} < \text{C}_2\text{H}_5\text{NH}_2 < (\text{C}_2\text{H}_5)_2\text{NH}$

141. The correct order of decreasing basic character of the three aliphatic primary amines is



- (a) (I) > (II) > (III) (b) (III) > (II) > (I) (c) (I) > (II) $\approx$ (III) (d) (I) = (II) $\equiv$ (III)

142. Which of the statement is true regarding the basicity of the following two primary amines?



- (a) Both are equally basic because both are 1° amines.
 (b) (I) > (II) because it is an aromatic amine.
 (c) (II) > (I) because it is an aliphatic amine
 (d) (I) < (II) because of difference in the nature of β -carbon.

143. The correct order of decreasing basic character is



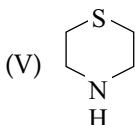
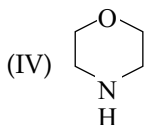
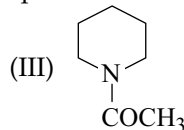
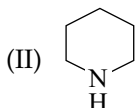
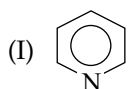
- (a) (II) > (I) > (III) > (IV) (b) (IV) > (II) > (I) > (III)
 (c) (IV) > (III) > (II) > (I) (d) (IV) > (II) > (III) > (I)

144. The correct order of increasing basicity is



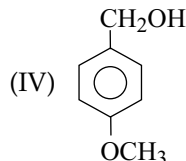
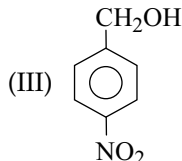
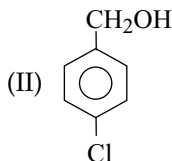
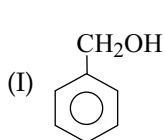
- (a) (II) < (III) < (I) (b) (I) $\approx$ (III) < (II) (c) (I) < (II) < (III) (d) (III) < (I) < (II)

145. The relative order of basic character of the following compound is



- (a) (II) > (I) > (III) > (IV) > (V) (b) (II) > (III) > (IV) > (V) > (I)
 (c) (II) > (V) > (IV) > (I) > (III) (d) (II) > (IV) > (V) > (III) > (I)

146. The basic character of the following alcohols is



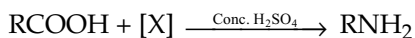
- (a) (IV) > (I) > (II) > (III) (b) (II) > (III) > (IV) > (I)
 (c) (IV) > (II) > (III) > (I) (d) (I) > (II) > (III) > (IV)

147. When aniline is treated with acetyl chloride in presence of anhydrous aluminium chloride, the main product is
 (a) o-aminoacetophenone (b) p-aminoacetophenone
 (c) Both (a) and (b) (d) m-aminoacetophenone

148. $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2 \xrightarrow[0^\circ\text{C}]{\text{NaNO}_2, \text{HCl}}$ P. P is
 (a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (b) $(\text{CH}_3)_2\text{CHCl}$
 (c) Both (a) and (b) (d) Reaction not possible

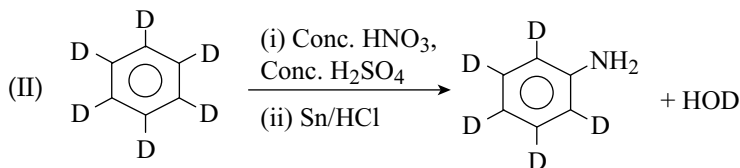
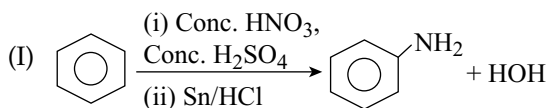
149. Benzenediazonium chloride when treated with phenols gives an azo dye, to get best result the pH of the medium should be
 (a) around 4 (b) around 8 (c) around 10 (d) 12

150. In the following reaction, the reagent X should be



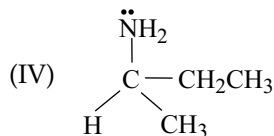
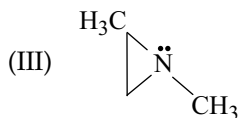
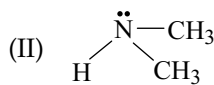
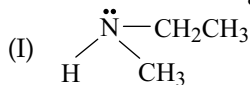
- (a) NH_3 (b) HN_3 (c) Either of the two (d) None of the two
151. Which of the following can undergo electrophilic substitution when treated with nitrous acid at 0°C ?
 (a) $\text{C}_6\text{H}_5\text{NH}_2$ (b) $\text{C}_6\text{H}_5\text{NHCH}_3$ (c) $\text{C}_6\text{H}_5\text{N}(\text{CH}_3)_2$ (d) None of these

152. Which of the two reactions proceed faster?

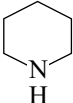
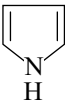


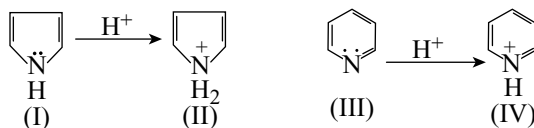
- (a) (I) (b) (II) (c) (I) = (II) (d) Not definite
153. Which of the following does not reduce $\text{C}_6\text{H}_5\text{NO}_2$ to aniline?
 (a) Sn/HCl (b) SnCl_2/HCl (c) Zn/HCl (d) LiAlH_4

154. Which of the following amines can be resolved into two enantiomers?



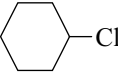
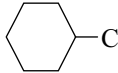
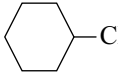
- (a) (I), (IV) (b) (I), (II) (c) (I), (III), (IV) (d) (III), (IV)

155. Which of the following statement is false?
 (a) Dimethyl amine as well as trimethyl amine are soluble in water.
 (b) Trimethyl amine forms hydrogen bond neither with itself nor with water.
 (c) Trimethyl amine can act as hydrogen bond acceptor only, while dimethyl amine can serve as both a hydrogen bond donor and acceptor.
 (d) All the three statements are false.
156. Ephedrine is a secondary amine. It is widely used in cold and allergy conditions in the form of its hydrochloride but not as such because
 (a) the amine itself has an unpleasant smell, while its salt is odourless.
 (b) the amine is insoluble in water, while the salt is soluble in water.
 (c) the amine is unstable and easily oxidized by air, while the salt is resistant to atmospheric oxidation.
 (d) of all the above facts
157. Benzamide and benzyl amine can be distinguished by
 (a) cold. dil. NaOH (b) cold dil. HCl
 (c) both (a) and (b) (d) NaNO_2 , HCl, 0°C , then β -naphthol
158. The correct order for the basic character of the compounds I to IV should be
 (I)  (II)  (III)  (IV) CH_3CN
 (a) $(\text{IV}) < (\text{III}) < (\text{I}) < (\text{II})$ (b) $(\text{IV}) < (\text{I}) < (\text{III}) < (\text{II})$
 (c) $(\text{IV}) < (\text{II}) < (\text{III}) < (\text{I})$ (d) $(\text{IV}) < (\text{III}) < (\text{II}) < (\text{I})$
159. Which of the following is true regarding basic character of pyridine and pyrrole?
 (a) Pyrrole is more basic because its non-bonding electrons occupy sp^3 orbital.
 (b) Pyridine is more basic because its non-bonding electrons are not part of aromatic sextet.
 (c) Both are equally basic.
 (d) Pyridine is less basic because it is a tertiary amine.
160. Pyrrole and pyridine both are basic and form salts with acids.

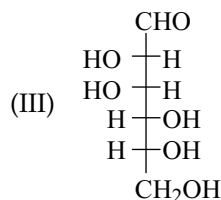
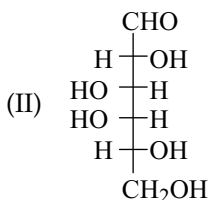
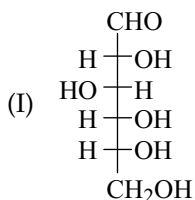


Which of the following statement is true regarding the aromatic character of the four species?

- (a) All the four are aromatic (b) (I), (III) and (IV) are aromatic
 (c) (I), (II) and (III) are aromatic (d) (I) and (III) are aromatic

161. Ethylene can be prepared in good yield by
 (a) $\text{CH}_3\text{CH}_2\text{N}^+(\text{CH}_3)_3\text{I}^- \xrightarrow{\text{Heat}} \text{CH}_2=\text{CH}_2 + (\text{CH}_3)_3\text{N} + \text{HI}$
 (b) $\text{CH}_3\text{CH}_2\text{N}^+(\text{CH}_3)_3\text{OH}^- \xrightarrow{\text{Heat}} \text{CH}_2=\text{CH}_2 + (\text{CH}_3)_3\text{N} + \text{H}_2\text{O}$
 (c) Both (a) and (b)
 (d) $\text{CH}_3\text{CH}_2\text{NH}_2 \xrightarrow{\text{Heat}} \text{CH}_2=\text{CH}_2 + \text{NH}_3$
162. Which one of the following is not an oxidation product of a primary amine?
 (a) A hydroxylamine (b) A nitroso compound
 (c) A nitro compound (d) None of these
163. Which of the following method is used for eliminating nitrogen of an amine present outside the ring?
 (a) Hofmann elimination (b) Cope elimination
 (c) Both (a) and (b) (d) Emde degradation
164. Which of the following does not react with nitrous acid?
 (a) $\text{C}_6\text{H}_5\text{NH}_2$ (b) $\text{C}_6\text{H}_5\text{NHCH}_3$ (c) $\text{C}_6\text{H}_5\text{N}(\text{CH}_3)_2$ (d) None of these
165. Which of the following leads to carbon-carbon double bond?
 (a) $1^\circ \text{ amine} + \text{RCHO} \longrightarrow$ (b) $2^\circ \text{ Amine} + \text{R}_2\text{CO} \longrightarrow$
 (c) $2^\circ \text{ amine} + \text{RCHO} \longrightarrow$ (d) Both (b) and (c)
166. Electrophilic aromatic substitution of pyridine resembles with
 (a) benzene (b) aniline (c) nitrobenzene (d) none of these
167. Arrange the following compounds in decreasing order of reactivity towards electrophilic substitution.
- (I)  (II)  (III) 
- (a) (I) > (II) > (III) (b) (I) > (II) = (III) (c) (III) > (II) > (I) (d) (III) > (I) > (II)
168. Which reaction seems to be incorrect?
 (a) $\text{Me}_3\text{CCl} \xrightarrow{\text{NH}_3} \text{Me}_3\text{CNH}_2$ (b) $\text{Me}_3\text{CCl} \xrightarrow{\text{NH}_3} \text{Me}_2\text{C}=\text{CH}_2$
 (c)  $\xrightarrow{\text{NH}_3}$  (d)  $\xrightarrow{\text{NH}_3}$ 

169. Which of the statement regarding following structure is true?



- (a) (I) and (II) are epimers (b) (I) and (III) are epimers
(c) Both (a) and (b) are true (d) All the three are epimers

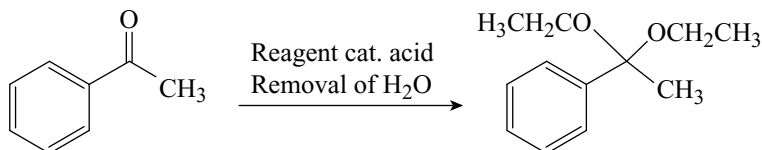
170. Which of the following pair represents an example of diastereoisomers

- (i) (+)-Tartaric acid and meso-tartaric acid
(ii) Maleic acid and fumaric acid
(iii) D(+)-Galactose and D(+)-mannose
(iv) (+)-Lactic acid and (–)-Lactic acid
(a) (I) and (III) (b) (I), (III) and (IV) (c) (I), (II) and (III) (d) (IV)

LEVEL 2

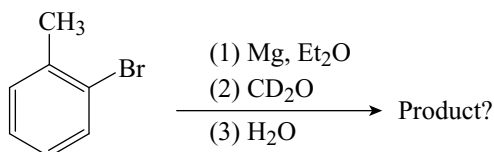
Single and Multiple-choice Type

1. What could be the reagent for the following reaction?



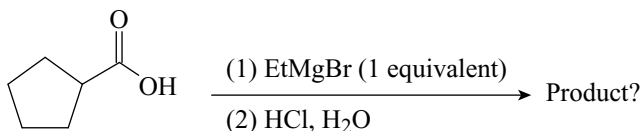
- (a) CH_3COCH_3 (b) CH_3OH (c) $\text{CH}_3\text{CH}_2\text{OH}$ (d) $\text{HOCH}_2\text{CH}_2\text{OH}$

2. What could be the product for the following reaction?



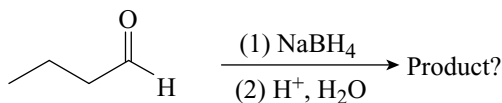
- (a) (b) (c) (d)

3. What could be the major product for the following reaction?



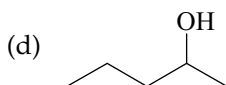
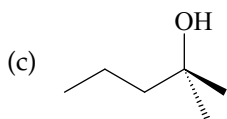
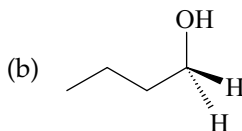
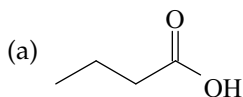
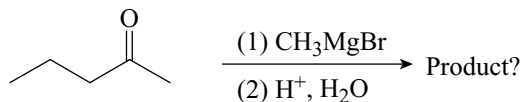
- (a) (b) (c) (d)

4. What could be the product for the following reaction?

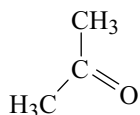


- (a) (b) (c) (d)

5. What could be the product for the following reaction?



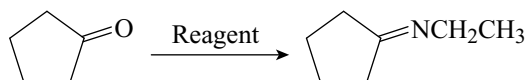
6. What is not the name for the following compound?



- (a) 2-Acetaldehyde
(c) acetone

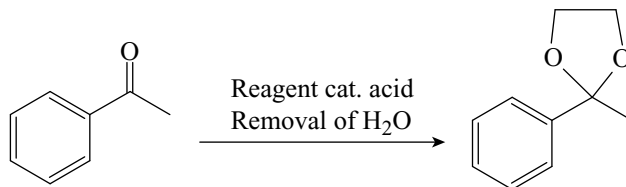
- (b) propanone
(d) dimethyl ketone

7. What could be the reagent for the following reaction?

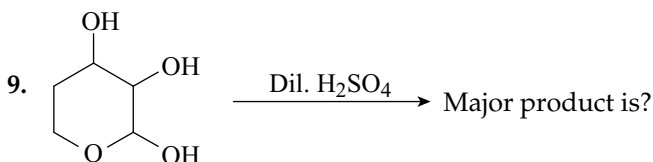


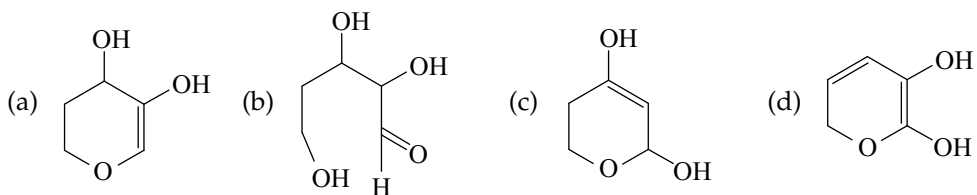
- (a) $\text{H}_2\text{N}-\text{Ph}$, removal of water
(b) $\text{H}_2\text{NCH}_2\text{CH}_3$, $\text{CH}_3\text{CO}_2\text{H}$, removal of water
(c) H_2NCH_3 , H_2O , removal of water
(d) $\text{H}_2\text{NCH}_2\text{CH}_2\text{CH}_3$, $\text{CH}_3\text{CO}_2\text{H}$, removal of water

8. What could be the reagent for the following reaction?

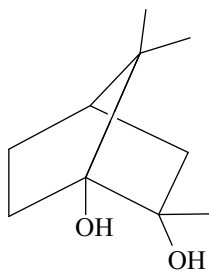


- (a) CH_3COCH_3 (b) $\text{HOCH}_2\text{CH}_2\text{OH}$ (c) $\text{CH}_3\text{CH}_2\text{OH}$ (d) CH_3OH

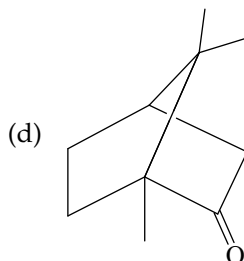
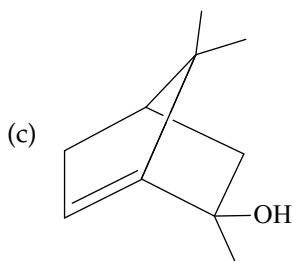
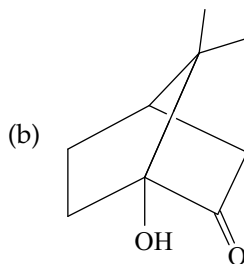
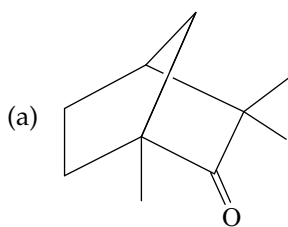




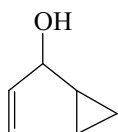
10.

Conc. H_2SO_4

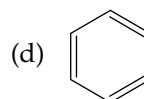
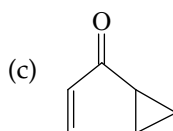
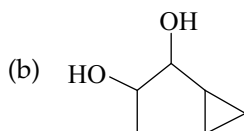
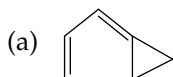
Major product is?

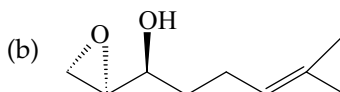
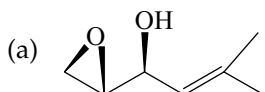
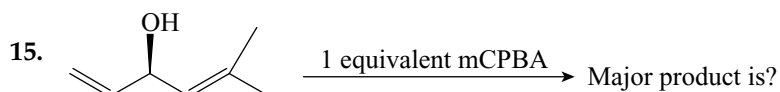
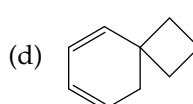
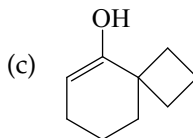
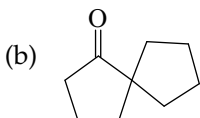
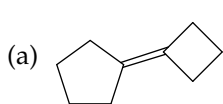
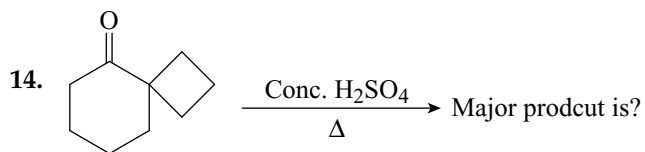
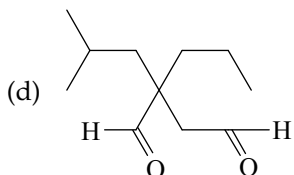
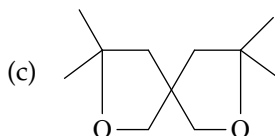
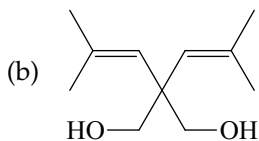
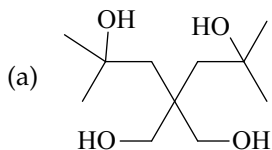
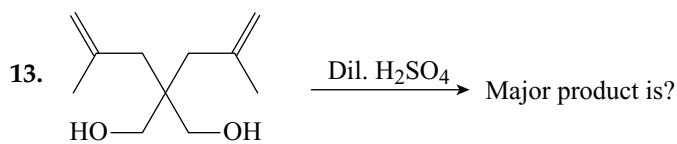
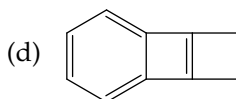
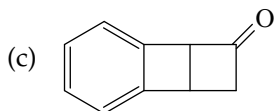
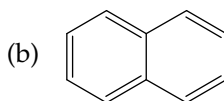
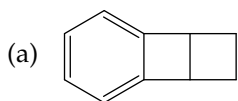
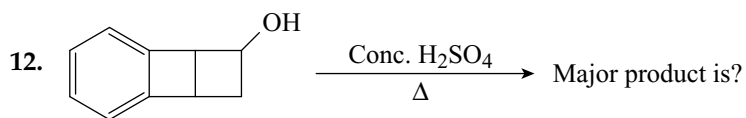


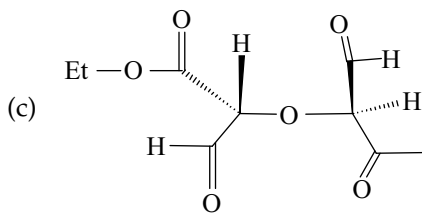
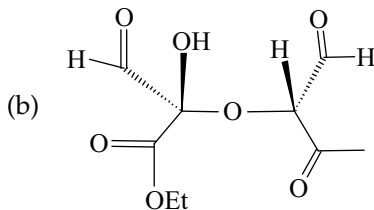
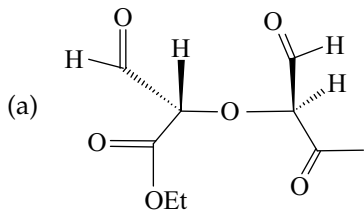
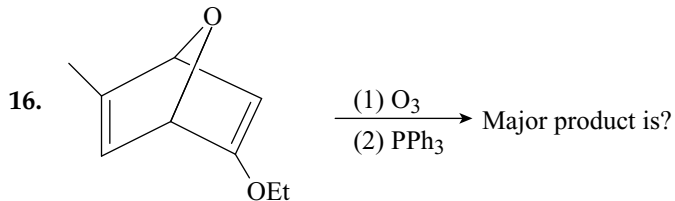
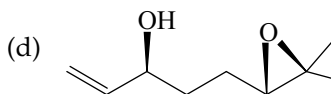
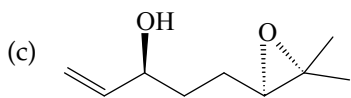
11.

Conc. H_2SO_4
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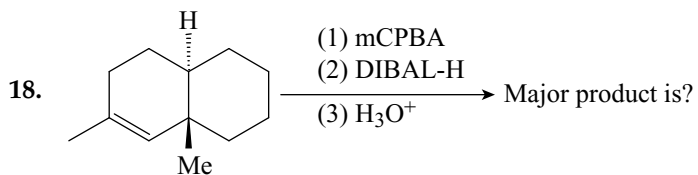
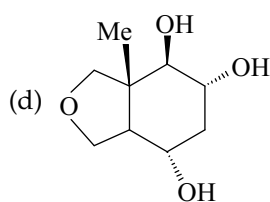
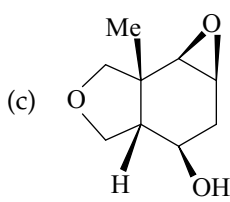
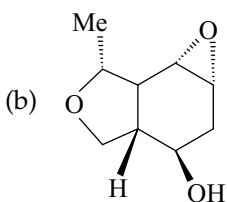
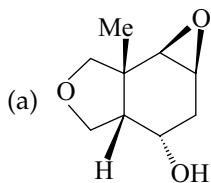
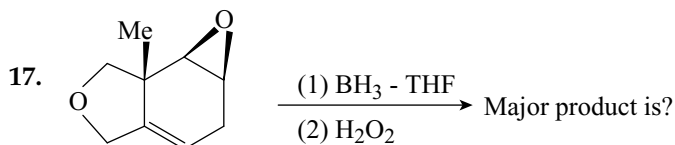
Major product is?

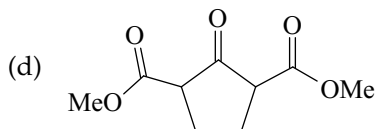
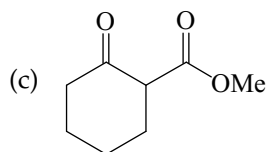
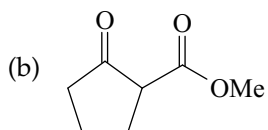
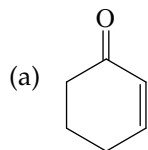
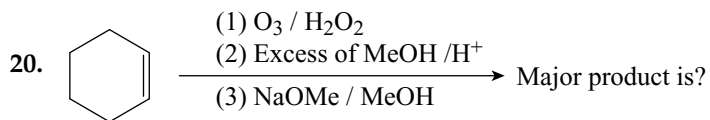
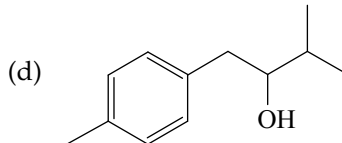
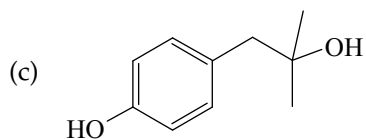
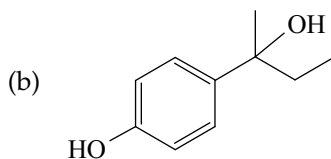
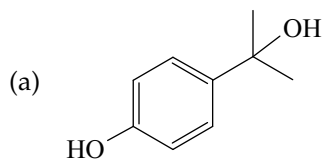
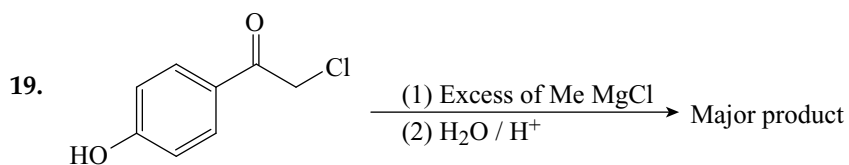
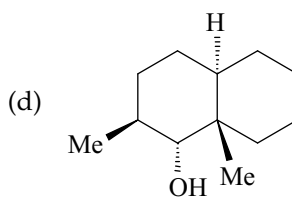
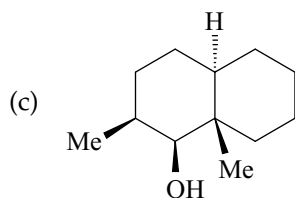
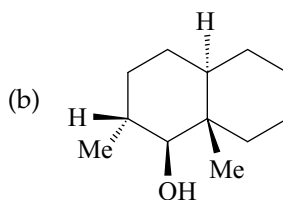
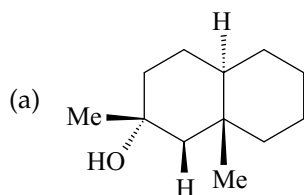


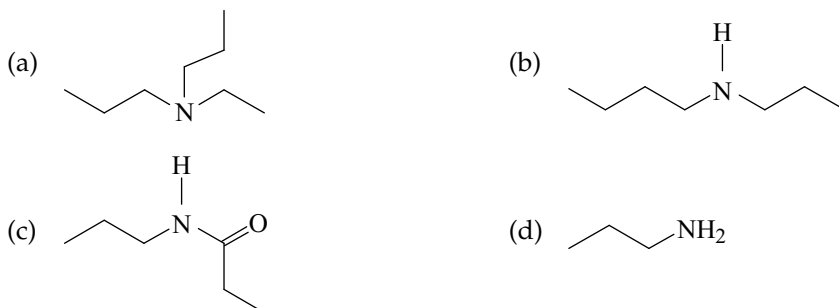
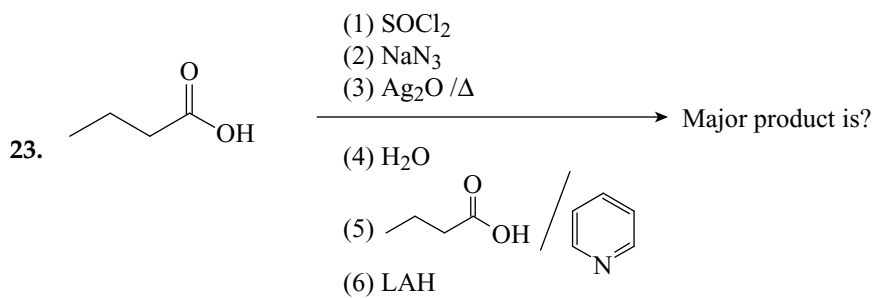
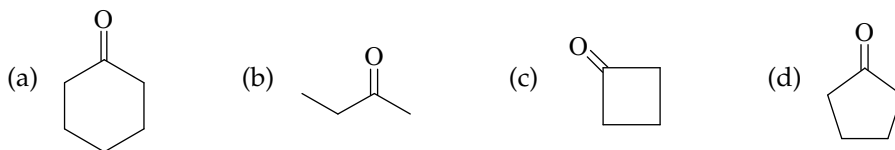
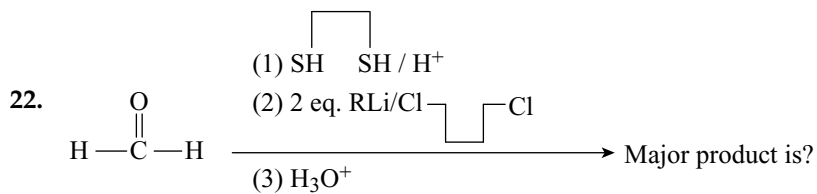
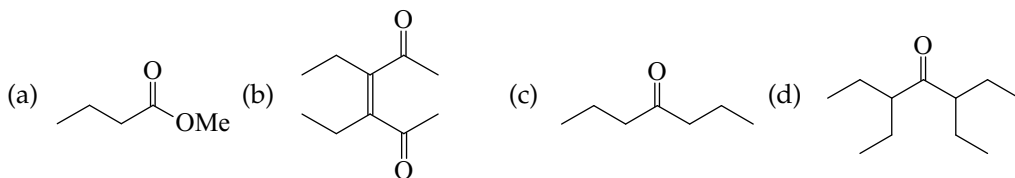
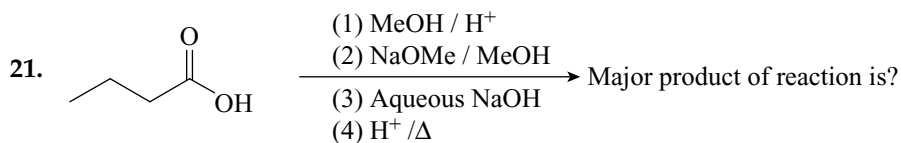


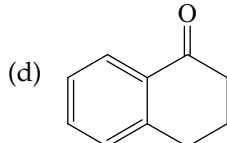
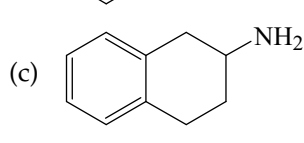
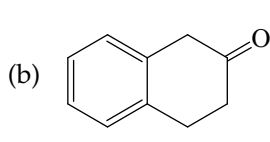
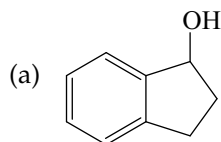
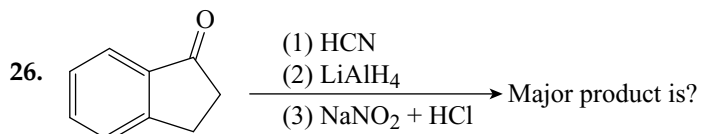
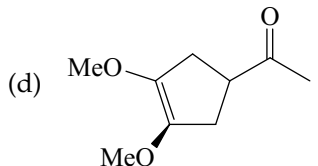
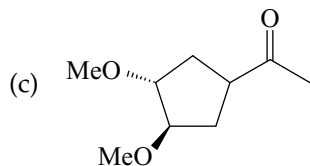
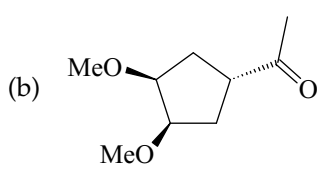
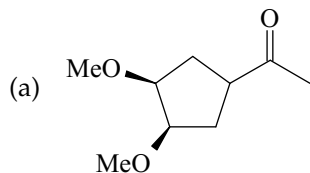
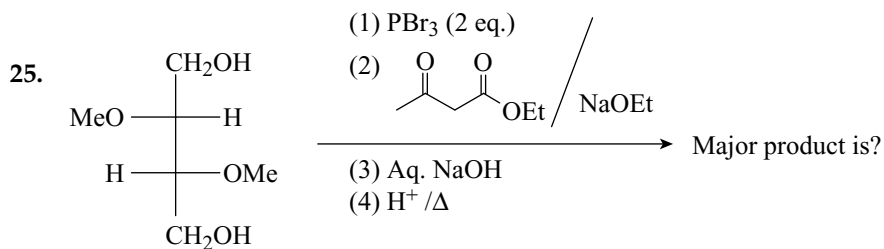
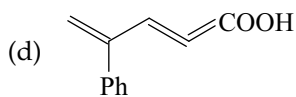
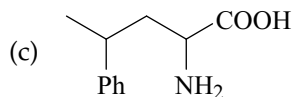
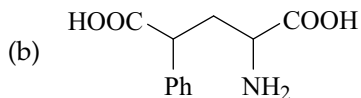
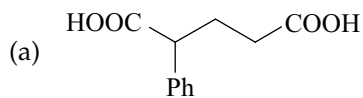
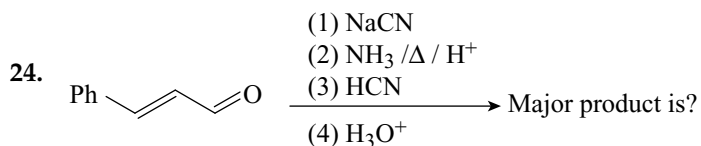


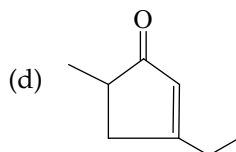
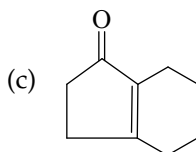
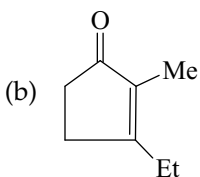
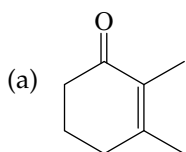
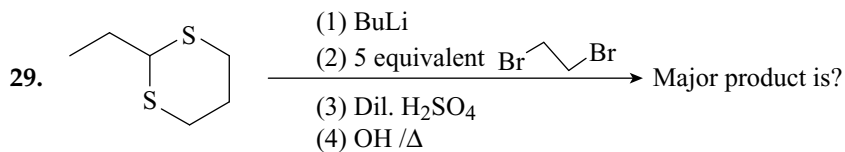
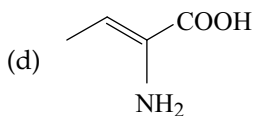
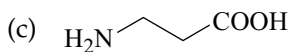
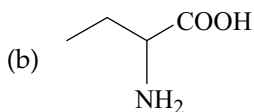
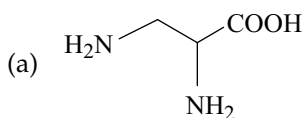
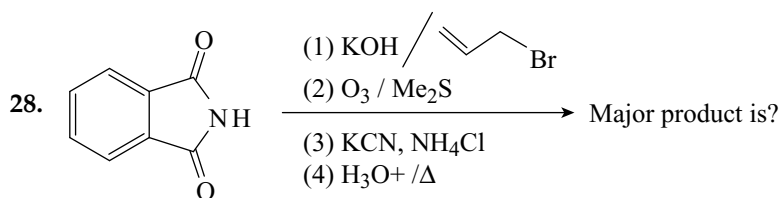
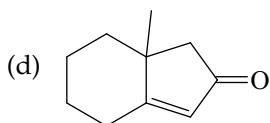
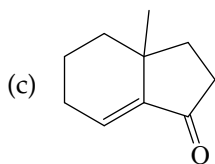
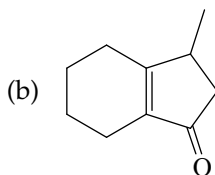
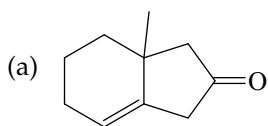
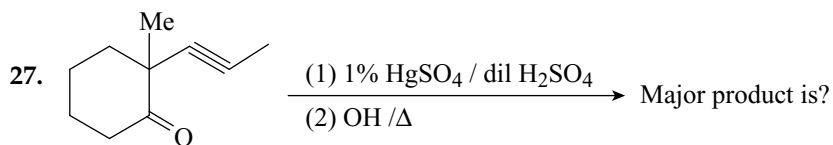
(d) None of these

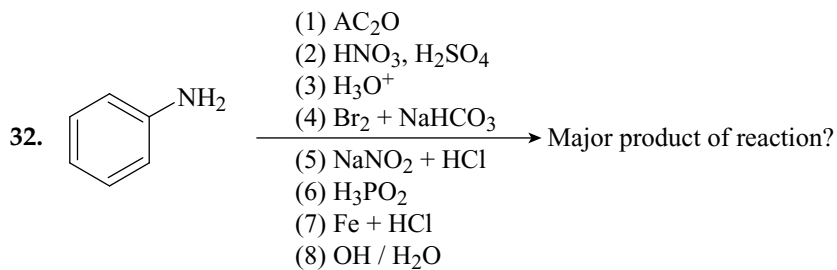
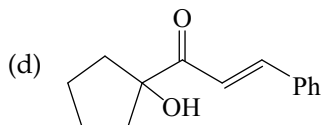
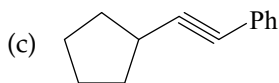
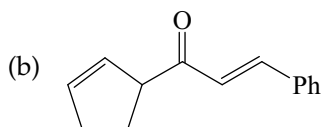
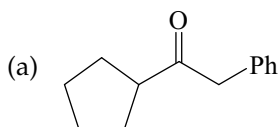
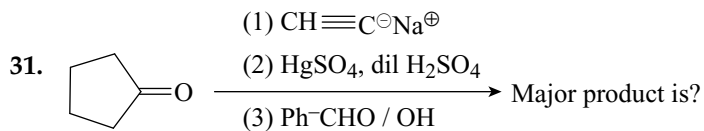
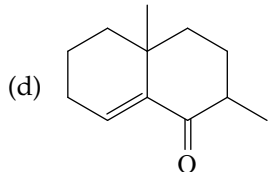
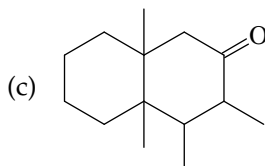
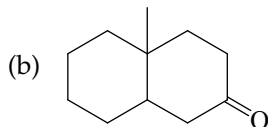
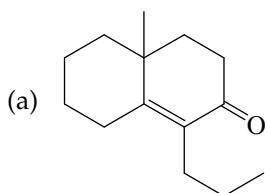
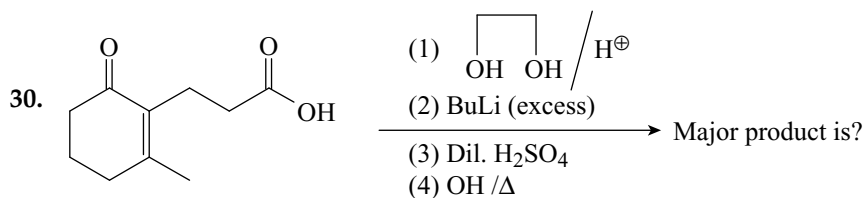


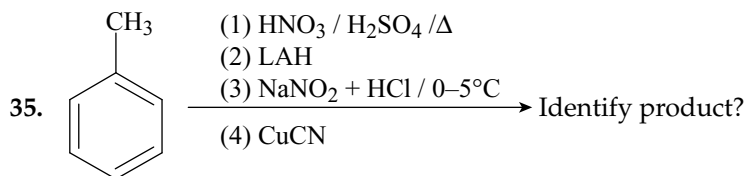
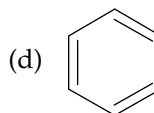
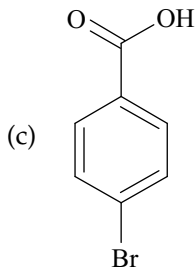
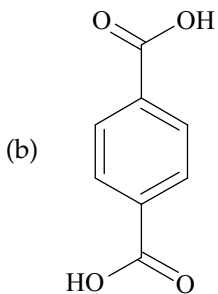
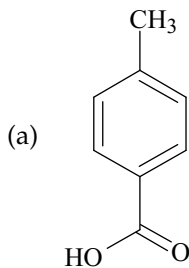
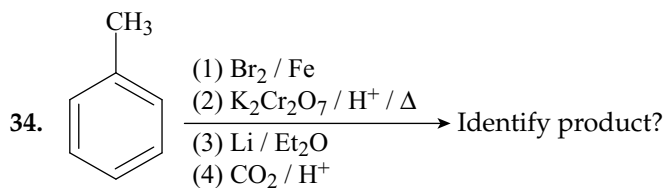
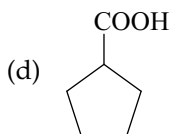
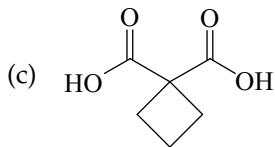
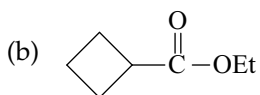
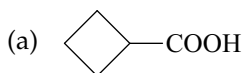
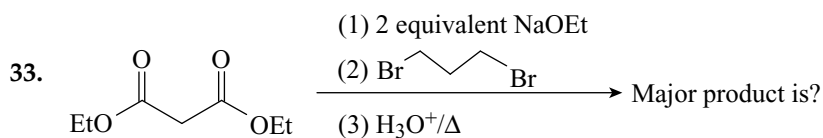
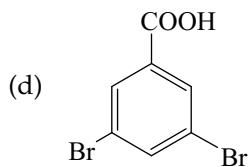
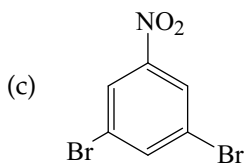
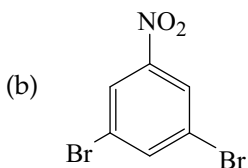
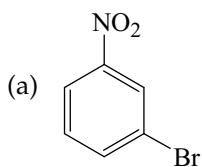


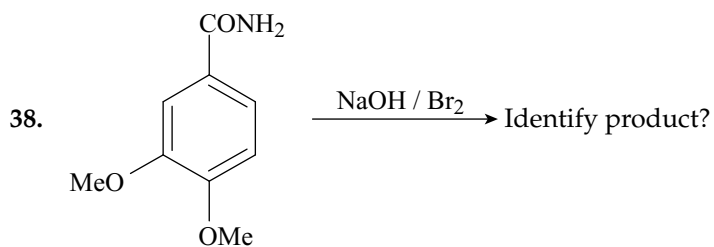
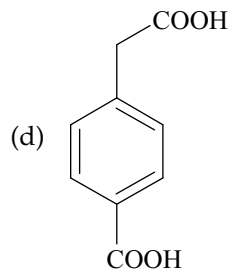
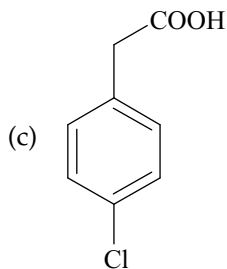
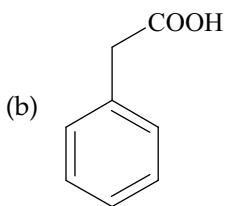
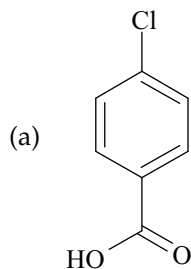
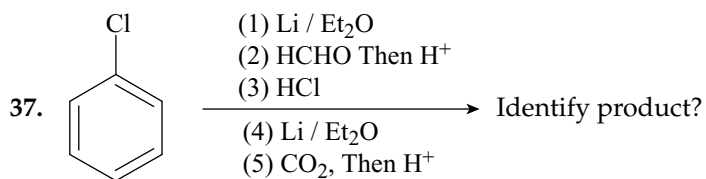
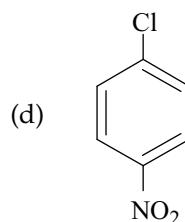
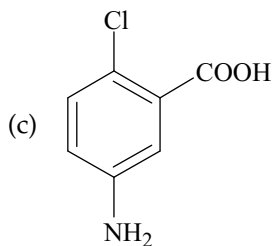
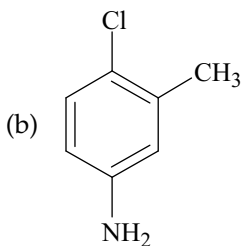
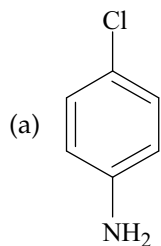
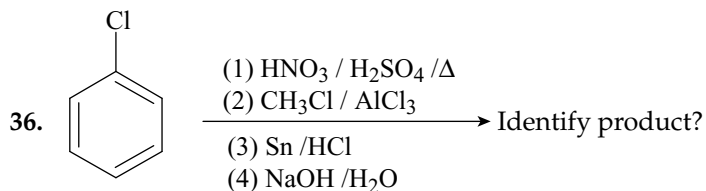
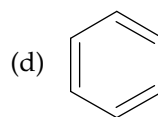
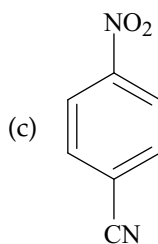
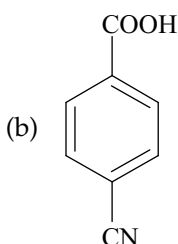
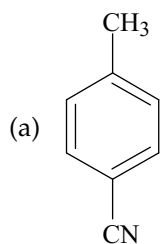


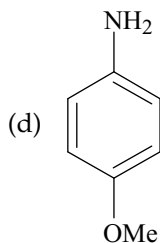
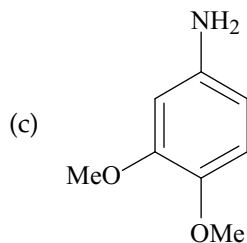
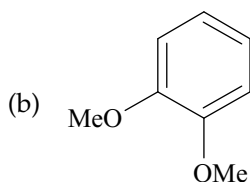
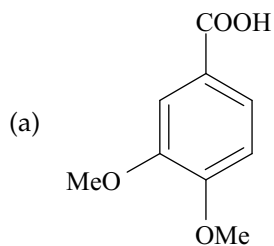




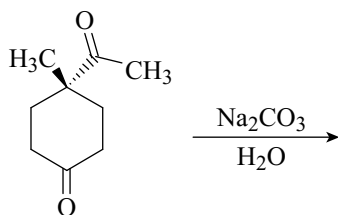




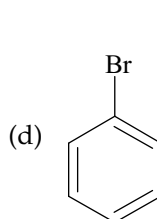
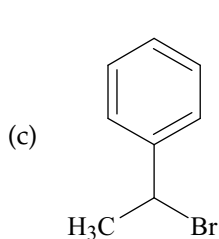
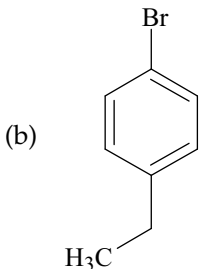
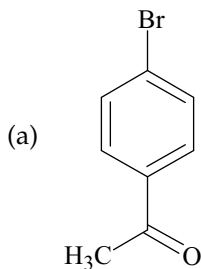
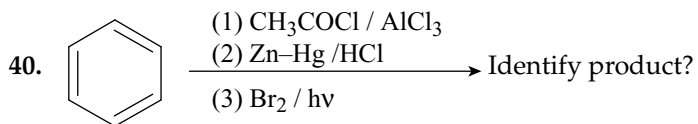
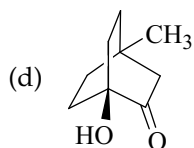
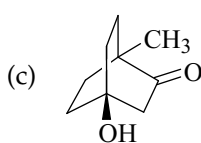
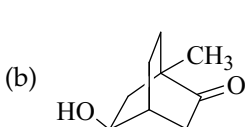
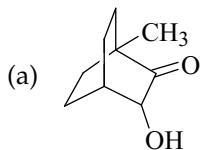


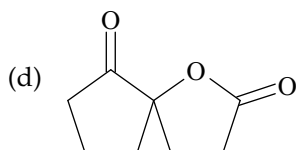
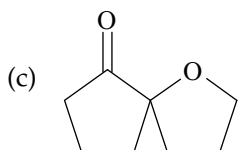
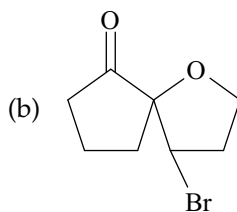
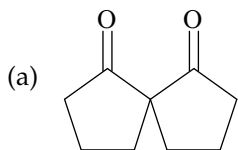
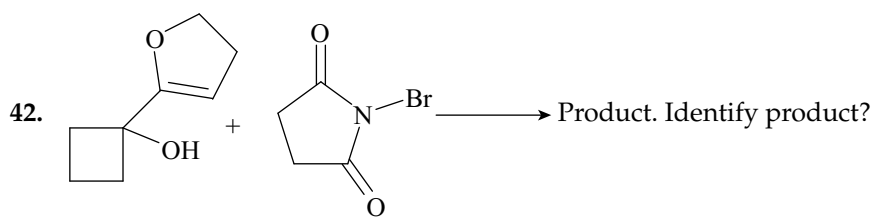
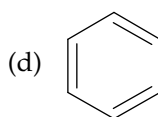
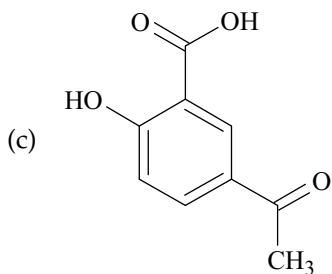
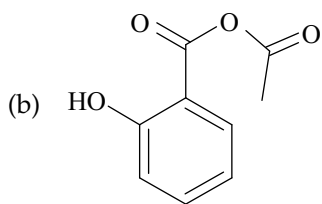
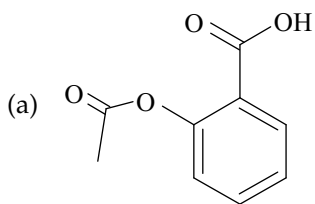
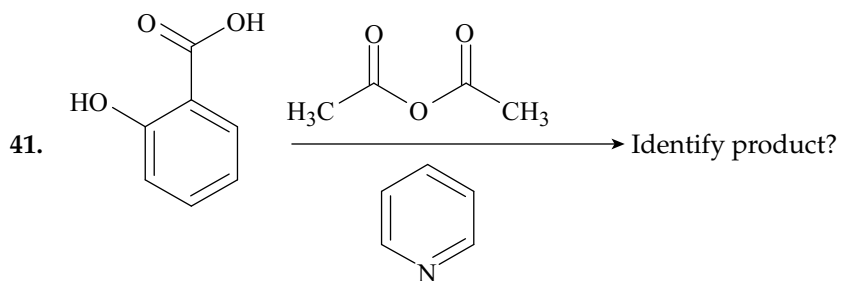


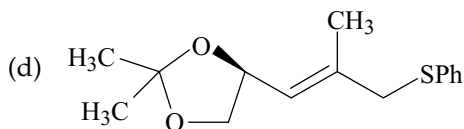
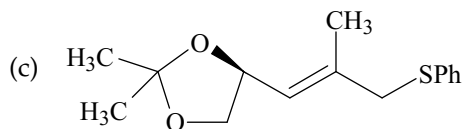
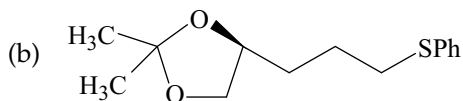
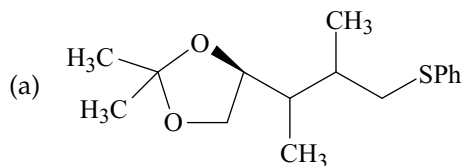
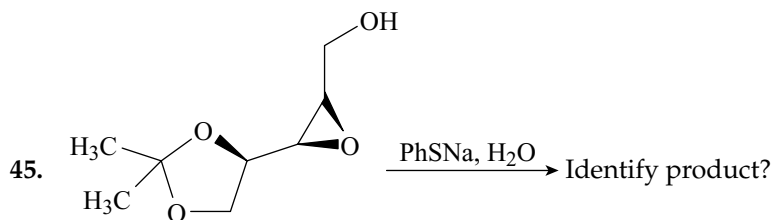
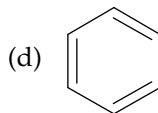
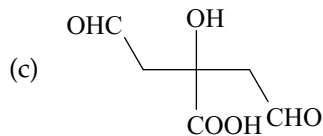
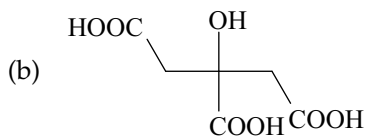
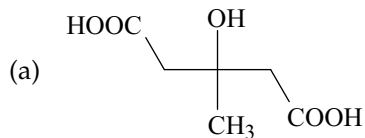
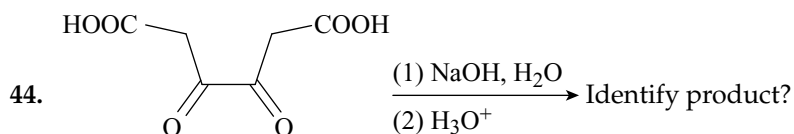
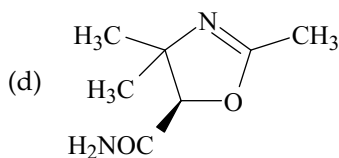
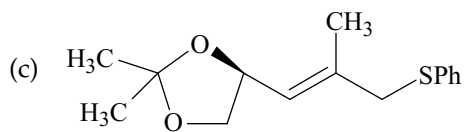
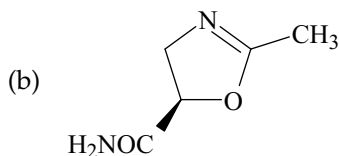
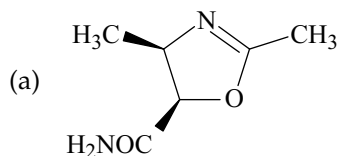
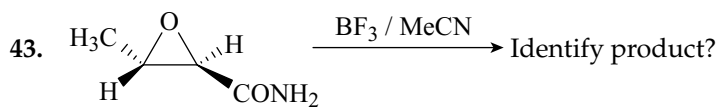
39. In the presence of a base, the compound below cyclizes to give a compound Y.

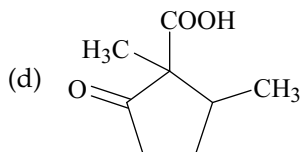
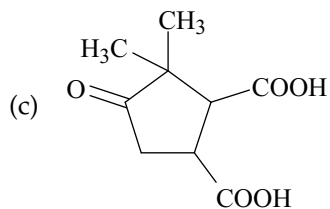
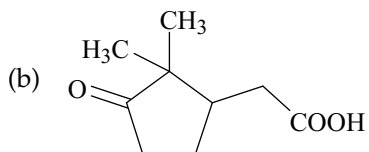
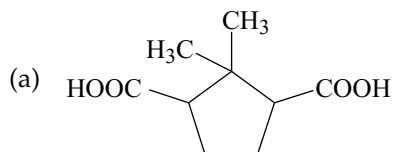
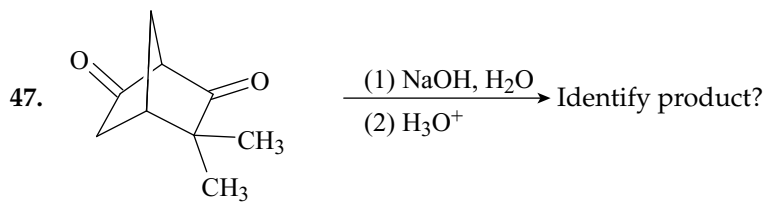
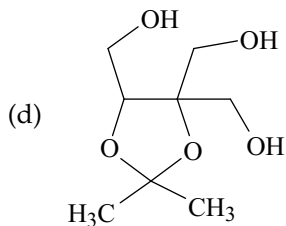
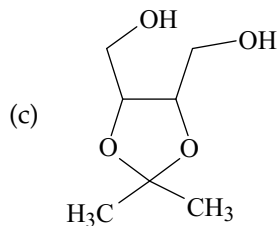
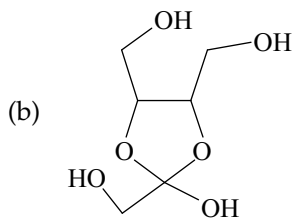
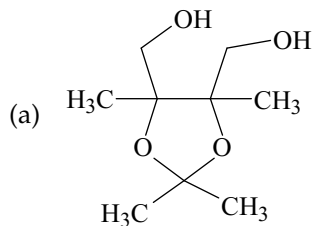
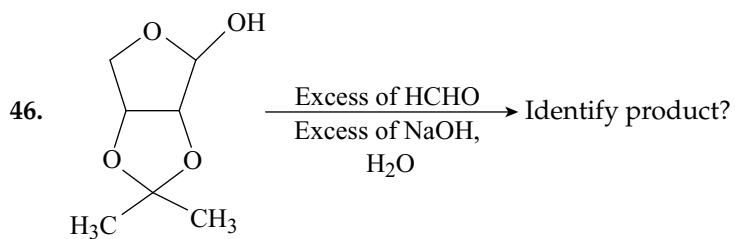


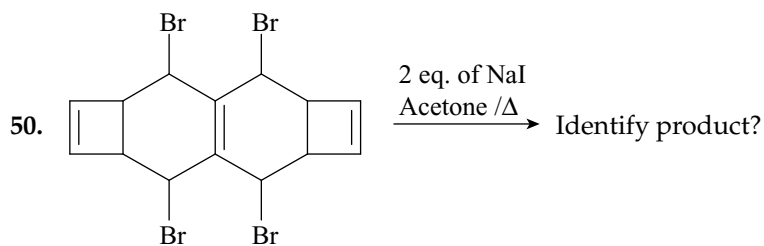
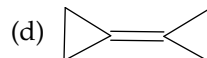
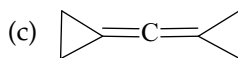
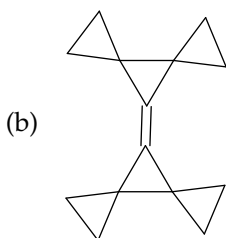
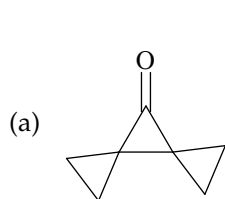
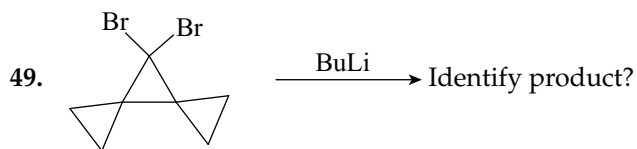
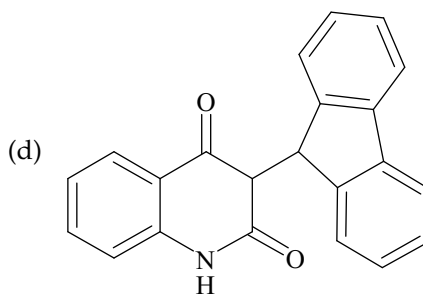
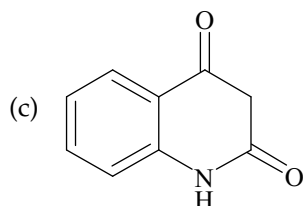
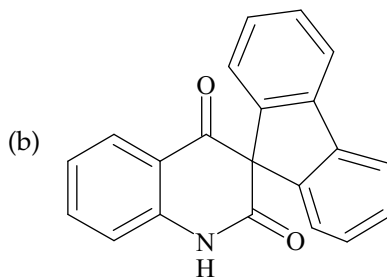
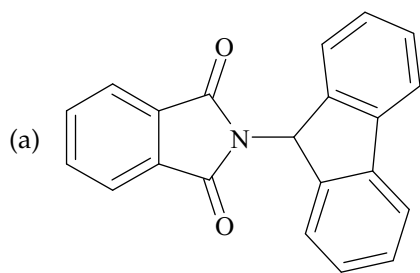
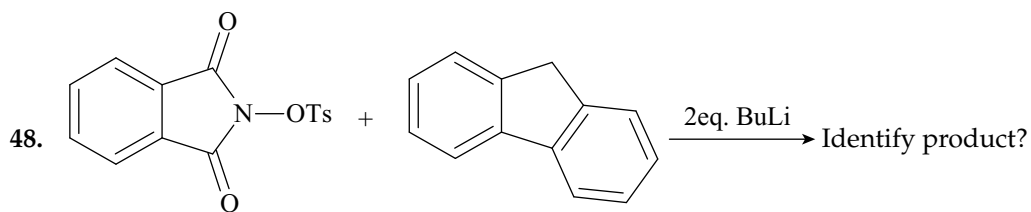
Identify the structure of compound Y.

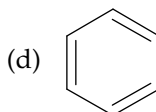
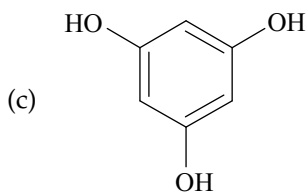
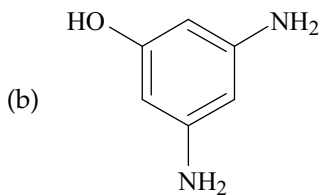
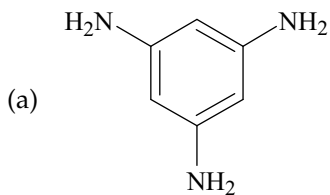
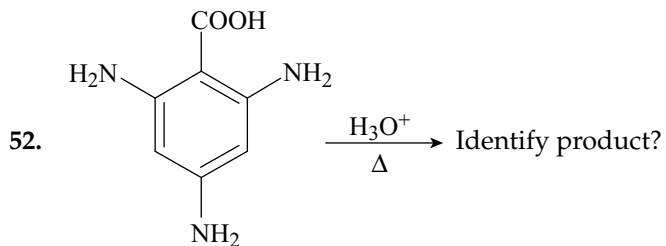
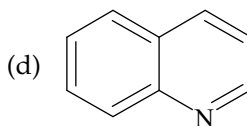
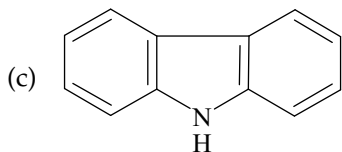
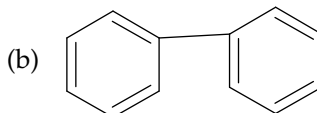
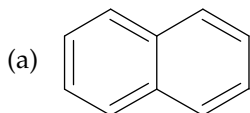
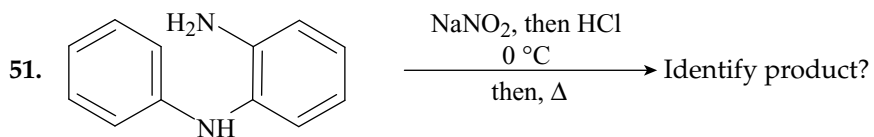
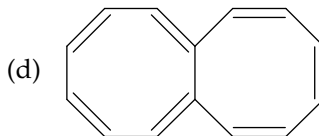
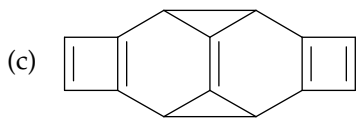
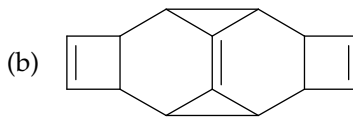
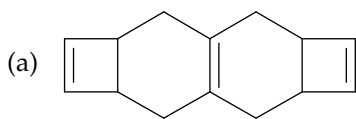


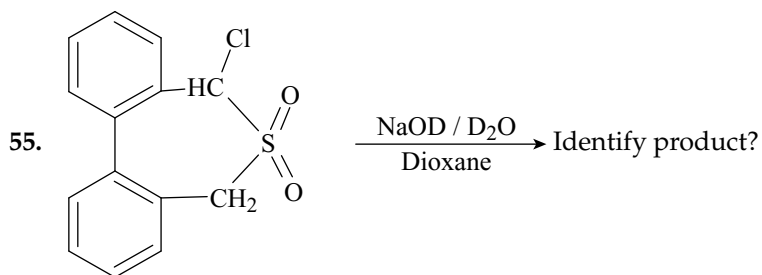
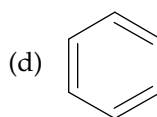
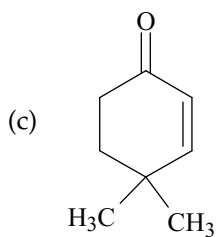
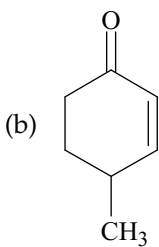
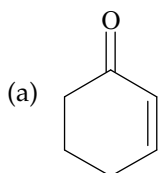
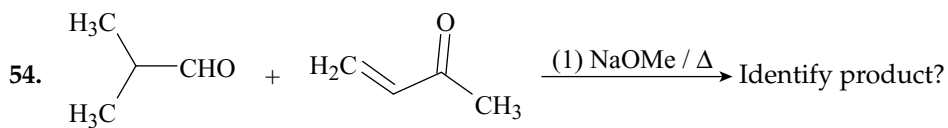
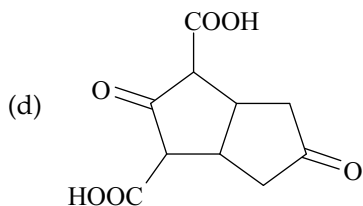
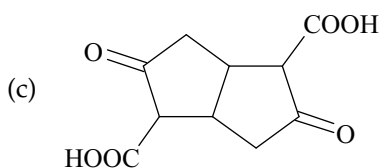
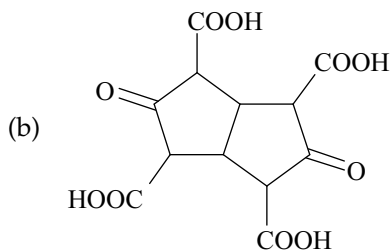
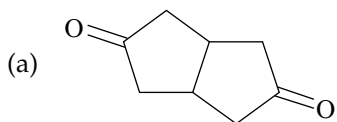
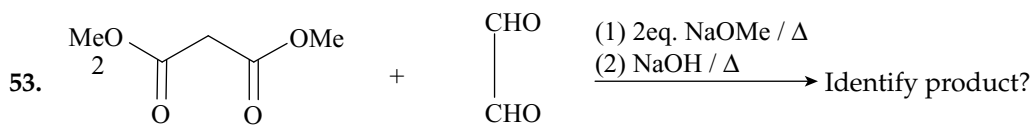


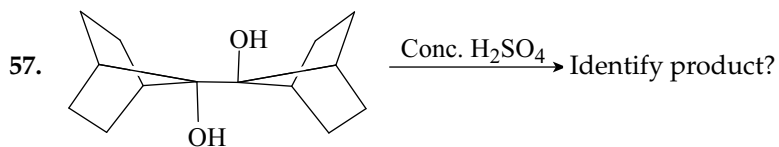
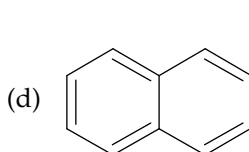
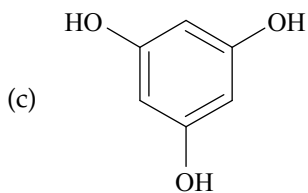
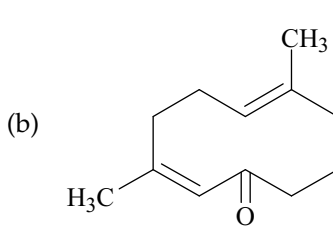
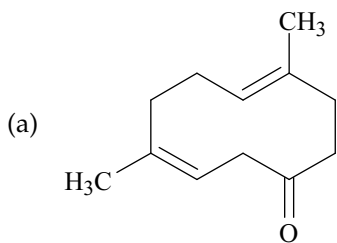
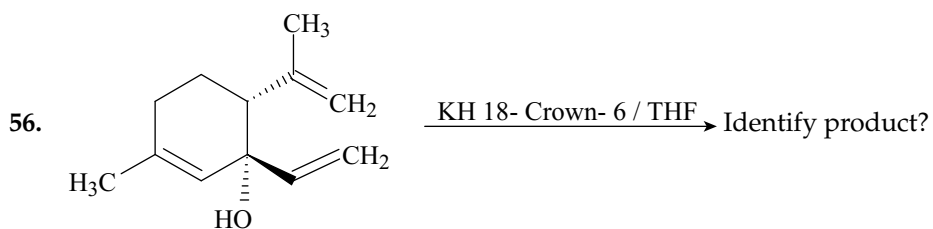
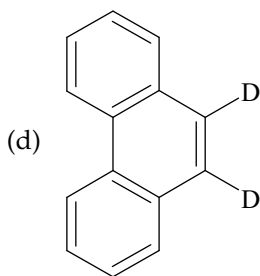
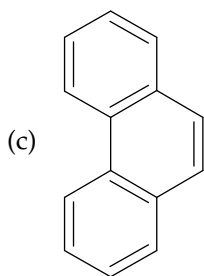
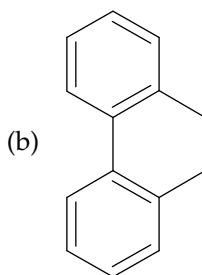
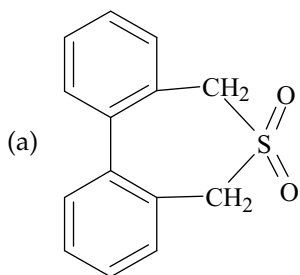


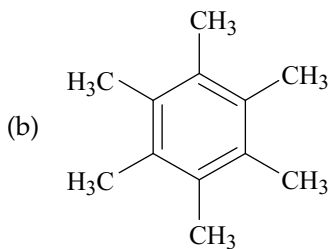
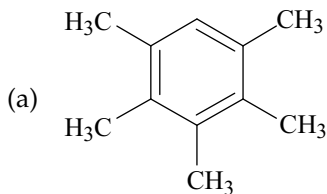
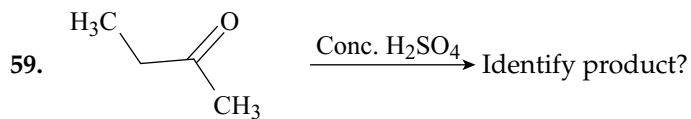
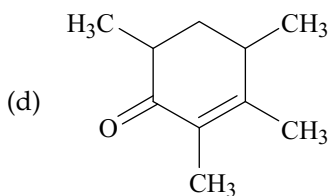
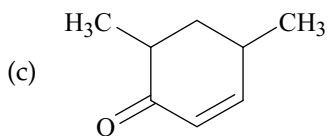
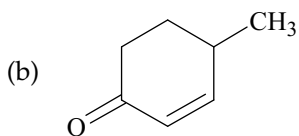
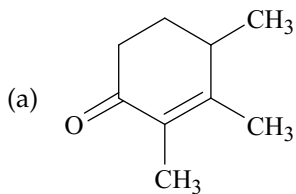
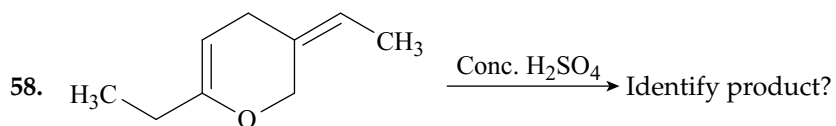
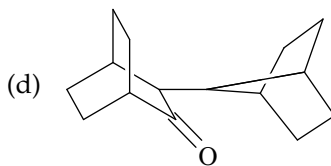
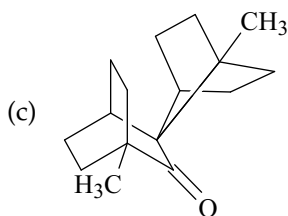
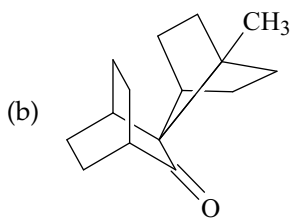
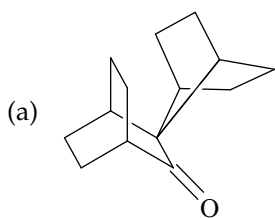


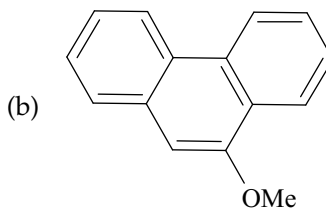
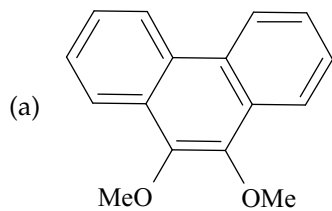
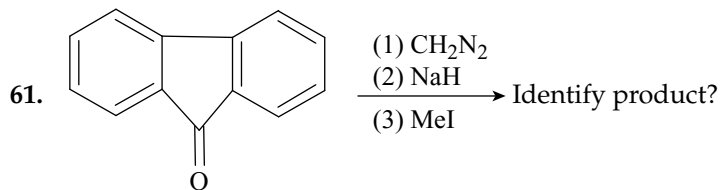
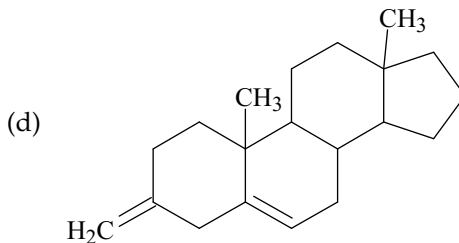
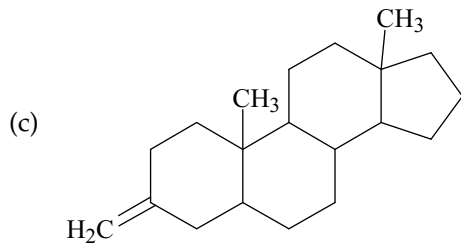
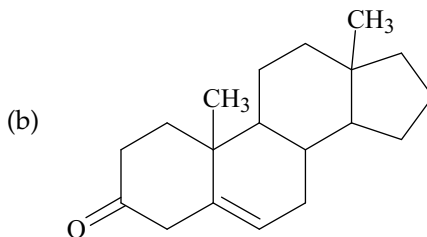
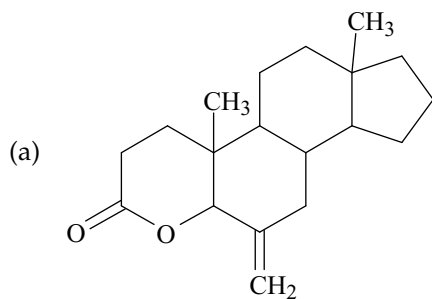
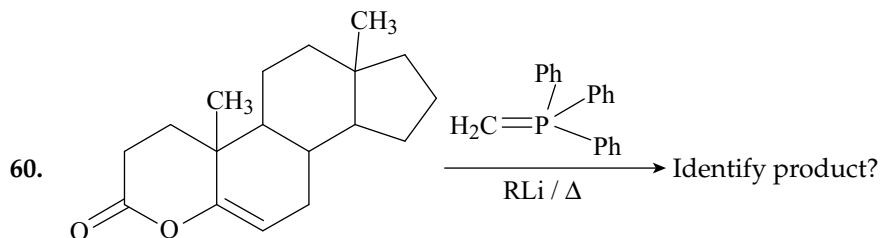
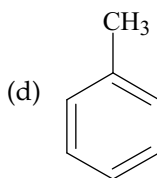
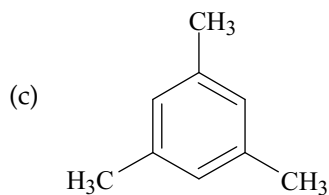


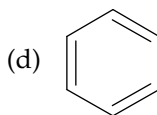
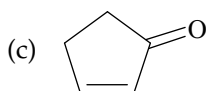
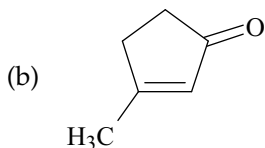
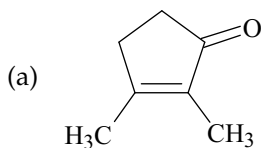
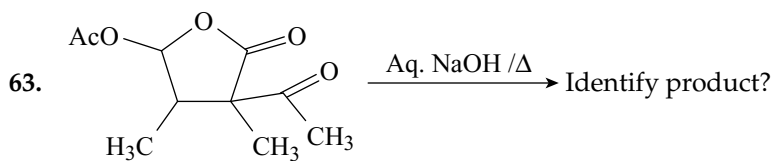
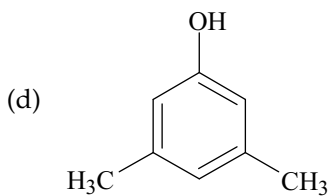
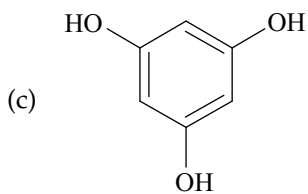
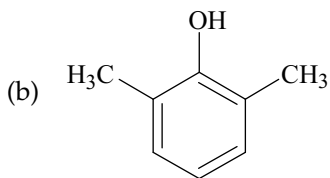
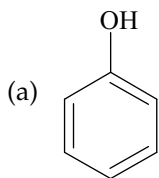
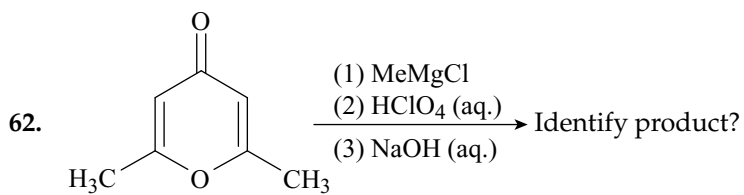
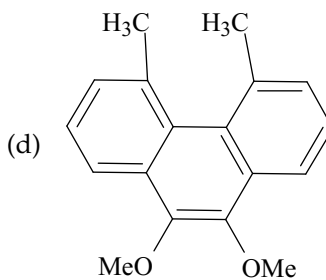
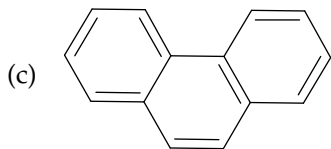


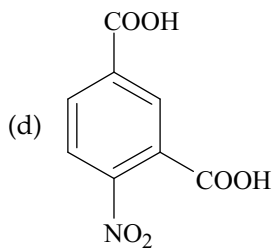
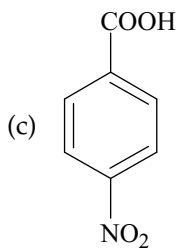
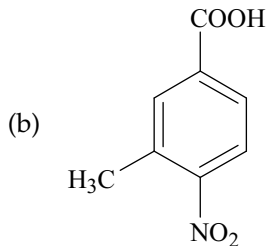
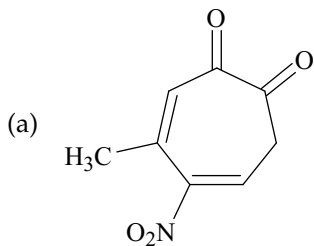
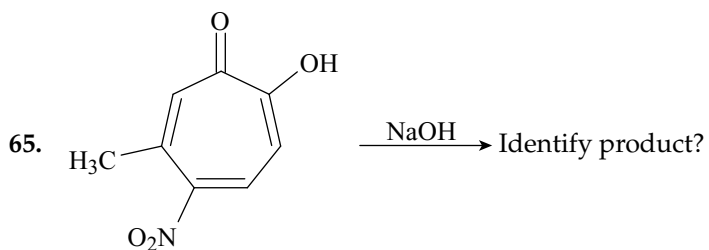
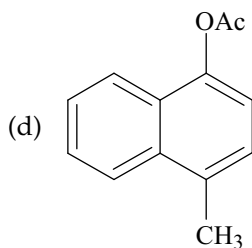
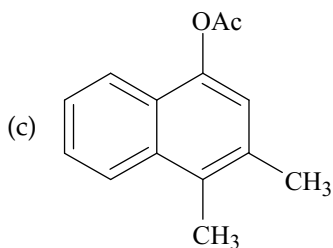
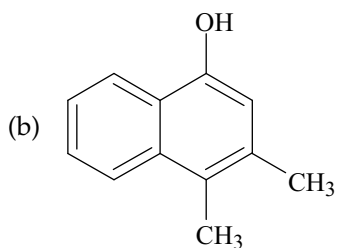
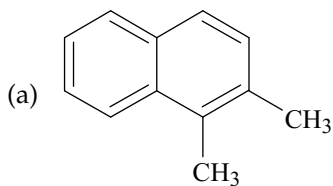
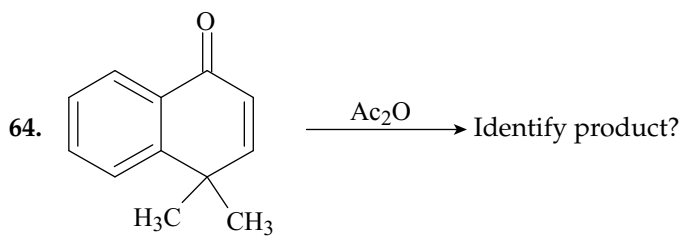


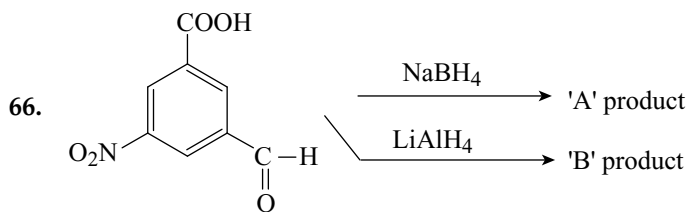




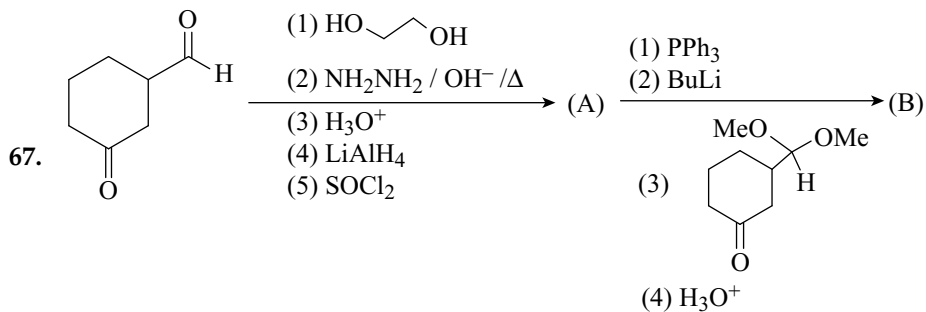
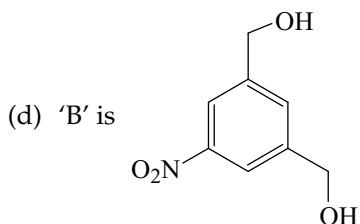
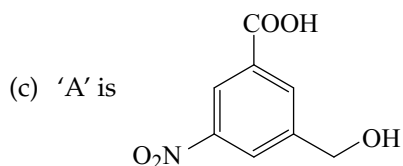
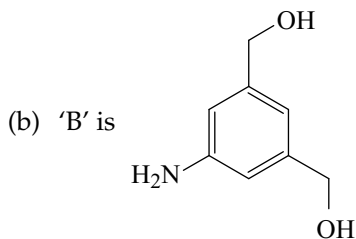
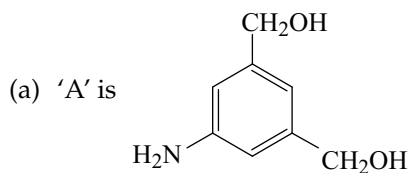




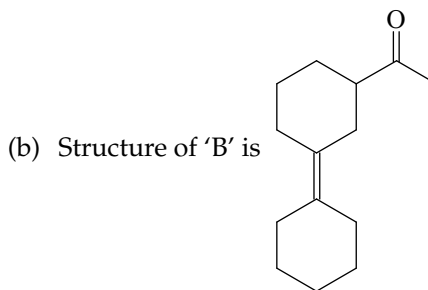
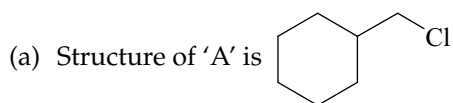




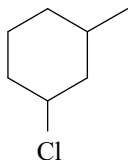
Identify true statement for A and B products.



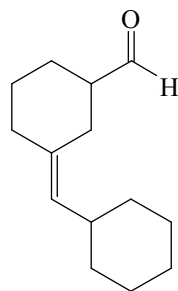
Identify correct statement for (A) and (B) products.



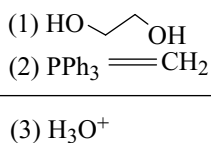
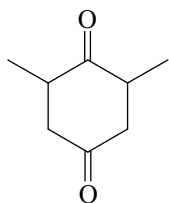
(c) Structure of 'A' is



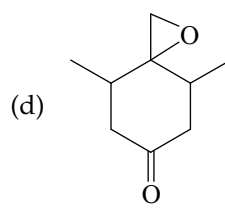
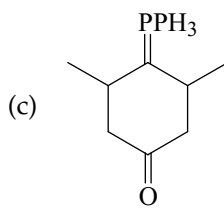
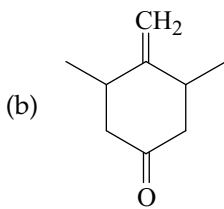
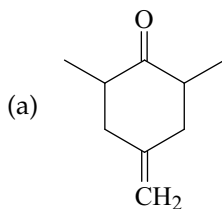
(d) Structure of 'B' is



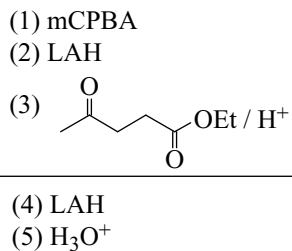
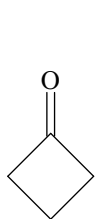
68.



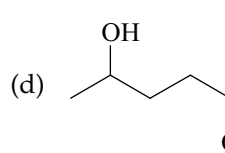
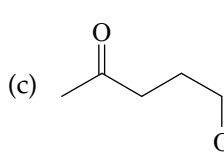
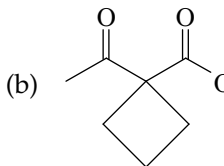
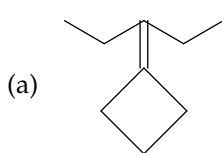
Major product is?



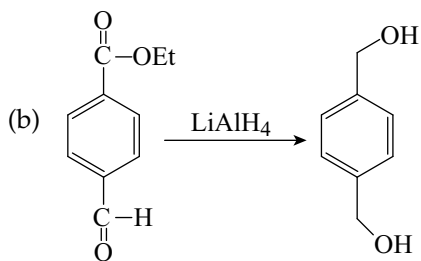
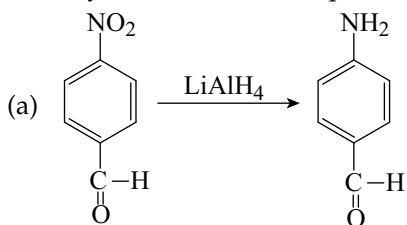
69.

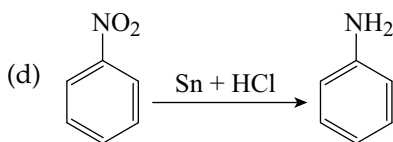
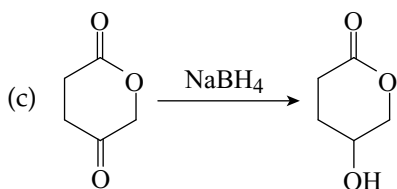


Major product is?

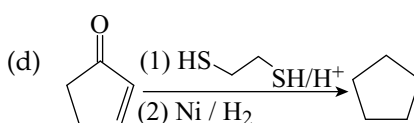
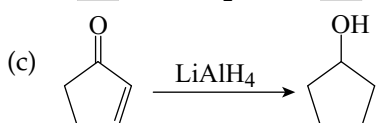
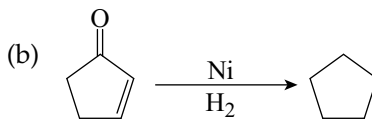
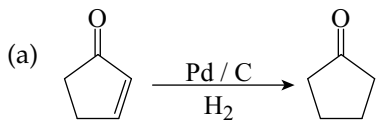


70. Identify correct reaction sequence

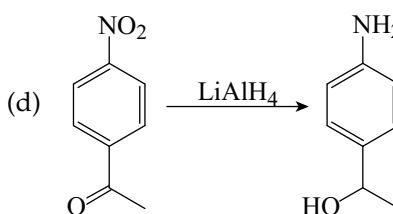
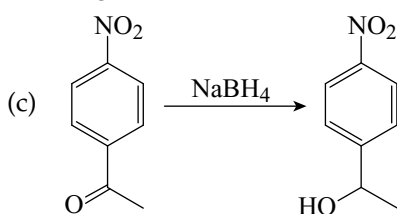
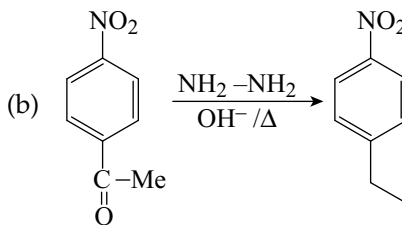
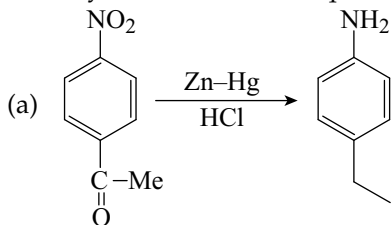




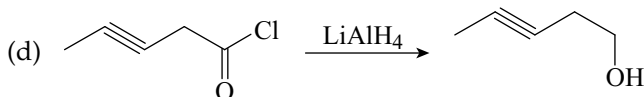
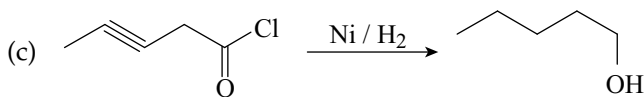
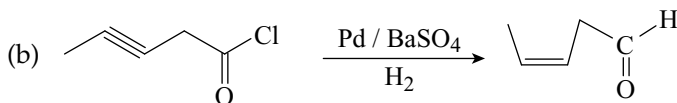
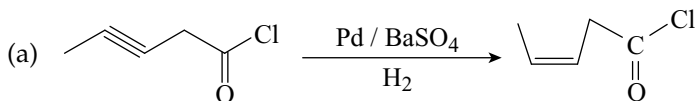
71. Identify correct reaction sequence

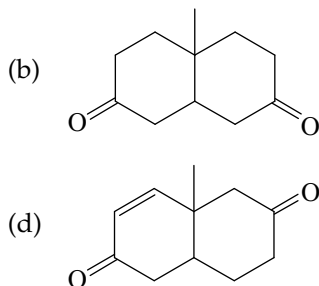
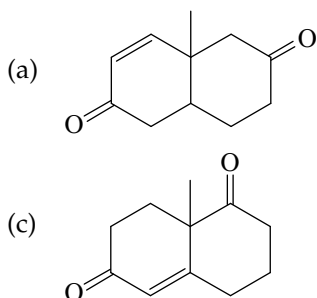
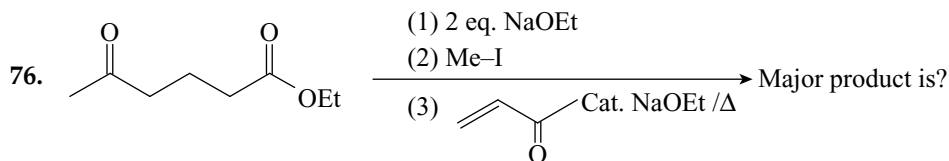
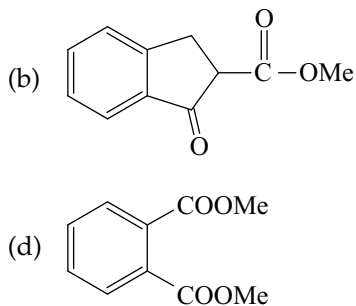
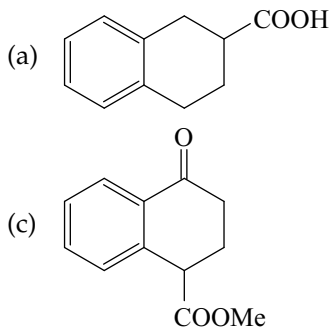
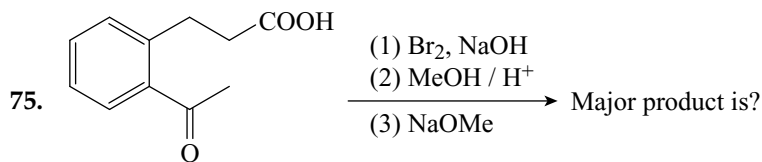
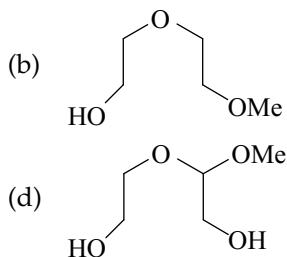
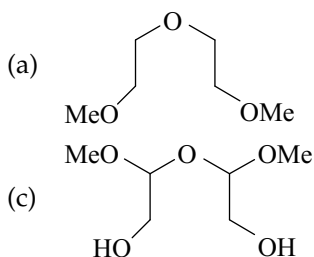
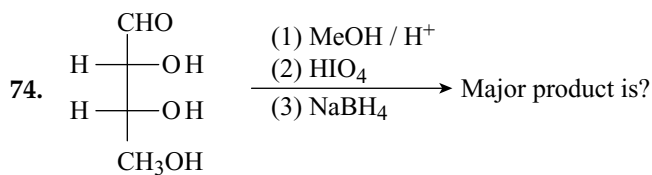


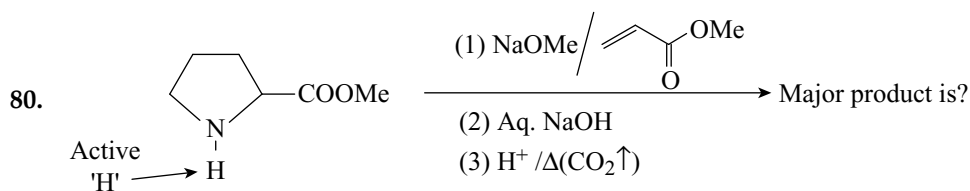
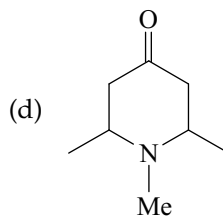
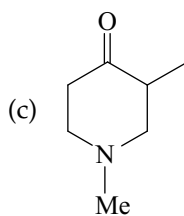
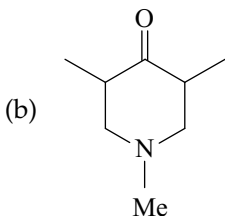
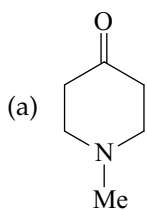
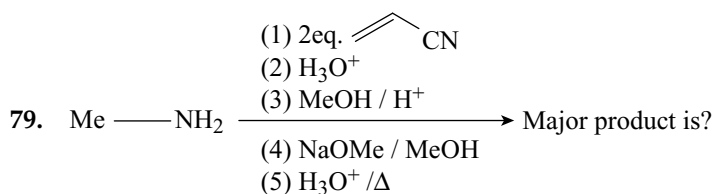
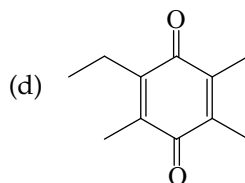
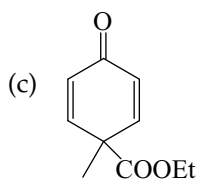
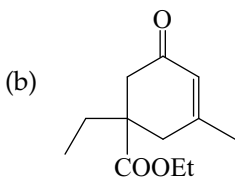
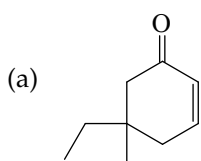
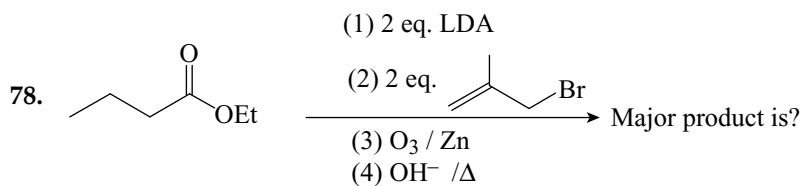
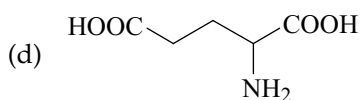
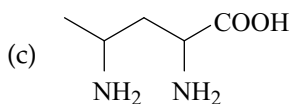
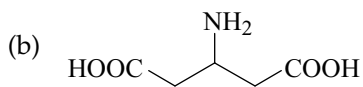
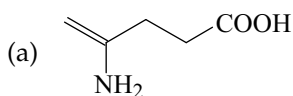
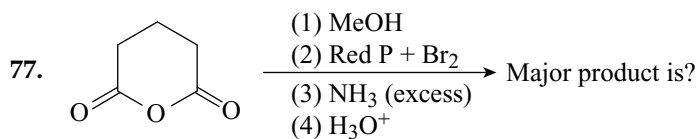
72. Identify correct reaction sequence

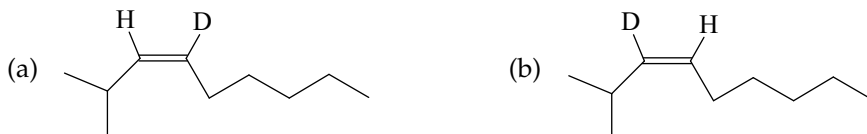
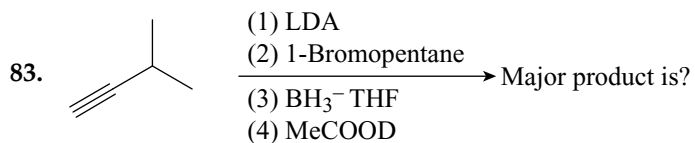
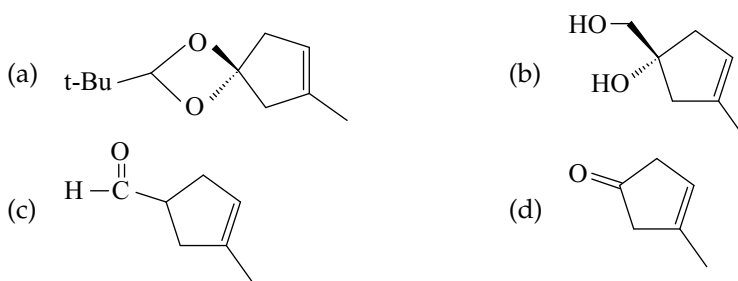
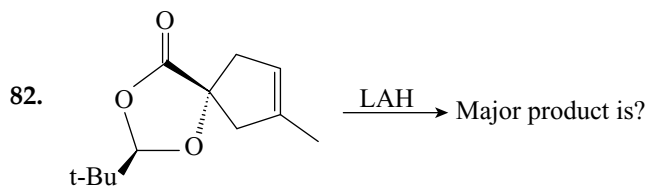
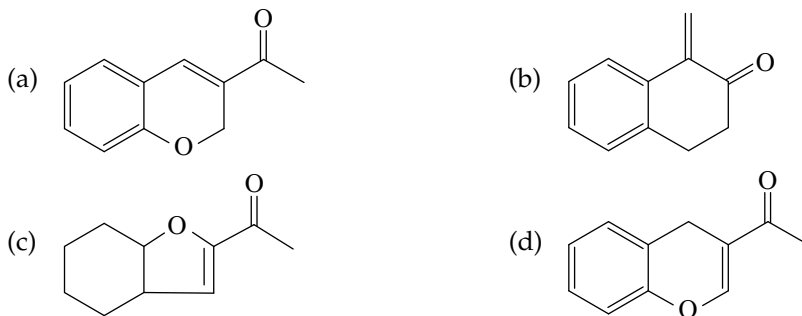
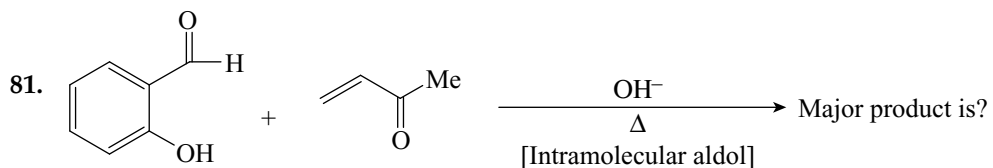
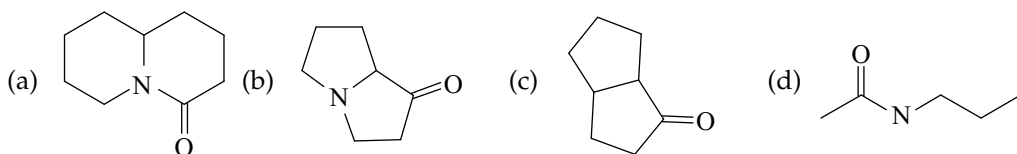


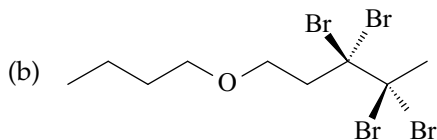
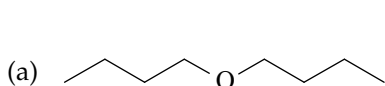
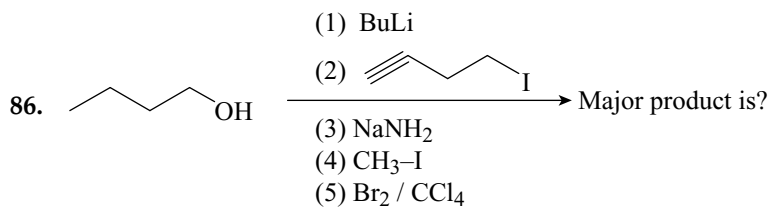
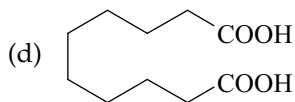
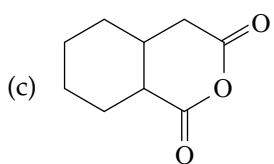
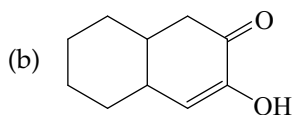
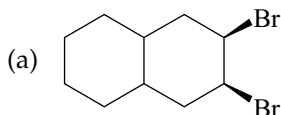
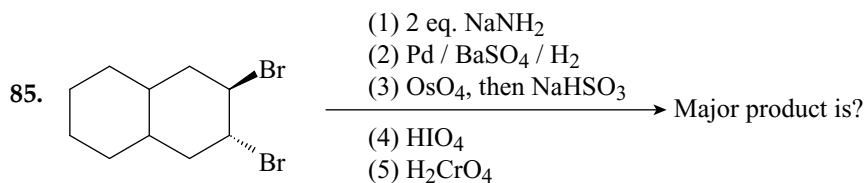
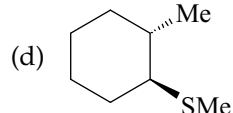
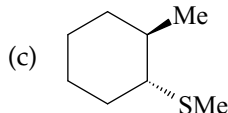
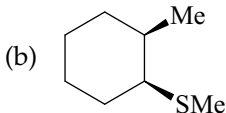
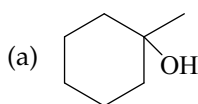
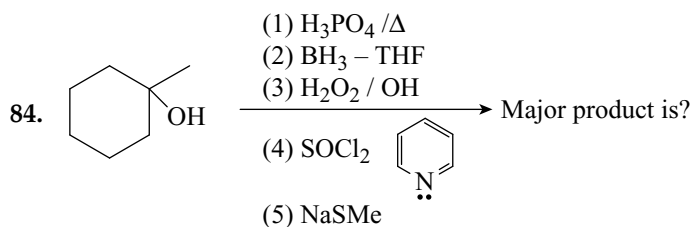
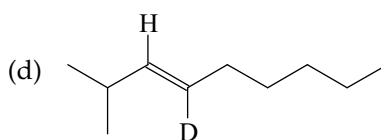
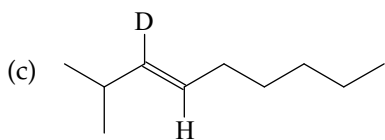
73. Identify correct reaction sequence

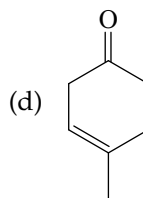
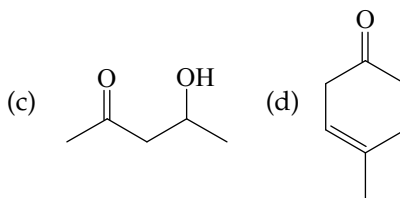
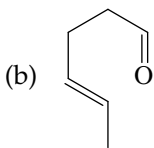
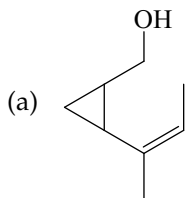
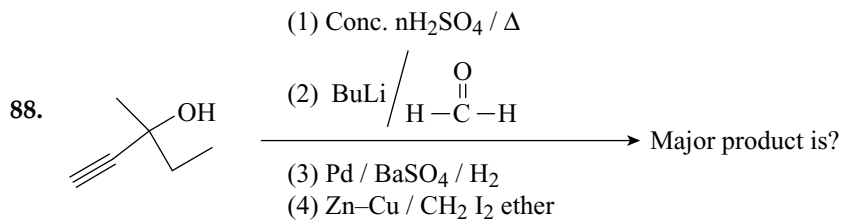
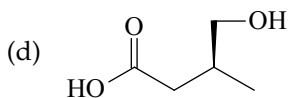
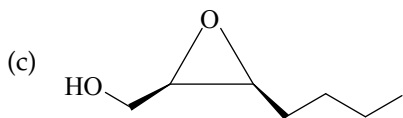
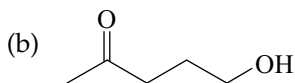
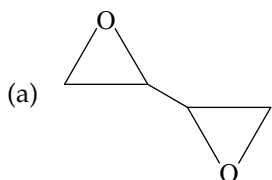
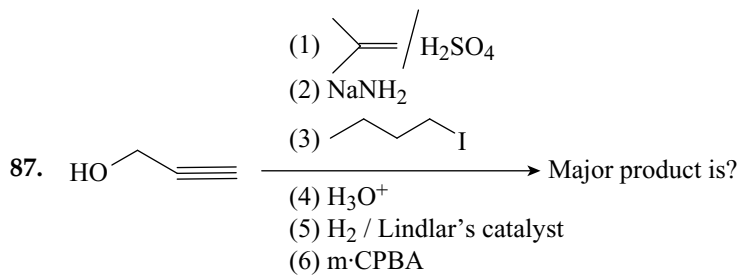
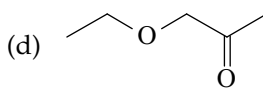
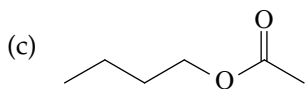




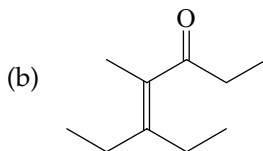
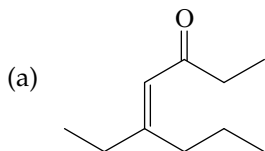


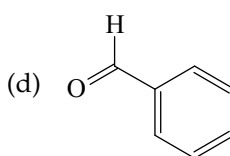
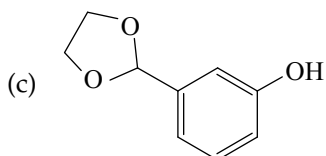
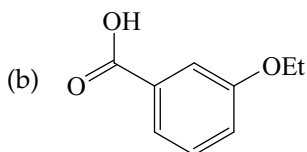
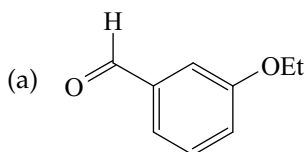
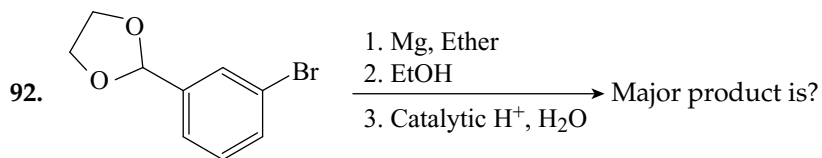
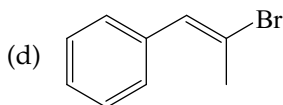
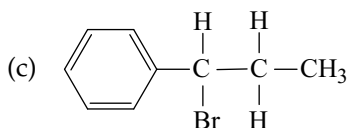
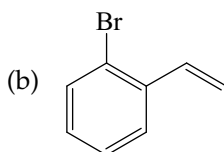
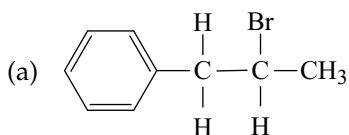
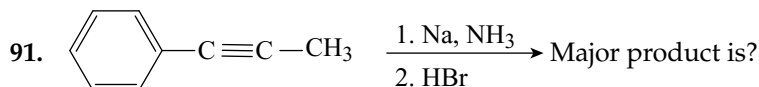
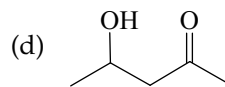
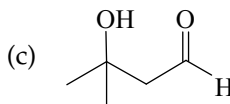
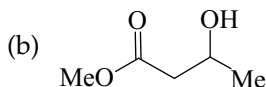
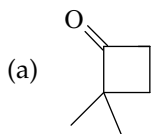
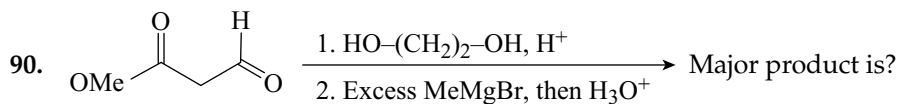
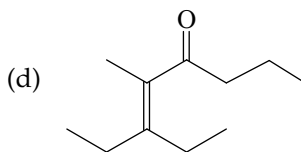
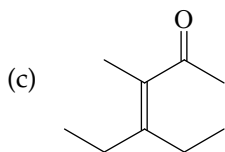




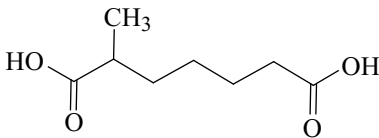
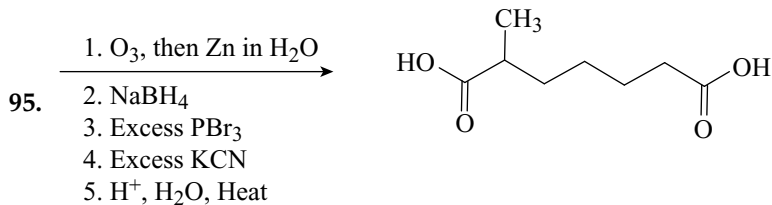
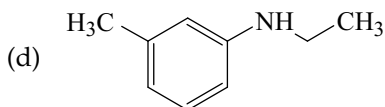
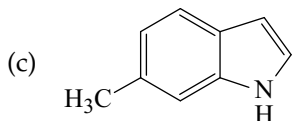
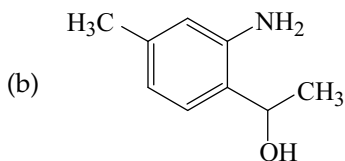
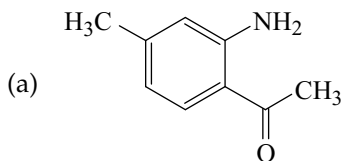
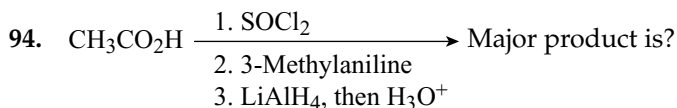


89. Which of the following structures is the product from the self-condensation of two molecules of pentan-3-one?

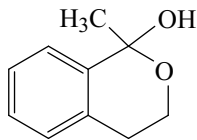
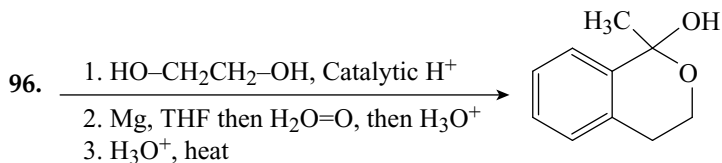
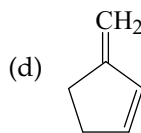
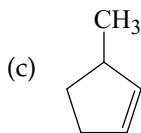
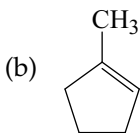
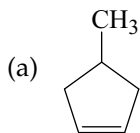




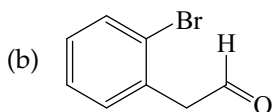
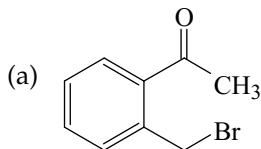
93. Select all of the following statements about the Aldol experiment that are true:
- The ketone was added to a solution containing the aldehyde and base.
 - The reaction can be catalyzed by an acid.
 - The electrophile is the enolate ion.
 - The aldehyde is used in excess.

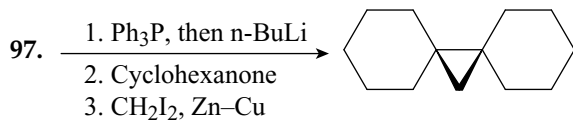
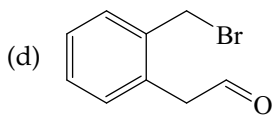
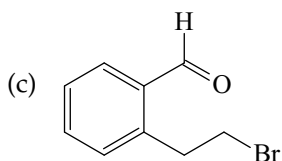


Identify reactant?

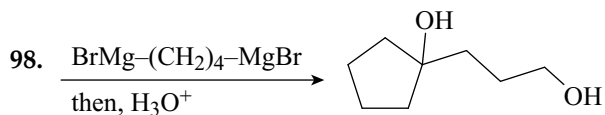
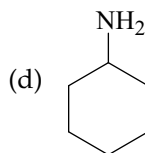
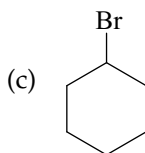
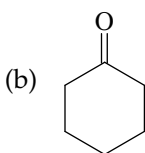
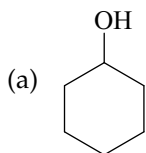


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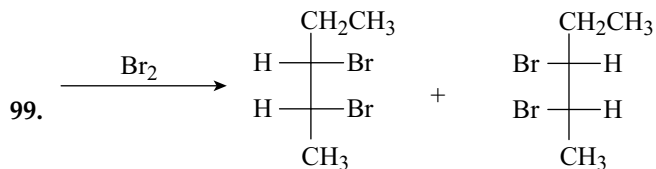
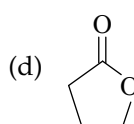
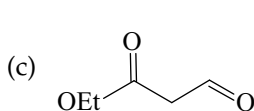
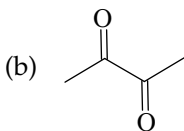
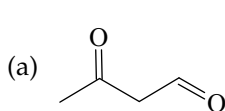




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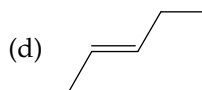
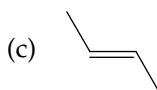
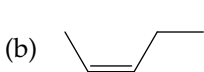
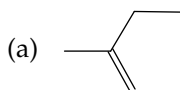


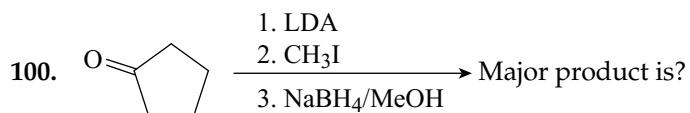
Identify reactant?

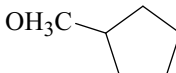
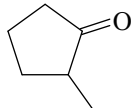
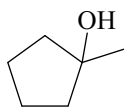
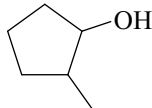


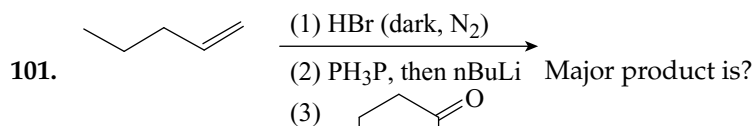
(50:50 of these enantiomers)

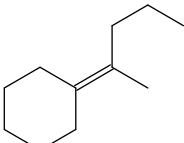
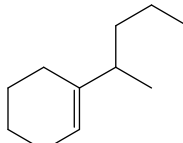
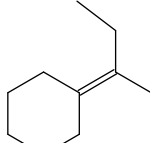
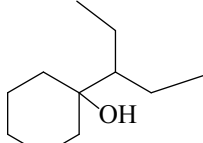
Identify reactant?

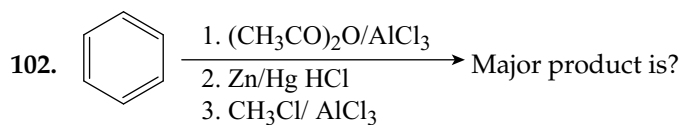


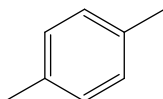
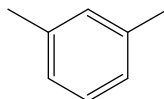
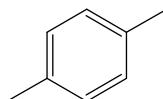
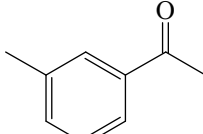


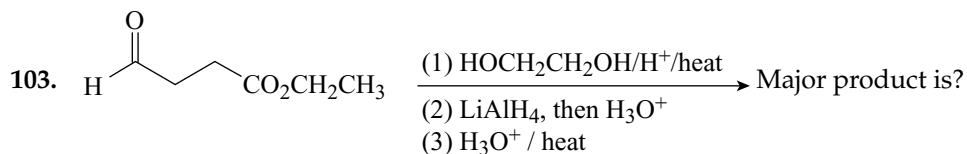
- (a)  (b)  (c)  (d) 

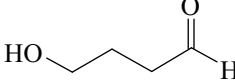
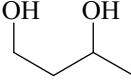
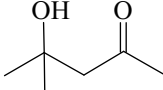
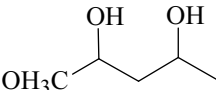


- (a)  (b)  (c)  (d) 



- (a)  (b)  (c)  (d) 

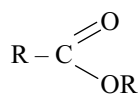


- (a)  (b)  (c)  (d) 

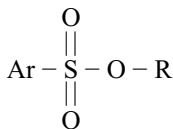
Comprehension Type

Passage 1

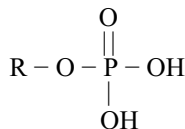
When the following three different types of esters are hydrolyzed in a basic medium,



Carboxylate



Sulphonate



Phosphate

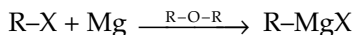
the hydroxide anion attacks the acyl carbon in carboxylates while it attacks the alkyl carbon in sulphonates leading to a difference in the site of cleavage. More interestingly, phosphate esters lie somewhat in between carboxylates and sulphonates in that cleavage can occur in either direction.

In an acidic solution, all the three types of phosphates (monoalkyl, dialkyl and trialkyl) are hydrolyzed to phosphoric acid, while in a basic solution only trialkyl phosphates undergo hydrolysis and only one alkoxy group is removed.

104. Which of the following factor explains the difference in attack of the nucleophile, OH^- on carboxylates and sulphonates?
- Sulphonate anions are weakly basic and hence good leaving groups.
 - Carboxylate anions are strongly basic and hence poor leaving groups.
 - Both (a) and (b)
 - None of the these
105. Competition between phosphorus and alkyl carbon to nucleophilic attack is due to the fact that
- Phosphorus can accept an additional pair of electrons.
 - Phosphoric acid lies between carboxylic acid and sulphonic acid.
 - Both (a) and (b)
 - None of the these
106. The rate of hydrolysis of monoalkyl phosphates tends to with increase in pH.
- Decrease
 - Increase
 - Remains unaffected
 - None of these
107. In an aqueous solution, a monoalkylphosphate ester can exist as
- A neutral ester
 - A monoanion and dianion
 - A monoanion, dianion and protonated ester
 - A monoanion, dianion, protonated ester and neutral ester

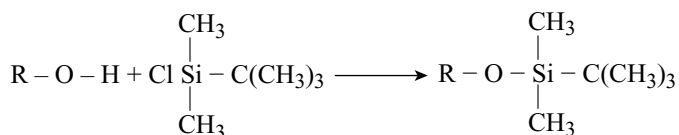
Passage 2

Grignard reagents (RMgX) are prepared by the reaction of an organic halide and magnesium metal in ether solvent.

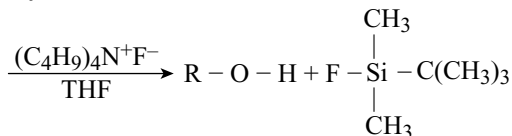


The solvent (usually diethyl ether or tetrahydrofuran) plays a crucial role in the formation of a Grignard reagent. Alkyl halides are more reactive than aryl and vinyl halides. Indeed, aryl and vinyl chlorides do not form Grignard reagent in diethyl ether.

However, an alkyl halide containing an alcoholic $-\text{OH}$ group can be converted to Grignard reagent by first protecting the $-\text{OH}$ group to tert-butyldimethylsilyl ether which is inert to Grignard reagent. The protecting group is finally liberated by treatment with fluoride ion.

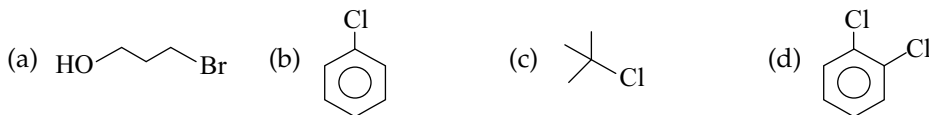


tert-butyldichlorodimethylsilane



108. Grignard reactions generally occur in dry ether because
- The stronger acid diethyl ether will displace the weaker RH acid from its salt.
 - The stronger acid H_2O will displace the weaker acid RH from its salt.
 - Water slows down the reaction.
 - Water mixes with ether preventing ether to perform its function.

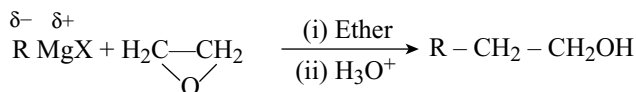
109. Grignard reagent cannot be prepared from



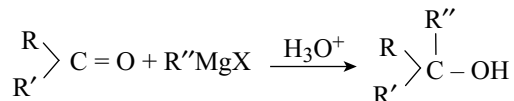
110. $\text{H}_2\text{N}(\text{CH}_2)_3\text{Br}$ cannot be converted into corresponding Grignard reagent because of
- Reaction between $-\text{NH}_2$ and $-\text{Br}$ groups present in the same molecule
 - Strong nucleophilic character of the Grignard reagent
 - Strong basic nature of the Grignard reagent
 - All the three factors
111. The function of tetrahydrofuran in the preparation of Grignard reagent is that it
- Acts as a solvent
 - Helps in maintaining the reactivity of magnesium
 - Both (a) and (b)
 - None of these

Passage 3

Grignard reagents are powerful nucleophiles and strong bases. They act as nucleophiles by attacking a variety of compounds including saturated and unsaturated carbon atoms. Examples of reaction on saturated carbon include oxiranes (epoxides) which form alcohols as the final product.



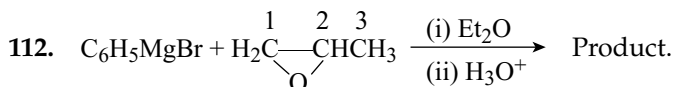
Examples of reaction on unsaturated carbon are attack on >C=O , $\text{-C}\equiv\text{N}$, >C=S , etc.



When R and R' = H, product is 1° alcohol.

When R and R' = Alkyl group, product is 3° alcohol.

When one R or R' is alkyl, product is 2° alcohol.



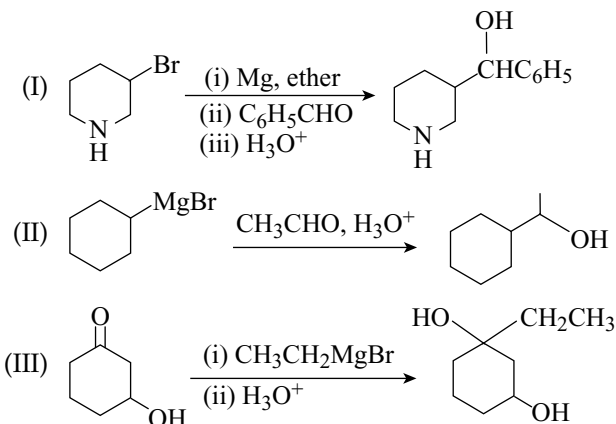
Here, the nucleophile C_6H_5^- attacks

- (a) C₁ (b) C₂ (c) C₃ (d) Any of the three

113. Epoxides react with Grignard reagent to form

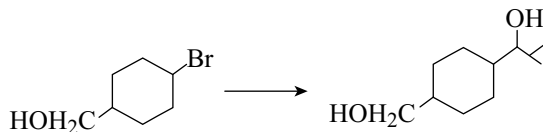
- (a) Primary alcohols (b) Secondary alcohols
(c) Tertiary alcohols (d) Any of the three

114. On the basis of the above two passages, predict which of the reaction is feasible.



- (a) Only (II) (b) (I) and (II) (c) All the three (d) None of three

115. On the basis of the above two passages, steps involved in the following conversion are



- (a) Protection of the -OH group, followed by reaction with $\text{H}_2\text{C}=\text{CHCH}_3$
 (b) Protection of the -OH group, followed by reaction with $\text{CH}_3\text{CH}_2\text{CHO}$
 (c) Protection of the -OH group, followed by reaction with $(\text{CH}_3)_2\text{CHCHO}$
 (d) Reaction is not feasible to give quantitative yield

Passage 4

Alkyl halides and alcohols easily undergo nucleophilic substitution either through $\text{S}_{\text{N}}1$ or $\text{S}_{\text{N}}2$ mechanism. The relative ease of these two processes depends upon the nature of the substrate (alkyl group as well as leaving group), nature of nucleophile and also upon the nature of solvent.

$\text{S}_{\text{N}}1$ mechanism involves the formation of carbocation as intermediate while $\text{S}_{\text{N}}2$ mechanism involves the formation of a transition pentavalent state. $\text{S}_{\text{N}}1$ is the main mechanism in 3° alkyl halides and alcohols, while $\text{S}_{\text{N}}2$ mechanism is the path adopted by most of the 1° alkyl halides and 2° alkyl halides may follow $\text{S}_{\text{N}}1$ as well as $\text{S}_{\text{N}}2$.

116. Which of the following solvent will give maximum yield for an alkyl halide undergoing $\text{S}_{\text{N}}1$ mechanism?
 (a) Water (b) Ethanol (c) Diethyl ether (d) n-hexane
117. Rearrangement of alkyl groups occur when hydrogen halides react with alcohols except with most primary alcohols. The best explanation is that
 (a) The 1° carbocations are unstable and hence are not formed.
 (b) The 1° carbocations are unable to undergo rearrangement.
 (c) Both (a) and (b) are true
 (d) Both (a) and (b) are false
118. Neopentyl alcohol, $\text{Me}_3\text{CCH}_2\text{OH}$, reacts with HX according to
 (a) $\text{S}_{\text{N}}1$ mechanism (b) $\text{S}_{\text{N}}2$ mechanism
 (c) Both (a) and (b) (d) None

Passage 5

119. A chemist treated a compound X with NaOH in presence of acetone as solvent. However, he recovered the starting material as such, and instead isolated a small amount of the product A. The product A was shown to have C, H and O and it had a molecular weight of 116 g/mol. It gave a positive iodoform test and was found to be identical with a compound obtained by the aldol self-condensation of acetone.

Although the product A did not discharge colour of bromine in CCl_4 , its dehydration product B with hot sulphuric acid discharged bromine dissolved in CCl_4 .

120. What is the molecular weight of a compound that undergoes an aldol self-condensation reaction and whose dehydrated product has a molecular weight of 70?
 (a) 35 (b) 44 (c) 49 (d) 58
121. The aldol self-condensation of acetone is in equilibrium that favours acetone over its corresponding product. Which of the following conditions is most likely to shift the position of equilibrium toward product A?
 (a) By using a catalytic amount of NaOH.
 (b) By using only a catalytic amount of acetone.
 (c) By removing product A as soon as it is formed.
 (d) By increasing reaction temperature.
122. Which of the following compounds will give a positive iodoform test?
 (a) Only compound A (b) Only compound B
 (c) Both (a) and (b) (d) None of these
123. The compound X can be
 (a) $\text{CH}_3\overset{\text{O}}{\parallel}\text{CH}$ (b) $\text{CH}_3-\overset{\text{O}}{\parallel}\text{C}-\text{CH}_3$
 (c) HCHO (d) $\text{n-CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$

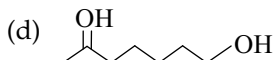
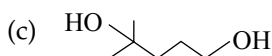
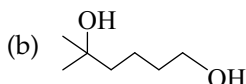
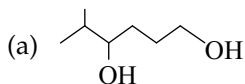
Passage 6

Amides undergo hydrolysis to yield carboxylic acid plus amine on heating in either aqueous acid or aqueous base. The conditions required for amide hydrolysis are more severe than those required for the hydrolysis of esters, anhydrides or acid chlorides, but the mechanism is similar (nucleophilic acyl substitution). Nucleophilic acyl substitutions involve a tetrahedral intermediate, hence these are quite different from alkyl substitution ($\text{RCH}_2\text{Br} \xrightarrow{\text{NaCN}} \text{RCH}_2\text{CN}$) which involves a pentavalent intermediate or transition state.

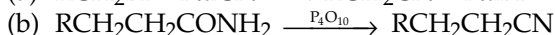
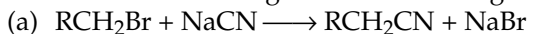
One of the important reactions of esters is their reaction with two equivalent of a Grignard reagent to give tertiary alcohols.

124. The mechanism involved during the hydrolysis of acid derivatives is
 (a) elimination-addition (b) addition-elimination
 (c) nucleophilic addition-elimination (d) electrophilic addition-elimination
125. Which of the following constitutes the best substrate during the acidic hydrolysis of amides?
 (a) $\text{R}-\overset{\text{O}}{\parallel}\text{C}-\text{NH}_2$ (b) $\text{R}-\overset{\text{O}}{\parallel}\text{C}-\overset{+}{\text{NH}}_3$ (c) $\text{R}-\overset{\text{OH}^+}{\parallel}\text{C}-\text{NH}_2$ (d) $\text{R}-\overset{\text{OH}^+}{\parallel}\text{C}-\overset{+}{\text{NH}}_3$
126. For which functional derivative of carboxylic acids, acidic hydrolysis is avoided?
 (a) Acid chlorides (b) Acid amides
 (c) Acid anhydrides (d) Esters

127. When  is treated with two equivalent of methyl magnesium iodide, the product that acidified the final product will be



128. Which of the following methods is more general for preparing nitriles?

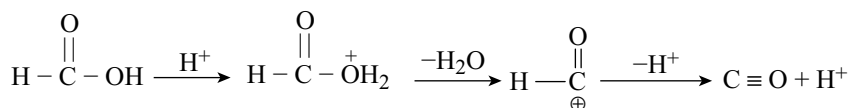


(c) Both (a) and (b)

(d) None of these

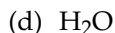
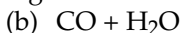
Passage 7

Methanoic acid, the first member of carboxylic acid series, when warmed with concentrated sulphuric acid decompose in the following way and evolve carbon monoxide

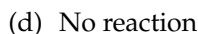
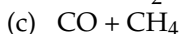
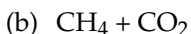
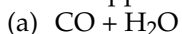


The driving force for this reaction lies in the fact that the $\text{HC} \equiv \text{O}^+$ ion is very unstable acid and thus easily loses H^+ .

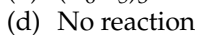
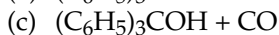
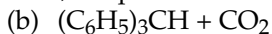
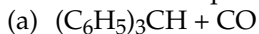
129. Formic acid on heating with conc. H_2SO_4 gives



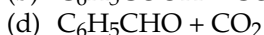
130. What happens when acetic acid is treated with conc. H_2SO_4 ?



131. If acetic acid is replaced by triphenylacetic acid, the product formed will be



132. If formic acid is replaced by benzoylformic acid, $\text{C}_6\text{H}_5\text{COCOOH}$, the product formed will be

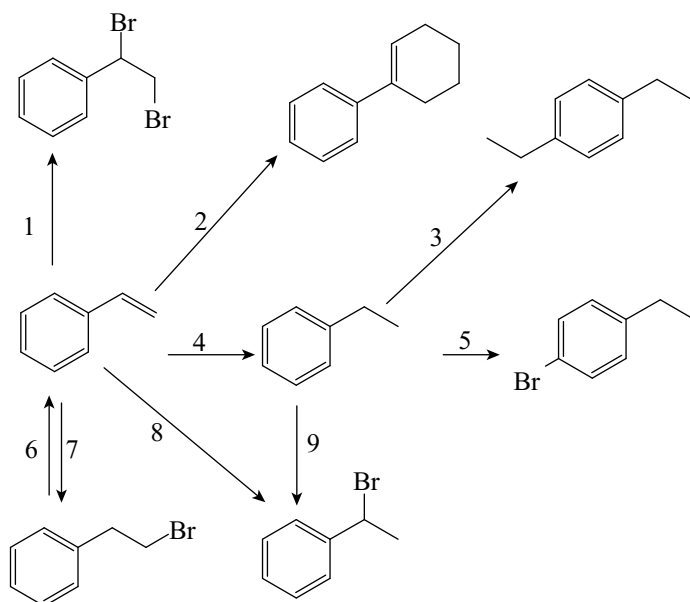


WORKBOOK EXERCISE 1

Matrix Type

Identify reagents (1 to 9) used in the following conversion from reagent present in the second Column II (A to L).

Column I (Conversion)

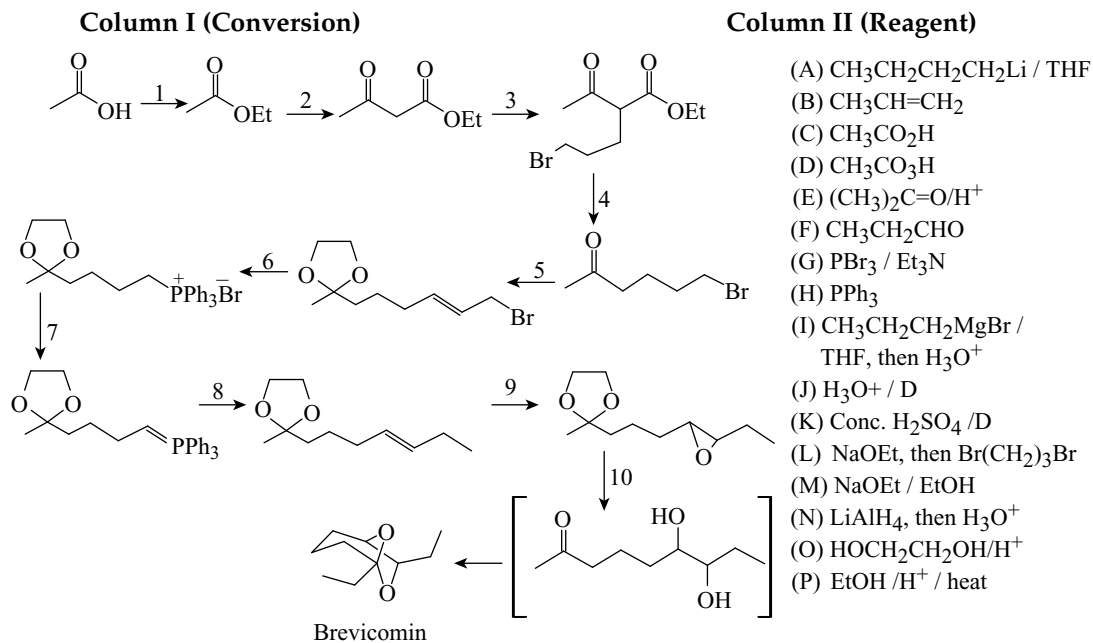


Column II (Reagent)

- (A) Br_2
- (B) NaBr / acetone
- (C) Br_2 / UV light
- (D) Br_2 / FeBr_3
- (E) HBr / dark
- (F) HBr / light
- (G) EtBr
- (H) EtBr / AlCl_3
- (I) KOH / EtOH / Δ
- (J) $\text{CH}_2\text{CHCHCH}_2$ / Δ
- (K) H_2SO_4
- (L) H_2 / Pd

WORKBOOK EXERCISE 2

Identify reagents (1 to 10) used in the following conversion from reagent present in the second Column II (A to P).

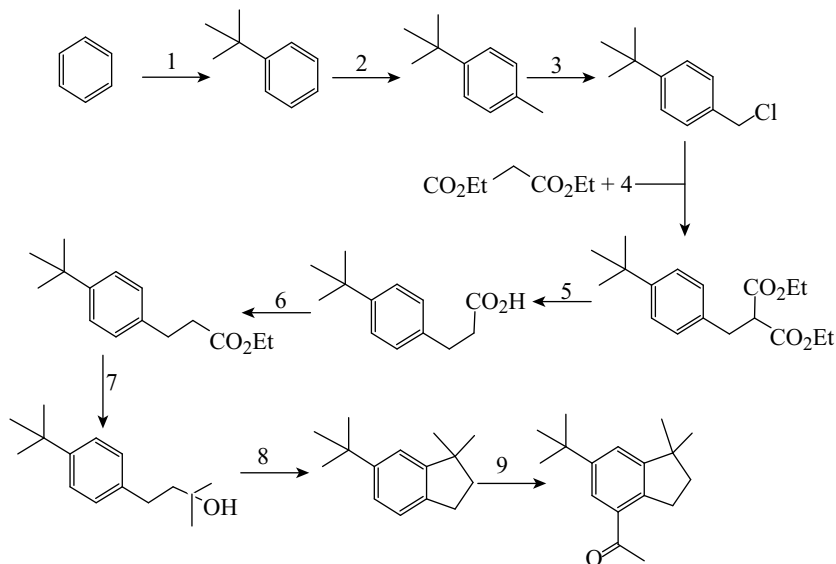


WORKBOOK EXERCISE 3

Identify reagents (1 to 9) used in the following conversion from reagent present in the second Column II (A to S).

Column I (Conversion)

Column II (Reagent)

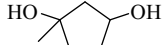
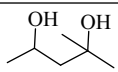
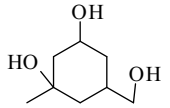
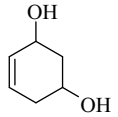
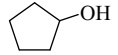
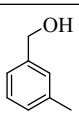
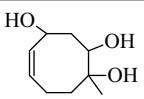


- (A) KOH / H₂O
- (B) NaOEt / EtOH
- (C) CH₃ONa / CH₃OH
- (D) CH₃CH₂OH / H⁺
- (E) CH₃OH / H⁺
- (F) Excess CH₃I
- (G) Conc. H₂SO₄
- (H) BH₃, then aq. NaOH / H₂O₂
- (I) CH₃Cl / AlCl₃
- (J) CH₃COCl / AlCl₃
- (K) Excess CH₃MgI, then H₃O⁺
- (L) Conc. HCl
- (M) NCS or Cl₂
- (N) LiAlH₄, then H₃O⁺
- (O) NaBH₄
- (P) Ph₃P=CH₂
- (Q) tBuOH / H⁺
- (R) tBuCl / KOH
- (S) H₂O / HgSO₄ / H₂SO₄

WORKBOOK EXERCISE 4

Identify products in the following matrix.

Reagents $\longrightarrow$

Compounds	PCC	K ₂ Cr ₂ O ₇	NaH/Mel	MnO ₂	Conc. KMnO ₄
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					

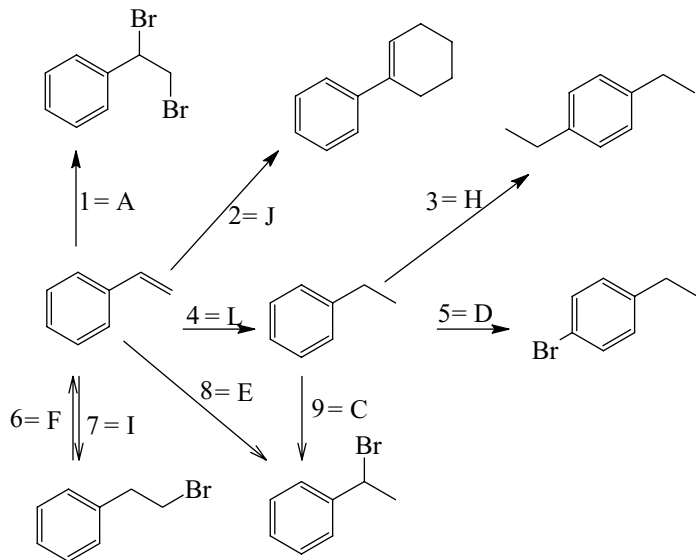
[illegible]

LEVEL 2

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
c	d	c	c	c	a	b	b	b	a	d	b	c	b	a
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
a	c	d	c	b	c	d	b	b	c	b	d	a	b	a
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
d	c	a	b	a	d	b	c	c	b	a	b	a	b	b
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
d	b	b	c	d	c	a	a	c	d	b	a	a	b	b
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
b	d	a	c	b	bc	d	b	c	bcd	ad	abcd	bcd	d	b
76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
c	d	b	a	b	a	b	a	c	d	b	c	a	b	c
91	92	93	94	95	96	97	98	99	100	1 01	1 02	1 03	1 04	105
c	d	abd	d	b	a	c	d	d	d	a	c	a	c	a
106	1 07	108	1 09	110	111	112	113	114	115	116	117	118	119	120
c	d	b	a	c	c	a	d	d	d	a	d	a	b	c
121	1 22	123	124	125	126	127	128	129	130	131	132			
c	d	c	c	a	b	b	b	d	c	c	c			

WORKBOOK EXERCISE 1

Column I (Conversion)



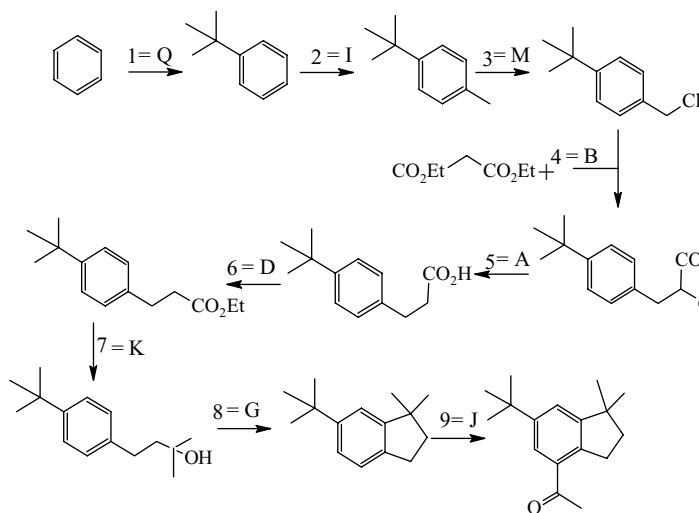
Column II (Reagent)

- (A) Br_2
- (B) NaBr / acetone
- (C) Br_2 / uv light
- (D) Br_2 / FeBr_3
- (E) HBr / dark
- (F) HBr / light
- (G) EtBr
- (H) EtBr / AlCl_3
- (I) KOH / EtOH/Δ
- (J) $\text{CH}_2\text{CHCHCH}_2/\Delta$
- (K) H_2SO_4
- (L) H_2 / Pd

WORKBOOK EXERCISE 3

Column I (Conversion)

Column II (Reagent)



- (A) $\text{KOH} / \text{H}_2\text{O} / \Delta$
 (B) $\text{NaOEt} / \text{EtOH}$
 (C) $\text{CH}_3\text{ONa} / \text{CH}_3\text{OH}$
 (D) $\text{CH}_3\text{CH}_2\text{OH} / \text{H}^+$
 (E) $\text{CH}_3\text{OH} / \text{H}^+$
 (F) Excess CH_3I
 (G) conc. H_2SO_4
 (H) BH_3 then aq. $\text{NaOH} / \text{H}_2\text{O}_2$
 (I) $\text{CH}_3\text{Cl} / \text{AlCl}_3$
 (J) $\text{CH}_3\text{COCl} / \text{AlCl}_3$
 (K) excess CH_3MgI then H_3O^+
 (L) conc. HCl
 (M) NCS or Cl_2
 (N) LiAlH_4 then H_3O^+
 (O) NaBH_4
 (P) $\text{Ph}_3\text{P}=\text{CH}_2$
 (Q) $\text{tBuOH} / \text{H}^+$
 (R) $\text{tBuCl} / \text{KOH}$
 (S) $\text{H}_2\text{O} / \text{HgSO}_4 / \text{H}_2\text{SO}_4$

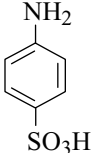
correction

WORKBOOK EXERCISE 4

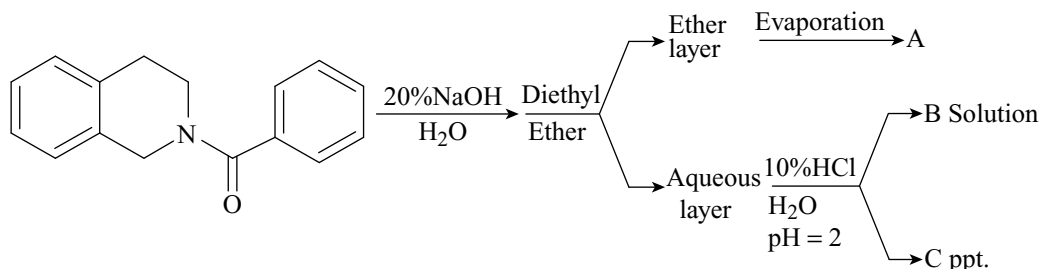
Compounds	Reagents $\longrightarrow$				
	PCC	$K_2Cr_2O_7$	NaH/MeI	MnO_2	Conc. $KMnO_4$
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					
					

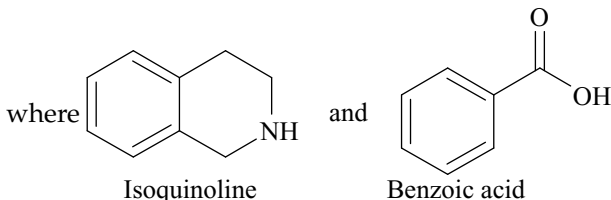
Question Bank

LEVEL 1

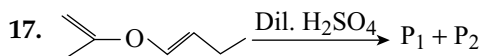
- Sodium nitroprusside $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]$ is used as reagent for detection of ____ and the compound formed is ____.
(a) Sulphur, $\text{Na}_4[\text{Fe}(\text{CN})_5\text{NOS}]$ (b) Nitrogen, $\text{Na}_4[\text{Fe}(\text{CN})_6]$
(c) Sulphur, $\text{Na}_2[\text{Fe}(\text{CN})_4\text{NOS}]$ (d) Sulphur, $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NOS}]$
- The prussian blue colouration obtained in the test for nitrogen in the organic compound is
(a) $\text{K}_4[\text{Fe}(\text{CN})_6]$ (b) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$ (c) $\text{Fe}[\text{Fe}(\text{CN})_6]$ (d) $\text{Fe}_3[\text{Fe}(\text{CN})_6]_2$
- If N and S both are present in an organic compound during Lassaigne's test, both will change into
(a) Na_2S and NaCN (b) NaSCN
(c) Na_2SO_3 and NaCN (d) Na_2S and NaCNO
- Which of the following will not give test for 'N' in sodium extract?
(a) $\text{C}_6\text{H}_5\text{NHNH}_2$ (b) NH_2CONH_2 (c) $\text{NH}_2\text{-NH}_2$ (d) 
- Which of the following will be blood red colour with FeCl_3 in sodium extract (Lassaigne extract)?
(a) NH_2CONH_2 (b) NH_2CSNH_2 (c) $\text{C}_6\text{H}_5\text{NHNH}_2$ (d) $\text{CH}_3\text{C}\equiv\text{N}$
- A mixture of acetone and CCl_4 can be separated by
(a) Azeotropic distillation (b) Fractional distillation
(c) Steam distillation (d) Vacuum distillation
- Phenol and benzoic acid can be separated by
(a) NaHCO_3 (b) NaOH solution (c) FeCl_3 solution (d) All of these
- Anthracene can be purified by
(a) Distillation (b) Sublimation
(c) Filtration (d) Fractional distillation
- KOH can be used as a drying agent for
(a) amines (b) phenols (c) acids (d) esters

10. Silver salt method is used to determine molecular weight of
 (a) organic acids (b) organic bases
 (c) both acids and bases (d) none of these
11. Which of the following observations is correct and is used in the identification of carboxylic acids?
 (a) Carboxylic acids liberate CO_2 gas from NaHCO_3 solution.
 (b) They produce fruity smell of esters when heated with alcohol in presence of Conc. H_2SO_4 .
 (c) Both (a) and (b)
 (d) Iodoform test
12. An organic compound is heated with HNO_2 at 0°C and then the resulting solution is added to a solution of β -naphthol whereby a brilliant red dye is produced. The observations indicate that the compound possesses
 (a) $-\text{NO}_2$ group (b) $-\text{CONH}_2$ group
 (c) aromatic NH_2 group (d) aliphatic NH_2 group
13. An organic compound contains C, H, N, S and Cl. For the detection of chlorine, the sodium extract of the compound is first heated with a few drops of concentrated HNO_3 and then AgNO_3 is added to get a white ppt. of AgCl . The digestion with HNO_3 before the addition of AgNO_3 is
 (a) to prevent the formation of NO_2
 (b) to create a common ion effect
 (c) to convert CN^- and S^{2-} to volatile HCN and H_2S , or else they will interfere with the test forming AgCN or Ag_2S
 (d) to prevent the hydrolysis of NaCN and Na_2S
14. Rectified spirit contains
 (a) 95.6 per cent ethanol and 4.4 per cent methanol
 (b) 100 per cent ethanol
 (c) 95.6 per cent ethanol and 4.4 per cent water
 (d) 95.6 per cent ethanol and 4.4 per cent benzene
15. Aniline can be separated from phenol using
 (a) NaHCO_3 (b) $\text{NaNO}_2 + \text{HCl}$ at 0°C
 (c) NaCl (d) Acidified KMnO_4
16. Identify correct statement for A, B and C in the following sequence.





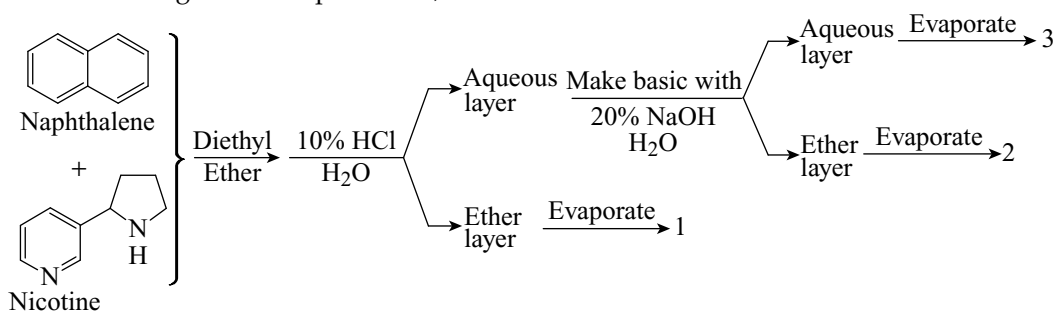
- (a) A = benzoic acid, B = sodium chloride and C = isoquinoline
(b) A = isoquinoline, B = benzoic acid and C = sodium chloride
(c) A = isoquinoline, B = sodium chloride and C = benzoic acid
(d) A = sodium chloride, B = isoquinoline and C = benzoic acid



P_1 and P_2 products are identify by

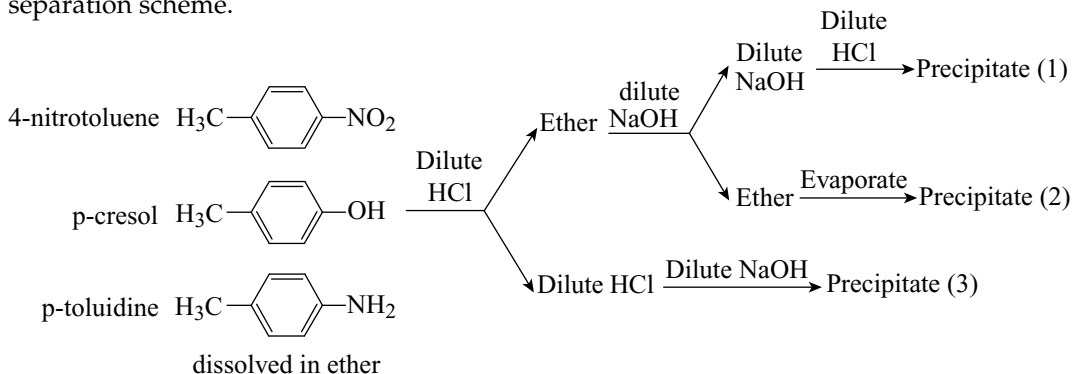
- (a) Tollen's reagent
(b) Iodoform test
(c) $\text{Br}_2 + \text{H}_2\text{O}$ test
(d) 1 per cent alkaline KMnO_4

18. In the following extraction procedure, choose the number where nicotine would be found.



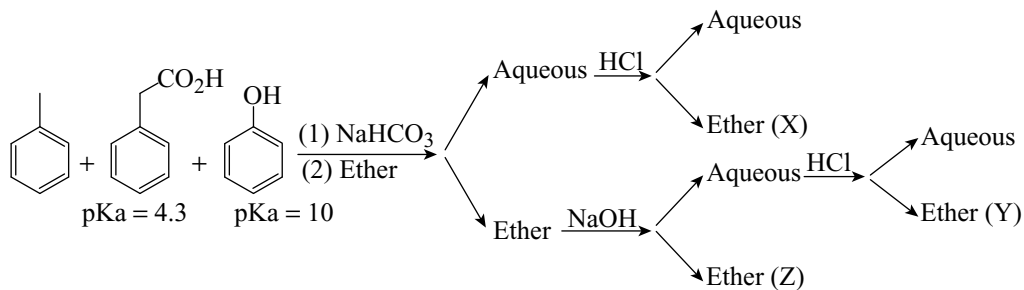
- (a) 1 = Nicotine
(b) 1 + 3 = Nicotine
(c) 2 = Nicotine
(d) 2 + 3 = Nicotine

19. Choose the answer that has the following compounds located correctly in the separation scheme.



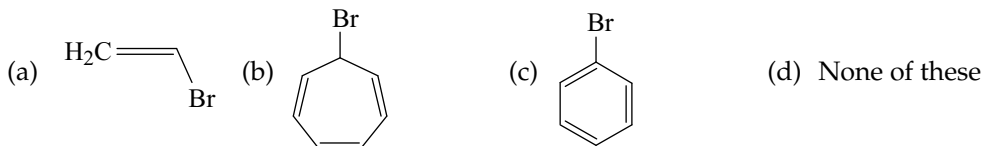
- (a) 1 = 4-nitrotoluene, 2 = p-cresol, 3 = p-toluidine
 (b) 1 = 4-nitrotoluene, 2 = p-toluidine, 3 = p-cresol
 (c) 1 = p-toluidine, 2 = 4-nitrotoluene, 3 = p-cresol
 (d) 1 = p-cresol, 2 = 4-nitrotoluene, 3 = p-toluidine

20. Choose the answer that has the following compounds located correctly in the separation scheme.



- (a) toluene is in (Y); phenylacetic acid is in (Y); phenol is in (Z)
 (b) toluene is in (Y); phenylacetic acid is in (X); phenol is in (Z)
 (c) toluene is in (Z); phenylacetic acid is in (Y); phenol is in (X)
 (d) toluene is in (Z); phenylacetic acid is in (X); phenol is in (Y)

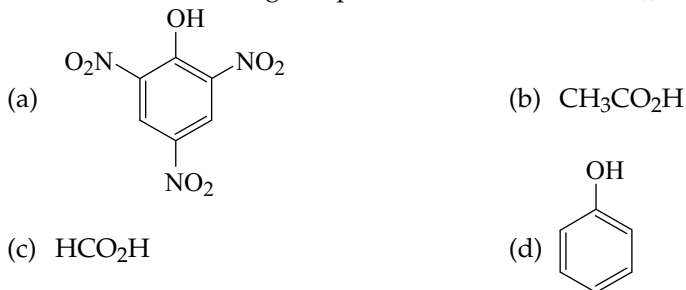
21. Among the following which compound gives precipitate with AgNO_3 solution?



22. Disodium pentacyanonitrosyl ferrate reagent gives purple colour when which of the following element is present?

- (a) N (b) Cl (c) F (d) S

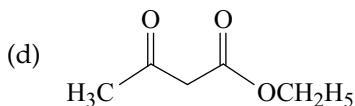
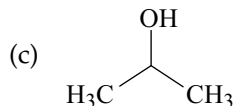
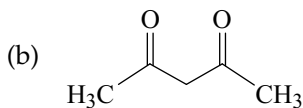
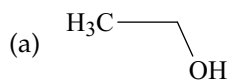
23. Which of the following compound cannot liberate CO_2 when treated with NaHCO_3 ?



24. Phenol + Phthalic anhydride $\xrightarrow[\text{fusion}]{\text{Conc. H}_2\text{SO}_4}$ B. B gives which of the following colour in alkaline medium?

- (a) Yellow (b) Colourless (c) Pink (d) Violet

25. Among the following, which will not respond to iodoform test?



26. Among the following, which statement is not correct?

- (a) $\text{H}_3\text{C}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$ will not respond to haloform test
 (b) Schiff's reagent and Schiff's base are different compounds
 (c) Fehling's solution is a good reagent to detect aromatic aldehydes
 (d) Both aldehyde and ketone can react with 2, 4-dinitrophenylhydrazine reagent

27. To separate a mixture of amines from each other, one should follow

- (a) Beckmann's method (b) Hinsberg's method
 (c) Zeisel's method (d) Victor Meyer's Method

28. Phenol can be distinguished from ethanol by

- (a) Tollen's reagent (b) Benedict's reagent
 (c) FeCl_3 (d) Schiff's base

29. $\text{p-Cl}-\text{C}_6\text{H}_4\text{NH}_2$ and $\text{PhNH}_3^+\text{Cl}^-$ can be distinguished by

- (a) NaOH (b) AgNO_3 (c) LiAlH_4 (d) Zn

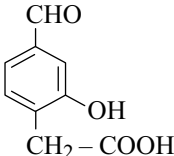
LEVEL 2

Single and Multiple-choice Type

- In Lassaigne's test, the organic compound is first fused with sodium metal. The sodium metal is used because
 - The melting point of sodium metal is low.
 - Sodium metal reacts with elements present in organic compounds to form inorganic compounds.
 - All sodium salts are soluble in water.
 - All sodium salts are not soluble in water.
- Molecular weight of acids can be determined by
 - Silver salt method
 - Volumetric method
 - Plants chloride method
 - Victor Meyer's method
- Ethanol and ethanal are distinguished by
 - Fehling's solution test
 - Tollen's reagent test
 - Iodoform test
 - Cerric ammonium nitrate
- Which of the following statements are correct?
 - An organic compound is pure if mixed melting point is same.
 - Ethanol and water can be separated by azeotropic distillation because it forms azeotrope.
 - Impure aniline is purified by steam distillation as it is steam volatile.
 - Glycerol is purified by vacuum distillation because it decomposes at its normal boiling point.
- Which of the following will respond to iodoform test?
 - $\text{CH}_3 - \overset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{COOH}$
 - $\text{CH}_3 - \overset{\text{OH}}{\underset{|}{\text{CH}}} - \text{COOH}$
 - $\text{CH}_3 - \overset{\text{OH}}{\underset{|}{\text{CH}}} - \text{CH}_3$
 - $\text{CH}_3\text{CH}_2\text{OH}$
- Which of the following will not show iodoform test?
 - $\text{CH}_3 - \overset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{CH}_3$
 - $\text{CH}_3 - \overset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{Cl}$
 - $\text{CH}_3 - \overset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{NH}_2$
 - $\text{CH}_3 - \text{COOH}$
- HCOOH and CH_3COOH can be distinguished by
 - Tollen's reagent
 - Fehling's solution
 - KMnO_4
 - NaHCO_3

8. The desiccants used for absorbing water during Liebig's method for estimation of carbon and hydrogen are

- (a) anhydrous CaCl_2 (b) anhydrous Na_2SO_4
 (c) $\text{Mg}(\text{ClO}_4)_2$ (d) $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$

9. An organic compound has the structure . It will give

- (a) ceric ammonium nitrate test
 (b) brick effervescence with sodium bicarbonate
 (c) a characteristic colouration with neutral ferric chloride after decarboxylation and reduction by Clemenson's method
 (d) Fehling's test

10. Which of the following organic compounds will give white precipitate with AgNO_3 ?

- (a) $\text{C}_6\text{H}_5\text{NH}_3^+\text{Cl}^-$ (b) NaCl
 (c) 2,4,6-trinitro chlorobenzene (d) Benzyl chloride

11. Which of the following reactions occur during the detection of nitrogen in organic substances by Lassaigne's test?

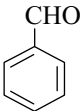
- (a) $\text{Na} + \text{C} + \text{N} \longrightarrow \text{NaCN}$
 (b) $\text{FeSO}_4 + 6\text{NaCN} \longrightarrow \text{Na}_4[\text{Fe}(\text{CN})_6] + \text{Na}_2\text{SO}_4$
 (c) $3\text{Na}_4[\text{Fe}(\text{CN})_6] + 2\text{Fe}_2(\text{SO}_4)_3 \longrightarrow \text{Fe}_4[\text{Fe}(\text{CN})_6]_3 + 6\text{Na}_2\text{SO}_4$
 (d) None of these

12. Compound A reacts with CHCl_3 and KOH and gives an offensive smelling compound. A can be

- (a) Primary aliphatic amine (b) Primary aromatic amine
 (c) Secondary amine (d) Tertiary amine

13. HCOOH and HCHO may be distinguished by

- (a) Tollen's test (b) Sodium bicarbonate test
 (c) 2,4-DNP test (d) Benedict's test

14.  and CH_3CHO can be distinguished by

- (a) Tollen's test (b) Benedict's test (c) Iodoform test (d) 2,4-DNP test

15. Acetic acid and CH_3COCl can be distinguished by

- (a) NaHCO_3 test (b) Na metal test
 (c) Ester formation test (d) $\text{Br}_2(\text{aq.})$ test

Comprehension Type

Passage 1

Steam distillation is used to purify a compound which is steam volatile and insoluble in water. The impurities should not be steam volatile. It is based on the principle that liquid will boil when partial vapour pressure of liquid and partial vapour pressure of steam both become equal to atmospheric pressure, $P = p_1 + p_2$. It reduces the boiling point of a liquid.

$$\frac{\text{Weight of water distilled}}{\text{Wt. of substance distilled}} = \frac{\text{M. Wt. of water} \times \text{VP of steam}}{\text{M. Wt. of substance} \times \text{VP of aniline}}$$

16. Isolation of essential oils from flowers, etc. is done by
 - (a) Steam distillation
 - (b) Distillation
 - (c) Fractional distillation
 - (d) Distillation under reduced pressure
17. Which of the following is steam volatile?
 - (a) o-nitrophenol
 - (b) p-nitrophenol
 - (c) p-hydroxy benzaldehyde
 - (d) Ethanol
18. Calculate weight of aniline distilled if weight of water distilled is 100 g when $P_{\text{organic compound}} = 100 \text{ mm}$ and $P_{\text{H}_2\text{O}} = 200 \text{ mm}$.
 - (a) 250 g
 - (b) 258 g
 - (c) 100 g
 - (d) 25.8 g
19. Steam distillation is based on
 - (a) Dalton's law of partial pressures
 - (b) Graham's law of diffusion
 - (c) Raoult's law of non-volatile solute
 - (d) None of these
20. Which of the following cannot be separated by steam distillation?
 - (a) Nitrobenzene
 - (b) Essential oil
 - (c) Aniline
 - (d) Glycerol

Passage 2

Test (Q). A compound X was fused with Na metal and the extract gave a white precipitate with AgNO_3 . The Lassaigne's extract gave a red colouration with neutral FeCl_3 .

Test (R). While compound Y when fused with Na metal and subsequent analysis on its Lassaigne's extract did not give any characteristic test.

Test (S). While compound Y on fusion with fusion mixture (sodium carbonate + potassium nitrate) or Na_2O_2 , followed by extraction, followed by addition of Conc. HNO_3 and ammonium molybdate gave a yellow precipitate.

21. Compound X contains
 - (a) N
 - (b) S
 - (c) N + S
 - (d) P
22. Compound X
 - (a) will contain halogens
 - (b) may contain halogens
 - (c) may contain only sulphur
 - (d) will contain only nitrogen

23. Compound Y contain
 (a) N (b) S (c) X (d) P
24. The chemical reaction taking place in Y, when it is fused with fusion mixture is
 (a) $3\text{Na} + \text{P} + 4\text{O} \longrightarrow \text{Na}_3\text{PO}_4$
 (b) $3\text{Na}_2\text{CO}_3 + 2\text{P} + [\text{O}] \longrightarrow 2\text{Na}_3\text{PO}_4 + 3\text{CO}_2$
 (c) $3\text{KNO}_3 + \text{P} + 3\text{O} \longrightarrow \text{K}_3\text{PO}_4 + 3\text{NO}_2 + \text{O}_2$
 (d) None of these
25. The formula of yellow precipitate is
 (a) $(\text{NH}_4)_3\text{PO}_4$ (b) $\text{Mo}(\text{PO}_4)$
 (c) $(\text{NH}_4)_3\text{PO}_4 \cdot 12\text{Mo}_3\text{O}_3$ (d) $\text{Mo}(\text{PO}_4)_2$

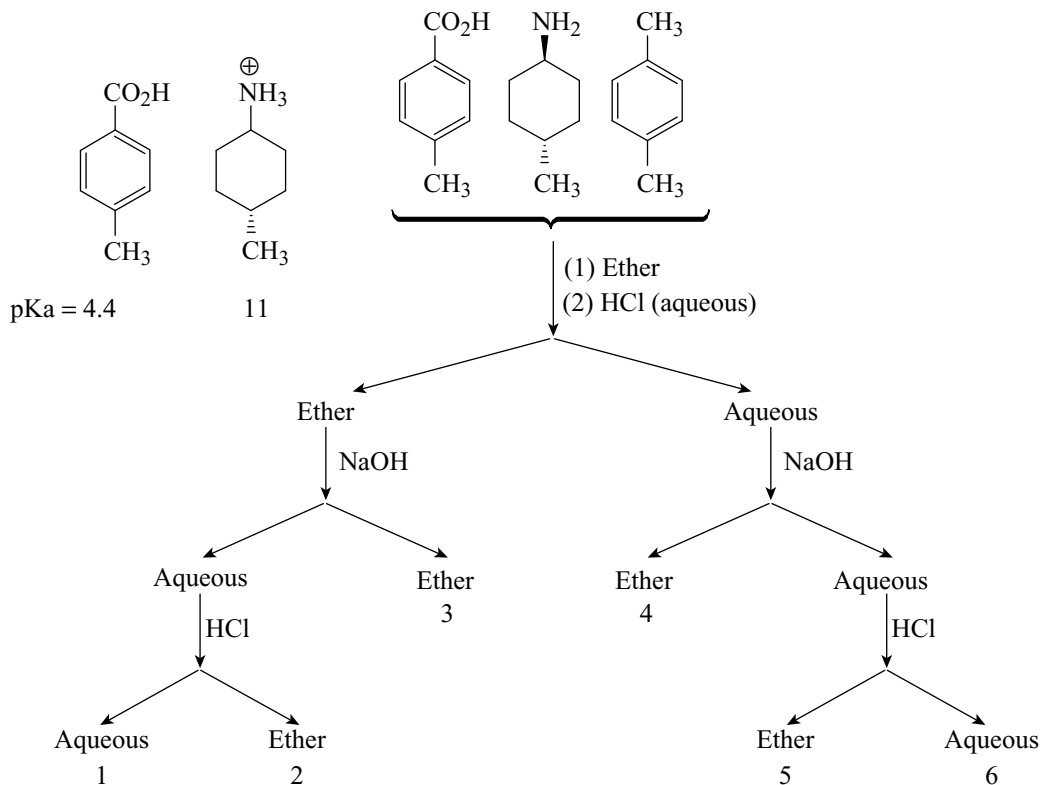
Passage 3

The 0.2 g of anhydrous organic acid gave on combustion 0.04 g of water and 0.195 g of CO_2 . The acid is a dibasic acid and 0.5 g of its silver salt leaves on ignition 0.355 g of silver.

26. The percentage of carbon in the compound is
 (a) 50 (b) 52 (c) 26.6 (d) 90
27. The percentage of hydrogen in the compound is
 (a) 5.6 (b) 2.22 (c) 4.44 (d) 10
28. The empirical weight of the compound is
 (a) 90 (b) 100 (c) 10 (d) 45
29. The molecular weight of the compound is
 (a) 90 (b) 100 (c) 10 (d) 45
30. The molecular formula of the compound is
 (a) CHO (b) CHO_2 (c) $\text{C}_2\text{H}_2\text{O}_4$ (d) $\text{C}_3\text{H}_6\text{O}_2$

Passage 4

A student in a lab had a mixture of three compounds, 4-methylbenzoic acid, 4-methylcyclohexylamine and 1,4-dimethylbenzene. In order to separate the three compounds the following extraction (separation) scheme was followed. At the end of the procedure the student had six separate flasks containing either an aqueous or an ether solution. Locate each compound by designating the flask expected to contain each compound. Some relevant pK_a data is given.

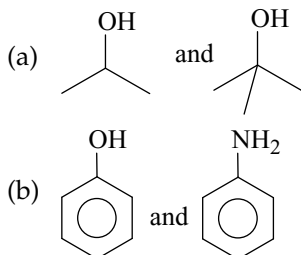


31. Which flask contains the 4-methylcyclohexylamine?
 (a) 1 (b) 2 (c) 3 (d) 4
32. Which flask contains the 4-methylbenzoic acid?
 (a) 1 (b) 2 (c) 3 (d) 4
33. Which flask contains the 1,4-dimethylbenzene?
 (a) 1 (b) 2 (c) 3 (d) 4

Matrix Type

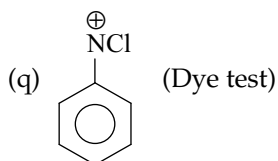
Match the columns:

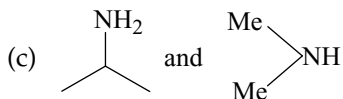
34. Column I (pair of compounds)



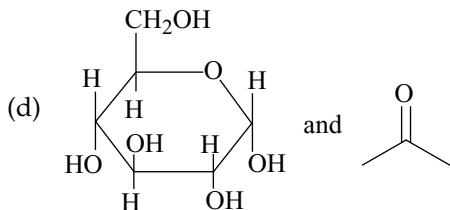
Column II (test to distinguish)

(p) Libermann nitroso test





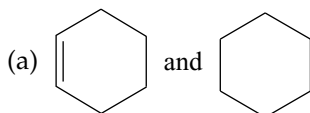
(r) Iodoform



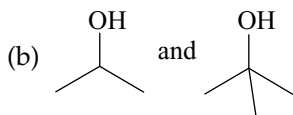
(s) Lucas

(t) NaHSO_3

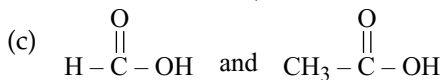
35. Column I
(pair of compounds)



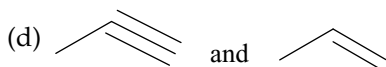
(p) Tollen's reagent test



(q) $\text{Br}_2 + \text{H}_2\text{O}$ test



(r) Lucas test



(s) Iodoform test

(t) Ammonical Cu_2Cl_2 test

36. Column I (reagents)

(a) Product of reaction of propyne with 1 per cent HgSO_4 and Dil. H_2SO_4

(b) Br_2 water test given by

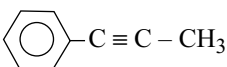
(c) Addition of HBr can be with

(d) Tollen's reagent give white ppt. with

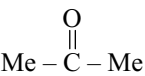
Column II

(p) $\text{Me}-\text{CH}=\text{CH}-\text{Me}$

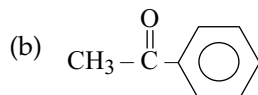
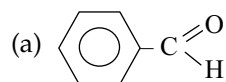
(q) $\text{Me}-\text{C}\equiv\text{C}-\text{H}$

(r) 

(s) $\text{Me}-\text{CH}=\text{CH}_2$

(t) 

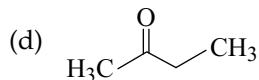
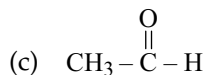
37. Column I (compound)



Column II (tests)

(p) 2,4-DNP test

(q) Yellow ppt. with $\text{NaOH} + \text{I}_2$



(r) Red ppt. with Fehling's solution

(s) Silver mirror with Tollen's reagent

38. Column I (reagents)

(a) Tollen's reagent give white ppt. with

(b) $\text{Br}_2 + \text{H}_2\text{O}$ test given by

(c) Product of reaction of acetylene with
1 per cent HgSO_4 and Dil. H_2SO_4

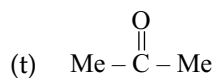
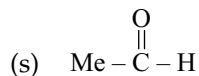
(d) Pd/H_2 reacts with

Column II

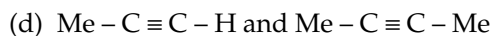
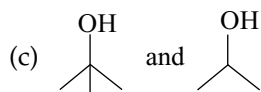
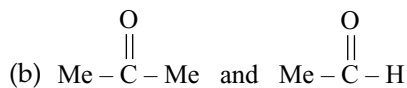
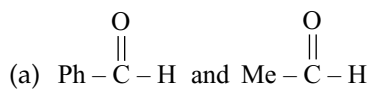
(p) $\text{Me} - \text{CH} = \text{CH} - \text{Me}$

(q) $\text{Me} - \text{C} \equiv \text{C} - \text{H}$

(r) $\text{CINH}_3^+ - \text{NH}_2$



**39. Column I
(pair of compounds)**



**Column II
(test of identification)**

(p) Tollen's reagent

(q) Fehling's reagent

(r) Iodoform test

(s) Victor Meyer's test

(t) Lucas test

Answer Keys

LEVEL 1

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
a	b	b	c	b	b	a	b	a	b	c	c	c	c	b
16	17	18	19	20	21	22	23	24	25	26	27	28	29	
b	ab	c	d	c	b	d	abc	c	abcd	c	b	c	b	

LEVEL 2

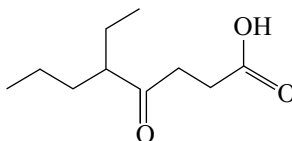
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
abc	ab	abd	abcd	abcd	bcd	abc	ac	be	abcd	abc	ab	be	be	ab
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
a	a	b	a	d	c	b	d	c	c	c	b	d	a	c
31	32	33	34(a)	34(b)	34(c)	34(d)	35(a)	35(b)	35(c)	35(d)	36(a)	36(b)	36(c)	36(d)
d	b	c	rs	pq	p	rt	q	rs	p	pt	t	pqrs	pqrs	q
37(a)	37(b)	37(c)	37(d)	38(a)	38(b)	38(c)	38(d)	39(a)	39(b)	39(c)	39(d)			
ps	pq	pqrs	pq	qr	pq	s	pqst	qr	pq	rst	p			

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Question Bank

Single and Multiple-choice Type

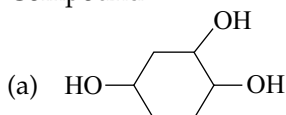
1. The IUPAC name of the compound shown here is



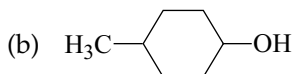
- (a) 5-Ethyl-1-hydroxyoctan-1,4-dioic acid
 (b) 2-Carboxyethyl-3-hexyl ketone
 (c) 3-(3-Hexylcarbonyl) propanoic acid
 (d) 5-Ethyl-4-keto-octanoic acid
2. Which of the following compounds are not correctly matched?

Compound

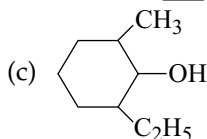
IUPAC name



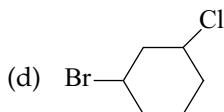
Cyclohexane-1,2,4-triol



4-Methylcyclohexanol

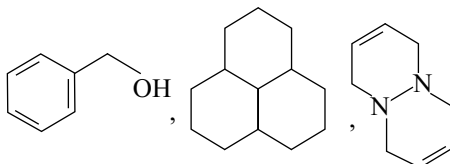


2-Ethyl-6-methylcyclohexanol



3-Bromo-1-chlorocyclohexane

3. The no. of secondary C-atoms present in the following compounds respectively is



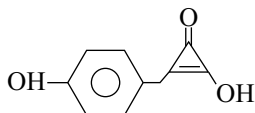
(a) 7, 9, 4

(b) 5, 12, 8

(c) 6, 9, 8

(d) 5, 9, 4

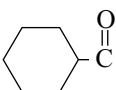
4. The organic compound (Cyclohex-3-enyl) 2-ethyl cyclobutanecarboxylate does not contain
- two 3° -carbon atoms
 - two 1° -carbon atoms
 - 9 secondary C-atoms
 - one 1° -carbon atoms
5. The IUPAC name of the compound given below is



- 2-Hydroxy-3-(4'-hydroxyphenyl) cycloprop-2-en-1-one
 - 2-Hydroxy-3-(4'-hydroxyphenylmethyl) cycloprop-2-en-1-one
 - 4-(2'-Hydroxy-3'-oxocycloprop-1'-enylmethyl) phenol
 - 4-(1'-Hydroxy-3'-oxocycloprop-1'-enylmethyl) phenol
6. The correct IUPAC name of the compound is

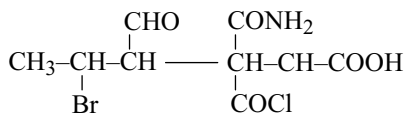


- 2-Bromopropyl-3-hydroxy-1-methylcyclopentane carboxylate
- 3'-Hydroxy-1'-methylcyclopentyl-2-bromopropanoate
- 1'-Hydroxy-3'-methylcyclopentyl-2-bromopropanoate
- None of these

7. The IUPAC name of  is

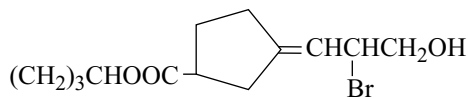
- N-Cyclohexylbenzamide
- N-Phenyl-N-cyclohexylmethanamide
- N-phenylcyclohexane carboxamide
- N-Cyclohexyl-N-phenylmethyl amide

8. The IUPAC name of the given compound is



- 2-Bromo-4-carbamoyl-5-chloroformyl-3-formylhexanoic acid
- 5-Bromo-3-carbamoyl-2-chloroformyl-4-formylhexanoic acid
- 4-Formyl-2-chloroformyl-3-carbamoyl-5-bromohexanoic acid
- 2-Chloroformyl-3-carbamoyl-4-formyl-5-bromohexanoic acid

9. The correct IUPAC name of the compound given below is



- (a) 3-(3'-isopropoxycarbonyl cyclopentylidene) propan-1-ol
- (b) 3-(2'-Bromo-3'-hydroxypropylidene) cyclopentanecarboxylate
- (c) Isopropyl-3-(2'-bromo-3'-hydroxypropylidenyl) cyclopentanecarboxylate
- (d) Isopropyl-3-(2'-bromo-3'-hydroxypropylidene) cyclopentanecarboxylate

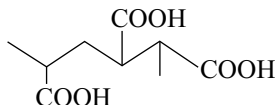
10. IUPAC name of isopentyl alcohol is

- (a) 1-Hydroxy-2-methyl pentane
- (b) 3-Methyl butan-1-ol
- (c) 2-Methyl propanol
- (d) 2-Methyl-2-butanol

11. Which of the following structure represent Dimethyl butanedioate?

- | | |
|--|---|
| (a) $\begin{array}{c} \text{CH}_2 - \text{CO} - \text{CH}_3 \\ \\ \text{CH}_2 - \text{CO} - \text{CH}_3 \end{array}$ | (b) $\begin{array}{c} \text{CH}_3 - \text{COOCH}_2 \\ \\ \text{CH}_3 - \text{COOCH}_2 \end{array}$ |
| (c) $\begin{array}{c} \text{CH}_3 - \text{OOC} - \text{CH}_2 \\ \\ \text{CH}_3 - \text{OOC} - \text{CH}_2 \end{array}$ | (d) $\begin{array}{c} \text{COOCO} \\ \quad \\ \text{CH}_3 \quad \text{CH}_2 - \text{CH}_3 \end{array}$ |

12. IUPAC name of the following compound is

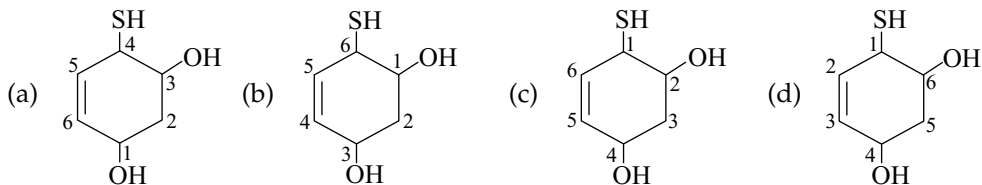


- (a) 2-(1-carboxyethyl)-4-methyl pentanoic acid
- (b) 3,5-Dicarboxy-2-methyl hexanoic acid
- (c) 2,4,5-Hexane tricarboxylic acid
- (d) 2,3,5-Hexane tricarboxylic acid

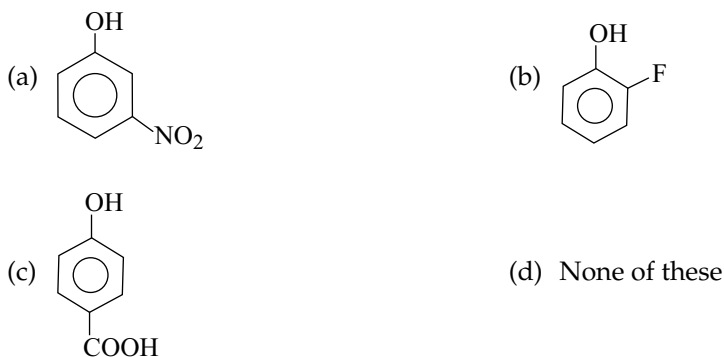
13. Which of the following compound has incorrect IUPAC nomenclature?

- (a) $\text{CH}_3\text{CH}_2 - \text{CH}_2 - \text{COOC}_2\text{H}_5$: Ethyl butanoate
- (b) $\begin{array}{c} \text{CH}_3 - \text{CH} - \text{CH} - \text{CH}_3 \\ | \quad | \\ \text{OH} \quad \text{CH}_3 \end{array}$: 2-Methyl butan-3-ol
- (c) $\begin{array}{c} \text{CH}_3 - \text{CH} - \text{C} - \text{CH}_2 - \text{CH}_3 \\ | \quad || \\ \text{CH}_3 \quad \text{O} \end{array}$: 2-Methyl pentan-3-one
- (d) $\begin{array}{c} \text{CH}_3 - \text{CH} - \text{CH}_2 - \text{CHO} \\ | \\ \text{CH}_3 \end{array}$: 3-Methyl butanal

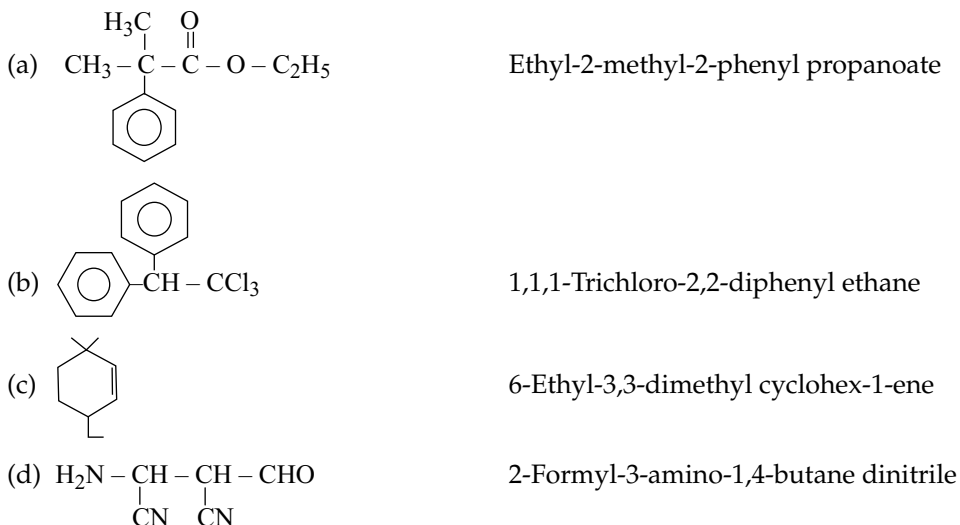
14. Select the structure with correct numbering for the IUPAC name of the given compound.



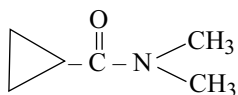
15. In which of the following H-bonding is responsible for low volatility?



16. Which of the following is not correctly matched?

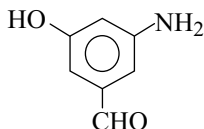


17. IUPAC name of the following compound is



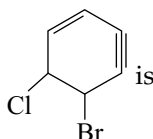
- (a) N,N-Dimethyl cyclopropanamide
- (b) N-Methyl cyclopropanamide
- (c) N,N-Dimethyl cyclopropane carboxamide
- (d) N,N-Dimethyl cyclopropolonamide

18. Correct IUPAC name of the given compound

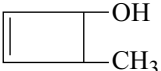


- (a) 5-amino-3-hydroxy benzene carbaldehyde
- (b) 3-amino-5-formyl benzenol
- (c) 3-amino-5-hydroxy benzaldehyde
- (d) 3-formyl-5-hydroxy aniline

19. The IUPAC name of the given structure



- (a) 5-Bromo-6-chloro cyclohex-1-ene-3-yne
- (b) 6-Bromo-5-chloro cyclohex-1-ene-3-yne
- (c) 6-Bromo-5-chloro cyclohex-3-ene-1-yne
- (d) 4-Bromo-3-chloro cyclohex-1-ene-5-yne

20. The IUPAC name of  is

- (a) 3-Methyl cyclobut-1-ene-2-ol
- (b) 2-Methyl cyclobut-3-ene-1-ol
- (c) 4-Methyl cyclobut-1-ene-3-ol
- (d) 4-Methyl cyclobut-2-ene-1-ol

21. Correct name for $(\text{C}_2\text{H}_5)_2\text{C} = \text{C}(\text{CH}_3)\text{CH}_2\text{CO}_2\text{H}$

- (a) 4, 4-Diethyl-3-methyl-3-butenic acid
- (b) 4-Ethyl-3-methyl-3-hexenoic acid
- (c) 3-Ethyl-4-methyl-3-hexenoic acid
- (d) 3-Ethyl-4-methyl-3-hexen-6-oic acid

22. Which statement is incorrect in the following?

- (a) Methyl acetate is the first higher homologue of methyl formate.
- (b) Propanoic acid and propionic acid are identical.
- (c) Vinyl alcohol is the lower homologue of propionaldehyde.
- (d) $\text{CH}_3\text{-NH-CH}_2\text{-CH}_3$ is the lower homologue of $\text{CH}_3\text{-}\underset{\text{CH}_3}{\text{CH}}\text{-NH-CH}_3$.

23. Match the following



(a) a - ii; b - iv; c - ii; d - iii

(b) a - iv; b - i; c - iii; d - ii

(c) a - iv; b - i; c - ii; d - iii

(d) a - iv; b - ii; c - i; d - iii

24. Number of C-atoms in first higher homologue of first member of ester family is

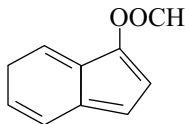
(a) 5

(b) 4

(c) 3

(d) 2

25. In a given compound if number of σ - and π -bonds are X and Y, respectively then (X + Y) will be



(a) 28

(b) 26

(c) 27

(d) 24

26. Which statement is correct in the following?

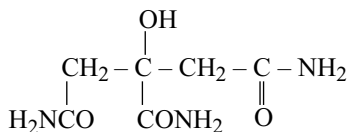
(a) Alicyclic compounds are acyclic compounds.

(b) In secondary amine, nitrogen is attached to 2° carbon atom.

(c) 1° , 2° , 3° amines are homologues of each other.

(d) HCOOCH_3 and CH_3COOH are not homologues of each other.

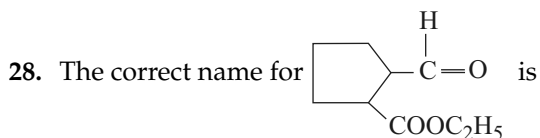
27. The correct systematic IUPAC name of the given compound is



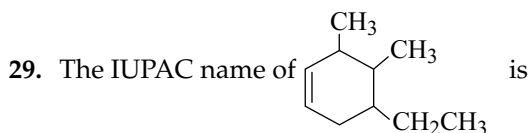
(a) 3-Carbamoyl-3-hydroxybutanediamide

(b) 2-Hydroxypropane-1,2,3-tricarbamoyl

- (c) 2-Hydroxypropane-1,2,3-tricarboxamide
 (d) 2-Bis(carbamoyl)-2-hydroxyethanamide



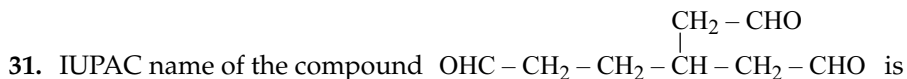
- (a) 2-Oxocyclopentanecarboxylate
 (b) 2-Formylcyclopentanecarboxylate
 (c) Ethyl-2-formylcyclopentanecarboxylate
 (d) Ethyl-2-oxocyclopentanecarboxylate



- (a) 2, 3-Methyl-1-ethylcyclohex-4-ene
 (b) 5-Ethyl-3, 4-dimethylcyclohex-1-ene
 (c) 4-Ethyl-5, 6-dimethylcyclohex-1-ene
 (d) 1-Ethyl-2, 3-dimethylcyclohex-4-ene

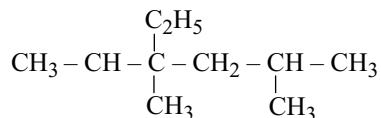
30. Which of the following names is correct?

- (a) 4, 4-Dimethyl-3-ethylpentane
 (b) 3-Methyl-4-oxobutanoic acid
 (c) 1-Ethyl-2-methylpent-4-ene
 (d) Bicyclo[1.2.3]heptane



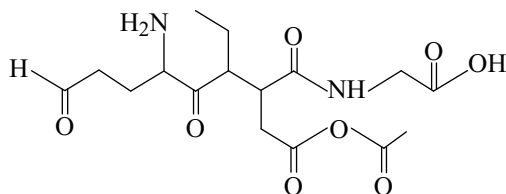
- (a) 4, 4-Di (formylmethyl) butanal
 (b) 2-(formylmethyl) butane-1, 4-dicarbaldehyde
 (c) Hexane-3-acetal-1, 6-dial
 (d) 3-(formylmethyl) hexanedial

32. The correct IUPAC name of the following compound is



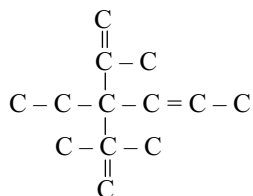
- (a) 3-Ethyl-3,5,-dimethylhexane
 (b) 4-Ethyl-2,4,-dimethylhexane
 (c) 2,4-Dimehtyl-4-ethylhexane
 (d) 4-Ethyl-2,4,-dimethylhexene

33. The number of functional groups present in the following compounds is



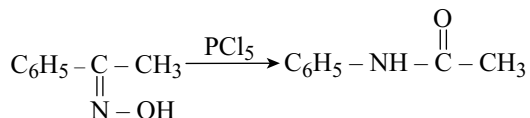
- (a) 5 (b) 7 (c) 6 (d) 8

34. Write the IUPAC name of the following compound



- (a) 3-(1,1-dimethylethenyl)-3-ethyl-2-methyl-1,4 hexadiene
 (b) 3-ethyl-2-methyl-3-(1,1-dimethylethenyl)-1,4 hexadiene
 (c) 3-ethyl-2-2-dimethyl-3-(1-methylethenyl)-1,4 hexadiene
 (d) 3-ethyl-3-(1-methylethenyl)-2-2-dimethyl-1,4 hexadiene

35. Hybridization of which atom changes in the following reaction?



- (a) C and O (b) C and N
 (c) N and O (d) No change in the hybridization of any atom

36. $\text{---}\overset{\oplus}{\underset{\text{a}|}{\text{C}}}\text{---}\overset{\cdot}{\underset{\text{b}|}{\text{C}}}\text{---}\overset{\cdot\cdot}{\underset{\text{c}|}{\text{C}}}\text{---}\overset{-}{\underset{\text{d}|}{\text{C}}}\text{---}$; carbon atoms a, b, c and d are in

- (a) sp^2 , sp^3 , sp^2 and sp^3 hybrid states respectively
 (b) sp , sp^2 , sp and sp^3 hybrid states respectively
 (c) sp^2 , sp , sp^2 and sp^3 hybrid states respectively
 (d) sp^2 , sp^2 , sp^2 and sp^3 hybrid states respectively

37. Ratio of π - and σ -bonds in Ph-COOH will be

- (a) 5:3 (b) 3:5
 (c) 3:16 (d) 4:15

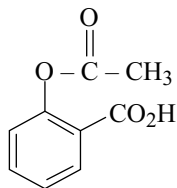
38. Which of the following compound has incorrect IUPAC nomenclature?

- (a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOC}_2\text{H}_5$: Ethyl butanoate
 (b) $\begin{array}{c} \text{CH}_3 - \text{CH} - \text{CH} - \text{CH}_3 \\ | \quad | \\ \text{OH} \quad \text{CH}_3 \end{array}$: 2-Methyl butan-3-ol
 (c) $\begin{array}{c} \text{CH}_3 - \text{CH} - \text{C} - \text{CH}_2 - \text{CH}_3 \\ | \quad || \\ \text{CH}_3 \quad \text{O} \end{array}$: 2-Methyl pentan-3-one
 (d) $\begin{array}{c} \text{CH}_3 - \text{CH} - \text{CH}_2 - \text{CHO} \\ | \\ \text{CH}_3 \end{array}$: 3-Methyl butanal

39. Identify incorrect matching in the following

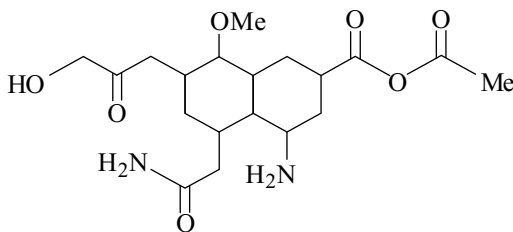
- (a) Allyl alcohol-Prop-2-en-1-ol (b) Vinyl alcohol-Ethenol
 (c) Propargyl alcohol-But-3-yn-1-ol (d) Acrylaldehyde-2-Propenal

40. IUPAC name of the following compound



- (a) Aspirin (b) 2-Acetyl benzoic acid
 (c) Acetyl salicylic acid (d) 2-Acetoxy benzoic acid

41. How many functional groups are present in the given molecule?



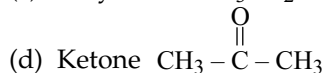
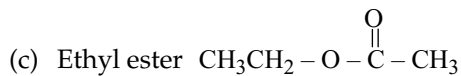
- (a) 4 (b) 5 (c) 6 (d) 7

42. What is the ratio of number of σ - and π -bonds in the molecule $\text{HC}(\text{O})\text{CH}_2\text{CH}_2\text{COOH}$?

- (a) 5 (b) 6 (c) 4 (d) none of these

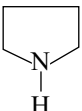
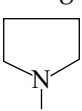
43. Which is not a first member of given homologous series?

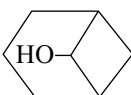
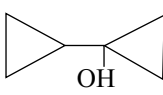
- (a) Alkadiene $\text{CH}_2=\text{C}=\text{CH}_2$
 (b) Alkenyne $\text{HC}\equiv\text{C}-\text{CH}=\text{CH}_2$

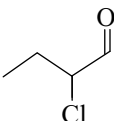
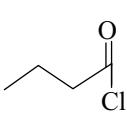


44. Consider the given statements

(I) Acetone and acetaldehyde are homologues.

(II)  is secondary amine while  is a tertiary amine.

(III)  is a secondary alcohol while  is a tertiary alcohol.

(IV)  and  both have different functional groups

Which is not incorrect?

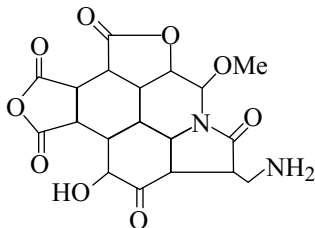
(a) I, II, III

(b) II, III, IV

(c) I, II, IV

(d) I, III, IV

45. Number of functional groups present in the given compound is



(a) 5

(b) 6

(c) 7

(d) 8

46. Which of the following compounds has isopropyl group?

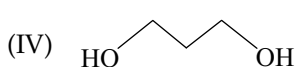
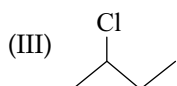
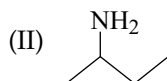
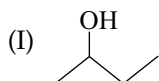
(a) 2,2,3,3-Tetramethyl pentane

(b) 2,2-Dimethyl pentane

(c) 2,2,3-Trimethyl pentane

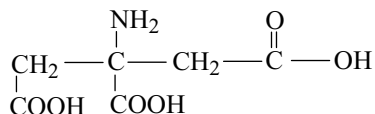
(d) 2-Methyl pentane

47. What is the order of solubility in water of the following compounds?



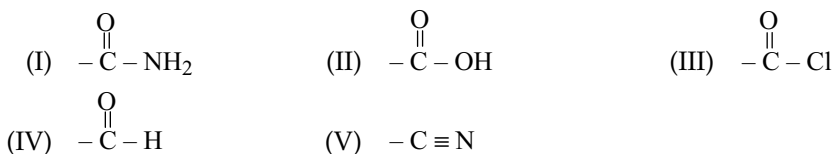
- (a) I > II > III > IV
 (b) III > IV > I > II
 (c) IV > I > III > II
 (d) IV > I > II > III

48. The correct IUPAC name of the given compound is



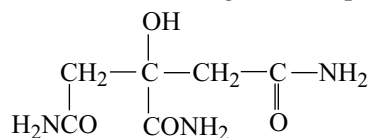
- (a) 3-Carboxy-3-hydroxy butane dioic acid
 (b) 2-Amino propane-1,2,3-tricarboxylic acid
 (c) 3-Amino butanedioic acid
 (d) 2-Bis(carboxymethyl)-2-hydroxy ethanoic acid
49. Total number of isomers of group C_5H_{11} is
 (a) 5 (b) 3 (c) 8 (d) 9
50. Write the correct name from the incorrect name 2,3-, dichloro-2,4-dibromohept-4-ene
 (a) 2, 4-dibromo-2,3-dichloro hept-4-ene
 (b) 4, 6-dibromo-5,6-dichloro hept-3-ene
 (c) 4, 6-dibromo-5,6-dichloro hept-4-ene
 (d) 5, 6-dichloro-4,6-dibromo hept-3-ene

51. In IUPAC nomenclature of compounds

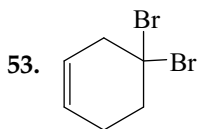


The priority order of group is/are

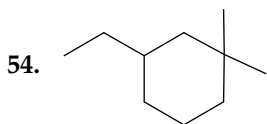
- (a) II > I > III > IV > V (b) II > III > I > V > IV
 (c) II > III > V > I > IV (d) II > III > I > IV > V
52. The correct systematic IUPAC name of the given compound is



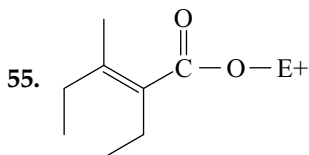
- (a) 3-Carbamoyl-3-hydroxybutanediamide
 (b) 2-Hydroxypropane-1,2,3-tricarbamoyl
 (c) 2-Hydroxypropane-1,2,3-tricarboxamide
 (d) 2-Bis(carbamoyl)-2-hydroxyethanamide



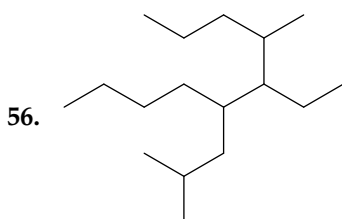
- (a) trans-1,1-dibromocyclohexene
- (b) 1,1-dibromocyclohex-3-ene
- (c) 3,3-dibromocyclohexene
- (d) 4,4-dibromocyclohexene



- (a) 1-Ethyl-3,3-dimethylcyclohexane
- (b) 3,3-Dimethyl-1-ethylcyclohexane
- (c) 3-Ethyl-1,1-dimethylcyclohexane
- (d) 1,1-Dimethyl-3-ethylcyclohexane

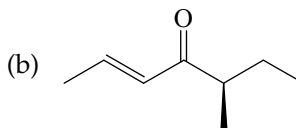
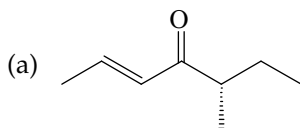


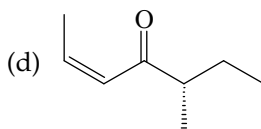
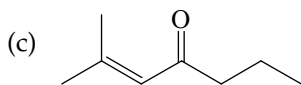
- (a) ethyl 2-ethyl-3-methyl-2-butenoate
- (b) O,2-diethyl-3-methyl-2-butenoate
- (c) 1-ethoxy-2-ethyl-3-methyl-2-butenal
- (d) 2-methyl-2-pentenyl propanoate



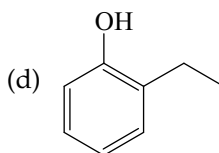
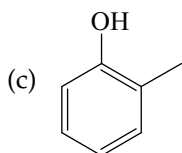
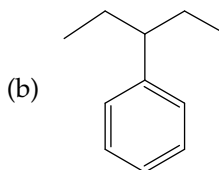
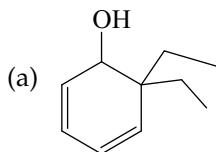
- (a) 4-Isobutyl-3-pentyl-2-methyl-2-octane
- (b) 5-Ethyl-4-methyl-6-(2-methylpropyl)decane
- (c) 6-Ethyl-7-methyl-5-(methylpropyl)decane
- (d) 4-Butyl-5-ethyl-2,6-dimethylnonane

57. (2E, 5R)-5-methylhept-2-en-4-one

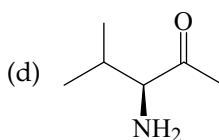
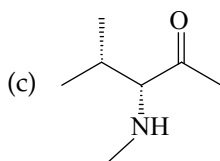
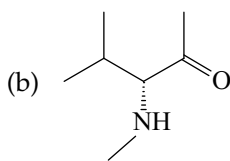
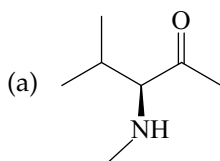




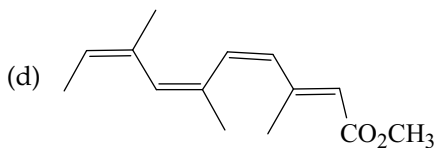
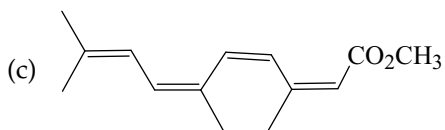
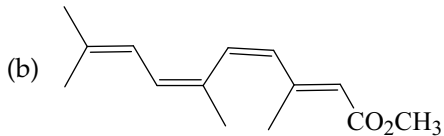
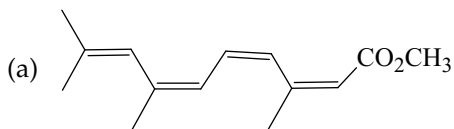
58. 2-Ethylphenol

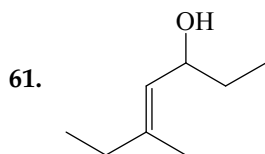


59. (S)-(N,4)-dimethyl-3-aminopentan-2-one

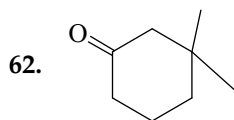


60. Methyl (2Z,4Z,6Z)-3,6,9-trimethyl-2,4,6,8-decatetraenoate

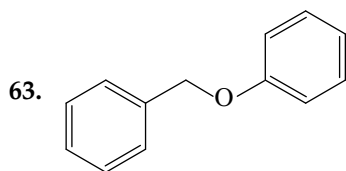




- (a) (E)-5-methyl-3-heptanal
(b) (Z)-3-methyl-3-hepten-5-al
(c) (E)-5-methyl-4-hepten-3-ol
(d) (Z)-5-methyl-4-hepten-3-ol

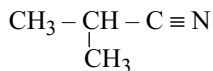


- (a) 5,5-Dimethylcyclohexanol
(b) 3,3-Dimethylcyclohexanone
(c) 1,1-Dimethylcyclohexanone
(d) 3,3-Dimethylcyclohexanal



- (a) Dibenzyl ether
(b) Benzyl phenyl ether
(c) Benzyl phenyl ester
(d) 3,3-Dibromocyclohexene

64. Which name is correct for the following compound?

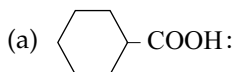


- (a) 2-Methyl propanenitrile
(b) Isobutyl cyanide
(c) Isobutyronitrile
(d) Isopropyl cyanide

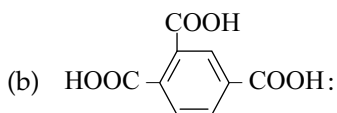
65. IUPAC name of which of the following compounds is correct?

Compound

IUPAC name



Cyclohexanoic acid



1,2,4-Benzene tricarboxylic acid



66. Name(s) of $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CN}$ is/are

- (a) Isopropyl cyanide
- (b) Sec-propyl cyanide
- (c) Isobutyro nitrile
- (d) 2-methyl propane nitrile

67. In Lassaigne's test, the organic compound is at first fused with sodium metal. The sodium metal is used because

- (a) The melting point of sodium is low; so, it is easily fused with organic substances.
- (b) Sodium is very much effective to bring about destructive reductions of organic compounds forming ionic inorganic salts such as NaCN , Na_2S and NaX .
- (c) All sodium salts are soluble in water.
- (d) None of these.

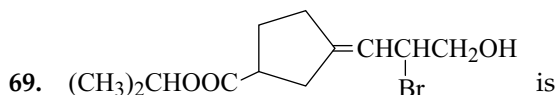
68. Chromatographic technique can be used for separation of

- (a) Volatile solids
- (b) Amino acids
- (c) Plant pigments
- (d) Sugars

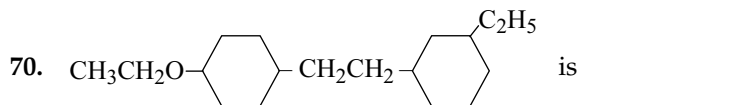
Comprehension Type

Passage 1

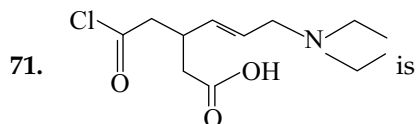
The IUPAC has set guidelines for logical and methodical naming of organic compounds. The complex substituents are written in brackets and their numbering is done separately. The bivalent radicals are named by adding 'idene' to the name of alkyl group. In polyfunctional compounds, all lower priority groups are written as a prefix. Now, name the following compounds.



- (a) 3-(3'-isopropoxycarbonyl cyclopentylidene) propane-1-ol
- (b) 3-(2'-bromo-3'-hydroxypropylidene) cyclopentane carboxylate
- (c) Iso-propyl-3-(2'-bromo-3'-hydroxypropylidenyl) cyclopentane carboxylate
- (d) Iso-propyl-3-(2'-bromo-3'-hydroxypropylidene) cyclopentane carboxylate



- (a) 2-(3'-Ethylcyclohexyl)-1-(4'-ethoxycyclohexyl) ethane
 (b) 1-Ethyl-3-(2'-(4''-ethoxycyclohexyl) ethyl) benzene
 (c) 1-(3'-Ethylcyclohexyl)-2-(4'-ethoxycyclohexyl) ethane
 (d) None of these



- (a) 3-Chlorocarbonyl-6-(N,N-diethylamino) hex-4-ene-1-oic acid
 (b) 4-Chlorocarbonyl-3-(N,N-diethylamino) butanoic acid
 (c) 3-Chlorocarbonyl-3-(3-N,N-diethylamino prop-1'-enyl) butane-1-oic acid
 (d) 3-Chlorocarbonyl methyl-6-(N,N-diethylamino) hex-4-en-1-oic acid

Passage 2

Rules for IUPAC nomenclature of compounds containing one functional group, multiple bonds and substituents are given hereunder.

- (a) Select the longest possible chain of carbon atoms containing the functional group and the maximum number of multiple bonds as the parent chain without caring whether it also denotes the longest possible carbon chain or not.
 (b) After selecting the parent chain, the numbering should be done in such a way that it gives lower possible number the functional group.
 (c) When a chain containing functional groups such as $-\text{CHO}$, $-\text{COOH}$, COOR , $-\text{COCl}$, etc. is present, it is always given number 1 and number 1 is usually omitted from the final name of the compound.
 (d) If the organic compound contains a functional group, multiple bond, side chain or substituent, the following order of preference should be followed.

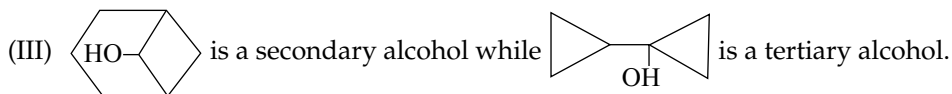
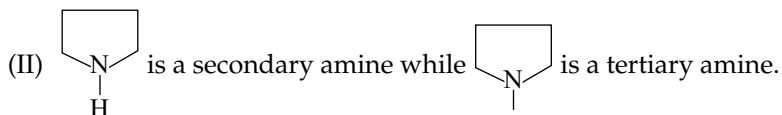
Functional group > Double bond > Triple bond > Substituent

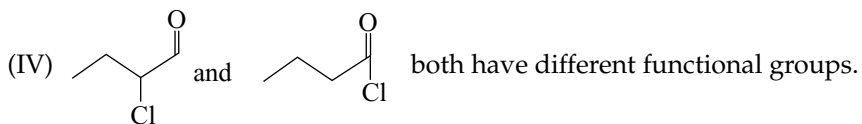
- (e) If a compound contains two or more like groups, the numerical prefixes di, tri, tetra, etc. are used and terminal 'e' from the primary suffix is retained.

On the basis of the above rules, give answers for the following questions:

72. Consider the given statements

- (I) Acetone and acetaldehyde are homologues.





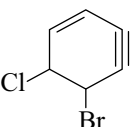
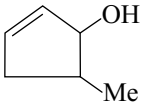
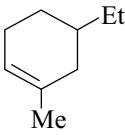
Which is not incorrect?

- (a) I, II, III (b) II, III, IV
(c) I, II, IV (d) I, III, IV

73. Correct IUPAC name from the incorrect name 4-amino-3-hydroxy-2-butene is

- (a) 1-Amino-2-hydroxy-2-butene
(b) 4-Amino-2-buten-3-ol
(c) 1-Amino-2-buten-2-ol
(d) 1-Amino-2-butenol

74. Which of the following has incorrect IUPAC name?

- (a)  5-Bromo-6-chloro cyclohex-1-en-3-yne
- (b)  2-Methyl cyclopent-4-en-1-ol
- (c)  5-Ethyl-1-methyl cyclohexene
- (d)  2,4-Dimethyl hexane

Passage 3

Rules for IUPAC nomenclature of compounds containing one functional group, multiple bonds and substituents are given hereunder.

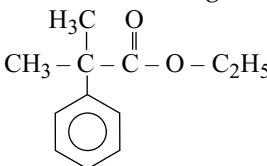
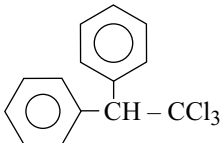
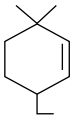
- Select the longest possible chain of carbon atoms containing the functional group and the maximum number of multiple bonds as the parent chain without caring whether it also denotes the longest possible carbon chain or not.
- After selecting the parent chain, the numbering should be done in such a way that it gives lower possible number for the functional group.
- When a chain containing functional groups such as $-\text{CHO}$, $-\text{COOH}$, COOR , $-\text{COCl}$, etc. is present, it is always given number 1 and number 1 is usually omitted from the final name of the compound.
- If the organic compound contains a functional group, multiple bond, side chain or substituent, the following order of preference should be followed.

Functional group > Multiple bond > Substituent

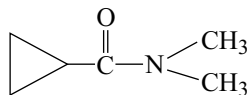
- (e) If a compound contains two or more like groups, the numerical prefixes di, tri, tetra, etc. are used and terminal 'e' from the primary suffix is retained.

On the basis of the above rules, give answers for the following questions:

75. Which of the following is not correctly matched?

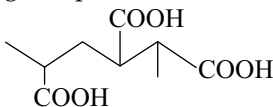
- (a)  Ethyl-2-methyl-2-phenyl propanoate
- (b)  1,1,1-Trichloro-2,2-diphenyl ethane
- (c)  6-Ethyl-3,3-dimethyl cyclohex-1-ene
- (d) $\text{H}_2\text{N}-\underset{\text{CN}}{\text{CH}}-\underset{\text{CN}}{\text{CH}}-\text{CHO}$ 2-Formyl-3-amino-1,4-butane dinitrile

76. IUPAC name of the following compound is



- (a) N,N-Dimethyl cyclopropanamide
 (b) N-Methyl cyclopropanamide
 (c) N,N-Dimethyl cyclopropane carboxamide
 (d) N,N-Dimethyl cyclopropolonamide

77. IUPAC name of the following compound is

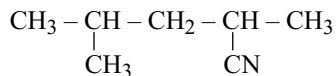


- (a) 2-(1-carboxyethyl)-4-methyl pentanoic acid
 (b) 3,5-Dicarboxy-2-methyl hexanoic acid
 (c) 2,4,5-Hexane tricarboxylic acid
 (d) 2,3,5-Hexane tricarboxylic acid

Passage 4

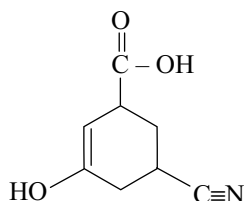
IUPAC system is the most rational and widely used system of nomenclature in organic chemistry. Any given molecule has only one IUPAC name and any given IUPAC name denotes only one molecular structure. Numbering the principal chain order is principal functional group > multiple bond > substituents.

78. IUPAC name of



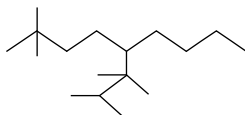
- (a) 2-Cyano-4-methylpentane (b) 2-Methyl-4-cyanopentane
(c) 2,4-Dimethyl pentane nitrile (d) 2-Dimethyl-4-cyanopentane

79. The IUPAC name of the following compound is



- (a) 4-Hydroxy-3-cyano cyclohex-5-ene carboxylic acid
(b) 3-Hydroxy-5-cyano cyclohex-5-enoic acid
(c) 5-Cyano-3-hydroxy cyclohex-2-ene carboxylic acid
(d) 3-Carboxy-5-hydroxycyclohex-5-ene carbonitrile

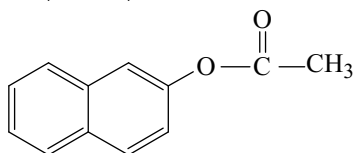
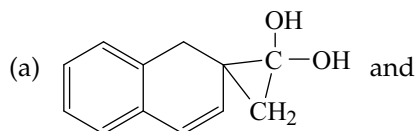
80. The correct IUPAC name of the following compound is



- (a) 5-(1',1'2'-trimethyl propyl)-2-2-dimethyl nonane
(b) 4-Butyl-2,3,3,7,7-pentamethylnonane
(c) 2,3,3,7, 7-Pentamethyl-4-butyloctane
(d) 2,2-Dimethyl-5-(1',1',2' trimethylpropyl)nonane

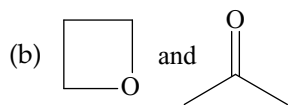
Matrix Type

81. Column I

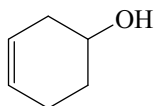
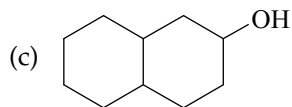


Column II

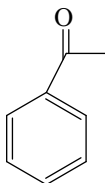
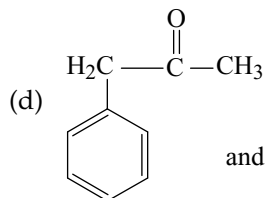
- (p) Functional groups are same pair of compound



(q) DBE is same



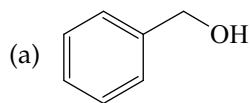
(r) Functional groups are different in pair of compound



(s) Even number of 2° carbon 'C' atom present in both compounds

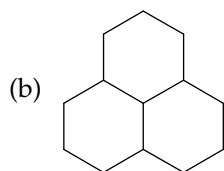
(t) Odd number of 2° 'C' atom present in pair of compound in both compounds

82. Column I

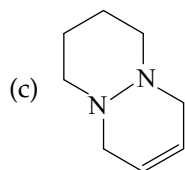


Column II

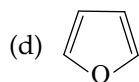
(p) Odd number of 3° 'C'



(q) Odd DBE (double bond equivalents)



(r) Even DBE (double bond equivalents)

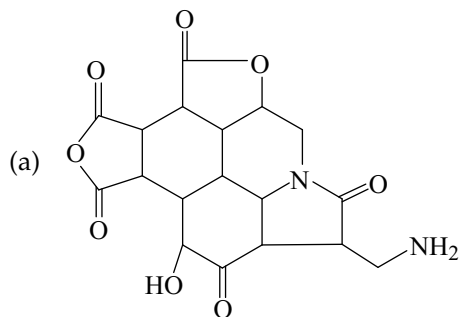


(s) Heterocyclic ring is present

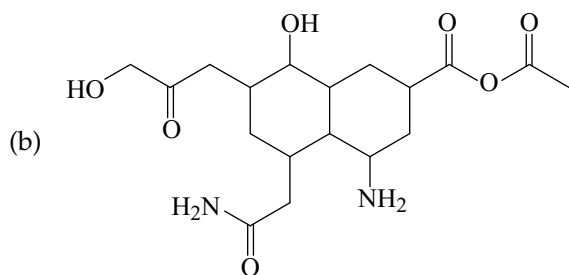
(t) Even number of 2° 'C' atom

83. Column I

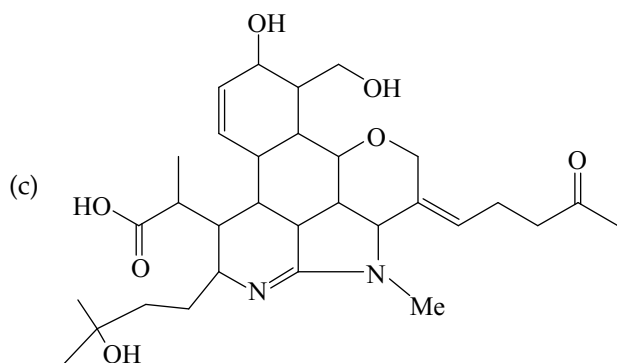
Column II



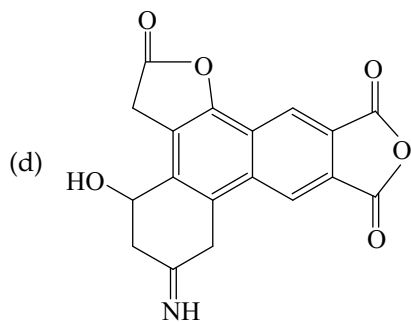
(p) DU/DBE is odd



(q) DBE/DU is even



(r) Functional groups are odd



(s) Functional groups are even

(t) Number of 2° 'C' even

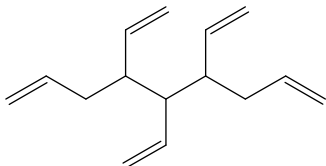
(u) Lactone is present as functional group

84. Column I

(a) Cyclohexa-1, 3-diene



(b) 4, 5, 6-Triethenlnona-1, 8-diene



(c) Cyclo octa-1, 3, 5-triene

(d) 1, 3, 5-trimethyl benzene

Column II

(p) DU is odd

(q) DU is even

(r) Compound has 1 : 1 : 1 ratio for 1°, 2° and 3° 'C' atoms

(s) Unsaturated compound

[illegible]